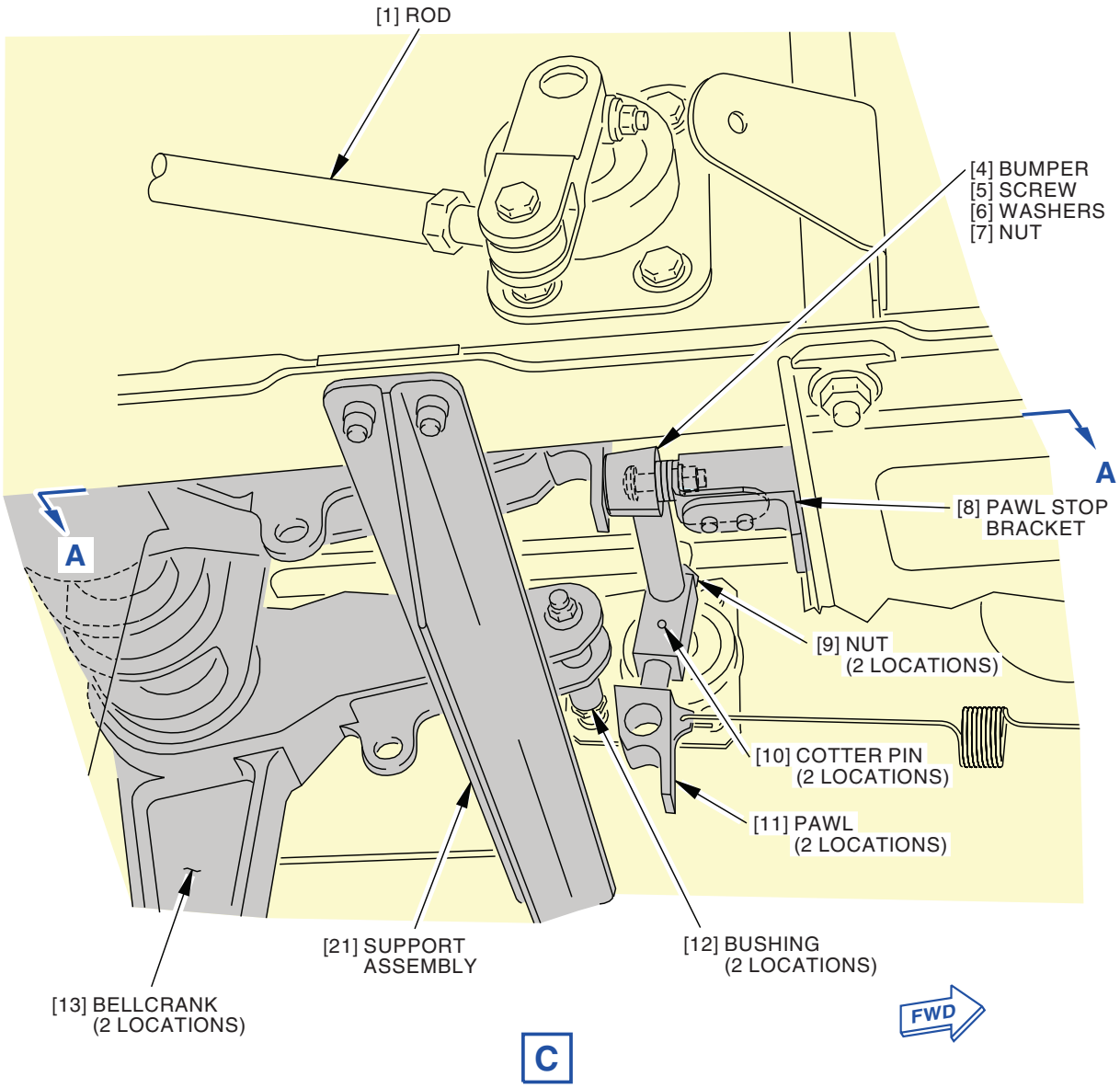
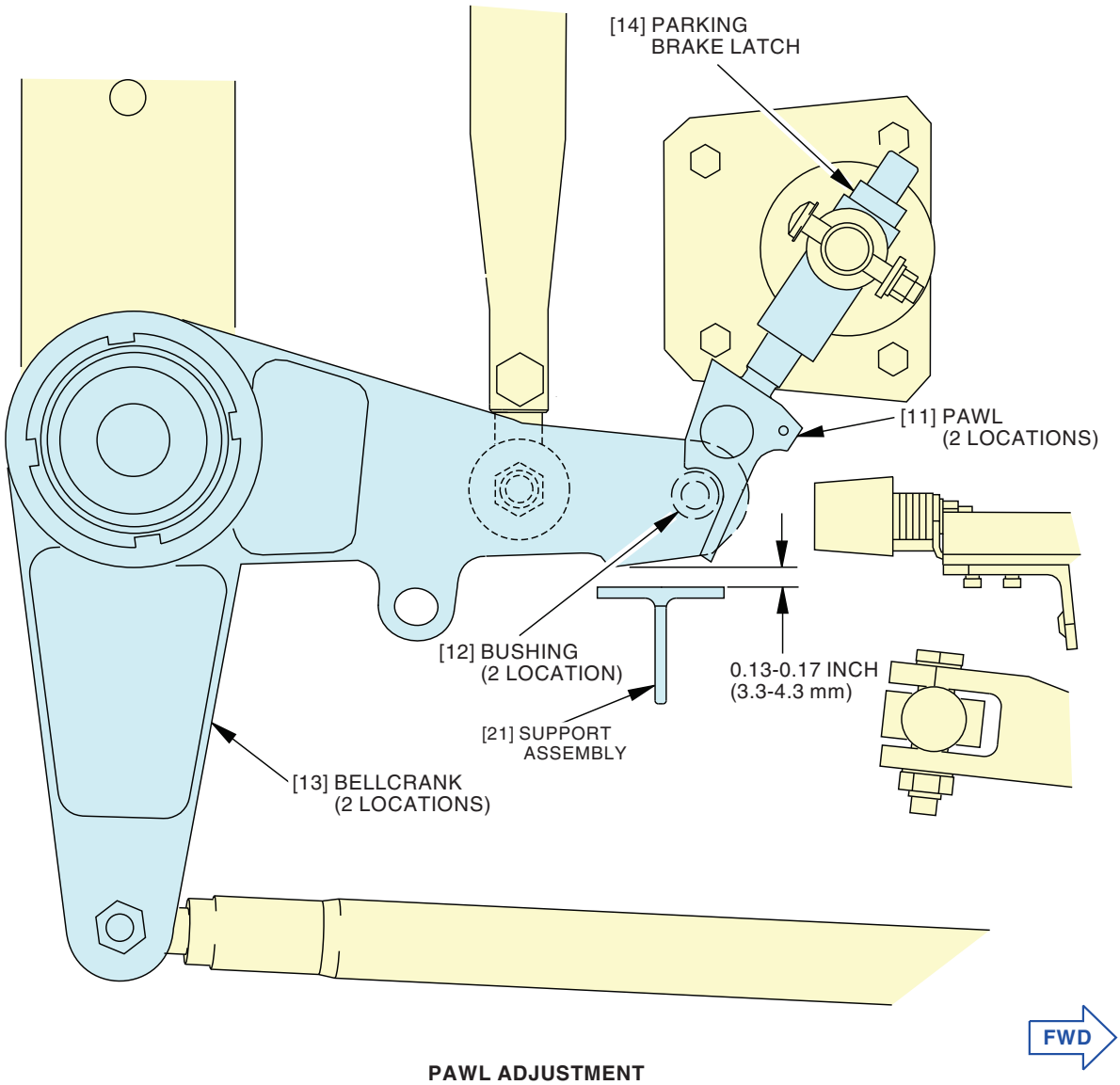


2414416 S00061535137_V2

Parking Brake System Adjustment
Figure 501/32-44-00-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



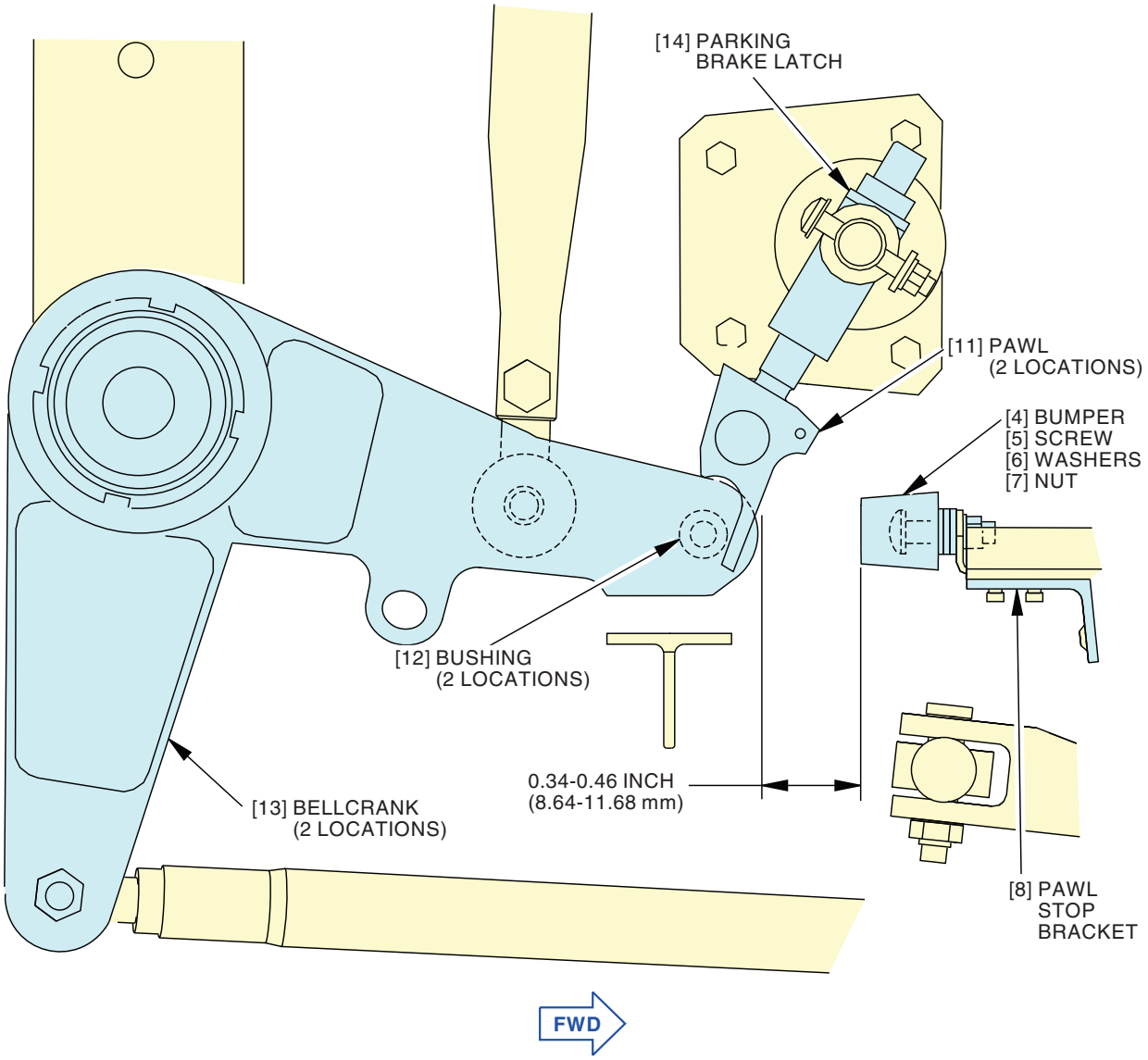
PAWL ADJUSTMENT
(PARKING BRAKE ''ON'' POSITION)

A-A

2414417 S00061535138_V2

Parking Brake System Adjustment
Figure 501/32-44-00-990-801 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



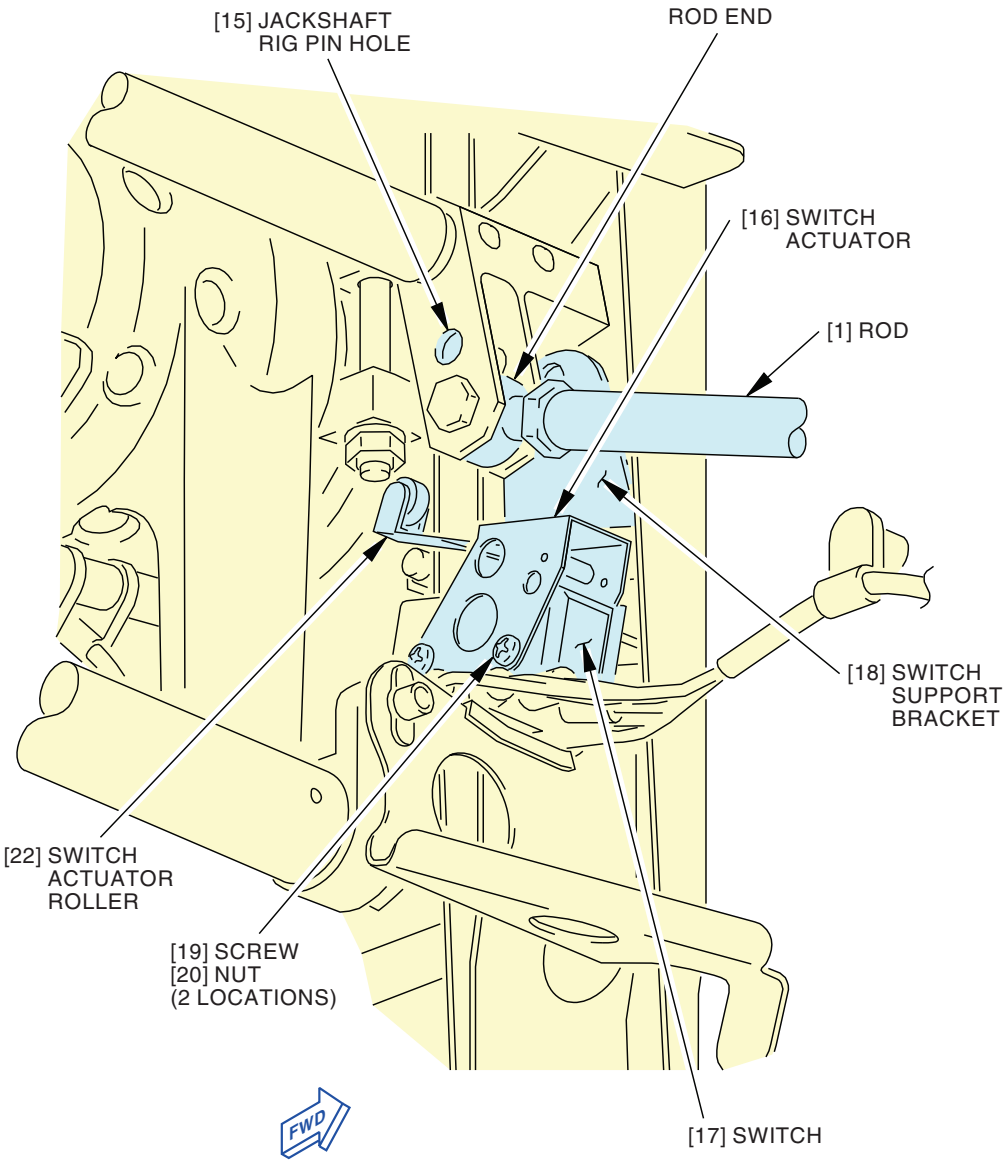
PAWL ADJUSTMENT
(PAWLS CONTACT BUSHINGS)

A-A

2414418 S00061535139_V2

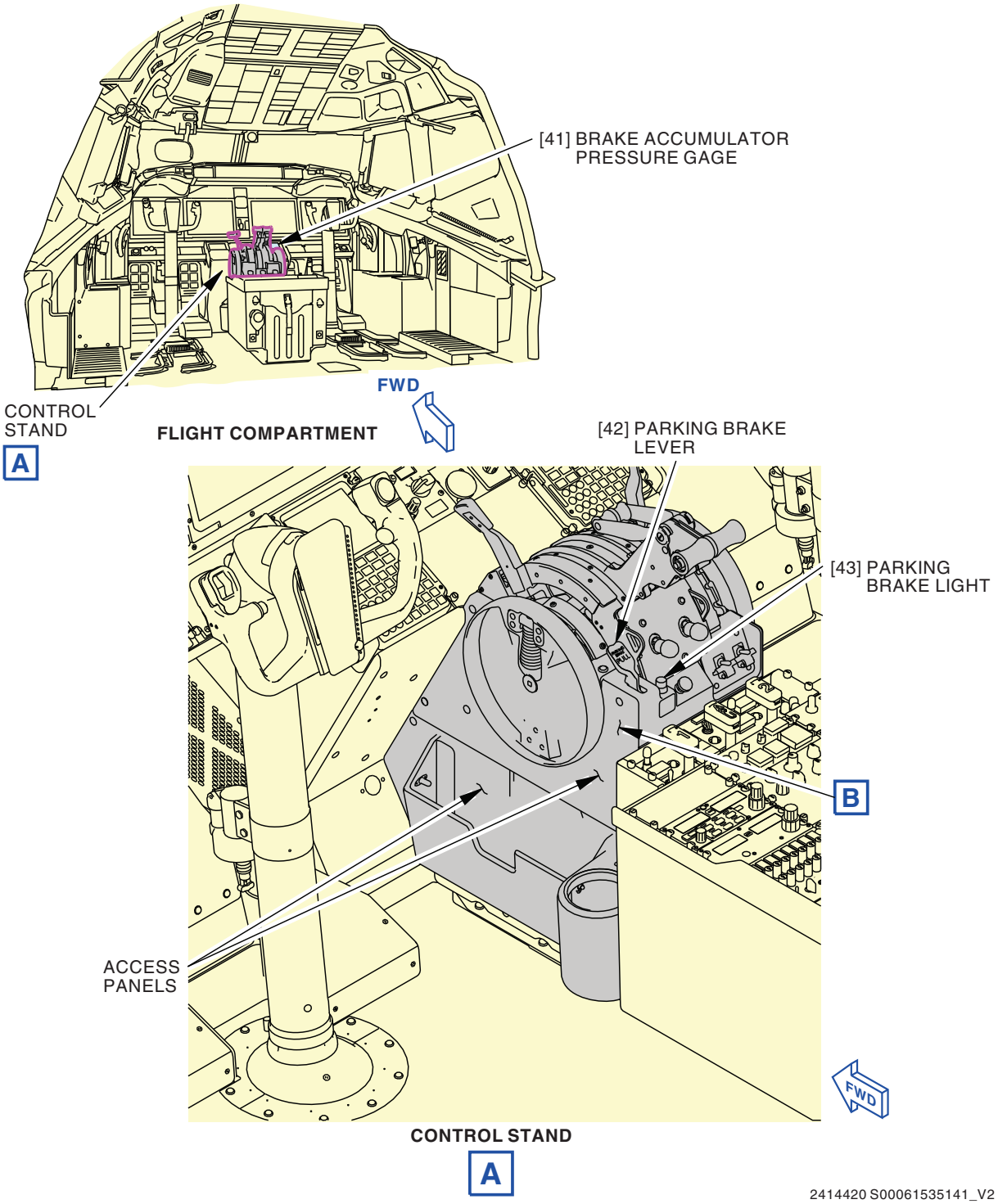
Parking Brake System Adjustment
Figure 501/32-44-00-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



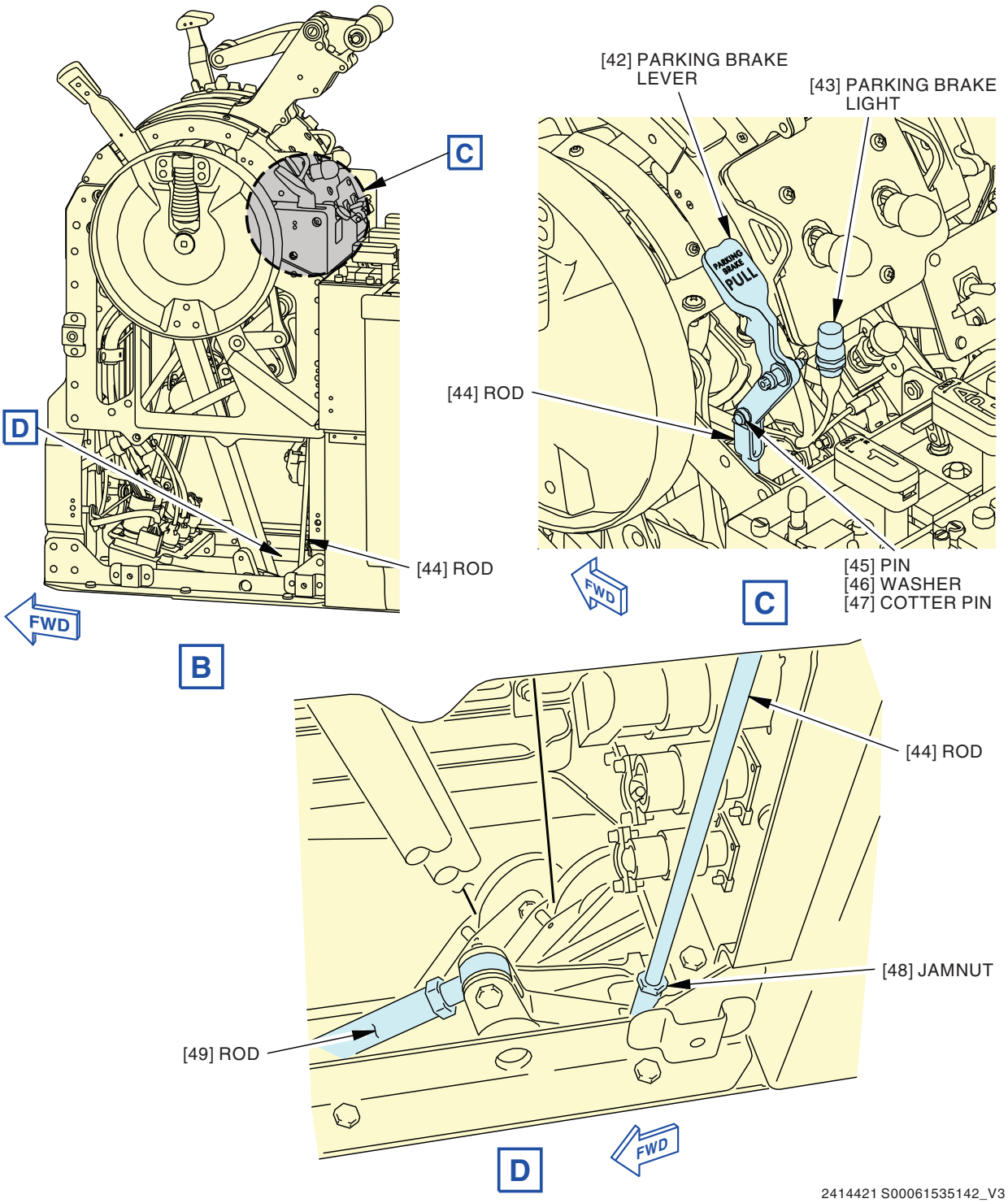
Parking Brake System Adjustment
Figure 501/32-44-00-990-801 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Parking Brake Lever Adjustment
Figure 502/32-44-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Parking Brake Lever Adjustment
Figure 502/32-44-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-00-710-801

3. Parking Brake System Operational Test

(Figure 501, Figure 502)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Test

SUBTASK 32-44-00-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-44-00-860-005

(2) For the system A and system B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-44-00-480-005

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-00-860-006

(4) Release the parking brake.

SUBTASK 32-44-00-840-001

(5) Fully push the brake pedals 12 times to remove hydraulic pressure from the brake accumulator.

NOTE: You can look at the brake pistons to see that the brakes no longer actuate when the brake pedals are pushed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-00-480-006

(6) Install rig pins R1 and R2, from rigging pin kit, SPL-1585, through the captain's and first officer's rudder pedal shaft rig pin holes [2] and into the structure.

NOTE: The placard on the inside of the box for the rigging pin kit, SPL-1585, identifies the different rigging pins.

E. Parking Brake System Operational Test

SUBTASK 32-44-00-790-001

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) For the B hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-44-00-720-001

(2) Make sure that the pressure shown on the HYD BRAKE PRESS gage is approximately the same as the B hydraulic system pressure.

NOTE: The HYD BRAKE PRESS gage is in the flight compartment on the main instrument panel for the first officer.

SUBTASK 32-44-00-710-001

(3) Do these steps for the captain's brake pedals:

(a) Push the two brake pedals fully to their stops and set the parking brake.

(b) Release the brake pedals.

(c) Make sure that the parking brake engages.

(d) Make sure that the parking brake light [43] on the control stand is on.

(e) Look at the pistons on each brake to see that the brakes are engaged.

(f) Release the parking brake.

(g) Make sure that the parking brake light [43] on the control stand is off.

(h) Look at the pistons on each brake to see that the brakes are not engaged.

(i) For the B hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(j) Fully push the captain's brake pedals twelve times to remove the pressure from the brake accumulator.

(k) Push the two brake pedals fully to their stops and set the parking brake.

(l) Release the brake pedals.

(m) Make sure that the parking brake engages.

(n) Make sure that the parking brake light [43] on the control stand is on.

(o) Look at the pistons on each brake to see that the brakes are not engaged.

(p) Release the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(q) For the B hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(r) Push the left brake pedal fully to the stop.

(s) Pull the parking brake lever [42] on the control stand.

(t) Make sure that the parking brake will not engage.

(u) Make sure that the parking brake light [43] on the control stand is off.

(v) Look at the pistons on each brake to see that the brakes on the left gear are engaged and the brakes on the right gear are not engaged.

(w) Release the left brake pedal.

(x) Push the right brake pedal fully to the stop.

(y) Pull the parking brake lever [42] on the control stand.

(z) Make sure that the parking brake will not engage.

(aa) Make sure that the parking brake light [43] on the control stand is off.

(ab) Look at the pistons on each brake to see that the brakes on the right gear are engaged and the brakes on the left gear are not engaged.

(ac) Release the right brake pedal.

SUBTASK 32-44-00-710-002

(4) Do these steps for the first officer's brake pedals:

(a) Push the two brake pedals fully to their stops and set the parking brake.

(b) Release the brake pedals.

(c) Make sure that the parking brake engages.

(d) Release the parking brake.

(e) Push the left brake pedal fully to the stop.

(f) Pull the parking brake lever [42] on the control stand.

(g) Make sure that the parking brake will not engage.

(h) Release the left brake pedal.

(i) Push the right brake pedal fully to the stop.

(j) Pull the parking brake lever [42] on the control stand.

(k) Make sure that the parking brake will not engage.

(l) Release the right brake pedal.

SUBTASK 32-44-00-080-001

(5) Remove rig pins R1 and R2, from rigging pin kit, SPL-1585, from the rudder pedal shaft rig pin hole [2].

SUBTASK 32-44-00-720-002

(6) Do this task: Parking Brake Hold Check, TASK 32-44-00-790-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-00-790-801

4. Parking Brake Hold Check

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task includes:

(a) Parking brake system bleed down test - Parking brake set

(b) Parking brake system bleed down test - Parking brake released.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Check

SUBTASK 32-44-00-480-010

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-44-00-480-008

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-00-860-007

(3) Release the parking brake.

SUBTASK 32-44-00-860-008

(4) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-44-00-870-001

(5) Push the Captain's or the First Officer's left and right toe brake pedals to the stops several times until the brakes no longer actuate.

NOTE: This will release the hydraulic pressure at the accumulator.

(a) You can look at the pistons on each brake to see that the brakes no longer actuate when the brake pedals are pushed.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-00-860-009

(6) Make sure that the brake accumulator Nitrogen charge is correct.

NOTE: You can find the correct charge pressure from the accumulator placard located on the rear bulkhead in the right main gear wheel well.

SUBTASK 32-44-00-790-002

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) To pressurize B hydraulic system to 3000 ±150 psig (20,684 ±1034 kPa), do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-44-00-860-010

(8) Set the parking brake.

NOTE: This step keeps the brake system pressurized with the accumulator.

SUBTASK 32-44-00-720-003

(9) Wait at least 10 minutes for the accumulator temperature to stabilize.

E. Parking Brake Hold Check - Parking Brake Set

SUBTASK 32-44-00-720-004

(1) Write down the pressure shown on the PRESS gage for the brake accumulator.

NOTE: The PRESS gage is located on the rear bulkhead in the right main gear wheel well.

SUBTASK 32-44-00-860-011

(2) For the B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

NOTE: This will leave the accumulator charged and pressurized.

(a) Record the initial time when the B hydraulic system pressure is removed.

SUBTASK 32-44-00-710-003

(3) Do a check of the brake pressure at the end of any of the time periods that follow (if the necessary pressure drop for one time period is not within the limits, you can use the next time period):

NOTE: It is permitted to read the brake pressures from the brake pressure gage in the cockpit, or the brake accumulator gage in the wheel well.

(a) After 10 minutes, the pressure drop should be less than 100 psi (689 kPa).

(b) After 30 minutes, the pressure drop should be less than 300 psi (2068 kPa).

(c) After 1 hour, the pressure drop should be less than 500 psi (3447 kPa).

(d) After 8 hours, the pressure should be more than 1200 psi (8274 kPa) minimum.

SUBTASK 32-44-00-780-001

(4) If the brake accumulator pressure drops faster than permitted, do a check for leaks in the accumulator system.

SUBTASK 32-44-00-780-002

(5) The troubleshooting of the brake system can be delayed until the next convenient maintenance period if the conditions that follow are met:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) With the parking brake set, the brake accumulator pressure drop must be less than 800 psi (5516 kPa) in the first 30 minutes.

(b) The system passes the Parking Brake Hold Check - Parking Brake Released.

SUBTASK 32-44-00-860-012

(6) Release the parking brake.

SUBTASK 32-44-00-720-005

(7) Push the Captain's or First Officer's left and right toe brake pedals to the stops 7 times while you monitor the brake pistons on all of the brakes.

NOTE: Wait approximately 3 seconds each time you apply the brakes.

(a) Make sure that the brake pistons compress and release each of the 7 times the pedals are pushed.

F. Parking Brake Hold Check - Parking Brake Released

SUBTASK 32-44-00-790-003

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Pressurize B hydraulic system to 3000 ±150 psig (20,684 ±1034 kPa), do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-44-00-720-006

(2) Wait at least 10 minutes for the accumulator temperature to stabilize.

SUBTASK 32-44-00-720-007

(3) Write down the pressure shown on the PRESS gage for the brake accumulator.

NOTE: The PRESS gage is located on the rear bulkhead in the right main gear wheel well.

SUBTASK 32-44-00-860-013

(4) For the B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

NOTE: This will leave the accumulator charged and pressurized.

(a) Record the initial time when the B hydraulic system pressure is removed.

SUBTASK 32-44-00-780-003

(5) Do a check of the brake pressure at 30 minutes after the test start.

(a) Make sure that the pressure decrease of the brake accumulator is less than 450 psi (3103 kPa).

SUBTASK 32-44-00-780-004

(6) If the brake accumulator pressure drops faster than permitted, do a check of the accumulator system for leaks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-44-00-790-004

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Pressurize B hydraulic system to 3000 ±150 psig (20,684 ±1034 kPa), do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-44-00-720-008

(2) Set the parking brake.

SUBTASK 32-44-00-860-014

(3) For the B hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PARKING BRAKE LINKAGE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the parking brake lever.

(2) Installation of the parking brake lever.

(3) Removal of the parking brake jackshaft.

(4) Installation of the parking brake jackshaft.

(5) Removal of the parking brake latch.

(6) Installation of the parking brake latch.

(7) Removal of the parking brake latch switch.

(8) Installation of the parking brake latch switch.

(9) Visual inspection of the parking brake mechanism.

TASK 32-44-11-000-801

2. Parking Brake Lever Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left

C. Prepare for the Removal

SUBTASK 32-44-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-44-11-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-11-860-001

(3) Release the parking brake.

SUBTASK 32-44-11-860-019

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-11-860-002

WARNING

STAY AWAY FROM THE STABILIZER TRIM WHEEL ON THE CONTROL STAND IF YOU USE THE STABILIZER TRIM SWITCH ON THE CONTROL WHEEL. THE STABILIZER TRIM SWITCH CAN CAUSE THE STABILIZER TRIM WHEEL TO TURN QUICKLY. INJURY TO PERSONS CAN OCCUR IF YOU TOUCH IT WHEN IT TURNS.

(5) Open the access panels on the left side of the control stand to get access to the parking brake linkage.

D. Parking Brake Lever Removal

SUBTASK 32-44-11-020-001

(1) Remove the cotter pin [5], washer [4], and pin [3] which hold the parking brake lever [1] to the rod [6].

SUBTASK 32-44-11-020-002

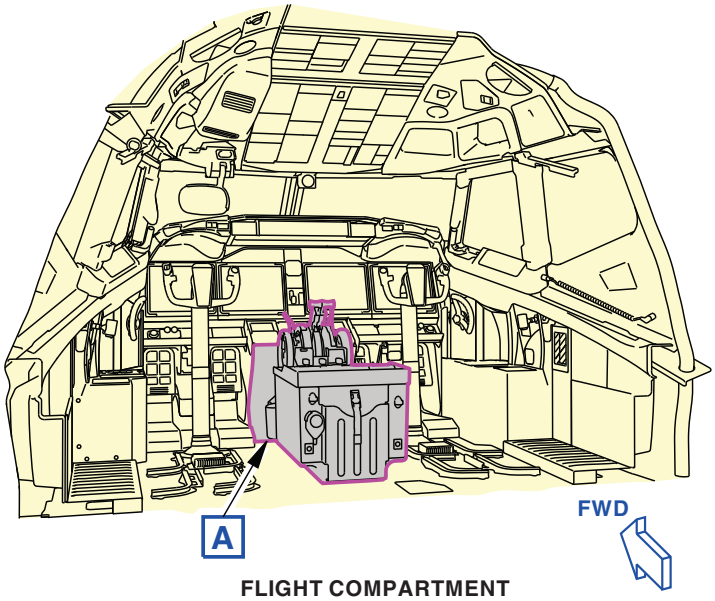
(2) Remove the bolt [8], washer [9] and bearing [10] which hold the parking brake lever [1] to the control stand.

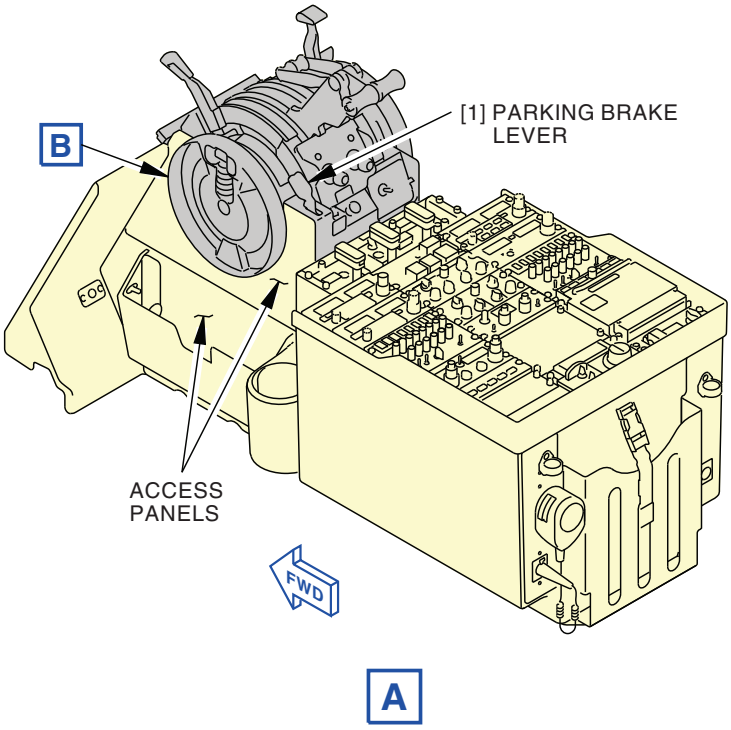
SUBTASK 32-44-11-020-003

(3) Remove the parking brake lever [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

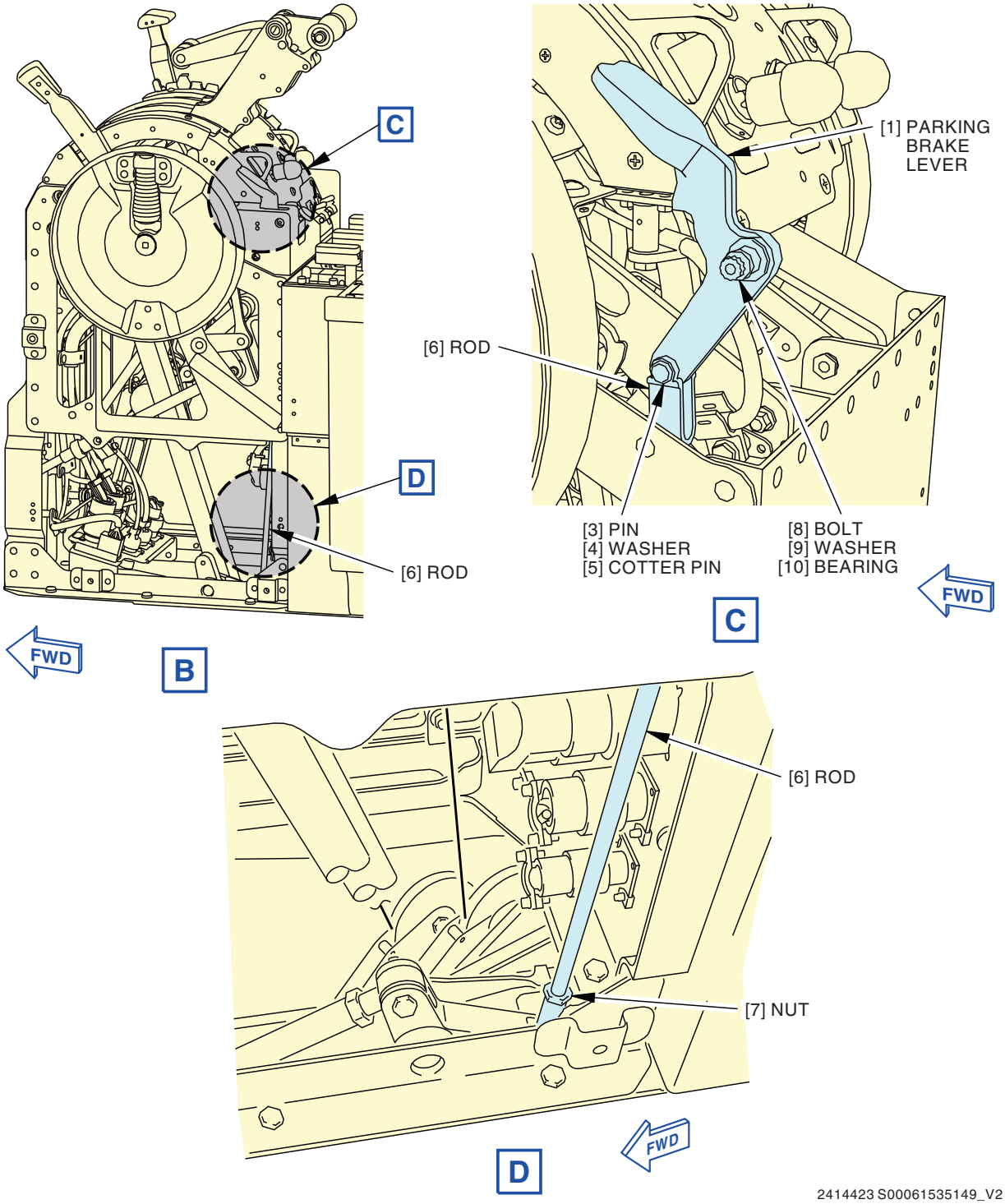




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Parking Brake Lever Installation
Figure 401/32-44-11-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Parking Brake Lever Installation
Figure 401/32-44-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-11-400-801

3. Parking Brake Lever Installation

(Figure 401)

A. References

Reference	Title
32-44-00-820-801	Parking Brake System Adjustment (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Parking brake lever	32-44-11-05-070	PTF ALL
5	Cotter pin	32-44-11-05-005	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Parking Brake Lever Installation

SUBTASK 32-44-11-420-001

(1) Install the parking brake lever [1] on the control stand.

SUBTASK 32-44-11-420-002

(2) Install the bearing [10], washer [9] and bolt [8] which hold the parking brake lever [1] to the control stand.

SUBTASK 32-44-11-020-004

(3) Install the pin [3] to hold the parking brake lever [1] to the rod [6].

SUBTASK 32-44-11-820-001

(4) If the stop surface of the parking brake lever [1] is not against the top of the control stand, adjust the length of the rod [6] (TASK 32-44-00-820-801).

SUBTASK 32-44-11-020-005

(5) Install the washer [4] and new cotter pin [5] to hold the pin [3] in the rod [6].

SUBTASK 32-44-11-410-001

WARNING

STAY AWAY FROM THE STABILIZER TRIM WHEEL ON THE CONTROL STAND IF YOU USE THE STABILIZER TRIM SWITCH ON THE CONTROL WHEEL. THE STABILIZER TRIM SWITCH CAN CAUSE THE STABILIZER TRIM WHEEL TO TURN QUICKLY. INJURY TO PERSONS CAN OCCUR IF YOU TOUCH IT WHEN IT TURNS.

(6) Close the access panels on the left side of the control stand.

SUBTASK 32-44-11-860-018

(7) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	10	C00207	FLIGHT CONTROL STAB TRIM CONT
D	10	C00840	FLIGHT CONTROL STAB TRIM ACTUATOR

SUBTASK 32-44-11-860-003

(8) Set the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the parking brake light comes on.

——— END OF TASK ———

TASK 32-44-11-000-802

4. Parking Brake Jackshaft Removal

(Figure 402)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 32-44-11-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-44-11-480-004

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-11-860-004

(3) Release the parking brake.

SUBTASK 32-44-11-860-005

(4) Open this access panel:

Number	Name/Location
112A	Forward Access Door

to get access to the parking brake jackshaft mechanism.

E. Parking Brake Jackshaft Removal

SUBTASK 32-44-11-020-006

(1) Remove the bolt [30], washer [31], and nut [32] to disconnect the rod [29] from the outboard jackshaft lever.

SUBTASK 32-44-11-020-007

(2) Remove the bolt [33], washer [34], and nut [35] to disconnect the rod [21] from the inboard jackshaft lever.

SUBTASK 32-44-11-020-008

(3) Do these steps to remove the jackshaft [28]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the cotter pin [23], nut [24], and washer [25] from the inboard end of the stud [22] that goes through the jackshaft [28].

(b) Remove the stud [22] from the other side of the jackshaft [28].

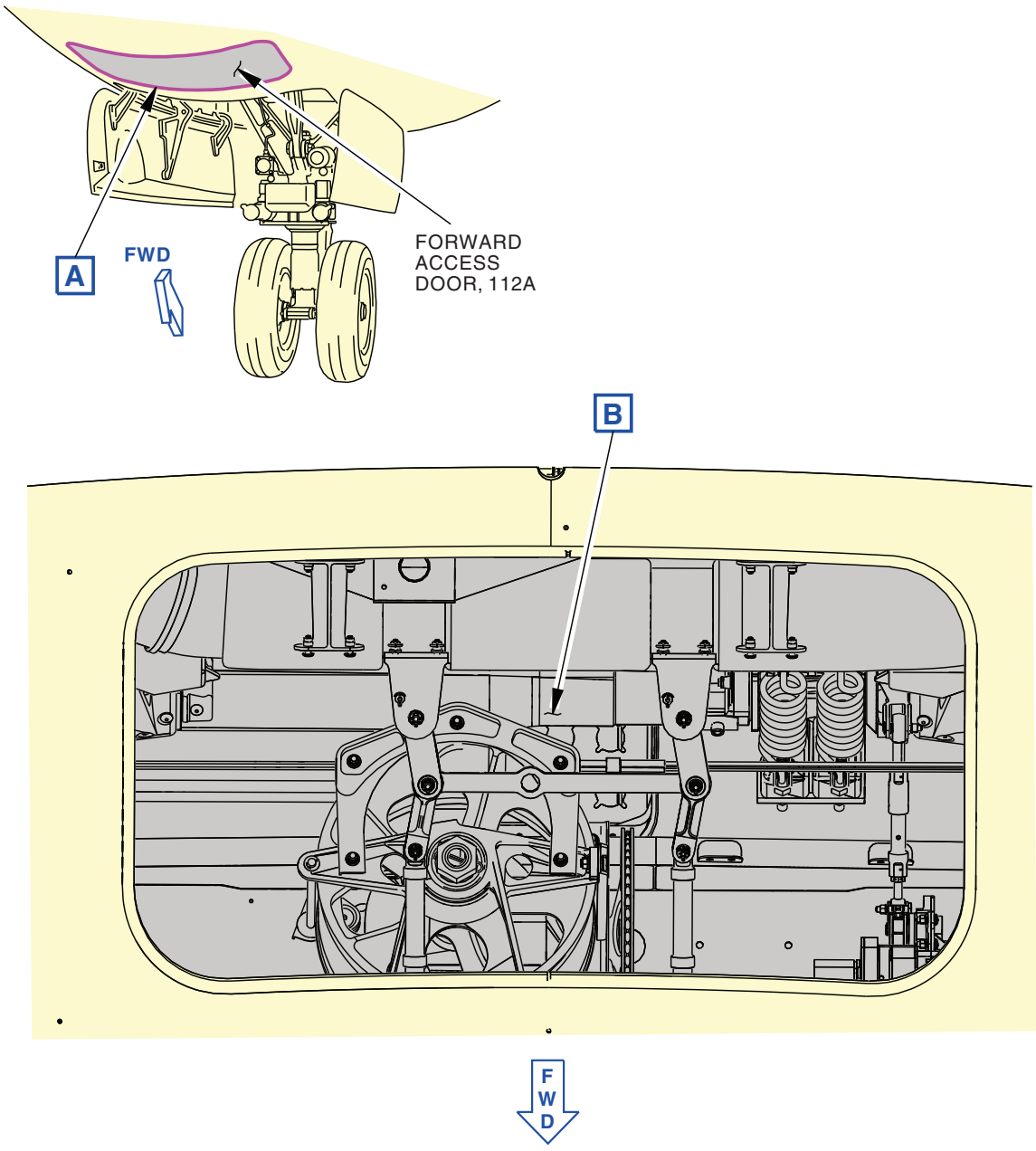
(c) Remove the jackshaft [28] from the airplane.

1) Remove the bearings [26] and the spacer [27] if it is necessary.

NOTE: One bearing is installed in each end of the jackshaft, and the spacer is installed between these bearings.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



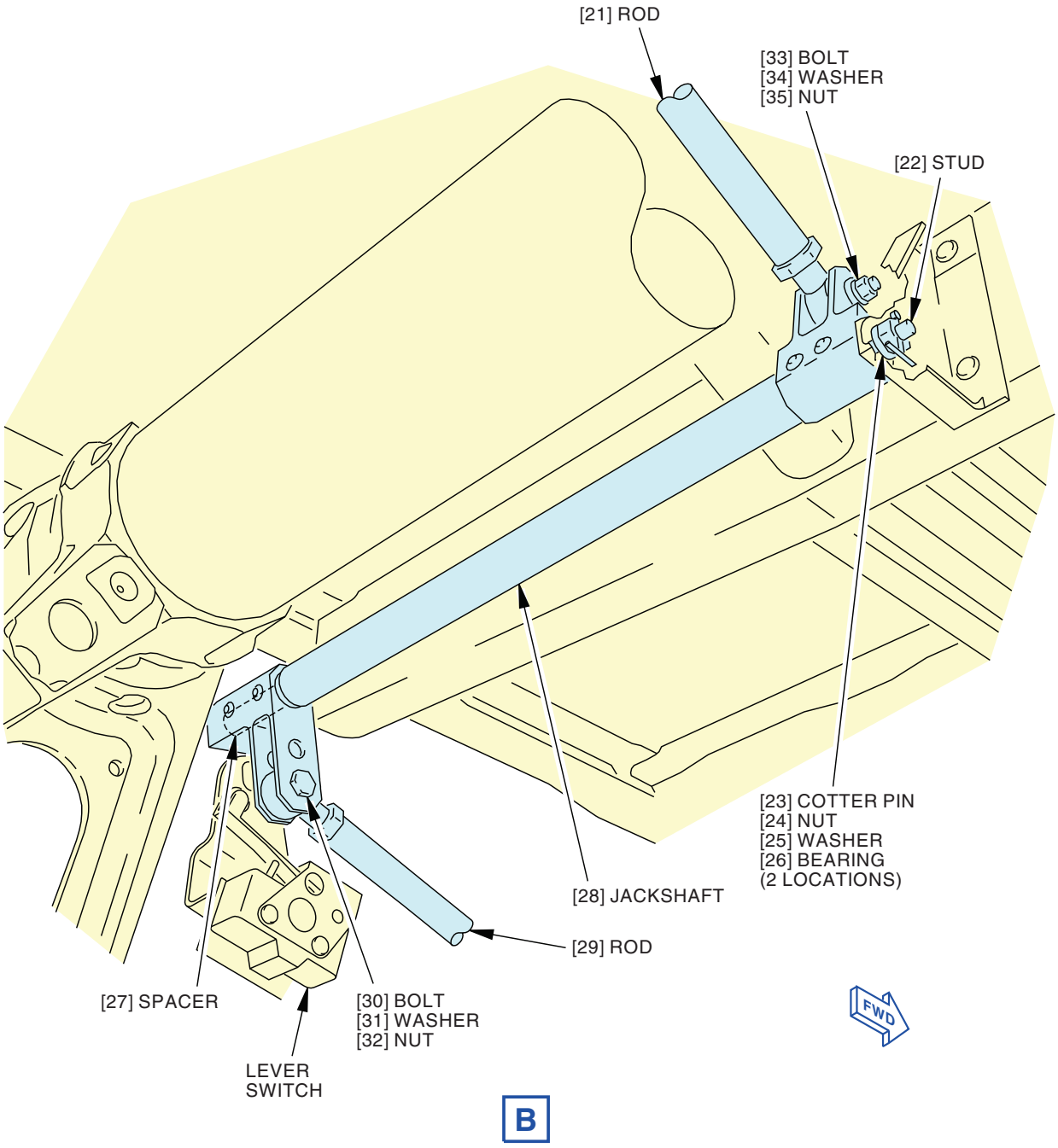
FORWARD ACCESS DOOR, 112A
(VIEW IN THE UP DIRECTION)

A

2414424 S00061535152_V1

Parking Brake Jackshaft Mechanism Installation
Figure 402/32-44-11-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Parking Brake Jackshaft Mechanism Installation
Figure 402/32-44-11-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-11-400-802

5. Parking Brake Jackshaft Installation

(Figure 402)

A. References

Reference	Title
32-44-00-820-801	Parking Brake System Adjustment (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
23	Cotter pin	32-44-11-01-115	PTF ALL
26	Bearing	32-44-11-01-140	PTF ALL
27	Spacer	32-44-11-01-145	PTF ALL
28	Jackshaft	32-44-11-01-150	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Parking Brake Jackshaft Installation

SUBTASK 32-44-11-420-003

(1) Install the new bearings [26] and new spacer [27] if they were removed.

NOTE: One bearing is installed in each end of the jackshaft, and the spacer is installed between these bearings.

SUBTASK 32-44-11-420-004

(2) Do these steps to install the jackshaft [28]:

(a) Put the jackshaft [28] in its position in the airplane.

1) Make sure that the jackshaft lever with the rig pin hole is above the lever switch.

(b) Install the stud [22] in the jackshaft [28].

(c) Install the washer [25], nut [24], and new cotter pin [23] on the end of the stud [22].

SUBTASK 32-44-11-420-005

(3) Install the bolt [30], washer [31], and nut [32] to connect the rod [29] to the outboard jackshaft lever.

SUBTASK 32-44-11-020-009

(4) Install the bolt [33], washer [34], and nut [35] to connect the rod [21] to the inboard jackshaft lever.

(a) Make sure that the end of the bolt [33] has clearance with the adjacent bracket.

SUBTASK 32-44-11-820-002

(5) If the length of any of the connecting rods were changed, do this task: Parking Brake System Adjustment, TASK 32-44-00-820-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-11-860-006

(6) Close this access panel that you opened to get access to the parking brake jackshaft mechanism:

Number	Name/Location
112A	Forward Access Door

END OF TASK

TASK 32-44-11-000-803

6. Parking Brake Latch Removal

(Figure 403)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 32-44-11-420-013

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-44-11-420-014

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-11-860-007

(3) Release the parking brake.

SUBTASK 32-44-11-860-008

(4) Open this access panel to get access to the parking brake latch:

Number	Name/Location
112A	Forward Access Door

E. Parking Brake Latch Removal

SUBTASK 32-44-11-020-010

(1) Do these steps to remove the arm [55]:

(a) Remove the nut [54], washer [52], washer [53], and bolt [51] that hold the rod [29] to the arm [55].

NOTE: This step is optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove the nut [59], washer [58], washer [57], and bolt [56] that hold the arm [55] to the parking brake latch [63].

(c) Remove the arm [55] from the parking brake latch [63].

SUBTASK 32-44-11-020-011

(2) Disconnect the return spring [64] from the pawl.

SUBTASK 32-44-11-020-012

(3) Do these steps to remove the inboard housing [66] from the parking brake latch [63]:

(a) Remove the nuts [62], washers [61], and bolts [60] that hold the inboard housing [66] to the airplane.

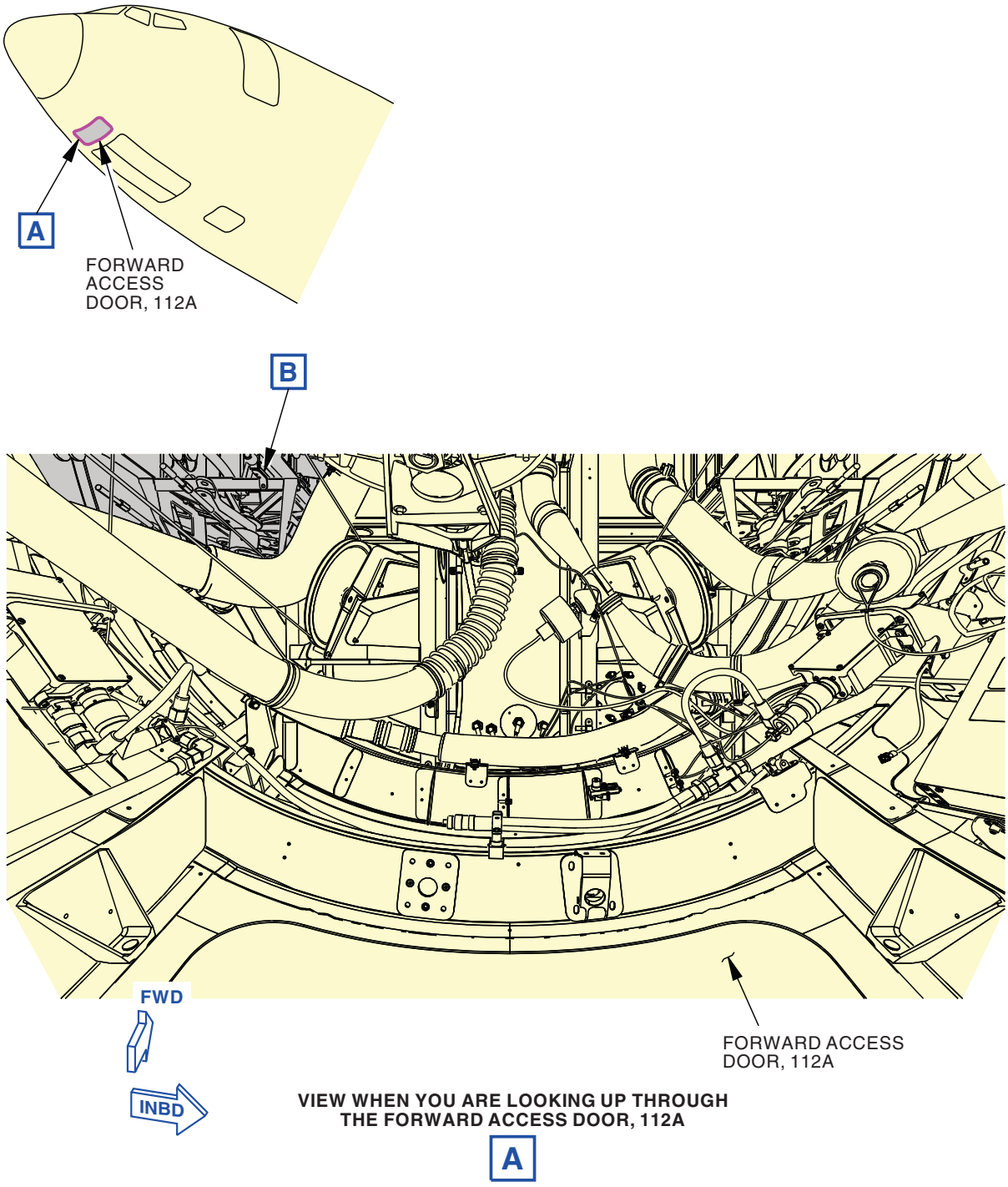
(b) Remove the inboard housing [66] with the bearing [67] from the parking brake latch [63].

SUBTASK 32-44-11-020-013

(4) Remove the parking brake latch [63] from the outboard housing [65].

——— END OF TASK ———

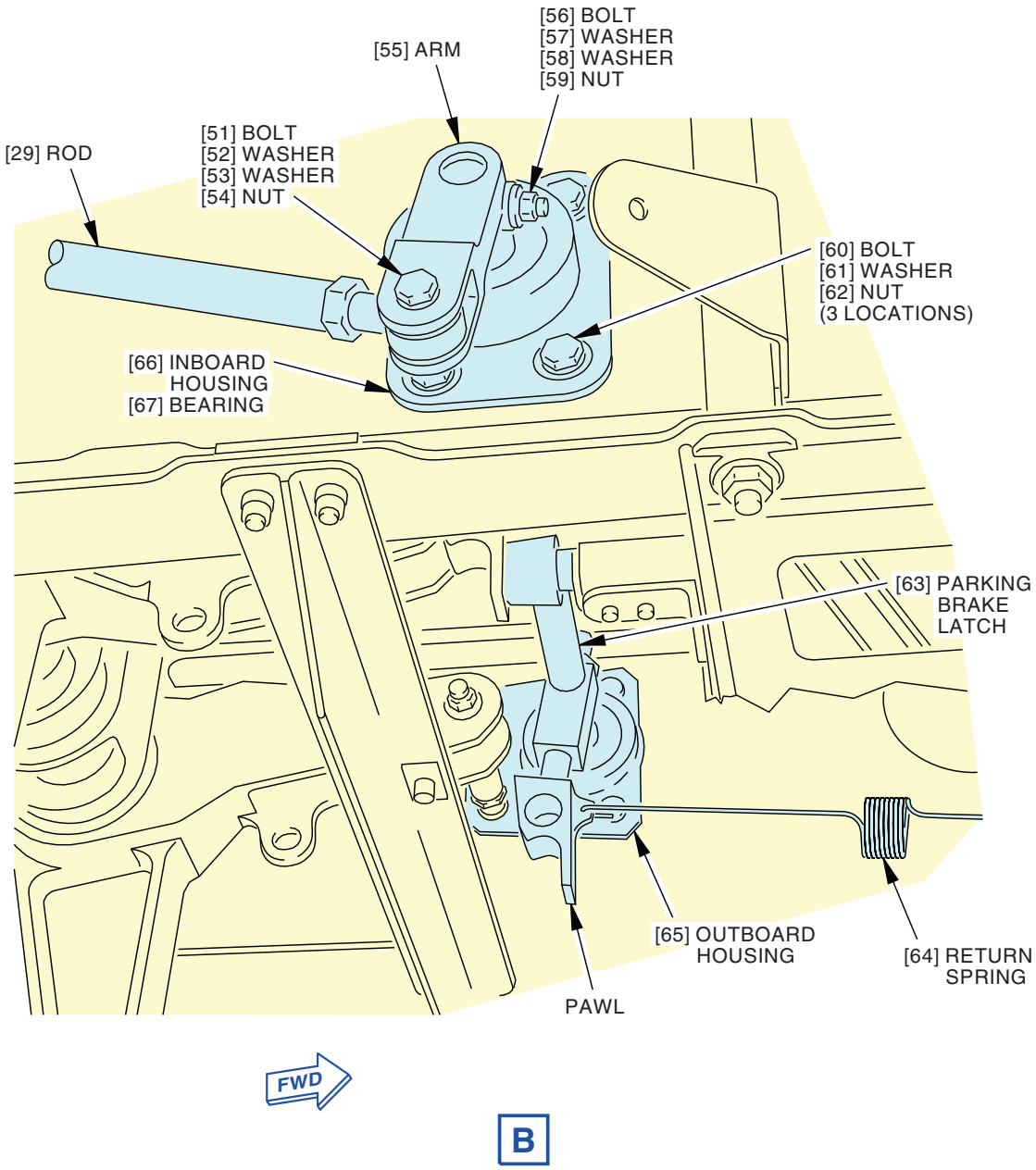
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414426 S00061535156_V1

Parking Brake Latch Mechanism Installation
Figure 403/32-44-11-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414427 S00061535157_V1

Parking Brake Latch Mechanism Installation
Figure 403/32-44-11-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-11-400-803

7. Parking Brake Latch Installation

(Figure 403)

A. References

Reference	Title
32-44-00-820-801	Parking Brake System Adjustment (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
63	Parking brake latch	32-44-11-02-003	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Parking Brake Latch Installation

SUBTASK 32-44-11-420-006

(1) Install the parking brake latch [63] in the outboard housing [65].

SUBTASK 32-44-11-420-007

(2) Do these steps to install the inboard housing [66] for the parking brake latch [63]:

(a) Install the inboard housing [66] and bearing [67] on the parking brake latch [63].

(b) Install the bolts [60], washers [61], and nuts [62] that hold the inboard housing [66] to the airplane.

SUBTASK 32-44-11-020-014

(3) Connect the return spring [64] to the pawl.

SUBTASK 32-44-11-420-008

(4) Do these steps to install the arm [55]:

(a) Install the arm [55] on the parking brake latch [63] such that the bolt holes in the arm [55] and the parking brake latch [63] are aligned.

1) Make sure that the flat surface of the arm [55] is on the side that is opposite the bearing and the yoke for the rod pointing down (Figure 403).

(b) Install the bolt [56], washer [57], washer [58], and nut [59] that hold the arm [55] to the parking brake latch [63].

(c) Install the bolt [51], washer [52], washer [53], and nut [54] that hold the rod [29] to the arm [55].

1) Install the thin washer [53] under the head of the bolt [51].

SUBTASK 32-44-11-820-003

(5) If you replaced the pawls, the length of the pawls was changed, or the length of the connecting rod was changed, do this task: Parking Brake System Adjustment, TASK 32-44-00-820-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-11-860-009

(6) Close this access panel that you opened to get access to the parking brake latch mechanism:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

TASK 32-44-11-000-804

8. Parking Brake Latch Switch Removal

(Figure 404)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 32-44-11-480-007

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-44-11-480-008

(2) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-11-860-010

(3) Release the parking brake.

SUBTASK 32-44-11-860-011

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-44-11-860-012

(5) Open this access panel to get access to the parking brake latch mechanism:

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Parking Brake Latch Switch Removal

SUBTASK 32-44-11-020-015

(1) Disconnect the wires [82] from the switch [81].

SUBTASK 32-44-11-020-016

(2) Remove the screw [83] and nut [84] which hold the switch [81] to the mounting bracket.

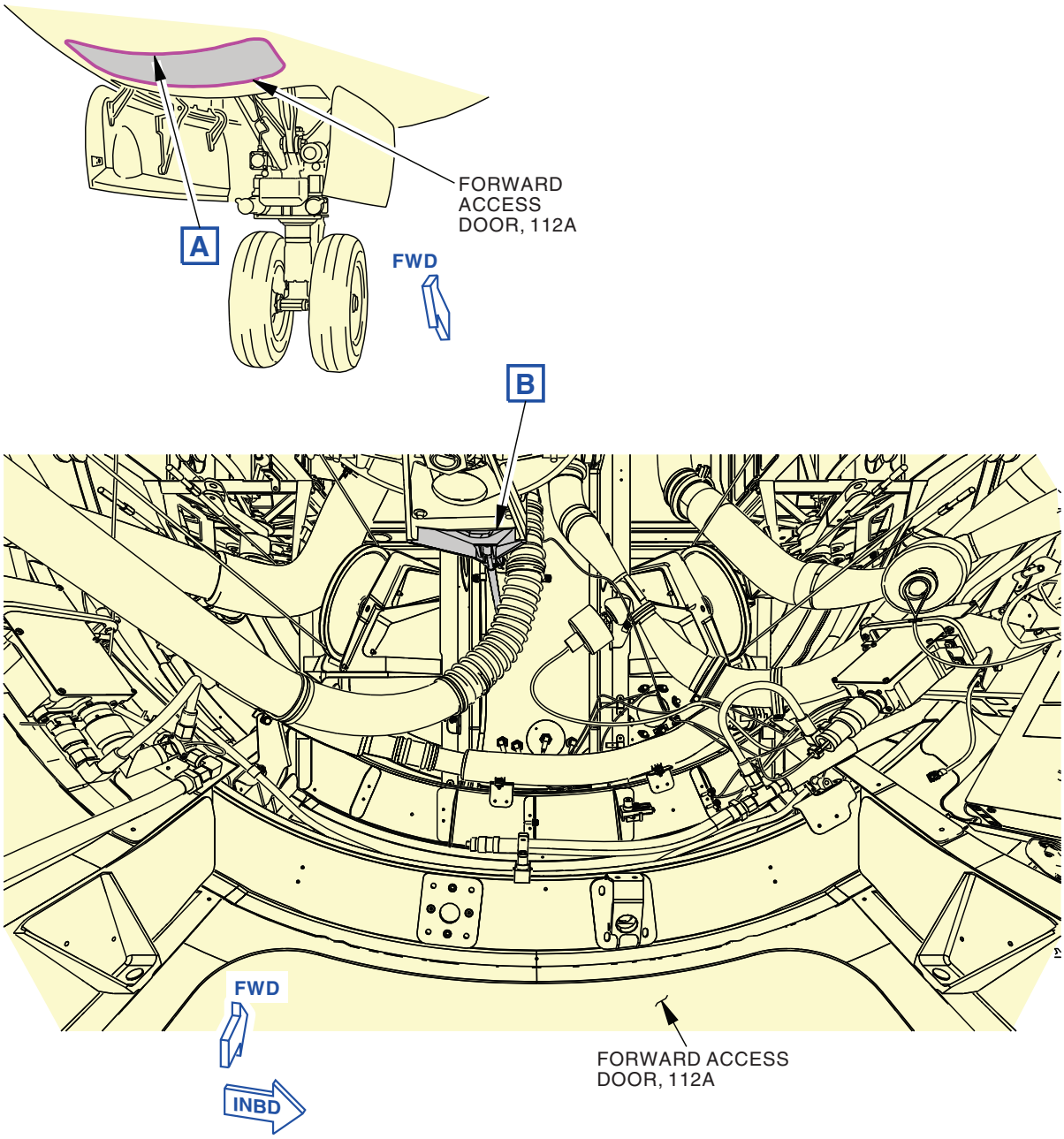
NOTE: The switch [81] and the switch actuator are held together by the mounting screws.

SUBTASK 32-44-11-020-017

(3) Remove the switch [81] from the airplane.

——— END OF TASK ———

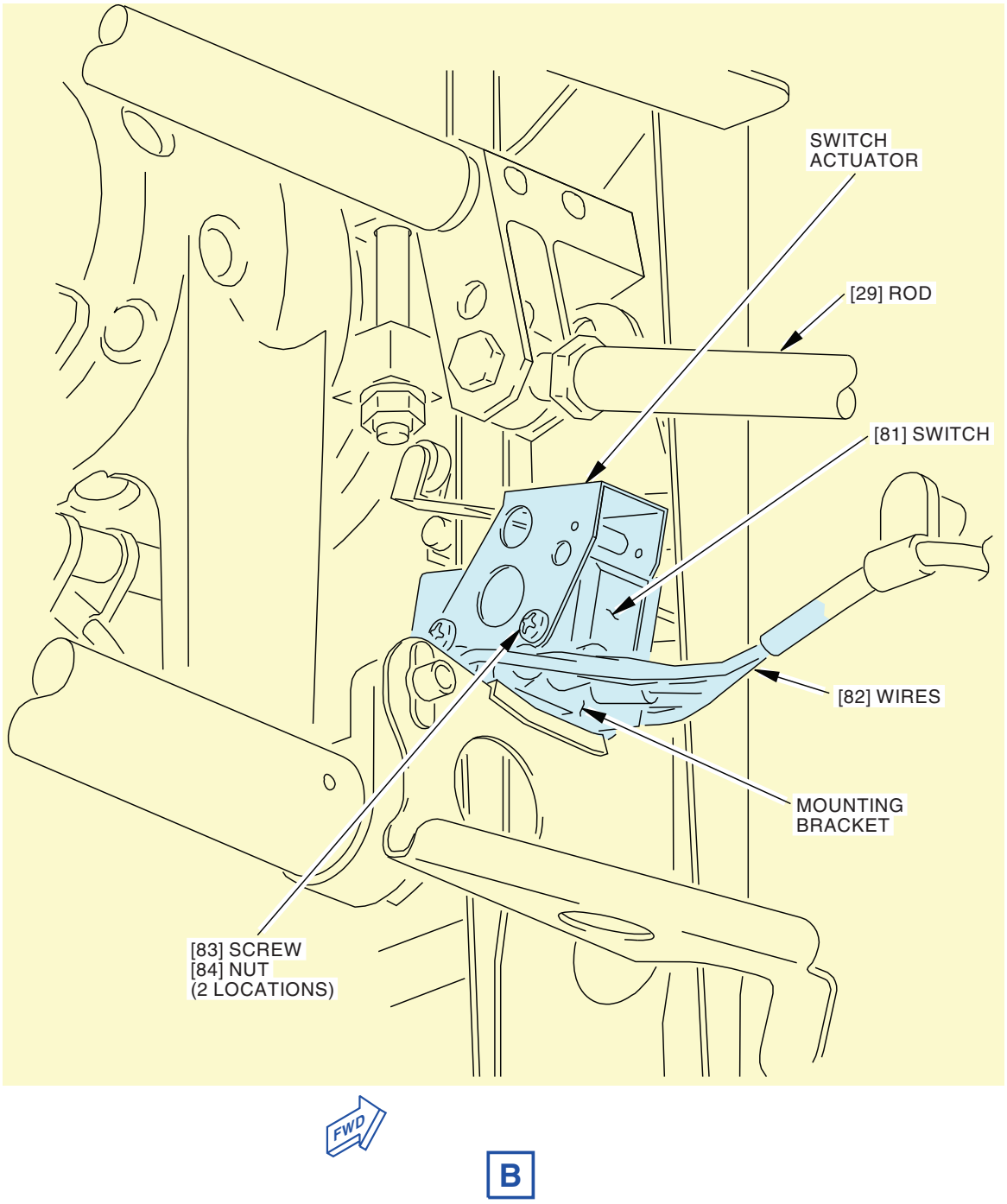
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414428 S00061535160_V1

Parking Brake Latch Switch Installation
Figure 404/32-44-11-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414429 S00061535161_V1

Parking Brake Latch Switch Installation
Figure 404/32-44-11-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-11-400-804

9. Parking Brake Latch Switch Installation

(Figure 404)

A. References

Reference	Title
32-44-00-820-801	Parking Brake System Adjustment (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
81	Switch	32-44-11-01-070	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Parking Brake Latch Switch Installation

SUBTASK 32-44-11-420-009

(1) Make sure that the switch actuator is correctly mated to the switch plunger on the switch [81].

SUBTASK 32-44-11-420-010

(2) Hold the switch [81] in its position on the airplane.

SUBTASK 32-44-11-420-011

(3) Install the screw [83] and nut [84] which hold the switch [81] to the mounting bracket.

SUBTASK 32-44-11-420-012

(4) Connect the wires [82] to the switch [81].

SUBTASK 32-44-11-860-013

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-44-11-820-004

(6) Do the applicable step to adjust the parking brake latch switch [81], do this task: TASK 32-44-00-820-801.

NOTE: Do not bend the switch actuator arm to adjust the switch [81].

SUBTASK 32-44-11-720-002

(7) Do a function check of the parking brake latch switch as follows:

(a) Release the parking brake.

(b) Set the parking brake.

(c) On the OMF, make sure that the maintenance messages related to the parking brake switches are not shown or are NOT ACTIVE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-11-410-004

(8) Close this access panel that you opened to get access to the parking brake latch mechanism:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

TASK 32-44-11-000-805

10. Parking Brake Mechanism Visual Inspection

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
211	Flight Compartment - Left

B. Access Panels

Number	Name/Location
112A	Forward Access Door

C. Parking Brake Mechanism Inspection

SUBTASK 32-44-11-860-015

(1) Open the access panels on the left side of the control stand to get access to the parking brake mechanism.

SUBTASK 32-44-11-860-016

(2) Open this access panel to get access to the parking brake mechanism:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-44-11-210-001

(3) Do a detail visual inspection of the mechanical control path for the parking brake mechanism.

SUBTASK 32-44-11-410-002

(4) Close the access panels on the left side of the control stand.

SUBTASK 32-44-11-860-017

(5) Close this access panel that you opened to get access to the parking brake latch mechanism:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PARKING BRAKE SHUTOFF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Parking Brake Shutoff Valve Removal

(2) Parking Brake Shutoff Valve Installation.

B. The parking brake shutoff valve is located on the aft wall of the main landing gear wheel well.

TASK 32-44-21-000-801

2. Parking Brake Shutoff Valve Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-7394	Cap - Dust, Electrical Connector

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 32-44-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-44-21-860-001

(2) Remove the pressure from the normal (system B) and alternate (system A) hydraulic systems (TASK 29-09-00-860-802).

SUBTASK 32-44-21-480-002

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-21-860-002

(4) Release the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-21-840-001

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: It will be necessary to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

SUBTASK 32-44-21-860-003

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

E. Parking Brake Shutoff Valve Removal

SUBTASK 32-44-21-020-001

(1) Disconnect the electrical connector [6] from the parking brake shutoff valve [5].

SUBTASK 32-44-21-020-002

(2) Disconnect the tee fitting from the reducer [3] on the parking brake shutoff valve [5].

SUBTASK 32-44-21-020-003

(3) Disconnect the elbow from the reducer [7] on the parking brake shutoff valve [5].

SUBTASK 32-44-21-480-003

(4) Install the dust caps, STD-7394, on the elbow and the tee fitting.

SUBTASK 32-44-21-020-004

(5) Remove the bolts [2] and washers [1] that hold the parking brake shutoff valve [5] to the airplane.

SUBTASK 32-44-21-020-005

(6) If the replacement parking brake shutoff valve [5] does not have hydraulic fittings installed, do these steps:

(a) Remove the reducer [3], reducer [7] and packings [4].

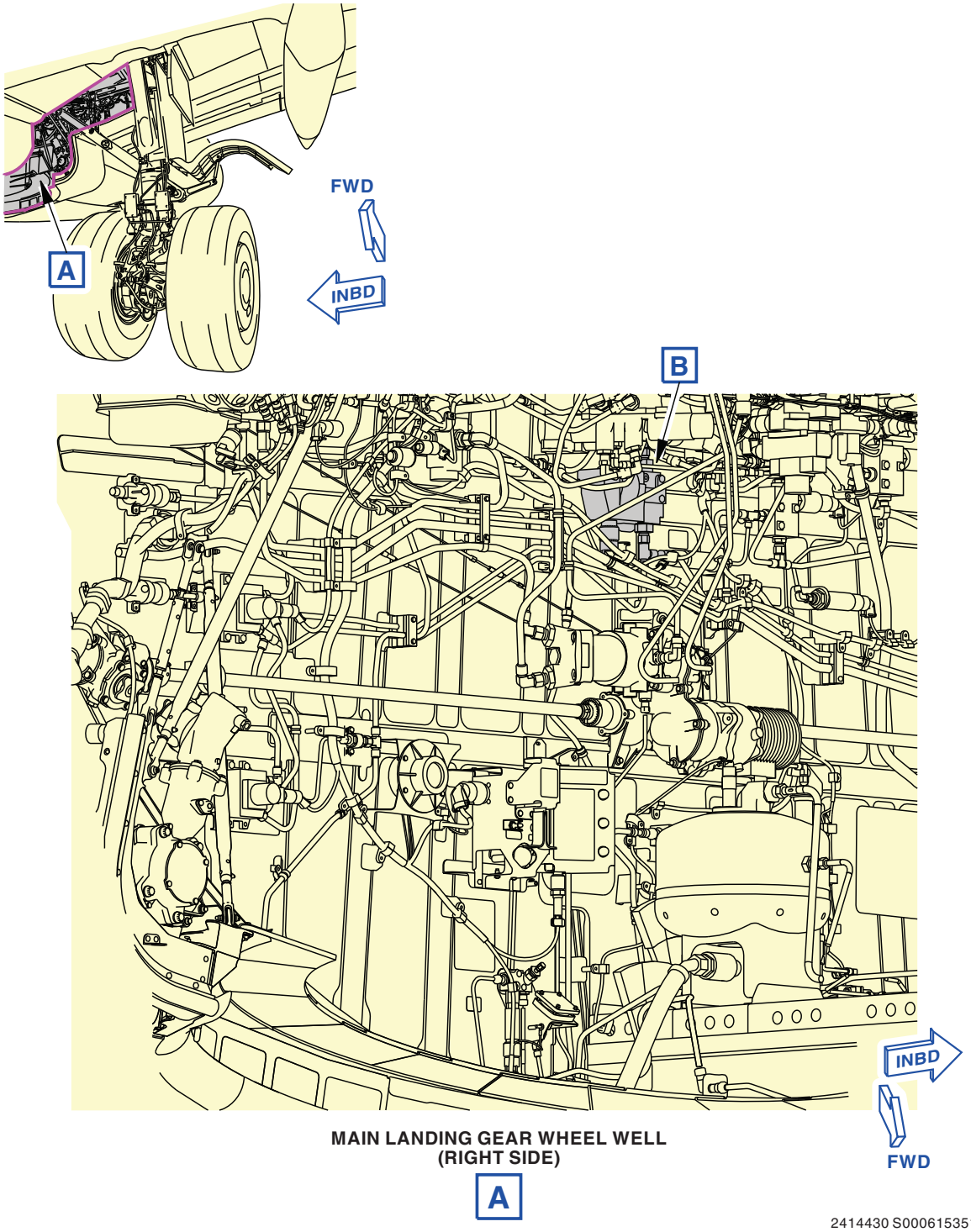
(b) Discard the packings [4].

SUBTASK 32-44-21-480-004

(7) Install the dust caps, STD-7394, in the ports of the parking brake shutoff valve [5].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

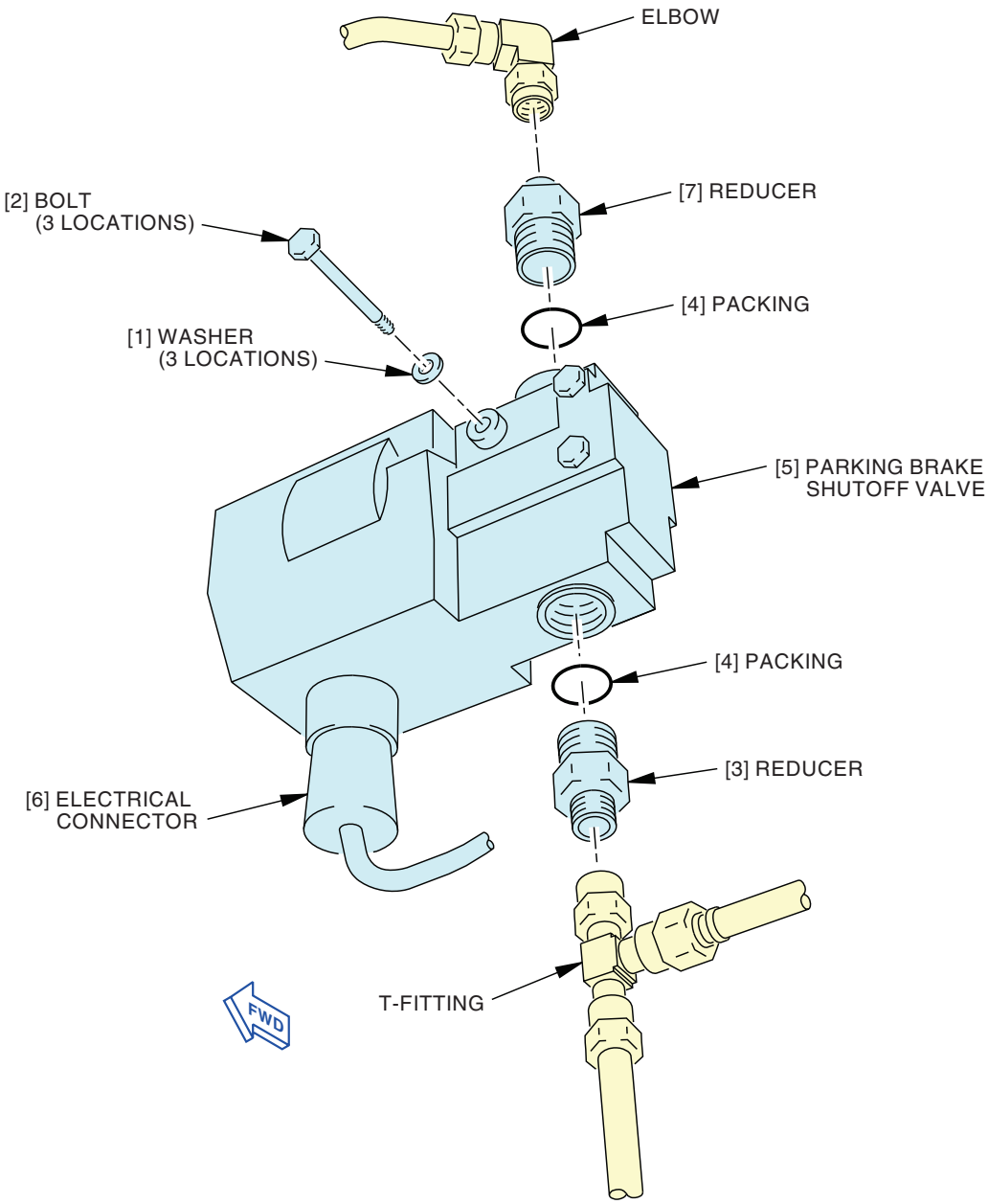


MAIN LANDING GEAR WHEEL WELL
(RIGHT SIDE)

2414430 S00061535167_V1

Parking Brake Shutoff Valve Installation
Figure 401/32-44-21-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



PARKING BRAKE SHUTOFF VALVE

B

2414431 S00061535168_V2

Parking Brake Shutoff Valve Installation
Figure 401/32-44-21-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-44-21-400-801

3. Parking Brake Shutoff Valve Installation

(Figure 401)

A. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-870-801	Hydraulic Brake System Upstream of the Antiskid Valves Bleeding (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-7394	Cap - Dust, Electrical Connector

C. Consumable Materials

Reference	Description	Specification
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01043	Cloth - Lint-free	
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Packing	32-44-21-01-033	PTF ALL
5	Parking brake shutoff valve	32-44-21-01-010	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Parking Brake Shutoff Valve Installation

SUBTASK 32-44-21-080-001

(1) Remove the dust caps from the ports of the parking brake shutoff valve [5].

SUBTASK 32-44-21-420-001

(2) If the replacement parking brake shutoff valve [5] does not have hydraulic fittings installed, do these steps:

(a) Lubricate the reducer [3], reducer [7], and new packings [4] for the parking brake shutoff valve [5] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) Install the packing [4] on the reducer [3].

(c) Install the reducer [3] in the parking brake shutoff valve [5].

1) Tighten the reducer [3] to 360 ±18 in-lb (41 ±2 N·m).

(d) Install the packing [4] on the reducer [7].

(e) Install the reducer [7] in the parking brake shutoff valve [5].

1) Tighten the reducer [7] to 360 ±18 in-lb (41 ±2 N·m).

SUBTASK 32-44-21-100-001

(3) To get a good electrical bond between the parking brake shutoff valve [5] and support bracket, do these steps:

(a) To fully clean the mating surface of the support bracket and parking brake shutoff valve [5], do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

(b) Chemically prepare the mating surface of the support bracket with coating, C00064.

SUBTASK 32-44-21-390-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(4) Apply a layer of corrosion inhibiting material, G50136, to the threads and shank of each bolt [2].

NOTE: Do not let corrosion inhibiting material, G50136, dry before the bolts installation.

(a) Install the bolts [2] when corrosion inhibiting material, G50136, is wet.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-21-390-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a layer of corrosion inhibiting material, G50136, to the mating surface of the mounting bracket.

NOTE: Do not let corrosion inhibiting material, G50136, dry before the bolts installation.

SUBTASK 32-44-21-420-002

(6) Hold the parking brake shutoff valve [5] in its position on the airplane.

SUBTASK 32-44-21-420-003

(7) Install the bolts [2] and washers [1] that hold the parking brake shutoff valve [5] to the airplane.

SUBTASK 32-44-21-210-001

(8) Make sure that there is squeeze-out around mating surfaces of the support bracket and parking brake shutoff valve [5].

SUBTASK 32-44-21-100-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(9) If it is necessary, remove the excess corrosion inhibiting material, G50136, with a dry lint-free cloth, G01043.

NOTE: The stains of the corrosion inhibiting material, G50136, remaining in crevices are acceptable.

SUBTASK 32-44-21-760-001

(10) Measure the electrical bonding resistance between the parking brake shutoff valve [5] and bracket (SWPM 20-20-00).

(a) Use an intrinsically safe approved bonding meter, COM-1550.

(b) Make sure that the electrical resistance is 0.0025 ohm (2.5 milliohms) or less.

1) If the measured resistance is not less than 0.0025 ohm (2.5 milliohms), clean the mating surfaces of the bracket and parking brake shutoff valve [5], do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

2) Apply coating, C00064, again.

SUBTASK 32-44-21-080-002

(11) Remove the dust caps, STD-7394, from the elbow and tee fitting.

SUBTASK 32-44-21-640-001

(12) Apply a thin layer of hydraulic fluid, D00153 on the threads of the elbow and tee fitting.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-21-420-004

(13) Connect the tee fitting to the reducer [3] on the parking brake shutoff valve [5].

(a) Tighten the B-nut on the tee fitting to 270.0 ±13.5 in-lb (30.5 ±1.5 N·m).

SUBTASK 32-44-21-420-005

(14) Connect the elbow to the reducer [7] on the parking brake shutoff valve [5].

(a) Tighten the B-nut on the elbow to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

SUBTASK 32-44-21-420-006

(15) Before the connection of the electrical connector [6] to the parking brake shutoff valve [5], examine the electrical connector [6] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [6] (SWPM 20-60-03).

SUBTASK 32-44-21-420-007

(16) Connect the electrical connector [6] to the parking brake shutoff valve [5].

SUBTASK 32-44-21-860-004

(17) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

G. Parking Brake Shutoff Valve Installation Test

SUBTASK 32-44-21-790-001

(1) Do these steps to do a leak check of the parking brake shutoff valve [5]:

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) For the normal (system B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Fully push the captain's brake pedals twelve times.

(c) Look for leaks around the parking brake shutoff valve [5].

SUBTASK 32-44-21-710-001

(2) Do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-44-21-710-002

(3) Do these steps to do a check of the parking brake shutoff valve [5] installation:

(a) Set the parking brake.

(b) Make sure that the parking brake light on the captain's control stand comes on.

(c) Release the parking brake.

(d) Make sure that the parking brake light is off.

SUBTASK 32-44-21-860-006

(4) Make sure that the ANTISKID INOP light, on the P2 panel, is off.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-44-21-860-005

(1) For the normal (system B) hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRES AND WHEELS - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Landing Gear Tire - Deflation

(2) Landing Gear Tire - Inflation

TASK 32-45-00-910-801

2. Landing Gear Tire - Deflation

(Figure 201)

A. General

(1) This procedure has instructions to deflate nose and main landing gear tires.

B. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4046	Deflator - Tire, Valve Core Removal and Installation (Small Bore) and Tire Deflation Part #: 320008B-SO Supplier: SFN11 Part #: 968RB Supplier: 5W9Y4 Opt Part #: 320008B Supplier: SFN11

D. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare to Deflate the Tire

SUBTASK 32-45-00-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-45-00-480-002

(2) Make sure that the nose and main wheel assembly is clear of the ground before you deflate the tire, do these tasks:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task: Lift the Airplane on Axle Jacks-Handling, TASK 07-10-03-580-801.

(b) Do this task: Lift the Nose Landing Gear Axle With Axle Jacks-Handling, TASK 07-10-03-580-803.

F. Deflate the Tire

SUBTASK 32-45-00-080-001

(1) Remove the valve cap [1] from the nose wheel assembly.

SUBTASK 32-45-00-080-002

(2) Remove the valve cap [2] from the main wheel assembly.

SUBTASK 32-45-00-840-001

WARNING DEFLATE THE TIRE BEFORE YOU REMOVE THE WHEEL AND TIRE ASSEMBLY. OR, DO AN INSPECTION OF THE WHEEL AND TIRE TO MAKE SURE THAT THEY ARE SAFE. AIR PRESSURE CAN CAUSE AN EXPLOSION OF THE DAMAGED WHEEL AND TIRE COMPONENTS. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT ALL PERSONNEL ARE CLEAR OF THE INFLATION VALVE ASSEMBLY PATH WHILE YOU REMOVE IT. IF THE INFLATION VALVE ASSEMBLY EJECTS WHILE YOU REMOVE IT, INJURY TO PERSONNEL CAN OCCUR.

WARNING BE CAREFUL WHEN DEFLATING TIRES, A FLYING INFLATION VALVE CAP AND CORE CAN CAUSE DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL.

CAUTION MAKE SURE THAT THE DEFLATED TIRE THAT IS ON THE AXLE DOES NOT TOUCH THE GROUND. IF THE TIRE DOES TOUCH THE GROUND, THE TIRE'S BEAD SEAT CAN MOVE OUT OF ITS POSITION AND BECOME UNSEATED. THE UNSEATED BEAD SEAT CAN CAUSE LEAKAGE WHICH THEN CAN CAUSE DAMAGE TO THE TIRE AND WHEEL.

(3) Deflate the tire with the small bore tire deflator, COM-4046.

NOTE: Always go near the wheels from the front or rear.

NOTE: If the tire is hot, let the tire cool before you deflate it.

WARNING DO NOT LOOSEN OR REMOVE THE VALVE BODY UNDER PRESSURE. THE VALVE CAN BLOW OUT CAUSING INJURY TO PERSONNEL.

(a) If you cannot use the usual procedure to deflate the tire because the valve core is damaged, do these steps:

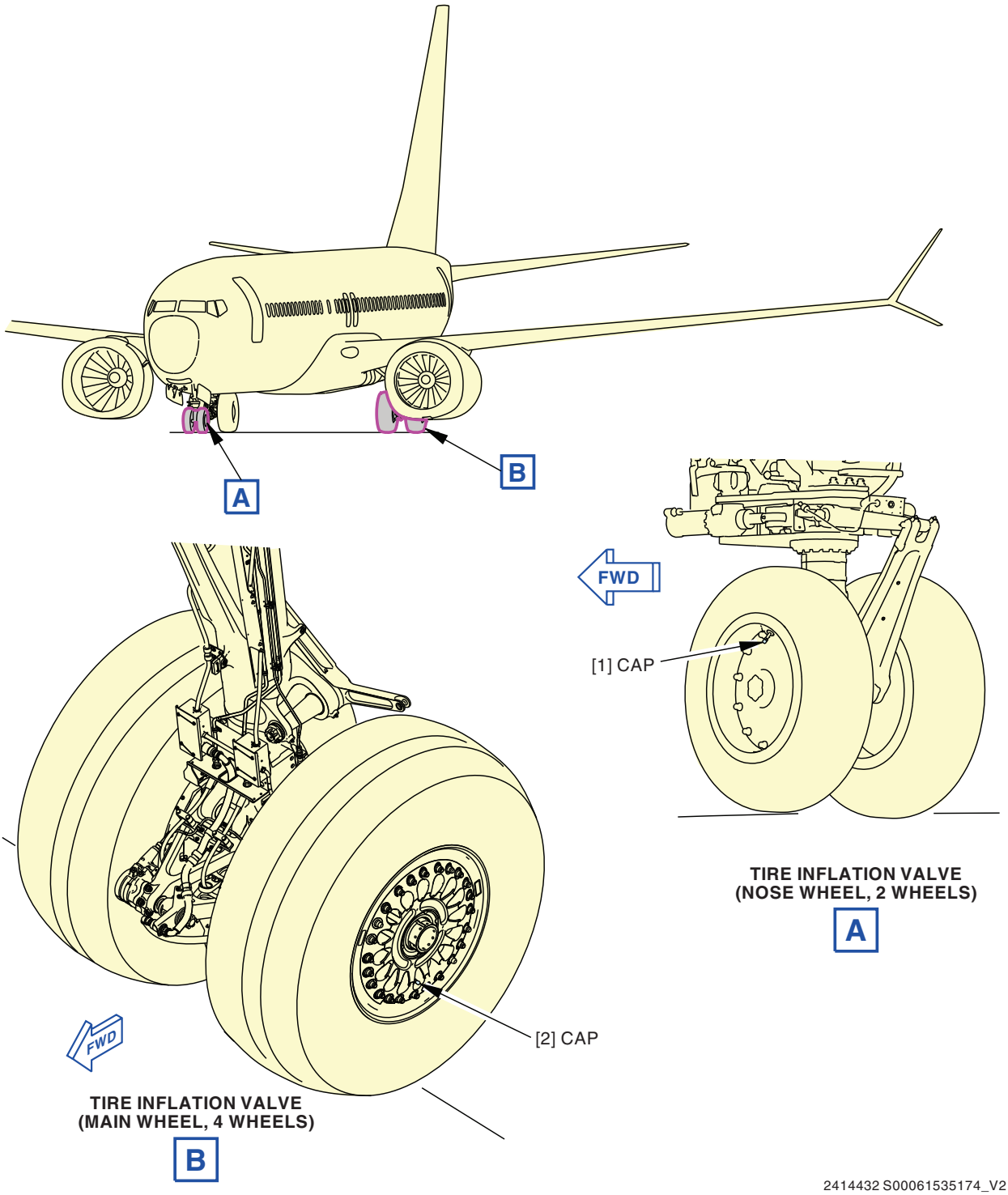
1) Turn the valve assembly slowly in a counterclockwise direction until there is gas leakage through the boss.

2) At the same time, push lightly on the valve assembly.

3) After all of the gas has been released, remove the valve assembly from the tire.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Landing Gear Tire Servicing
Figure 201/32-45-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-45-00-910-802

3. Landing Gear Tire - Inflation

(Figure 201)

A. General

(1) This procedure has instructions to inflate nose and main landing gear tires.

B. References

Reference	Title
12-15-51-610-802	Add Nitrogen or Air to the Tire (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare to Inflate the Tire

SUBTASK 32-45-00-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. Inflate the Tire

SUBTASK 32-45-00-080-003

(1) Remove the valve cap [1] from the nose wheel assembly.

SUBTASK 32-45-00-080-004

(2) Remove the valve cap [2] from the main wheel assembly.

SUBTASK 32-45-00-910-001

WARNING USE A PRESSURE SOURCE WITH A REGULATOR TO INFLATE THE TIRES. A PRESSURE SOURCE WITHOUT A REGULATOR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Do this task: Add Nitrogen or Air to the Tire, TASK 12-15-51-610-802.

SUBTASK 32-45-00-480-004

(4) Install the valve cap [1] to the nose wheel assembly.

SUBTASK 32-45-00-480-005

(5) Install the valve cap [2] to the main wheel assembly.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRES AND WHEELS - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Wheels Fast Check (Wheel Installed on the Airplane)

(2) Wheels Inspection (Wheel Removed from the Airplane)

(3) Landing Gear Tire Inspection

TASK 32-45-00-700-801

2. Wheels Fast Check (Wheel Installed on the Airplane)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do a fast check of the installed wheel on the airplane.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Procedure

SUBTASK 32-45-00-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-45-00-210-001

(2) Look at the wheels for these types of damage:

(a) Corrosion

(b) Cracks

(c) Flaked or blistered paint

(d) Damage to the heat shield

(e) Tiebolts and tiebolt nuts that are loose, damaged, or missing

(f) Damage from heat

(g) Chafing on the rim of the wheel due to low tire pressure

NOTE: It is not necessary to remove the hubcap to examine the wheels for damage unless the examination reveals a condition that makes a more detailed check of the area hidden by the hubcap necessary.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-45-00-020-001

(3) If you found one or more of these, replace the wheel-tire assembly:

(a) There is a loose, or damaged tiebolt.

(b) A tiebolt is missing.

SUBTASK 32-45-00-210-002

(4) Make sure that the hubcaps are not loose.

(a) Tighten or replace hubcaps if it is necessary.

——— END OF TASK ———

TASK 32-45-00-700-802

3. Wheels Inspection (Wheel Removed from the Airplane)

A. General

(1) This task gives the instructions to do the wheel inspection when removed from the airplane.

B. References

Reference	Title
05-51-07-210-801	High Energy Stop (P/B 201)

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Procedure

SUBTASK 32-45-00-210-003

(1) Examine the wheel for:

(a) Breaks

(b) Corrosion

(c) Cracks

(d) Distortion

(e) Flaked or blistered paint

(f) Heat shield damage

(g) Loose, damaged, or missing tiebolts and tiebolt nuts

(h) Nicks or gouges

(i) Overheat damage

(j) Scratches along the rim of a wheel that show the wheel rolled without a tire

(k) Worn plating

(l) Other damage.

SUBTASK 32-45-00-210-004

(2) Examine the grease seals for:

(a) Nicks

(b) Gouges

(c) Overheat damage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-45-00-210-005

(3) Examine the bearing and bearing surfaces (Wheel Bearings Inspection, TASK 32-45-00-211-801).

SUBTASK 32-45-00-210-006

(4) Examine the brake torque bars for wear.

SUBTASK 32-45-00-210-007

(5) Examine the wheel heat shield for:

NOTE: Dents, scratches and discoloration are permitted.

(a) Cracks

(b) Tears.

SUBTASK 32-45-00-220-001

(6) If a wheel has a blown thermal fuse, do this task: High Energy Stop, TASK 05-51-07-210-801.

——— END OF TASK ———

TASK 32-45-00-700-803

4. Landing Gear Tire Inspection

(Figure 601 and Figure 602)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives the instructions to do the inspection of the aircraft radial ply tires.

B. References

Reference	Title
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Landing Gear Tire Inspection

SUBTASK 32-45-00-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-45-00-220-002

(2) Do this task: Landing Gear Tire Pressure Check and Tire Servicing, TASK 12-15-51-780-801.

SUBTASK 32-45-00-210-008

(3) Examine the tires as follows (Figure 601 and Figure 602):

(a) Examine the tires for air leaks, abrasions, unusual worn areas, cuts, and flat spots.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Examine the tires for the presence of contaminants.

1) Keep the tires clean of contaminants such as oils, fuels, hydraulic fluids, aircraft cleaning agents, and greases.

a) Cover the tires if these or other potentially harmful chemicals can spill on the tires during maintenance.

b) Clean the tire with a soapy solution if the tire becomes contaminated.

2) The tire should be removed from service as soon as practical if the surface of the tire appears soft, spongy, or there are bulges present.

(c) Remove the tires that have the conditions that follow:

1) Cuts or weather cracks in the grooves, the tread, shoulders or sidewalls that exceed the limits (Figure 602).

2) Blisters, bulges, or other signs of ply separation in the tread, shoulder or sidewall area.

3) Tires with a flat spot which shows the tread reinforcement/cut protector (radial).

NOTE: If the reinforcement/cut protector (radial) shows, the tire should be replaced as soon as possible. If necessary, the tire may be used for a small number of landings until it is replaced. However, you may not be able to retread the tire if you leave the tire in service too long with this condition.

4) Other types of damage which can cause tire problems.

(d) Examine the tires for worn areas.

1) Measure the depth of the tire tread groove at three points that are equally apart.

2) If the average depth of any groove is 1/32 inch (0.79 mm) or less, the tire must be replaced at the next convenient maintenance opportunity.

3) If the tread belt ply (radial) shows at any location, the tire is not serviceable and must be replaced.

4) If the tread is worn so that the reinforcement/cut protector (radial) shows at any location, the tire must be replaced at the next convenient maintenance opportunity.

NOTE: If the reinforcement/cut protector (radial) shows, the tire should be replaced as soon as possible. If necessary, the tire may be used for a small number of landings until it is replaced. However, you may not be able to retread the tire if you leave the tire in service too long with this condition.

E. Tire Tread Conditions and Service

SUBTASK 32-45-00-220-003

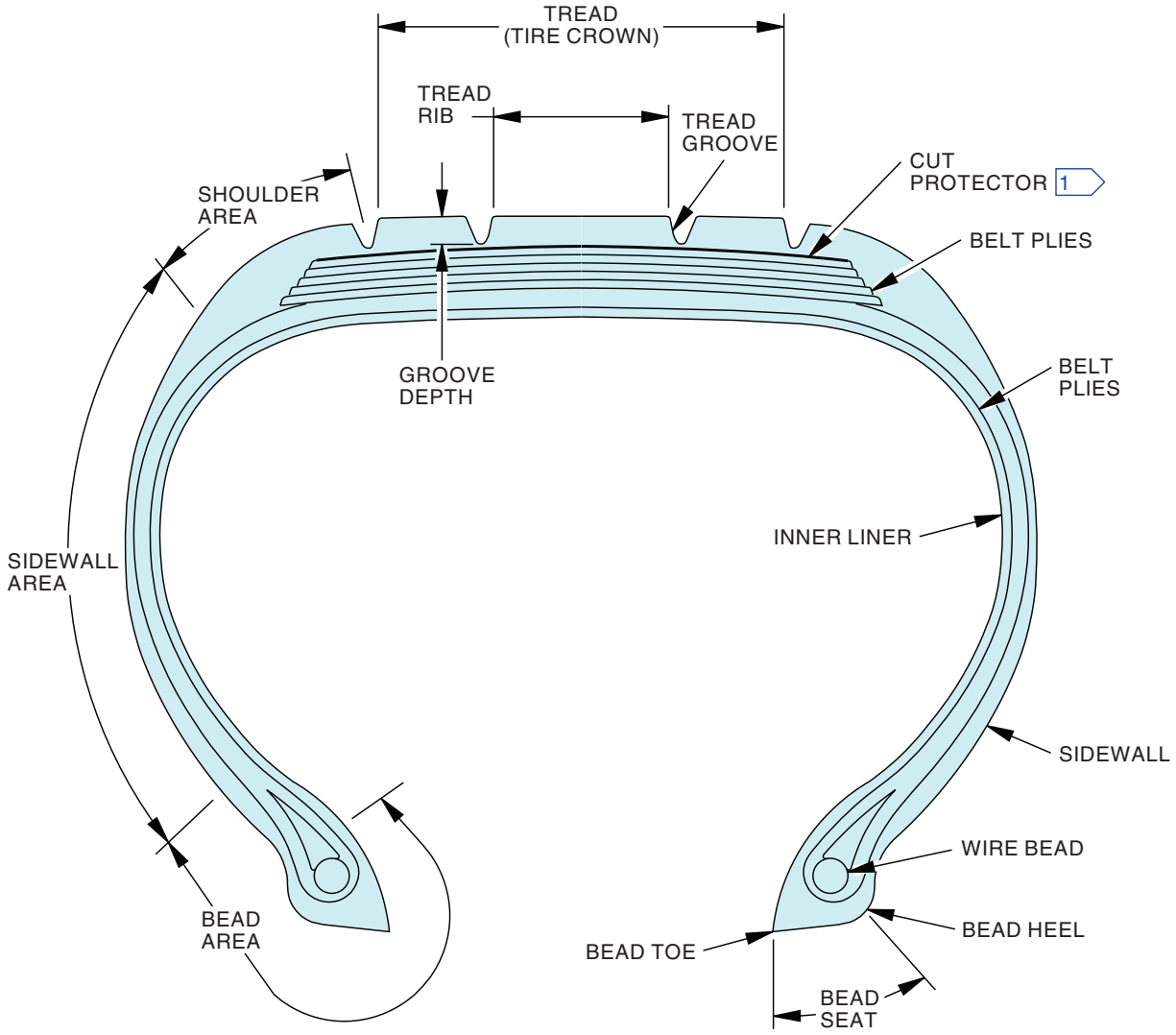
WARNING

DO NOT PUSH ON A DAMAGED TIRE WITH TOOLS. DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(1) Check the tire tread conditions (Figure 602).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



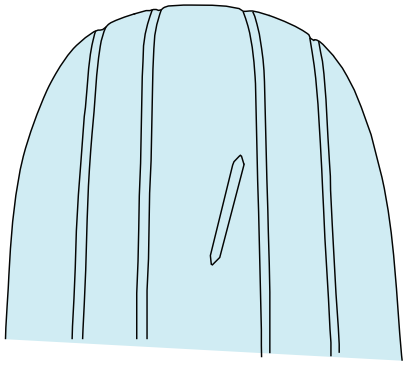
RADIAL TIRE

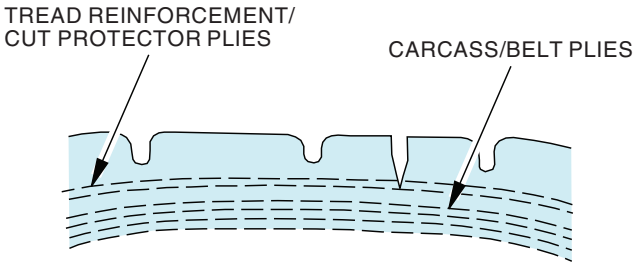
1 THE TREAD REINFORCEMENT/CUT PROTECTOR CAN BE METAL OR KEVLAR.

2414433 S00061535179_V4

Tire Nomenclature (Example)
Figure 601/32-45-00-990-802

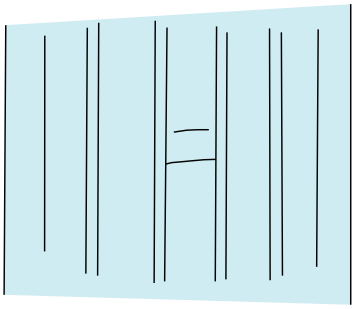
ECCN 9E991 BOEING PROPRIETARY - See title page for details





CIRCUMFERENTIAL CUTS

CIRCUMFERENTIAL CUTS THAT PENETRATE ANY OF THE FABRIC PLIES ARE NOT SERVICEABLE.



TRANSVERSE CUTS

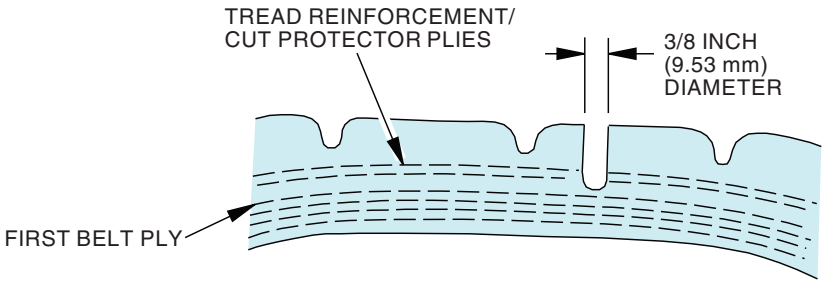
(A) TRANSVERSE CUTS THAT ARE CONTAINED WITHIN A RIB AND PENETRATE INTO THE CARCASS PLIES (BIAS) OR BELT PLIES (RADIAL) ARE NOT SERVICEABLE.

(B) CUTS THAT EXTEND ACROSS A RIB FROM GROOVE TO GROOVE WHICH ARE DEEPER THAN THE GROOVE ARE NOT SERVICEABLE.

F98055 S0006575654_V3

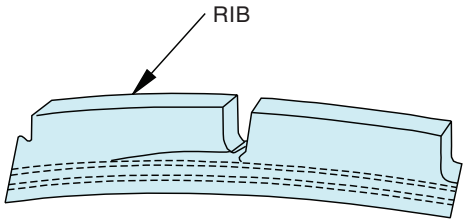
Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



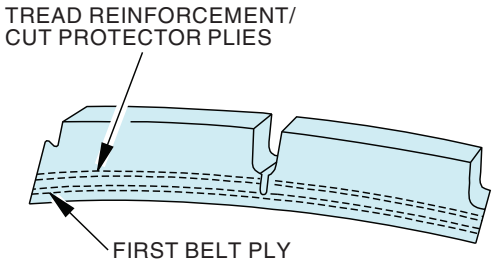
PUNCTURE

TIRES WITH HOLES LARGER THAN 3/8 INCH (9.53 mm) DIAMETER OR HOLES THAT PENETRATE INTO THE BELT PLIES (RADIAL) ARE NOT SERVICEABLE.



RIB UNDERCUTS

TIRES WITH A CRACK THAT EXTENDS BELOW A TREAD RIB ARE NOT SERVICEABLE.



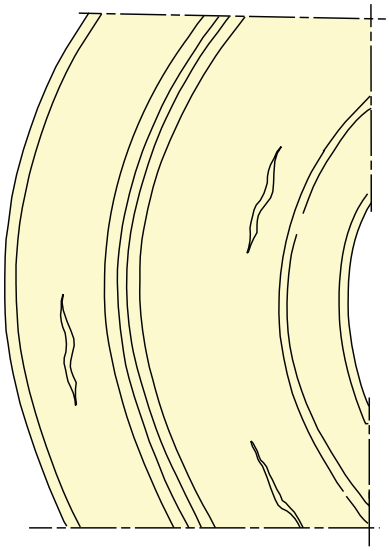
GROOVE CRACKS

TIRES WITH CIRCUMFERENTIAL CRACKS AT THE BOTTOM OF THE GROOVES ARE NOT SERVICEABLE IF THE CRACKS EXPOSE ANY FABRIC.

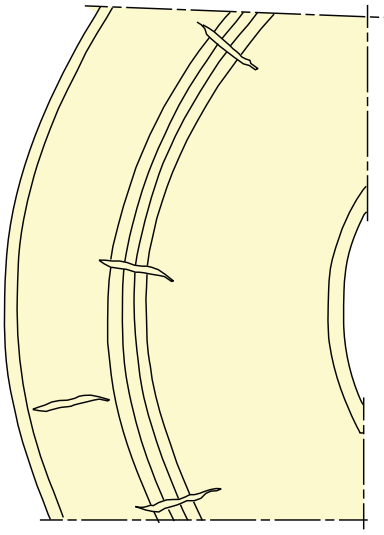
2414436 S00061535182_V2

Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

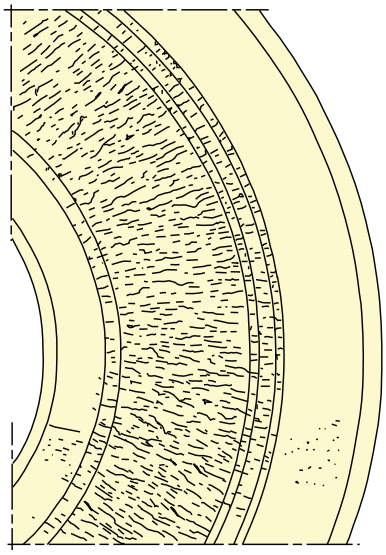


CIRCUMFERENTIAL SIDEWALL CUTS OR CRACKS



RADIAL SIDEWALL CUTS OR CRACKS

TIRES WITH CIRCUMFERENTIAL OR RADIAL CUTS OR CRACKS IN EITHER THE SIDEWALL OR SHOULDER AREA, WHICH EXPOSE THE FABRIC, SHOULD BE IMMEDIATELY REMOVED FROM THE AIRCRAFT. TIRES WITH CIRCUMFERENTIAL OR RADIAL CUTS IN EITHER THE SIDEWALL OR SHOULDER AREA, WHICH ARE DEEP ENOUGH TO REACH THE FABRIC, ARE NOT SERVICEABLE.



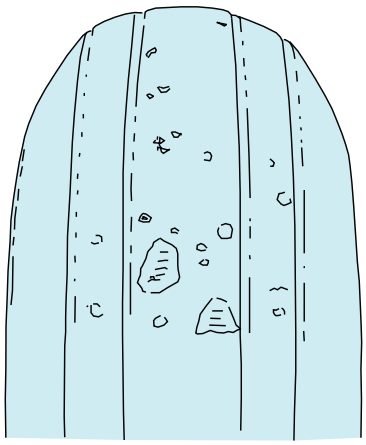
OZONE AND/OR WEATHER CUTS OR CRACKS

TIRES WITH CUTS OR CRACKS IN THE SIDEWALL WHICH ARE CAUSED BY AGE/WEATHER DETERIORATION ARE NOT SERVICEABLE IF THE FABRIC IS EXPOSED.

2414437 S00061535183_V2

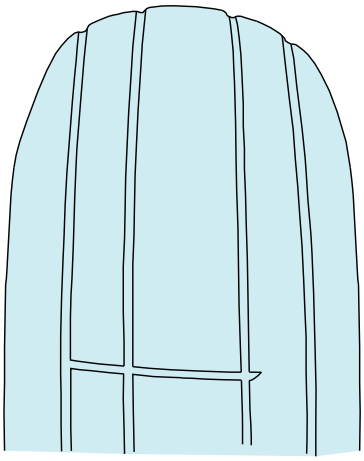
Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



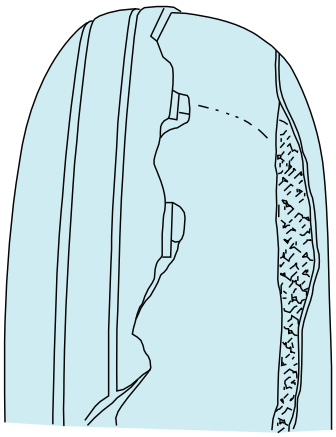
TREAD FLAKING, CHIPPING OR CHUNKING

TIRES WITH FLAKING, CHIPPING OR CHUNKING TREADS ARE NOT SERVICEABLE IF ANY FABRIC IS EXPOSED.



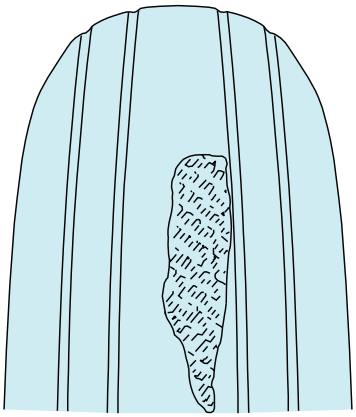
OPEN TREAD SPLICE

TIRES WITH AN OPEN TREAD SPLICE ARE NOT SERVICEABLE. A CLOSED TREAD SPLICE WILL APPEAR AS A THIN LINE THAT CROSSES THE TREAD RIBS.



THROWN TREAD

TIRES WITH THROWN TREADS ARE NOT SERVICEABLE.



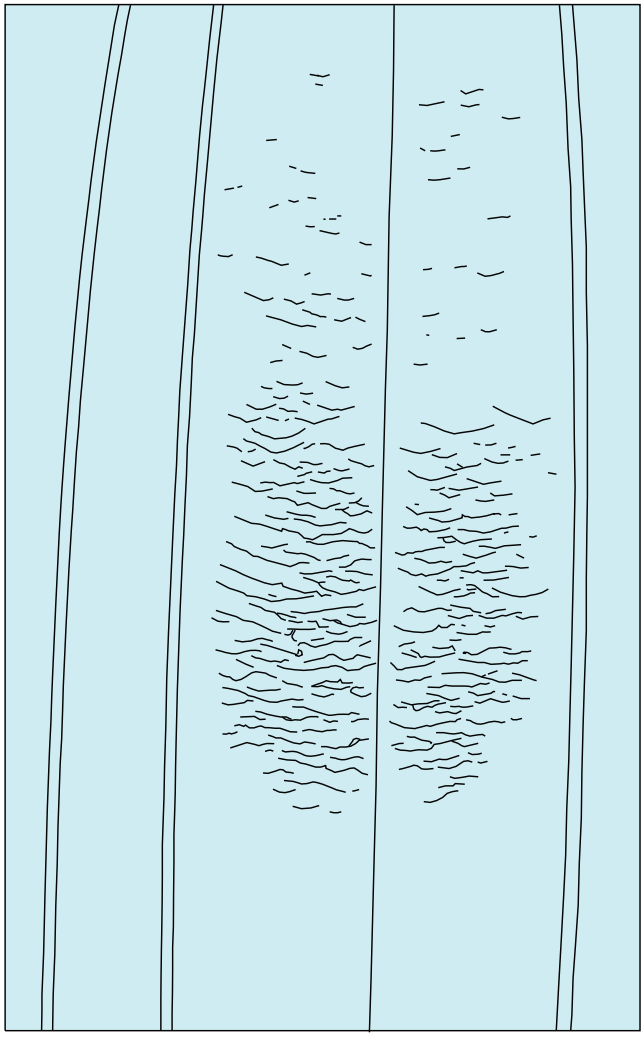
PEELED RIB

TIRES WITH PEELED RIBS ARE NOT SERVICEABLE.

2414438 S00061535184_V1

Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



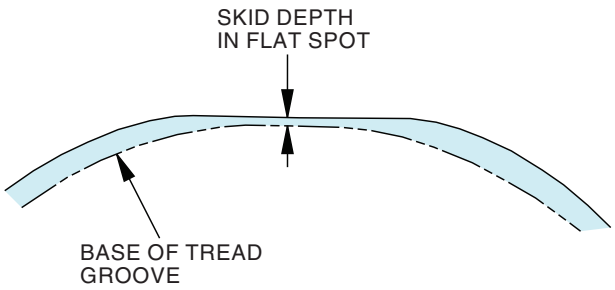
CHEVRON CUTS

TIRES WITH CHEVRON CUTS IN THE TREAD ARE NOT SERVICEABLE IF ANY OF THE SINGLE CUT LIMITS ARE EXCEEDED OR CHUNKING OCCURS WHICH EXPOSES THE FABRIC.

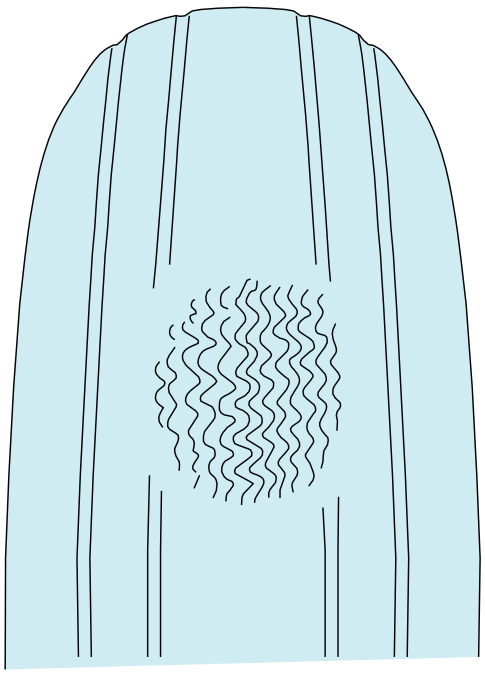
2414439 S00061535185_V1

Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

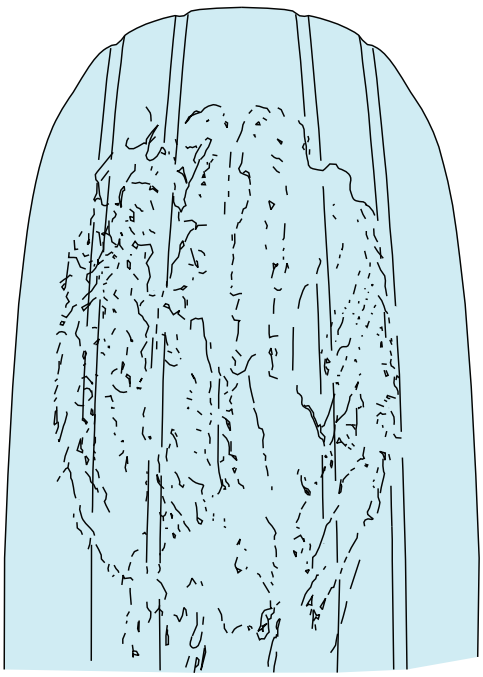


TREAD FLAT SPOTS



FLAT SPOT

TIRES WITH FLAT SPOTS ARE NOT SERVICEABLE IF THE FLAT SPOT EXPOSES THE TREAD REINFORCEMENT OR CUT PROTECTOR PLIES OR TIRE BALANCE IS AFFECTED.



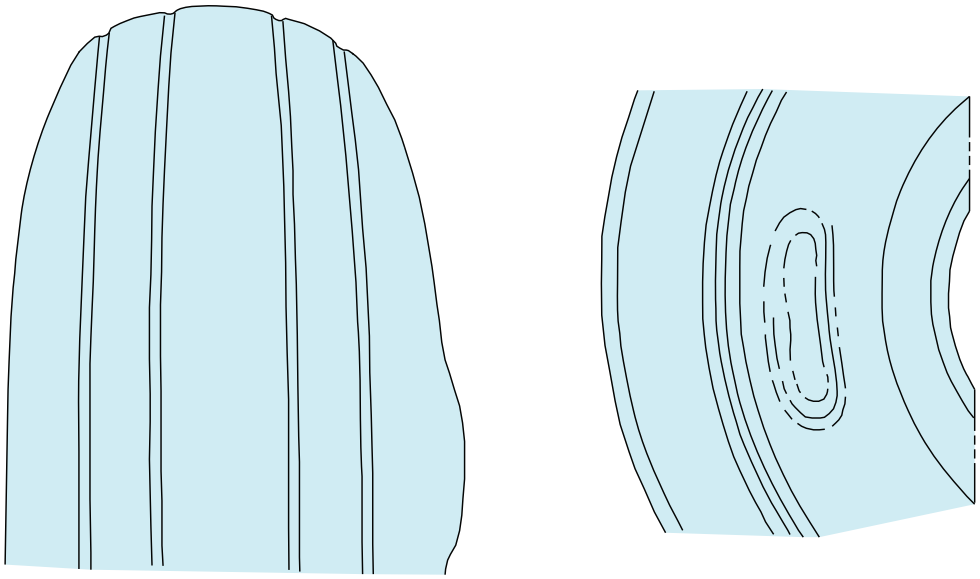
ICE SKID BURN/TREAD RUBBER REVERSION

TIRES WITH ICE BURN/TREAD RUBBER REVERSION ARE NOT SERVICEABLE IF THE DAMAGE EXPOSES THE TREAD REINFORCEMENT OR CUT PROTECTOR PLIES OR TIRE BALANCE IS AFFECTED.

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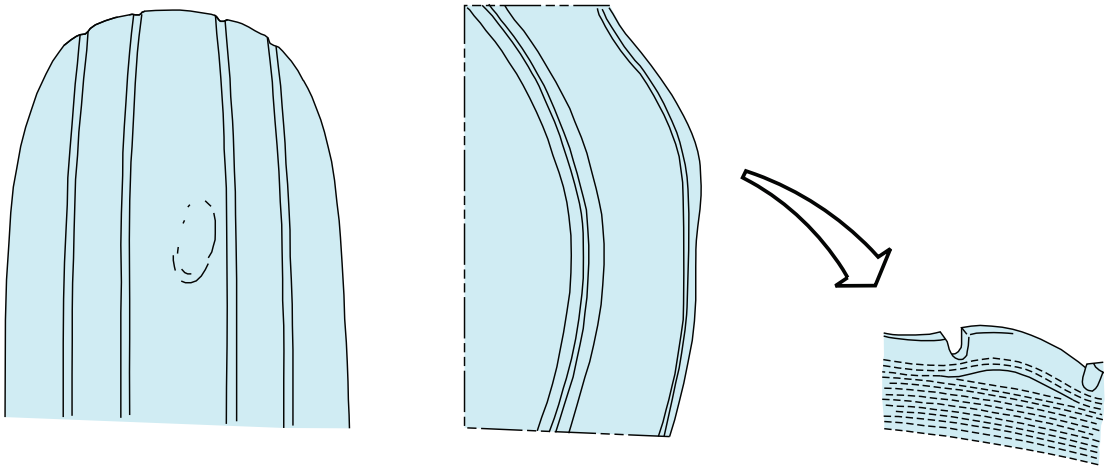
Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SIDEWALL SEPARATION

TIRES WITH BLISTERS OR BULGES IN THE SIDEWALL ARE NOT SERVICEABLE. REMOVE TIRE IMMEDIATELY.



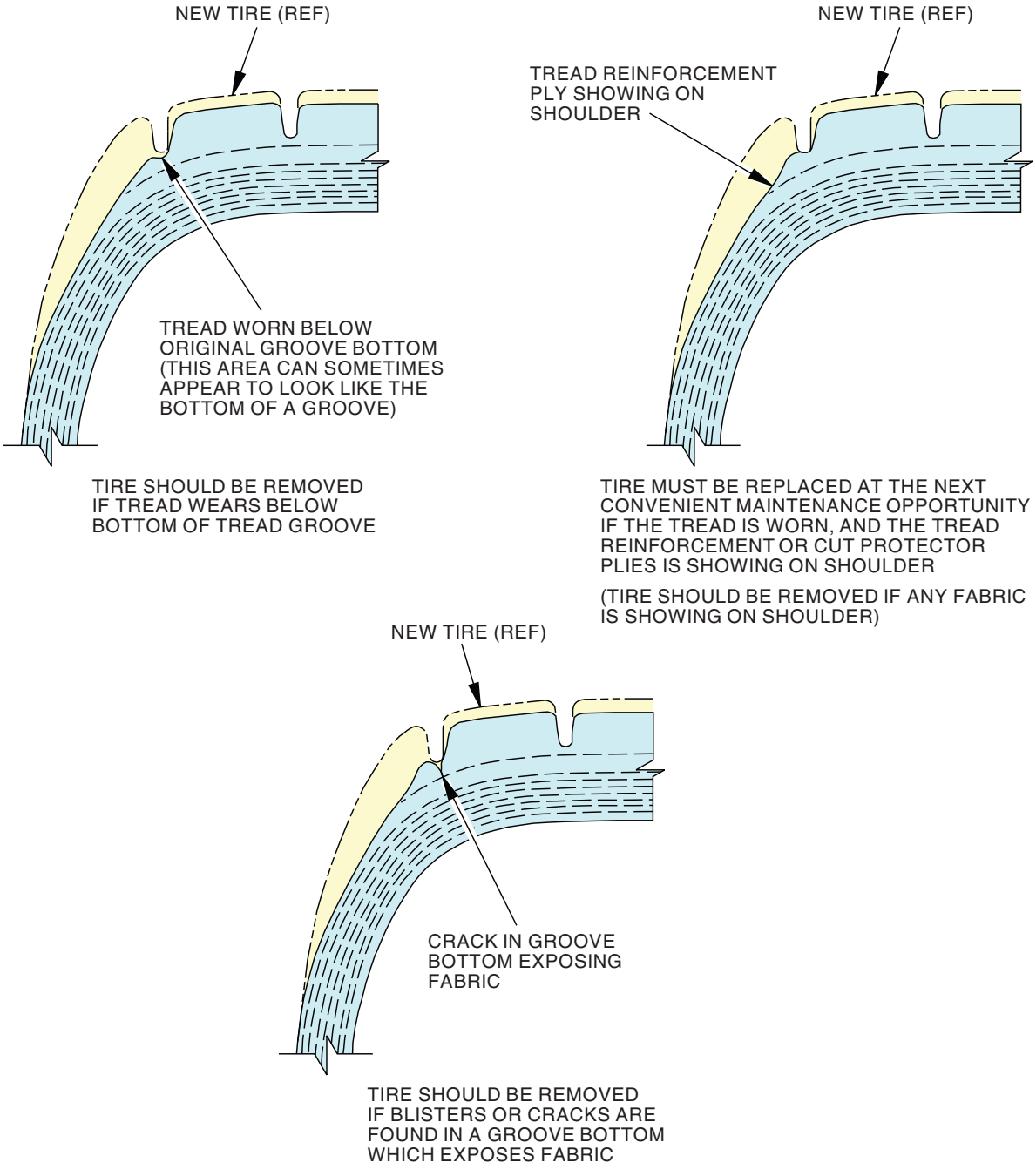
TREAD SEPARATION

TIRES WITH BLISTERS OR BULGES IN THE TREAD ARE NOT SERVICEABLE. REMOVE TIRE IMMEDIATELY.

2414441 S00061535187_V1

Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tire Wear Conditions (Example)
Figure 602/32-45-00-990-803 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-45-00-211-801

5. Wheel Bearings Inspection

(Figure 603)

A. General

(1) This procedure will inspect the internal and external areas of the wheel bearings.

B. References

Reference	Title
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)
32-45-21-000-801	Nose Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-45-21-400-801	Nose Landing Gear Wheel and Tire Assembly Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00233	Grease - Aircraft, General Purpose, Wide Temperature - Mobil 28	MIL-PRF-81322
D00378	Grease - Aircraft, General Purpose, Wide Temperature - Aeroshell 22	MIL-PRF-81322
D50005	Grease - Wheel Bearing - Mobil Aviation Grease SHC 100	
G00834	Cloth - Lint-free Cotton	
G50492	Pad - 3M Scotch Brite 7448 Ultra Fine Pad	MIL-A-9962A Type III Grade AAA
G51224	Pad - Abrasive, Scotch-Brite, Very Fine, Maroon 7447	A-A-58054 Type I Grade A

D. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Inspection

SUBTASK 32-45-00-020-002

(1) If not previously done, perform the following tasks as applicable:

(a) For the Main Landing Gear: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

(b) For the Nose Landing Gear: Nose Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-21-000-801.

F. Wheel Bearings Inspection

SUBTASK 32-45-00-020-003

(1) Remove the bearing cone from the bearing cup (Figure 603 (Sheet 1)).

(a) Put the bearing cone in a clean area.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that no unwanted materials get into the bearing cone while setting it aside.

SUBTASK 32-45-00-210-009

(2) Examine the internal and external surfaces of the bearing cup and cone for water or any other unwanted material in the grease.

(a) If water or unwanted material is found in the grease, remove and replace the entire wheel assembly.

NOTE: It is normal for the grease color to change from red or light brown to dark brown, gray, or black. Grease color is not a cause to reject unless the grease color is white. White color is an indication that grease contains unwanted water.

SUBTASK 32-45-00-210-010

(3) Examine the internal and external surfaces of the bearing cup and cone for any obvious signs of cracking, flat spots or damage.

(a) If cracking, flat spots or damage is found, remove and replace the entire wheel assembly, including the wheel bearing cone.

SUBTASK 32-45-00-100-001

(4) Wipe off the grease only from the cup race with a lint-free cloth, G00834.

NOTE: Remove only enough grease to get a clear view of bearing cup race surface.

NOTE: Do not use solvents to clean the bearing cup race surface.

SUBTASK 32-45-00-210-011

(5) Visually examine the cup race with a flashlight for marks, discoloration or corrosion. See Figure 603 (Sheet 2) for examples of marks and discolorations.

(a) If no defects are found, do the steps:

1) Re-apply a thin layer of bearing grease Aeroshell 22 grease, D00378, Mobil 28 grease, D00233 or Mobil Aviation Grease SHC 100 grease, D50005 to the surface of the cup race.

2) Re-insert the wheel bearing cone into the bearing cup.

(b) If discoloration, texture or surface corrosion is found, attempt to remove it by gently using Scotch-Brite 7447 pad, G51224 or Scotch-Brite 7448 pad, G50492. Perform this task for no longer than 2 minutes.

NOTE: Equivalent abrasive papers or cloths to the ones given inhere (with a grade between 600 to 1000) can be used instead.

NOTE: Do not use wire metal brushes, rotary brushes or flappers of any kind to perform this task.

(c) After the area has been reworked, slowly slide a ball point pen, a non-metallic scraper/pick or a bare fingernail along the whole cup race surface to feel for abnormalities:

1) If texture is felt, remove and replace the entire wheel assembly including the wheel bearing cone.

2) If corrosion is still present, remove and replace the entire wheel assembly including the wheel bearing cone.

3) If no texture is felt, do the steps:

a) Re-apply a thin layer of bearing grease Aeroshell 22 grease, D00378, Mobil 28 grease, D00233 or Mobil Aviation Grease SHC 100 grease, D50005 to the surface of the cup race.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Re-insert the wheel bearing cone into the bearing cup.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-45-00-840-002

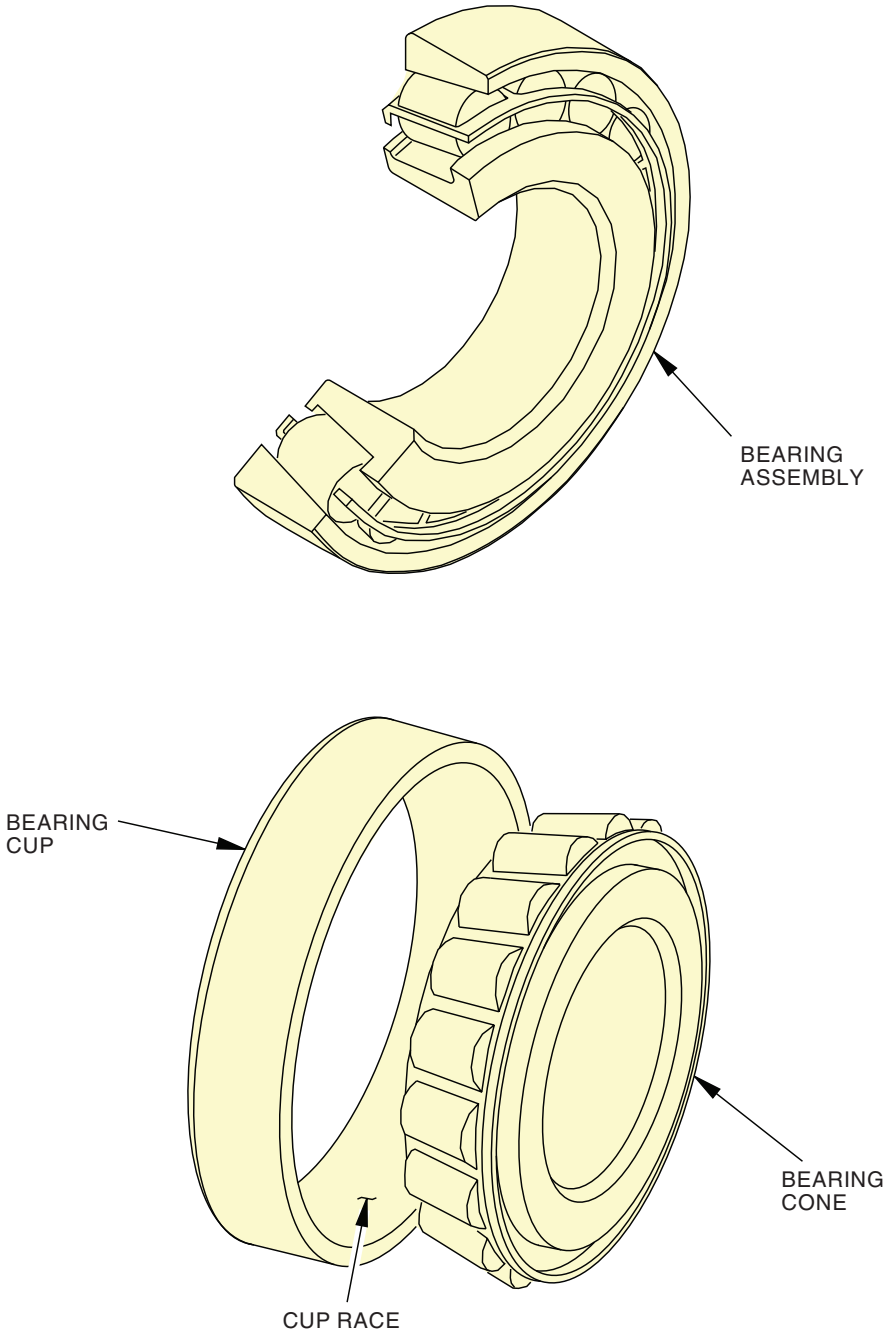
(1) Do as applicable:

(a) For the Main Landing Gear:Main Landing Gear Wheel and Tire Assembly - Installation, TASK 32-45-11-400-801.

(b) For the Nose Landing Gear:Nose Landing Gear Wheel and Tire Assembly Installation, TASK 32-45-21-400-801.

——— END OF TASK ———

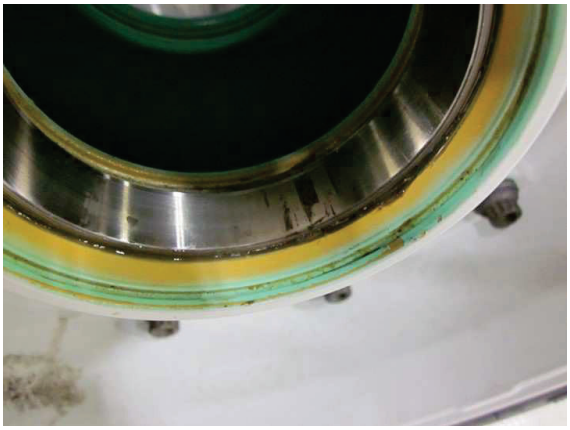
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Wheel Bearings Inspection
Figure 603/32-45-00-990-804 (Sheet 1 of 2)

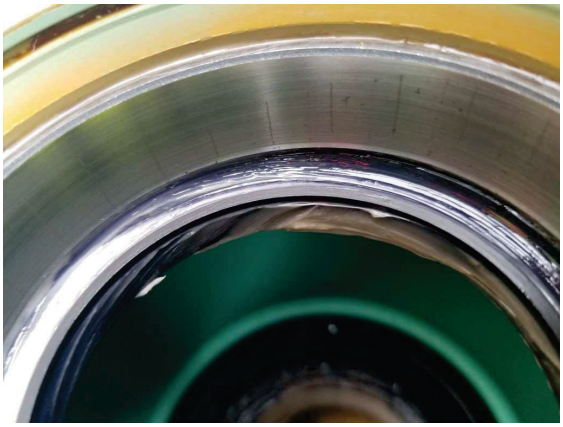
ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE WHEEL BEARING CUP DISCOLORATION



MAIN WHEEL BEARING CUP DISCOLORATION



BEARING CUP MARKS

2932996 S0000710029_V1

Wheel Bearings Inspection
Figure 603/32-45-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRE INFLATION VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the tire inflation valve

(2) An installation of the tire inflation valve.

B. These tasks are for both the main and nose landing gear tires.

C. The procedures for removal and installation of the tire pressure sensor/fill valve are given in PAGEBLOCK 32-49-03/401.

TASK 32-45-08-000-801

2. Tire Inflation Valve Removal

(Figure 401)

A. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4046	Deflator - Tire, Valve Core Removal and Installation (Small Bore) and Tire Deflation Part #: 320008B-SO Supplier: SFN11 Part #: 968RB Supplier: 5W9Y4 Opt Part #: 320008B Supplier: SFN11

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Tire Inflation Valve Removal

SUBTASK 32-45-08-420-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-45-08-420-002

(2) Install chocks around all tires (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-45-08-580-001

(3) Raise the axle on the jack so that there is clearance between the wheel and tire assembly and the ground (TASK 07-10-03-580-801).

SUBTASK 32-45-08-840-001

WARNING MAKE SURE THAT YOU DEFLATE THE TIRE BEFORE YOU REMOVE THE VALVE. IF YOU DO NOT DEFLATE THE TIRE, VALVE PARTS CAN COME OUT OF THE VALVE AT A HIGH VELOCITY WHEN YOU TRY TO REMOVE THE VALVE. THIS CAN CAUSE INJURIES TO PERSONNEL.

WARNING MAKE SURE THAT ALL PERSONNEL ARE CLEAR OF THE INFLATION VALVE ASSEMBLY PATH WHILE YOU REMOVE IT. IF THE INFLATION VALVE ASSEMBLY EJECTS WHILE YOU REMOVE IT, INJURY TO PERSONNEL CAN OCCUR.

WARNING BE VERY CAREFUL WHEN DEFLATING TIRES (606) TO PREVENT INJURY TO PERSONNEL FROM A FLYING INFLATION VALVE CAP AND/OR VALVE CORE.

CAUTION MAKE SURE THAT THE DEFLATED TIRE THAT IS ON THE AXLE DOES NOT TOUCH THE GROUND. IF THE TIRE DOES TOUCH THE GROUND, THE TIRE'S BEAD SEAT CAN MOVE OUT OF ITS POSITION AND BECOME UNSEATED. THE UNSEATED BEAD SEAT CAN CAUSE LEAKAGE WHICH THEN CAN CAUSE DAMAGE TO THE TIRE AND WHEEL.

(4) Deflate the tire with a small bore tire deflator, COM-4046.

SUBTASK 32-45-08-020-003

(5) If it is necessary, remove lockwire from the inflation valve [2].

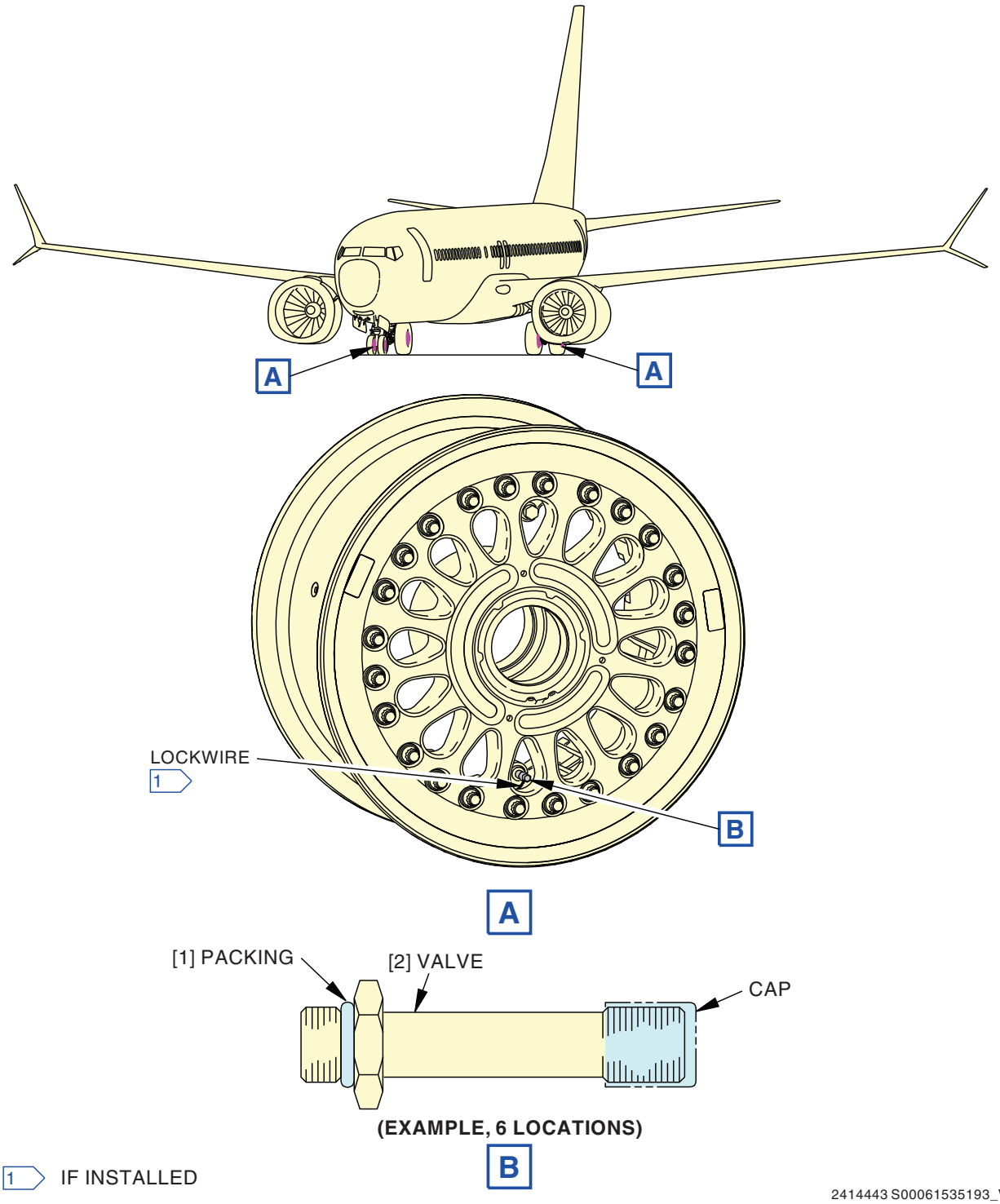
SUBTASK 32-45-08-020-001

(6) Remove the tire inflation valve [2] from the wheel.

(a) Remove and discard the packing [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tire Inflation Valve Installation
Figure 401/32-45-08-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-45-08-400-801

3. Tire Inflation Valve Installation

(Figure 401)

A. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
12-15-51-610-802	Add Nitrogen or Air to the Tire (P/B 301)

B. Tools/Equipment

Reference	Description
STD-7044	Wrench - Socket

C. Consumable Materials

Reference	Description	Specification
A50584	Adhesive - Threadlocking, Loctite 242 (Telair ES406-13)	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
A50585	Adhesive - Threadlocking, Loctite 243 (Telair ES406-13)	
D00504	Grease - Petrolatum	VV-P-236
D50113	Grease - Silicone (PolySi Technologies Inc. - PST 525)	SAE AS8660 (Supercedes MIL-S-8660)
D50395	Vaseline - Mineral, Lubricating	AIR 3565
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50135	Leak Detector - Liquid, Non-Corrosive Soap Compound	MIL-PRF-25567

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Packing	32-45-11-02-023	PTF ALL
		32-45-21-02-018	PTF ALL
		32-49-03-01-045	PTF ALL
		32-49-03-01-047	PTF ALL
2	Inflation valve	32-45-11-02-020	PTF ALL
		32-45-21-02-015	PTF ALL
		32-49-03-01-035	PTF ALL
		32-49-03-01-036	PTF ALL

E. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Tire Inflation Valve Installation

SUBTASK 32-45-08-210-001

(1) Do a check of the port in the wheel half for cracks, burrs, or scratches propagating from the port.

(a) If you find cracks, burrs, or scratches, remove the wheel and the tire.

SUBTASK 32-45-08-420-004

(2) If the main and nose wheel assemblies used the tire pressure sensor as an inflation valve [2], do these steps:

NOTE: There are two configurations available for the tire inflation valve:

- A standard valve
- A combination inflation valve/pressure sensor.

(a) For the Goodrich wheel assembly, do these steps:

1) Apply a thin layer of grease, D00504, for the orange or red packing [1] or PST 525 silicone grease, D50113, for the black packing [1].
2) Prevent contact packing [1] with the thread of the inflation valve [2].
3) Use a socket wrench, STD-7044, to install the inflation valve [2] on the wheel.
4) Tighten the tire inflation valve [2] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).
5) Make sure that the red mark on the tire is aligned with the inflation valve [2].

NOTE: The red dot identifies the lightest area of the tire.

6) If it is necessary, install MS20995C32 lockwire, G01048, between the inflation valve [2] and the outboard wheel.

(b) For the Safran wheel assembly, do these steps:

1) Apply a thin layer of vaseline, D50395, for the red packing [1] or PST 525 silicone grease, D50113, for the black packing [1].
2) Apply Loctite 242 adhesive, A50584, or Loctite 243 adhesive, A50585, to the inflation valve [2] threads.

NOTE: It is necessary to install and tighten the inflation valve within 1 minute of the application of retaining compound. If it is tightened after 1 minute of the application of retaining compound, the bond of the retaining compound will not be satisfactory and inflation valve must be removed and installed again.

3) Do not let the primer or retaining compound touch the packing [1].
4) Use a socket wrench, STD-7044, to install the inflation valve [2] on the wheel.
5) Tighten the tire inflation valve [2] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).
   a) Make sure that the inflation valve [2] was tightened within 1 minute after application of compound.
6) Make sure that the red mark on the tire is aligned with the inflation valve [2].

NOTE: The red dot identifies the lightest area of the tire.

SUBTASK 32-45-08-400-001

(3) If the inflation valve [2] is not a combination valve and tire pressure sensor, do these steps to install the inflation valve [2]:

(a) Install the inflation valve [2] to the main wheel assembly.

1) For Goodrich main wheel assembly, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Apply a thin layer of grease, D00504, for the orange or red packing [1] or PST 525 silicone grease, D50113, for the black packing [1].

b) Prevent contact packing [1] with the thread of the inflation valve [2].

c) Tighten the tire inflation valve [2] to 210 in-lb (23.7 N·m).

d) Install MS20995C32 lockwire, G01048, between the inflation valve [2] and the outer wheel.

2) For the Safran main wheel assembly, do these steps:

a) Apply a thin layer of vaseline, D50395, for the red packing [1] or PST 525 silicone grease, D50113, for the black packing [1].

b) Apply Loctite 242 adhesive, A50584, or Loctite 243 adhesive, A50585, to the inflation valve [2] threads.

c) Prevent contact packing [1] with the thread of the inflation valve [2].

d) Tighten the tire inflation valve [2] to 182.7 in-lb (20.6 N·m) - 197.9 in-lb (22.4 N·m).

(b) Install the inflation valve [2] to the nose wheel assembly:

1) For Goodrich nose wheel assembly, do these steps:

a) Apply a thin layer of grease, D00504, for the orange or red packing [1] or PST 525 silicone grease, D50113, for the black packing [1].

b) Prevent contact packing [1] with the thread of the inflation valve [2].

c) Tighten the tire inflation valve [2] to 175 in-lb (19.8 N·m).

d) Install MS20995C32 lockwire, G01048, between the inflation valve [2] and lockwire anchor bolt.

2) For the Safran nose wheel assembly, do these steps:

a) Apply a thin layer of vaseline, D50395, for the red packing [1] or PST 525 silicone grease, D50113, for the black packing [1].

b) Prevent contact packing [1] with the thread of the inflation valve [2].

c) Tighten the tire inflation valve [2] to 143.4 in-lb (16.2 N·m) - 175.2 in-lb (19.8 N·m).

3) For the Honeywell nose wheel assembly, tighten the tire inflation valve [2] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

SUBTASK 32-45-08-580-002

(4) Inflate the tire (TASK 12-15-51-610-802).

G. Tire Inflation Valve Installation Test

SUBTASK 32-45-08-210-002

(1) Do a check for leaks around base of the tire inflation valve [2] with leak detector, G50135.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-45-08-580-003

(1) Lower the aircraft (TASK 07-10-03-580-801).

SUBTASK 32-45-08-020-002

(2) If it is necessary, remove the chocks.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Main Landing Gear Wheel and Tire Assembly Removal

(2) Main Landing Gear Wheel and Tire Assembly Installation

TASK 32-45-11-000-801

2. Main Landing Gear Wheel and Tire Assembly Removal

(Figure 401 or Figure 402 or Figure 403)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to remove the main landing gear wheel and tire assembly.

B. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-45-00-910-801	Landing Gear Tire - Deflation (P/B 201)
32-49-03-000-802	Main Landing Gear Hubcap Assembly Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1818	Dolly - Wheel/Brake Change Part #: 15F590 Supplier: 56535 Part #: 17500M Supplier: 94861 Part #: 175S-2 Supplier: 94861 Part #: 227A-JMS Supplier: CD856 Part #: 9092-016 Supplier: 00994 Part #: IA3201A-500 Supplier: 053H3 Part #: MH13-003 Supplier: D2029 Part #: STC90475-101 Supplier: 07132 Opt Part #: 175M Supplier: 94861 Opt Part #: 8436-012 Supplier: 00994 Opt Part #: 9092-010 Supplier: 00994 Opt Part #: 9092-015 Supplier: 00994 Opt Part #: PF32-002-1 Supplier: 3D5B2 Opt Part #: TB900 Supplier: 56535

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(Continued)

Reference	Description
COM-4046	Deflator - Tire, Valve Core Removal and Installation (Small Bore) and Tire Deflation Part #: 320008B-SO Supplier: SFN11 Part #: 968RB Supplier: 5W9Y4 Opt Part #: 320008B Supplier: SFN11
SPL-1865	Removal/Installation Equipment - Retention Nut, Wheel MLG and NLG Part #: C32012-12 Supplier: 81205
SPL-1876	Thread Protector Equipment - Axle, MLG Part #: C32022-38 Supplier: 81205 Opt Part #: C32022-15 Supplier: 81205 Opt Part #: C32022-17 Supplier: 81205 Opt Part #: C32022-23 Supplier: 81205
SPL-14320	Thread Protector Eqpt - MLG Axle (1/8 inch undersized threads) Part #: C32022-39 Supplier: 81205 Opt Part #: C32022-32 Supplier: 81205
STD-1132	Gauge - Tire Pressure, 0-300 PSIG (0-2069 KPa), +/- 3 psig accuracy

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-45-11-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-45-11-480-002

(2) Make sure that the tires have chocks installed around them (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-45-11-860-001

(3) Make sure that the parking brake is released.

SUBTASK 32-45-11-580-001

(4) Make sure to lift the wheel and tire assemblies with jacks until they are clear of the ground (TASK 07-10-03-580-801).

SUBTASK 32-45-11-860-002

(5) Make sure that the parking brake is set.

NOTE: When you set the parking brake, the brake rotors will be aligned for wheel installation.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-45-11-840-001

WARNING DEFLATE THE TIRE BEFORE YOU REMOVE THE WHEEL AND TIRE ASSEMBLY. OR, DO AN INSPECTION OF THE WHEEL AND TIRE TO MAKE SURE THAT THEY ARE SAFE. AIR PRESSURE CAN CAUSE AN EXPLOSION OF THE DAMAGED WHEEL AND TIRE COMPONENTS. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING REMOVE WHEEL AND TIRE ASSEMBLIES FROM THE LANDING GEAR ON ISOLATED AXLES. MAKE SURE THAT YOU DO NOT REMOVE MORE THAN ONE WHEEL AND A TIRE ASSEMBLY FROM THE LANDING GEAR. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT AND INJURIES TO PERSONNEL CAN OCCUR.

(6) Do the following steps to examine the tire and wheel assembly to see if it is safe to remove without tire deflation or deflate the tire (TASK 32-45-00-910-801).

NOTE: If you will not install the same wheel and tire assembly, deflate the tire to prevent transporting an inflated tire.

(a) Inspect the outboard wheel and tire assembly [1] and/or the inboard wheel and tire assembly [17] to see if it has the conditions that follow:

1) Inspect the tires and make sure that there are no air leaks, abrasions, unusual worn areas, cuts or flat spots, do this task: Landing Gear Tire Inspection, TASK 32-45-00-700-803.

2) Inspect the wheels and make sure that there is no evidence of wheel damage, including corrosion, loose or damaged or missing tire bolts or tire nuts, overheat damage or cracks, do this task: Wheels Fast Check (Wheel Installed on the Airplane), TASK 32-45-00-700-801.

3) If the wheel and tire assembly has any of the conditions in the above steps, you must deflate the tire.

WARNING MAKE SURE THAT ALL PERSONNEL ARE CLEAR OF THE INFLATION VALVE ASSEMBLY PATH WHILE YOU REMOVE IT. IF THE INFLATION VALVE ASSEMBLY EJECTS WHILE YOU REMOVE IT, INJURY TO PERSONNEL CAN OCCUR.

CAUTION MAKE SURE THAT THE DEFLATED TIRE THAT IS ON THE AXLE DOES NOT TOUCH THE GROUND. IF THE TIRE DOES TOUCH THE GROUND, THE TIRE'S BEAD SEAT CAN MOVE OUT OF ITS POSITION AND BECOME UNSEATED. THE UNSEATED BEAD SEAT CAN CAUSE LEAKAGE WHICH THEN CAN CAUSE DAMAGE TO THE TIRE AND WHEEL.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE TOO MUCH FORCE ON THE VALVE CORE. THE CORE PARTS CAN MOVE AND THE SLEEVE AND THE LOWER END OF THE POPPET WILL STAY IN THE VALVE STEM. IF PARTS STAY IN THE VALVE STEM, DAMAGE CAN OCCUR TO THE VALVE STEM.

(b) Deflate the tire with the small bore tire deflator, COM-4046.

NOTE: If the wheel and tire are not damaged, it is acceptable to leave approximately 50 psi (345 kPa) or 25% of the service pressure in the tire while deflating. Leaving approximately 50 psi (345 kPa) or 25% of the service pressure in the tire can prevent damage to the tire when the wheel and tire are transported.

1) Use the 0-300 psig (0-2069 kpa) tire pressure gauge, STD-1132, to measure the tire pressure (TASK 12-15-51-780-801).

(c) If you cannot use the usual procedure to deflate the tire because the valve core is damaged, do these steps:

1) Turn the valve assembly slowly in a counterclockwise direction until there is gas leakage through the boss.

2) At the same time, push lightly on the valve assembly.

3) After all of the gas has been released, remove the valve assembly from the tire.

SUBTASK 32-45-11-020-004

(7) Do this task: Main Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-802.

F. Main Landing Gear Wheel and Tire Assembly Removal

SUBTASK 32-45-11-020-002

(1) Remove the outboard wheel and tire assembly [1] or inboard wheel and tire assembly [17], do these steps for the applicable wheel:

(a) Remove the two nuts [8], two washers [9], and two bolts [10] that hold the axle nut [16] to the axle.

WARNING

DO NOT PUT FINGERS IN THE LIGHTENING HOLES OF THE WHEEL RETAINING NUT SOCKET TOOL WHEN YOU REMOVE OR INSTALL THE WHEEL RETAINING NUT. THIS CAN CAUSE INJURIES TO PERSONS.

(b) Remove the nut [16] with the socket, SPL-1865.

(c) Remove the retention washer [15].

(d) Install the thread protector assembly, of the thread protector, SPL-1876, or the thread protector, SPL-14320, (for undersized threads) on the axle threads.

(e) Use the wheel/brake changer dolly, COM-1818, to lift and remove the wheel.

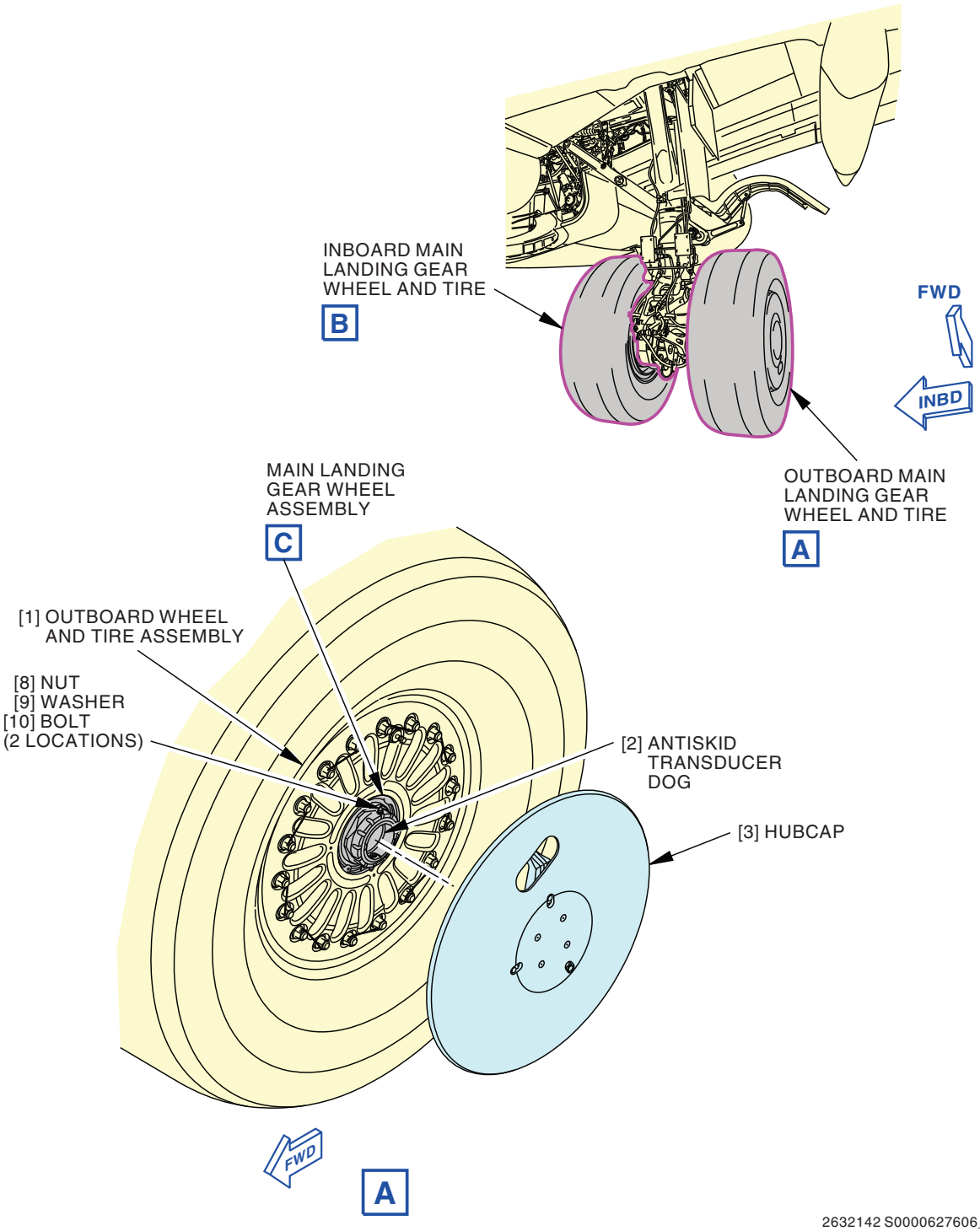
(f) Mark the reason for the tire removal on the tire to aid the inspectors when they examine the tire.

(g) Remove the wheel and tire assembly.

(h) Install the axle protector assembly, of the thread protector, SPL-1876, on the axle.

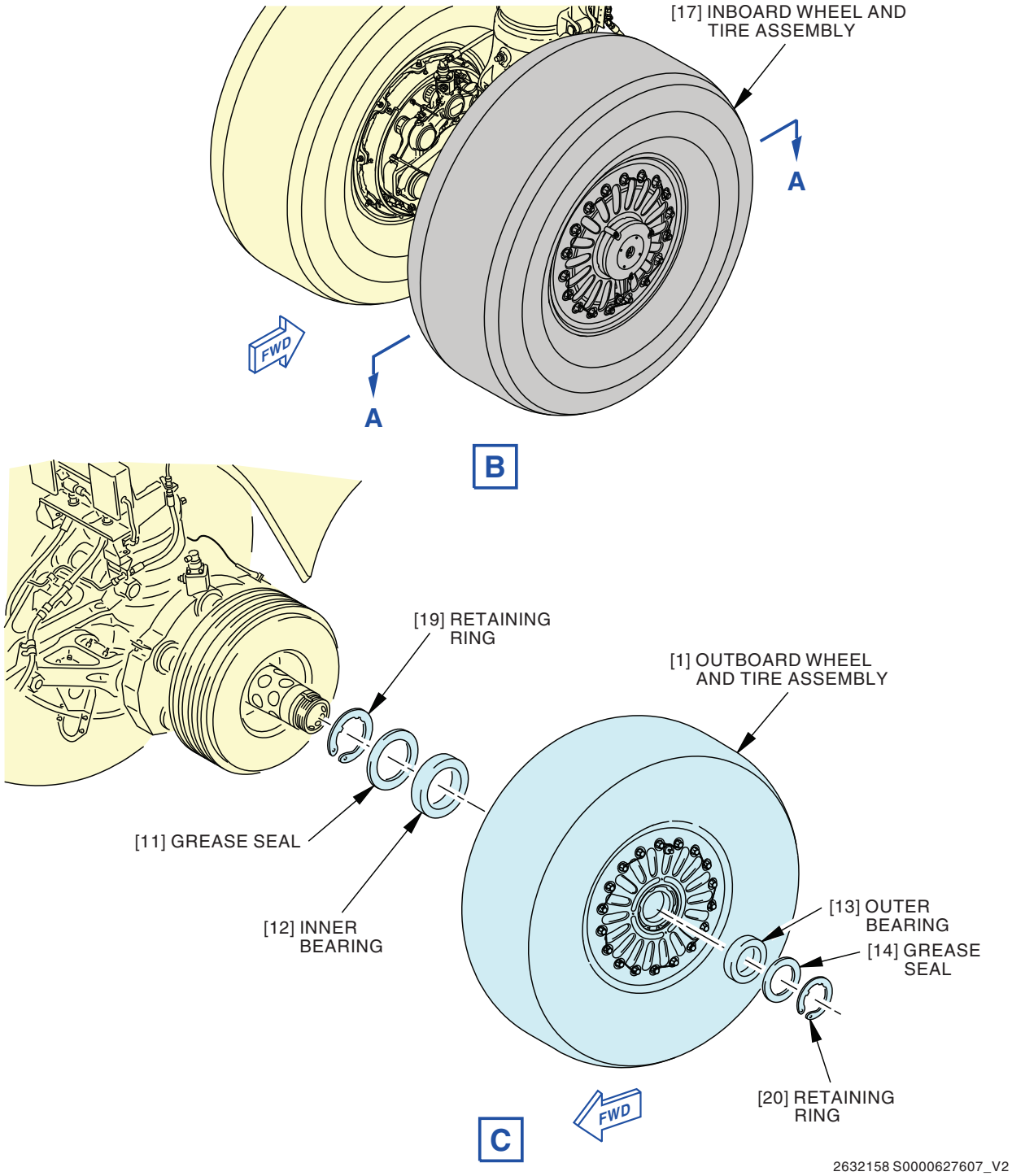
——— END OF TASK ———

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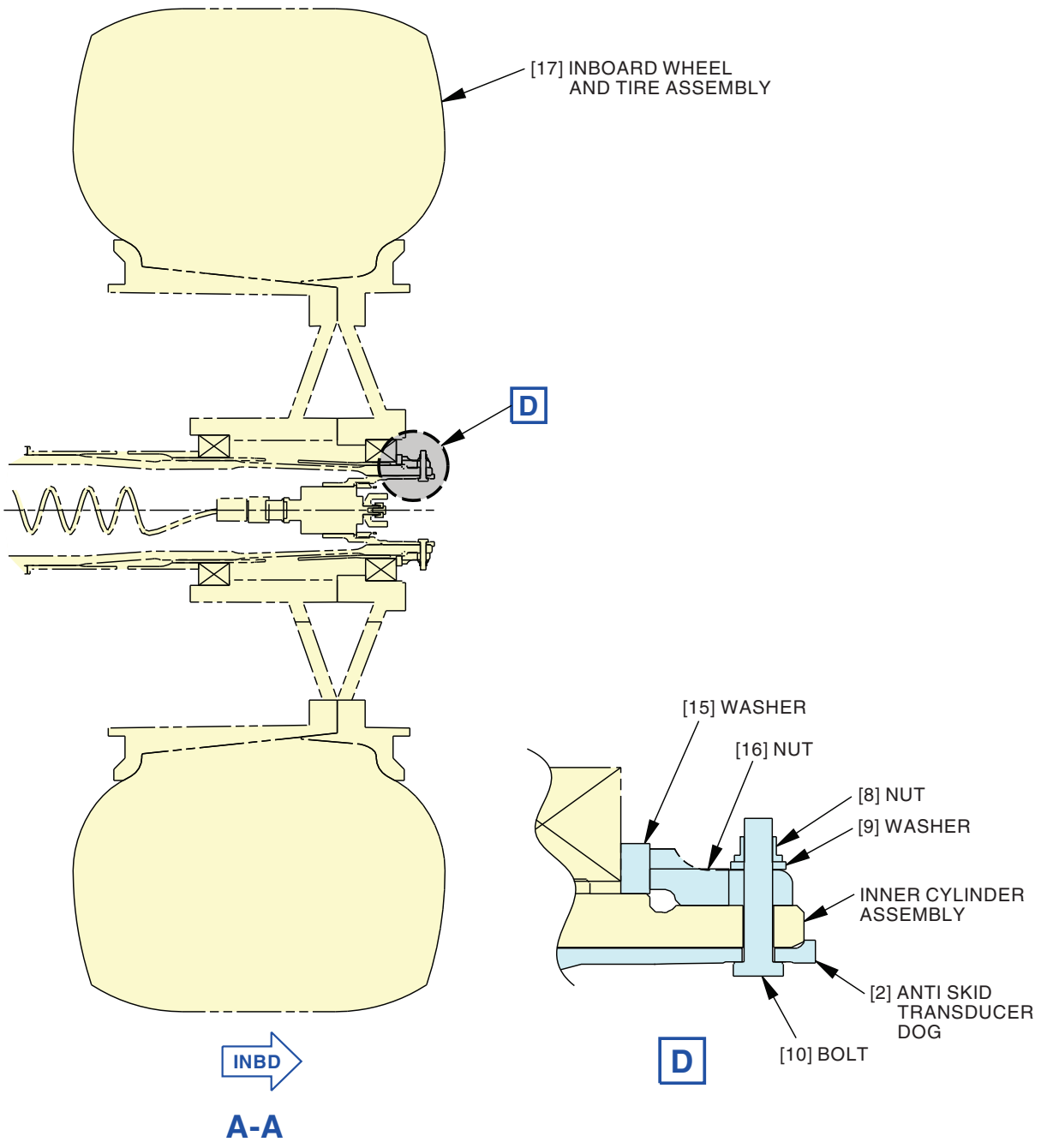
Main Landing Gear Wheel and Tire Assembly Installation
Figure 401/32-45-11-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



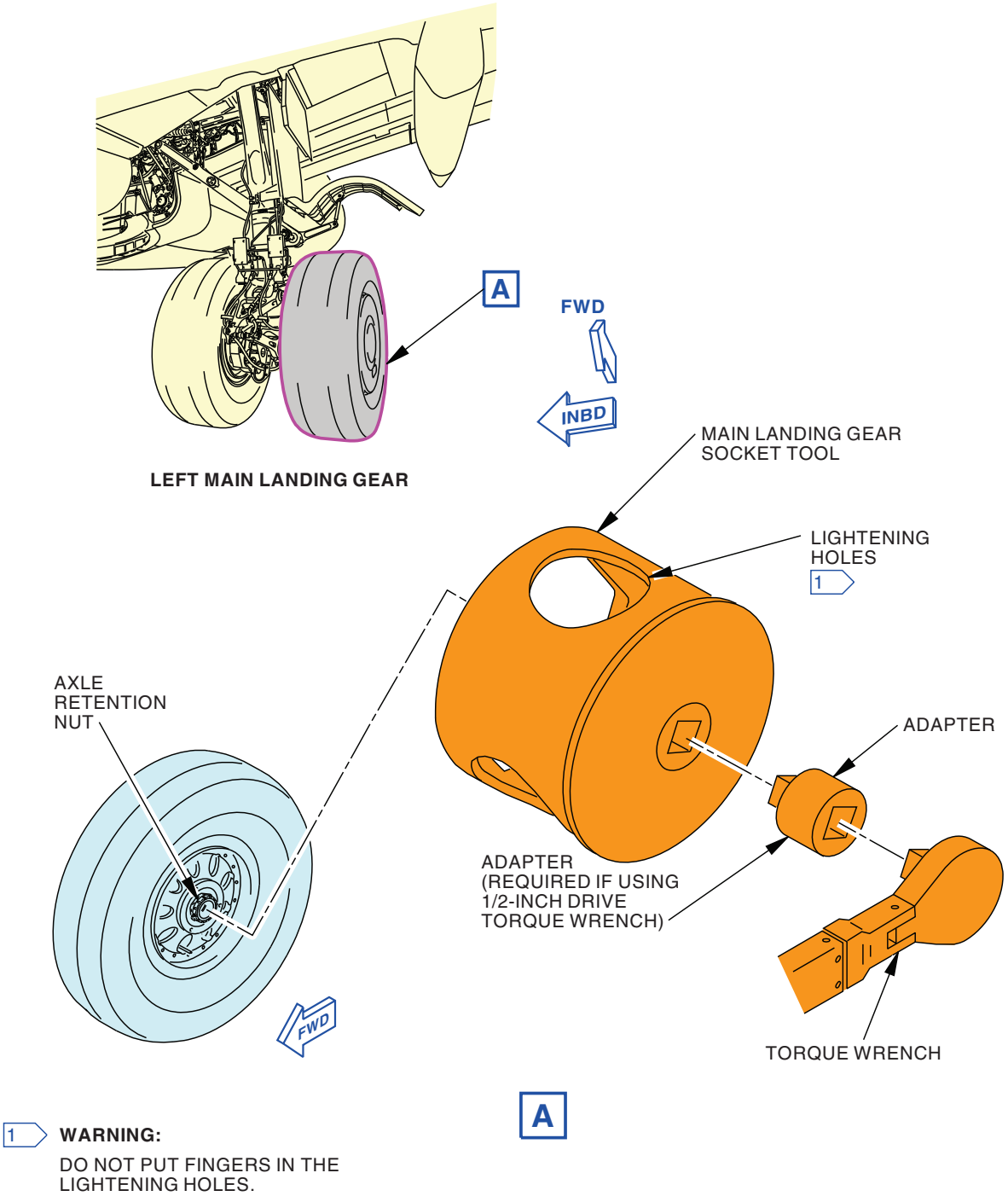
Main Landing Gear Wheel and Tire Assembly Installation
Figure 401/32-45-11-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Wheel and Tire Assembly Installation
Figure 401/32-45-11-990-801 (Sheet 3 of 3)

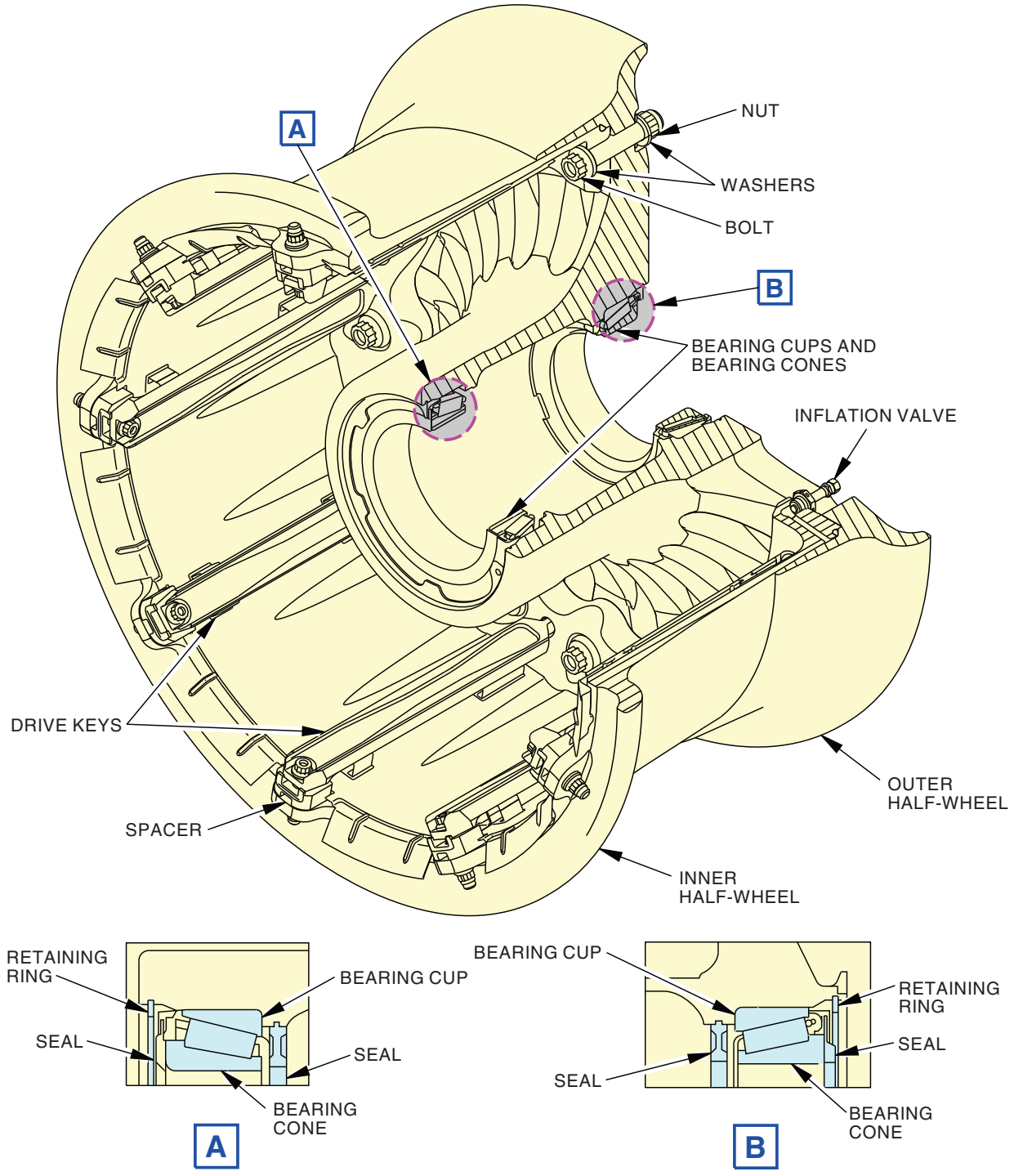
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Main Landing Gear Axle Nut Socket Tool
Figure 402/32-45-11-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Wheel for Carbon Brakes
Figure 403/32-45-11-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-45-11-400-801

3. Main Landing Gear Wheel and Tire Assembly - Installation

(Figure 401 or Figure 402 or Figure 403)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to install main landing gear wheel and tire assembly.

B. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
20-50-11-910-801	Standard Torque Values (P/B 201)
32-41-41-700-803	Main Landing Gear Brake Check (Wheel Removed from the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-49-03-400-802	Main Landing Gear Hubcap Assembly Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1818	Dolly - Wheel/Brake Change Part #: 15F590 Supplier: 56535 Part #: 17500M Supplier: 94861 Part #: 175S-2 Supplier: 94861 Part #: 227A-JMS Supplier: CD856 Part #: 9092-016 Supplier: 00994 Part #: IA3201A-500 Supplier: 053H3 Part #: MH13-003 Supplier: D2029 Part #: STC90475-101 Supplier: 07132 Opt Part #: 175M Supplier: 94861 Opt Part #: 8436-012 Supplier: 00994 Opt Part #: 9092-010 Supplier: 00994 Opt Part #: 9092-015 Supplier: 00994 Opt Part #: PF32-002-1 Supplier: 3D5B2 Opt Part #: TB900 Supplier: 56535
SPL-1865	Removal/Installation Equipment - Retention Nut, Wheel MLG and NLG Part #: C32012-12 Supplier: 81205
SPL-1876	Thread Protector Equipment - Axle, MLG Part #: C32022-38 Supplier: 81205 Opt Part #: C32022-15 Supplier: 81205 Opt Part #: C32022-17 Supplier: 81205 Opt Part #: C32022-23 Supplier: 81205
SPL-14320	Thread Protector Eqpt - MLG Axle (1/8 inch undersized threads) Part #: C32022-39 Supplier: 81205 Opt Part #: C32022-32 Supplier: 81205
STD-1021	Wrench - Torque, 100 to 600 ft-lbs, +/-5% accuracy

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D. Consumable Materials

Reference	Description	Specification
D00233	Grease - Aircraft, General Purpose, Wide Temperature - Mobil 28	MIL-PRF-81322
D00378	Grease - Aircraft, General Purpose, Wide Temperature - Aeroshell 22	MIL-PRF-81322
D50005	Grease - Wheel Bearing - Mobil Aviation Grease SHC 100	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Wheel and tire assembly	32-45-11-02-010	PTF ALL
11	Grease seal	Not Specified	
14	Grease seal	Not Specified	

F. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-45-11-840-002

(1) Do these steps to prepare for installation:

(a) Remove the axle protector assembly, of the thread protector, SPL-1876, from the axle.

(b) Remove all the old grease from the axle.

(c) Remove all the old grease from the inner diameter of the brake.

(d) Do an inspection of the brake (TASK 32-41-41-700-803).

NOTE: Do this inspection with special attention to the whole circumference of each rotor for the signs of a broken rotor drive clip(s).

(e) Do an inspection of the axle.

(f) Examine the antiskid transducer drive components in the axle and hubcap [3], hubcap [18] for damage.

(g) If the wheel is not new, inspect the bearings, seals, and other parts of the wheel assembly.

(h) If the tire is not new, do a tire inspection (TASK 32-45-00-700-803).

(i) If it is necessary to install the wheel inboard and outboard bearing assemblies, do these steps:

1) Apply Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to all of the parts of the bearing assemblies.

CAUTION

MAKE SURE TO INSTALL THE OUTSIDE BEARING ON THE WHEEL, INSTALLATION OF THE INSIDE BEARING ON THE OUTSIDE OF THE WHEEL WILL DAMAGE THE WHEEL.

2) Install the inner bearing [12] in the inner side of the wheel and the outer bearing [13] in the outer side of the wheel.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION: MAKE SURE TO INSTALL THE OUTSIDE GREASE SEAL ON THE WHEEL, INSTALLATION OF THE INSIDE GREASE SEAL IN THE OUTSIDE GREASE SEAL LOCATION WILL DAMAGE THE SEAL.

3) Install the grease seal [11] in the inner side of the wheel and the grease seal [14] in the outer side of the wheel.

4) If part of the assembly, install retention ring [19] in the inner side of the wheel and the retention ring [20] in the outer side of the wheel.

NOTE: Some wheel assemblies may not have the retention rings retaining rings as part of the configuration, or they may look different from those illustrated.

H. Main Landing Gear Wheel and Tire Assembly Installation

SUBTASK 32-45-11-420-001

(1) Install the outboard wheel and tire assembly [1] or inboard wheel and tire assembly [17], do these steps for the applicable wheel:

CAUTION: DO NOT APPLY GREASE TO THE AREA BETWEEN AXLE BEARINGS. HIGH TEMPERATURES CAN OCCUR IN THIS AREA DURING A LANDING, DAMAGE TO THE WHEELS, TIRES AND BRAKES CAN OCCUR.

(a) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to the inner diameter surface of the two bearings where they will touch the axle.

(b) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to the faces of the washer [15] and nut [16], and to the axle threads.

(c) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, on the axle where the axle will touch the wheel bearings.

(d) If you need to align the brake disks, do these steps:

1) Release the parking brake.

CAUTION: MAKE SURE BRAKE ROTORS ARE ALIGNED. DRIVE KEYS ON THE WHEELS CAN DAMAGE THE ROTORS.

2) Align the brake disks.

3) Set the parking brake.

NOTE: This will keep the brake rotors aligned.

(e) Make sure that the thread protector assembly, of the thread protector, SPL-1876, or thread protector, SPL-14320, (for undersized threads) is installed.

(f) Install the wheel with the wheel/brake changer dolly, COM-1818.

1) Make sure that the rotor drive keys on the wheel fully engage the key slots in the brake rotors.

(g) Remove the thread protector assembly, of the thread protector, SPL-1876, or thread protector, SPL-14320, (for undersized threads) from the axle threads.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE TO PUSH WHEEL AND LOCK WASHER ON THE AXLE UNTIL THE WHEEL IS SEATED BEFORE INSTALLING THE AXLE NUT. DAMAGE TO THE WHEEL AND THE AIRCRAFT CAN OCCUR.

CAUTION MAKE SURE THE KEY LOCK IS FIRMLY ENGAGED IN THE KEYWAY GROOVE IN THE AXLE. DAMAGE TO THE WHEEL AND AIRCRAFT CAN OCCUR.

CAUTION DO NOT USE THE AXLE NUT TO PUSH THE WHEEL AND LOCK WASHER ONTO THE AXLE. DAMAGE TO THE WHEEL AND AIRCRAFT CAN OCCUR.

(h) Install the washer [15].

(i) Install the nut [16].

(j) Release the parking brake.

WARNING DO NOT PUT FINGERS IN THE LIGHTENING HOLES OF THE WHEEL RETAINING NUT SOCKET TOOL WHEN YOU REMOVE OR INSTALL THE WHEEL RETAINING NUT. THIS CAN CAUSE INJURIES TO PERSONS.

(k) Do these steps to tighten the nut [16] with the socket, SPL-1865, and torque wrench, STD-1021:

1) While you turn the wheel, tighten the nut [16] to 500 ft-lb (677.9 N·m) - 600 ft-lb (813.5 N·m).
2) Stop the wheel.
3) Loosen the nut [16] to 10 ft-lb (14 N·m) - 30 ft-lb (41 N·m).
4) While you turn the wheel, tighten the nut [16] to 150 ft-lb (203 N·m).
5) If the nut [16] does not align with the holes of the bolts [10], continue to tighten the nut [16] to the first available locking position.

NOTE: There are locking positions every 6 degrees.

(l) Do these steps to install the two bolts [10] to lock the nut [16] to the wheel:

1) Install the two bolts [10] with the heads to the inside of the axle.
2) Install the washers [9] and nuts [8].
3) Check the self locking nuts (TASK 20-50-11-910-801).
4) Tighten the nuts [8] to 63 in-lb (7.1 N·m) - 67 in-lb (7.6 N·m).

NOTE: This is per Standard Torque Values, TASK 20-50-11-910-801 to have the lowest maximum value when both nuts and the bolts can be tightened.

SUBTASK 32-45-11-420-003

(2) Do this task: Main Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-802.

SUBTASK 32-45-11-610-002

(3) Make sure that the landing gear tire pressure is within the servicing limits (TASK 12-15-51-780-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-45-11-580-002

WARNING MAKE SURE THE AREA AROUND THE TIRES IS CLEAR OF PERSONS AND EQUIPMENT. IF THE AREA AROUND THE TIRES IS NOT CLEAR, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Lower the axle and remove the jacks (TASK 07-10-03-580-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR WHEEL AND TIRE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the nose landing gear wheel and tire assembly

(2) An installation of the landing gear wheel and tire assembly.

B. When a replacement of the nose wheel and tire assembly is accomplished, it is recommended to replace the wheel and tire assembly on the opposite side. The differences in diameter between the new and worn tires can contribute to the nose landing gear vibration. To prevent the nose landing gear vibration, the installed tires should have equal pressure, nearly the same inflated diameters, and matching supplier part numbers.

TASK 32-45-21-000-801

2. Nose Landing Gear Wheel and Tire Assembly Removal

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has steps to remove the nose landing gear wheel and tire assembly.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-45-00-700-801	Wheels Fast Check (Wheel Installed on the Airplane) (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-45-00-910-801	Landing Gear Tire - Deflation (P/B 201)
32-49-03-000-801	Nose Landing Gear Hubcap Assembly Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
COM-1818	Dolly - Wheel/Brake Change Part #: 15F590 Supplier: 56535 Part #: 17500M Supplier: 94861 Part #: 175S-2 Supplier: 94861 Part #: 227A-JMS Supplier: CD856 Part #: 9092-016 Supplier: 00994 Part #: IA3201A-500 Supplier: 053H3 Part #: MH13-003 Supplier: D2029 Part #: STC90475-101 Supplier: 07132 Opt Part #: 175M Supplier: 94861 Opt Part #: 8436-012 Supplier: 00994 Opt Part #: 9092-010 Supplier: 00994 Opt Part #: 9092-015 Supplier: 00994 Opt Part #: PF32-002-1 Supplier: 3D5B2 Opt Part #: TB900 Supplier: 56535
COM-4046	Deflator - Tire, Valve Core Removal and Installation (Small Bore) and Tire Deflation Part #: 320008B-SO Supplier: SFN11 Part #: 968RB Supplier: 5W9Y4 Opt Part #: 320008B Supplier: SFN11
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1817	Socket, C32012-3 (included in C32012 Equipment) Part #: C32012-12 Supplier: 81205
SPL-1865	Removal/Installation Equipment - Retention Nut, Wheel MLG and NLG Part #: C32012-12 Supplier: 81205
SPL-1877	Thread Protector Equipment - Axle, NLG Part #: C32022-20 Supplier: 81205 Opt Part #: C32022-2 Supplier: 81205
SPL-1880	Downlock Equipment - NLG and MLG Part #: C32026-15 Supplier: 81205
SPL-14582	Thread Protector Eqpt - NLG Axle (1/16" undersized threads) Part #: C32022-35 Supplier: 81205
STD-1132	Gauge - Tire Pressure, 0-300 PSIG (0-2069 KPa), +/- 3 psig accuracy

D. Location Zones

Zone	Area
713	Nose Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 32-45-21-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the landing gear downlock downlock equipment, SPL-1880, NLG towing lever pin, SPL-1499, are installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-45-21-480-002

(2) Make sure that the tires have chocks installed around them (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-45-21-860-001

(3) Make sure that the parking brake is set.

SUBTASK 32-45-21-580-001

(4) Make sure to lift the nose landing gear and wheel and tire assembly with jacks until they are clear of the ground (TASK 07-10-03-580-803).

SUBTASK 32-45-21-840-001

WARNING DEFLATE THE TIRE BEFORE YOU REMOVE THE WHEEL AND TIRE ASSEMBLY. OR, DO AN INSPECTION OF THE WHEEL AND TIRE TO MAKE SURE THAT THEY ARE SAFE. AIR PRESSURE CAN CAUSE AN EXPLOSION OF THE DAMAGED WHEEL AND TIRE COMPONENTS. INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING REMOVE ONE WHEEL AND THE TIRE ASSEMBLY FROM THE NOSE LANDING GEAR AT A TIME. IF YOU REMOVE THE TWO WHEEL AND THE TIRE ASSEMBLIES AT THE SAME TIME, THE AIRPLANE CAN FALL. STRUCTURAL DAMAGE AND INJURY TO PERSONNEL CAN OCCUR.

(5) Do the following steps to examine the tire and wheel assembly to see if it is safe to remove without tire deflation, or deflate the tire (TASK 32-45-00-910-801):

NOTE: If you will not install the same tire and wheel assembly, deflate the tire to prevent transporting an inflated tire.

(a) Inspect the wheel and tire assembly [1] to see if it has the conditions that follow:

1) Inspect the tires and make sure that there are no air leaks, abrasions, unusual worn areas, cuts or flat spots (TASK 32-45-00-700-803).

2) Inspect the wheels and make sure that there is no evidence of wheel damage, including corrosion, loose or damaged or missing tie bolts or tie nuts, overheat damage or cracks (TASK 32-45-00-700-801).

3) If the wheel and tire assembly [1] has any of the conditions in the above steps, you must deflate the tire.

WARNING MAKE SURE THAT ALL PERSONNEL ARE CLEAR OF THE INFLATION VALVE ASSEMBLY PATH WHILE YOU REMOVE IT. IF THE INFLATION VALVE ASSEMBLY EJECTS WHILE YOU REMOVE IT, INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING BE CAREFUL WHEN DEFLATING TIRES, A FLYING INFLATION VALVE CAP AND CORE CAN CAUSE DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL.

CAUTION MAKE SURE THAT THE DEFLATED TIRE THAT IS ON THE AXLE DOES NOT TOUCH THE GROUND. IF THE TIRE DOES TOUCH THE GROUND, THE TIRE'S BEAD SEAT CAN MOVE OUT OF ITS POSITION AND BECOME UNSEATED. THE UNSEATED BEAD SEAT CAN CAUSE LEAKAGE WHICH THEN CAN CAUSE DAMAGE TO THE TIRE AND WHEEL.

(b) Deflate the tire with the small bore tire deflator, COM-4046.

1) Use the 0-300 psig (0-2069 kpa) tire pressure gauge, STD-1132, to measure the tire pressure (TASK 12-15-51-780-801).

(c) If you cannot use the usual procedure to deflate the tire because the valve core is damaged, do these steps:

1) Turn the valve assembly slowly in a counterclockwise direction until there is gas leakage through the boss.

2) At the same time, push lightly on the valve assembly.

3) After all of the gas has been released, remove the valve assembly from the tire.

SUBTASK 32-45-21-020-002

(6) Do this task: Nose Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-801.

F. Nose Landing Gear Wheel and Tire Assembly Removal

SUBTASK 32-45-21-020-001

(1) Do these steps to remove the wheel and tire assembly [1]:

(a) Remove the nuts [13], washers [14], and bolts [17] that hold the axle nut [16] to the axle.

(b) Remove the axle nut [16] with the NLG socket, C32012-3, SPL-1817, from kit socket, SPL-1865.

(c) Remove the retention washer [15].

(d) Install the axle thread protector, SPL-1877, or the thread protector, SPL-14582 (if the axle has undersized threads).

1) If it is necessary, use tape to hold the thread protector to the axle.

CAUTION MAKE SURE TO KEEP CLEARANCE BETWEEN THE NOSE-WHEEL INNER BEARING AND PAINTED SURFACES OF THE NOSE GEAR AXLE DURING REMOVAL AND INSTALLATION. FRICTION CAN CAUSE DAMAGE TO NOSE-WHEEL AXLE PAINT AND OTHER EQUIPMENT.

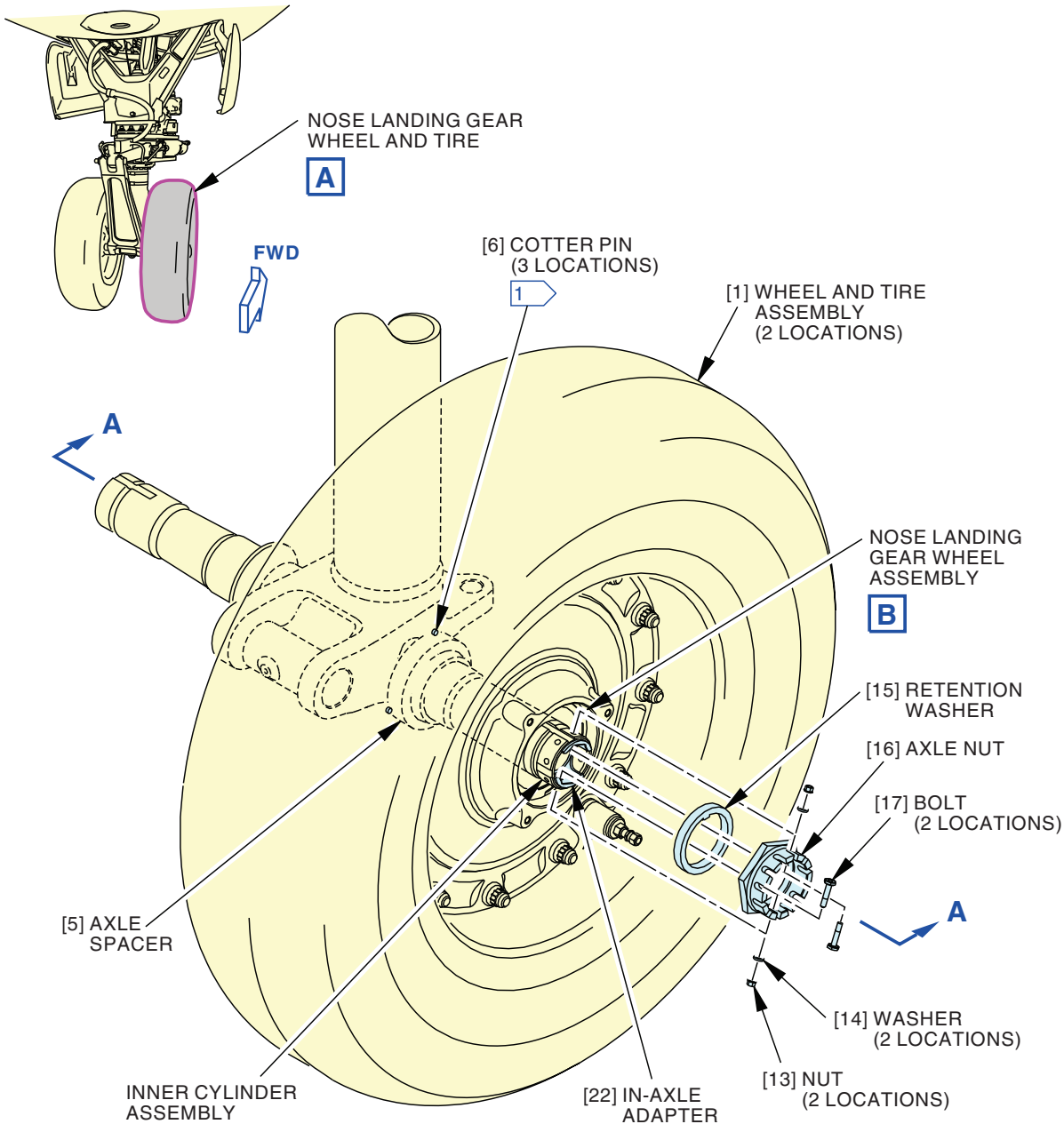
(e) Use the wheel/brake changer dolly, COM-1818, to lift and remove the wheel and tire assembly [1].

(f) Remove the wheel and tire assembly [1].

(g) Mark the reason for the tire removal on the tire to aid the inspectors when they examine the tire.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE LANDING GEAR WHEEL AND TIRE (EXAMPLE)

A

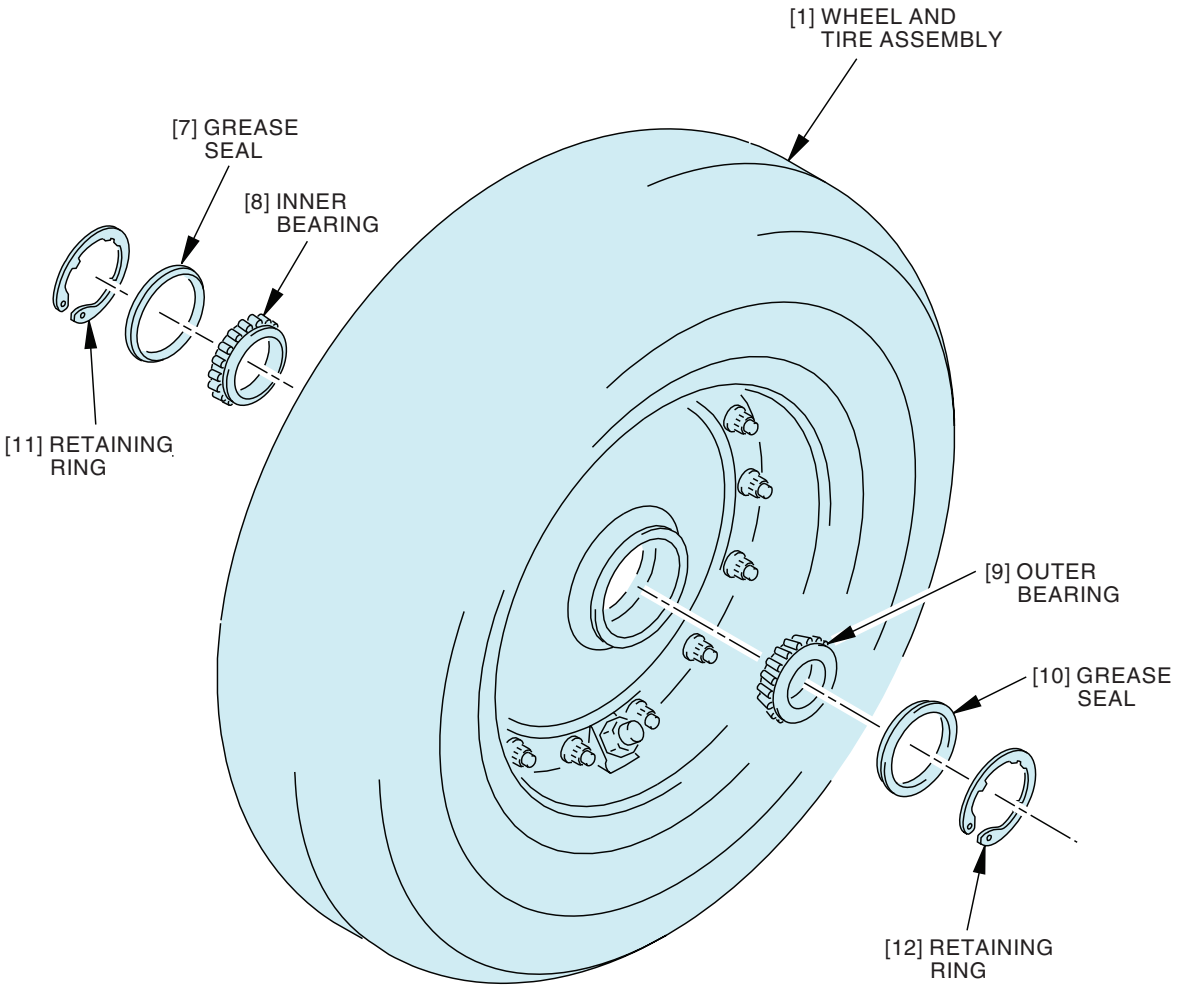
NOTE:

1 SPACERS WITH SIX HOLES USE LOCKWIRE

2818504 S0000650797_V3

Nose Landing Gear Wheel and Tire Assemblies Installation
Figure 401/32-45-21-990-803 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE LANDING GEAR WHEEL ASSEMBLY

B

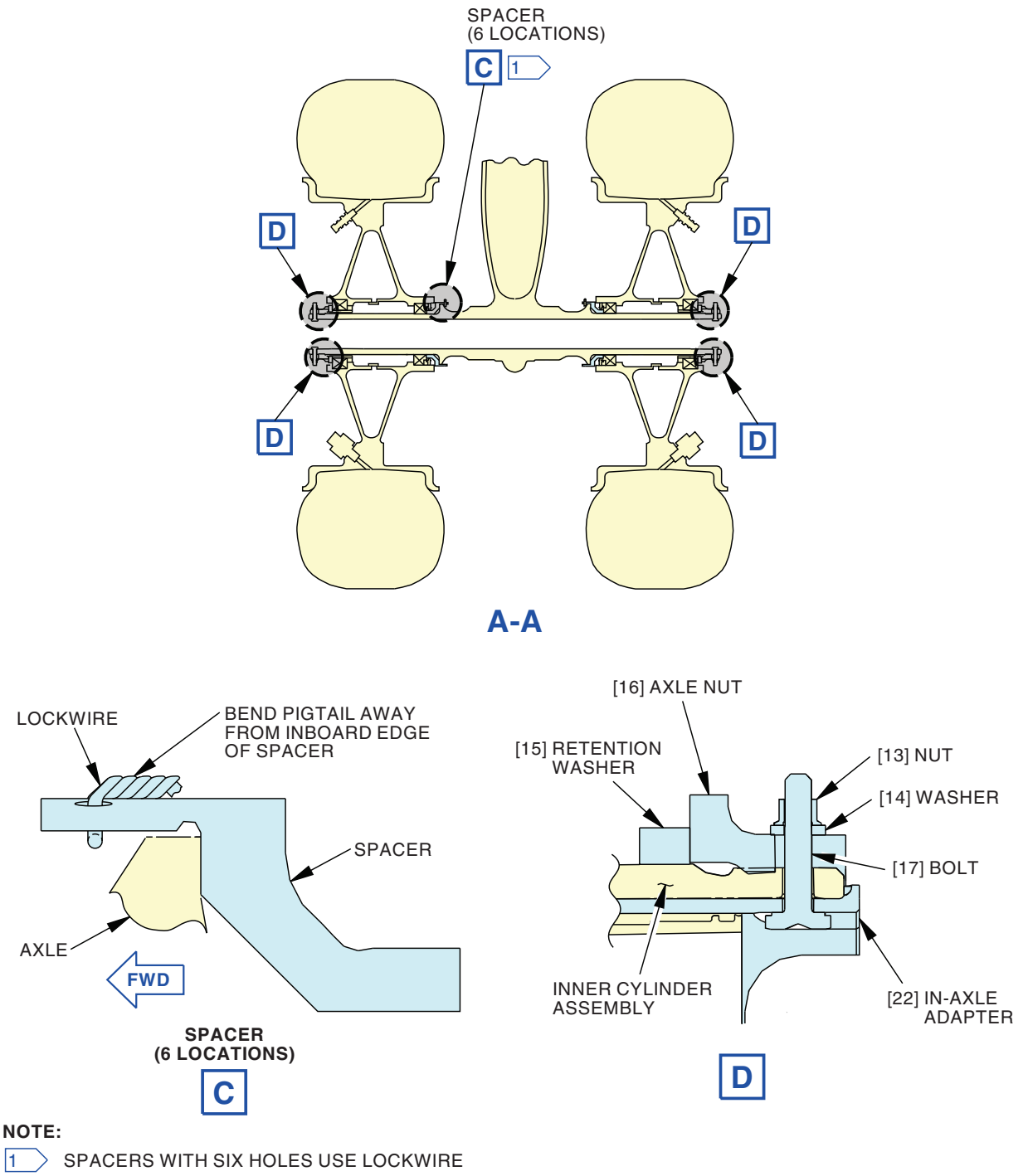
NOTE:

RETAINING RINGS CAN LOOK DIFFERENT.

2414450 S00061535207_V1

Nose Landing Gear Wheel and Tire Assemblies Installation
Figure 401/32-45-21-990-803 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



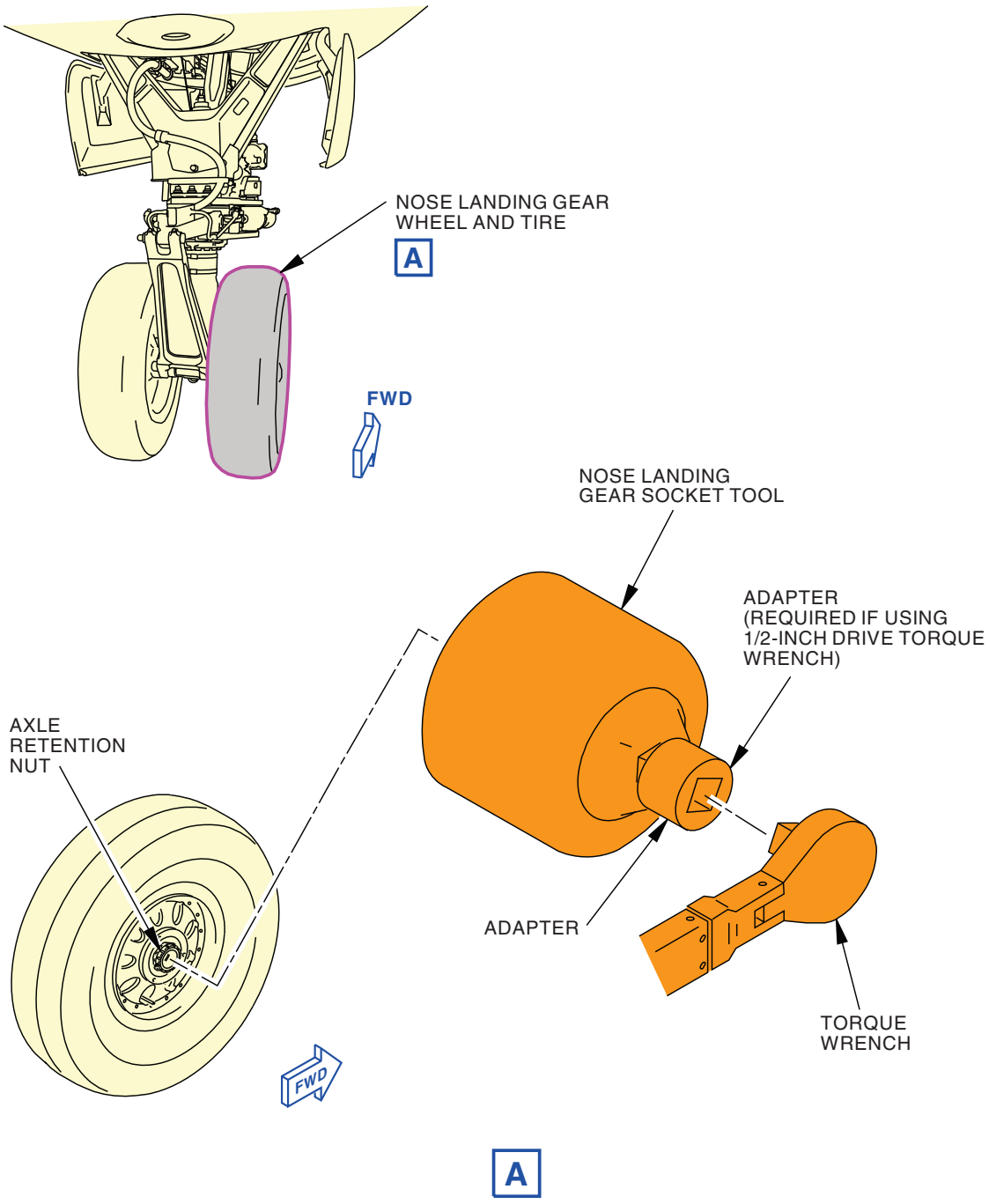
NOTE:

1 SPACERS WITH SIX HOLES USE LOCKWIRE

2818648 S0000651055_V3

Nose Landing Gear Wheel and Tire Assemblies Installation
Figure 401/32-45-21-990-803 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414452 S00061535209_V1

Nose Landing Gear Axle Nut Socket Tool
Figure 402/32-45-21-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-45-21-400-801

3. Nose Landing Gear Wheel and Tire Assembly Installation

(Figure 401, Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has the steps to install the nose landing gear wheel and tire assembly.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
20-50-11-910-801	Standard Torque Values (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-21-71-200-801	Nose Gear Axle Detailed Inspection for Corrosion (P/B 601)
32-45-00-700-803	Landing Gear Tire Inspection (P/B 601)
32-49-03-400-801	Nose Landing Gear Hubcap Assembly Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1818	Dolly - Wheel/Brake Change Part #: 15F590 Supplier: 56535 Part #: 17500M Supplier: 94861 Part #: 175S-2 Supplier: 94861 Part #: 227A-JMS Supplier: CD856 Part #: 9092-016 Supplier: 00994 Part #: IA3201A-500 Supplier: 053H3 Part #: MH13-003 Supplier: D2029 Part #: STC90475-101 Supplier: 07132 Opt Part #: 175M Supplier: 94861 Opt Part #: 8436-012 Supplier: 00994 Opt Part #: 9092-010 Supplier: 00994 Opt Part #: 9092-015 Supplier: 00994 Opt Part #: PF32-002-1 Supplier: 3D5B2 Opt Part #: TB900 Supplier: 56535
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1817	Socket, C32012-3 (included in C32012 Equipment) Part #: C32012-12 Supplier: 81205
SPL-1865	Removal/Installation Equipment - Retention Nut, Wheel MLG and NLG Part #: C32012-12 Supplier: 81205
SPL-1877	Thread Protector Equipment - Axle, NLG Part #: C32022-20 Supplier: 81205 Opt Part #: C32022-2 Supplier: 81205

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Description
SPL-14582	Thread Protector Eqpt - NLG Axle (1/16" undersized threads) Part #: C32022-35 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00233	Grease - Aircraft, General Purpose, Wide Temperature - Mobil 28	MIL-PRF-81322
D00378	Grease - Aircraft, General Purpose, Wide Temperature - Aeroshell 22	MIL-PRF-81322
D50005	Grease - Wheel Bearing - Mobil Aviation Grease SHC 100	
G00440	Lockwire - MS20995C41, Corrosion Resistant Steel - 0.041 Inch (1.0414 mm) Diameter	NASM20995

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Wheel and tire assembly	32-45-21-02-005	PTF ALL
7	Grease seal	32-45-21-16-010	PTF ALL
10	Grease seal	32-45-21-16-025	PTF ALL

F. Location Zones

Zone	Area
713	Nose Landing Gear

G. Prepare for the Installation

SUBTASK 32-45-21-840-002

(1) Do these steps to prepare for installation:

(a) Remove the axle thread protector, SPL-1877, or thread protector, SPL-14582, from the axle.

(b) Remove all the old grease from the axle.

(c) If the wheel is not new, inspect the bearings, the seals, and other parts of the wheel assembly for damage.

(d) If the tire is not new, do a tire inspection (TASK 32-45-00-700-803).

(e) Do these steps if the component of the nose wheel assembly (the bearings, seals, and other parts) need to be installed:

1) Apply Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to the inner bearing [8], outer bearing [9], grease seal [7], and grease seal [10].

CAUTION

MAKE SURE TO INSTALL THE OUTSIDE BEARING ON THE WHEEL, INSTALLATION OF THE INSIDE BEARING ON THE OUTSIDE OF THE WHEEL WILL DAMAGE THE WHEEL.

2) Install the inner bearing [8] on the inner side of the wheel and outer bearing [9] on the outer side of the wheel.

a) Check that the inner bearing [8] is installed on the wheel side opposite inflation valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE TO INSTALL THE OUTSIDE GREASE SEAL ON THE WHEEL, INSTALLATION OF THE INSIDE GREASE SEAL IN THE OUTSIDE GREASE SEAL LOCATION WILL DAMAGE THE SEAL.

3) Install the grease seal [7] on the inner side of the wheel and grease seal [10] on the outer side of the wheel.
   a) Check that tab of grease seal fits indentation of cutout in wheel housing.
4) Install the retaining ring [11] on the inner side of the wheel and retaining ring [12] on the outer side of the wheel.

H. Nose Landing Gear Wheel and Tire Assembly Installation

SUBTASK 32-45-21-420-001

(1) Do these steps to install the wheel and tire assembly [1]:
   (a) Inspect the axles for nicks, gouges, and damaged chrome plate.
   (b) If the axle spacer is removed, do a general visual inspection of the axles for corrosion (TASK 32-21-71-200-801).

CAUTION MAKE SURE THAT THE SPACER IS CORRECTLY INSTALLED ON THE AXLE. THE SPACER CAN MOVE WHEN THE WHEEL AND TIRE ASSEMBLY IS REMOVED. IF IT IS NOT IN ITS CORRECT POSITION, THE WHEEL AND TIRE ASSEMBLY WILL NOT INSTALL CORRECTLY. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

   (c) Make sure that the axle spacer [5] is installed on the axle.

   NOTE: The axle spacer is inboard of the wheel on the axle. The inner bearing will touch the spacer when the wheel is installed.

   1) For airplanes that use the cotter pins, make sure that the cotter pins [6] are installed to hold the spacer on the axle.

      NOTE: An axle spacer that uses cotter pins has 3 holes.

      a) Make sure that there is no interference between the cotter pins [6] installed on the spacer and lower torsion link lug during wheel installation.

   2) For airplanes that use the lockwire, make sure that you install MS20995C41 lockwire, G00440.

      a) Make sure that the lockwire ends pigtails are bent away from the inboard edge of the spacer (Figure 401).

      b) Examine the completed installation to be sure that the lockwire ends do not touch the axle or torsion link lugs.

      c) Adjust the installed lockwire as necessary to be sure there will be no interference with the landing gear parts during landing gear operations.

         NOTE: An axle spacer that uses lockwire has six holes.

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WARNING DO NOT APPLY GREASE TO THE AREA BETWEEN THE AXLE BEARINGS. HIGH TEMPERATURES IN THIS AREA DURING A LANDING CAN CAUSE ALL GREASE IN THIS AREA TO BURN. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(d) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to the face and outer diameter of the axle spacer [5].

(e) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to the inner diameter surface of the two bearings where they will touch the axle.

(f) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, to the faces of the retention washer [15], axle nut [16], and axle threads.

(g) Apply a thin layer of Aeroshell 22 grease, D00378, Mobil 28 grease, D00233, or Mobil Aviation Grease SHC 100 grease, D50005, on the axle where the axle will touch the wheel bearings.

(h) Install the axle thread protector, SPL-1877, or thread protector, SPL-14582 (if the axle has undersized threads).

CAUTION MAKE SURE TO KEEP CLEARANCE BETWEEN THE NOSE-WHEEL INNER BEARING AND PAINTED SURFACES OF THE NOSE GEAR AXLE DURING REMOVAL AND INSTALLATION. FRICTION CAN CAUSE DAMAGE TO NOSE-WHEEL AXLE PAINT AND OTHER EQUIPMENT.

(i) Use the wheel/brake changer dolly, COM-1818, to install the wheel and tire assembly [1].

(j) Install the wheel and tire assembly [1].

(k) Remove the axle thread protector, SPL-1877, or thread protector, SPL-14582, from the axle.

(l) Align the key of the washer with the keyway of the axle and install the retention washer [15].

(m) Install the axle nut [16].

(n) Do these steps to tighten the axle nut [16] with the NLG socket, C32012-3, SPL-1817, from kit socket, SPL-1865:

1) While you turn the wheel, tighten the axle nut [16] to 80 ft-lb (108.5 N·m) - 100 ft-lb (135.6 N·m).
2) Stop the wheel.
3) Loosen the axle nut [16] to 5 ft-lb (6.8 N·m) - 15 ft-lb (20.3 N·m) torque.
4) While you turn the wheel, tighten the axle nut [16] to 30 ft-lb (40.7 N·m).
5) Make sure that the axle nut [16] locking holes align.
6) If it is necessary, tighten the axle nut [16] to the first available locking position.

NOTE: There are locking positions every 15 degrees.

(o) Install the lock bolts [17] to lock the axle nut [16] to the in-axle adapter [22] and inner cylinder assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Install the lock bolts [17] through the holes in the in-axle adapter [22], inner cylinder assembly, and axle nut [16] with the heads to the inside of the inner cylinder assembly.
2) Install the washers [14] and nuts [13].
3) Check the self locking nut (TASK 20-50-11-910-801).
4) Tighten the nuts [13] to 34 in-lb (3.8 N·m) - 36 in-lb (4.1 N·m).

SUBTASK 32-45-21-580-002

(2) Make sure that the tire is inflated to at least half of the normal inflation pressure before you remove the jacks (TASK 12-15-51-780-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-45-21-580-003

(1) Lower the axle and remove the jacks (TASK 07-10-03-580-803).

SUBTASK 32-45-21-420-002

(2) Do this task: Nose Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-801.

SUBTASK 32-45-21-840-003

(3) Remove the NLG towing lever pin, SPL-1499, from the nose wheel steering mechanism (TASK 32-00-01-080-801).

SUBTASK 32-45-21-610-001

(4) Do this task: Landing Gear Tire Pressure Check and Tire Servicing, TASK 12-15-51-780-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE WHEEL SPIN BRAKE LINING - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks for the nose wheel spin brake lining.

(1) Removal of the nose wheel spin brake lining

(2) Installation of the nose wheel spin brake lining

(3) Inspection of the nose wheel spin brake lining.

B. There are two nose wheel spin brake linings, one for each nose landing gear tire.

TASK 32-45-31-000-801

2. Nose Wheel Spin Brake Lining Removal

(Figure 201)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

C. Nose Wheel Spin Brake Lining Removal

SUBTASK 32-45-31-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-45-31-860-001

(2) Set the parking brake.

SUBTASK 32-45-31-020-001

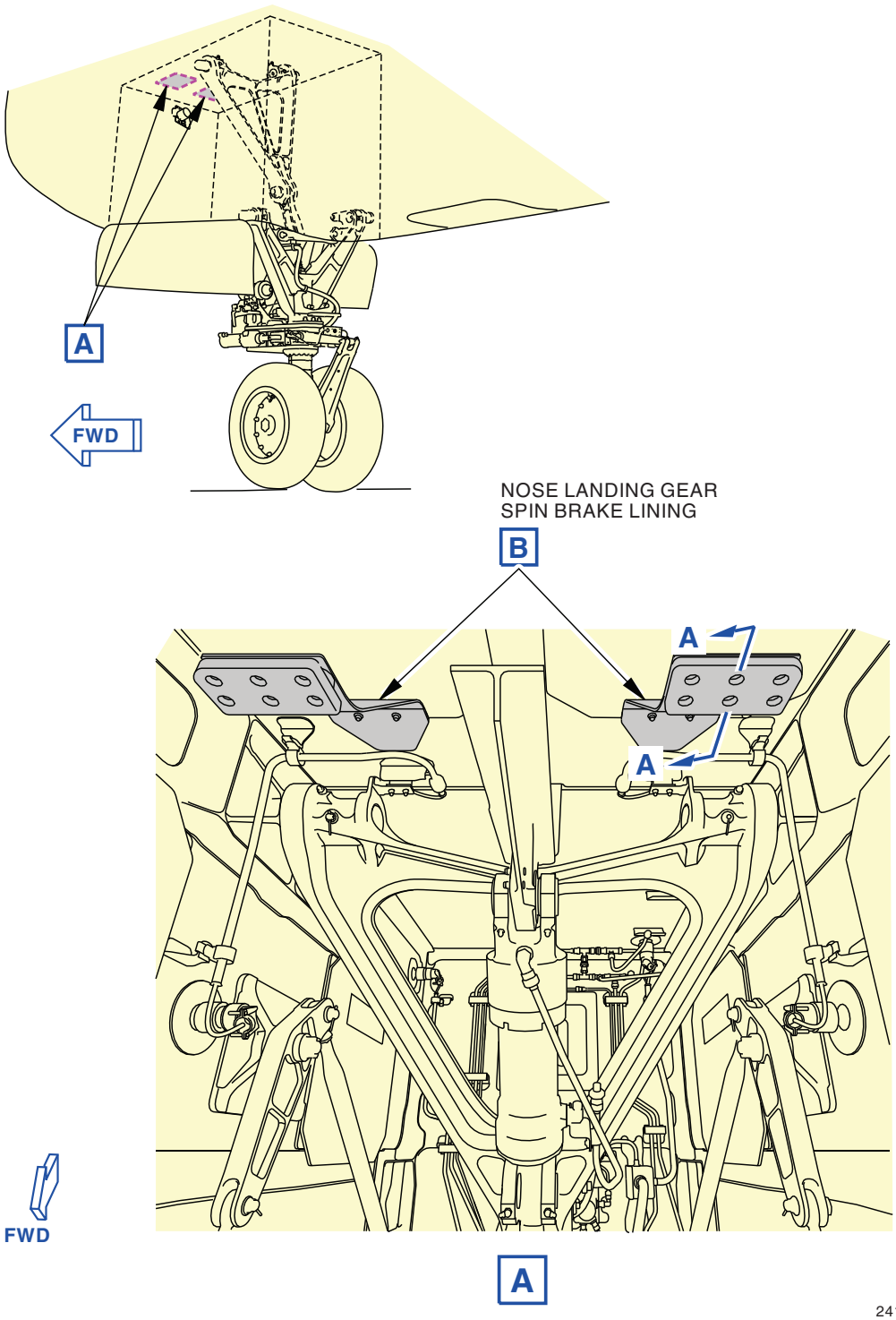
(3) Remove the screws [6], washers [1], and nuts [2] which hold the spin brake lining [5] to the spin brake spring [3].

SUBTASK 32-45-31-020-002

(4) Remove the spin brake lining [5] and the spacers [4].

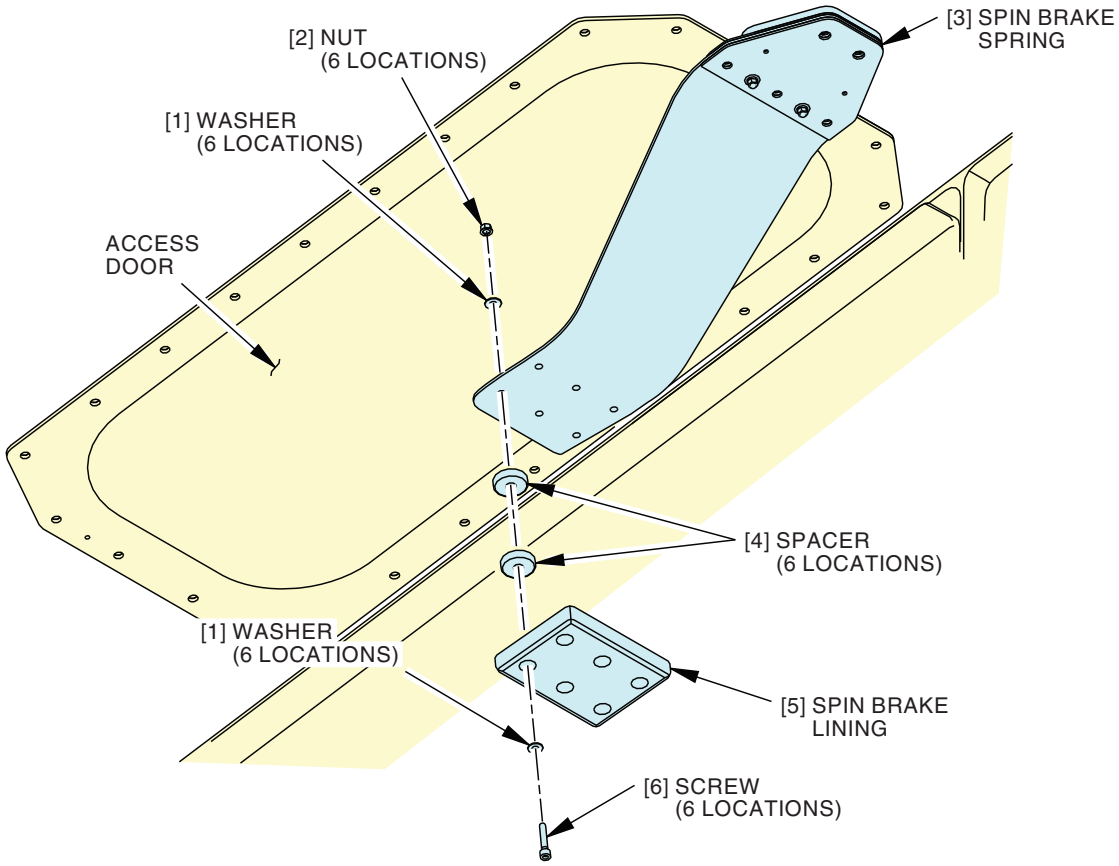
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



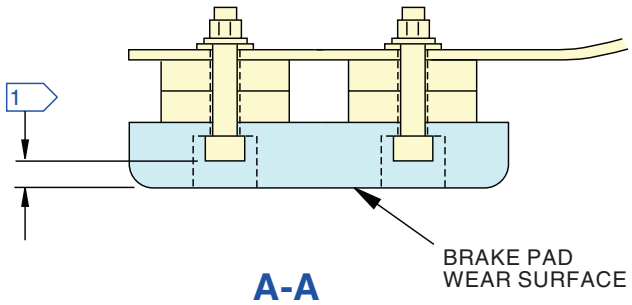
Nose Landing Gear Spin Brake Lining Installation
Figure 201/32-45-31-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE LANDING GEAR SPIN BRAKE LINING
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

B



A-A

1 REPLACE BRAKE PAD IF THICKNESS IS EQUAL TO OR LESS THAN 0.060 INCH (1.524 mm).

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Nose Landing Gear Spin Brake Lining Installation
Figure 201/32-45-31-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-45-31-400-801

3. Nose Wheel Spin Brake Lining Installation

(Figure 201)

A. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Spin brake lining	32-22-11-02-120	PTF ALL

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

D. Nose Wheel Spin Brake Lining Installation

SUBTASK 32-45-31-420-001

(1) Apply sealant, A00247, to the six locations on the spin brake spring [3] where you will install the spacers [4].

SUBTASK 32-45-31-420-002

(2) Hold the spin brake lining [5] in its position on the spin brake spring [3].

SUBTASK 32-45-31-420-003

(3) Do these steps for each screw [6]:

(a) Install the two spacers [4] between the spin brake lining [5] and spin brake spring [3] where you will install the screw [6].

(b) Install the screw [6] and washer [1] through the spin brake lining [5], spacers [4], and spin brake spring [3].

(c) Install the washer [1] and nut [2] on the screw [6].

——— END OF TASK ———

TASK 32-45-31-700-801

4. Nose Wheel Spin Brake Lining Inspection

(Figure 201)

A. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

B. Nose Wheel Spin Brake Lining Inspection

SUBTASK 32-45-31-220-001

(1) Measure the distance from the head of each screw [6] to the wear surface of the spin brake lining [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-45-31-350-001

(2) If the measured distance is equal to or less than 0.060 in. (1.524 mm), replace that spin brake lining [5] (Nose Wheel Spin Brake Lining Removal, TASK 32-45-31-000-801, Nose Wheel Spin Brake Lining Installation, TASK 32-45-31-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRE AND BRAKE MONITORING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has this task:

(1) A system test of the tire and brake monitoring system.

TASK 32-46-00-730-801

2. Tire and Brake Monitoring System - System Test

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Test

SUBTASK 32-46-00-860-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

D. System Test

SUBTASK 32-46-00-410-001

(1) Open and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

(a) Make sure the second circuit breaker is closed within 2.5 seconds after the first circuit breaker.

NOTE: This will make sure that the BRAKE TEMP light and TIRE PRESS light come on.

SUBTASK 32-46-00-730-004

(2) Make sure that the BRAKE TEMP light and the TIRE PRESS light, on the P9 panel, comes on for approximately 2 seconds.

SUBTASK 32-46-00-730-009

(3) Wait for a minimum of 15 seconds.

NOTE: After power application, the Tire and Brake Monitoring System (TBMS) performs an internal check and a system check that takes 5 to 15 seconds.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-46-00-730-003

(4) Select the CAPT SELECTOR or FO SELECTOR switch to display on the left/center or center/right Multi Function Display (MFD).

NOTE: The default screen element will be auto-highlighted.

NOTE: Navigate through the selectable screen elements by rotating the tabber clock-wise or counter-clockwise until the desired screen element is highlighted. Press the center selection "SEL" button to active the screen element's intended behavior.

SUBTASK 32-46-00-730-005

(5) Select the SYS button on the Multi Function Panel (MFP).

SUBTASK 32-46-00-860-002

(6) Make sure that the landing gear synoptic page on the MFD shows the value of "0.0" to "0.1" in all 4 brake temperature readouts.

NOTE: This test assumes that the main landing gear brakes are below 100°F (37.8°C) in temperature.

SUBTASK 32-46-00-730-006

(7) Select MAINT on the MFD.

SUBTASK 32-46-00-730-007

(8) Select STATUS on the MFD.

(a) Make sure that the status messages BRAKE TEMP SYS or TIRE PRESS SYS do not show on the display.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE TEMPERATURE SENSOR - REMOVAL/ INSTALLATION

1. General

A. This procedure contains these tasks:

(1) A removal of a brake temperature sensor

(2) An installation of a brake temperature sensor.

TASK 32-46-21-000-801

2. Brake Temperature Sensor Removal

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)

B. Location Zones

Zone	Area
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Removal

SUBTASK 32-46-21-860-002

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

SUBTASK 32-46-21-420-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

D. Brake Temperature Sensor Removal

SUBTASK 32-46-21-020-001

(1) Disconnect the electrical connector [2] from the brake temperature sensor [1].

SUBTASK 32-46-21-020-004

(2) Remove lockwire from the bolts [3].

SUBTASK 32-46-21-020-002

(3) Remove the bolts [3] and washers [4] that hold the brake temperature sensor [1] to the brake.

SUBTASK 32-46-21-020-003

(4) Remove the brake temperature sensor [1] from the airplane.

(a) If the brake temperature sensor [1] cannot be removed, do these tasks:

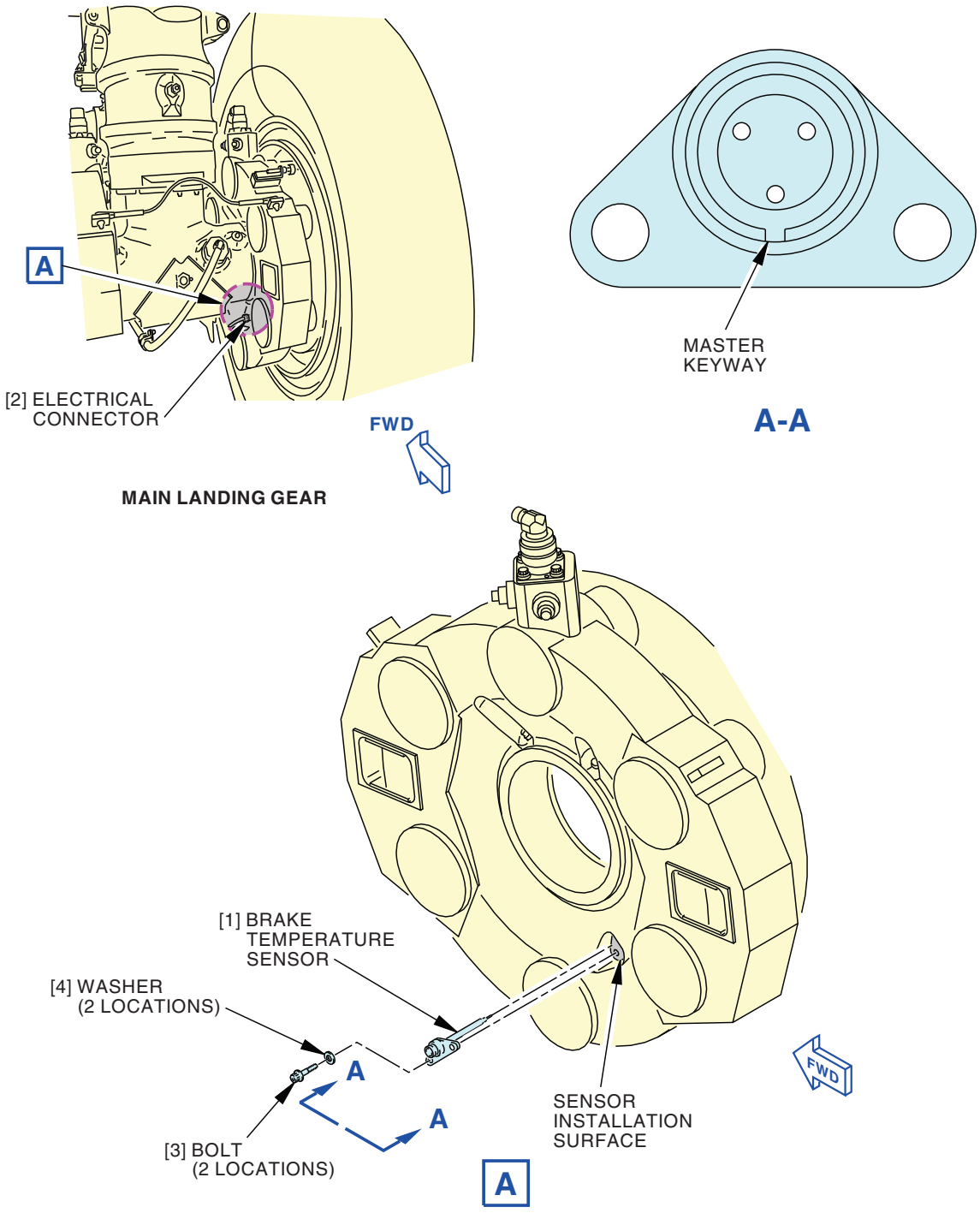
ECCN 9E991 BOEING PROPRIETARY - See title page for details

- Main Landing Gear Brake Removal, TASK 32-41-41-000-801
- Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: The conduit bracket at the bottom of the shock strut can interfere with the removal or installation of the brake temperature sensor.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Temperature Sensor Installation
Figure 401/32-46-21-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-46-21-400-801

3. Brake Temperature Sensor Installation

(Figure 401)

A. References

Reference	Title
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Brake temperature sensor	32-46-21-01-010	PTF ALL

D. Location Zones

Zone	Area
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Brake Temperature Sensor Installation

SUBTASK 32-46-21-010-001

(1) If it is necessary, remove and discard the screws and washers from the holes where the brake temperature sensor [1] will be installed.

NOTE: New brake assembly may be equipped with screws and washers that protect the holes where the brake temperature sensor is installed from the water, dirt, and moisture ingestion.

SUBTASK 32-46-21-420-002

(2) Install the brake temperature sensor [1] into its position in the brake.

(a) If the brake temperature sensor [1] cannot be installed, do these tasks:

- Main Landing Gear Brake Removal, TASK 32-41-41-000-801
- Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: The conduit bracket at the bottom of the shock strut can interfere with the removal and installation of the brake temperature sensor.

SUBTASK 32-46-21-420-003

(3) Install the bolts [3] and washers [4] that hold the brake temperature sensor [1] to the brake.

(a) Tighten the bolts [3] to 65 in-lb (7 N·m) - 100 in-lb (11 N·m).

SUBTASK 32-46-21-420-004

(4) Install MS20995C32 lockwire, G01048, on the bolts [3] with double twist method.

SUBTASK 32-46-21-420-005

(5) Connect the correct electrical connector [2] to the brake temperature sensor [1] (Table 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Table 401/32-46-21-993-801

TEMPERATURE SENSOR	ELECTRICAL CONNECTOR
T455 L INBD	D3322
T454 L OUTBD	D3320
T456 R INBD	D3324
T457 R OUTBD	D3326

SUBTASK 32-46-21-860-003

(6) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F. Brake Temperature Sensor Installation Test

SUBTASK 32-46-21-860-001

(1) Do this task: Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRE AND BRAKE MONITORING UNIT - REMOVAL/ INSTALLATION

1. General

A. This procedure contains these tasks:

(1) Removal of the tire and brake monitoring unit

(2) Installation of the tire and brake monitoring unit.

B. The tire and brake monitoring unit is located in the electronic equipment rack, E1-3, which is located on the left side of the main equipment center.

TASK 32-46-31-000-801

2. Tire and Brake Monitoring Unit Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Tire and Brake Monitoring Unit (TBMU) Removal

SUBTASK 32-46-31-040-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-46-31-420-001

(2) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 32-46-31-420-002

(3) Get access to the main equipment center:

(a) Open this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-46-31-420-003

(4) Remove the tire and brake monitoring unit [1] from the electrical equipment rack E1-3 (TASK 20-10-07-000-801).

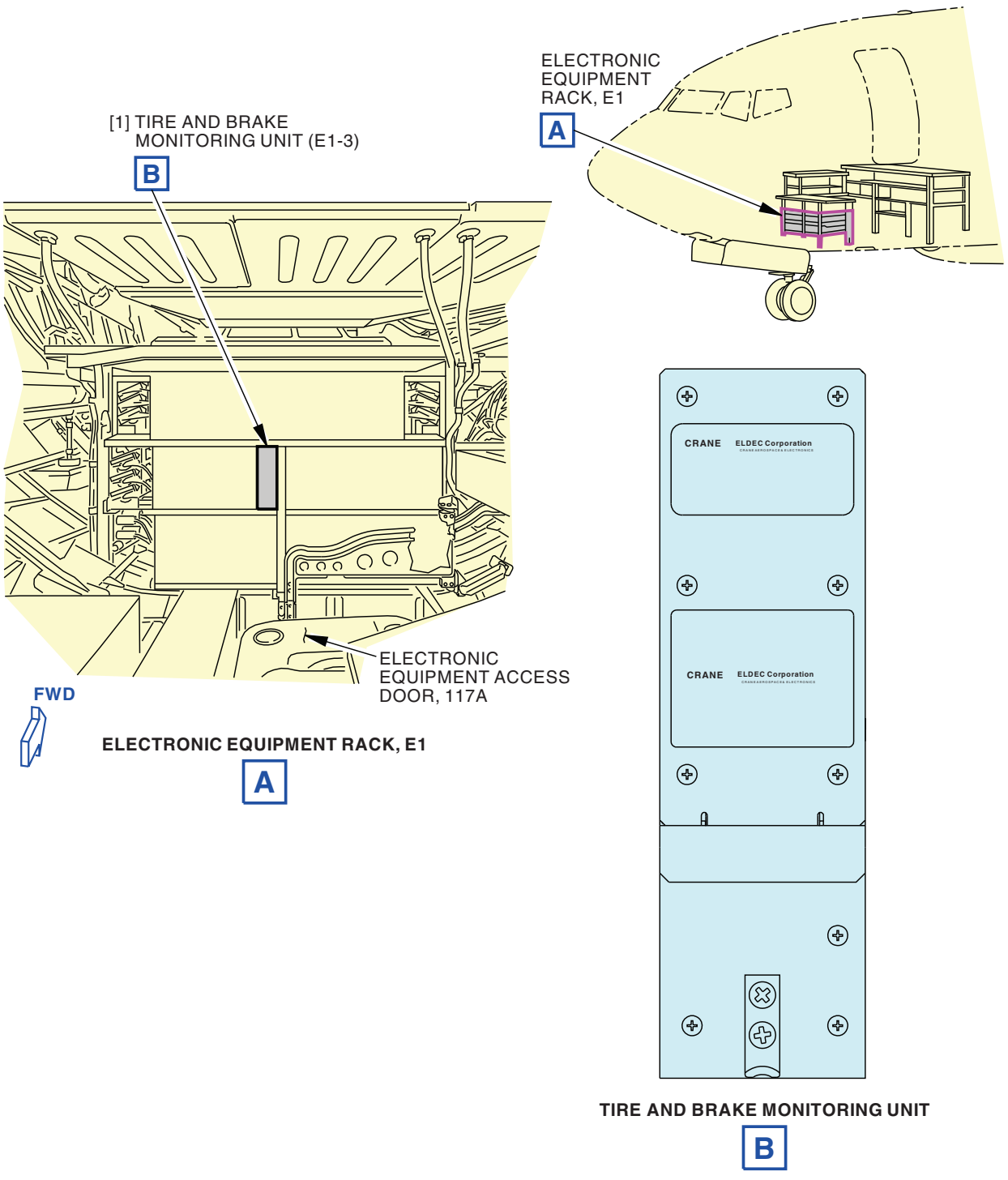
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-46-31-420-004

(5) Do this task: Conductive Dust Cap and Connector Cover Installation, TASK 20-40-12-400-803.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tire and Brake Monitoring System
Figure 401/32-46-31-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-46-31-400-801

3. Tire and Brake Monitoring Unit Installation

(Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-000-803	Conductive Dust Cap and Connector Cover Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Tire and brake monitoring unit	32-46-31-01-010	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Tire and Brake Monitoring Unit (TBMU) Installation

SUBTASK 32-46-31-840-001

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 32-46-31-840-002

(2) Do this task: Conductive Dust Cap and Connector Cover Removal, TASK 20-40-12-000-803.

SUBTASK 32-46-31-420-005

(3) Install the tire and brake monitoring unit [1] in the E1-3 electrical equipment rack (TASK 20-10-07-400-801).

F. Tire and Brake Monitoring Unit Installation Test

SUBTASK 32-46-31-420-007

(1) Close this access door:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-46-31-440-001

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-46-31-730-001

(3) When you apply power, make sure that the Crew Alert BRAKE TEMP and TIRE PRESS lights illuminate for approximately 2 seconds, and then go off.

(a) Wait a minimum of 15 seconds.

NOTE: When you apply power, the TBMU performs an internal check and system check which takes 5-15 seconds.

SUBTASK 32-46-31-710-001

(4) Do this task: Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.

SUBTASK 32-46-31-710-002

(5) Do this task: Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Tire Pressure Indication System - Adjustment/Test

1. General

A. This procedure has this task:

(1) A functional check of the tire pressure sensor.

TASK 32-49-00-720-801

2. Tire Pressure Sensor Functional Test

A. References

Reference	Title
12-15-51-610-802	Add Nitrogen or Air to the Tire (P/B 301)
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Prepare for the Functional Test

SUBTASK 32-49-00-480-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

D. Tire Pressure Sensor Functional Test Procedure

SUBTASK 32-49-00-610-001

(1) Make sure that you service the tires to the correct pressure (TASK 12-15-51-780-801).

SUBTASK 32-49-00-710-001

(2) Push the SYS button on the multi-function display select panel (MFD).

SUBTASK 32-49-00-710-002

(3) Make sure that the MFD Landing Gear Synoptic Page shows tire pressures for each of the main landing gear and nose landing gear tires.

(a) If the tire pressures are not correct, repeat (Landing Gear Tire Pressure Check and Tire Servicing, TASK 12-15-51-780-801).

SUBTASK 32-49-00-970-001

(4) Record the initial MFD Landing Gear Synoptic Page display readouts of the main landing gear and nose landing gear tire pressures for comparison with later steps.

SUBTASK 32-49-00-870-001

(5) For each tire, do the steps that follow:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Bleed the tire pressure of the left outboard main wheel [1] until the displayed pressure on the MFD changes.

(b) Make sure that the MFD Landing Gear Synoptic Page shows pressure change for wheel #1.

1) If the pressure changes for a wheel other than #1, check for crossed wiring (SWPM 20).

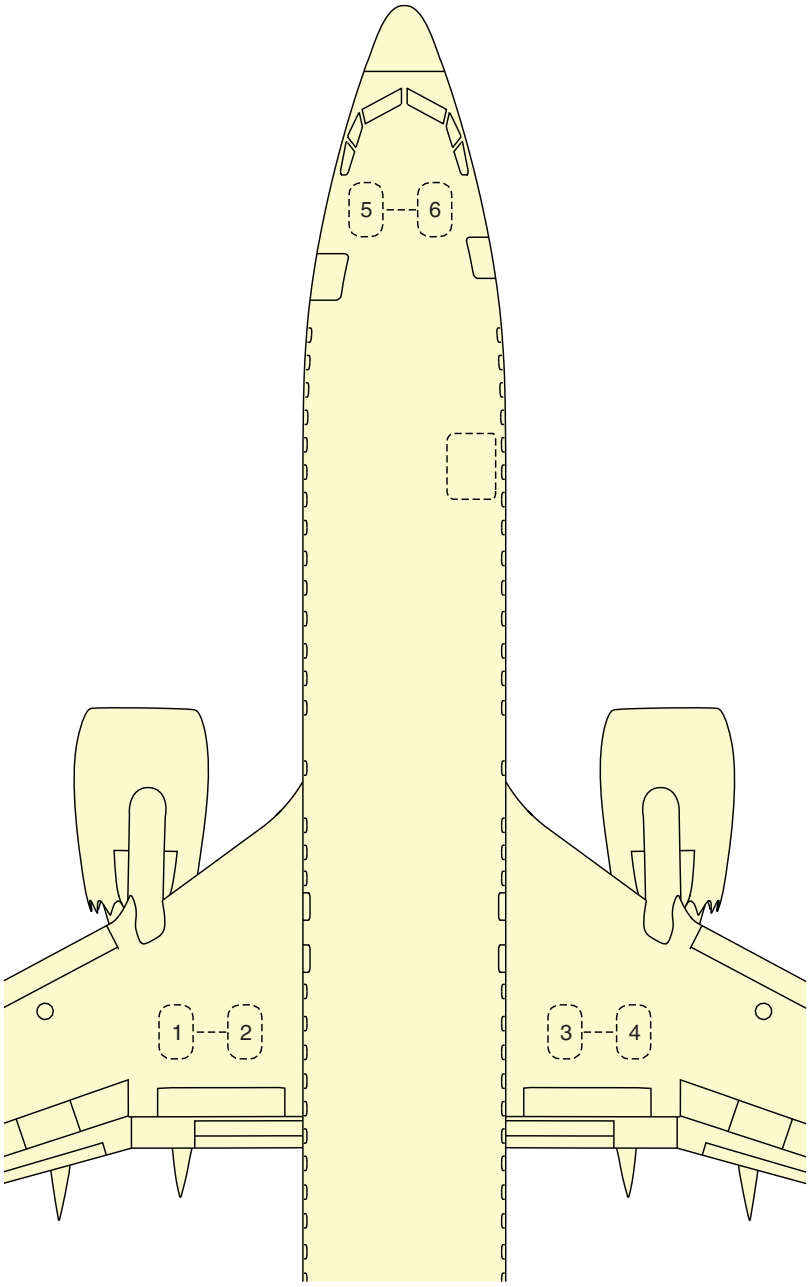
(c) Repeat these steps for the remaining wheels.

SUBTASK 32-49-00-710-003

(6) If one or more of the tires show a pressure below the range specified in Landing Gear Tire Pressure Check and Tire Servicing, TASK 12-15-51-780-801, do this task: Add Nitrogen or Air to the Tire, TASK 12-15-51-610-802.

——— END OF TASK ———

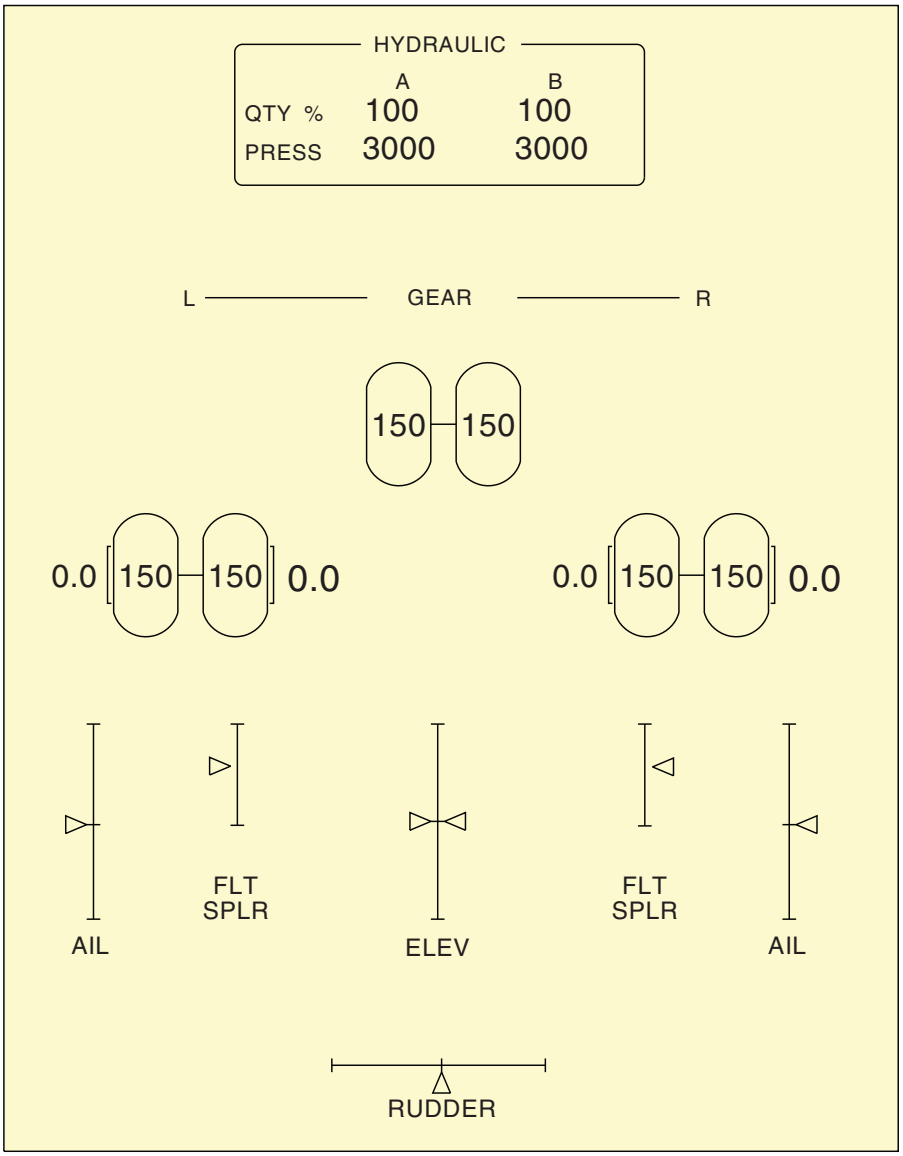
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2474054 S0000579946_V1

Tire Pressure Sensor
Figure 501/32-49-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MULTI-FUNCTION DISPLAY
(LANDING GEAR SYNOPTIC PAGE)
(EXAMPLE)

2474057 S0000580161_V2

Tire Pressure Sensor
Figure 501/32-49-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TIRE PRESSURE INDICATION SYSTEM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose Landing Gear Hubcap Assembly Removal

(2) Nose Landing Gear Hubcap Assembly Installation

(3) Main Landing Gear Hubcap Assembly Removal

(4) Main Landing Gear Hubcap Assembly Installation

(5) Nose Landing Gear In-Axle Adapter Removal

(6) Nose Landing Gear In-Axle Adapter Installation

(7) Main Landing Gear In-Axle Adapter Removal

(8) Main Landing Gear In-Axle Adapter Installation

(9) Tire Pressure Sensor/Fill Valve Removal

(10) Tire Pressure Sensor/Fill Valve Installation.

TASK 32-49-03-000-801

2. Nose Landing Gear Hubcap Assembly Removal

(Figure 401 and Figure 402)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
713	Nose Landing Gear

C. Prepare for the Removal

SUBTASK 32-49-03-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-49-03-480-002

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-49-03-860-001

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

D. Nose Landing Gear Hubcap Assembly Removal

SUBTASK 32-49-03-000-001

(1) Remove the lockwire from the bolts [2] and tire pressure sensor/fill valve.

SUBTASK 32-49-03-000-002

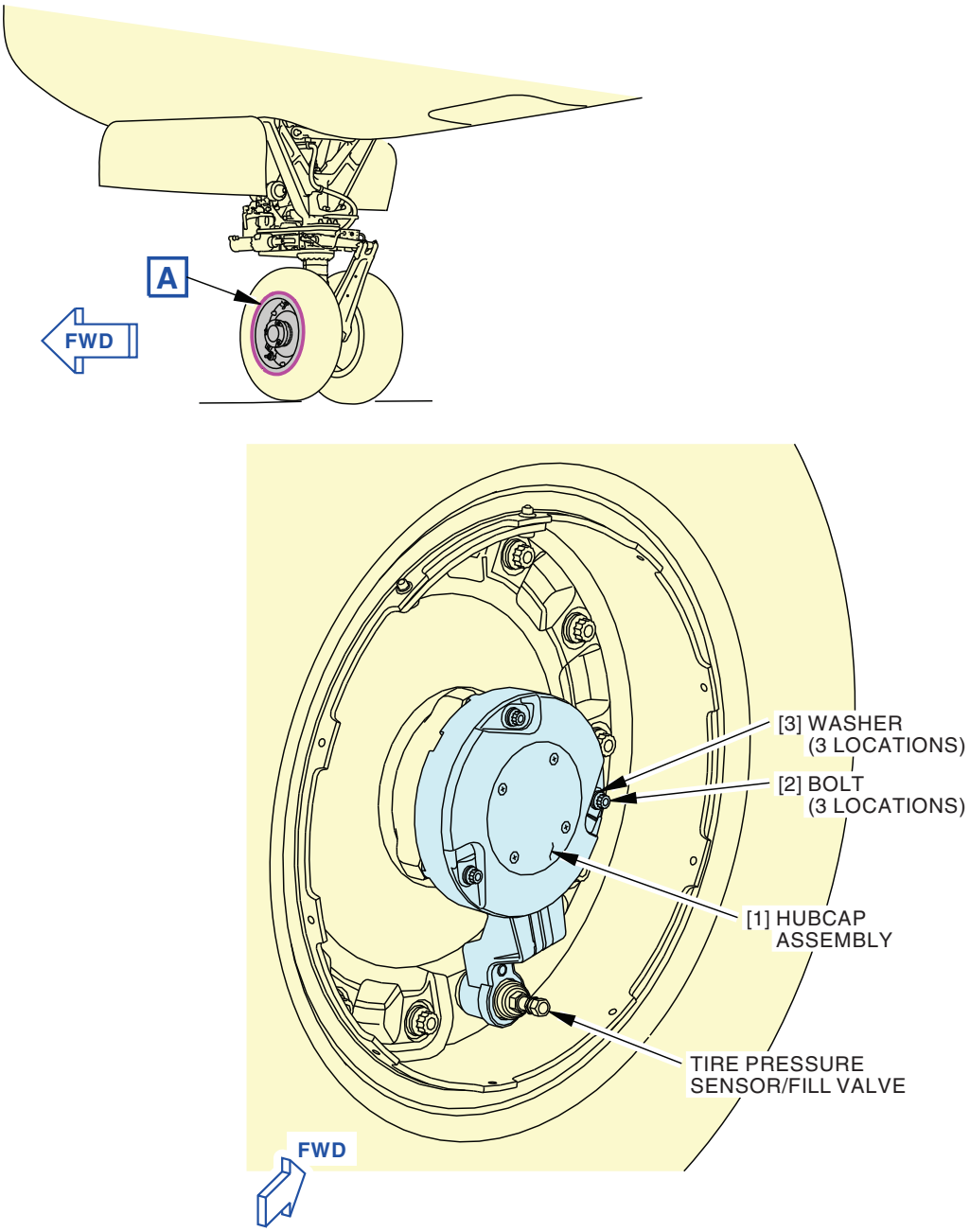
(2) Remove the bolts [2] and washers [3] from the hubcap assembly [1].

SUBTASK 32-49-03-000-003

(3) Disconnect the hubcap assembly [1] from the wheel assembly.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



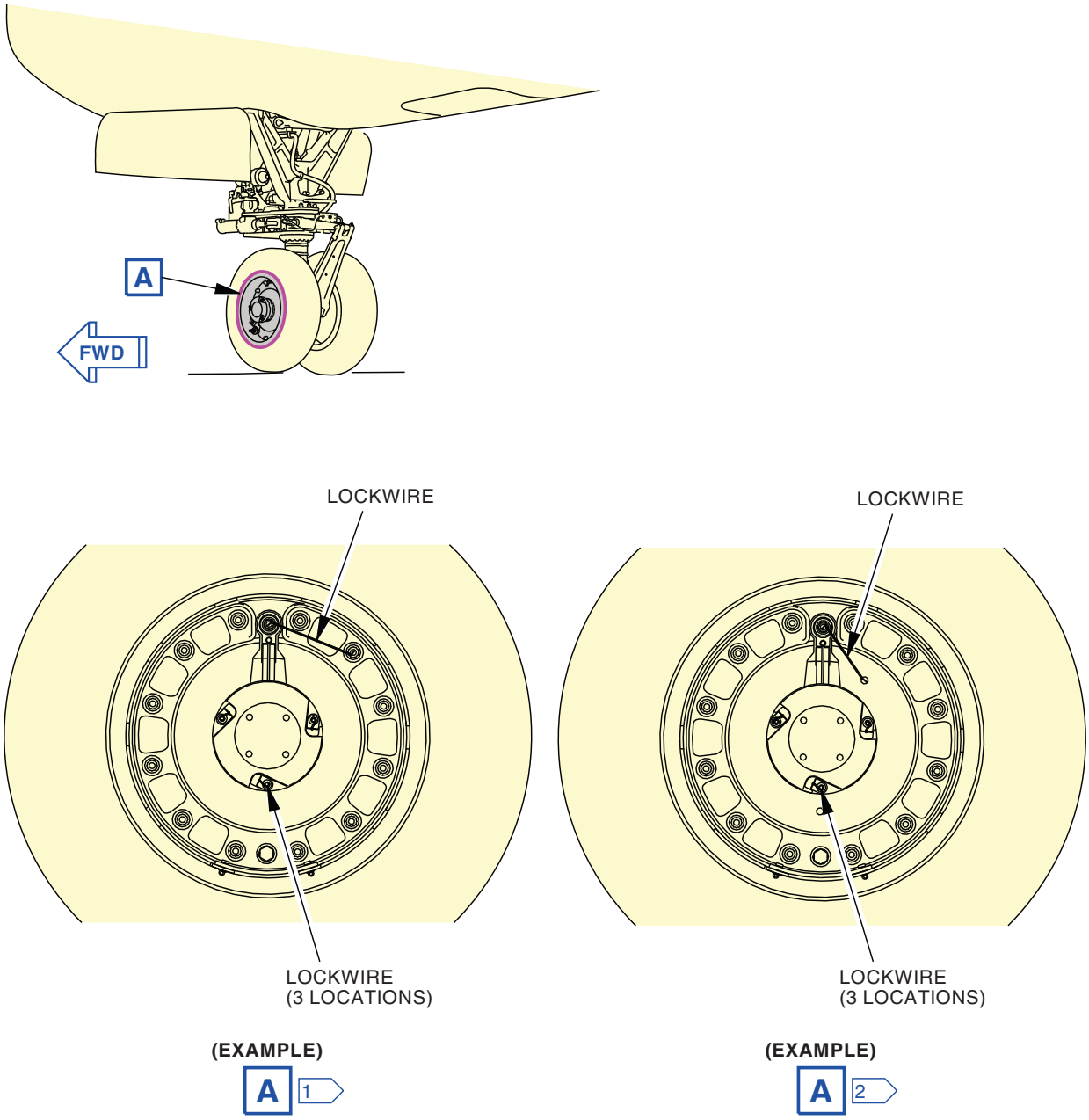
NOSE LANDING GEAR HUBCAP ASSEMBLY
(2 LOCATIONS)
(EXAMPLE)

A

2583506 S0000623060_V1

Nose Landing Gear Hubcap Assembly Installation
Figure 401/32-49-03-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 AIRPLANES WITH MESSIER-BUGATTI DOWTY WHEELS
2 AIRPLANES WITH GOODRICH WHEELS

2937667 S0000713510_V2

Nose Landing Gear Lockwire Installation
Figure 402/32-49-03-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-49-03-400-801

3. Nose Landing Gear Hubcap Assembly Installation

(Figure 401 and Figure 402)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G51827	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 6 Inches Long	BACC13AT3K6

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Hubcap assembly	32-42-50-02-020	PTF ALL

D. Location Zones

Zone	Area
713	Nose Landing Gear

E. Nose Landing Gear Hubcap Assembly Installation

SUBTASK 32-49-03-600-001

(1) Apply a thin layer of corrosion inhibiting material, G50136, on the shanks, threads, and under the heads of the bolts [2].

SUBTASK 32-49-03-420-001

(2) Install the bolts [2] and washers [3] to attach the hubcap assembly [1] to the wheel assembly.

SUBTASK 32-49-03-420-002

CAUTION

DO NOT TIGHTEN THE HUBCAP BOLTS TOO MUCH. MAKE SURE THAT THE HUBCAP IS IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE BOLTS. TOO MUCH TORQUE WILL CAUSE DAMAGE TO THE BOLTS AND THE HUBCAP.

(3) Tighten the bolts [2] to 75 ±2 in-lb (8.5 ±0.3 N·m).

SUBTASK 32-49-03-160-001

(4) Remove any excess corrosion inhibiting material, G50136.

SUBTASK 32-49-03-420-003

(5) Install MS20995C32 lockwire, G01048, with the double twist method, or safety cable kit, G51827, on the bolts [2] (TASK 20-10-44-400-801).

SUBTASK 32-49-03-420-004

(6) Install MS20995C32 lockwire, G01048, with the double twist method, or safety cable kit, G51827, on the tire pressure sensor/fill valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-49-03-860-002

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-720-001

(2) Do this task: Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.

——— END OF TASK ———

TASK 32-49-03-000-802

4. Main Landing Gear Hubcap Assembly Removal

(Figure 403 and Figure 404)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Removal

SUBTASK 32-49-03-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-49-03-480-004

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-49-03-860-003

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

D. Main Landing Gear Hubcap Assembly Removal

SUBTASK 32-49-03-000-005

(1) Remove the lockwire from the bolts [6].

SUBTASK 32-49-03-000-006

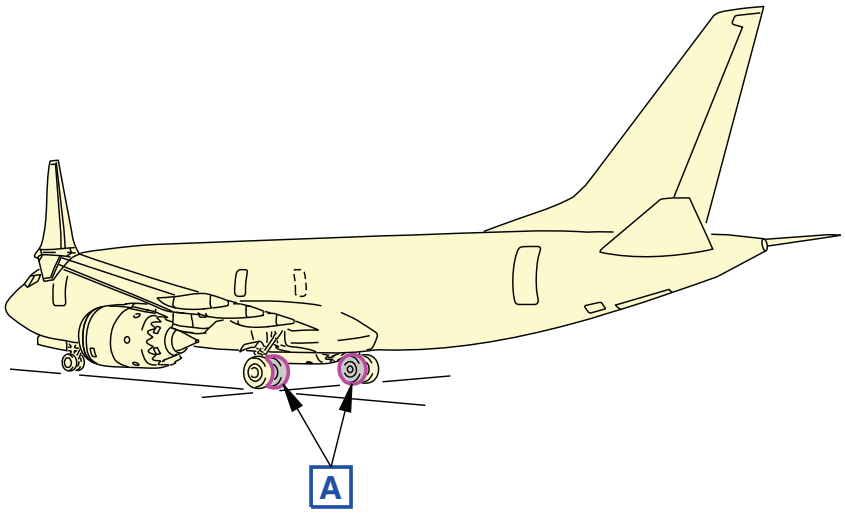
(2) Remove the bolts [6] and washers [5] from the hubcap assembly [4].

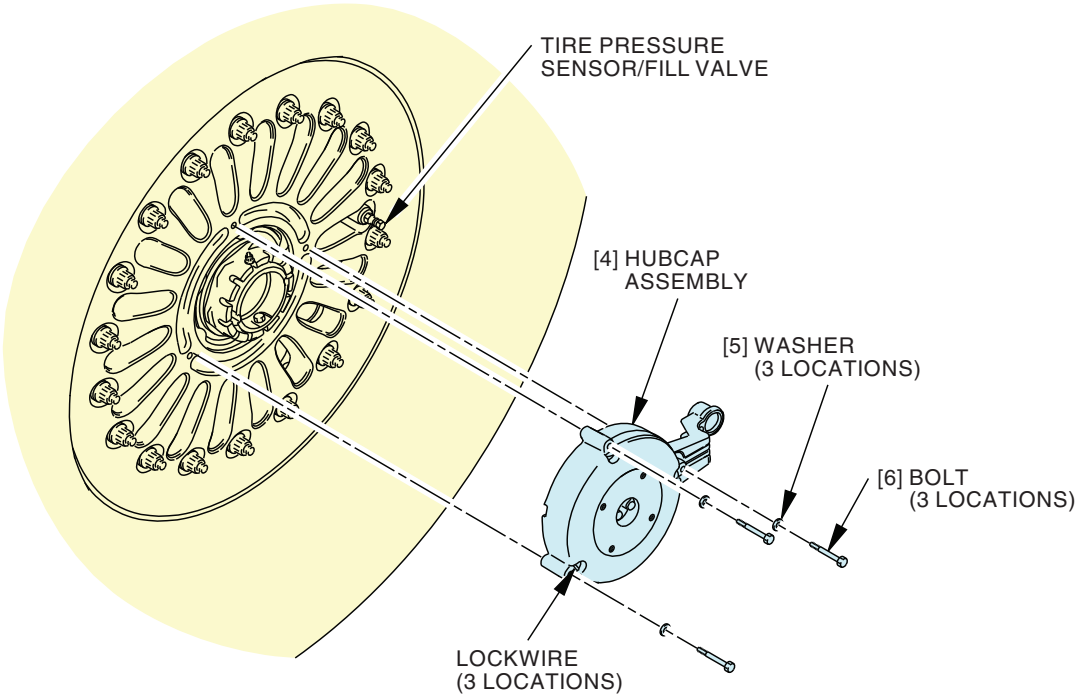
SUBTASK 32-49-03-000-007

(3) Disconnect the hubcap assembly [4] from the wheel assembly.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





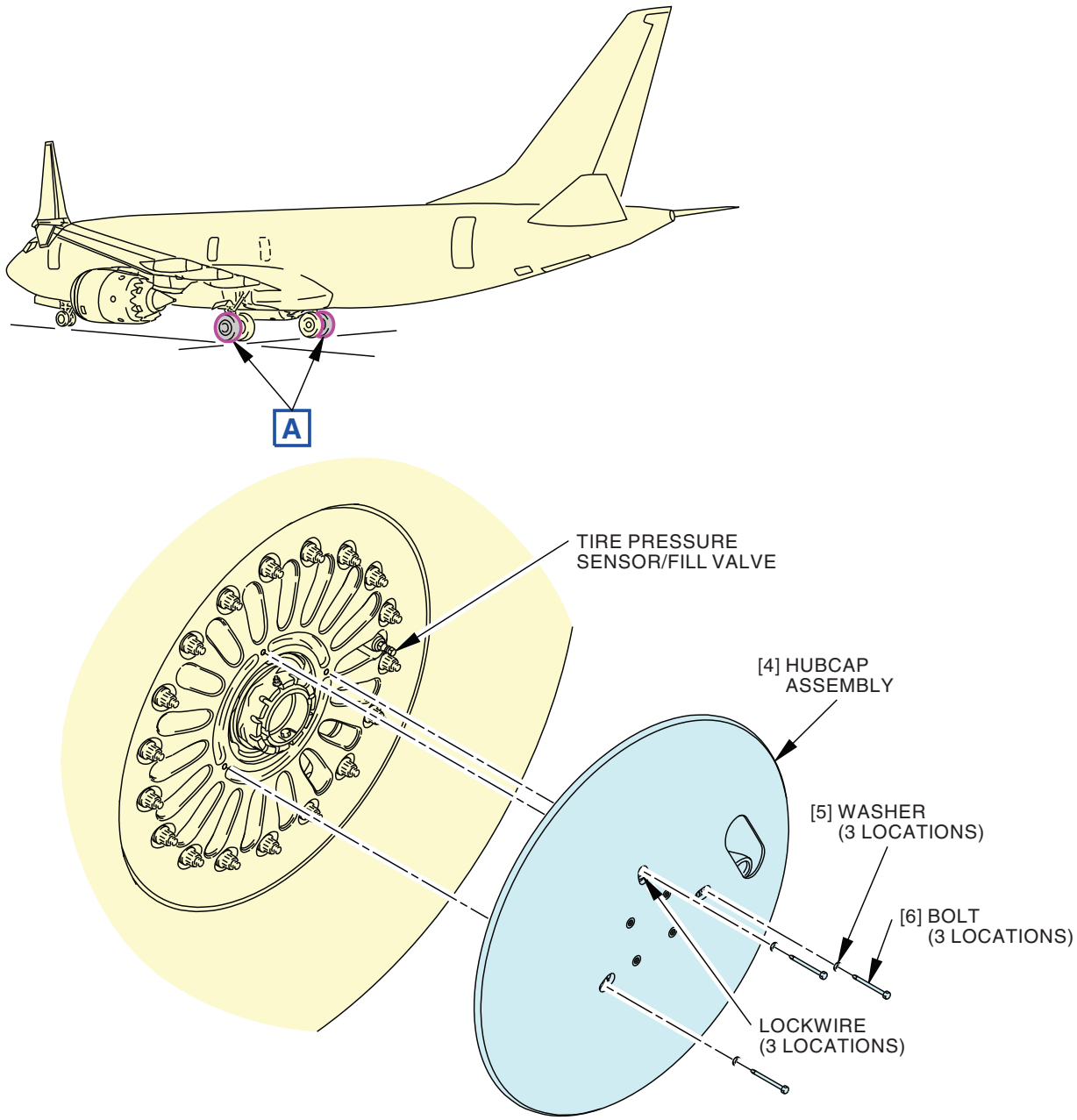
INBOARD MAIN LANDING GEAR HUBCAP ASSEMBLY
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)
(EXAMPLE)

A

2937668 S0000713759_V1

Inboard Main Landing Gear Hubcap Assembly Installation
Figure 403/32-49-03-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details



OUTBOARD MAIN LANDING GEAR HUBCAP ASSEMBLY
(LEFT SIDE IS SHOWN, RIGHT SIDE IS OPPOSITE)
(EXAMPLE)

A

2937669 S0000713760_V1

Outboard Main Landing Gear Hubcap Assembly Installation
Figure 404/32-49-03-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-49-03-400-802

5. Main Landing Gear Hubcap Assembly Installation

(Figure 403 and Figure 404)

A. References

Reference	Title
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G51827	Kit - Safety Cable, 321 CRES - 0.032 Inch (0.81 mm) Diameter, (Contains both Cable and Ferrule), 6 Inches Long	BACC13AT3K6

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Hubcap assembly	32-42-50-03-030	PTF ALL
		32-42-50-03-035	PTF ALL

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Main Landing Gear Hubcap Assembly Installation

SUBTASK 32-49-03-940-001

(1) Turn the drive coupling on the in-axle until its coupling fingers lock with the coupling fingers on the hubcap assembly [4].

SUBTASK 32-49-03-160-002

(2) Apply a thin layer of corrosion inhibiting material, G50136, on the shanks, threads, and under the heads of the bolts [6].

SUBTASK 32-49-03-420-005

(3) Install the bolts [6] and washers [5] to attach the hubcap assembly [4] to the wheel assembly.

SUBTASK 32-49-03-420-006

CAUTION

DO NOT TIGHTEN THE HUBCAP BOLTS TOO MUCH. MAKE SURE THAT THE HUBCAP IS IN THE CORRECT POSITION BEFORE YOU TIGHTEN THE BOLTS. TOO MUCH TORQUE WILL CAUSE DAMAGE TO THE BOLTS AND THE HUBCAP.

(4) Tighten the bolts [6] to 75 ±2 in-lb (8.5 ±0.3 N·m).

SUBTASK 32-49-03-160-003

(5) Remove any excess corrosion inhibiting material, G50136.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-49-03-420-007

(6) Install MS20995C32 lockwire, G01048, with the double twist method, or safety cable kit, G51827, on the bolts [6] (TASK 20-10-44-400-801).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-49-03-860-005

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-720-002

(2) Do this task: Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.

——— END OF TASK ———

TASK 32-49-03-000-803

6. Nose Landing Gear In-Axle Adapter Removal

(Figure 405)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
713	Nose Landing Gear

C. Prepare for the Removal

SUBTASK 32-49-03-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-49-03-480-006

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-49-03-860-006

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-860-022

(4) Do this task: Nose Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-801.

D. Nose Landing Gear In-Axle Adapter Removal

SUBTASK 32-49-03-000-008

(1) Remove the bolts [7], nuts [8], and washers [9] that hold the in-axle adapter [11] to the wheel-retention nut and inner cylinder.

SUBTASK 32-49-03-860-023

(2) Carefully pull the in-axle adapter [11] from the wheel assembly.

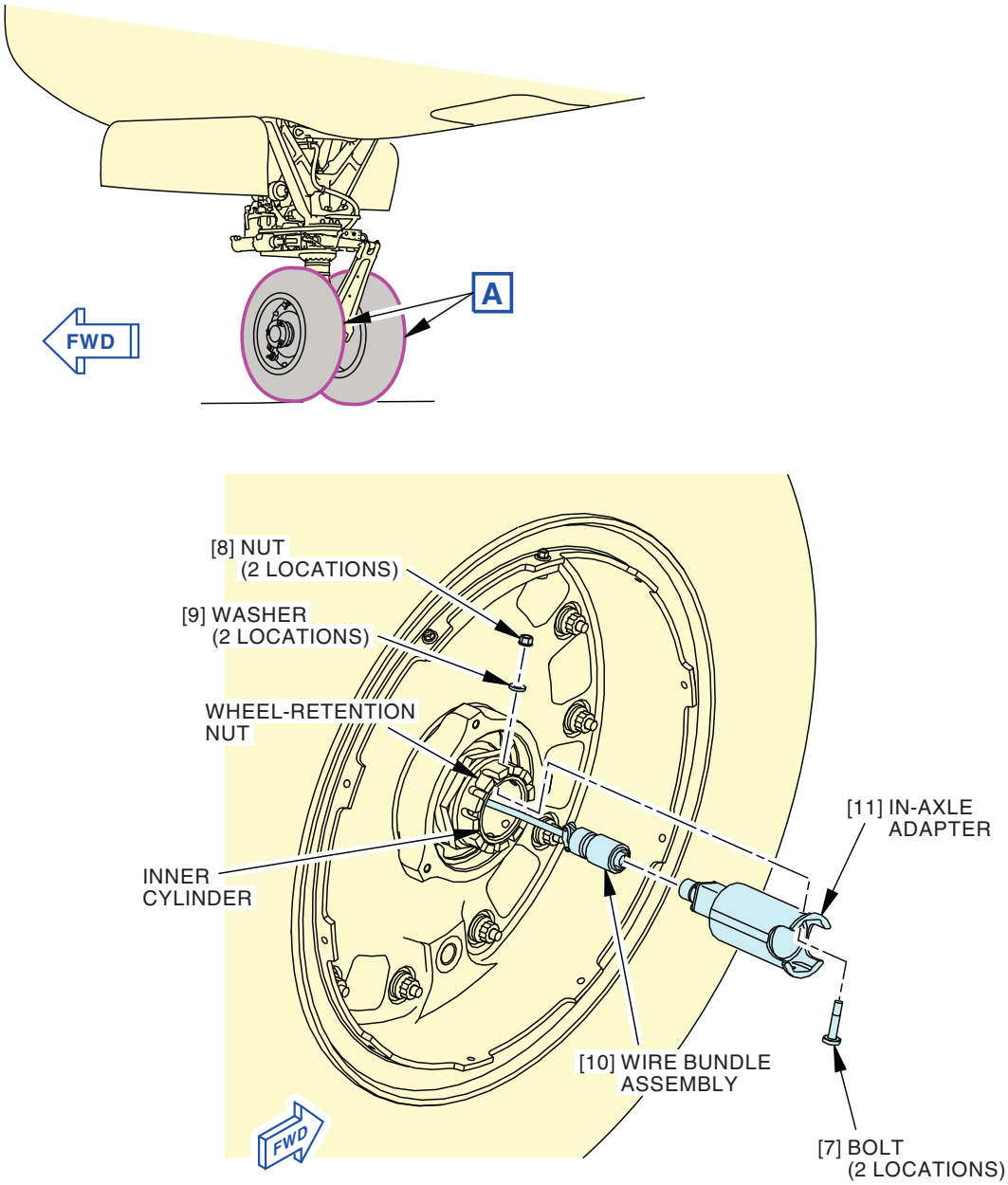
NOTE: The in-axle adapter is connected to a wire bundle assembly in the inner cylinder.

SUBTASK 32-49-03-000-009

(3) Disconnect the in-axle adapter [11] from the wire bundle assembly [10].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE LANDING GEAR IN-AXLE ADAPTER
(2 LOCATIONS)
(EXAMPLE)

A

NOTE:

HUBCAP ASSEMBLY NOT SHOWN.

2584051 S0000623064_V1

Nose Landing Gear In-Axle Adapter Installation
Figure 405/32-49-03-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-49-03-400-803

7. Nose Landing Gear In-Axle Adapter Installation

(Figure 405)

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	In-axle adapter	32-21-51-20-035	PTF ALL

C. Location Zones

Zone	Area
713	Nose Landing Gear

D. Nose Landing Gear In-Axle Adapter Installation

SUBTASK 32-49-03-100-001

(1) Do this task for the wire bundle assembly [10]: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

SUBTASK 32-49-03-420-009

(2) Connect the in-axle adapter [11] to the wire bundle assembly [10].

SUBTASK 32-49-03-860-024

(3) Carefully insert the in-axle adapter [11] inside the axle.

SUBTASK 32-49-03-420-010

(4) Install the bolts [7], nuts [8], and washers [9] that hold the in-axle adapter [11] to the wheel-retention nut and inner cylinder.

(a) Make sure to install the bolts [7] with the heads to the inside of the axle.

SUBTASK 32-49-03-860-008

(5) Do this task: Nose Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-801.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-49-03-860-009

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-720-003

(2) Do this task: Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-49-03-000-804

8. Main Landing Gear In-Axle Adapter Removal

(Figure 406)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-42-11-000-801	Antiskid Wheel Speed Transducer Removal (P/B 401)

B. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Removal

SUBTASK 32-49-03-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-49-03-480-008

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-49-03-860-010

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-860-026

(4) Do this task: Main Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-802.

D. Main Landing Gear In-Axle Adapter Removal

SUBTASK 32-49-03-000-011

(1) Remove the bolts [16], nuts [15], and washers [14] that hold the in-axle adapter [13] to the wheel-retention nut and axle.

SUBTASK 32-49-03-860-027

(2) Carefully pull the in-axle adapter [13] from the wheel assembly.

NOTE: The in-axle adapter and antiskid wheel speed transducer are connected to a wire bundle assembly inside the axle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-49-03-000-012

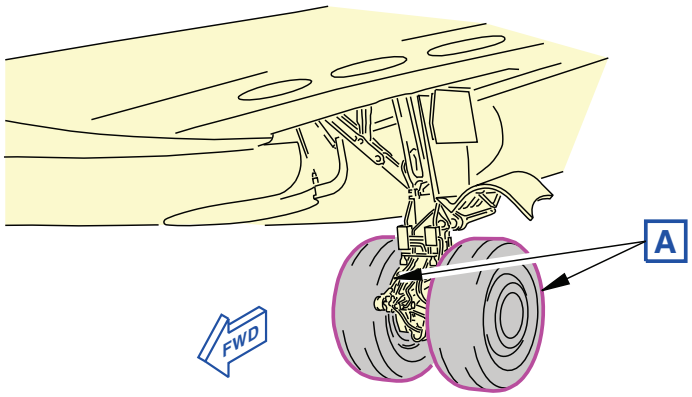
(3) Disconnect the in-axle adapter [13] and transducer [20] from the wire bundle assembly [12].

SUBTASK 32-49-03-020-001

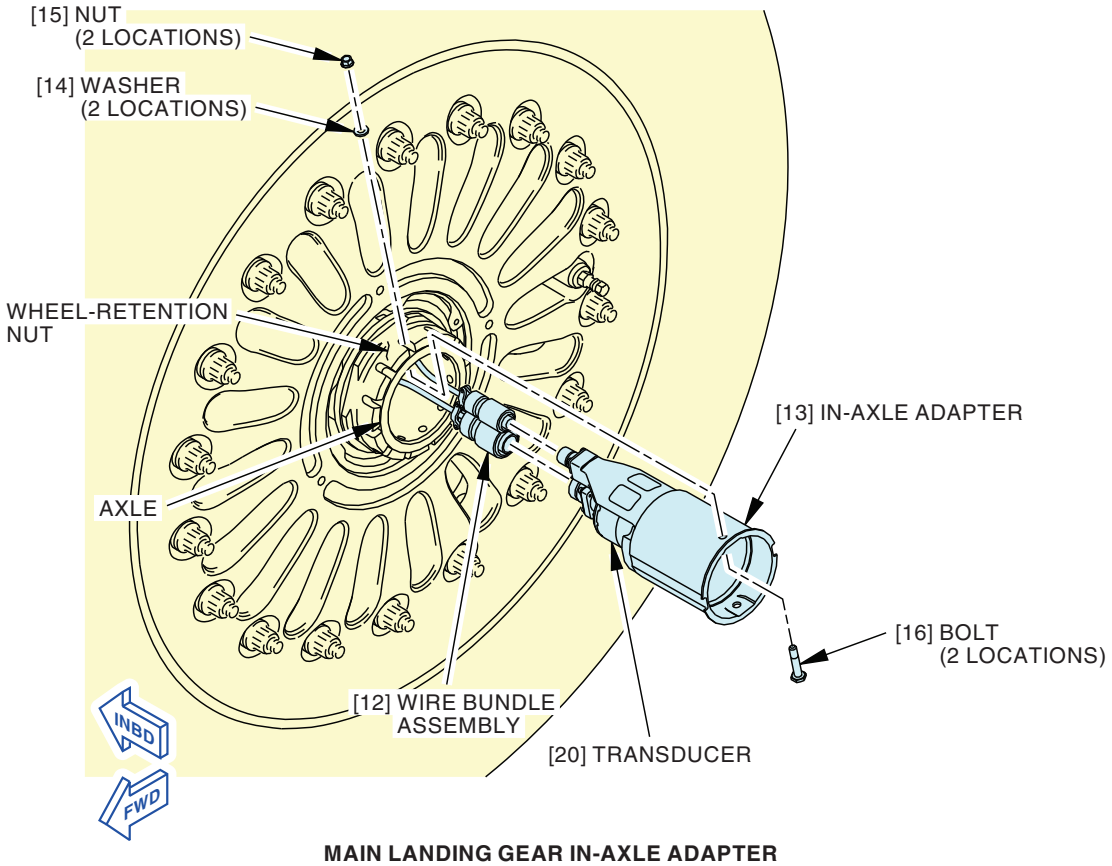
(4) Remove the transducer [20] from the in-axle adapter [13] (TASK 32-42-11-000-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(LEFT MAIN LANDING GEAR IS SHOWN, RIGHT MAIN LANDING GEAR IS OPPOSITE)



MAIN LANDING GEAR IN-AXLE ADAPTER (4 LOCATIONS) (EXAMPLE)

NOTE:

HUBCAP ASSEMBLY NOT SHOWN.

A

2584020 S0000623063_V2

Main Landing Gear In-Axle Adapter Installation
Figure 406/32-49-03-990-806

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-49-03-400-804

9. Main Landing Gear In-Axle Adapter Installation

(Figure 406)

A. References

Reference	Title
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
32-42-11-400-801	Antiskid Wheel Speed Transducer Installation (P/B 401)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	In-axle adapter	32-42-11-01-070	PTF ALL

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Main Landing Gear In-Axle Adapter Installation

SUBTASK 32-49-03-420-019

(1) Install the transducer [20] to the in-axle adapter [13] (TASK 32-42-11-400-801).

SUBTASK 32-49-03-100-002

(2) Do this task for the wire bundle assembly [12]: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

SUBTASK 32-49-03-420-012

(3) Connect the in-axle adapter [13] and transducer [20] to the wire bundle assembly [12].

SUBTASK 32-49-03-860-025

(4) Carefully insert the in-axle adapter [13] inside the axle.

SUBTASK 32-49-03-420-013

(5) Install the bolts [16], nuts [15], and washers [14] that hold the in-axle adapter [13] to the wheel-retention nut and axle.

(a) Make sure to install the bolts [16] with the heads to the inside of the axle.

SUBTASK 32-49-03-420-014

(6) Tighten the nuts [15] to 65 ±2 in-lb (7.3 ±0.3 N·m).

SUBTASK 32-49-03-860-012

(7) Do this task: Main Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-802.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-49-03-860-013

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-720-004

(2) Do this task: Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.

——— END OF TASK ———

TASK 32-49-03-000-805

10. Tire Pressure Sensor/Fill Valve Removal

(Figure 407)

A. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4046	Deflator - Tire, Valve Core Removal and Installation (Small Bore) and Tire Deflation Part #: 320008B-SO Supplier: SFN11 Part #: 968RB Supplier: 5W9Y4 Opt Part #: 320008B Supplier: SFN11
STD-24796	Deep Socket, Hex 7/16 Inch

C. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-49-03-480-009

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-49-03-480-010

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-49-03-860-014

(3) Make sure that the parking brake is set.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-49-03-860-015

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-480-011

(5) Do this task: Lift the Airplane on Axle Jacks-Handling, TASK 07-10-03-580-801.

NOTE: Lift only the axle where you remove the tire pressure sensor/fill valve assembly.

E. Tire Pressure Sensor/Fill Valve Removal

SUBTASK 32-49-03-860-016

(1) For the nose landing gear tire pressure sensor/fill valve [18], do this task: Nose Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-801.

SUBTASK 32-49-03-860-017

(2) For the main landing gear tire pressure sensor/fill valve [18], do this task: Main Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-802.

SUBTASK 32-49-03-840-001

WARNING YOU MUST FULLY DEFLATE THE TIRE BEFORE YOU REMOVE THE WHEEL AND TIRE ASSEMBLY. A DEFECTIVE WHEEL AND TIRE CAN EXPLODE IF YOU DO NOT DEFLATE THE TIRE BEFORE YOU REMOVE THE ASSEMBLY. INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING MAKE SURE THAT ALL PERSONNEL ARE CLEAR OF THE INFLATION VALVE ASSEMBLY PATH WHILE YOU REMOVE IT. IF THE INFLATION VALVE ASSEMBLY EJECTS WHILE YOU REMOVE IT, INJURY TO PERSONNEL CAN OCCUR.

WARNING BE VERY CAREFUL WHEN DEFLATING TIRES (606) TO PREVENT INJURY TO PERSONNEL FROM A FLYING INFLATION VALVE CAP AND/OR VALVE CORE.

CAUTION MAKE SURE THAT THE DEFLATED TIRE THAT IS ON THE AXLE DOES NOT TOUCH THE GROUND. IF THE TIRE DOES TOUCH THE GROUND, THE TIRE'S BEAD SEAT CAN MOVE OUT OF ITS POSITION AND BECOME UNSEATED. THE UNSEATED BEAD SEAT CAN CAUSE LEAKAGE WHICH THEN CAN CAUSE DAMAGE TO THE TIRE AND WHEEL.

(3) Do these steps to deflate the tire:

(a) Remove the cap [17].

(b) Deflate the tire with the small bore tire deflator, COM-4046.

(c) If you cannot use the usual procedure to deflate the tire because the valve core is damaged, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If installed, remove lockwire from the tire pressure sensor/fill valve [18].
2) Turn the tire pressure sensor/fill valve [18] slowly in a counterclockwise direction until there is a gas leakage through the boss.
3) At the same time, push lightly on the valve assembly.
4) After all of the gas has been released, use the 7/16 inch socket, STD-24796, to remove the tire pressure sensor/fill valve [18] from the tire.

SUBTASK 32-49-03-000-014

(4) Remove the O-ring [19] from the tire pressure sensor/fill valve [18].
   (a) Discard the O-ring [19].

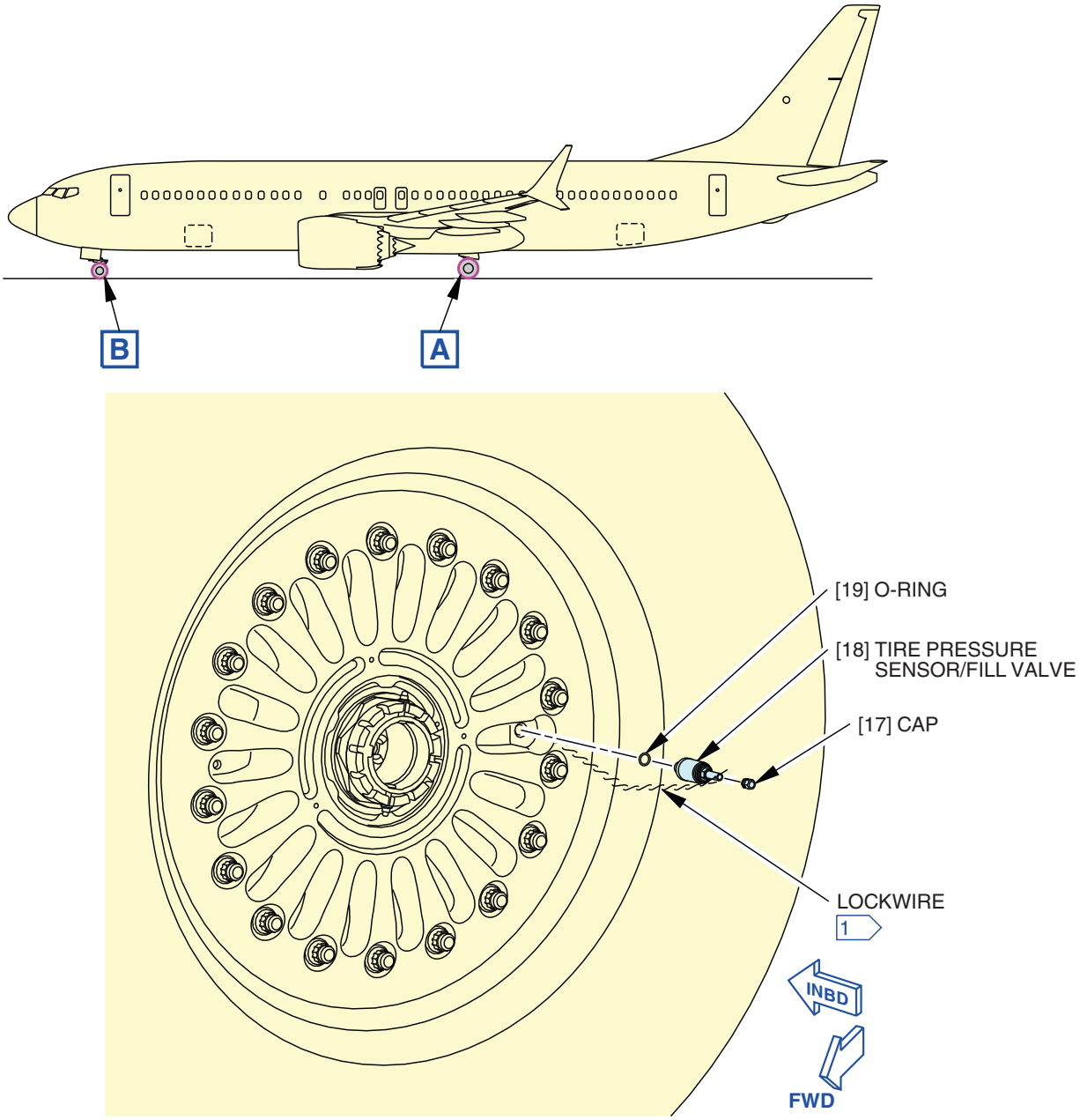
F. Inspection of the Tire Pressure Sensor/Fill Valve

SUBTASK 32-49-03-200-001

(1) Inspect the tire pressure sensor/fill valve [18].
   (a) Replace the tire pressure sensor/fill valve [18] if the two staked halves of the tire pressure valve are loose or misaligned.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR TIRE PRESSURE SENSOR/FILL VALVE (4 LOCATIONS) (EXAMPLE)

A

NOTE:

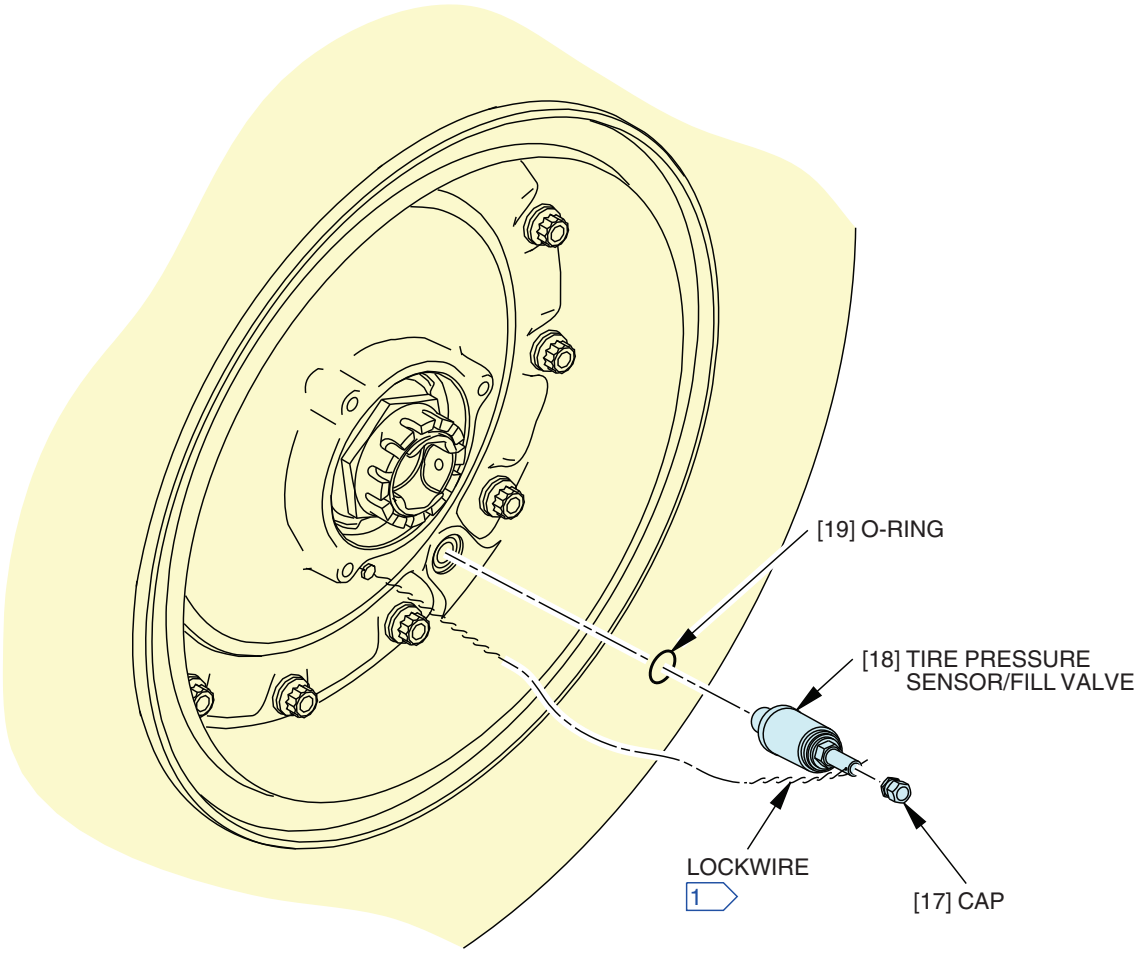
HUBCAP ASSEMBLY NOT SHOWN.

1 IF INSTALLED

2583377 S0000622795_V2

Tire Pressure Sensor/Fill Valve Installation
Figure 407/32-49-03-990-807 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE LANDING GEAR TIRE PRESSURE SENSOR/FILL VALVE (2 LOCATIONS) (EXAMPLE)

B

1 IF INSTALLED

2583380 S0000622796_V2

Tire Pressure Sensor/Fill Valve Installation
Figure 407/32-49-03-990-807 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-49-03-400-805

11. Tire Pressure Sensor/Fill Valve Installation

(Figure 407)

A. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
32-49-00-720-801	Tire Pressure Sensor Functional Test (P/B 501)

B. Tools/Equipment

Reference	Description
STD-24796	Deep Socket, Hex 7/16 Inch

C. Consumable Materials

Reference	Description	Specification
A50129	Primer - Loctite 7649	ASTM D5363
A50584	Adhesive - Threadlocking, Loctite 242 (Telair ES406-13)	ASTM D5363 Grp 3 Cl 2 Gd 1 (SUPERSEDES MIL-S-46163)
C50088	Primer - Adhesive (MIL-S-46163 Grade F)	MIL-S-46163
D00016	Grease - Aircraft, General Purpose, Wide Temperature Range	MIL-PRF-81322
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
18	Tire pressure sensor/fill valve	32-45-11-02-020	PTF ALL
		32-45-21-02-015	PTF ALL
19	O-ring	32-45-11-02-023	PTF ALL
		32-45-21-02-018	PTF ALL

E. Location Zones

Zone	Area
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Tire Pressure Sensor/Fill Valve Installation

SUBTASK 32-49-03-400-001

CAUTION

MAKE SURE THAT THE DEFLATED TIRE THAT IS ON THE AXLE DOES NOT TOUCH THE GROUND. IF THE TIRE DOES TOUCH THE GROUND, THE TIRE'S BEAD SEAT CAN MOVE OUT OF ITS POSITION AND BECOME UNSEATED. THE UNSEATED BEAD SEAT CAN CAUSE LEAKAGE WHICH THEN CAN CAUSE DAMAGE TO THE TIRE AND WHEEL.

(1) Make sure that the deflated tire that is on the axle does not touch the ground.

SUBTASK 32-49-03-200-002

(2) Do a visual inspection of a reused tire pressure sensor/fill valve [18].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Replace the reused tire pressure sensor/fill valve [18] if the two staked halves of the tire pressure sensor/fill valve [18] are loose or misaligned.

SUBTASK 32-49-03-420-020

(3) To install the tire pressure sensor/fill valve [18] on the Goodrich Main and Nose Wheel assemblies, do these steps:

(a) Lubricate the new O-ring [19] with grease, D00016, and install it on tire pressure sensor/fill valve [18].

NOTE: MIL-PRF-81322 qualified lubricants should be used on the vendor specified O-rings for the tire pressure sensor/fill valve for the TBMS.

1) Avoid contact with the threads of the tire pressure sensor/fill valve [18].

(b) Use the 7/16 inch socket, STD-24796, to install the tire pressure sensor/fill valve [18] on the wheel.

(c) Tighten the tire pressure sensor/fill valve [18] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

(d) Make sure that the red mark on the tire is aligned with the tire pressure sensor/fill valve [18].

NOTE: The red dot identifies the lightest area of the tire.

(e) Install MS20995C32 lockwire, G01048, between the tire pressure sensor/fill valve [18] and outboard wheel.

SUBTASK 32-49-03-420-021

(4) To install the tire pressure sensor/fill valve [18] on the Safran Main Wheel assemblies, do these steps:

(a) Install the new O-ring [19] on the tire pressure sensor/fill valve [18].

(b) Apply Loctite 7649 Primer, A50129, or adhesive primer, C50088, and Loctite 242 adhesive, A50584, to the tire pressure sensor/fill valve [18] threads.

NOTE: It is necessary to install and tighten the inflation valve within 1 minute of the application of retaining compound. If it is tightened after 1 minute of the application of retaining compound, the bond of the retaining compound will not be satisfactory and inflation valve must be removed and installed again.

(c) Do not let the primer or retaining compound touch the O-ring [19].

(d) Use the 7/16 inch socket, STD-24796, to install the tire pressure sensor/fill valve [18] on the wheel.

(e) Tighten the tire pressure sensor/fill valve [18] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

1) Make sure that the tire pressure sensor/fill valve [18] was tightened within 1 minute after application of compound.

(f) Make sure that the red mark on the tire is aligned with the tire pressure sensor/fill valve [18].

NOTE: The red dot identifies the lightest area of the tire.

SUBTASK 32-49-03-420-022

(5) To install the tire pressure sensor/fill valve [18] on the Safran Nose Wheel assemblies, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Lubricate the new O-ring [19] with grease, D00016, and install it on tire pressure sensor/fill valve [18].

NOTE: MIL-PRF-81322 qualified lubricants should be used on the vendor specified O-rings for the tire pressure sensor/fill valve for the TBMS.

1) Avoid contact with the threads of the tire pressure sensor/fill valve [18].

(b) Use the 7/16 inch socket, STD-24796, to install the tire pressure sensor/fill valve [18] on the wheel.

(c) Tighten the tire pressure sensor/fill valve [18] to 150 in-lb (16.9 N·m) - 200 in-lb (22.6 N·m).

(d) Make sure that the red mark on the tire is aligned with the tire pressure sensor/fill valve [18].

NOTE: The red dot identifies the lightest area of the tire.

SUBTASK 32-49-03-860-018

(6) For the nose landing gear tire pressure sensor/fill valve [18], do this task: Nose Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-801.

SUBTASK 32-49-03-860-019

(7) For the main landing gear tire pressure sensor/fill valve [18], do this task: Main Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-802.

SUBTASK 32-49-03-860-020

(8) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	15	C01094	LANDING GEAR BRAKE TEMP MONITOR

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	21	C04034	TIRE PRESSURE INDICATION

SUBTASK 32-49-03-860-021

(9) Do this task: Landing Gear Tire Pressure Check and Tire Servicing, TASK 12-15-51-780-801.

NOTE: The tire must be inflated to at least half of the normal tire inflation pressure before you remove the jacks.

SUBTASK 32-49-03-420-018

(10) Install the cap [17] after the tire has been serviced.

SUBTASK 32-49-03-720-005

(11) Do this task: Tire Pressure Sensor Functional Test, TASK 32-49-00-720-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-49-03-580-001

WARNING MAKE SURE THE AREA AROUND THE TIRES IS CLEAR OF PERSONS AND EQUIPMENT. IF THE AREA AROUND THE TIRES IS NOT CLEAR, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Remove the appropriate jack (TASK 07-10-03-580-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE WHEEL STEERING SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains these tasks:

(1) A general check of the nose wheel steering system adjustment

(2) An adjustment check of the rudder pedal interconnect mechanism

(3) An adjustment of the rudder pedal interconnect mechanism

(4) An adjustment check of the nose wheel steering system

(5) An adjustment of the nose wheel steering system

(6) An adjustment of the nose wheel steering system (adjusting steering metering valve input rod method)

(7) A test of the nose wheel steering system (airplane on ground)

(8) A test of the nose wheel steering system (airplane on jacks).

B. Make sure that you do the two parts of the nose wheel steering system adjustment in the correct sequence. The first part is an adjustment of the interconnect mechanism for the rudder pedals that includes the nose gear piston position (NGPP) cable loop. The second part is an adjustment of the cable loop for the nose wheel steering (NWS) and rudder input. In general, it is necessary to adjust the cables when the rig loads disagree with the loads given in Table 501 by more than 15 lb (7 kg)/-5 lb (-2 kg).

C. Hydraulic pressure to the nose wheel steering system is necessary for this adjustment procedure.

D. The rig load of the cables change with the ambient temperature given in Table 501.

E. Use the shock strut centering cam to get the centered position for the nose gear. Manual aid can be necessary to make sure that the cam is correctly centered.

F. When you adjust the cables to the correct rig loads, make sure that you can engage all the rig pins in their rig pin holes. Make sure that you can remove the rig pins freely, without binding.

TASK 32-51-00-820-801

2. General Check of the Nose Wheel Steering System Adjustment

(Figure 501)

A. General

(1) This procedure gives instructions to do a general check of the nose wheel steering system adjustment. If the cable tension is within limits, the rig pin fit is good, and the system test is satisfactory, then the nose wheel steering system is fully operable.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-5505	Spring Scale

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

F. Prepare for the General Check of the Nose Wheel Steering System Adjustment

SUBTASK 32-51-00-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-00-860-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-010-001

(3) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

G. General Check of the Nose Wheel Steering System Adjustment

SUBTASK 32-51-00-820-001

(1) Make sure that the tension in the NGPPA and NGPPB cables is such that the load necessary to deflect the cable 0.50 ±0.05 in. (12.70 ±1.27 mm) downward at the turnbuckle is 19 ±3 lbf (85 ±14 N) (as measured with a spring scale, STD-5505).

SUBTASK 32-51-00-820-002

(2) Make sure that the tension in the NWS-A and NWS-B cables is in tolerance (Table 501).

SUBTASK 32-51-00-480-002

(3) Install these rig pins, from rigging pin kit, SPL-1585:

(a) Rig pin NS2 in the Captain's control wheel drum.

(b) Rig pin NS1 in the interconnect quadrant.

(c) Rig pin NS5 in the interconnect eccentric drum:

1) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

2) Make sure that the interconnect eccentric drum rig pin hole and bracket rig pin slot line up.

3) Install rig pin NS5.

(d) Rig pin R3 in the rudder forward quadrant.

(e) Rig pin NS4 in the summing mechanism bracket at the steering metering valve.

SUBTASK 32-51-00-820-003

(4) Make sure that all the rig pins have a free fit.

NOTE: The rig pins have a free fit if you can install or remove them with two fingers. Some resistance is permitted.

SUBTASK 32-51-00-080-001

(5) Remove all the rig pins.

SUBTASK 32-51-00-080-002

(6) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 32-51-00-730-001

(7) With the control wheel in the center position, make sure that the steering angle is 0 ±1.5 degrees on the decal that is on the nose gear shock strut.

NOTE: If the decal is not on the nose gear shock strut, then use the illustration in Figure 501 to make a decal with tape.

SUBTASK 32-51-00-210-001

(8) Examine all of the adjustable rod ends to make sure that there is correct thread engagement.

(a) Make sure that the end of the rod end covers a minimum of 50 percent of the inspection hole.

SUBTASK 32-51-00-210-002

(9) After the cables are adjusted, the cable cannot touch the pulley or quadrant flanges for the full travel of the cables.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the cables do not rub on the sides of the pulleys or quadrant.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-410-001

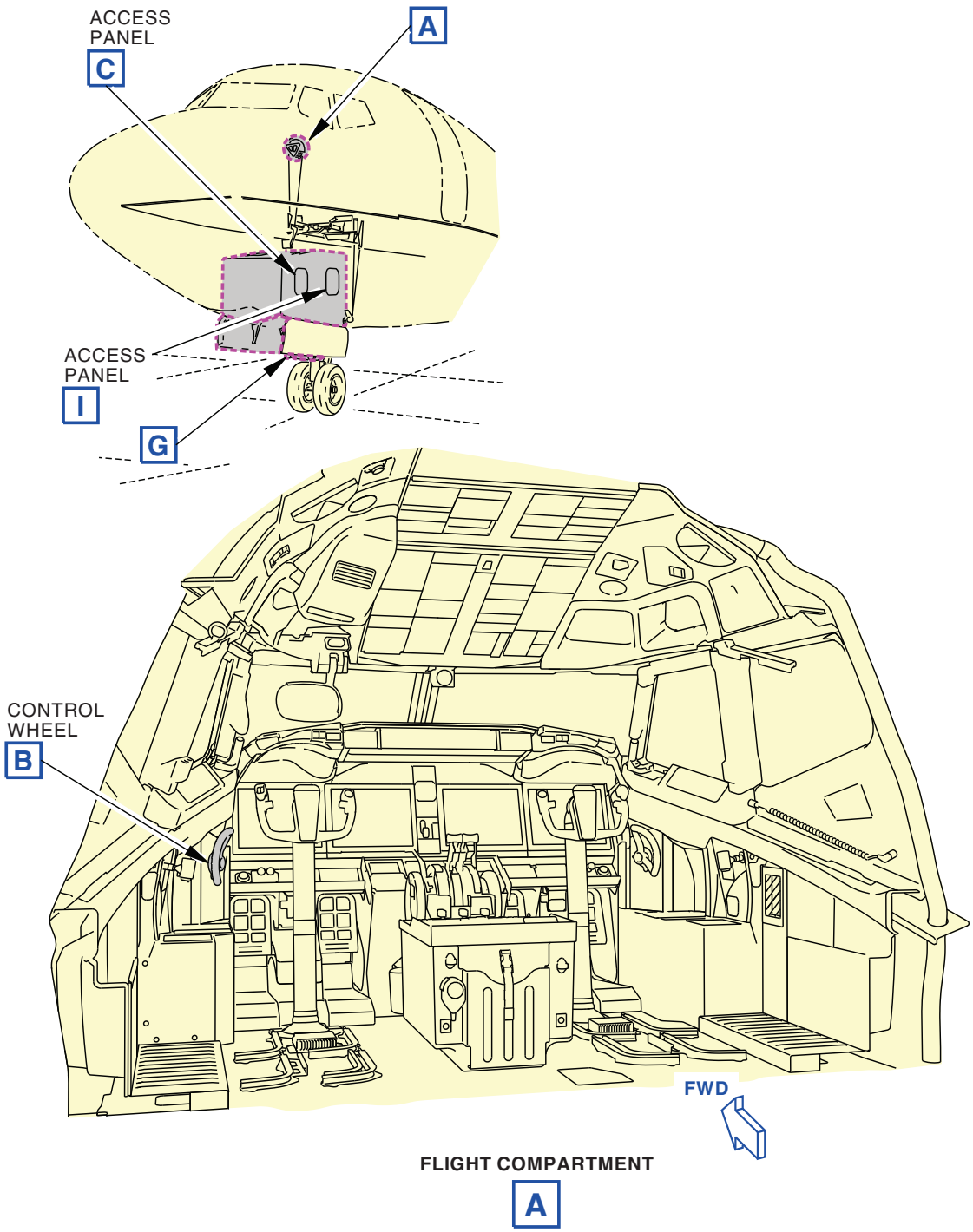
(1) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

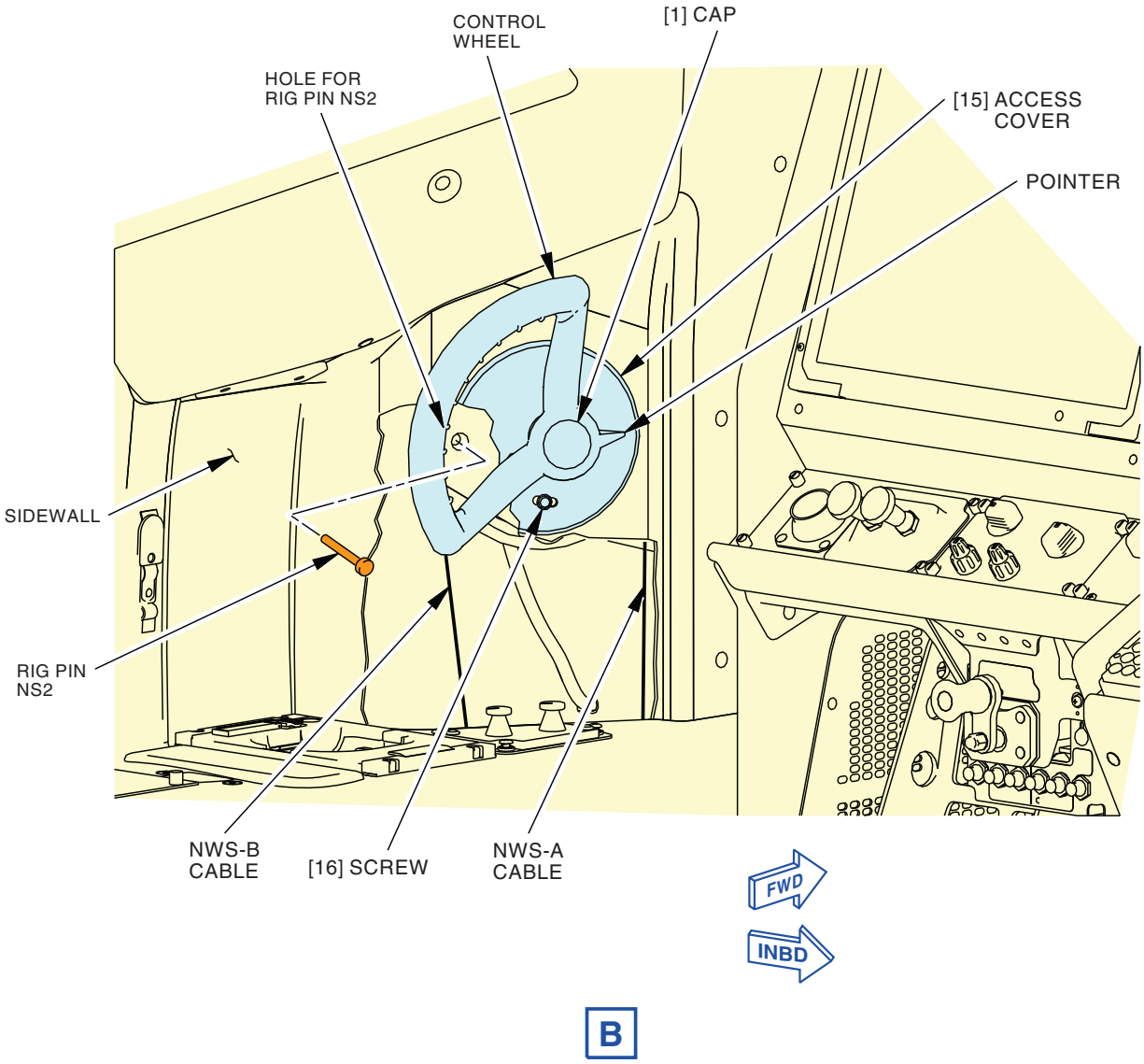
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 1 of 8)

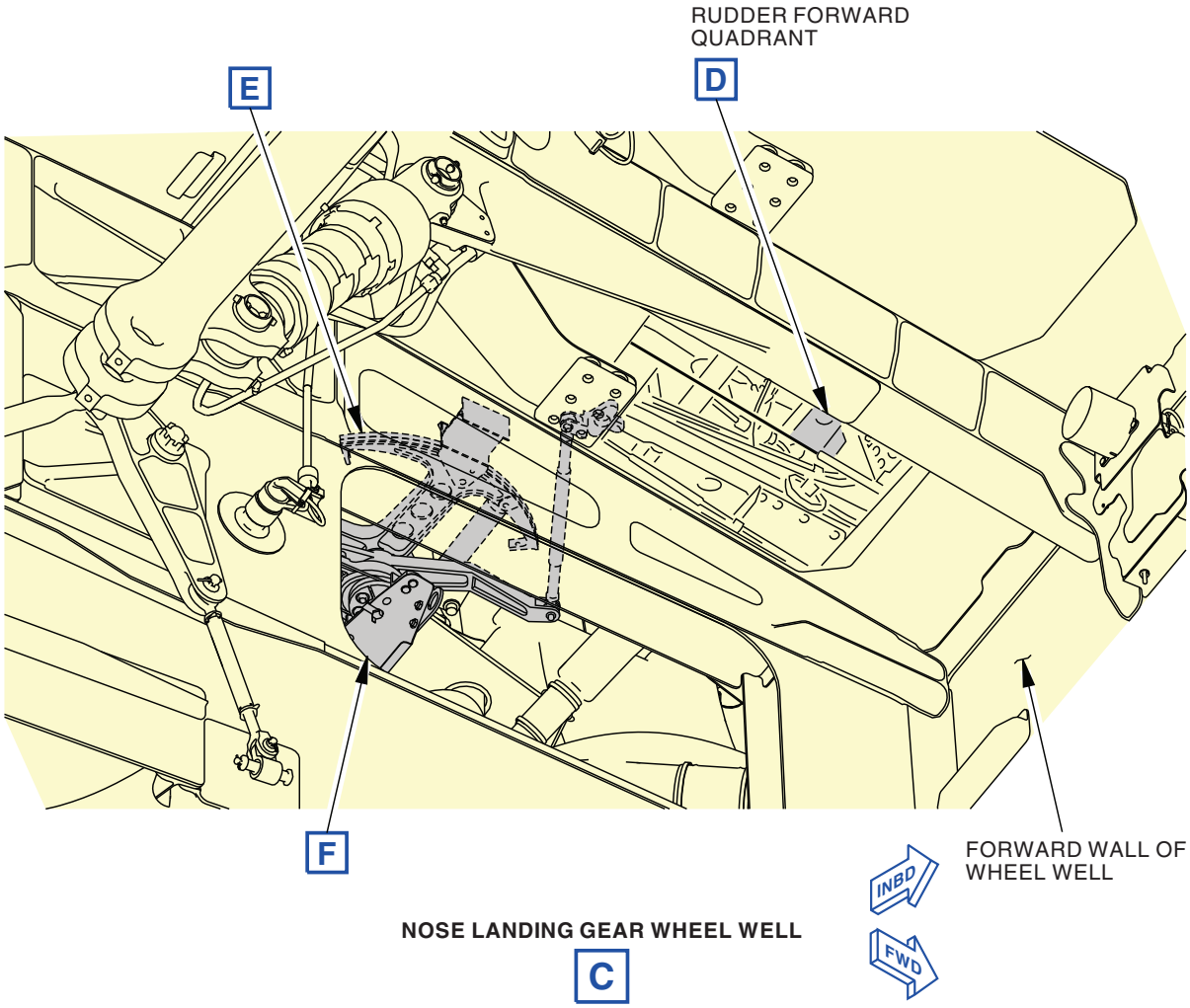
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414456 S00061535223_V2

Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 2 of 8)

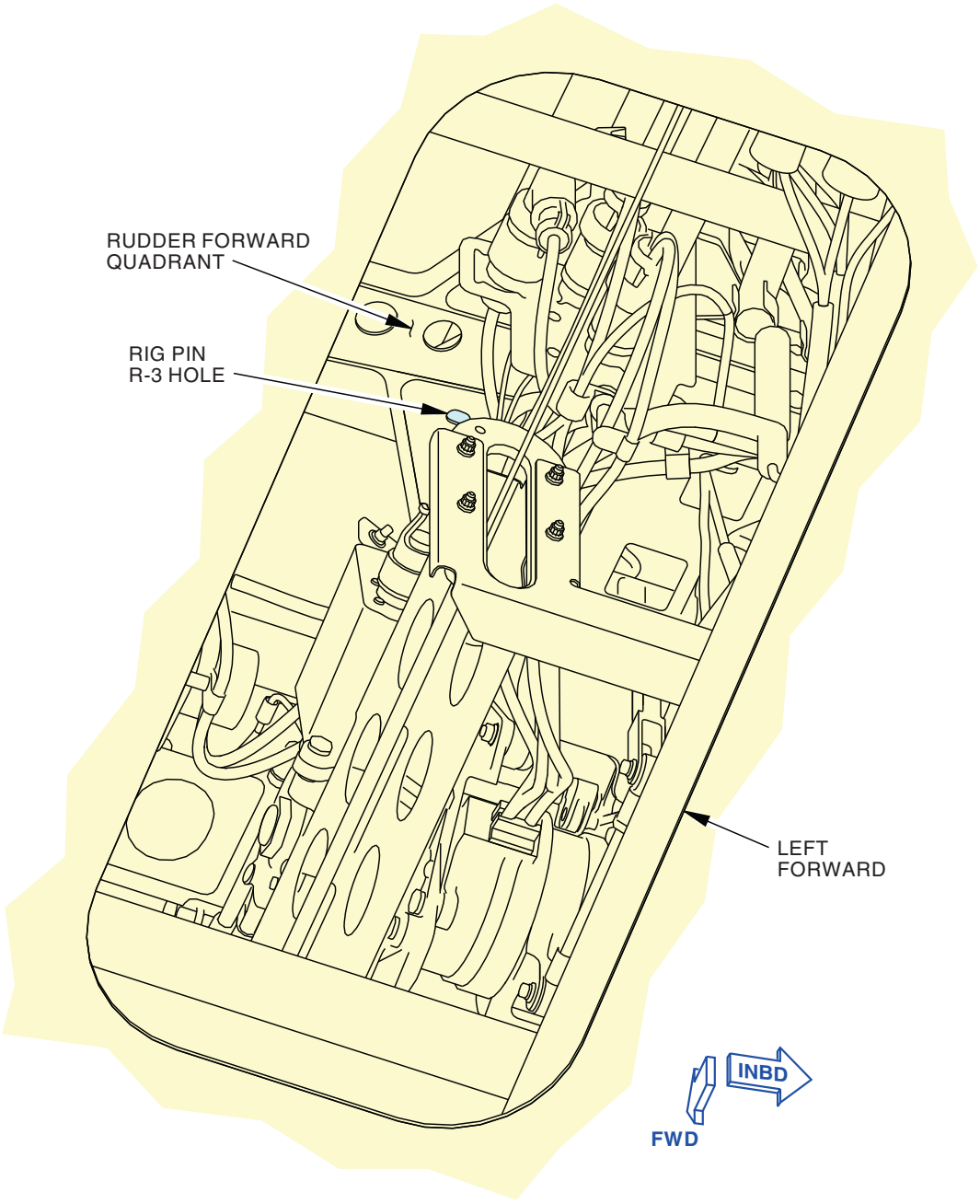
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414457 S00061535224_V2

Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



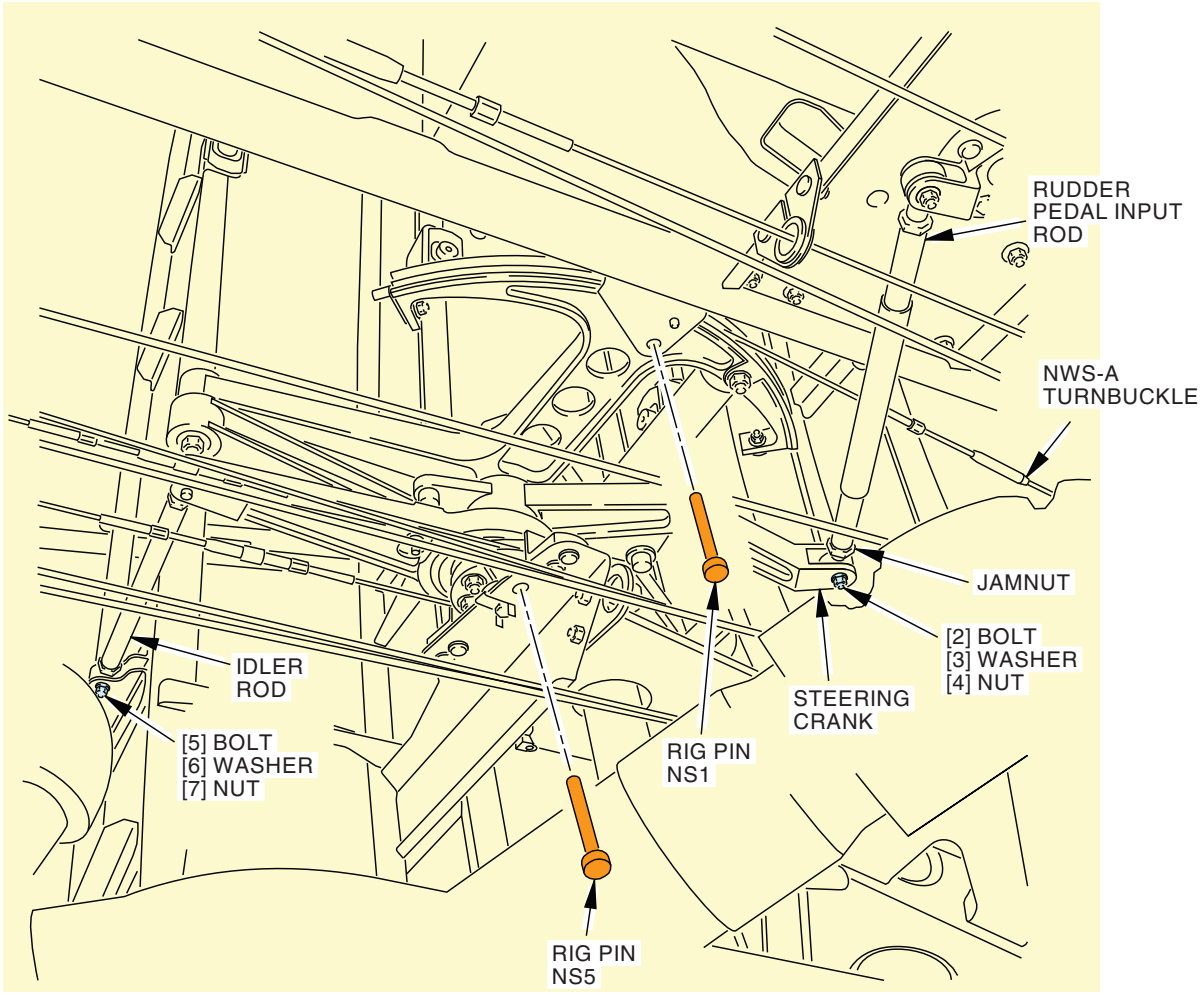
CAPTAIN'S RUDDER FORWARD QUADRANT

D

2414458 S00061535225_V3

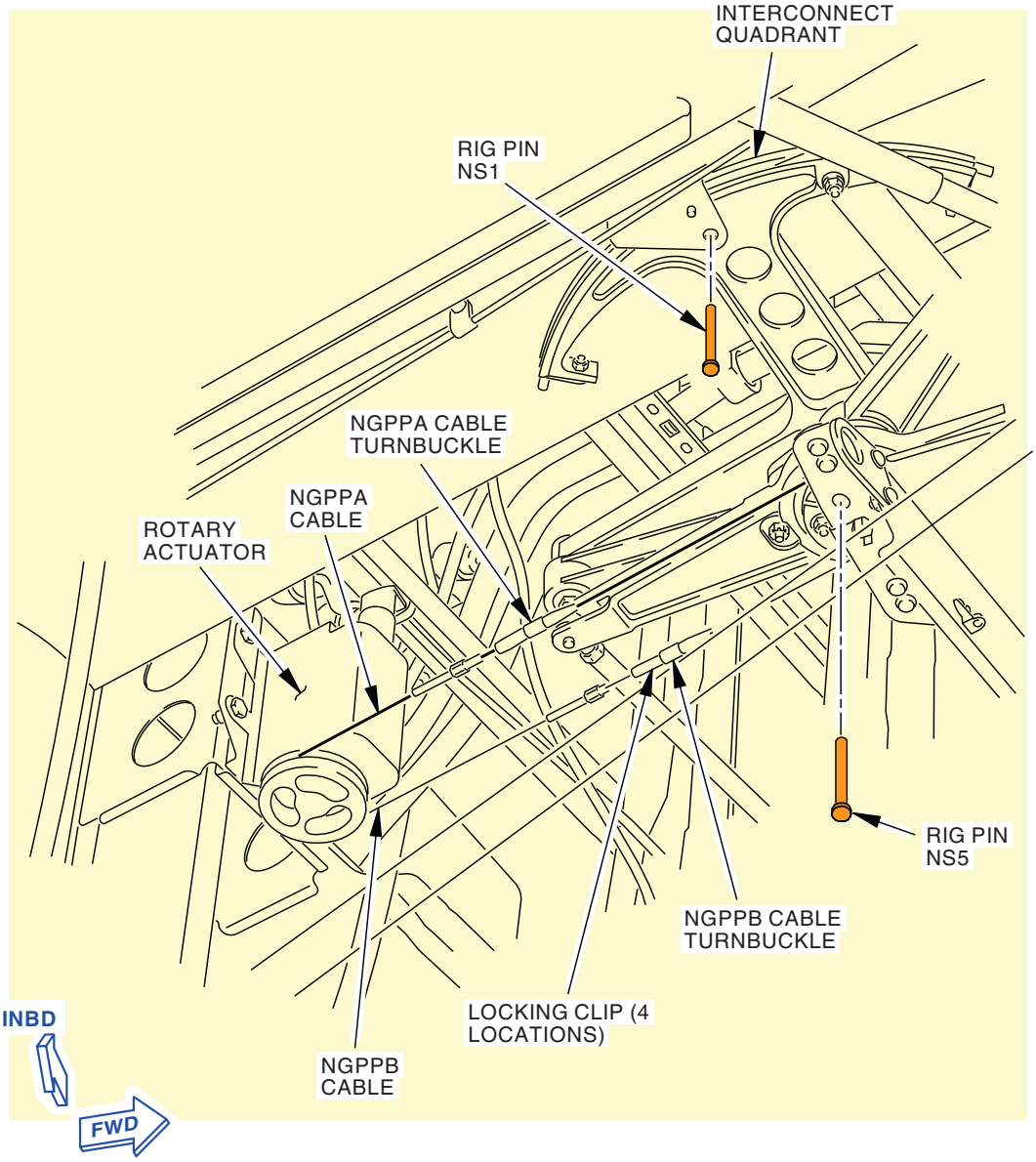
Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



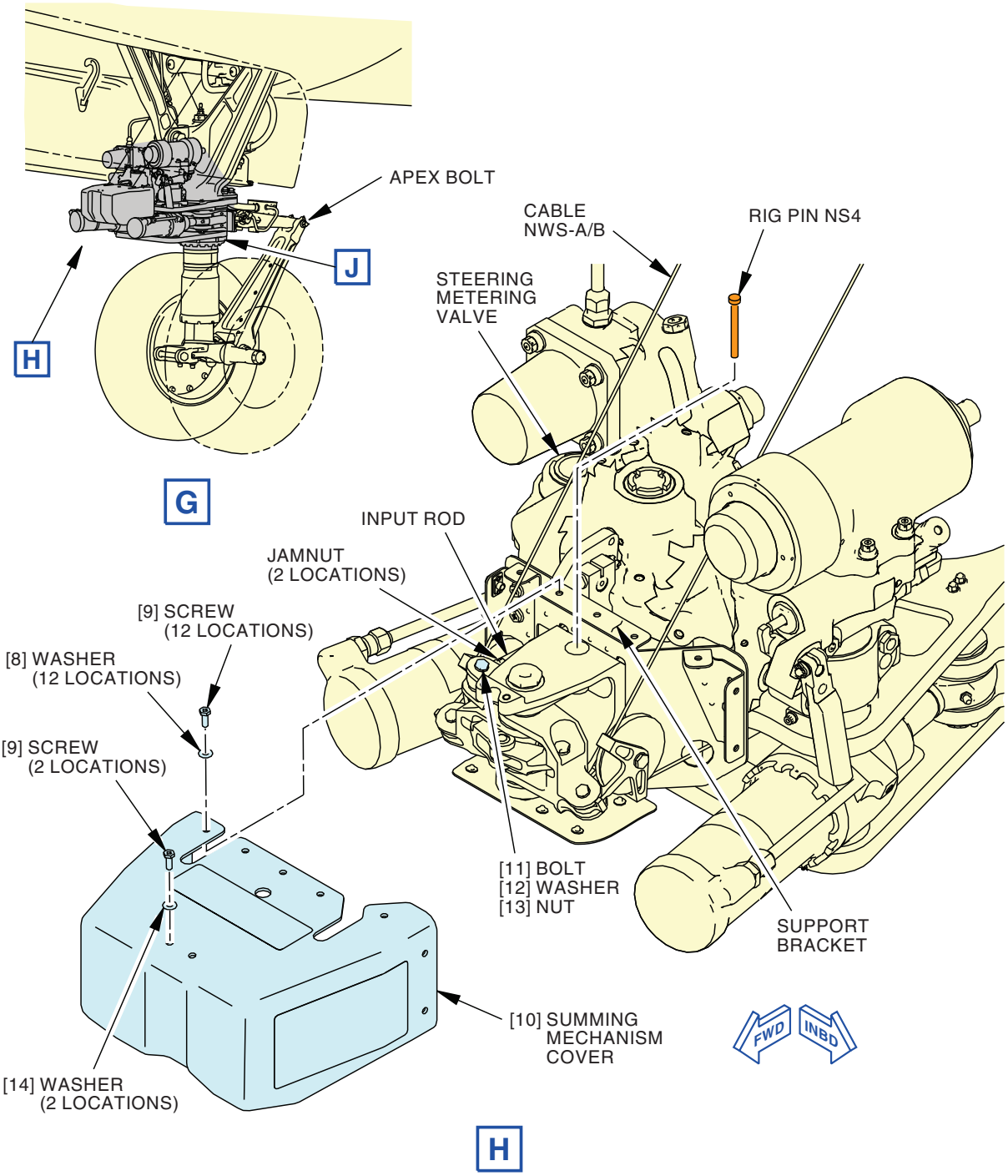
RUDDER PEDAL STEERING INTERCONNECT MECHANISM AND ROTARY ACTUATOR

F

2414460 S00061535227_V1

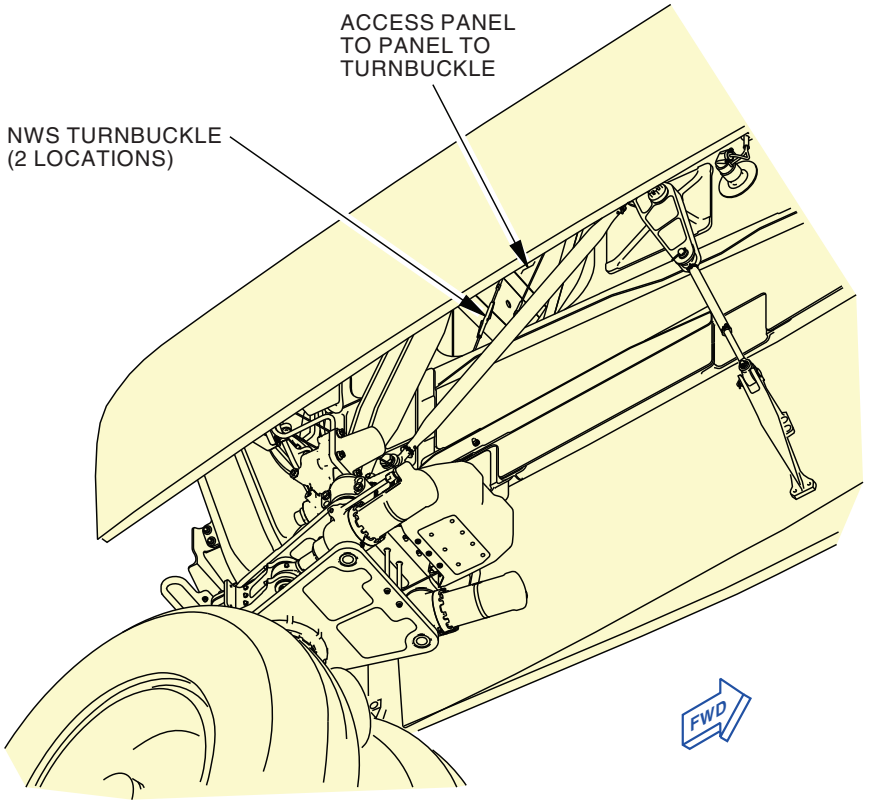
Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414462 S00061535229_V1

Nose Wheel Steering System Adjustment
Figure 501/32-51-00-990-801 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-00-820-802

3. Rudder Pedal Interconnect Mechanism Adjustment Check

(Figure 501)

A. General

(1) This procedure gives instructions to do an adjustment check of the rudder pedal interconnect mechanism.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-2529	Gauge - Force, Push/Pull/Spring Scale, 50 Pound Capacity Part #: FDIX 50 Supplier: 0BFD9 Opt Part #: FDI 50 Supplier: 0BFD9 Opt Part #: FDV 50 Supplier: 0BFD9
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare to Do the Adjustment Check

SUBTASK 32-51-00-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: TASK 32-00-01-480-801.

SUBTASK 32-51-00-860-003

WARNING REMOVE HYDRAULIC POWER FOR THE NOSE WHEEL STEERING BEFORE YOU DO THE CHECK. HYDRAULIC POWER CAN CAUSE THE NOSE WHEELS TO MOVE. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-010-002

(3) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

G. Rudder Pedal Interconnect Mechanism Adjustment Check

SUBTASK 32-51-00-820-004

(1) Make sure that the conditions of this check are satisfactory.

(a) If they are not, do this task: Rudder Pedal Interconnect Mechanism Adjustment, TASK 32-51-00-820-803.

SUBTASK 32-51-00-820-005

(2) Make sure that you can install the rig pin NS1, from rigging pin kit, SPL-1585, in the interconnect quadrant.

NOTE: The rig pins have a free fit if you can install or remove them with two fingers. Some resistance is permitted.

SUBTASK 32-51-00-820-006

(3) Make sure that you can install the rig pin NS5, from rigging pin kit, SPL-1585, in the interconnect eccentric drum:

NOTE: The rig pins have a free fit if you can install or remove them with two fingers. Some resistance is permitted.

(a) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(b) Make sure that the interconnect eccentric drum rig pin hole and bracket rig pin slot line up.

(c) Install rig pin NS5.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-00-080-003

(4) Remove the rig pins NS1 and NS5.

SUBTASK 32-51-00-840-001

(5) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 32-51-00-820-007

(6) Make sure that the tension in the NGPPA cable and NGPPB cable is such that the load necessary to move the cable 0.50 ±0.05 in. (12.70 ±1.27 mm) at the turnbuckle is 19 ±3 lbf (85 ±14 N) (as measured with a spring gauge, COM-2529).

(a) If you cannot get the correct tension, make sure that the cable terminal ends are started equally into the threads of the turnbuckle barrel.

SUBTASK 32-51-00-210-003

(7) Examine all of the adjustable rod ends to make sure that there is correct thread engagement.

(a) Make sure that the end of the rod end covers a minimum of 50 percent of the inspection hole.

SUBTASK 32-51-00-210-004

(8) After the cables are adjusted, make sure that the cable does not touch the pulley or quadrant flanges for the full travel of the cables.

(a) Make sure that the cables do not rub on the sides of the pulleys or quadrant.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-410-002

(1) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

——— END OF TASK ———

TASK 32-51-00-820-803

4. Rudder Pedal Interconnect Mechanism Adjustment

(Figure 501)

A. General

(1) This procedure gives instructions to do the adjustment of the rudder pedal interconnect mechanism.

B. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-5505	Spring Scale

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

F. Prepare for the Rudder Pedal Interconnect Mechanism Adjustment

SUBTASK 32-51-00-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-00-580-001

(2) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-51-00-860-004

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-00-860-005

(4) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-51-00-860-006

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Move the towing lever to the tow position and install the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-820-008

(6) Make sure that the nose wheels are in the center position.

NOTE: The inner cylinder centering cam will put the nose wheels in the center position when the airplane is lifted on jacks. Manual aid may be necessary to make sure the nose gear and cam is in the center position.

SUBTASK 32-51-00-010-003

(7) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

G. Rudder Pedal Interconnect Mechanism Adjustment

SUBTASK 32-51-00-020-001

(1) Remove the turnbuckle locking clips from the turnbuckles on the NGPPA and NGPPB cables.

SUBTASK 32-51-00-020-002

(2) Remove the nut [7], washer [6], and bolt [5] to disconnect the idler rod on the rudder pedal interconnect mechanism from the structure.

SUBTASK 32-51-00-220-001

(3) Make sure that the rotary actuator is turned in the counterclockwise direction with the index marks aligned to show that it is in the air mode.

SUBTASK 32-51-00-480-005

(4) Install the rig pins NS1 and NS5:

(a) Install the rig pin NS1, from rigging pin kit, SPL-1585.

(b) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

1) Make sure that the interconnect eccentric drum rig pin hole and bracket rig pin slot line up.

(c) Install rig pin NS5.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-00-820-010

(5) Adjust the tension in the NGPPA and NGPPB cables until the load necessary to move the cable 0.50 ±0.05 in. (12.70 ±1.27 mm) downward at the turnbuckle is 19 ±3 lbf (85 ±14 N) (as measured with a spring scale, STD-5505).

NOTE: It is not necessary to use the temperature because the NGPPA cable and NGPPB cable are very short.

SUBTASK 32-51-00-080-006

(6) Remove the rig pin NS5.

SUBTASK 32-51-00-710-001

(7) Operate the rotary actuator five times from the air mode (Put the Airplane in the Air Mode, TASK 32-09-00-860-801), back to the ground mode (Return the Airplane to the Ground Mode, TASK 32-09-00-860-802), and back again.

(a) Make sure that the rotary actuator changes from the air mode to the ground mode, and back again each time.

SUBTASK 32-51-00-210-005

(8) Make sure that you attach the rig pin NS5 freely:

(a) Do this task: Put the Airplane in the Air Mode, TASK 32-09-00-860-801.

(b) Make sure that the interconnect eccentric drum rig pin hole and bracket rig pin slot line up.

(c) Make sure that you can remove and install the rig pin NS5 freely.

SUBTASK 32-51-00-820-011

(9) Make sure that the tension in the NGPPA cable and NGPPB cable is such that the load necessary to move the cable 0.50 ±0.05 in. (12.70 ±1.27 mm) at the turnbuckle is 19 ±3 lbf (85 ±14 N) (as measured with a spring scale).

(a) If necessary, adjust the cable turnbuckles until the rig pin NS5 is free and the tension in the cables is correct.

SUBTASK 32-51-00-820-009

(10) Adjust the length of the idler rod, if necessary. Make sure that you attach the bolt [5] freely when you install it through the rod end and the structure.

(a) Install the bolt [5], washer [6], and nut [7] to attach the idler rod to the structure.

SUBTASK 32-51-00-080-004

(11) Make sure that you attach the rig pin NS1 freely.

SUBTASK 32-51-00-080-005

(12) Remove rig pin NS1.

SUBTASK 32-51-00-420-001

(13) Install the turnbuckle locking clips on the turnbuckles for the NGPPA cable and NGPPB cable.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-410-003

(1) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-860-007

(2) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

——— END OF TASK ———

TASK 32-51-00-820-804

5. Nose Wheel Steering System Adjustment Check

(Figure 501)

A. General

(1) This procedure gives instructions to do an adjustment check of the nose wheel steering system.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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E. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

F. Prepare for the Adjustment Check

SUBTASK 32-51-00-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: TASK 32-00-01-480-801.

SUBTASK 32-51-00-010-004

(2) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(3) Lift the airplane on jacks, until the nose gear shock strut is fully extended and the wheels do not touch the ground (TASK 07-10-01-580-801).

SUBTASK 32-51-00-860-008

(4) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 32-51-00-860-009

(5) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-860-010

(6) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-51-00-860-011

(7) Make sure that the NOSE WHEEL STEERING ALTERNATE switch is in the NORM position.

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SUBTASK 32-51-00-860-012

(8) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
D	17	C01027	LANDING GEAR NOSE GEAR STEER

G. Nose Wheel Steering System Adjustment Check

SUBTASK 32-51-00-820-012

(1) Make sure that the conditions of this adjustment check are satisfactory.

(a) If they are not, do this task: Nose Wheel Steering System Adjustment, TASK 32-51-00-820-805.

SUBTASK 32-51-00-730-002

(2) Make sure that the steering angle is 0 ±1.5 degrees on the decal that is on the nose gear shock strut.

NOTE: The inner cylinder centering cam will put the nose wheels in the center position when the airplane is lifted on jacks. Manual aid may be necessary to make sure the nose gear and cam is in the center position.

(a) If the decal is not on the nose gear shock strut, then use Figure 501 to make a decal with tape.

SUBTASK 32-51-00-860-013

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-730-003

(4) Make sure that the nose gear stays centered.

NOTE: If the nose gear moves when the hydraulic system is pressurized, the nose gear steering is not adjusted correctly. There is a disagreement between the steering system and the mechanical centerline position of the nose gear centering cams.

SUBTASK 32-51-00-820-013

(5) Make sure that the tension in the cables NWS-A and NWS-B is in tolerance (Table 501).

SUBTASK 32-51-00-010-005

(6) Remove the screws [9], washers [8], washers [14], and the summing mechanism cover [10].

SUBTASK 32-51-00-010-020

(7) Do these steps that follow to get access to the Captain's rig pin hole for the control wheels:

(a) Remove the screw [16] from the access cover [15].

(b) Turn the access cover [15] counterclockwise to get access to the rig pin hole.

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SUBTASK 32-51-00-480-007

(8) Make sure that you can install these rig pins, from rigging pin kit, SPL-1585:

NOTE: The steering system has a 1 - 2 degree dead band in the center position.

(a) Rig pin NS2 in the Captain's control wheel drum.

(b) Rig pin NS1 in the interconnect quadrant.

(c) Rig pin R3 in the rudder forward quadrant.

(d) Rig pin NS4 in the summing mechanism bracket at the steering metering valve.

SUBTASK 32-51-00-820-014

(9) Make sure that all the rig pins have a free fit.

NOTE: The rig pins have a free fit if you can install or remove them with two fingers. Some resistance is permitted.

SUBTASK 32-51-00-080-007

(10) Remove all the rig pins.

SUBTASK 32-51-00-410-022

(11) Do these steps that follow to cover the Captain's rig pin hole for the control wheels:

(a) Turn the access cover [15] clockwise.

1) Make sure that the pointer on the Captain's control wheel is within the CENTER band on the access cover [15] marker.

(b) Install the screw [16] to attach the access cover [15].

H. Travel Operation of the Nose Wheel to the Center with the Hydraulic Power On Check

SUBTASK 32-51-00-860-014

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-480-008

(2) Make sure that the greased plates are under the nose wheels.

NOTE: Plates made of steel are recommended. Plates made of a soft material, such as aluminum, can be destroyed by the test.

SUBTASK 32-51-00-580-003

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

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(CAUTION PRECEDES)

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(3) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-51-00-730-004

(4) Make sure that the Captain's control wheel operates correctly:

(a) Turn the Captain's control wheel to operate the nose wheel steering through at least 6 complete cycles, and make sure that it operates freely and smoothly.

NOTE: One cycle of the system is from center to full right, then to full left, then back to center.

(b) Make sure that there is no leakage of hydraulic fluid from the nose gear steering tubing and fittings from the landing gear selector valve to the valves on the steering actuators.

(c) Turn the nose wheels with the Captain's control wheel from the center position to the LEFT (CCW) stop position.

1) Make sure that the steering angle is 78 ±2 degrees on the decal that is on the nose gear shock strut.

(d) Release the Captain's control wheel and let it return to the center position.

1) Make sure that the steering angle is at 0 ±1.5 degrees.

(e) Turn the nose wheels with the Captain's control wheel from the center position to the RIGHT (CW) stop position.

1) Make sure that the steering angle is 78 ±2 degrees on the decal that is on the nose gear shock strut.

(f) Release the Captain's control wheel and let it return to the center position.

1) Make sure that the steering angle is at 0 ±1.5 degrees.

SUBTASK 32-51-00-210-006

(5) Examine all of the adjustable rod ends to make sure that there is correct thread engagement.

(a) Make sure that the end of the rod end covers a minimum of 50 percent of the inspection hole.

SUBTASK 32-51-00-210-007

(6) After the cables are adjusted, make sure that the cable does not touch the pulley or quadrant flanges for the full travel of the cables.

(a) Make sure that the cables do not rub on the sides of the pulleys or quadrant.

SUBTASK 32-51-00-410-004

(7) Install the summing mechanism cover [10] with screws [9], washers [8], and washers [14].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-860-015

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-580-014

(2) Lift the airplane on jacks, until the nose gear shock strut is fully extended and the wheels do not touch the ground (TASK 07-10-01-580-801).

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SUBTASK 32-51-00-020-005

(3) Remove the greased plates from under the nose wheels.

SUBTASK 32-51-00-580-015

(4) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-51-00-860-016

(5) Do this task: Remove Electrical Power, TASK 24-22-00-860-802.

SUBTASK 32-51-00-410-005

(6) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

——— END OF TASK ———

TASK 32-51-00-820-805

6. Nose Wheel Steering System Adjustment

(Figure 501)

A. General

(1) This task gives the instructions to do the adjustment of the nose wheel steering system.

(a) There are three methods to do this task:

1) Method 1: Airplane On Ground/Off Jacks

a) Method 1 requires removal of the torsion link apex pin to disconnect the steering mechanism from the wheel assembly.

2) Method 2: Airplane On Jacks

a) Method 2 requires the Nose Landing Gear (NLG) to be compressed while the airplane is jacked.

3) Method 3: Airplane On Grease Plate.

a) Method 3 does not need removal of the torsion pin. The grease plate will prevent tire scuffing when the NLG is turned left or right.

B. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
09-10-00-580-801	Nose Gear Maintenance Towing (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)
32-51-11-400-801	Steering Metering Valve Installation (P/B 401)

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(Continued)

Reference	Title
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

F. Prepare for the Nose Wheel Steering System Adjustment

SUBTASK 32-51-00-480-009

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: TASK 32-00-01-480-801.

SUBTASK 32-51-00-860-017

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-51-00-010-006

(3) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles.

(a) Open these access panels:

(Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801)

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-860-081

(4) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-51-00-860-082

(5) Make sure that the rotary actuator is in ground mode for all methods.

NOTE: When the rotary actuator is in ground mode, the rotary actuator will move the cable drum clockwise. The witness line on the cable drum shall move to a tolerance angle of 102 degrees - 112 degrees with respect to the witness lines on the housing. This is used to determine the position of the rotary actuator.

SUBTASK 32-51-00-580-004

(6) For Method 1, do these steps:

(a) Tow the airplane at least 12.0 ft (3.7 m) in a straight line before it is parked (TASK 09-10-00-580-801).

(b) Remove the clamps, brackets, and conduit that attached to the NLG/Taxi Light.

NOTE: Clamps on torsion links are removed to prevent damage to the conduit.

(c) Disconnect the torsion link apex pin (TASK 32-21-31-000-803).

SUBTASK 32-51-00-580-016

(7) For Method 2, do these steps:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

NOTE: Opening the circuit breaker will prevent steering disengaging during the test.

(b) Lift the airplane on jacks until the nose gear shock strut is fully extended and that the wheels do not touch the ground, do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

NOTE: The nose of the airplane is lifted on jacks to find the center position of the nose landing gear and the steering collar. The internal centering cam of the shock strut is used to do this.

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CAUTION MAKE SURE THAT THE SHOCK STRUT IS COMPRESSED. THE CENTERING CAMS WILL BE ENGAGED IF THE SHOCK STRUT IS NOT COMPRESSED THE MINIMUM AMOUNT. THIS CAN CAUSE DAMAGE TO THE AIRCRAFT.

(c) Make sure that the shock strut for the NLG is compressed a minimum of 3.5 in. (88.9 mm).

SUBTASK 32-51-00-580-017

(8) For Method 3, tow the airplane at least 12.0 ft (3.7 m) in a straight line before it is parked on the grease plate (TASK 09-10-00-580-801).

SUBTASK 32-51-00-860-019

(9) Make sure that the steering angle is 0 ±1.5 degrees on the decal that is on the nose gear shock.

SUBTASK 32-51-00-860-021

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(10) Move the towing lever to the TOW position and install the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-820-015

(11) Adjust the cable guard to clear the steering collar pulley by 0.015 in. (0.381 mm) - 0.045 in. (1.143 mm).

(a) If it is necessary, loosen the attach bolt and adjust the cable guard in its slotted hole.

SUBTASK 32-51-00-020-003

(12) Remove the nut [4], washer [3], and bolt [2] to disconnect the rudder pedal input rod from the steering crank.

SUBTASK 32-51-00-730-005

(13) Make sure that the steering angle is 0 ±1.5 degrees on the decal that is on the nose gear shock.

NOTE: The inner cylinder centering cam will put the nose wheels in the center position when the airplane is lifted on jacks. Manual assistance may be necessary to make sure that the nose gear and cam is in the center position.

(a) If the decal is not on the nose gear shock strut, use the illustration in Figure 501 to make a decal with tape.

SUBTASK 32-51-00-010-007

(14) Remove the screws [9], washers [8], washers [14], and summing mechanism cover [10].

SUBTASK 32-51-00-010-008

(15) Do the steps that follow to get access to the Captain's rig pin hole for the control wheels:

(a) Remove the screw [16] from the access cover [15].

(b) Turn the access cover [15] counterclockwise to get access to the rig pin hole.

G. Nose Wheel Steering System Adjustment

SUBTASK 32-51-00-820-016

(1) Disconnect the steering valve input rod from the steering metering valve summing mechanism.

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SUBTASK 32-51-00-820-025

(2) Install the rig pin NS4 in the summing mechanism bracket at the steering metering valve.

NOTE: This step does a check of the rig pin NS4 against only the centering springs inside the steering valve. A step will follow that does a check of the rig pin against the steering cables. These steps must be done in this order.

(a) If the removed bolt does not fit freely, do these steps to adjust the length of the steering valve input rod such that you can install the bolt freely:

1) Loosen the jamnuts to disengage the locking keys.

2) Adjust the length of the steering valve input rod.

3) Tighten the jamnuts and install safety wire.

4) Make sure that the safety wire tail is on the aft end of the steering valve input rod when installed.

SUBTASK 32-51-00-840-002

(3) Connect the steering valve input rod from the steering metering valve summing mechanism.

SUBTASK 32-51-00-480-010

(4) Install rig pin NS2 from rigging pin kit, SPL-1585, in the Captain's control wheel drum.

SUBTASK 32-51-00-490-001

(5) Install the rig pin NS1 in the interconnect quadrant.

SUBTASK 32-51-00-020-004

(6) Remove the locking clips from the turnbuckles on the NWS-A cable and NWS-B cable.

SUBTASK 32-51-00-820-017

(7) For new cables only, do these steps:

(a) Use the low tension cable tensiometer, COM-1553, to adjust the load on NWS-A cable and NWS-B cable to 85 ±5 lbf (378 ±22 N).

(b) Remove all rig pins.

(c) For Method 1, make sure that the upper torsion link and other components are supported in a way that it does not contact the tires or doors when the system is operated.

CAUTION

MAKE SURE THAT THE SHOCK STRUT IS COMPRESSED. THE CENTERING CAMS WILL BE ENGAGED IF THE SHOCK STRUT IS NOT COMPRESSED THE MINIMUM AMOUNT. THIS CAN CAUSE DAMAGE TO THE AIRCRAFT.

(d) Make sure that the shock strut for the NLG is compressed a minimum of 3.5 in. (88.9 mm).

(e) For Method 3, make sure that the grease plates are under the nose wheels.

NOTE: Plates made of steel are suggested. Plates made of a soft material such as aluminum can be destroyed by the test.

(f) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(g) Remove the NLG towing lever pin, SPL-1499, and move the towing lever to the "PRESSURE ON" position.

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CAUTION

HOLD THE UPPER TORSION LINK TO SET THE AIRPLANE NOSE TO THE GROUND MODE. IF YOU DO NOT OBEY, DAMAGE TO THE CONDUIT CAN OCCUR.

(h) For Method 1, hold the upper torsion link to set airplane nose to ground mode.

(i) Use the steering tiller to operate the steering system of the nose wheel through 25 cycles (center-full left-full right-center).

WARNING

USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(j) Move the towing lever to the TOW position and install the NLG towing lever pin, SPL-1499.

(k) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(l) Make sure that the gear is steered to the straight ahead position before rigging.

(m) Install the rig pins NS1, NS2, and NS4.

SUBTASK 32-51-00-820-018

(8) Adjust the tension of all the segments of the NWS-A cable and NWS-B cable until the rig pins NS1, NS2, and NS4 fit freely (Table 501).

Table 501/32-51-00-993-801

TEMPERATURE	RIG LOAD*[1]
130°F (54°C)	56 lbf (249 N)
110°F (43°C)	51 lbf (227 N)
90°F (32°C)	46 lbf (205 N)
70°F (21°C)	40 lbf (178 N)
50°F (10°C)	35 lbf (156 N)
30°F (-1°C)	29 lbf (129 N)
10°F (-12°C)	24 lbf (107 N)
-10°F (-23°C)	18 lbf (80 N)
-22°F (-30°C)	15 lbf (67 N)*[2]
-30°F (-34°C)	13 lbf (58 N)*[2]
-40°F (-40°C)	10 lbf (44 N)*[2]

*[1] Cable loads must be within 0 +10 / -0 lbf (0 +44 / -0 N) of table values when you adjust the system. After initial adjustment any deviation from table values 0 +15 / -5 lbf (0 +67 / -22 N), at -22°F (-30°C) or above, shall require that the system adjusted again.

*[2] At temperatures below -22°F (-30°C), the minimum cable load shall be 10 lbf (44 N).

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SUBTASK 32-51-00-420-008

(9) Make sure that the rotary actuator is in ground mode for all methods.

NOTE: When the rotary actuator is in ground mode, the rotary actuator will move the cable drum clockwise. The witness line on the cable drum shall move to a tolerance angle of 102 degrees - 112 degrees with respect to the witness lines on the housing. This is used to determine the position of the rotary actuator.

SUBTASK 32-51-00-420-002

(10) Install rig pin R3 in the rudder forward quadrant.

NOTE: This step is necessary to make sure that rudder system rigging will not affect the steering system rigging.

NOTE: Rig pin R3 will have a flat end, with a small chamfer.

(a) If you are unable to install R3 rig pin, then the rudder system needs to be adjusted to align this rig pin hole to avoid a possible steering issue in the future.

1) One method of adjusting the rudder system is to check if the rudder trim indication is zero, and positioning the rudder manually over the index plate.

SUBTASK 32-51-00-820-019

CAUTION

MAKE SURE THE END OF THE INPUT ROD WILL COVER AT LEAST ONE HALF OF THE INSPECTION HOLE. A MINIMUM OF THREE THREADS MUST BE OPEN OUTSIDE THE LOCKNUT ON THE CONTROL ROD AT THE CRANK END. INCORRECT THREAD ENGAGEMENT OF THE INPUT ROD CAN CAUSE AN INTERFERENCE AND A POSSIBLE FAILURE OF THE INPUT ROD CAN OCCUR.

(11) Do the steps below:

(a) Adjust the length of the rudder pedal input rod and make sure that the bolt [2], which attaches the rudder pedal input rod to the steering crank fits freely.

(b) Install the bolt [2], washer [3], and nut [4].

(c) Tighten the jamnut on the rudder pedal input rod.

SUBTASK 32-51-00-080-008

(12) Remove all of the rig pins.

SUBTASK 32-51-00-212-001

(13) For Method 3, make sure that the grease plates are under the nose wheels.

NOTE: Plates made of steel are recommended. Plates made of a soft material such as aluminum can be destroyed by the test.

SUBTASK 32-51-00-580-018

CAUTION

MAKE SURE THAT THE SHOCK STRUT IS COMPRESSED. THE CENTERING CAMS WILL BE ENGAGED IF THE SHOCK STRUT IS NOT COMPRESSED THE MINIMUM AMOUNT. THIS CAN CAUSE DAMAGE TO THE AIRCRAFT.

(14) For Method 2, make sure that the shock strut for the NLG is compressed a minimum of 3.5 in. (88.9 mm).

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SUBTASK 32-51-00-860-022

CAUTION HOLD THE UPPER TORSION LINK TO SET THE AIRPLANE NOSE TO THE GROUND MODE. IF YOU DO NOT OBEY, DAMAGE TO THE CONDUIT CAN OCCUR.

(15) For Method 1, hold the upper torsion link up to set the airplane nose to ground mode.

SUBTASK 32-51-00-420-010

(16) Make sure that the rotary actuator is in ground mode for all methods.

NOTE: When the rotary actuator is in ground mode, the rotary actuator will move the cable drum clockwise. The witness line on the cable drum shall move to a tolerance angle of 102 degrees - 112 degrees with respect to the witness lines on the housing. This is used to determine the position of the rotary actuator.

SUBTASK 32-51-00-860-023

(17) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-860-024

(18) Remove the NLG towing lever pin, SPL-1499, and let the towing lever move to the "PRESSURE ON" position.

SUBTASK 32-51-00-820-020

(19) Do a check of the steering adjustment.

(a) Use the steering tiller to operate the steering system of the nose wheel full left and full right.

NOTE: The steering system has a 1 - 2 degree dead band in the center position.

NOTE: For Method 1, be careful to avoid damage to the conduit during operation.

(b) If it is not within the dead band, do this task: Rudder Pedal Interconnect Mechanism Adjustment, TASK 32-51-00-820-803.

(c) If it is necessary, adjust the length of the rudder pedal input rod until the steering system returns equally on the two sides of the "0" mark after turns to the right and left.

SUBTASK 32-51-00-420-003

(20) Make sure that the rudder pedal input rod is correctly connected to the steering crank.

SUBTASK 32-51-00-420-004

WARNING MAKE SURE THAT THE STEERING WHEEL DOES NOT MOVE. IF IT DOES, IT CAN CAUSE ACCIDENTAL MOVEMENT OF THE RUDDER-PEDAL STEERING INTERCONNECTED-MECHANISM. THIS CAN CAUSE INJURY TO PERSONNEL.

(21) Install the locking clips on the turnbuckles of the NWS-A cable and NWS-B cable.

NOTE: If you are completing this task as a result of the installation of the steering metering valve Steering Metering Valve Installation, TASK 32-51-11-400-801, the subsequent steps do not apply and it is not necessary that they are completed.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-410-017

(1) For Method 1, do these steps:

(a) Connect the torsion link apex pin (TASK 32-21-31-400-803).

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(b) Install the clamps, brackets, and conduit on the NLG/Taxi Light.

NOTE: These components were removed to disconnect the upper torsion link.

SUBTASK 32-51-00-580-019

(2) For Method 2, do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-51-00-580-020

(3) For Method 3, move the airplane off the grease plate.

SUBTASK 32-51-00-410-015

(4) Do the steps that follow to cover the Captain's rig pin hole for the control wheels:

(a) Turn the access cover [15] clockwise.

1) Make sure that the pointer on the Captain's control wheel is within the CENTER band on the access cover [15] marker.

(b) Install the screw [16] to attach the access cover [15].

SUBTASK 32-51-00-410-016

(5) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles.

(a) Close these access panels:

(Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801)

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-860-080

(6) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-51-00-730-007

(7) Do the system test for the nose wheel steering system (TASK 32-51-00-700-801).

——— END OF TASK ———

TASK 32-51-00-820-806

7. Nose Wheel Steering System Adjustment (Adjusting Steering Metering Valve Input Rod Method)

(Figure 501)

A. General

(1) This procedure is to be used only after rigging in accordance with the usual rigging procedure (TASK 32-51-00-820-805).

(2) This task gives a procedure for adjusting the steering system rigging based on the recorded quantity of rudder trim necessary to counteract a steering pulling condition as measured during ground maneuvers.

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(3) This is a procedure for adjusting steering system only if necessary to improve steering centering characteristics. Prior to making steering adjustments, pilot evaluation of the rudder trim necessary to center the steering system, must be recorded. Fewer than 2 units of rudder trim is necessary for acceptable centering. One (1) unit of rudder trim is equivalent to ¼ degree of steered angle or 0.02 in. (0.51 mm) on the steering marker. The rudder trim that produces satisfactory centering characteristics is used to readjust the steering control system. Initial rigging requires the steering to center equally about the “0” mark when returning from turns from the right and left. After completion of this procedure, the steering does not have to center equally about the “0” mark.

B. References

Reference	Title
12-15-51-780-801	Landing Gear Tire Pressure Check and Tire Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Prepare for the Adjustment

SUBTASK 32-51-00-480-011

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

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SUBTASK 32-51-00-200-001

(2) Make sure that all of the tires are correctly inflated (TASK 12-15-51-780-801).

SUBTASK 32-51-00-000-001

(3) Put the nose landing gear on grease plates or disconnect the nose gear torsion links at the apex bolt for this adjustment procedure (TASK 32-21-31-000-803).

(a) Make sure that you tie up or support the upper torsion link during rigging off jacks at an angle that will not let the torque link touch the tires or doors when the system is operated.

F. Nose Wheel Steering System Adjustment (Adjusting Steering Metering Valve Input Rod Method)

SUBTASK 32-51-00-860-025

(1) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-080-009

(2) Remove the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-210-008

(3) Make sure that the rudder trim is set to zero (“0”) (± 1/4°).

SUBTASK 32-51-00-710-002

(4) With the rudder pedals, cycle the system to the left and right and let it return to center from each direction.

NOTE: Note the relationship on the indicator mark on the back of the steering collar to the zero (“0”) mark on the decal. There will be a difference in this relationship when returning from the left or returning from the right of up to 1 degree, 0.06 in. (1.52 mm) on the decal, due to the dead band in the steering valve.

SUBTASK 32-51-00-860-026

(5) Set the rudder trim (± 1/4°) to the setting the pilot noted for acceptable centering characteristics.

SUBTASK 32-51-00-710-003

(6) With the rudder pedals, cycle the system to the left and right and let it return to center from each direction.

NOTE: Note these relationships of the indicator mark on the back of the steering collar to the zero (“0”) mark on the decal. These are the positions to which the steering must return after adjustments are made.

SUBTASK 32-51-00-860-027

(7) Set the rudder trim back to zero (“0") (± 1/4°).

SUBTASK 32-51-00-860-028

(8) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-860-029

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(9) Move the towing lever to the TOW position and install the NLG towing lever pin, SPL-1499.

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SUBTASK 32-51-00-820-021

(10) Make sure that you can install these rig pins from the rigging pin kit, SPL-1585, and that the rig pins have a free fit:

NOTE: The rig pins must have a free fit when you install or remove them. Some resistance is permitted.

(a) Rig pin NS2 in the Captain's control wheel drum

(b) Rig pin NS4 in the summing mechanism bracket at the steering metering valve.

SUBTASK 32-51-00-000-002

(11) Loosen the jamnuts on the input rod.

SUBTASK 32-51-00-820-022

(12) Based on the setting the pilot noted for acceptable centering characteristics, adjust the length of the input rod to correct the steering pulling condition.

NOTE: Turning the hexagonal barrel between the rod ends in the clockwise direction when facing aft, lengthens the control rod. One complete revolution in the clockwise direction changes the steering angle approximately equivalent to one unit of left rudder trim. One complete revolution in the counterclockwise direction changes the steering angle approximately equivalent to one unit of right rudder trim.

SUBTASK 32-51-00-420-005

(13) Tighten the jamnuts.

SUBTASK 32-51-00-860-030

(14) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-860-031

(15) Remove the NLG towing lever pin, SPL-1499, from the towing lever.

SUBTASK 32-51-00-710-004

(16) Cycle the system to the left and right with the rudder pedals and let it return to center from each direction.

SUBTASK 32-51-00-200-002

(17) Compare the centering to what was noted earlier in the task when the rudder was cycled around the trim point the pilot had set.

SUBTASK 32-51-00-860-032

(18) If it is satisfactory, do this task to remove hydraulic system A pressure: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-420-006

(19) Install the safety wire on the input rod.

SUBTASK 32-51-00-200-003

(20) Make sure that the rig pin NS2 in the Captain's control wheel drum does not bind.

NOTE: It is not necessary for rig pin NS4 in the summing mechanism bracket at the steering metering valve to fit freely.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-420-007

(1) If it was disconnected, reconnect the nose gear torsion links at the apex bolt (TASK 32-21-31-400-803).

END OF TASK

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TASK 32-51-00-700-801

8. Nose Wheel Steering System Test (Airplane on Ground)

(Figure 501)

A. General

(1) This procedure gives instructions to do a test of the nose wheel steering system while the airplane is on the ground on grease plates.

(2) The test of the nose wheel steering system includes the tests that follow:

(a) A steering centering test to make sure that the steering metering valve is in the null position when the nose gear is mechanically centered

(b) A normal steering response test

(c) A tiller control wheel torque test

(d) A test of the nose gear turn radius and response to full rudder pedal input

(e) A tow test of the nose gear steering to make sure that the nose gear will turn freely with the tow lever in the tow position

(f) A test of the alternate steering system

(g) An air/ground mode test.

(3) The nose wheel steering system must be adjusted correctly before you do the system test.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)
32-51-11 P/B 401	STEERING METERING VALVE - REMOVAL/INSTALLATION

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-7306	Extension Bar, Steering Test - Torque Link, NLG Part #: C32020-1 Supplier: 81205

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(Continued)

Reference	Description
STD-1035	Wrench - Dial Torque,0 to 500 in-lbs (0 to 56.5 N-m), +/-5% accuracy

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
117A	Electronic Equipment Access Door

F. Prepare to Test the Steering System of the Nose Wheel

SUBTASK 32-51-00-860-076

(1) Make sure that you park the airplane on a flat surface with no crosswinds.

NOTE: Crosswinds can caster the nose gear when the hydraulic system is depressurized.

SUBTASK 32-51-00-480-019

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-51-00-610-001

(3) Service the hydraulic system (TASK 12-12-00-610-801).

SUBTASK 32-51-00-580-006

(4) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-51-00-860-034

(5) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-860-035

(6) Make sure that the LANDING GEAR control lever is in the DN position.

SUBTASK 32-51-00-860-036

(7) Make sure that the NOSE WHEEL STEER switch is in the NORM position.

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SUBTASK 32-51-00-860-037

(8) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	17	C01027	LANDING GEAR NOSE GEAR STEER

G. Steering Centering Test

SUBTASK 32-51-00-860-038

(1) Remove the NLG towing lever pin, SPL-1499, from the towing lever and let the lever move to the PRESSURE ON position.

SUBTASK 32-51-00-010-013

(2) To get access to the rudder quadrant, do this step:

(a) Open this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-860-039

(3) Make sure that the rig pin R3, from the rigging pin kit, SPL-1585, fits freely in the rudder quadrant.

NOTE: This will show that the rudder pedals are in the neutral position.

SUBTASK 32-51-00-730-009

(4) Make sure that the steering angle is 0 ±1.5 degrees on the decal that is on the nose gear shock.

NOTE: The inner cylinder centering cam will put the nose wheels in the center position when the airplane is lifted on jacks. Manual aid may be necessary to make sure the nose gear and cam is in the center position.

(a) If the decal is not on the nose gear shock strut, then make a decal with tape (Figure 501).

SUBTASK 32-51-00-860-040

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-730-010

(6) Make sure that the nose gear stays centered.

NOTE: If the nose gear moves when the hydraulic system is pressurized, the nose gear steering is not adjusted correctly. There is a disagreement between the steering system and the mechanical centerline position of the nose gear centering cams.

SUBTASK 32-51-00-860-041

(7) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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H. Do the Test of the Normal Steering Response

SUBTASK 32-51-00-480-013

(1) Make sure that the greased plates are under the nose wheels.

SUBTASK 32-51-00-580-010

(2) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-51-00-860-042

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-860-058

(4) Make sure that one or more main landing gear is in ground mode.

NOTE: The air/ground system needs two inputs to transition into the ground mode.

(a) On the MAX Display System (MDS), go to the 32 PSEU display in the MAINT DATA PGS (TASK 31-65-00-970-801).

1) Make sure that the AIR/GND STATE value for the LEFT GEAR or RIGHT GEAR is NEAR.

SUBTASK 32-51-00-730-011

(5) Make sure that the Captain's tiller control wheel operates correctly:

(a) Turn the Captain's tiller control wheel to operate the nose wheel steering through a minimum of 6 full cycles. Make sure that it operates freely and smoothly.

NOTE: One cycle of the system is from center to full right, then to full left, then back to center.

(b) Make sure that there is no leakage of hydraulic fluid from the nose gear steering tubing and fittings from the landing gear selector valve to the valves on the steering actuators.

(c) Turn the nose wheels with the Captain's tiller control wheel from the CENTER position to the LEFT (CCW) stop position.

1) Make sure that the steering angle is 78 ±1 degrees on the decal that is on the nose gear shock strut.

(d) Release the Captain's tiller control wheel and let it return to the CENTER position.

1) Make sure that the steering angle is at 0 ±1.5 degrees.

(e) Turn the nose wheels with the Captain's tiller control wheel from the CENTER position to the RIGHT (CW) stop position.

1) Make sure that the steering angle is 78 ±1 degrees on the decal that is on the nose gear shock strut.

(f) Release the Captain's tiller control wheel and let it return to the CENTER position.

1) Make sure that the steering angle is at 0 ±1.5 degrees.

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I. Do the Test for Tiller Control Wheel Torque

SUBTASK 32-51-00-010-009

(1) Remove the cap [1] from the center of the tiller control wheel.

SUBTASK 32-51-00-480-014

(2) Install a dial torque wrench, STD-1035, on the center of the tiller control wheel.

SUBTASK 32-51-00-730-012

(3) Measure the torque that is necessary to turn the tiller control wheel:

(a) Use the dial torque wrench, STD-1035, to turn the tiller control wheel LEFT (CCW, 78 degrees). Stop just before you touch the tiller control wheel stop.

(b) Make sure that the torque is in the limit of 60 ±15 in-lb (7 ±2 N·m).

(c) Let the tiller control wheel move back to the CENTER (neutral) position.

(d) Use the dial torque wrench, STD-1035, to turn the tiller control wheel RIGHT (CW, 78 degrees). Stop just before you touch the tiller control wheel stop.

(e) Make sure that the torque is in the limit of 60 ±15 in-lb (7 ±2 N·m).

(f) Let the tiller control wheel move back to the CENTER (neutral) position.

SUBTASK 32-51-00-080-010

(4) Remove the dial torque wrench, STD-1035, from the tiller control wheel.

SUBTASK 32-51-00-080-011

(5) Install the cap [1] on the tiller control wheel.

J. Do the Test of Nose Gear Turn Radius and Response to Full Rudder Pedal Input

SUBTASK 32-51-00-710-005

WARNING

BEFORE YOU MOVE THE RUDDER PEDALS, MAKE SURE THAT THE AREA AROUND THE RUDDER IS CLEAR OF ALL PERSONS AND EQUIPMENT. THIS WILL PREVENT INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Push the Captain's left rudder pedal fully.

(a) Make sure that the steering angle is 7 +1/-2 degrees left turn, on the decal that is on the nose gear shock strut.

SUBTASK 32-51-00-730-016

(2) Release the rudder pedals.

(a) Make sure that the steering angle is at the center, 0 ±1.5 degrees, on the decal that is on the nose gear shock strut.

SUBTASK 32-51-00-730-017

WARNING

BEFORE YOU MOVE THE RUDDER PEDALS, MAKE SURE THAT THE AREA AROUND THE RUDDER IS CLEAR OF ALL PERSONS AND EQUIPMENT. THIS WILL PREVENT INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Push the Captain's right rudder pedal fully.

(a) Make sure that the steering angle is 7 +1/-2 degrees right turn, on the decal that is on the nose gear shock strut.

SUBTASK 32-51-00-730-018

(4) Release the rudder pedals.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the steering angle is at the center, 0 ±1.5 degrees, on the decal that is on the nose gear shock strut.

K. Do the Nose Gear Steering Tow Test

SUBTASK 32-51-00-860-043

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Move the towing lever to the "TOW" position and install the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-860-044

(2) Remove the apex pin from the torsion links.

SUBTASK 32-51-00-730-019

(3) Move the steering system manually with the upper torsion link, approximately 10 degrees to the left and right. Return it to the center position.

(a) If necessary, install the extension bar, SPL-7306, on the nose landing gear to rotate the steering system.

(b) If the towing valve does not remove pressure from the steering valve. Replace the steering valve, (PAGEBLOCK 32-51-11/401).

NOTE: If the towing valve does not remove pressure from the steering valve, you cannot move the steering system manually with the upper torsion link.

SUBTASK 32-51-00-080-012

WARNING MAKE SURE THAT THE NOSE WHEEL STEERING SYSTEM IS IN THE CENTER POSITION. WHEN HYDRAULIC POWER IS ON AND THE TOWING LEVER LOCKPIN IS REMOVED, THE STEERING COLLAR AND TOP TORSION LINK WILL TURN QUICKLY TO CENTER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) If installed, remove the extension bar, SPL-7306, from the nose landing gear.

SUBTASK 32-51-00-860-045

(5) Install the apex pin in the torsion links (TASK 32-21-31-400-803).

SUBTASK 32-51-00-080-013

(6) Remove the NLG towing lever pin, SPL-1499, from the towing lever. Let the lever move to the PRESSURE ON position.

SUBTASK 32-51-00-730-020

(7) Make sure that the steering system operates normally with the tiller control wheel.

L. Do the Test of the Alternate Steering System

SUBTASK 32-51-00-860-046

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-010-019

(2) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-00-860-047

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-51-00-860-048

(4) Make sure that the hydraulic system B reservoir is full (TASK 12-12-00-610-801).

SUBTASK 32-51-00-860-049

(5) For hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-860-050

(6) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

SUBTASK 32-51-00-730-021

(7) On the P2 panel, put the NOSE WHEEL STEER switch to the ALT position.

SUBTASK 32-51-00-730-022

(8) Make sure that the steering system moves correctly with the tiller control wheel.

SUBTASK 32-51-00-730-023

(9) On the P2 panel, put the NOSE WHEEL STEER switch to the NORM position.

SUBTASK 32-51-00-730-024

(10) Make sure that the steering system does not move with the tiller control wheel.

M. Do the Test of Air/Ground Mode and Normal Steering Response

SUBTASK 32-51-00-580-007

(1) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-51-00-730-025

(2) With the nose gear shock strut fully extended, make sure that the steering rotary actuator is in the air mode:

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(a) Make sure that the index marks on the rotary actuator pulley and the housing are aligned.

(b) Make sure that the pulley on the rotary actuator is turned fully in the counterclockwise direction.

SUBTASK 32-51-00-730-026

(3) With the airplane on jacks (TASK 07-10-03-580-803), use the nose gear axle jack to compress the nose gear shock strut 2.00 ±0.10 in. (50.80 ±2.54 mm).

NOTE: The dimensions given in this test are all measured from the bottom of the inner diameter of the outer cylinder.

NOTE: The nose gear axle jack is necessary only for testing while the airplane is jacked.

(a) Make sure that the rotary actuator stays in air mode.

SUBTASK 32-51-00-730-027

(4) Continue to compress the nose gear shock strut until the rotary actuator changes to the ground mode.

SUBTASK 32-51-00-730-028

(5) Extend the nose gear shock strut back to 2.00 ±0.10 in. (50.80 ±2.54 mm) of compression.

(a) Make sure that the rotary actuator moves back to the air mode in 8 seconds or less.

SUBTASK 32-51-00-730-029

(6) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-51-00-730-030

(7) Make sure that the rotary actuator changes to the ground mode.

N. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-860-051

(1) For hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-860-052

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-51-00-410-007

(3) Install the summing mechanism cover [10] with screws [9], washers [8], and washers [14].

SUBTASK 32-51-00-410-013

(4) Close this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

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SUBTASK 32-51-00-860-054

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Move the towing lever to the TOW position and install the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-860-055

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-51-00-410-021

(7) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-51-00-580-008

(8) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-51-00-080-014

(9) Remove the greased plates from under the nose wheels.

SUBTASK 32-51-00-580-009

(10) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

——— END OF TASK ———

TASK 32-51-00-700-802

9. Nose Wheel Steering System Test (Airplane on Jacks)

(Figure 501)

A. General

(1) This procedure gives steps to do a test of the nose wheel steering system while the airplane is on jacks.

(2) The test of the nose wheel steering system includes the tests that follow:

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(a) A steering centering test to make sure that the steering metering valve is in the null position when the nose gear is mechanically centered

(b) A normal steering response test

(c) A tiller control wheel torque test

(d) A test of the nose gear turn radius and response to full rudder pedal input

(e) A tow test of the nose gear steering to make sure that the nose gear will turn freely with the tow lever in the tow position

(f) A test of the alternate steering system

(g) An air/ground mode test.

(3) You must correctly adjust the nose wheel steering system before you do the system test.

B. References

Reference	Title
07-10-03-580-803	Lift the Nose Landing Gear Axle With Axle Jacks-Handling (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-00-970-801	System Maintenance Data Pages Function (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)
32-51-11 P/B 401	STEERING METERING VALVE - REMOVAL/INSTALLATION

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-7306	Extension Bar, Steering Test - Torque Link, NLG Part #: C32020-1 Supplier: 81205
STD-1035	Wrench - Dial Torque,0 to 500 in-lbs (0 to 56.5 N-m), +/-5% accuracy

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left

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(Continued)

Zone	Area
116	Nose Landing Gear Wheel Well - Right
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
117A	Electronic Equipment Access Door

F. Prepare to Test the Steering System of the Nose Wheel

SUBTASK 32-51-00-860-077

(1) Make sure that you park the airplane on a flat surface with no crosswinds.

NOTE: Crosswinds can caster the nose gear when the hydraulic system is depressurized.

SUBTASK 32-51-00-480-020

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-51-00-610-002

(3) Service the hydraulic system (TASK 12-12-00-610-801).

SUBTASK 32-51-00-580-011

(4) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-51-00-730-033

(5) Make sure that the torsion links are connected at the apex.

SUBTASK 32-51-00-860-059

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(6) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-860-060

(7) On the P2 panel, make sure that the LANDING GEAR control lever is in the DN position.

SUBTASK 32-51-00-860-061

(8) Make sure that the NOSE WHEEL STEER switch is in the NORM position.

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G. Do the Steering Centering Test

SUBTASK 32-51-00-860-062

(1) Remove the NLG towing lever pin, SPL-1499, from the towing lever and let the lever move to the PRESSURE ON position.

SUBTASK 32-51-00-010-014

(2) To get access to the rudder quadrant, do this step:

(a) Open this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-860-063

(3) Make sure that the rig pin R3, from the rigging pin kit, SPL-1585, fits freely in the rudder quadrant.

NOTE: This will show that the rudder pedals are in the neutral position.

SUBTASK 32-51-00-730-034

(4) Make sure that the steering angle is 0 ±1.5 degrees on the decal that is on the nose gear shock.

NOTE: The inner cylinder centering cam will put the nose wheels in the center position when the airplane is lifted on jacks. Manual assistance may be necessary to make sure that the nose gear and cam is in the center position.

(a) If the decal is not on the nose gear shock strut, then make a decal with tape (Figure 501).

SUBTASK 32-51-00-860-064

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-730-035

(6) Make sure that the nose gear stays centered.

NOTE: If the nose gear moves when the hydraulic system is pressurized, the nose gear steering is not adjusted correctly. There is a disagreement between the steering system and the mechanical centerline position of the nose gear centering cams.

SUBTASK 32-51-00-860-065

WARNING

MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(7) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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H. Do the Test of the Normal Steering Response

SUBTASK 32-51-00-480-021

CAUTION MAKE SURE THAT YOU PUT THE TORSION LINK TO KEEP CLEARANCE AT THE GEAR DOORS AND UPPER STEERING SLEEVE. IF YOU DO NOT OBEY, THE TORSION LINK CAN CAUSE DAMAGE TO THE AFT OF THE GEAR DOORS AND THE PRIMARY/SECONDARY AIR/GROUND SENSORS.

(1) Do these steps to disconnect the torsion link apex joint:

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Install the NLG towing lever pin, SPL-1499, through the towing lever.

NOTE: The NLG towing lever pin, SPL-1499, holds the towing lever in the TOW position.

(b) Disconnect the torsion link apex joint, do this task: Nose Landing Gear Torsion Link Disconnection, TASK 32-21-31-000-803

1) Put the airplane in ground mode.

2) Secure the torsion link assembly to avoid damage during the operation.

(c) Remove the NLG towing lever pin, SPL-1499, from the towing lever.

SUBTASK 32-51-00-860-066

WARNING MAKE SURE THAT PERSONS AND EQUIPMENT ARE CLEAR OF ALL CONTROL SURFACES BEFORE YOU SUPPLY HYDRAULIC POWER. THE AILERONS, RUDDER, ELEVATORS, FLAPS, SLATS, SPOILERS, LANDING GEAR, AND THRUST REVERSERS CAN MOVE QUICKLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-51-00-210-010

(3) Make sure that one or more main landing gear is in ground mode.

NOTE: The air/ground system needs two inputs to transition into the ground mode.

(a) On the MAX Display System (MDS), go to the 32 PSEU display in the MAINT DATA PGS (TASK 31-65-00-970-801).

1) Make sure that the AIR/GND STATE value for the LEFT GEAR or RIGHT GEAR is NEAR.

SUBTASK 32-51-00-730-036

(4) Make sure that the Captain's tiller control wheel operates correctly:

(a) Turn the Captain's tiller control wheel to operate the nose wheel steering through a minimum of 6 full cycles. Make sure that it operates freely and smoothly.

NOTE: One cycle of the system is from center to full right, then to full left, then back to center.

1) Make sure that the tiller control wheel operates freely and smoothly.

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(b) Make sure that there is no leakage of hydraulic fluid from the nose gear steering tubing and fittings from the landing gear selector valve to the valves on the steering actuators.

(c) Turn the nose wheels with the Captain's tiller control wheel from the CENTER position to the LEFT (CCW) stop position.

1) Make sure that the steering angle is 78 ±2 degrees on the decal that is on the nose gear shock strut.

(d) Release the Captain's tiller control wheel and let it return to the CENTER position.

1) Make sure that the steering angle is at 0 ±1.5 degrees.

(e) Turn the nose wheels with the Captain's tiller control wheel from the CENTER position to the RIGHT (CW) stop position.

1) Make sure that the steering angle is 78 ±2 degrees on the decal that is on the nose gear shock strut.

(f) Release the Captain's tiller control wheel and let it return to the CENTER position.

1) Make sure that the steering angle is at 0 ±1.5 degrees.

I. Do the Test for Tiller Control Wheel Torque

SUBTASK 32-51-00-010-015

(1) Remove the cap [1] from the center of the tiller control wheel.

SUBTASK 32-51-00-480-022

(2) Install a dial torque wrench, STD-1035, on the center of the tiller control wheel.

SUBTASK 32-51-00-730-038

(3) Measure the torque that is necessary to turn the tiller control wheel:

(a) Use the dial torque wrench, STD-1035, to turn the tiller control wheel LEFT (CCW, 78 degrees) and stop just before you touch the tiller control wheel stop.

(b) Make sure that the torque is in the limit of 60 ±15 in-lb (7 ±2 N·m).

(c) Let the tiller control wheel move back to the CENTER (neutral) position.

(d) Use the dial torque wrench, STD-1035, to turn the tiller control wheel RIGHT (CW, 78 degrees). Stop just before you touch the tiller control wheel stop.

(e) Make sure that the torque is in the limit of 60 ±15 in-lb (7 ±2 N·m).

(f) Let the tiller control wheel move back to the CENTER (neutral) position.

SUBTASK 32-51-00-080-015

(4) Remove the dial torque wrench, STD-1035, from the tiller control wheel.

SUBTASK 32-51-00-410-011

(5) Install the cap [1] on the tiller control wheel.

J. Do the Test of Nose Gear Turn Radius and Response to Full Rudder Pedal Input

SUBTASK 32-51-00-480-023

(1) Make sure that the airplane nose gear is in ground mode.

NOTE: The air/ground system needs two inputs to transition into the ground mode. Both of the nose gear sensors are in ground mode by installing actuator (target) on ground proximity sensor or holding the upper torsion link in the ground position.

(a) Make sure that a minimum of one of the main gears is in ground mode.

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WARNING BEFORE YOU MOVE THE RUDDER PEDALS, MAKE SURE THAT THE AREA AROUND THE RUDDER IS CLEAR OF ALL PERSONS AND EQUIPMENT. THIS WILL PREVENT INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Push the Captain's left rudder pedal fully.

(a) Make sure that the steering angle is 7 +1/-2 degrees left turn, on the decal that is on the nose gear shock strut.

SUBTASK 32-51-00-730-039

(3) Release the rudder pedals.

(a) Make sure that the steering angle is at the center, 0 ±1.5 degrees, on the decal that is on the nose gear shock strut.

SUBTASK 32-51-00-730-040

WARNING BEFORE YOU MOVE THE RUDDER PEDALS, MAKE SURE THAT THE AREA AROUND THE RUDDER IS CLEAR OF ALL PERSONS AND EQUIPMENT. THIS WILL PREVENT INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) Push the Captain's right rudder pedal fully.

(a) Make sure that the steering angle is 7 +1/-2 degrees right turn, on the decal that is on the nose gear shock strut.

SUBTASK 32-51-00-730-041

(5) Release the rudder pedals.

(a) Make sure that the steering angle is at the center, 0 ±1.5 degrees, on the decal that is on the nose gear shock strut.

K. Do the Nose Gear Steering Tow Test

SUBTASK 32-51-00-860-067

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Move the towing lever to the "TOW" position and install the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-730-054

(2) Move the steering system manually with the upper torsion link, approximately 10 degrees to the left and right. Then return it to the center position.

(a) If necessary, install the extension bar, SPL-7306, on the nose landing gear to rotate the steering system.

(b) If the towing valve does not remove pressure from the steering valve, replace the steering valve, (PAGEBLOCK 32-51-11/401)

NOTE: If the towing valve does not remove pressure from the steering valve, you cannot move the steering system manually with the upper torsion link.

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SUBTASK 32-51-00-730-043

WARNING

MAKE SURE THAT THE NOSE WHEEL STEERING SYSTEM IS IN THE CENTER POSITION. WHEN HYDRAULIC POWER IS ON AND THE TOWING LEVER LOCKPIN IS REMOVED, THE STEERING COLLAR AND TOP TORSION LINK WILL TURN QUICKLY TO CENTER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) If installed, remove the extension bar, SPL-7306, from the nose landing gear.

SUBTASK 32-51-00-080-016

(4) Remove the NLG towing lever pin, SPL-1499, from the towing lever. Let the lever move to the PRESSURE ON position.

SUBTASK 32-51-00-730-044

(5) Make sure that the steering system operates normally with the tiller control wheel.

L. Do the Test of the Alternate Steering System

SUBTASK 32-51-00-860-068

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-010-018

(2) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-51-00-860-069

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(3) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-51-00-610-003

(4) Make sure that the hydraulic system B reservoir is full (TASK 12-12-00-610-801).

SUBTASK 32-51-00-860-070

(5) For hydraulic system B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

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SUBTASK 32-51-00-860-071

(6) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

SUBTASK 32-51-00-730-045

(7) On the P1 captain's panel, put the NOSE WHEEL STEER switch to the ALT position.

SUBTASK 32-51-00-730-046

(8) Make sure that the steering system moves correctly with the tiller control wheel.

SUBTASK 32-51-00-730-047

(9) On the P1 captain's panel, put the NOSE WHEEL STEER switch to the NORM position.

SUBTASK 32-51-00-730-048

(10) Make sure that the steering system does not move with the tiller control wheel.

M. Do the Test of Air/Ground Mode and Normal Steering Response

SUBTASK 32-51-00-480-024

(1) Do these steps to connect the torsion link apex joint:

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Install the NLG towing lever pin, SPL-1499, through the towing lever.

NOTE: The NLG towing lever pin, SPL-1499, holds the towing lever in the TOW position.

(b) Connect the torsion link apex joint, do this task: Nose Landing Gear Torsion Link Connection, TASK 32-21-31-400-803

(c) Remove the NLG towing lever pin, SPL-1499, from the towing lever.

SUBTASK 32-51-00-730-049

(2) With the nose gear shock strut fully extended, make sure that the steering rotary actuator is in the air mode:

(a) Make sure that the index marks on the rotary actuator pulley and the housing are aligned.

(b) Make sure that the pulley on the rotary actuator is turned fully in the counterclockwise direction.

SUBTASK 32-51-00-730-050

(3) With the airplane on jacks (TASK 07-10-03-580-803), use the nose gear axle jack to compress the nose gear shock strut 2.00 ±0.10 in. (50.80 ±2.54 mm). Make sure that the rotary actuator stays in the air mode.

NOTE: The dimensions given in this test are all measured from the bottom of the inner diameter of the outer cylinder.

NOTE: The nose gear axle jack is necessary only for testing while the airplane is jacked.

SUBTASK 32-51-00-730-051

(4) Continue to compress the nose gear shock strut until the rotary actuator changes to the ground mode.

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SUBTASK 32-51-00-730-052

(5) Extend the nose gear shock strut back to 2.00 ±0.10 in. (50.80 ±2.54 mm) of compression. Make sure that the rotary actuator moves back to the air mode in 8 seconds or less.

SUBTASK 32-51-00-580-012

(6) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-51-00-730-053

(7) Make sure that the rotary actuator changes to the ground mode.

N. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-860-072

(1) For hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-00-860-073

(2) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-51-00-410-012

(3) Install the summing mechanism cover [10] with screws [9], washers [8], and washers [14].

SUBTASK 32-51-00-410-014

(4) Close this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-860-074

WARNING USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Move the towing lever to the TOW position and install the NLG towing lever pin, SPL-1499.

SUBTASK 32-51-00-860-075

WARNING WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-51-00-410-020

(7) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-51-00-080-017

(8) If used, remove the actuator (target) from the proximity sensor.

SUBTASK 32-51-00-580-013

(9) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE WHEEL STEERING SYSTEM - INSPECTION/CHECK

1. General

A. This task has the data for a detailed visual inspection of the nose wheel steering system mechanism.

TASK 32-51-00-200-801

2. Nose Wheel Steering System Mechanism Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

D. Prepare for the Inspection

SUBTASK 32-51-00-480-017

WARNING MAKE SURE THAT THE LANDING GEAR DOWNLOCK PINS ARE INSTALLED. THIS WILL PREVENT ACCIDENTAL RETRACTION OF THE LANDING GEAR. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT OR INJURY TO PERSONNEL CAN OCCUR.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-00-860-057

WARNING REMOVE HYDRAULIC POWER FOR THE NOSE WHEEL STEERING BEFORE YOU DO THE CHECK. HYDRAULIC POWER CAN CAUSE THE NOSE WHEELS TO MOVE. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Nose Wheel Steering System Mechanism Inspection

SUBTASK 32-51-00-010-010

(1) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-010-011

(2) Remove the Map/Chart Light Control Module.

NOTE: It may be necessary to use an inspection mirror to do the inspection through the Map/Chart Light Control Module opening.

SUBTASK 32-51-00-010-012

(3) Remove the screws [1], washers [2], washers [3], and summing mechanism cover [4].

SUBTASK 32-51-00-210-009

(4) Do a detailed visual inspection of the mechanical control path for the nose wheel steering system mechanism.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-00-410-008

(1) Install the summing mechanism cover [4] with the washers [2], washers [3], and screws [1].

SUBTASK 32-51-00-410-009

(2) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

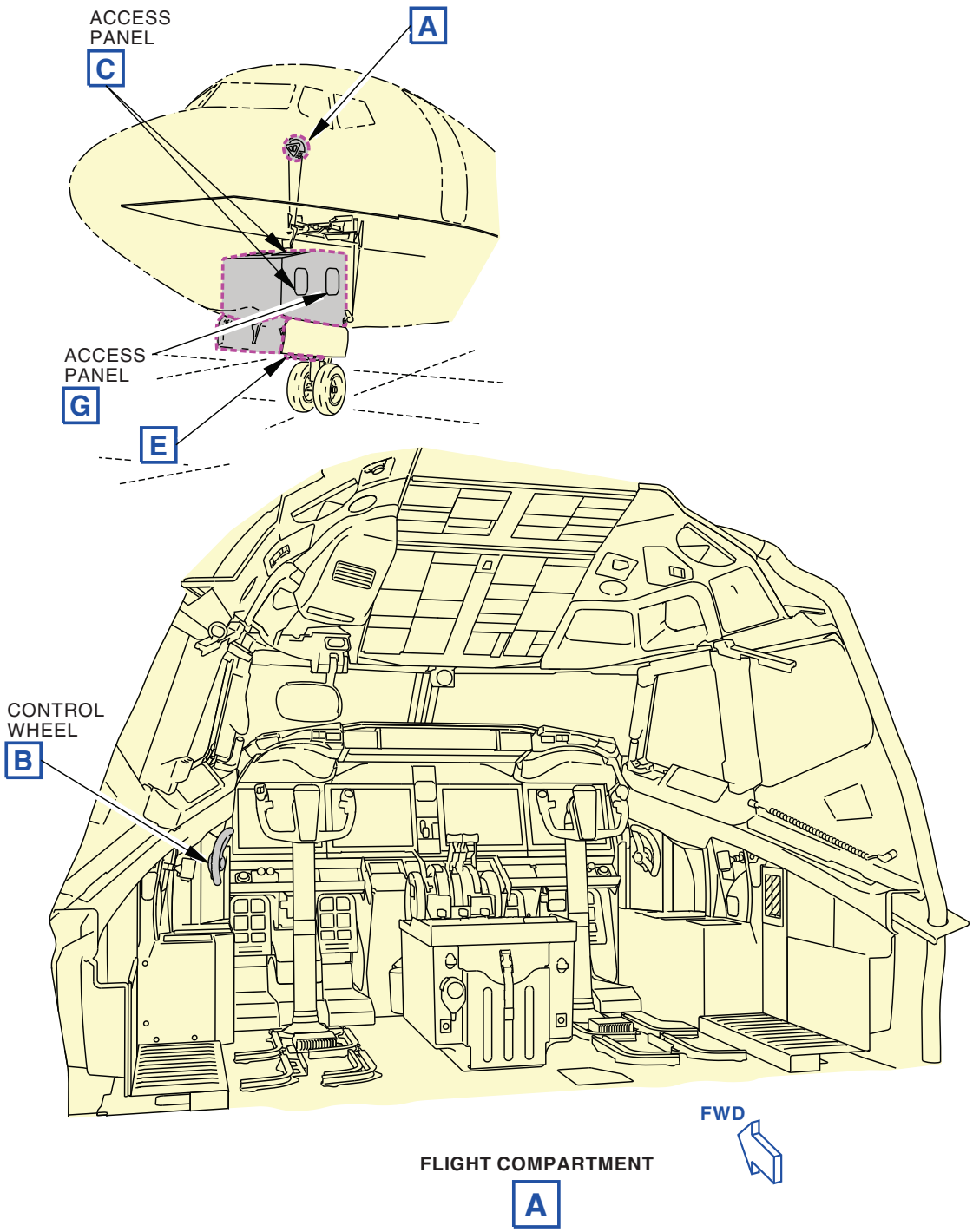
Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-00-410-010

(3) Install the Map/Chart Light Control Module.

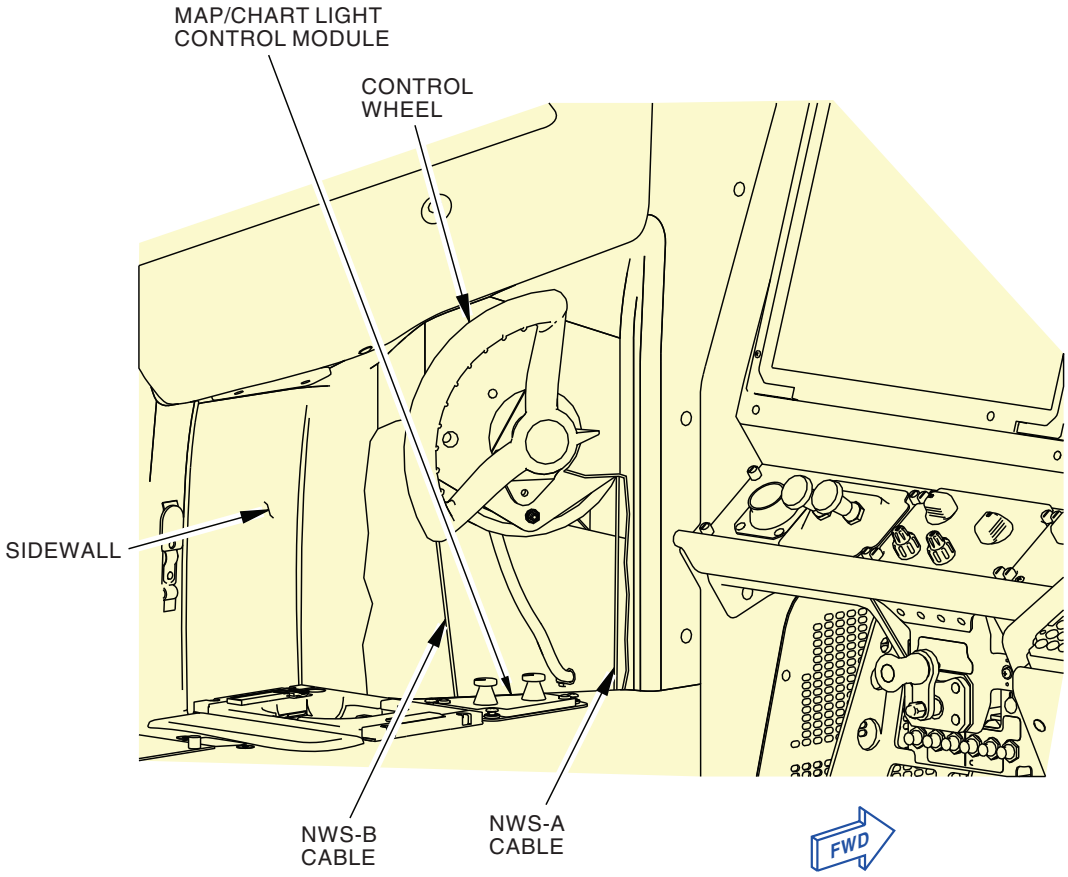
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Wheel Steering System Mechanism Inspection
Figure 601/32-51-00-990-803 (Sheet 1 of 6)

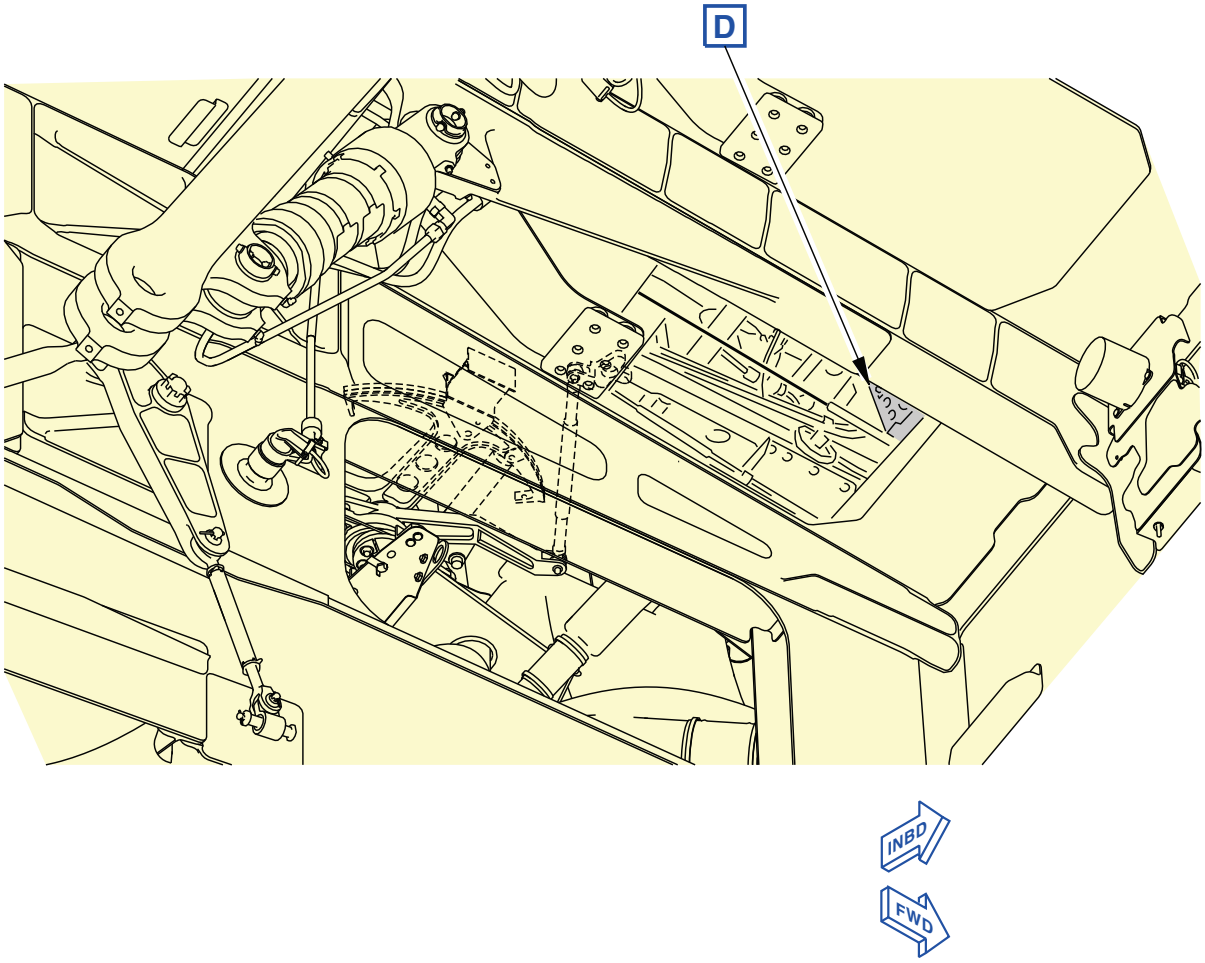
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Wheel Steering System Mechanism Inspection
Figure 601/32-51-00-990-803 (Sheet 2 of 6)

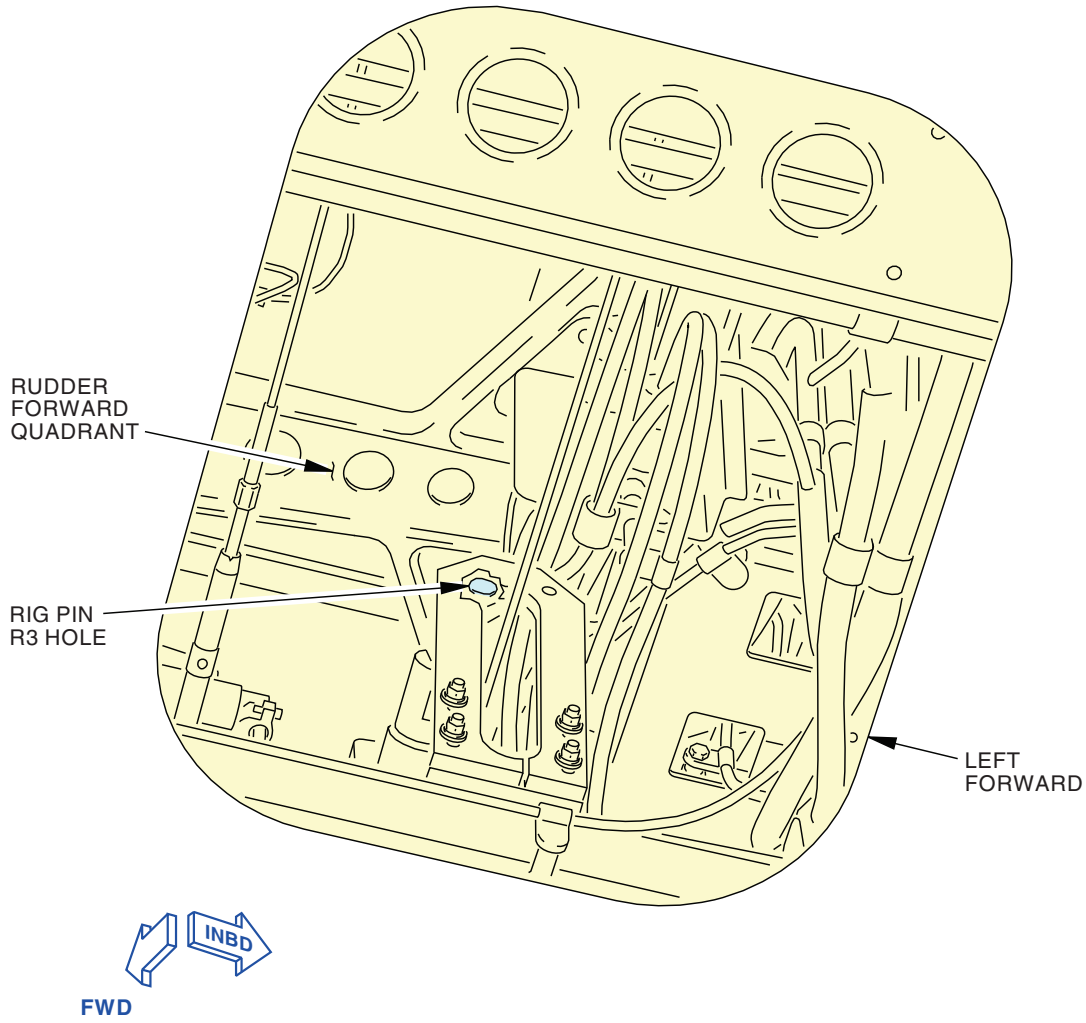
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Wheel Steering System Mechanism Inspection
Figure 601/32-51-00-990-803 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



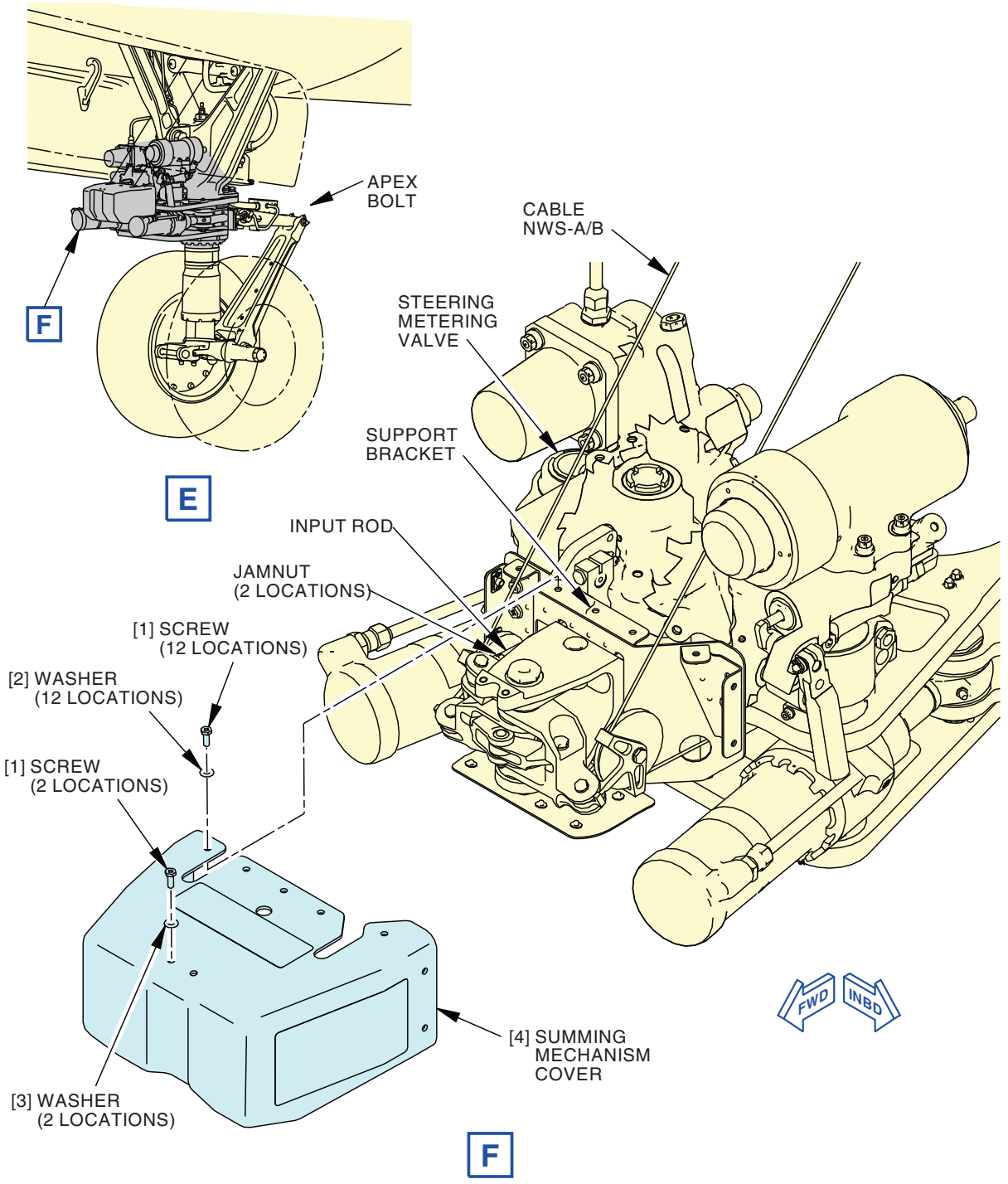
CAPTAIN'S RUDDER FORWARD QUADRANT

D

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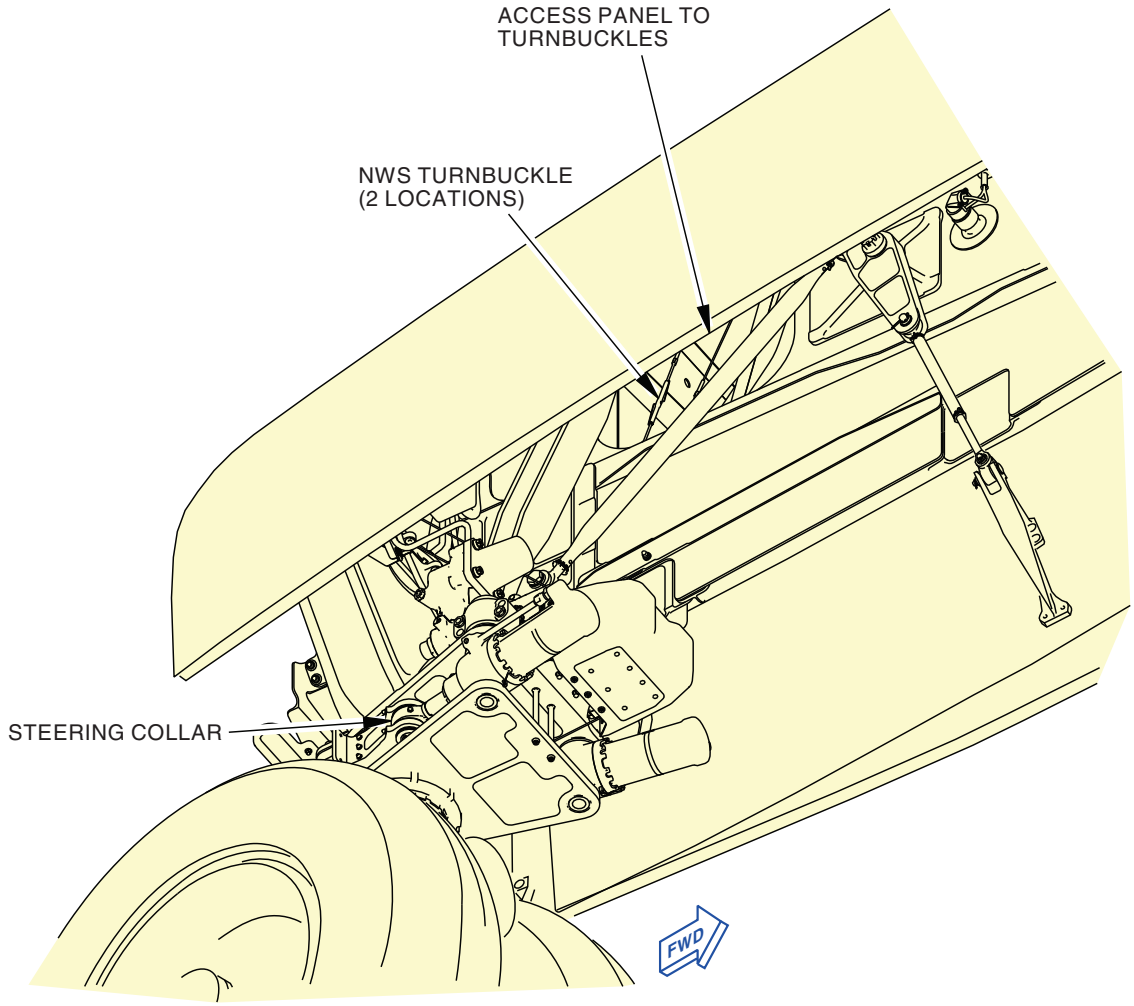
Nose Wheel Steering System Mechanism Inspection
Figure 601/32-51-00-990-803 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Wheel Steering System Mechanism Inspection
Figure 601/32-51-00-990-803 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Wheel Steering System Mechanism Inspection
Figure 601/32-51-00-990-803 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE WHEEL STEERING - MAINTENANCE PRACTICES

1. General

A. This procedure gives instructions to clean the nose wheel steering towing lever [1], crank [2], and plunger [3].

TASK 32-51-11-100-801

2. Clean the Exterior Surfaces of the Towing Lever Assembly

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)

D. Prepare for Cleaning

SUBTASK 32-51-11-480-014

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

E. Clean the Exterior Surfaces of the Towing Lever Assembly

SUBTASK 32-51-11-100-001

CAUTION DO NOT TRY TO DISASSEMBLE THE TOWING LEVER PLUNGER TO CLEAN IT. DAMAGE TO INTERNAL COMPONENTS WILL OCCUR.

(1) Use a clean cotton wiper, G00034, moistened in alcohol, B00130, to clean the towing lever [1], crank [2], and plunger [3] to remove dirt, grease, and grime.

NOTE: alcohol, B00130, is the most effective cleaner that will not damage the valve seals and finishes.

(a) If it is necessary, disassemble the crank [2] and towing lever [1]:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove the cotter pin [7], nut [6], washers [5], and bolt [4] from the crank [2].
2) Remove the towing lever [1].
3) Remove the crank [2] and washer [8] from the valve housing.
4) Clean the bushings in the valve housing and the area between them.
5) Install the crank [2] into the valve housing.
6) Install the washer [8] and towing lever [1] onto the crank [2] splines.
   a) Orient the splines so that the bolt [4] can be installed though the hole in the towing lever [1].
   b) Use a rig pin, from the rigging pin kit, SPL-1585, to line up the holes in the manifold and crank [2].
7) Install the bolt [4] with washer [5] under head through the towing lever [1] and crank [2].

   NOTE: Installation direction of the bolt is optional.
8) Install the washer [5] and nut [6].
   a) Install a rig pin, from the rigging pin kit, SPL-1585, to hold the towing lever [1] in position.
9) Tighten the nut [6] to 42.5 ±2.5 in-lb (4.8 ±0.3 N·m), as necessary to line up the slot in the nut [6] with the hole in the bolt [4].
10) Install the cotter pin [7].

SUBTASK 32-51-11-212-001

(2) Do a check of the operation of the towing lever [1] and plunger [3].

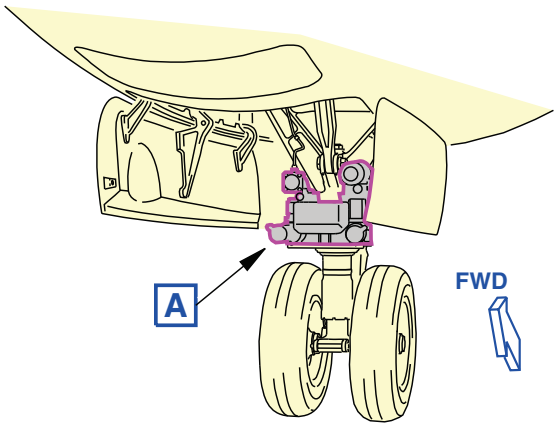
NOTE: Lubrication of the towing lever or plunger is not recommended as this will attract dirt and grime.

(a) Make sure that the towing lever [1] and plunger [3] return to the normal position when released.

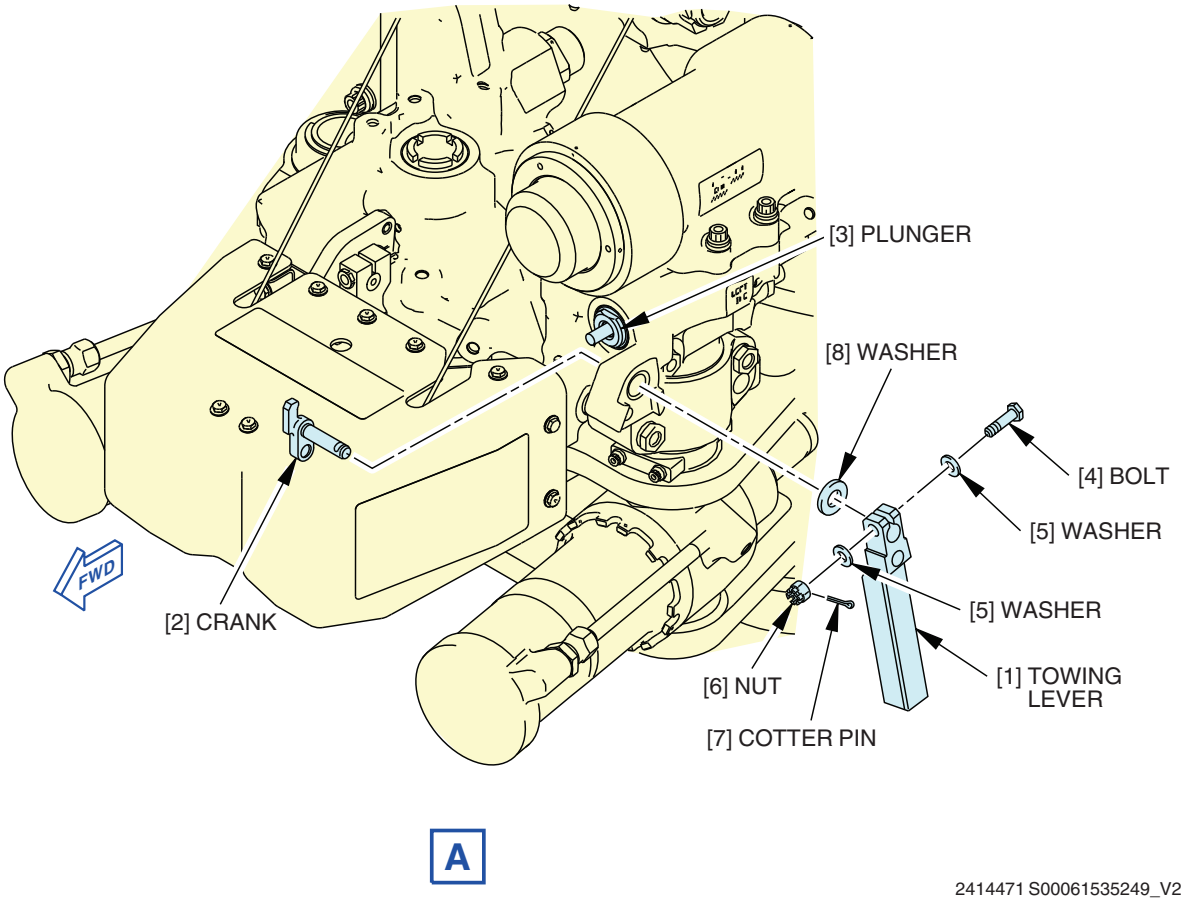
(b) Make sure that there is no external leakage or evidence of instability, sticking or binding of the towing lever [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOSE LANDING GEAR



Nose Wheel Steering System - Towing Lever Assembly
Figure 201/32-51-11-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEERING METERING VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the steering metering valve

(2) An installation of the steering metering valve.

TASK 32-51-11-000-801

2. Steering Metering Valve Removal

(Figure 401)

A. References

Reference	Title
20-10-20-800-801	Cap and Plug Hydraulic Fittings (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-51-11-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-11-860-001

(2) Make sure that the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-11-480-003

(3) Install this rig pin from the rigging pin kit, SPL-1585:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Rig pin 4 in the summing mechanism.

SUBTASK 32-51-11-860-002

(4) Remove the pressure from hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-51-11-860-011

(5) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-51-11-860-003

(6) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

SUBTASK 32-51-11-860-004

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Install the NLG towing lever pin, SPL-1499, through the towing lever.

NOTE: You use the NLG Towing Lever Lock Pin to hold the towing lever in the TOW position.

E. Steering Metering Valve Removal

SUBTASK 32-51-11-020-020

(1) Disconnect the lower nose gear door actuating rod from the left nose gear door center hinge.

NOTE: This will allow the door to swing outboard and provide clearance to remove the steering valve.

SUBTASK 32-51-11-010-002

(2) Remove the screws [4], washers [5], and washers [37] that hold the summing mechanism cover [10] to the summing mechanism bracket.

SUBTASK 32-51-11-010-003

(3) Remove the screws [9] and washers [5] that hold the support bracket to the steering metering valve [7].

SUBTASK 32-51-11-020-002

(4) Remove the nut [23], washer [5], bushing [24], and bolt [25] to disconnect the link assembly [6] from the input lever on the steering metering valve [7].

SUBTASK 32-51-11-020-003

(5) Disconnect and remove the hydraulic lines [11] from the steering metering valve [7] and swivel at the right gear trunnion.

SUBTASK 32-51-11-480-004

(6) Install caps/plugs, STD-14686, on the hydraulic lines [11], swivel, and ports on the steering metering valve [7] (TASK 20-10-20-800-801).

SUBTASK 32-51-11-020-004

(7) Remove the lockwire from the bolts [20].

SUBTASK 32-51-11-020-005

(8) Remove the bolts [20] and washers [18] that hold the steering metering valve [7] to the top and bottom support plates.

SUBTASK 32-51-11-020-006

(9) Remove the nuts [14], washers [15], washers [18], and bolts [19] that hold the steering metering valve [7] to the top support plate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-11-020-007

CAUTION

THE STEERING METERING VALVE IS TOP HEAVY. THE VALVE WILL FLIP OUT OF YOUR HOLD. BE CAREFUL NOT TO DROP AND DAMAGE THE STEERING METERING VALVE.

(10) Remove the nuts [16], bolts [22], washers [21], and clamps [17] that attach the swivel valves to the steering actuator trunnions.

(a) Make sure that the area around and in the trunnions is clean before the hydraulic ports are open to prevent contamination.

NOTE: The hydraulic ports will be open when you remove the steering metering valve from the actuator trunnions.

(b) Lift off the steering metering valve [7].

1) Pass the valve under the steering cables to the left side of the aircraft.

(c) Be careful when you remove the steering metering valve [7] to prevent contamination of the open hydraulic ports.

(d) Install lockwire in the top of the swivel valves to keep them in the steering metering valve [7].

CAUTION

DO NOT REMOVE THE SWIVEL VALVES FROM THE STEERING METERING VALVE. IF YOU DO NOT OBEY, DAMAGE TO THE MATING SURFACES OF THE SWIVEL VALVES AND THE STEERING METERING VALVE CAN OCCUR.

1) Do not remove the swivel valves from the steering metering valve [7].

SUBTASK 32-51-11-480-005

(11) Install caps/plugs, STD-14686, in the open hydraulic ports for the swivel valves (that are part of the steering metering valve [7]) and actuator trunnions (TASK 20-10-20-800-801).

SUBTASK 32-51-11-020-008

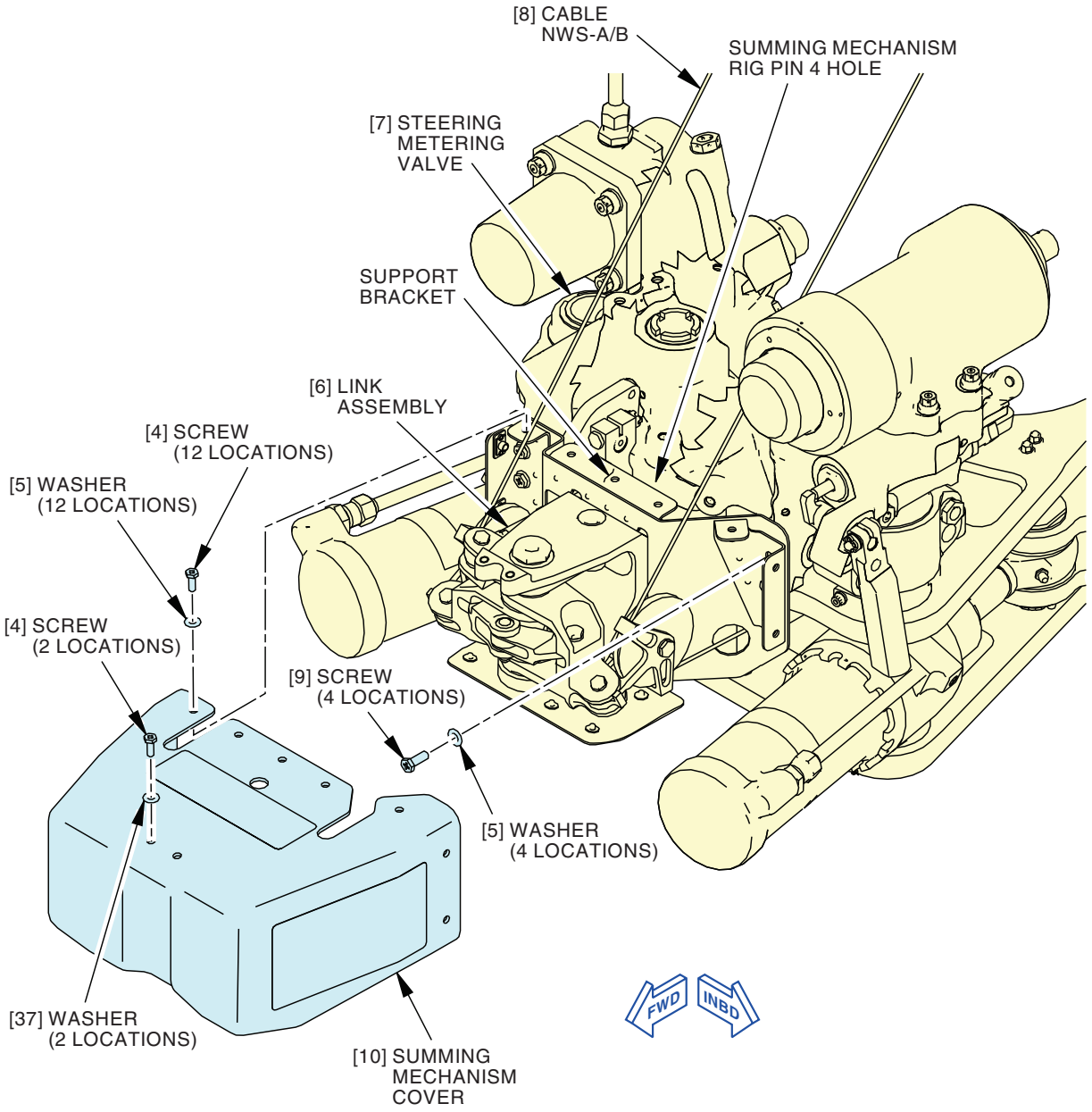
(12) Remove the union [12] and packings [13] from the steering metering valve [7].

SUBTASK 32-51-11-480-006

(13) Install caps/plugs, STD-14686, in the ports (TASK 20-10-20-800-801).

——— END OF TASK ———

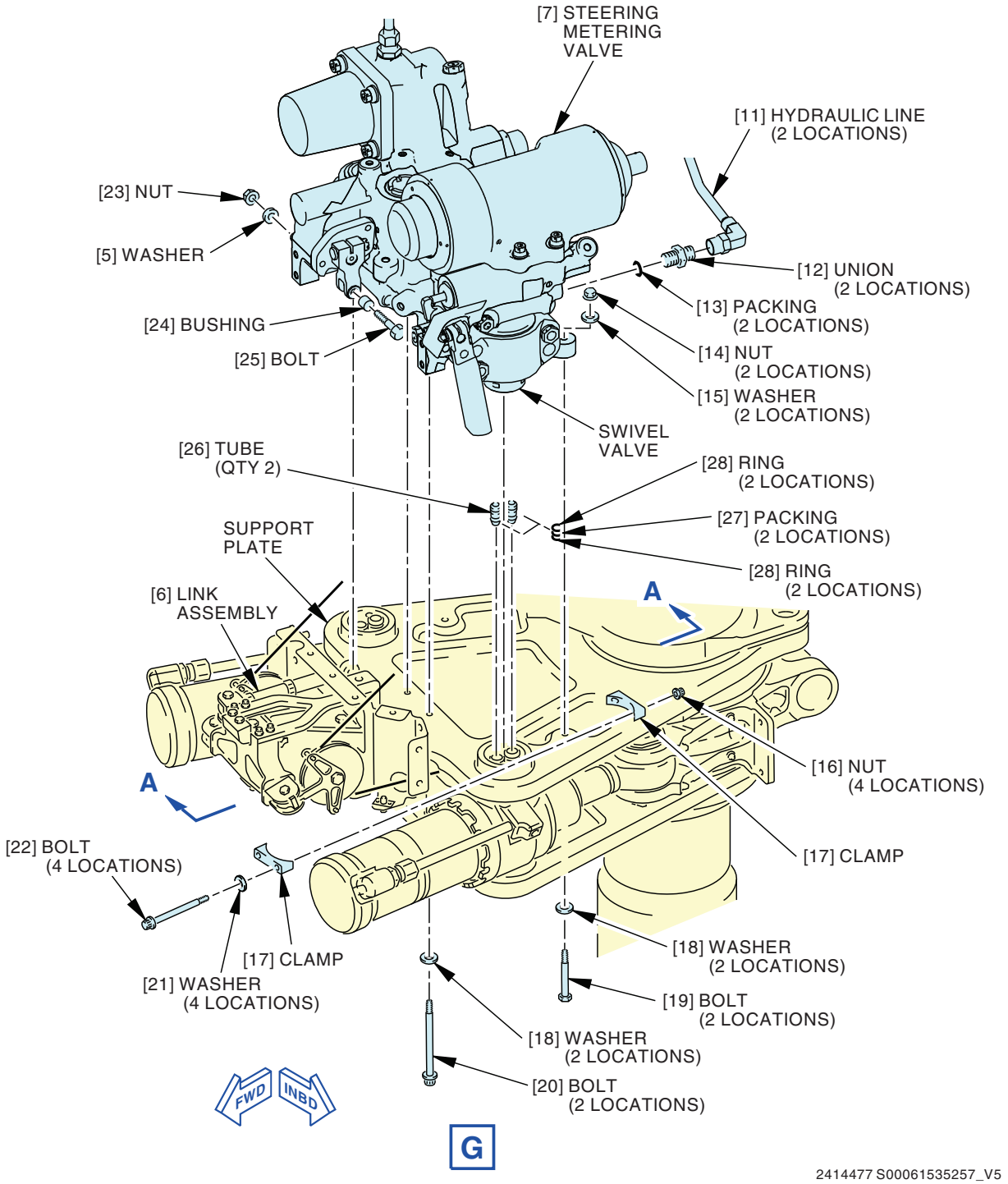
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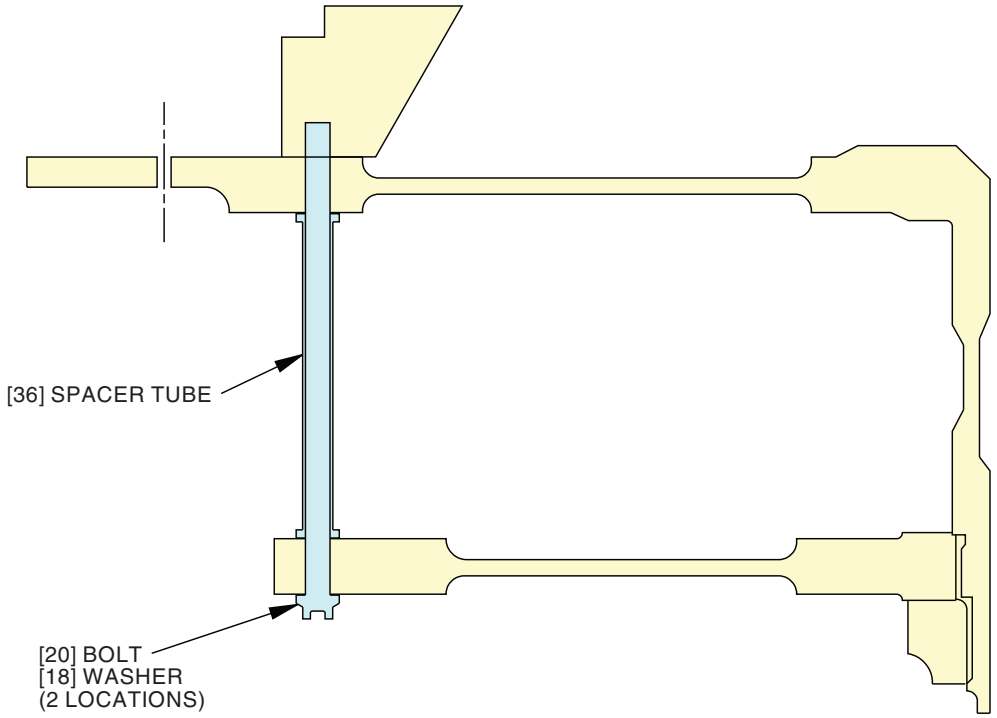
Steering Metering Valve Installation
Figure 401/32-51-11-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Steering Metering Valve Installation
Figure 401/32-51-11-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Steering Metering Valve Installation
Figure 401/32-51-11-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-11-400-801

3. Steering Metering Valve Installation

(Figure 401)

A. References

Reference	Title
20-10-20-800-801	Cap and Plug Hydraulic Fittings (P/B 201)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G02166	Lockwire - MS20995NC20, Monel - 0.020 Inch (0.508 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Steering metering valve	32-51-11-01-070	PTF ALL
13	Packing	32-51-11-01-065	PTF ALL
27	Packing	32-51-51-01-175	PTF ALL
28	Ring	32-51-51-01-180	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Steering Metering Valve Installation

SUBTASK 32-51-11-600-001

CAUTION

THE STEERING METERING VALVE IS TOP HEAVY. THE VALVE WILL FLIP OUT OF YOUR HOLD. BE CAREFUL NOT TO DROP AND DAMAGE THE STEERING METERING VALVE.

(1) Do these steps to prepare the steering metering valve [7] for installation:

(a) Lubricate the new packings [13] and unions [12] with hydraulic fluid, D00153.

(b) Install the new packings [13] and unions [12] in the steering metering valve [7].

(c) Apply hydraulic fluid, D00153, to the transfer tubes in the actuator trunnions.

(d) Install the new rings [28] and new packing [27] on both of the tubes [26] between the steering metering valve [7] and steering cylinders.

SUBTASK 32-51-11-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply Cor-Ban 27L Compound, G50237 (preferred), or corrosion inhibiting material, G50136 (alternate), to the faying surfaces of all four mounting feet on the steering metering valve [7].

SUBTASK 32-51-11-620-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Apply a thin layer of the Cor-Ban 27L Compound, G50237 (preferred), or corrosion inhibiting material, G50136 (alternate),, to these items:

(a) The shank of the bolts [19]

(b) The shank, threads, and thread reliefs of the bolts [20]

(c) The threads of the nuts [14]

(d) The faces of the washers [15] and washers [18].

(e) The threads of the bolts [22].

SUBTASK 32-51-11-420-019

(4) Install the metering valve.

(a) The valve can be installed by passing it below the steering cables from the left side of the aircraft.

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SUBTASK 32-51-11-020-009

CAUTION

MAKE SURE THAT THE OFFSET HOLE PATTERN IN THE SWIVEL VALVES FOR THE VALVE ASSEMBLY ARE CORRECTLY ALIGNED WITH THE MATING HOLE PATTERN IN THE ACTUATOR TRUNNION. IF YOU DO NOT OBEY, IT IS POSSIBLE THAT THE STEERING SYSTEM WILL NOT OPERATE CORRECTLY AND YOU CAN CAUSE DAMAGE TO THE STEERING SYSTEM.

(5) Put the swivel valves for the steering metering valve [7] into the steering actuator trunnions.

(a) Make sure that the offset hole pattern in the swivel valves is aligned with the mating hole pattern in the left and right trunnions.

SUBTASK 32-51-11-420-001

(6) Install the clamps [17], washers [21], bolts [22], and nuts [16] to attach the steering metering valve [7] to the actuator trunnions.

(a) Remove the lockwire from the swivel valves, if it is installed.

(b) Tighten the bolts [22] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 32-51-11-420-002

(7) Install the bolts [19], washers [15], washers [18], and nuts [14] to attach the steering metering valve [7] to the top support plate.

SUBTASK 32-51-11-420-003

(8) Install the bolts [20], spacer tubes [36], and washers [18] to attach the steering metering valve [7] to the top and bottom support plates.

(a) Make sure that the spacer tubes [36] have been installed.

SUBTASK 32-51-11-420-020

(9) Install MS20995NC20 lockwire, G02166, between the bolts [20] (TASK 20-10-44-400-801).

SUBTASK 32-51-11-420-004

(10) Install the screws [9] and washers [5] that hold the support bracket to the steering metering valve [7].

SUBTASK 32-51-11-480-007

(11) Make sure that the rig pin from the rigging pin kit, SPL-1585, is installed.

(a) Rig pin 4 in the summing mechanism.

SUBTASK 32-51-11-420-005

(12) Install the bushing [24], bolt [25], washer [5], and nut [23] to attach the link assembly [6] to the input lever on the steering metering valve [7].

(a) Adjust the link assembly [6] until you can easily install the bolt [25].

(b) Remove the rig pin 4 from the summing mechanism.

SUBTASK 32-51-11-080-001

(13) Remove the caps/plugs, STD-14686, from the hydraulic lines [11] and ports on the steering metering valve [7] (TASK 20-10-20-800-801).

SUBTASK 32-51-11-420-006

(14) Connect the hydraulic lines [11] to the steering metering valve [7].

(a) Tighten the B-nuts on hydraulic lines [11] to 265 ±9 in-lb (30 ±1 N·m).

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G. Steering Metering Valve Installation Test

SUBTASK 32-51-11-420-008

(1) Do these procedures in this task to do a test of the nose wheel steering system: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801

- General
- Prepare to Test the Steering System of the Nose Wheel
- Do the Test of the Normal Steering Response
- Do the Test for Control Wheel Torque
- Do the Test of Nose Gear Turn Radius and Response to Full Rudder Pedal Input
- Do the Nose Gear Steering Tow Test
- Put the Airplane Back to Its Usual Condition.

(a) Make sure that there is no leakage.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-11-420-009

(1) Put the summing mechanism cover [10] in its position.

(a) Install the screws [4], washers [5], and washers [37].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER PEDAL STEERING MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the interconnect spring for the rudder pedal steering mechanism.

(2) Installation of the interconnect spring for the rudder pedal steering mechanism.

(3) Removal of the rudder pedal steering mechanism and the nose gear position cables .

(4) Installation of the rudder pedal steering mechanism and the nose gear position cables.

(5) Visual inspection of the rudder pedal steering mechanism.

TASK 32-51-21-000-801

2. Interconnect Spring Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

C. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

D. Prepare for the Removal

SUBTASK 32-51-21-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-21-860-001

(2) Make sure the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-21-860-002

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Interconnect Spring Removal

SUBTASK 32-51-21-010-001

(1) Open the access panels on the left sidewall and the top of the nose wheel well to get access to the rudder pedal steering mechanism [1] and cable turnbuckles (Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-21-010-002

(2) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-21-020-001

(3) Remove the spring [23].

(a) Remove the two bolts [24], washers [25], spacers [26], washers [27] and nuts [28] to disconnect the spring [23] (Figure 401).

——— END OF TASK ———

TASK 32-51-21-400-801

3. Interconnect Spring Installation

(Figure 401)

A. References

Reference	Title
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
23	Spring	32-51-21-01-175	PTF ALL

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

D. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

E. Interconnect Spring Installation

SUBTASK 32-51-21-420-001

(1) Install the spring [23].

(a) Install the two bolts [24], washers [25], spacers [26], washers [27] and nuts [28] to reconnect the spring [23] (Figure 401).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-21-410-001

(1) Close the access panels in the wheel well that you opened to get access to the rudder pedal steering mechanism [1] and turnbuckles (Figure 401).

SUBTASK 32-51-21-410-002

(2) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

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(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

——— END OF TASK ———

TASK 32-51-21-000-802

4. Rudder Pedal Steering Mechanism Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

D. Prepare for the Removal

SUBTASK 32-51-21-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-21-860-003

(2) Make sure the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-21-860-004

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Rudder Pedal Steering Mechanism Removal

SUBTASK 32-51-21-010-003

(1) Open the access panels to get access to the rudder pedal steering mechanism [1] and cable turnbuckles (TASK 53-14-01-020-801).

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(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-21-020-002

(2) Remove the clips from the turnbuckles.

SUBTASK 32-51-21-020-003

(3) Loosen the NWS-A cable, the cable NGPP-A [19], and the cable NGPP-B [20] at the turnbuckles.

SUBTASK 32-51-21-020-004

(4) Remove the nut [10], washer [11], and bolt [12] to disconnect the idler rod at the structure.

SUBTASK 32-51-21-020-005

(5) Remove the nuts [2], washers [3], and screws [5] on the ends of the quadrant for the rudder pedal steering mechanism [1].

SUBTASK 32-51-21-480-003

(6) Remove the end plates [4] from the cable ends on the quadrant to disconnect the NWS-A cable.

SUBTASK 32-51-21-020-006

(7) Disconnect the cable NGPP-A [19] and the cable NGPP-B [20] at the turnbuckles.

SUBTASK 32-51-21-020-007

(8) Remove the lockwire, shaft [15], washers [14], and bolts [13] to disconnect the rudder pedal steering mechanism [1] from the rudder pedal jack shaft assembly.

SUBTASK 32-51-21-020-008

(9) Remove the nut [9] and washer [8] from the bolt [6] on the upper support bracket.

SUBTASK 32-51-21-020-009

(10) Hold the bolt [6], straighten the tabs on the retainer [7], and remove the bolt [6].

SUBTASK 32-51-21-000-001

(11) Remove the rudder pedal steering mechanism [1] complete with the cable NGPP-A [19] and the cable NGPP-B [20].

SUBTASK 32-51-21-020-010

(12) Remove and discard the retainer [7].

SUBTASK 32-51-21-020-011

(13) Remove the cable NGPP-A [19] and the cable NGPP-B [20] from the eccentric [18] on the rudder pedal steering mechanism [1]:

(a) Remove the nut [21] and washer [22] from the shaft [15].

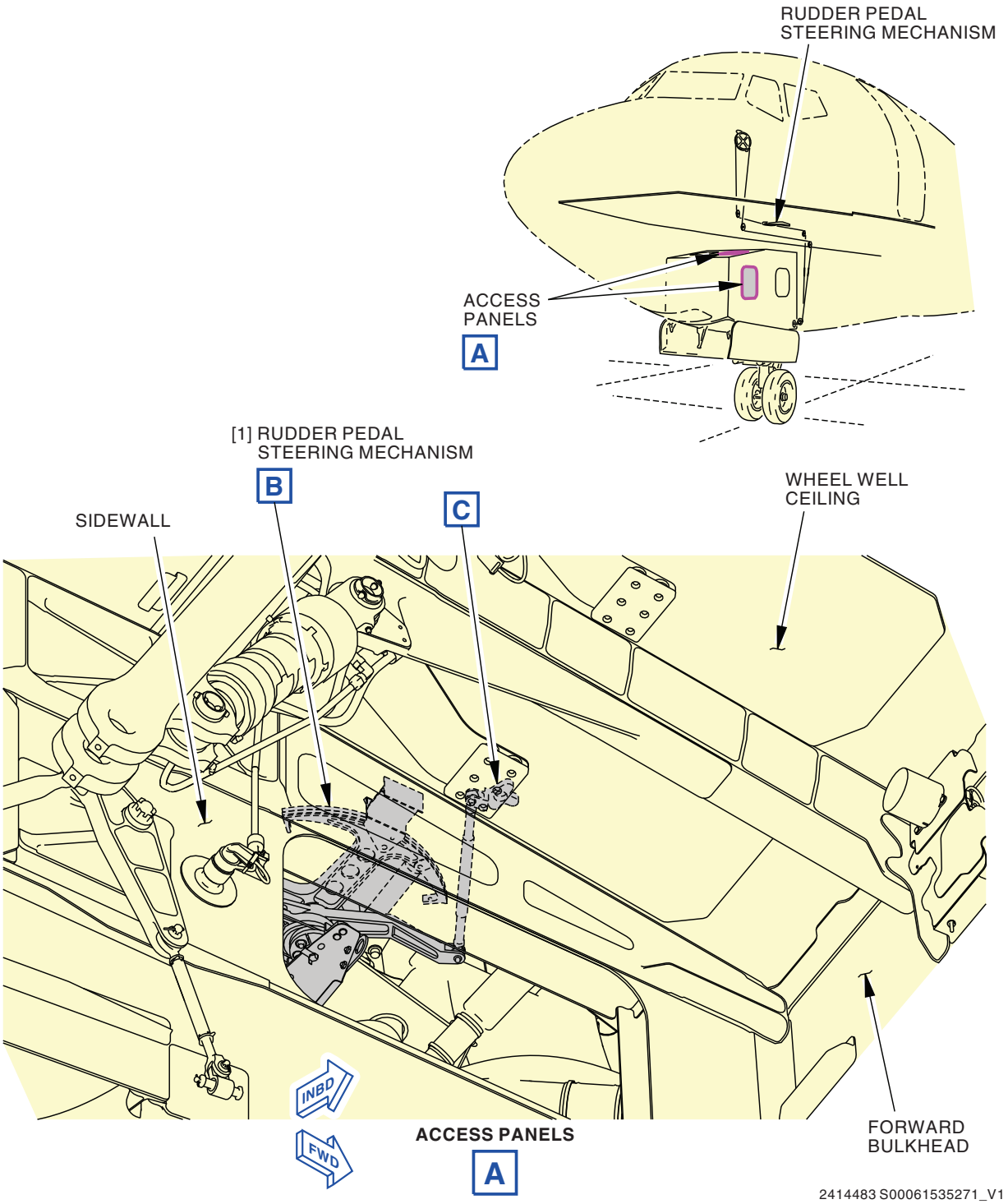
(b) Move the shaft [15] out far enough to get access to the bolt [16] and nut [17].

(c) Remove the nut [17] and bolt [16].

(d) Move the eccentric [18] out and remove the cable NGPP-A [19] and the cable NGPP-B [20].

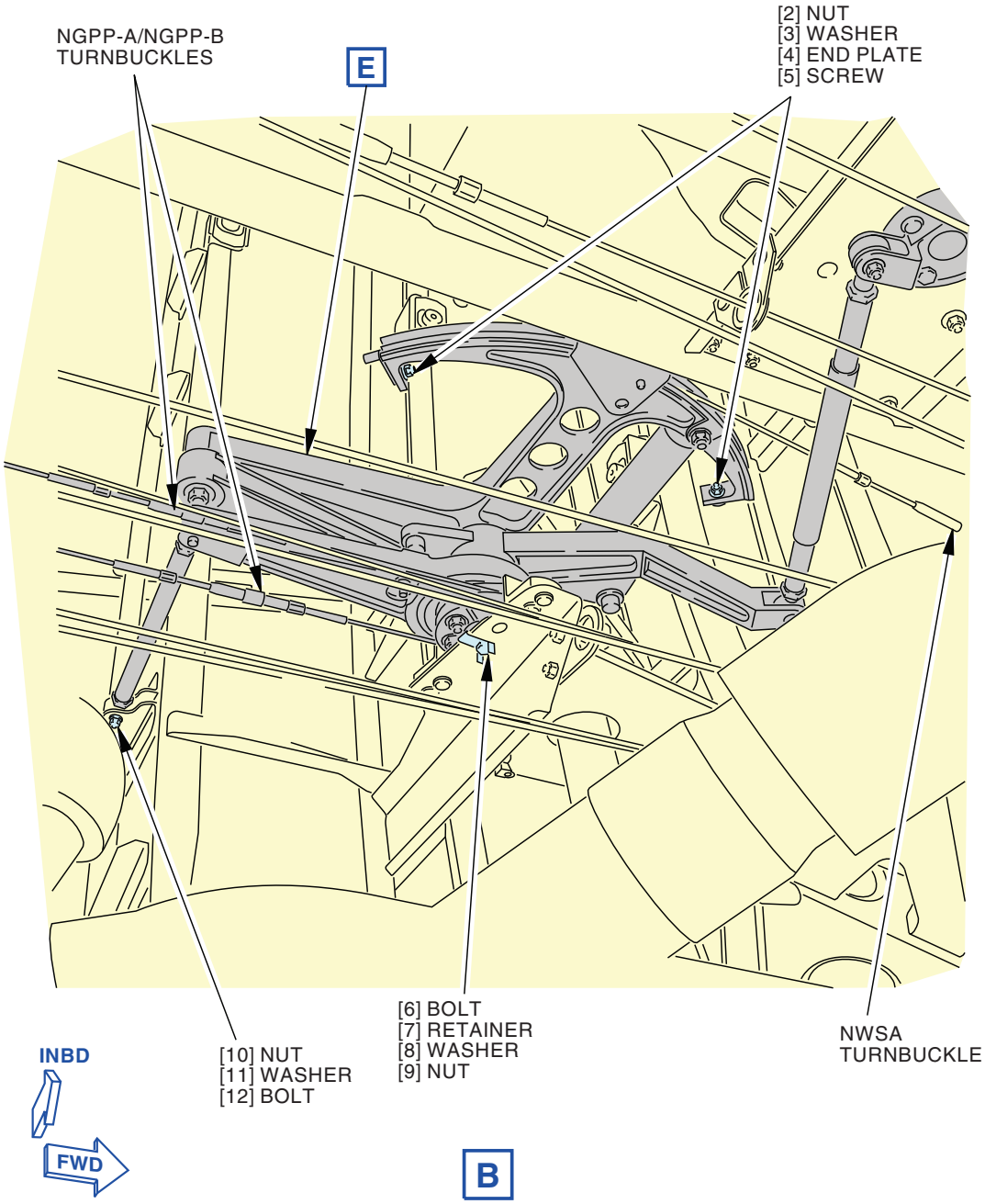
END OF TASK

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Rudder Pedal Steering Mechanism
Figure 401/32-51-21-990-801 (Sheet 1 of 4)

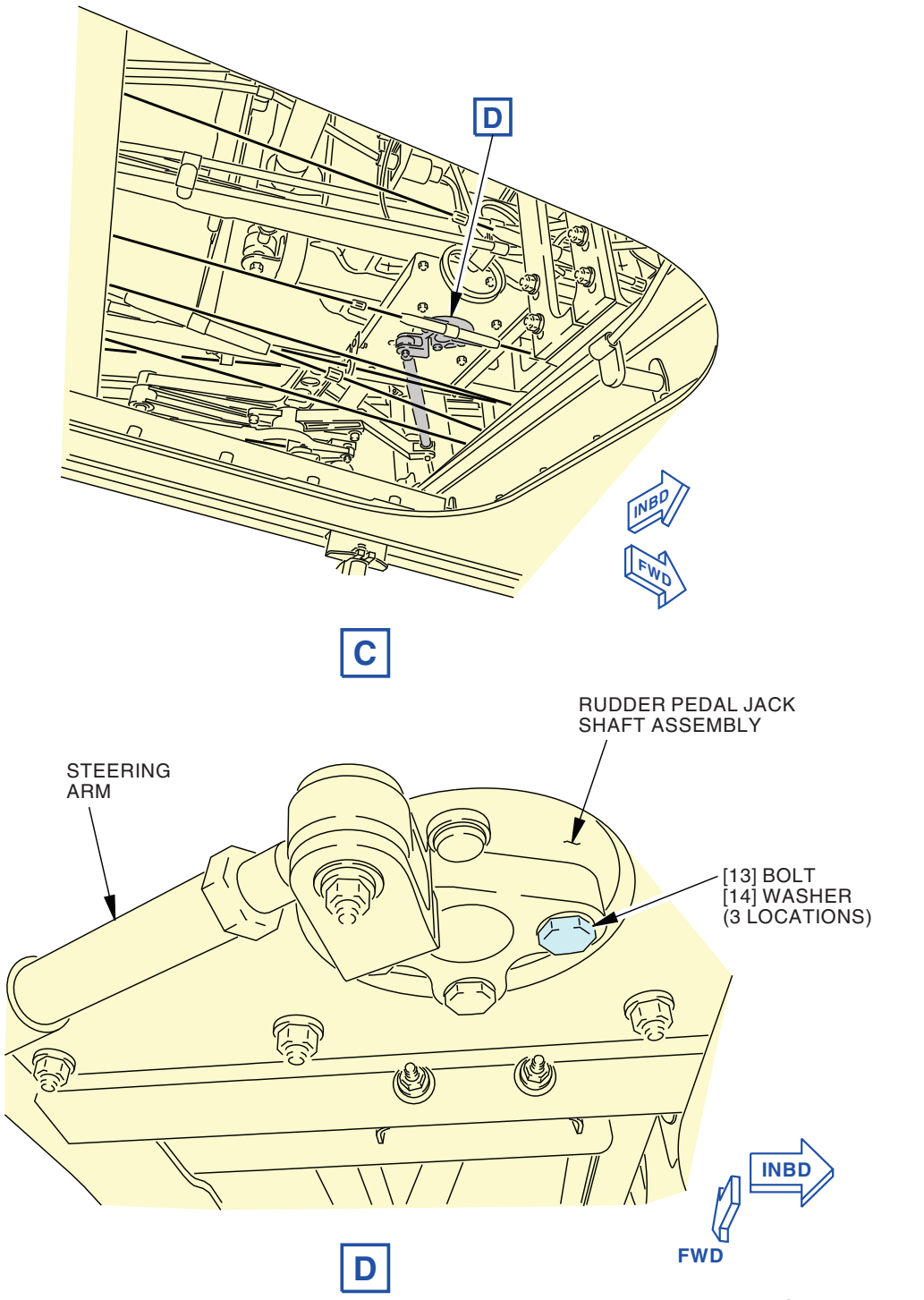
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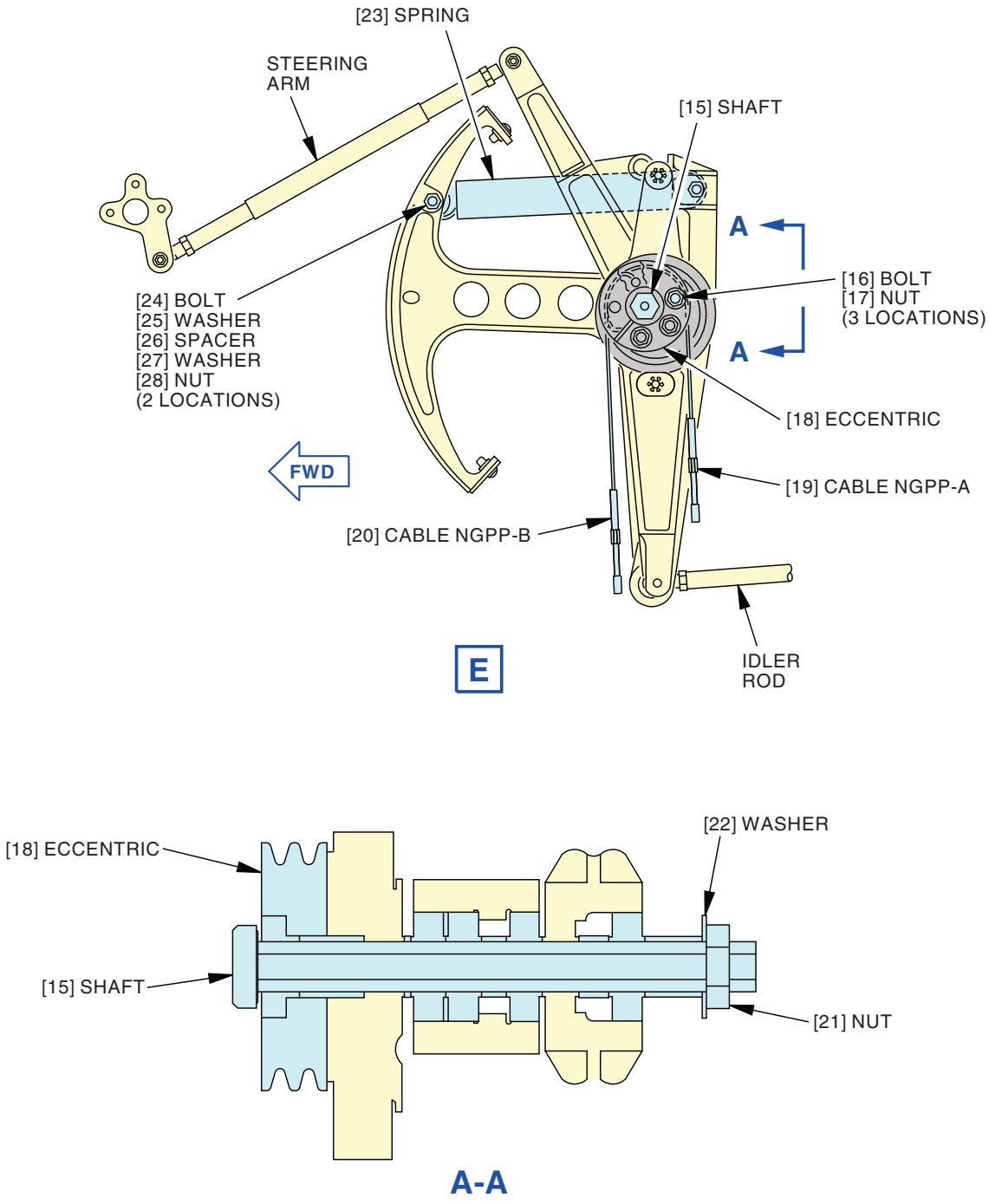
Rudder Pedal Steering Mechanism
Figure 401/32-51-21-990-801 (Sheet 2 of 4)

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Rudder Pedal Steering Mechanism
Figure 401/32-51-21-990-801 (Sheet 3 of 4)

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Rudder Pedal Steering Mechanism
Figure 401/32-51-21-990-801 (Sheet 4 of 4)

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TASK 32-51-21-400-802

5. Rudder Pedal Steering Mechanism Installation

(Figure 401)

A. References

Reference	Title
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-00-820-803	Rudder Pedal Interconnect Mechanism Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Rudder pedal steering mechanism	32-51-21-01-170	PTF ALL
7	Retainer	32-51-21-01-030	PTF ALL

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

E. Rudder Pedal Steering Mechanism Installation

SUBTASK 32-51-21-420-002

(1) Install the cable NGPP-A [19] and cable NGPP-B [20] on the eccentric [18] on the rudder pedal steering mechanism [1]:

(a) Move the eccentric [18] out and install the cable NGPP-A [19] and cable NGPP-B [20].

(b) Move the eccentric [18] into its position and install the nuts [17] and bolts [16].

(c) Install the shaft [15].

(d) Install the washer [22] and nut [21] on the shaft [15].

1) Tighten the nut [21] to 100 in-lb (11.3 N·m) above the run-on torque.

SUBTASK 32-51-21-420-003

(2) Install a new retainer [7].

SUBTASK 32-51-21-420-004

(3) Put the rudder pedal steering mechanism [1], complete with the cable NGPP-A [19] and cable NGPP-B [20], in position.

(a) Make sure that there is clearance between the clutch crank and the end of the bolt that holds the rudder pedal steering quadrant to the clutch arm.

SUBTASK 32-51-21-420-005

(4) Install the bolt [6], washer [8], and nut [9] through the rudder pedal steering mechanism [1] and the upper support bracket.

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(a) Tighten the nut [9] to approximately 65 in-lb (7.3 N·m) above the run-on torque.

SUBTASK 32-51-21-420-006

(5) Bend the tabs on the retainer [7] to hold the bolt [6].

SUBTASK 32-51-21-420-007

(6) Put the cable NGPP-A [19] and cable NGPP-B [20] through the fairleads and connect them at the turnbuckles.

SUBTASK 32-51-21-420-008

(7) Put the NWS-A cable in the end plates [4] and install the screws [5], washers [3], and nuts [2] to attach the cable ends to the quadrant.

SUBTASK 32-51-21-420-009

(8) Install the bolt [12], washer [11], and nut [10] to attach the idler rod to the structure.

SUBTASK 32-51-21-420-010

(9) Install the bolts [13] and washers [14] to attach the steering arm to the rudder pedal jack shaft assembly.

SUBTASK 32-51-21-420-011

(10) Install lockwire on the bolt [13].

SUBTASK 32-51-21-820-001

(11) Tighten the NWS-A cable, cable NGPP-A [19], and cable NGPP-B [20] and adjust the tension (TASK 32-51-00-820-803).

SUBTASK 32-51-21-420-012

(12) Install the clips on the turnbuckles.

F. Rudder Pedal Steering Mechanism Installation Test

SUBTASK 32-51-21-700-001

(1) Do a test of the steering system of the nose wheel (TASK 32-51-00-700-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-21-410-003

(1) Close the access panels to the rudder pedal steering mechanism [1] and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

——— END OF TASK ———

TASK 32-51-21-000-803

6. Rudder Pedal Steering Mechanism Visual Inspection

(Figure 402, Figure 403)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

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B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Access Panels

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

D. Steering Mechanism Inspection

SUBTASK 32-51-21-010-004

(1) Open the access panels on the left sidewall and the top of the nose wheel well to get access to the rudder pedal steering mechanism and cable turnbuckles (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-21-210-001

(2) Do a detail visual inspection of the rotary actuator and mechanical control path for the rudder pedal steering mechanism.

SUBTASK 32-51-21-211-001

(3) Do a visual inspection to make sure that nut, that supplies retention of the rudder pedal steering mechanism to the upper support bracket, is seated correctly.

NOTE: A small telescoping mirror will be necessary to view this area via the forward EE bay

SUBTASK 32-51-21-410-004

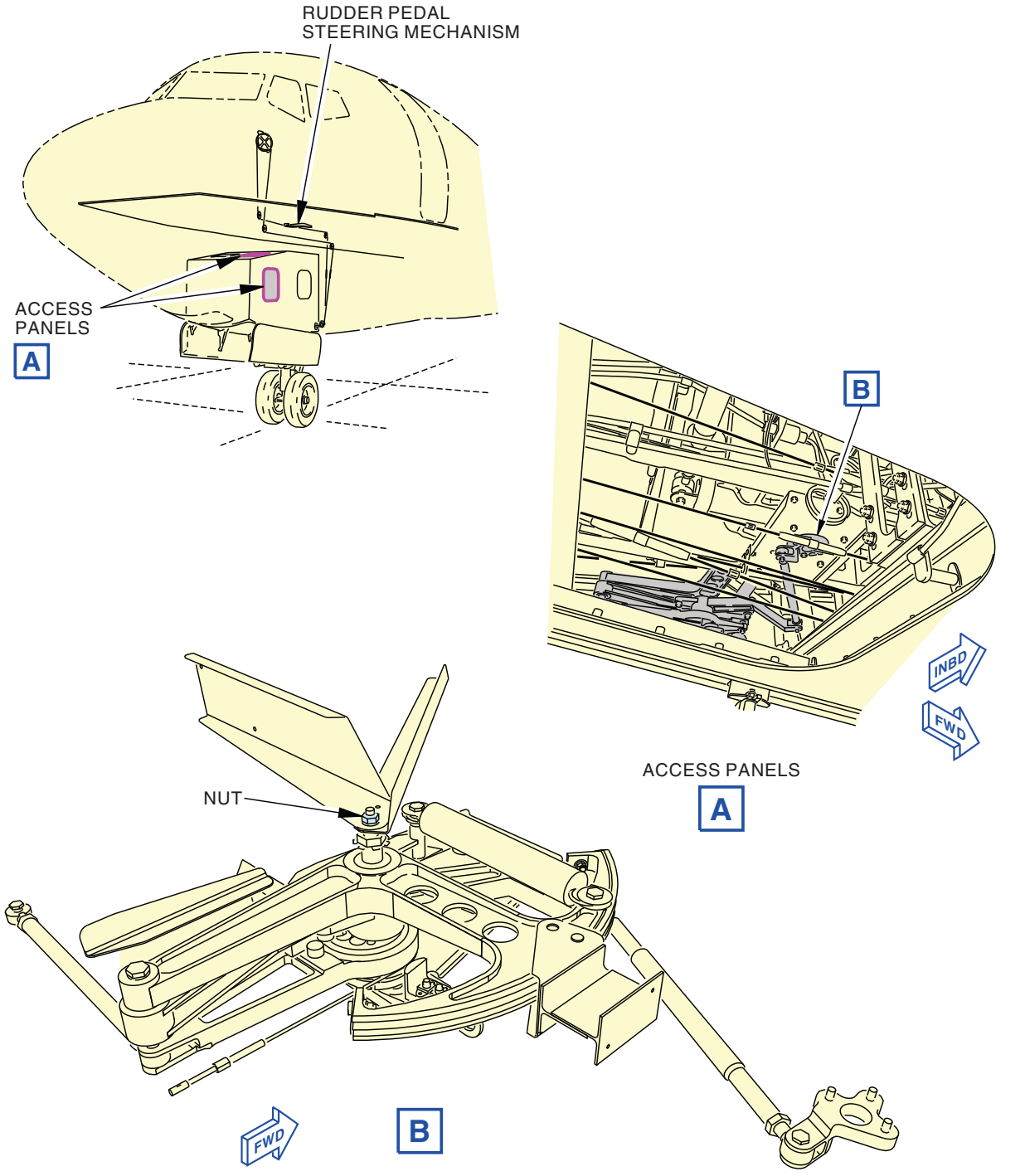
(4) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AC	Fwd Nose Wheel Well Upper Access Panel
113AW	Forward Nose Wheel Well Panel

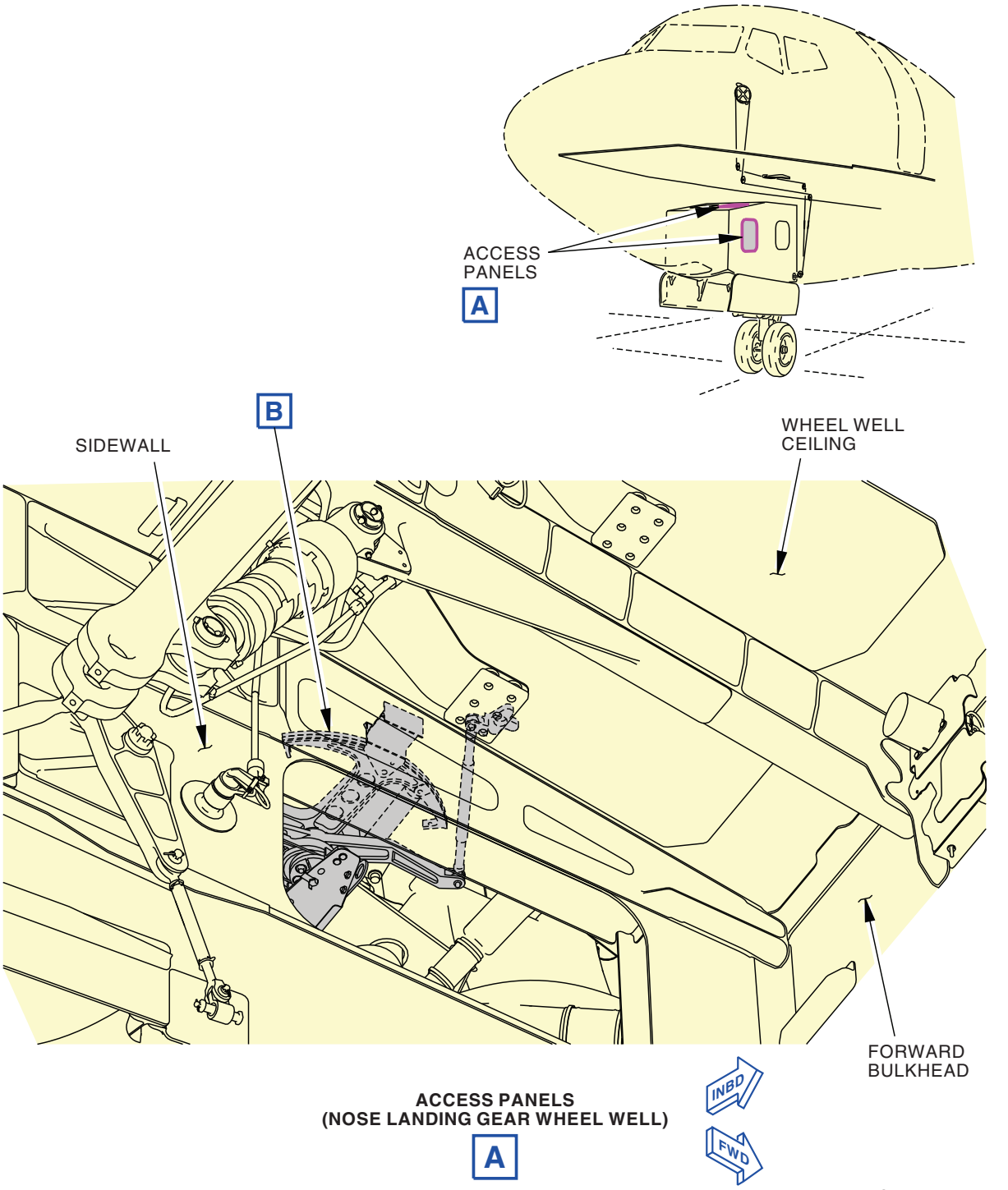
END OF TASK

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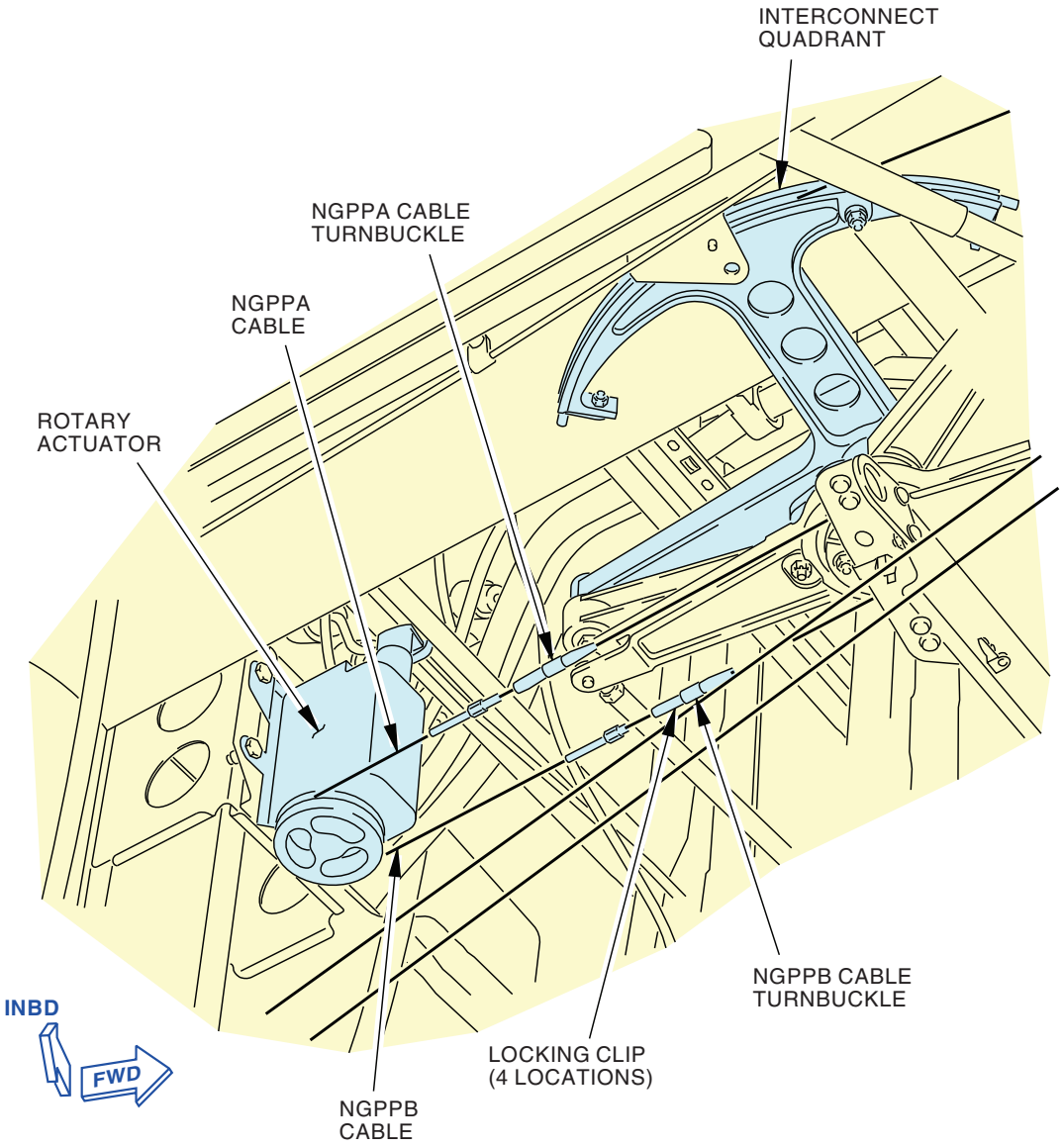
Rudder Pedal Steering Retainer Nut Inspection
Figure 402/32-51-21-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Rudder Pedal Steering Mechanism Control Path Inspection
Figure 403/32-51-21-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Rudder Pedal Steering Mechanism Control Path Inspection
Figure 403/32-51-21-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

STEERING SYSTEM CABLES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the position cables for the nose gear steering, referred to as NGPP-A cable and NGPP-B cable

(2) Installation of the position cables for the nose gear steering

(3) Removal of the steering system cables for the nose gear, referred to as NWS-A(A) cable, NWS-A(B) cable, NWS-A(C) cable, NWS-B cable, and NWS-A/B cable

(4) Installation of the steering system cables for the nose gear.

B. The position cables have four segments. This procedure can replace two of the segments attached to the rotary actuator pulley. You must use the procedure: RUDDER PEDAL STEERING MECHANISM - REMOVAL/INSTALLATION, PAGEBLOCK 32-51-21/401, to replace the two segments attached to the drum on the rudder pedal steering mechanism.

C. The steering system cables have five segments. This procedure can replace one or all of these segments. The steering cables, referred to as NWS-A(A) cable and NWS-B cable are attached to the drum that is part of the control wheel assembly. You must remove the control wheel and the drum to replace the steering cables.

TASK 32-51-31-000-801

2. Nose Wheel Steering Position Cables Removal

(Figure 401)

A. General

(1) This task gives instructions to remove the nose wheel steering position cables.

B. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
20-10-91-000-801	Control Cables Removal (P/B 401)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for Removal

SUBTASK 32-51-31-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: TASK 32-00-01-480-801.

SUBTASK 32-51-31-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-31-580-001

(3) Lift the nose of the airplane with jacks until the nose gear tires are clear of the ground (TASK 07-10-21-580-801).

SUBTASK 32-51-31-010-001

(4) Remove the forward access panel from the left sidewall of the nose wheel well to get access to the turnbuckles [2] of the NGPP-A cable [4] and NGPP-B cable [3] (TASK 53-14-01-020-801).

(a) Open this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-31-860-004

(5) Make sure that the airplane is in AIR mode (TASK 32-09-00-860-801).

NOTE: The rig pin NS5 cannot be installed unless the rotary actuator is in AIR mode.

SUBTASK 32-51-31-480-002

(6) Install the rig pin NS5 in the drum on the interconnect quadrant.

F. Nose Wheel Steering Position Cables Removal

SUBTASK 32-51-31-010-002

(1) Remove the NGPP-B cable [3] and NGPP-A cable [4] (TASK 20-10-91-000-801).

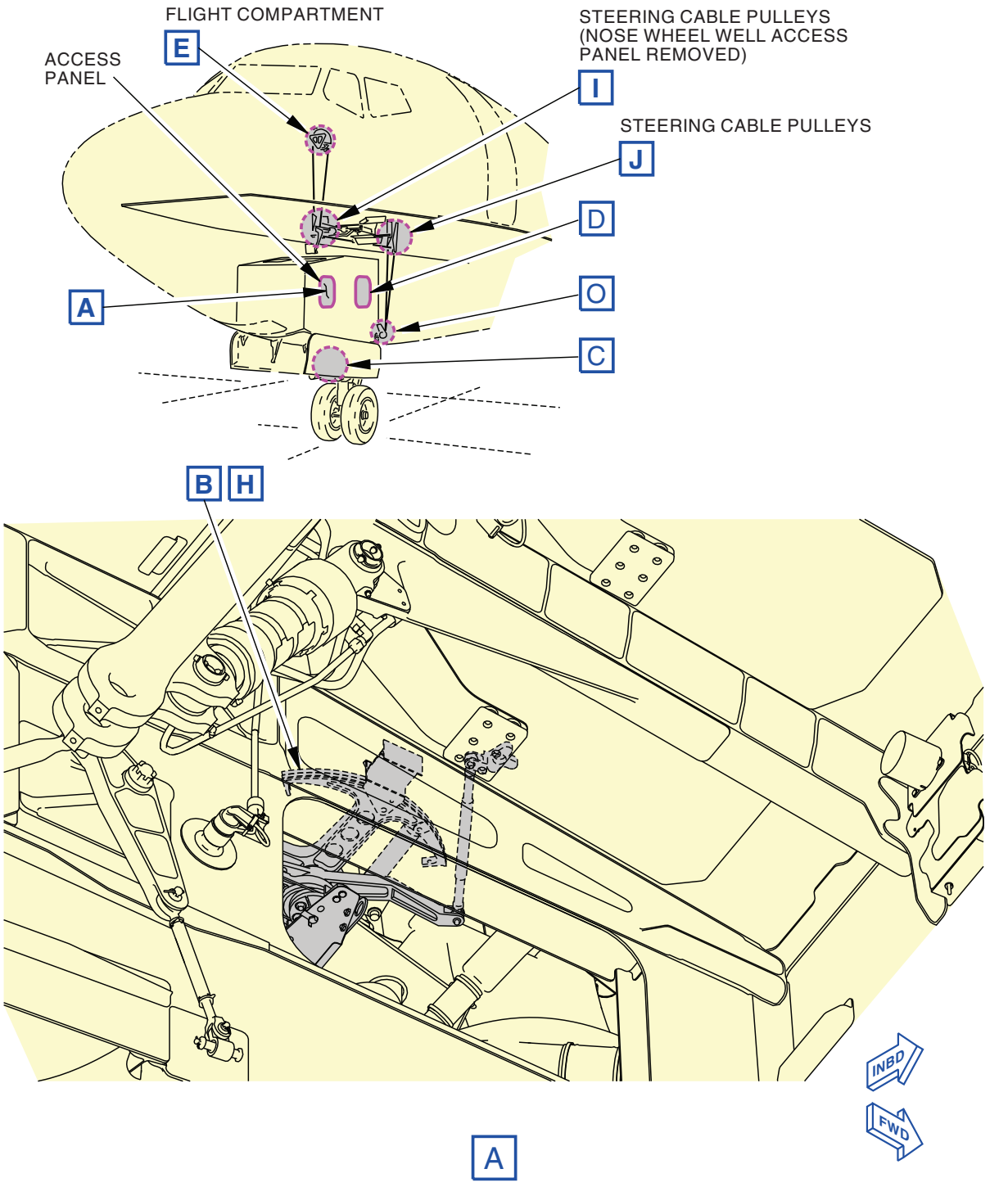
(a) Remove the clips [1] from the turnbuckles [2].

(b) Disconnect the NGPP-B cable [3] and NGPP-A cable [4] at the turnbuckles [2].

(c) Remove the pins [5] to disconnect the NGPP-B cable [3] and NGPP-A cable [4] from the rotary actuator pulley.

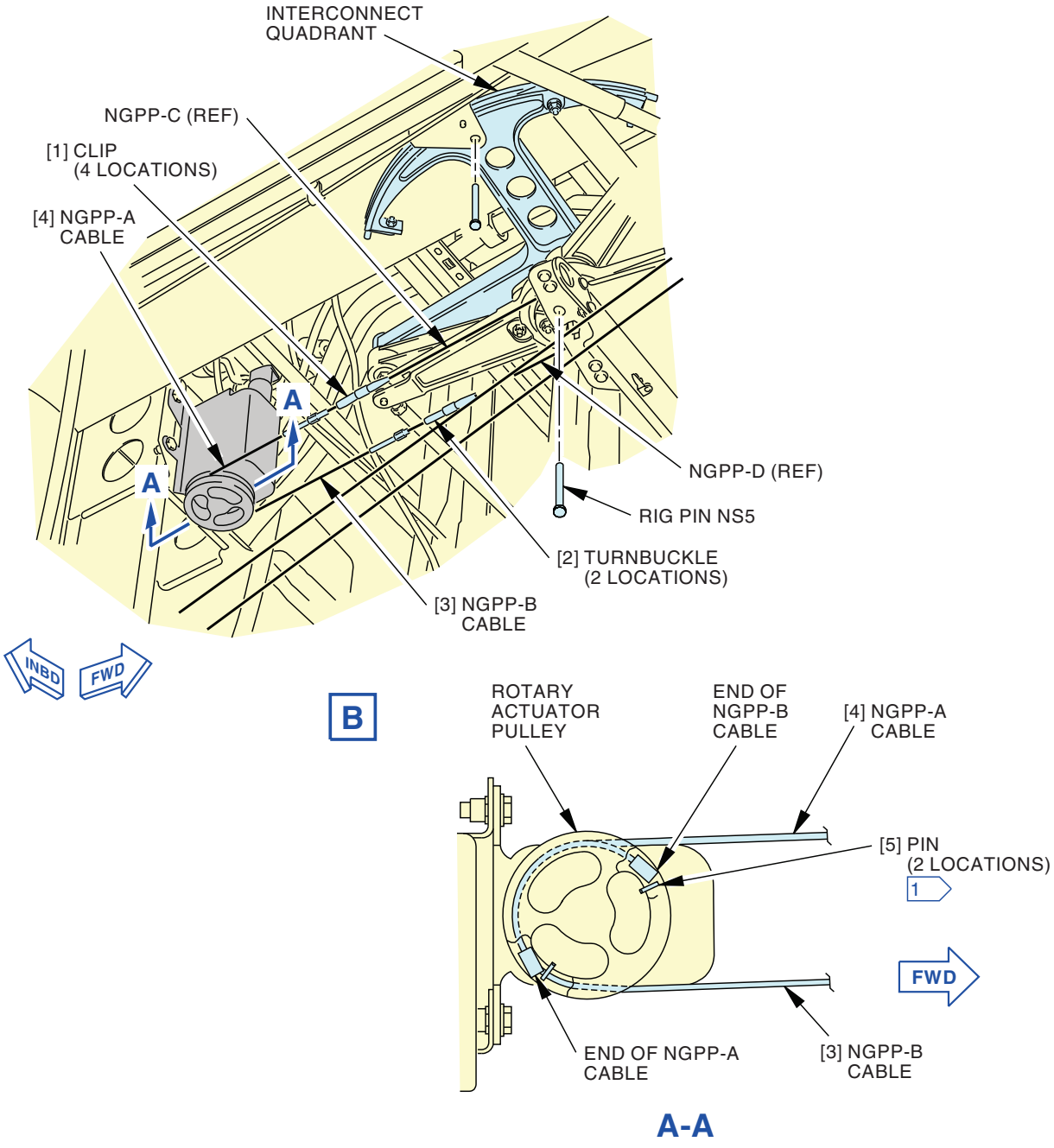
——— END OF TASK ———

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Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 1 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

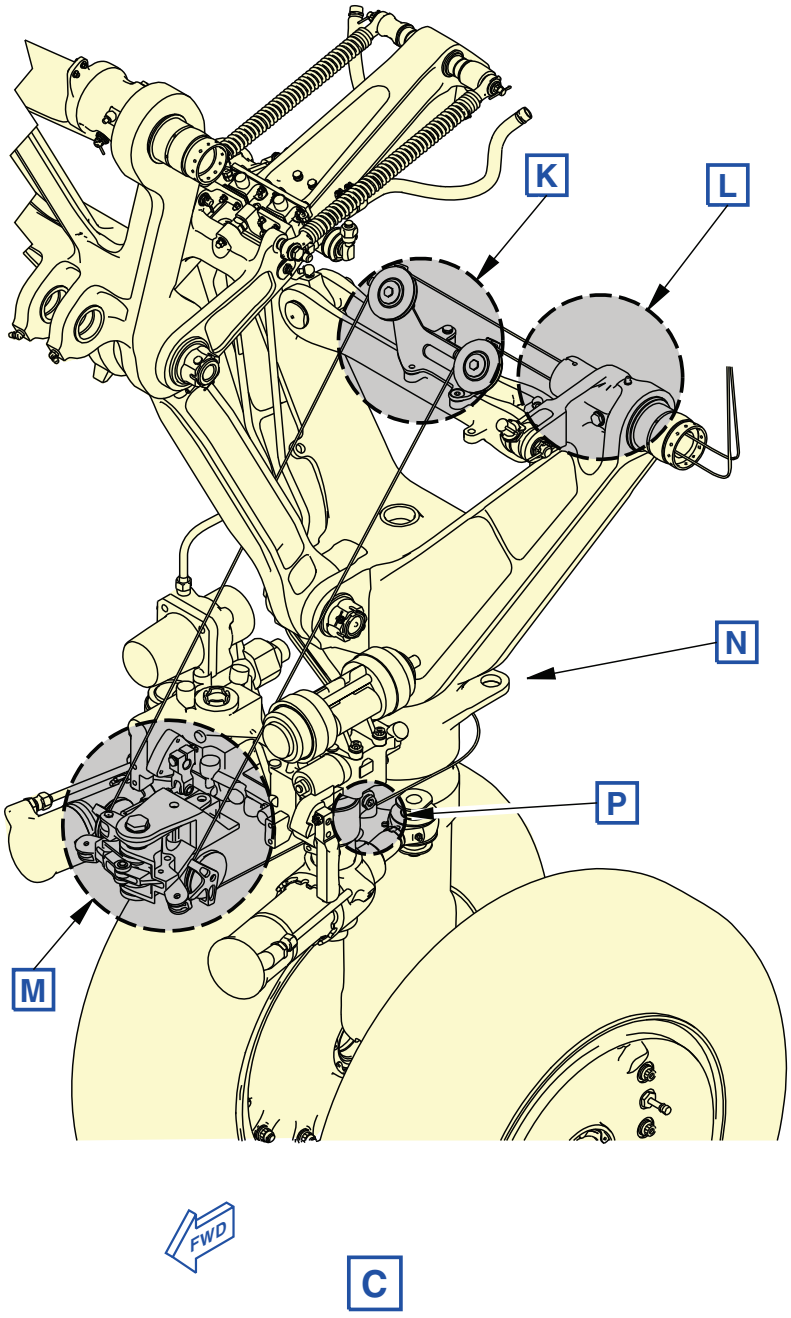


1 **CAUTION:**
MAKE SURE THE ENDS OF THE PINS [5] DO NOT TOUCH THE HOUSING OF THE ROTARY ACTUATOR AFTER INSTALLATION. IF THE PINS TOUCH THE HOUSING, DAMAGE TO EQUIPMENT CAN OCCUR.

2411042 S00061526781_V3

Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 2 of 12)

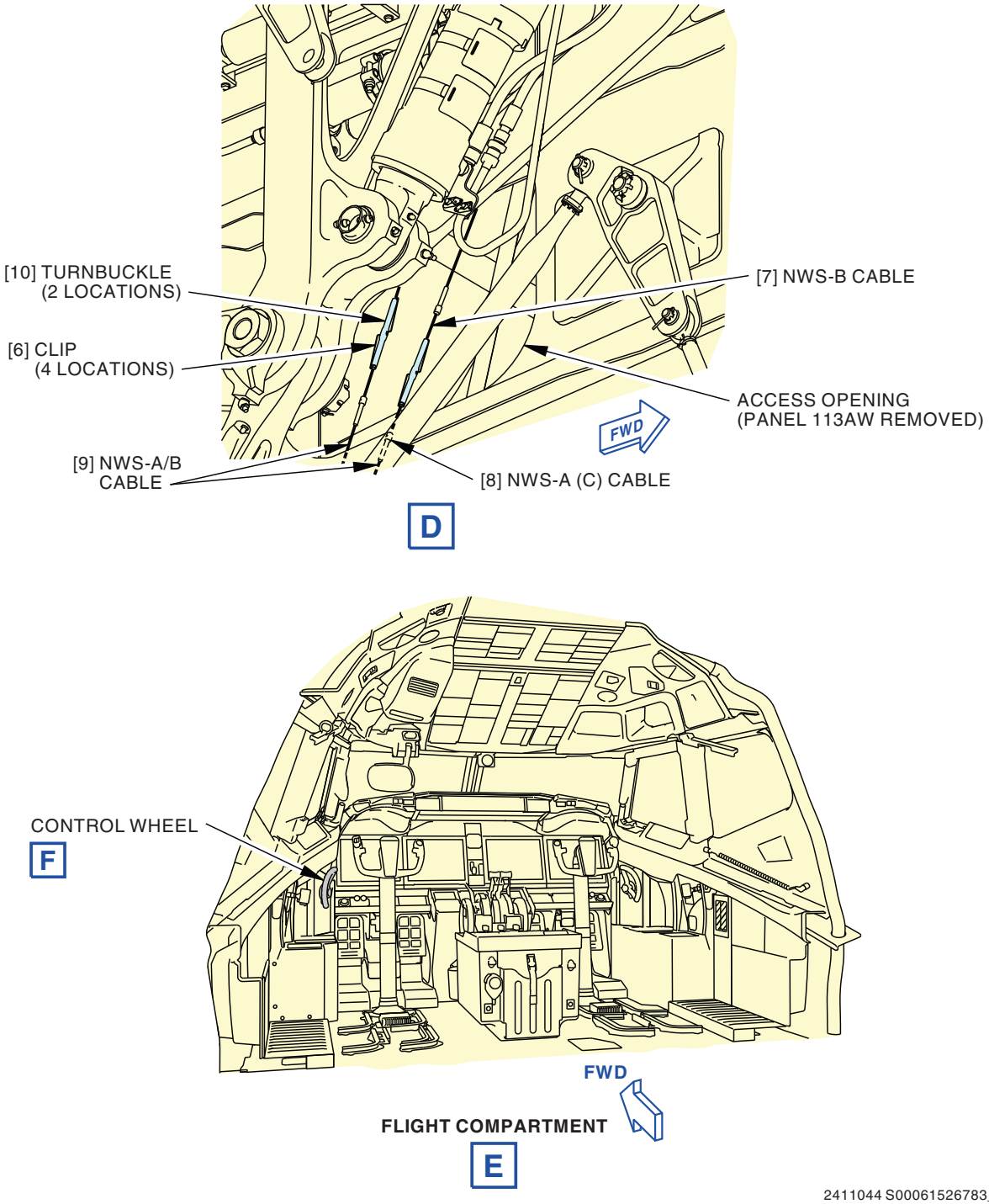
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411043 S00061526782_V2

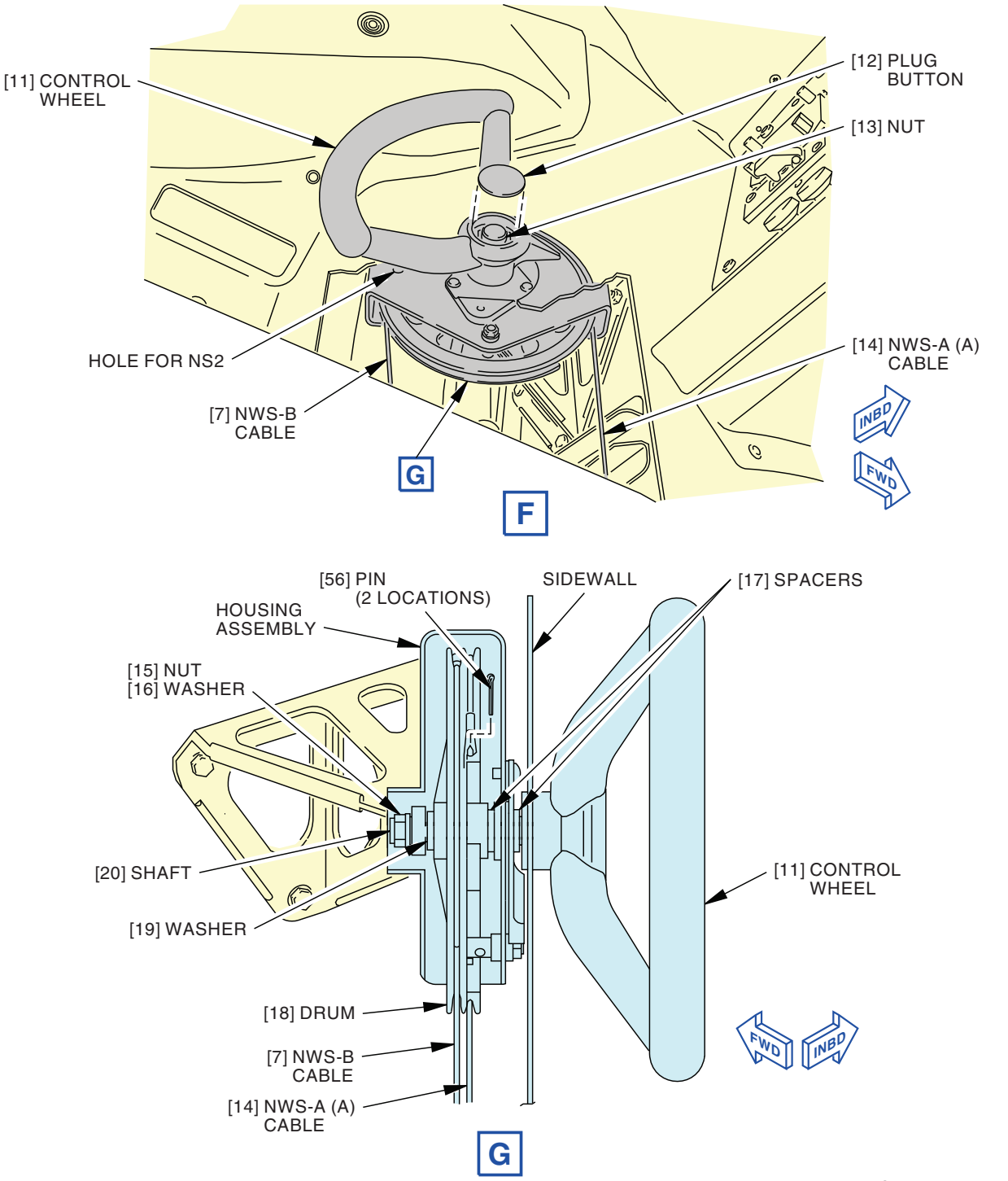
Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 3 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



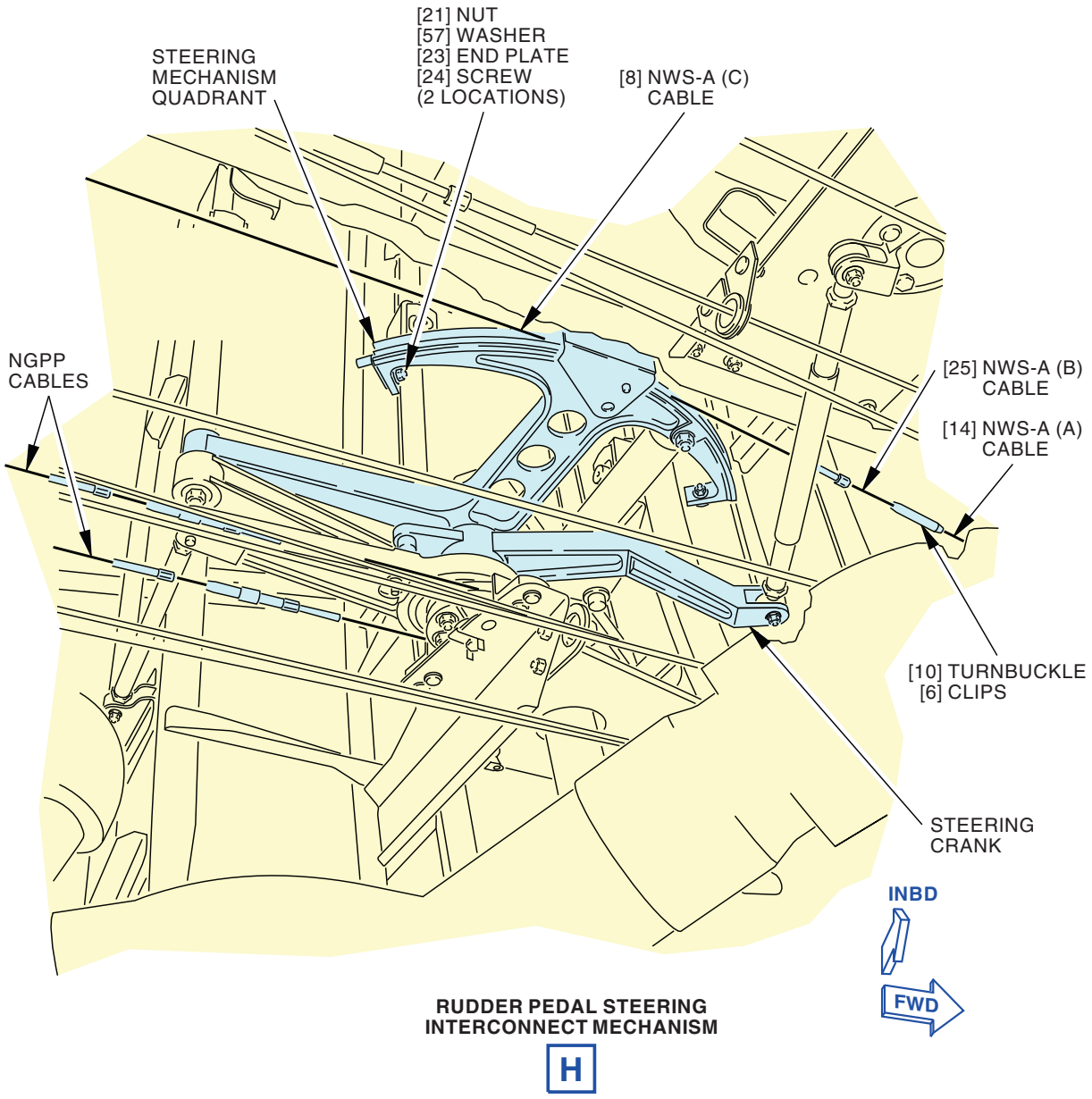
Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 4 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 5 of 12)

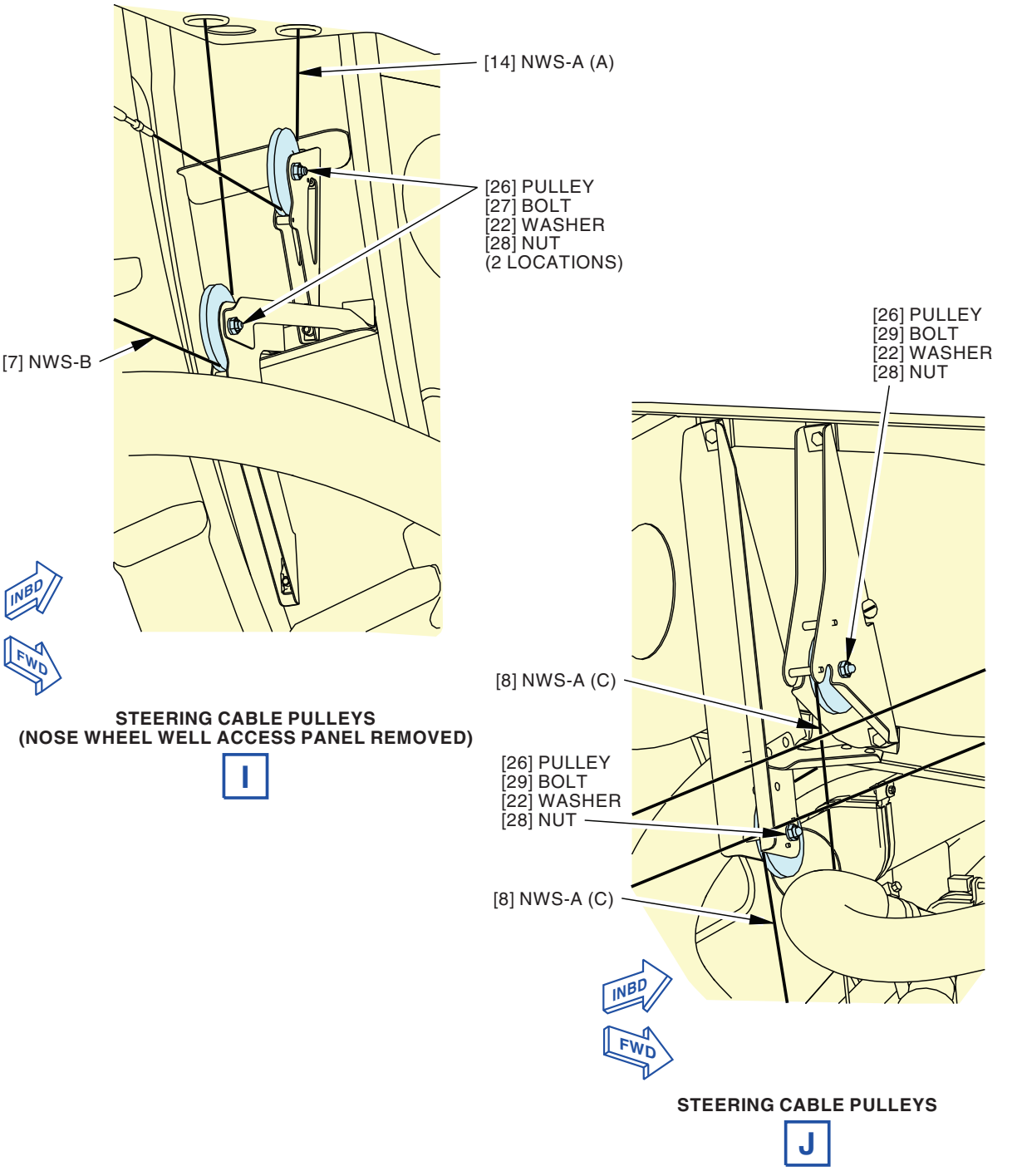
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411046 S00061526785_V2

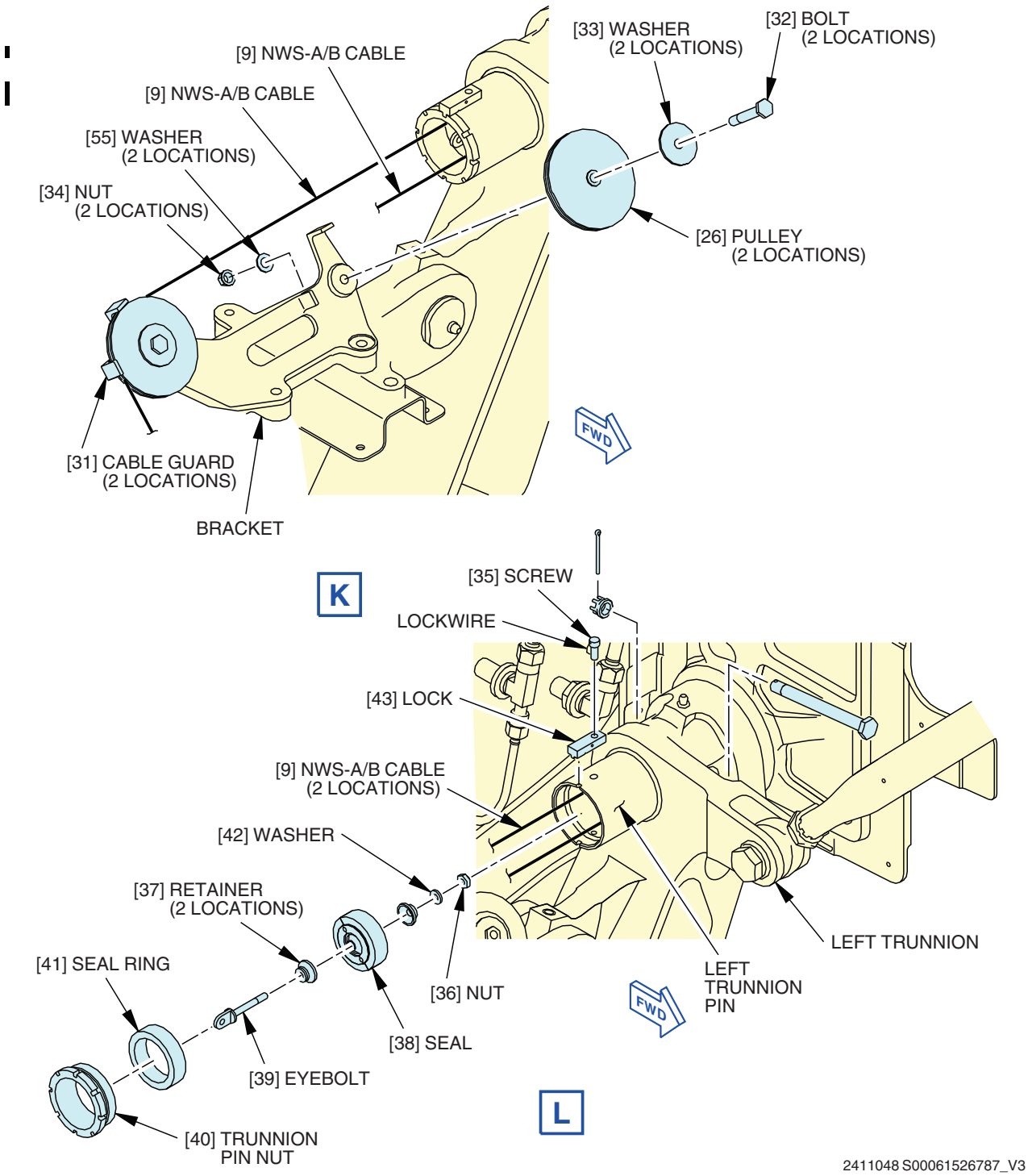
Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 6 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



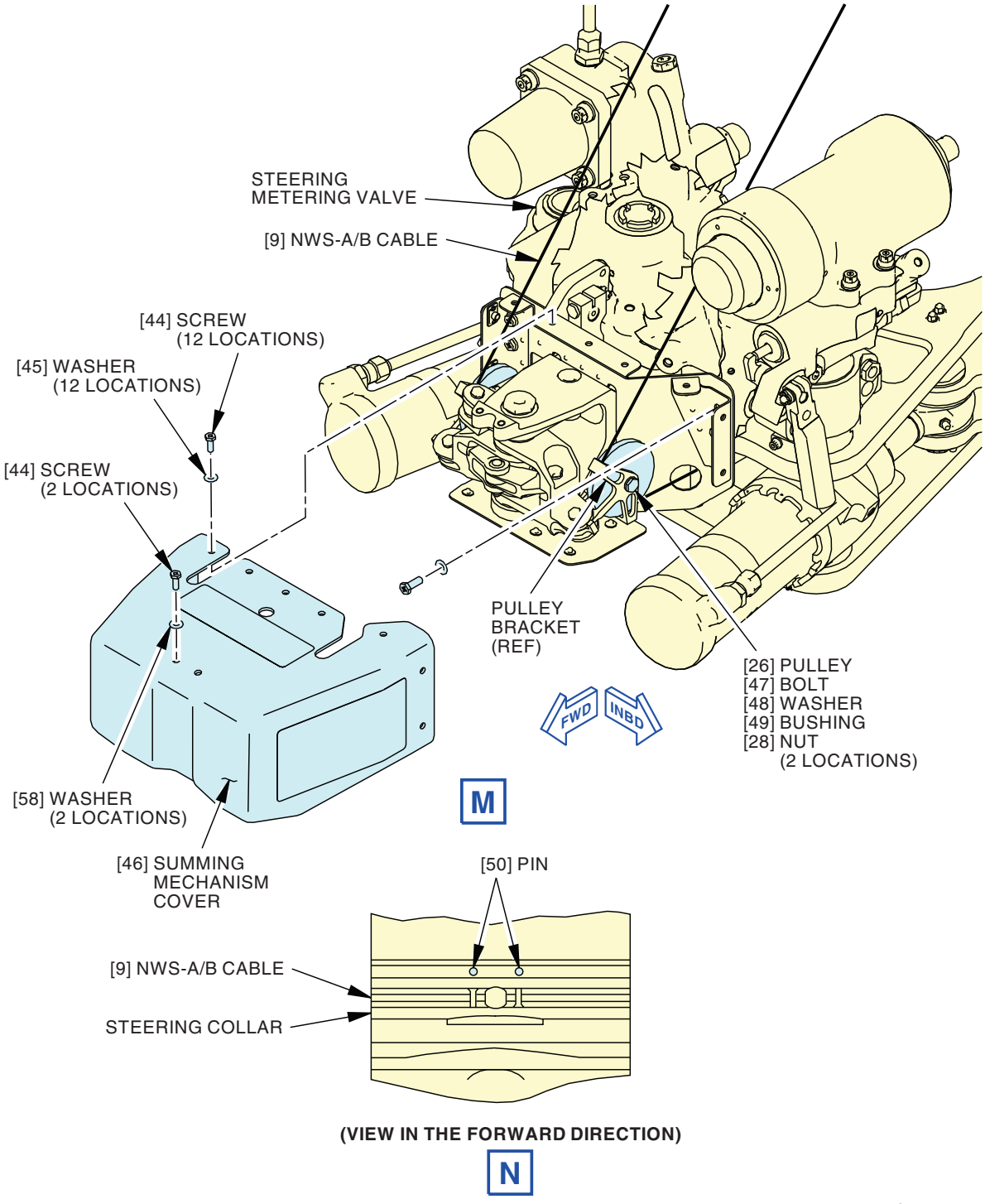
Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 7 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



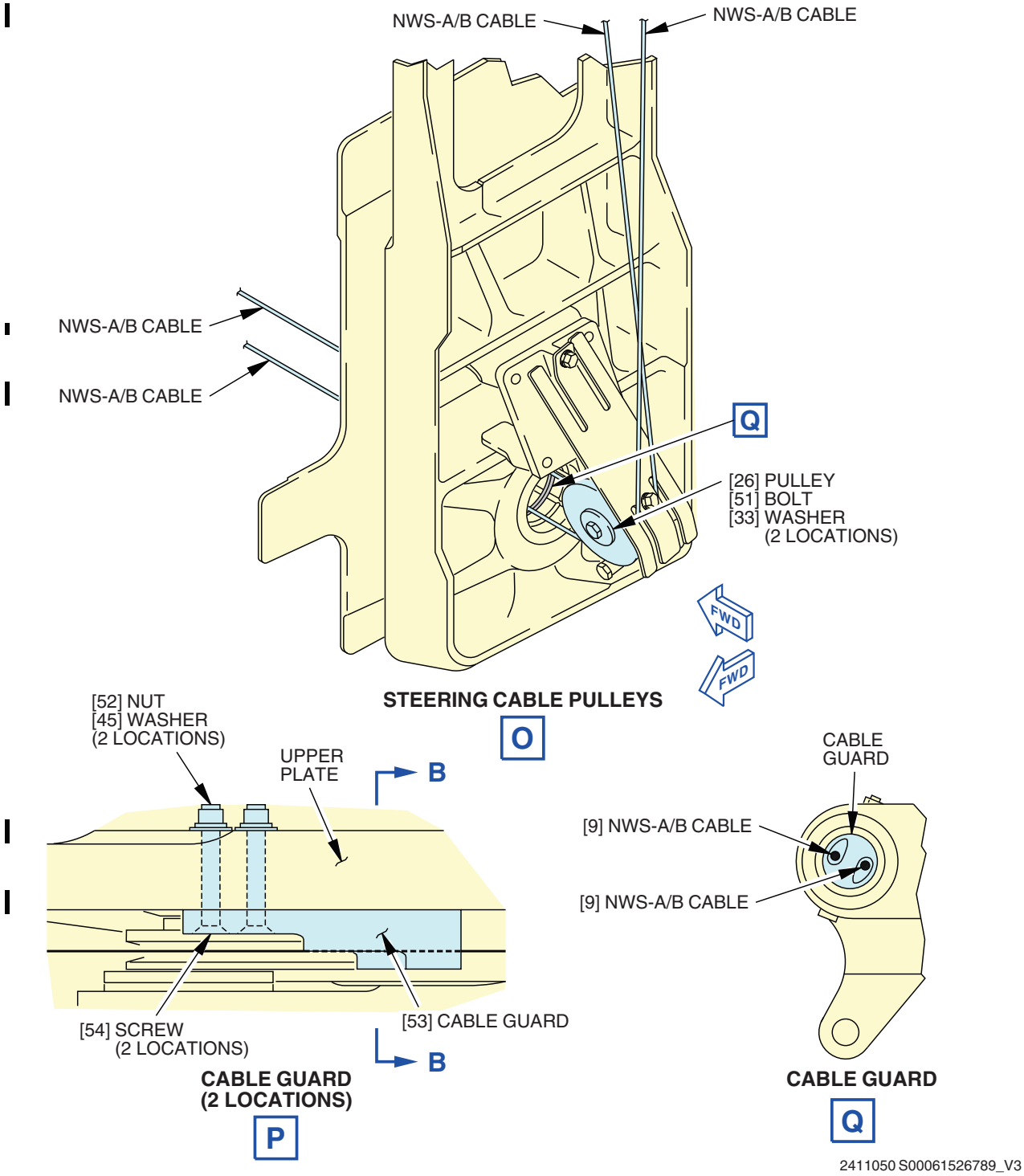
Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 8 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



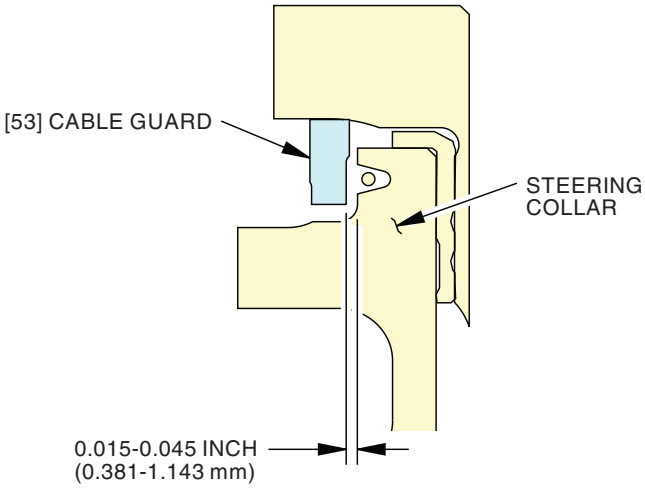
Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 9 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 10 of 12)

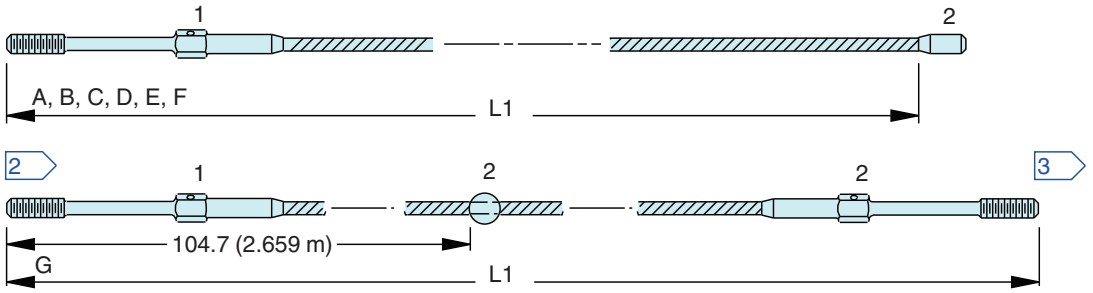
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 11 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABLE REFERENCE	FUNCTION	NUMBER REQUIRED	LENGTH INCHES 1	FITTINGS 1	
				1	2
A 5	NWSA (A)	1	L1 52.3 (1.328 m)	MS21260L3LH	BACT14A3
B 5	NWSA (B)	1	L1 16.8 (0.427 m)	MS21260L3RH	BACT14A3
C 5	NGPPB	1	L1 13.9 (0.353 m)	MS21260S3LH	BACT14A3
D 5	NGPPA	1	L1 14.1 (0.358 m)	MS21260S3RH	BACT14A3
E 5	NWSA (C)	1	L1 68.0 (1.727 m)	MS21260L3RH	BACT14A3
F 5	NWSB	1	L1 127.2 (3.231 m)	MS21260L3LH	BACT14A3
G 4	NWSA/B	1	L1 201.8 (5.126 m)	MS21260L3RH/LH	BACT14B3

1 REFERENCE ONLY. REFER TO CURRENT IPC FOR SPECIFIC PART NUMBER, LENGTH, MATERIAL AND END FITTING ASSEMBLY FABRICATION TENSIONING LOADS, AND TOLERANCES.

2 THIS END TAKES A RIGHT HAND THREADED FITTING.

3 THIS END TAKES A LEFT HAND THREADED FITTING.

4 CABLE MATERIAL: 3/32 CRES 7X7 PER BMS7-265, TYPE 1, COMPOSITION B.

5 CABLE MATERIAL: 3/32 TIN OVER ZINC CARBON STEEL, 7X7 PER BMS 7-265, TYPE 1, COMPOSITION A.

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Steering System Cables Installation
Figure 401/32-51-31-990-801 (Sheet 12 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-31-400-801

3. Nose Gear Steering Position Cables Installation

(Figure 401)

A. General

(1) This task gives instructions to install nose gear steering position cables.

B. References

Reference	Title
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
20-10-91-400-801	Control Cables Installation (P/B 401)
32-51-00-820-802	Rudder Pedal Interconnect Mechanism Adjustment Check (P/B 501)
32-51-00-820-803	Rudder Pedal Interconnect Mechanism Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
C00308	Compound - Corrosion Preventive, Petrolatum Hot Application	MIL-C-11796
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	NGPP-B cable	32-51-31-02-030	PTF ALL
4	NGPP-A cable	32-51-31-02-035	PTF ALL
5	Pin	32-51-31-02-005	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

G. Nose Gear Steering Position Cables Installation

SUBTASK 32-51-31-420-001

(1) Install the NGPP-B cable [3] and the NGPP-A cable [4] (TASK 20-10-91-400-801).

NOTE: The cable length and fittings are shown in (Figure 401).

(a) Apply a thin layer of corrosion preventive compound, C00308, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckles [2].

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SUBTASK 32-51-31-420-002

(2) Connect the NGPP-A cable [4] and the NGPP-B cable [3] to the rotary actuator pulley.

CAUTION

DO NOT LET THE ENDS OF THE PINS TOUCH THE HOUSING OF THE ACTUATOR WHEN THE CABLES ARE INSTALLED. DAMAGE TO THE ROTARY ACTUATOR CAN OCCUR.

(a) Install the pins [5] in the rotary actuator pulley.

SUBTASK 32-51-31-420-003

(3) Connect the NGPP-A cable [4] and the NGPP-B cable [3] at the turnbuckles [2].

SUBTASK 32-51-31-820-001

(4) Adjust the NGPP-A cable [4] and the NGPP-B cable [3] (TASK 32-51-00-820-802).

SUBTASK 32-51-31-820-002

(5) Do the adjustment for the rudder pedal interconnect mechanism (TASK 32-51-00-820-803).

SUBTASK 32-51-31-420-004

(6) Install the clips [1] on the turnbuckles [2].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-31-580-002

(1) Lower the nose of the airplane and remove the jacks (TASK 07-10-21-580-802).

SUBTASK 32-51-31-410-001

(2) Install the forward access panel on the left sidewall of the nose wheel (TASK 53-14-01-420-801).

(a) Close this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-31-080-001

(3) Remove the rig pin NS5 from the drum on the interconnect quadrant.

END OF TASK

TASK 32-51-31-000-802

4. Nose Gear Steering System Cables Removal

(Figure 401)

A. General

(1) This task gives instructions to remove nose gear steering system cables.

B. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
25-11-22-000-801	Flight Compartment Forward Side Panel Removal (P/B 401)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-51-00-820-801	General Check of the Nose Wheel Steering System Adjustment (P/B 501)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1821	Adjustable Spanner Wrench (0.75 to 2.00 Inch Dia. Retainer, 0.12 x 0.12 Key Arm) Part #: F72959-40 Supplier: 81205 Opt Part #: F72959-4 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

F. Prepare for the Removal

SUBTASK 32-51-31-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-31-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-31-580-003

(3) If it is necessary, lift the airplane with jacks, until the nose gear shock strut is fully extended and wheels do not touch the ground (TASK 07-10-21-580-801).

NOTE: The nose of the airplane is lifted with jacks to find the center position of the nose landing gear and steering collar. The internal centering cam of the shock strut is used to find the center position of the nose landing gear and steering collar. If the jacks are not available you can use grease plates under the nose wheels to put the nose gear and steering collar in the center position.

NOTE: To put the nose landing gear and steering collar in the center position without jacks, tow the airplane a minimum of 12 ft (4 m) in a straight line before it is parked.

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SUBTASK 32-51-31-480-004

(4) Install the rig pin NS2 in the drum for the captain's control wheel (TASK 32-51-00-820-801).

NOTE: Once the nose is lifted off the ground, the shock strut centering cams center the gear to let you insert rig pin NS2.

SUBTASK 32-51-31-010-003

(5) Remove the forward access panel from the left sidewall of the nose wheel well to get access to the turnbuckle [10] of the NWS-A(A) cable [14] and NWS-A(B) cable [25] (TASK 53-14-01-020-801).

(a) Remove this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-31-010-004

(6) Remove the aft access panel from the left sidewall of the nose wheel well to get access to the turnbuckles [10] for the NWS-B cable [7], NWS-A(C) cable [8], and NWS-A/B cable [9] (TASK 53-14-01-020-801).

(a) Remove this access panel:

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

G. Nose Gear Steering System Cables Removal

SUBTASK 32-51-31-020-001

(1) Remove the NWS-A(C) cable [8], NWS-A(A) cable [14], and NWS-A(B) cable [25].

(a) Remove the clips [6] from the turnbuckles [10].

(b) Disconnect the NWS-A(C) cable [8], NWS-A(A) cable [14], and NWS-A(B) cable [25] at the turnbuckles [10].

(c) Do these steps to disconnect the NWS-A(A) cable [14] from the drum [18]:

1) Remove the plug button [12] from the control wheel [11].
2) Remove the nut [13] that holds the control wheel [11] to the tiller assembly.
3) Remove the control wheel [11].
4) Remove the sidewall to get access to the drum [18] on the tiller assembly (TASK 25-11-22-000-801).
5) Remove the nut [15] and washer [16] from the shaft [20].
6) Remove the shaft [20], washer [19], and spacers [17].
7) Remove the drum [18].
8) Remove the pins [56] from the drum [18].

(d) Remove the nut [28], washer [22], bolt [27], and pulley [26] from the bracket.

(e) Remove the NWS-A(A) cable [14].

(f) Remove the nuts [21], washers [57], end plates [23], and screws [24] that attach the NWS-A(B) cable [25] and NWS-A(C) cable [8] to the quadrant for the steering mechanism (View H, Figure 401).

(g) Remove the NWS-A(B) cable [25].

(h) Remove the nut [28], washer [22], bolt [29], and pulley [26] from the bracket (Figure 401, View J).

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(i) Remove the NWS-A(C) cable [8].

SUBTASK 32-51-31-020-002

(2) Remove the NWS-B cable [7].

(a) Remove the clips [6] from the turnbuckles [10] (View D, Figure 401).

(b) Disconnect the NWS-B cable [7] and NWS-A/B cable [9] at the turnbuckle [10].

(c) Do these steps to disconnect the NWS-B cable [7] at the drum [18] (View G, Figure 401):

1) Remove the plug button [12] from the control wheel [11] (View F, Figure 401).

2) Remove the nut [13] that holds the control wheel [11] to the tiller assembly.

3) Remove the control wheel [11].

4) Remove the sidewall to get access to the drum [18] (TASK 25-11-22-000-801).

5) Remove the nut [15] and washer [16] from the shaft [20] (View G, Figure 401).

6) Remove the shaft [20], washer [19], and spacers [17].

7) Remove the drum [18].

8) Remove the pin [56] from the drum [18].

(d) Remove the nut [28], washer [22], bolt [27], and pulley [26] from the bracket.

(e) Remove the nut [28], washer [22], bolt [30], and pulley [26] from the bracket.

(f) Remove the NWS-B cable [7].

SUBTASK 32-51-31-020-003

(3) Remove the NWS-A/B cable [9].

(a) Remove the screws [44], washers [45], washers [58], and summing mechanism cover [46] (View M, Figure 401).

(b) Remove the pins [50] from the steering collar (View N, Figure 401).

(c) Remove the nuts [52], washers [45], and screws [54] to remove the cable guard [53] (View P, Figure 401).

(d) Remove the nuts [34], washers [55], washers [33], bolts [32], cable guards [31], and pulleys [26] from their brackets (View K, Figure 401).

(e) Remove the bolts [51], washers [33], and pulleys [26] from the bracket behind the trunnion (View O, Figure 401).

(f) Remove the nuts [28], bolts [47], washers [48], bushings [49], and pulleys [26] from the pulley brackets on the summing mechanism (View M, Figure 401).

(g) Remove the seal [38] from the left trunnion of the nose gear.

1) Remove the lockwire from the screw [35] and lock [43] (View L, Figure 401).

2) Remove the screw [35] that holds the lock [43] to the left trunnion pin.

3) Remove the lock [43] from the left trunnion pin.

4) Use a spanner wrench, SPL-1821, to remove the trunnion pin nut [40] from the left trunnion pin.

5) Pull the seal ring [41] and seal [38] from the left trunnion pin.

a) If it is difficult to pull the seal [38], hold the eyebolt [39] and pull the seal [38] out of the left trunnion pin.

6) Remove the nut [36], washer [42], and inboard retainer [37] from the eyebolt [39].

7) Remove the eyebolt [39] from the left trunnion pin.

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8) Remove the outboard retainer [37] from the eyebolt [39].

9) Separate the segments of the seal [38] from the NWS-A/B cable [9].

10) Pull the NWS-A/B cable [9] through the trunnion pin nut [40], seal ring [41], and left trunnion pin.

(h) Remove the NWS-A/B cable [9].

——— END OF TASK ———

TASK 32-51-31-400-802

5. Nose Gear Steering System Cables Installation

(Figure 401)

A. General

(1) This task gives instructions to install nose gear steering system cables.

B. References

Reference	Title
20-10-91-400-801	Control Cables Installation (P/B 401)
20-20-31-200-801	Inspection of the Control Cable Wire Rope (P/B 601)
25-11-22-400-801	Flight Compartment Forward Side Panel Installation (P/B 401)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-51-00-820-801	General Check of the Nose Wheel Steering System Adjustment (P/B 501)
32-51-00-820-802	Rudder Pedal Interconnect Mechanism Adjustment Check (P/B 501)
32-51-00-820-805	Nose Wheel Steering System Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1821	Adjustable Spanner Wrench (0.75 to 2.00 Inch Dia. Retainer, 0.12 x 0.12 Key Arm) Part #: F72959-40 Supplier: 81205 Opt Part #: F72959-4 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
C00308	Compound - Corrosion Preventive, Petrolatum Hot Application	MIL-C-11796
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

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E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	NWS-B cable	32-51-31-01-015	PTF ALL
8	NWS-A(C) cable	32-51-31-02-040	PTF ALL
9	NWS-A/B cable	32-51-31-02-045	PTF ALL
14	NWS-A(A) cable	32-51-31-01-010	PTF ALL
25	NWS-A(B) cable	32-51-31-02-045	PTF ALL
56	Pin	32-51-41-02-045	PTF ALL

F. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

G. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

H. Prepare for the Installation

SUBTASK 32-51-31-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed in the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-51-31-860-003

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-31-480-006

(3) Make sure that the rig pin NS2 is in the drum for the captain's control wheel (TASK 32-51-00-820-801).

I. Nose Gear Steering System Cables Installation

CAUTION MAKE SURE THAT THE STEERING CABLES ARE IN THE GROOVES OF THE PULLEYS WHEN YOU ASSEMBLE THEM. CABLES THAT ARE NOT IN THE CABLE GUARDS CAN CAUSE DAMAGE.

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(CAUTION PRECEDES)

CAUTION

DO NOT APPLY SOLVENTS, GREASE, OR OIL TO STAINLESS STEEL CONTROL CABLES. THESE MATERIALS CAN COLLECT CONTAMINATION THAT CAN CAUSE DAMAGE TO THE INTERNAL SURFACES OF THE CRES CABLE STRANDS. THIS CAN DECREASE THE SERVICE LIFE OF THE CABLE.

SUBTASK 32-51-31-420-005

(1) Install the NWS-A(C) cable [8], NWS-A(A) cable [14], and NWS-A(B) cable [25] (TASK 20-10-91-400-801).

NOTE: Cable length and fittings are shown in Figure 401.

(a) Apply a thin layer of corrosion preventive compound, C00308, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckles [10].

(b) Do these steps to install the NWS-A(A) cable [14] on the drum [18]:

1) Put the end of the NWS-A(A) cable [14] in the groove on the drum [18].
2) Install the new pin [56] in the drum [18].
3) Install the drum [18] on the steering tiller assembly.
4) Install the shaft [20], washer [19], and spacers [17].
   a) Install the shaft [20] with the missing tooth aligned in the drum [18].
5) Install the nut [15] and washer [16] on the shaft [20].
6) Install the control wheel [11].
7) Install the nut [13] that holds the control wheel [11] to the tiller assembly.
8) Tighten the nut [13] to 100 in-lb (11.3 N·m) - 150 in-lb (16.9 N·m) above the running torque.
9) Install the plug button [12] on the control wheel [11].
10) Install the sidewall you removed to get access to the drum [18] on the tiller assembly (TASK 25-11-22-400-801).

(c) Connect the NWS-A(A) cable [14] to the pulley [26].

(d) Install the bolt [27], washer [22], nut [28], and pulley [26] on the bracket.

(e) Install the NWS-A(B) cable [25].

(f) Connect the NWS-A(C) cable [8] and NWS-A(B) cable [25] to the steering mechanism quadrant.

1) Install the end plates [23], screws [24], washers [57], and nuts [21].

(g) Install the NWS-A(C) cable [8] in the groove of the pulley [26].

(h) Install the bolt [29], washer [22], nut [28], and pulley [26], on the bracket.

SUBTASK 32-51-31-420-006

(2) Install the NWS-B cable [7] (TASK 20-10-91-400-801).

NOTE: Cable length and fittings, are shown in Figure 401.

(a) Apply a thin layer of corrosion preventive compound, C00308, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckles [10].

(b) Do these steps to install the NWS-B cable [7] on the drum [18]:

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1) Put the end of NWS-B cable [7] in the groove on the drum [18].
2) Install the new pin [56] in the drum [18].
3) Install the drum [18] in the steering tiller assembly.
4) Install the shaft [20], washer [19], and spacers [17].
   a) Install the shaft [20] with the missing tooth aligned in the drum [18].
5) Install the nut [15] and washer [16] on the shaft [20].
6) Install the control wheel [11].
7) Install the nut [13] that holds the control wheel [11] to the tiller assembly.
8) Tighten the nut [13]] to 100 in-lb (11.3 N·m) - 150 in-lb (16.9 N·m) above the running torque.
9) Install the plug button [12] on the control wheel [11].
10) Install the sidewall you removed to get access to the drum [18] on the tiller assembly (TASK 25-11-22-400-801).

(c) Put the NWS-B cable [7] in the groove of the pulleys [26].

(d) Install the bolt [27], washer [22], nut [28], and pulley [26] on the bracket.

(e) Install the bolt [30], washer [22], nut [28], and pulley [26] on the bracket.

SUBTASK 32-51-31-420-007

(3) Install the NWS-A/B cable [9] (TASK 20-10-91-400-801).

NOTE: Cable length and fittings are shown in Figure 401.

(a) Apply a thin layer of corrosion preventive compound, C00308, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckles [10].

(b) Install the seal [38] and NWS-A/B cable [9] through the left trunnion of the nose gear.
1) Install the NWS-A/B cable [9] through the trunnion pin nut [40], seal ring [41], left trunnion pin, pulleys [26] outside the left trunnion, pulley brackets on the summing mechanism, and around the steering collar.
   a) Make sure that the ends of the NWS-A/B cable [9] are 1 in. (25 mm) or less apart when they are pulled tight where they come out of the left trunnion pin.
2) Install the segments of the seal [38] on the NWS-A/B cable [9].
3) Install the outboard retainer [37] on the eyebolt [39].
4) Install the inboard retainer [37], washer [42], and nut [36] on the eyebolt [39].
   a) Make sure that one NWS-A/B cable [9] goes over the trunnion pin bolt and one goes under the trunnion pin bolt where the NWS-A/B cables [9] go through the left trunnion.
5) Tighten the nut [36] sufficiently to clamp the seal [38].
6) Loosen the nut [36] approximately 1/4 turn until you can turn the eyebolt [39] with your fingers.
7) Put the seal [38] into the left trunnion pin until it touches the stop.

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WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

8) Apply corrosion inhibiting material, G50136, to the threads of the trunnion pin nut [40], seal ring [41], and seal [38].

9) Put the seal ring [41] and seal [38] into the left trunnion pin.

10) Use the spanner wrench, SPL-1821, to install the trunnion pin nut [40] on the left trunnion pin.

11) Tighten the trunnion pin nut [40] to 125 ±25 in-lb (14 ±3 N·m), then loosen the trunnion pin nut [40] to align the slots on the trunnion pin nut [40] with the bolt holes.

12) Install the lock [43] on the left trunnion pin.

13) Install the screw [35] to hold the lock [43] on the left trunnion pin.

14) Install MS20995C32 lockwire, G01048, on the screw [35] and lock [43].

(c) Install the pin [50] on the steering collar.

(d) For the pulleys [26] on the top of the outer cylinder, do these steps:

1) Put the NWS-A/B cable [9] in the groove on the pulleys [26].

2) Install the pulleys [26], cable guards [31], bolts [32], washers [55], washers [33], and nuts [34] on their brackets on the top of the outer cylinder.

(e) For the pulleys [26] on the bracket behind the trunnion, do these steps:

1) Put the NWS-A/B cable [9] in the groove on the pulleys [26].

2) Install the pulleys [26], bolts [51], and washers [33] on the bracket behind the trunnion.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

a) Install the bolts [51] with corrosion inhibiting material, G50136.

b) Tighten the bolts [51] to 27.5 ±2.5 in-lb (3.1 ±0.3 N·m).

(f) For the pulleys [26] on the summing mechanism do these steps:

1) Put the NWS-A/B cable [9] in the groove on the pulleys [26].

2) Install the nuts [28], bolts [47], washers [48], bushings [49], and pulleys [26] on the pulley brackets on the summing mechanism.

a) Tighten the nuts [28] to 50 in-lb (5.6 N·m) - 75 in-lb (8.5 N·m) above run-on torque.

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(g) Install the nuts [52], washers [45], and screws [54] to attach the cable guard [53] to the upper plate.

1) Make sure that the clearance between the cable guard [53] and steering collar is 0.015 in. (0.381 mm) - 0.045 in. (1.143 mm) (View B-B, Figure 401).

2) Tighten the nuts [52] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

(h) Connect the NWS-A(C) cable [8], NWS-B cable [7], and NWS-A/B cable [9] at the turnbuckles [10].

SUBTASK 32-51-31-200-001

(4) Do this task: Inspection of the Control Cable Wire Rope, TASK 20-20-31-200-801.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-31-210-001

(1) Visually examine the steering cables.

(a) Make sure that the cables are routed correctly around the pulleys and are not outside the cable guards.

NOTE: Incorrect cable routing can cause binding in the steering system resulting in the airplane becoming difficult to steer

SUBTASK 32-51-31-820-003

(2) Do the adjustment for the rudder pedal interconnect mechanism (TASK 32-51-00-820-802).

SUBTASK 32-51-31-820-004

(3) Do the adjustment for the nose wheel steering system (TASK 32-51-00-820-805).

SUBTASK 32-51-31-420-008

(4) Install the clips [6] on the turnbuckles [10].

SUBTASK 32-51-31-410-002

(5) Install the summing mechanism cover [46].

(a) Install the screws [44], washer [45], and washers [58].

SUBTASK 32-51-31-410-003

(6) Close the access panels to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

——— END OF TASK ———

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STEERING SYSTEM CABLES - INSPECTION/CHECK

1. General

A. This procedure has a task to do an inspection of the nose gear steering system cable. There are no steps for access, removal, or installation of the cables. Refer to the procedure: STEERING SYSTEM CABLES - REMOVAL/INSTALLATION, PAGEBLOCK 32-51-31/401, for procedures to do these steps.

TASK 32-51-31-220-801

2. Nose Gear Steering System Cables Inspection

A. References

Reference	Title
12-26-00-600-801	Control Cable Lubrication (P/B 301)
20-20-31-200-801	Inspection of the Control Cable Wire Rope (P/B 601)
20-20-31-200-802	Inspection of the Control Cable Fittings (P/B 601)
20-20-31-200-803	Inspection of Pulleys (P/B 601)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-51-31-000-802	Nose Gear Steering System Cables Removal (P/B 401)
32-51-31-990-801	Figure: Steering System Cables Installation (P/B 401)
32-51-52-020-801	Summing Mechanism Removal (Without Using Cable Clamp Assembly) (P/B 401)
32-51-52-420-801	Summing Mechanism Installation (Without Using Cable Clamp Assembly) (P/B 401)
32-51-52-990-801	Figure: Nose Landing Gear Steering Summing Mechanism Installation (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right

C. Prepare for the Inspection

SUBTASK 32-51-31-840-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed for the nose and main landing gear, do this task: TASK 32-00-01-480-801.

SUBTASK 32-51-31-010-005

(2) Do the steps to remove the summing mechanism cover for the nose landing gear steering as shown in this task: (TASK 32-51-52-020-801), (Figure 32-51-52-990-801).

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D. Nose Gear Steering System Cables Inspection

SUBTASK 32-51-31-210-002

CAUTION

DO NOT APPLY SOLVENTS, GREASE, OR OIL TO STAINLESS STEEL CONTROL CABLES. THESE MATERIALS CAN COLLECT CONTAMINATION THAT CAN CAUSE DAMAGE TO THE INTERNAL SURFACES OF THE CRES CABLE STRANDS. THIS CAN DECREASE THE SERVICE LIFE OF THE CABLE.

(1) Examine the interior of the left side trunnion pin, the area around the pressure seal, the steering cable and cable guard for grease and dirt.

(a) If dirt or grease is found, do this task: Control Cable Lubrication, TASK 12-26-00-600-801.

SUBTASK 32-51-31-210-003

(2) Examine the steering cables for correct routing, broken wires, fraying and corrosion (TASK 20-20-31-200-801).

NOTE: In order to examine the part of the cables in the trunnion pin, it will be necessary to turn the wheels fully in both directions.

(a) Refer to the task: Nose Gear Steering System Cables Removal, TASK 32-51-31-000-802, Figure 32-51-31-990-801, to find where the cables run.

SUBTASK 32-51-31-210-004

(3) Examine the control cable fittings and cable seals for these conditions: deterioration; out-of-round wear; mounting rings that are not secure (TASK 20-20-31-200-802).

(a) Refer to the task: Nose Gear Steering System Cables Removal, TASK 32-51-31-000-802, Figure 32-51-31-990-801, to find where the cable fittings and seals are located.

(b) Remove all grease that you find on the pressure seals.

SUBTASK 32-51-31-210-005

(4) Examine the cable pulley brackets, pulleys and fairleads for these conditions: cracks; distortion; loose mounting, cracks or misalignment (TASK 20-20-31-200-803).

(a) Refer to the task: Nose Gear Steering System Cables Removal, TASK 32-51-31-000-802, Figure 32-51-31-990-801, to find where the cable brackets, pulleys and fairleads are located.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-31-010-006

(1) Do the steps to install the summing mechanism cover for the nose landing gear steering as shown in the task: (TASK 32-51-52-420-801), (Figure 32-51-52-990-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR CONTROL WHEEL AND TILLER ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the nose landing gear steering control wheel

(2) An installation of the nose landing gear steering control wheel

(3) A removal of the nose landing gear tiller assembly

(4) An installation of the nose landing gear tiller assembly.

TASK 32-51-41-000-801

2. Nose Landing Gear Steering Control Wheel Removal

(Figure 401)

A. General

(1) This task includes the steps to remove the control wheel.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
211	Flight Compartment - Left
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-51-41-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

E. Nose Landing Gear Steering Control Wheel Removal

SUBTASK 32-51-41-020-001

(1) Remove the control wheel [7] from the tiller assembly [13].

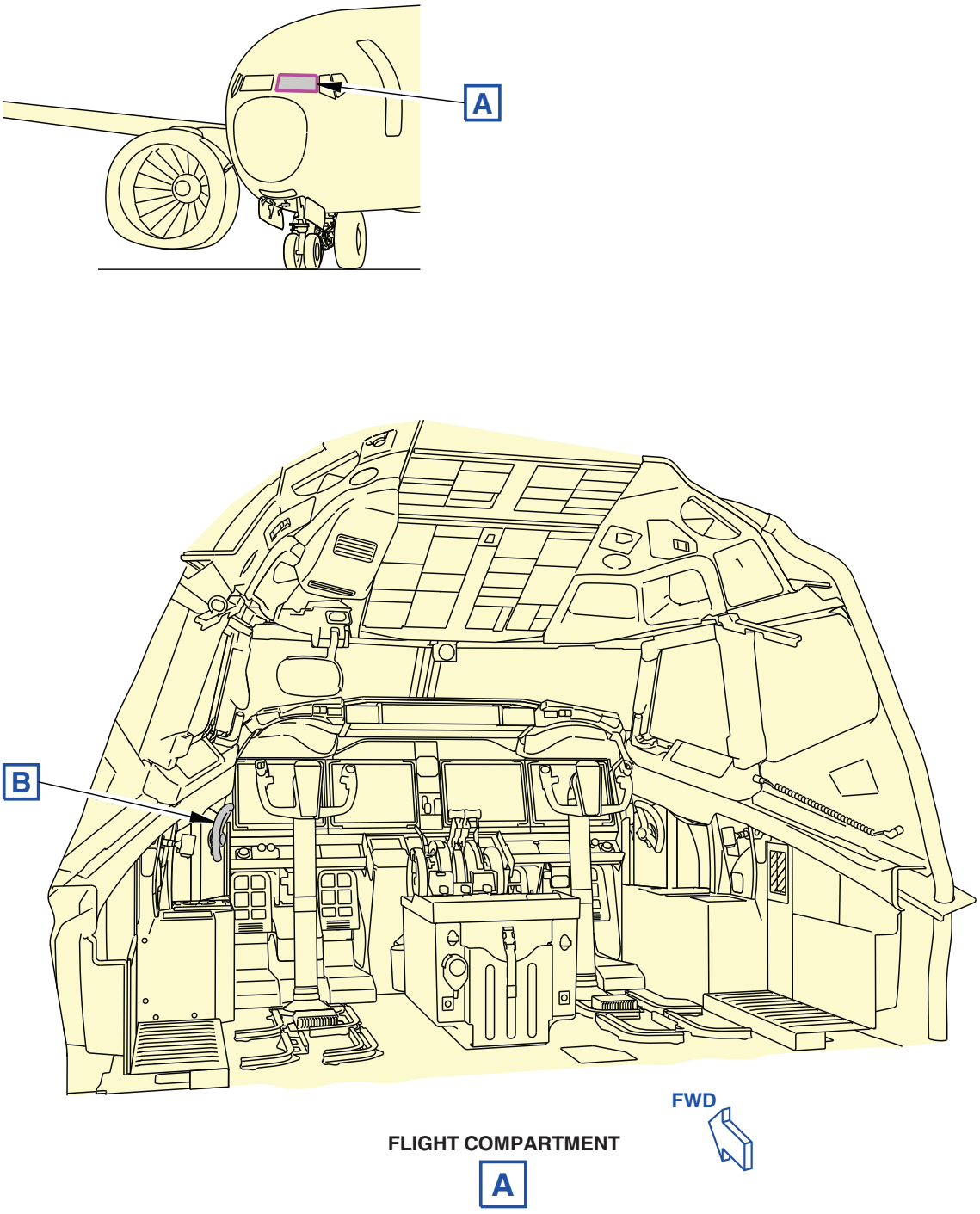
(a) Remove the plug button [1] from the control wheel [7].

(b) Remove the nut [2] and washer [19] that attach the control wheel [7] to the tiller assembly [13].

(c) Remove the control wheel [7] from the airplane.

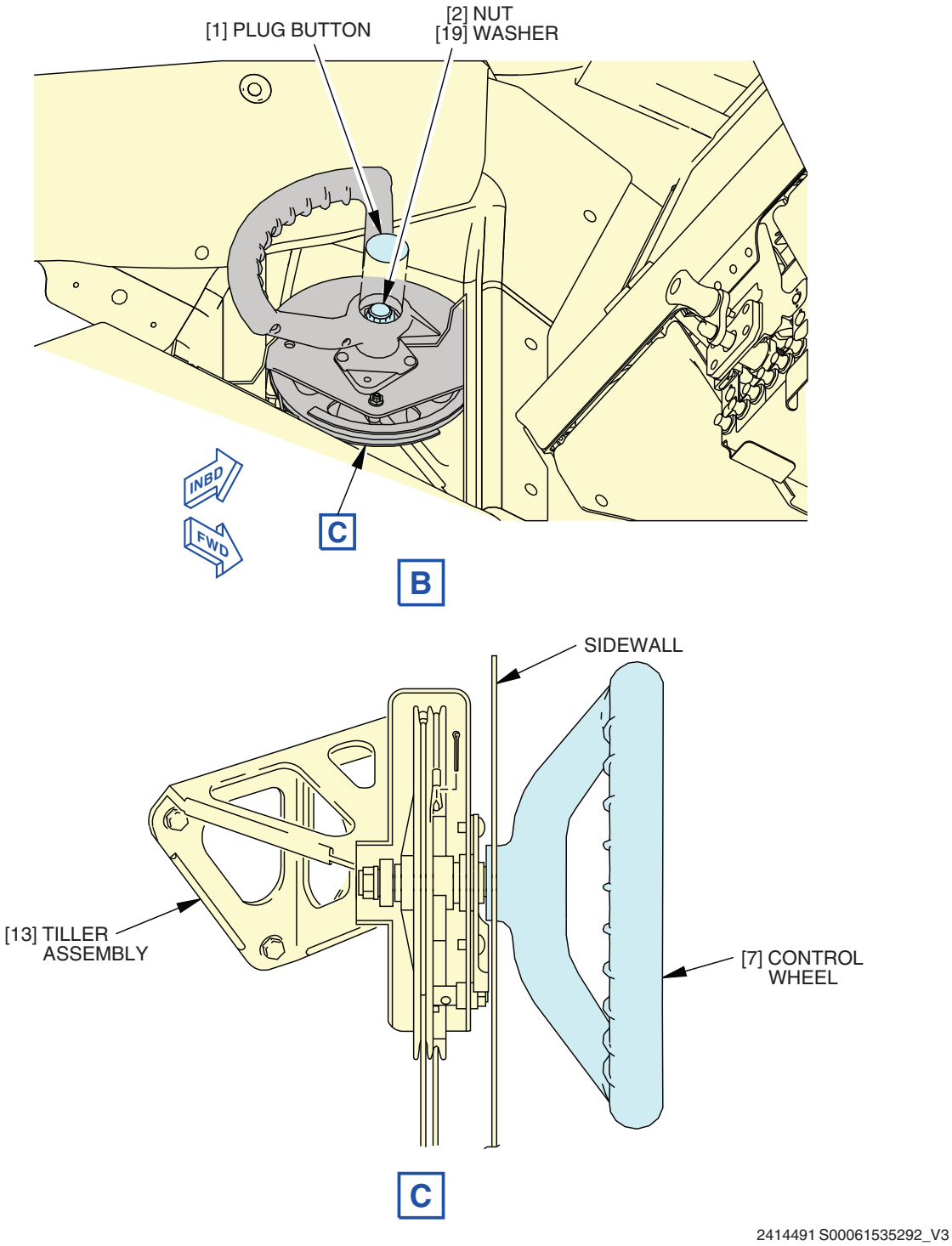
END OF TASK

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Nose Landing Gear Steering Control Wheel Installation
Figure 401/32-51-41-990-801 (Sheet 1 of 2)

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Nose Landing Gear Steering Control Wheel Installation
Figure 401/32-51-41-990-801 (Sheet 2 of 2)

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TASK 32-51-41-400-801

3. Nose Landing Gear Steering Control Wheel Installation

(Figure 401)

A. General

(1) This task includes the steps to install the control wheel.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-00-820-805	Nose Wheel Steering System Adjustment (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
A50060	Adhesive - Low Odor, Synthetic Rubber Cement	BMS5-30 Type III

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Control wheel	32-51-41-02-030	PTF ALL

F. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
211	Flight Compartment - Left
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-51-41-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

H. Nose Landing Gear Control Wheel Installation

SUBTASK 32-51-41-420-001

(1) Install the control wheel [7] on the tiller assembly [13].

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(a) Install rig pin NS2, from rigging pin kit, SPL-1585, in the captain's control wheel drum.

(b) Put the control wheel [7] in its position for attachment to the tiller assembly [13].

NOTE: With the rig pin NS2 installed, the arrow on the control wheel must align with the arrow on the sidewall when you install the control wheel.

(c) Install the nut [2] and washer [19] that attach the control wheel [7] to the tiller assembly [13].

(d) Make sure that the nut [2] is seated.

(e) Loosen the nut [2] approximately two flats.

(f) Find the peak torque that is necessary to seat the nut [2] again.

(g) Tighten the nut [2] to 100 in-lb (11.3 N·m) - 150 in-lb (16.9 N·m) above the observed peak torque.

1) Make sure that the full chamfer of the threaded shaft is above the nut [2].

(h) Install the plug button [1] on the control wheel [7].

1) Apply three drops of adhesive, A50060, evenly spaced to the underside of the plug button [1].

2) Hold the plug button [1] in position with low adhesive masking tape.

SUBTASK 32-51-41-480-003

(2) Remove the rig pin from the control wheel drum.

I. Nose Landing Gear Control Wheel Installation Test

SUBTASK 32-51-41-820-001

(1) Adjust the steering system on the nose wheel (TASK 32-51-00-820-805).

NOTE: It is not necessary to do the adjustment and system test of the nose wheel steering system if no other part has been disturbed.

SUBTASK 32-51-41-820-002

(2) Do a test of the steering system on the nose wheel (TASK 32-51-00-700-801).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-41-410-001

(1) Install the aft access panel in the wheel well for the nose landing gear.

——— END OF TASK ———

TASK 32-51-41-000-802

4. Nose Landing Gear Tiller Assembly Removal

(Figure 402)

A. General

(1) This procedure gives instructions to remove the tiller assembly [13].

B. References

Reference	Title
25-11-22-000-801	Flight Compartment Forward Side Panel Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left

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(Continued)

Zone	Area
211	Flight Compartment - Left
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-51-41-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-41-010-001

(2) Remove the aft access panel from the wheel well for the nose landing gear to get access to the turnbuckles for the NWS-A cable and the NWS-B cable.

SUBTASK 32-51-41-020-002

(3) Remove the control wheel [7] (TASK 32-51-41-000-801).

SUBTASK 32-51-41-020-003

(4) Remove the sidewall [5] to get access to the tiller assembly [13] (TASK 25-11-22-000-801).

E. Nose Landing Gear Tiller Assembly Removal

SUBTASK 32-51-41-020-004

(1) Disconnect the NWS-A cable and the NWS-B cable from the drum [8]:

(a) Remove the nut [16] and washer [17].

(b) Loosen the tension at the turnbuckles for the NWS-A cable and the NWS-B cable.

(c) Remove the shaft [14], washer [18], and spacers [6].

(d) Remove the drum [8] from the housing assembly [3].

(e) Remove the pins [4] that hold the cable ends to the drum [8].

(f) Attach identification tags on the NWS-A cable and the NWS-B cable.

(g) Isolate the NWS-A cable and the NWS-B from the tiller assembly [13].

SUBTASK 32-51-41-020-005

(2) Remove the tiller assembly [13]:

(a) Remove the nuts [12] and washers [11] from the bolt [9] and bolts [15].

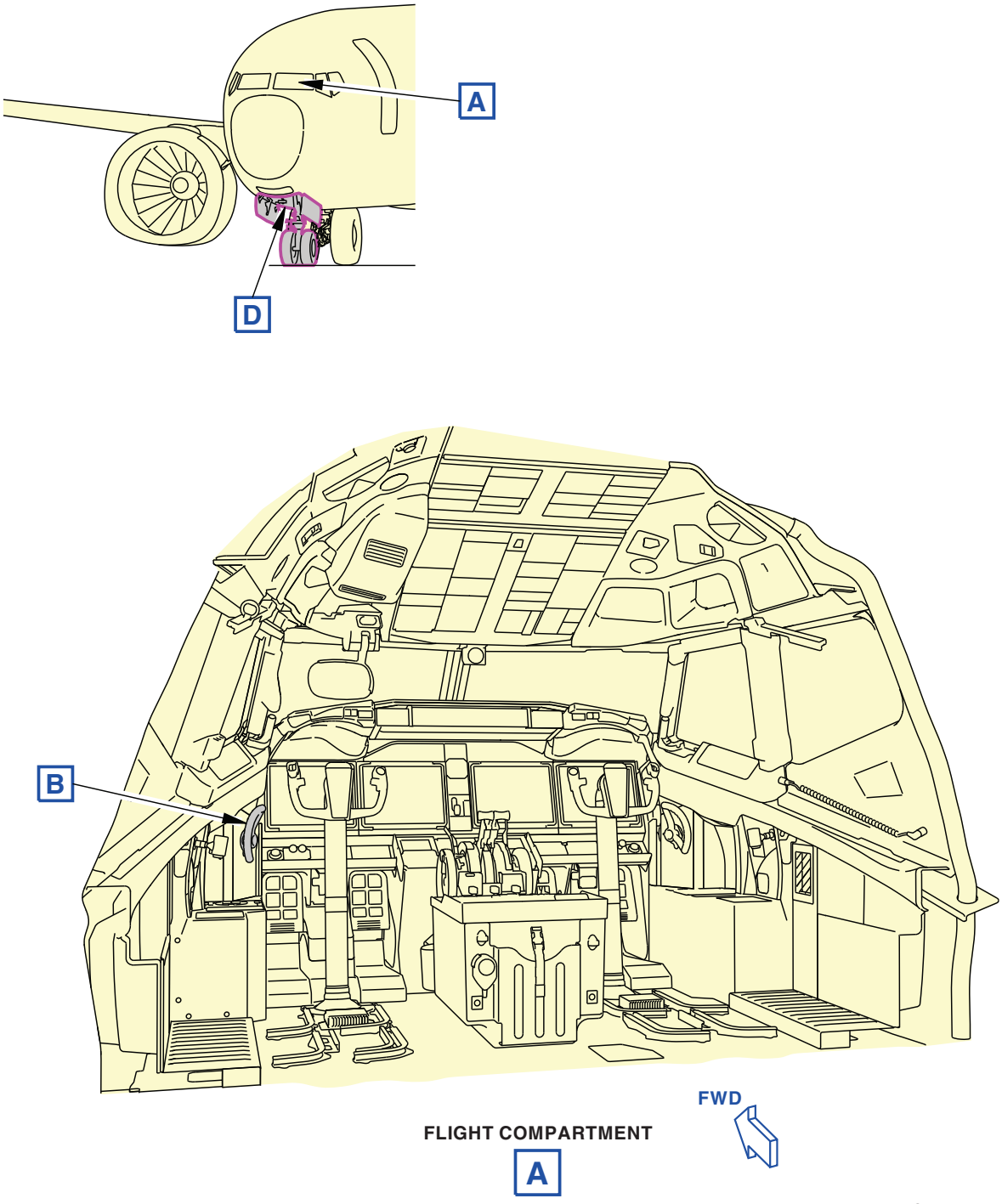
(b) Remove the bolt [9] and bolts [15] that attach the tiller assembly [13] to the mating structure.

(c) Remove the washers [10] from the bolt [9] and bolts [15].

(d) Remove the tiller assembly [13] from the airplane.

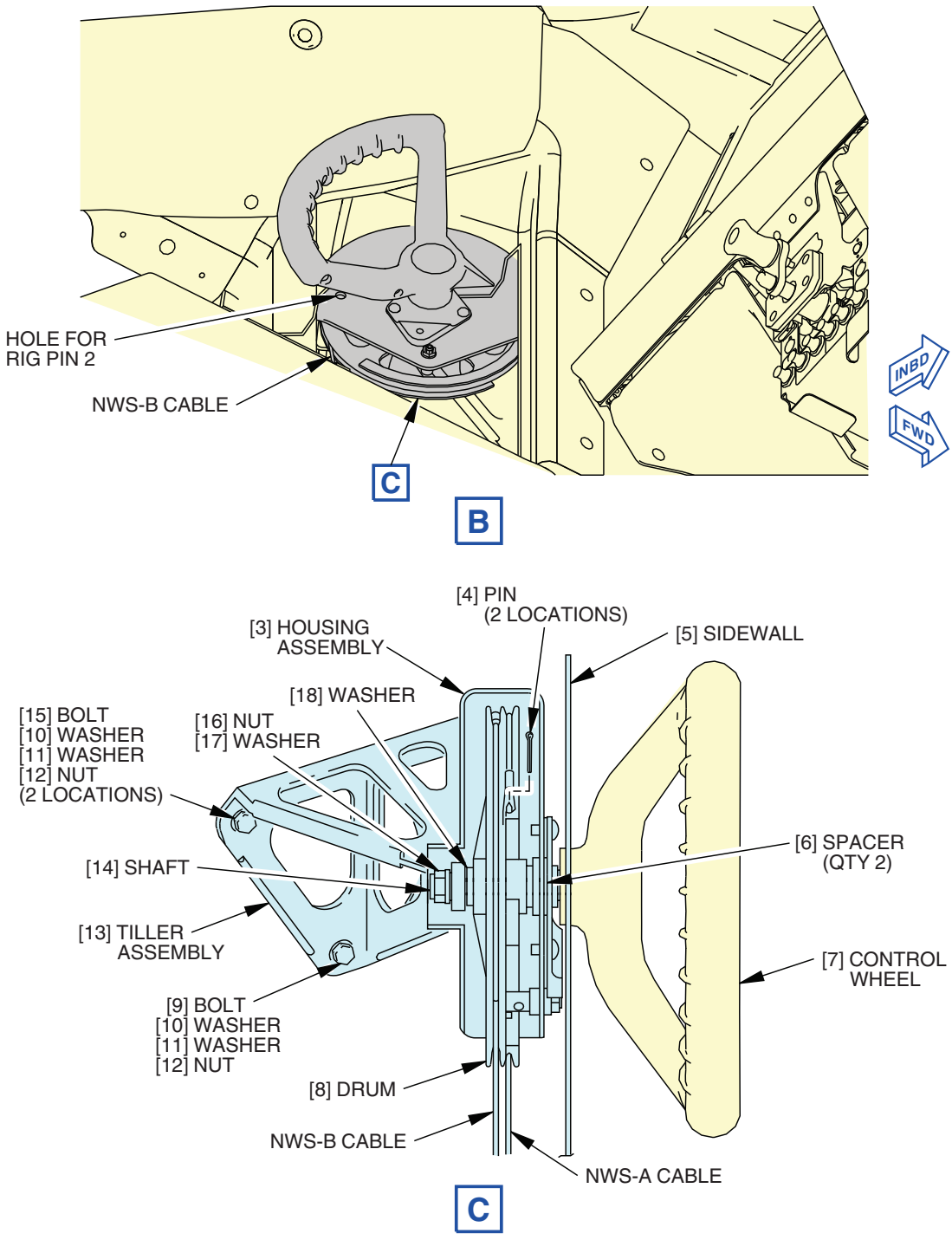
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



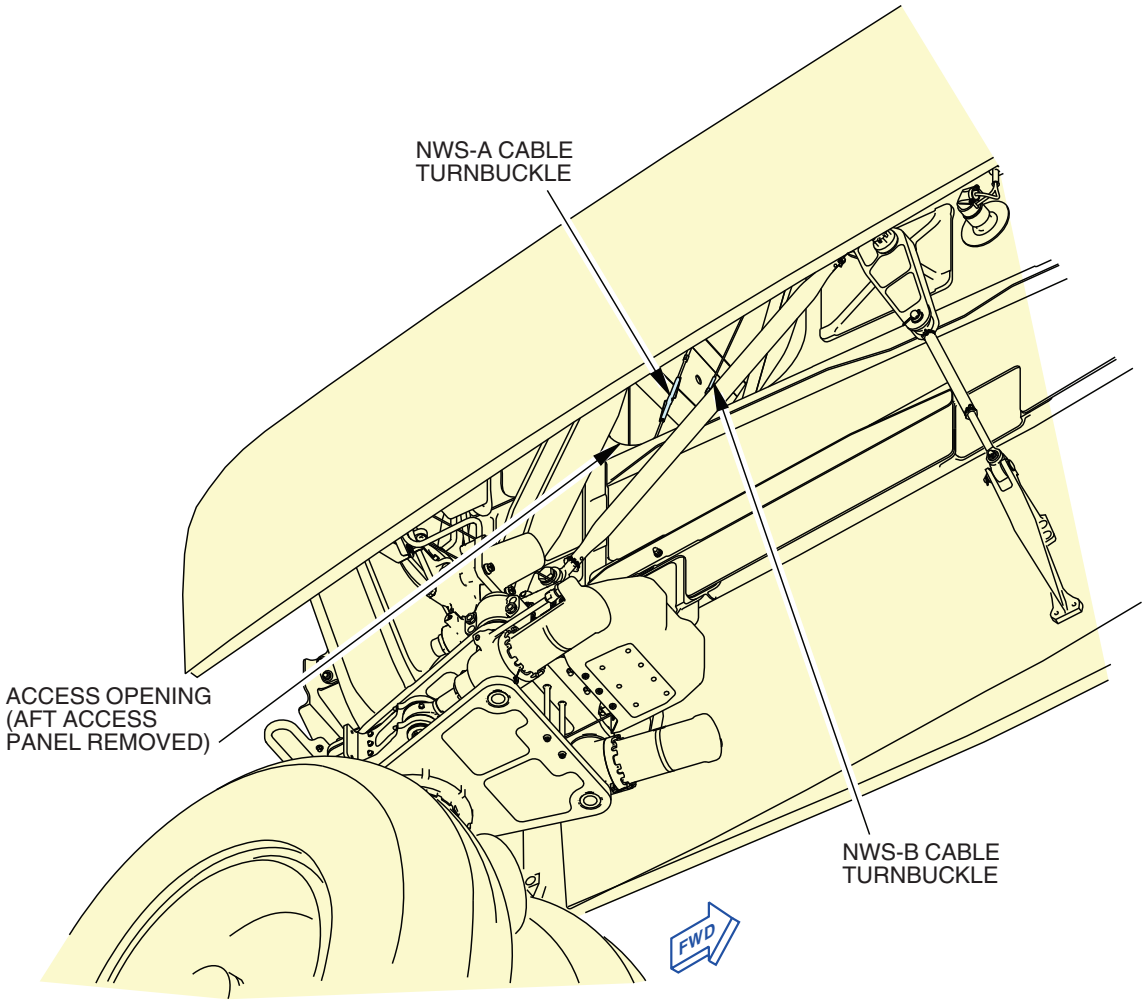
Nose Landing Gear Tiller Assembly Installation
Figure 402/32-51-41-990-802 (Sheet 1 of 3)

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Nose Landing Gear Tiller Assembly Installation
Figure 402/32-51-41-990-802 (Sheet 2 of 3)

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D

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Nose Landing Gear Tiller Assembly Installation
Figure 402/32-51-41-990-802 (Sheet 3 of 3)

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TASK 32-51-41-400-802

5. Nose Landing Gear Tiller Assembly Installation

(Figure 402)

A. General

(1) This procedure gives instructions to install the tiller assembly [13]y.

B. References

Reference	Title
25-11-22-400-801	Flight Compartment Forward Side Panel Installation (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00013	Grease - Aircraft And Instrument Grease	MIL-PRF-23827 (NATO G-354) (Supersedes MIL-G-23827)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
13	Tiller assembly	32-51-41-02-040	PTF ALL

E. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
211	Flight Compartment - Left
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Prepare for the Installation

SUBTASK 32-51-41-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

G. Nose Landing Gear Tiller Assembly Installation

SUBTASK 32-51-41-420-002

(1) Install the tiller assembly [13]:

(a) Put the tiller assembly [13] in its position for attachment to the mating structure.

(b) Align the holes in the tiller assembly [13] with the holes in the mating structure.

(c) Install the washers [10] on the bolt [9] and bolts [15].

(d) Put the inboard bolt [15] through the hole in the mating structure and the hole in the tiller assembly [13].

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(e) Put the outboard bolt [9] and bolt [15] through the holes in the tiller assembly [13] and the holes in the mating structure.

(f) Install the washers [11] and nuts [12] on the bolt [9] and bolts [15].

(g) Apply sealant, A00247, around the area of the bolt [9] and bolts [15] to seal them.

SUBTASK 32-51-41-420-003

(2) Connect the NWS-A cable and the NWS-B cable to the drum [8]:

(a) Put the cable fairleads of the NWS-A cable and the NWS-B cable in their position for attachment to the drum [8].

(b) Install the pins [4] to hold the cable ends to the drum [8].

(c) Remove the identification tags attached to the NWS-A cable and the NWS-B cable.

(d) Apply grease, D00013, to the bearings in the housing assembly [3].

(e) Install the drum [8] in the housing assembly [3].

(f) Install the shaft [14], spacers [6], and washer [18].

1) Install the shaft [14] with the missing tooth aligned in the drum [8].

(g) Install the washer [17] and nut [16].

(h) Tighten the nut [16].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-41-410-002

(1) Install the sidewall [5] (TASK 25-11-22-400-801).

SUBTASK 32-51-41-420-004

(2) Install the control wheel [7] (TASK 32-51-41-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR STEERING ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the nose gear steering actuator

(2) Installation of the nose gear steering actuator.

(3) Removal of the nose gear hydraulic swivel.

(4) Installation of the nose gear hydraulic swivel.

B. There are two identical nose gear steering actuators installed on the nose landing gear, just above the wheels.

C. The nose gear steering actuator weighs 22 lb (10 kg) when it is filled with hydraulic fluid.

TASK 32-51-51-000-801

2. Nose Gear Steering Actuator Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1901	Spanner Wrench - Steering Collar Retainer, NLG Part #: C32040-6 Supplier: 81205 Opt Part #: C32040-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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D. Prepare for the Removal

SUBTASK 32-51-51-860-001

WARNING

MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-51-860-002

(2) If the wheels of the nose landing gear are not in the centered position, do this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

SUBTASK 32-51-51-860-003

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-51-860-004

(4) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

SUBTASK 32-51-51-860-005

WARNING

ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Install the NLG towing lever pin, SPL-1499, through the towing lever.

NOTE: You use the NLG towing lever pin, SPL-1499, to hold the towing lever in the TOW position.

E. Nose Gear Steering Actuator Removal

(Figure 401)

SUBTASK 32-51-51-020-001

(1) Disconnect the torsion links at the apex joint (TASK 32-21-31-000-803).

SUBTASK 32-51-51-980-001

(2) Turn the steering collar far enough to permit access to the rod end of the actuator assembly [3] that you will remove.

SUBTASK 32-51-51-020-002

(3) Remove the cotter pin [6], nut [5], washer [4], bushings [2], anti-rotation washer [22] [22] (not on all airplanes), and pin [1] to disconnect the rod end of the actuator assembly [3] from the steering collar.

SUBTASK 32-51-51-020-003

(4) Remove the lockwire from the bolts [9] that attach the upper sleeve and the lower steering plate [18].

SUBTASK 32-51-51-020-004

(5) Remove the bolts [9], washers [7], and standoffs [8] that hold the two front feet of the metering valve assembly to the upper sleeve and lower steering plate [18].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-51-020-005

(6) Remove the nuts [13], bolts [11], washers [12], and clamps [10] that attach the trunnion of the actuator assembly [3] to the bushed hole in the upper steering sleeve and the steering valve swivel.

NOTE: The hydraulic ports will be open when you remove the trunnion of the actuator assembly [3] from the steering valve swivel.

(a) Make sure the area around trunnion is clean before the hydraulic ports are open to atmosphere, to prevent contamination.

SUBTASK 32-51-51-020-006

(7) Remove the lockwire from the bolts [14].

SUBTASK 32-51-51-020-007

(8) Remove the bolts [14], washers [15], and lock tab [16] from the steering collar retainer nut [17].

SUBTASK 32-51-51-800-001

(9) Hold thelower steering plate [18] in its position while you loosen the steering collar retainer nut [17] with a spanner wrench, SPL-1901, and lower it down the outer cylinder.

SUBTASK 32-51-51-010-001

(10) Hold the actuator assembly [3] in its position and lower the lower steering plate [18] down the outer cylinder.

SUBTASK 32-51-51-020-008

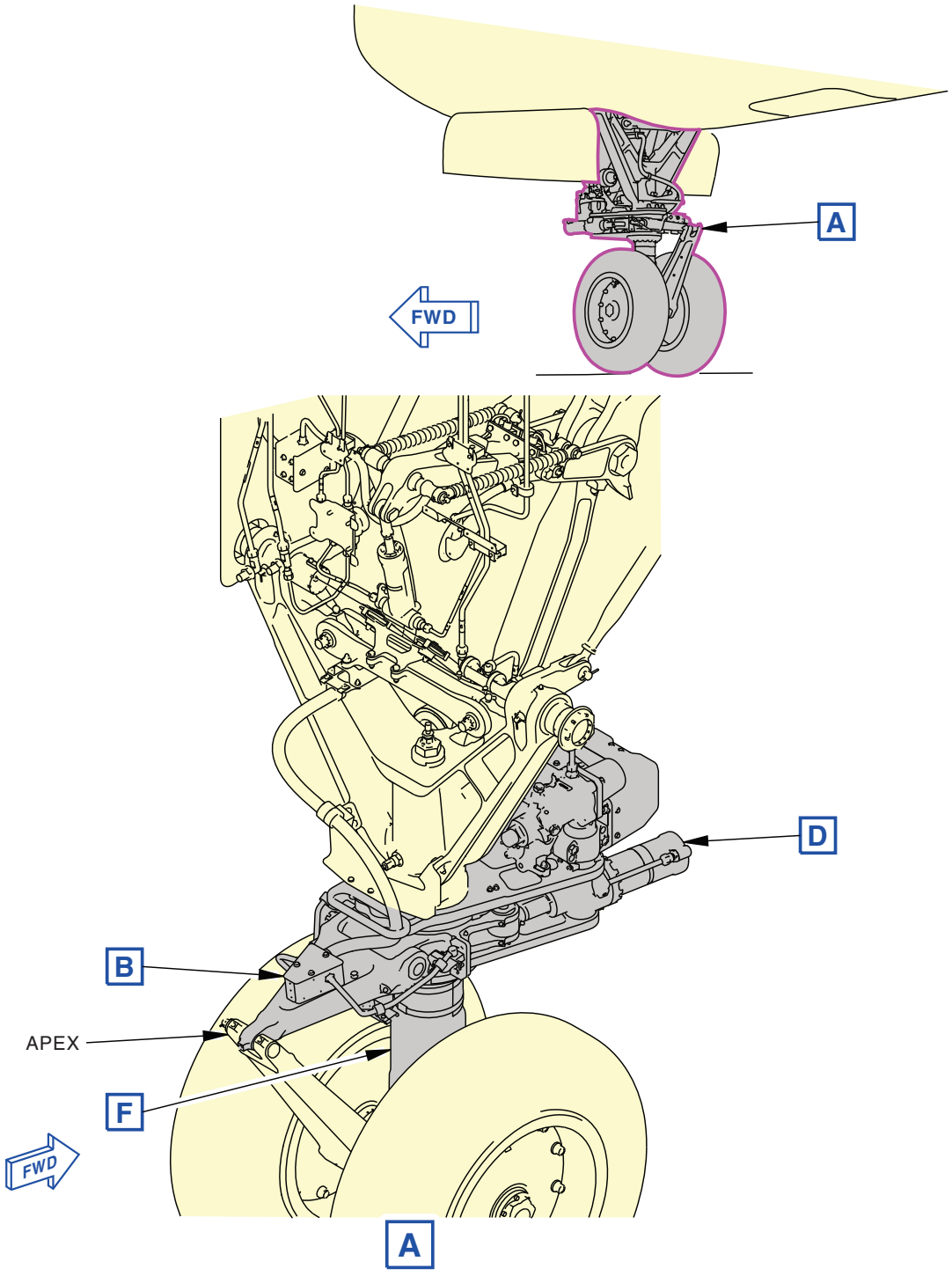
(11) Remove the actuator assembly [3] from the cavity in the steering valve swivel and the bushed hole in the upper steering sleeve.

SUBTASK 32-51-51-480-001

(12) Install caps/plugs, STD-14686, in the open hydraulic ports of the steering valve swivel and the trunnion of the actuator assembly [3].

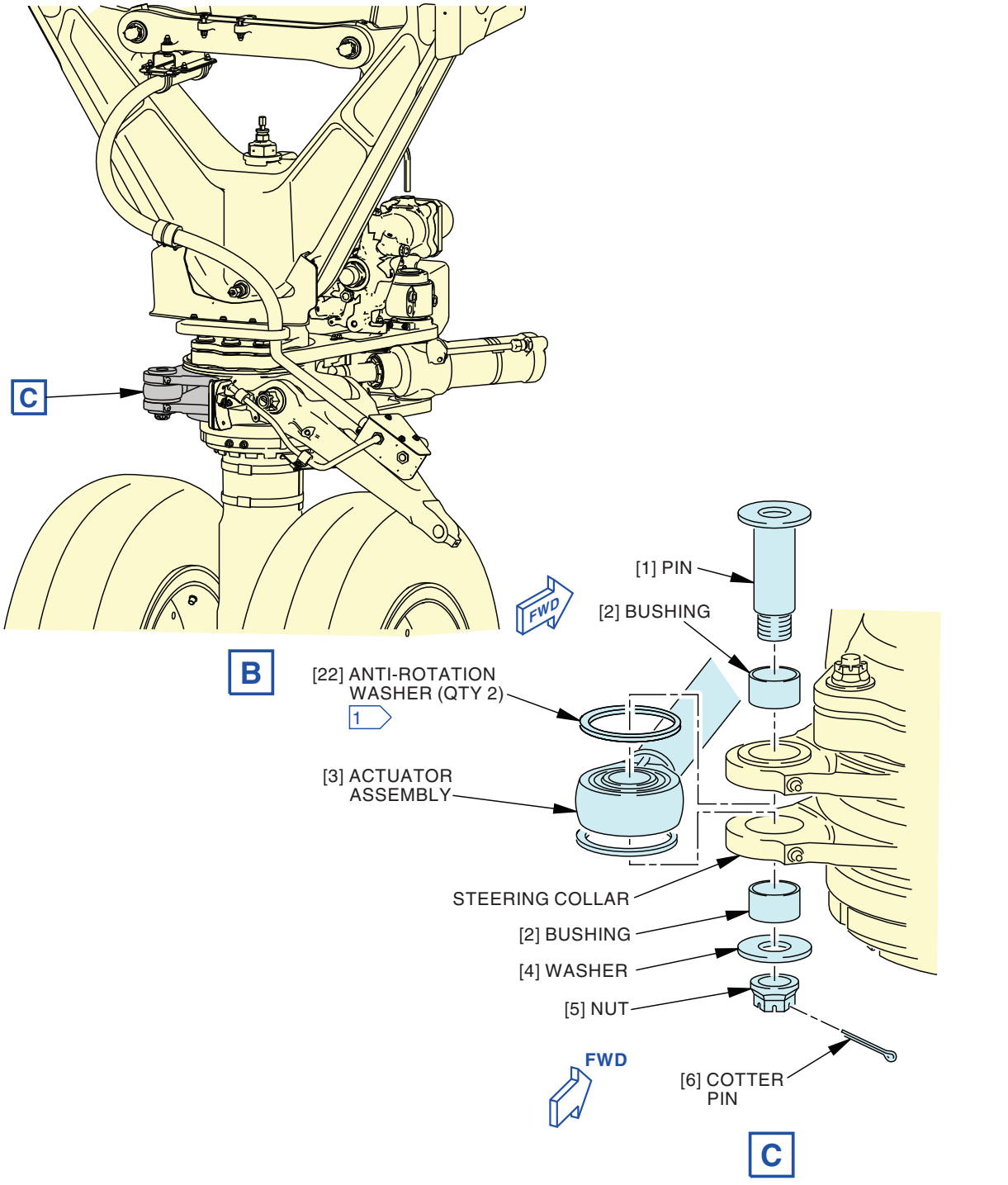
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



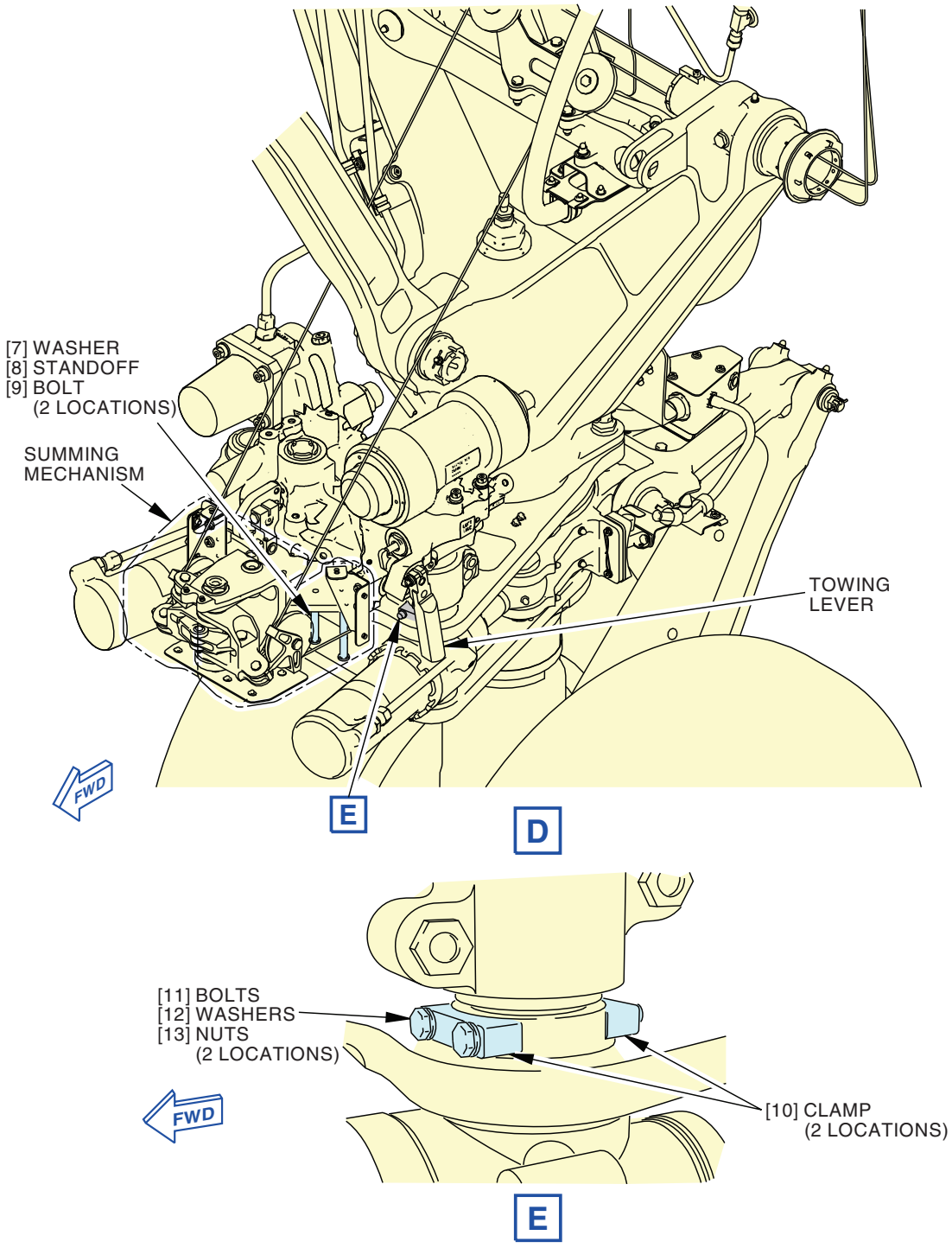
Nose Gear Steering Actuator Installation
Figure 401/32-51-51-990-801 (Sheet 1 of 5)

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Nose Gear Steering Actuator Installation
Figure 401/32-51-51-990-801 (Sheet 2 of 5)

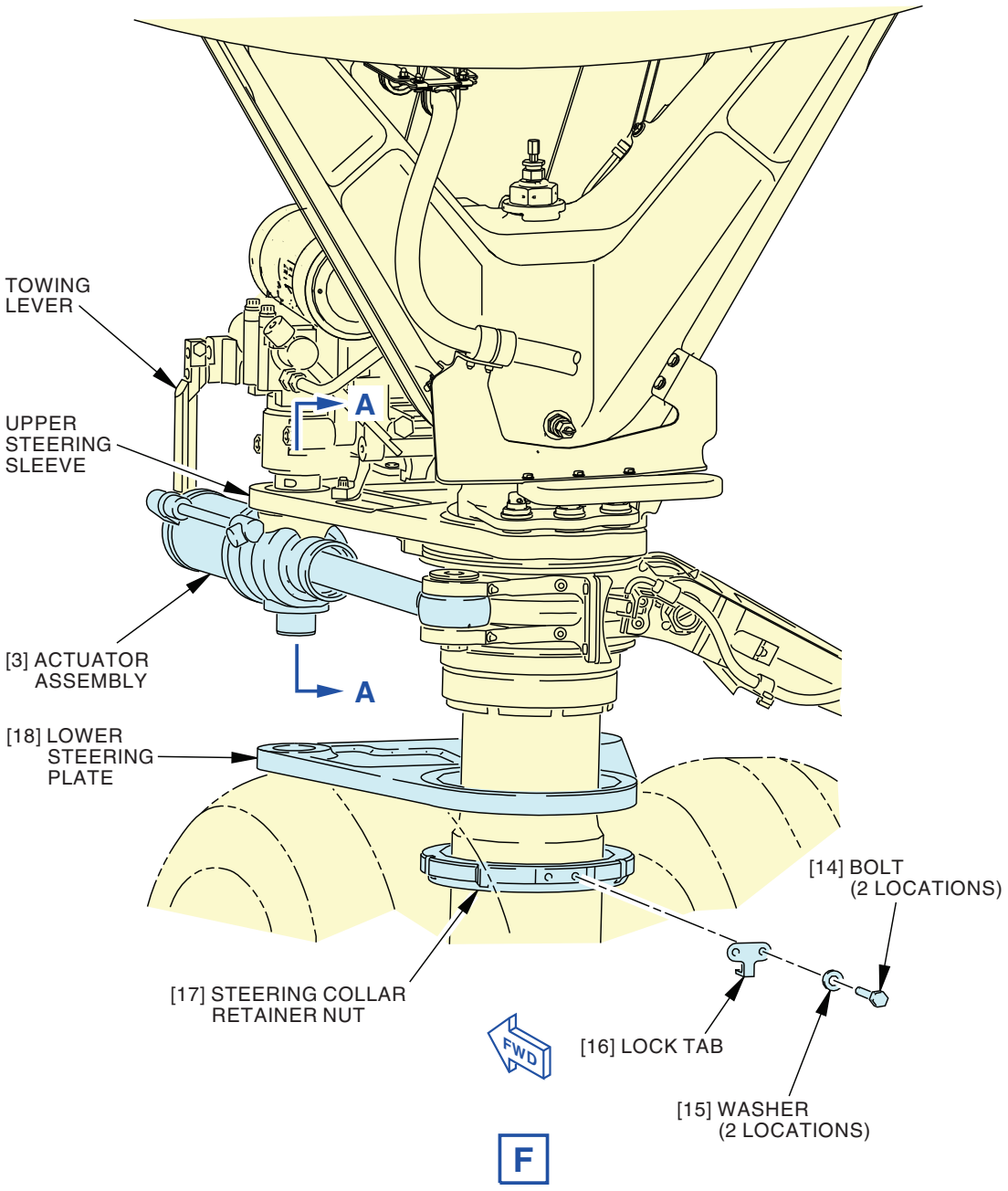
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Nose Gear Steering Actuator Installation
Figure 401/32-51-51-990-801 (Sheet 3 of 5)

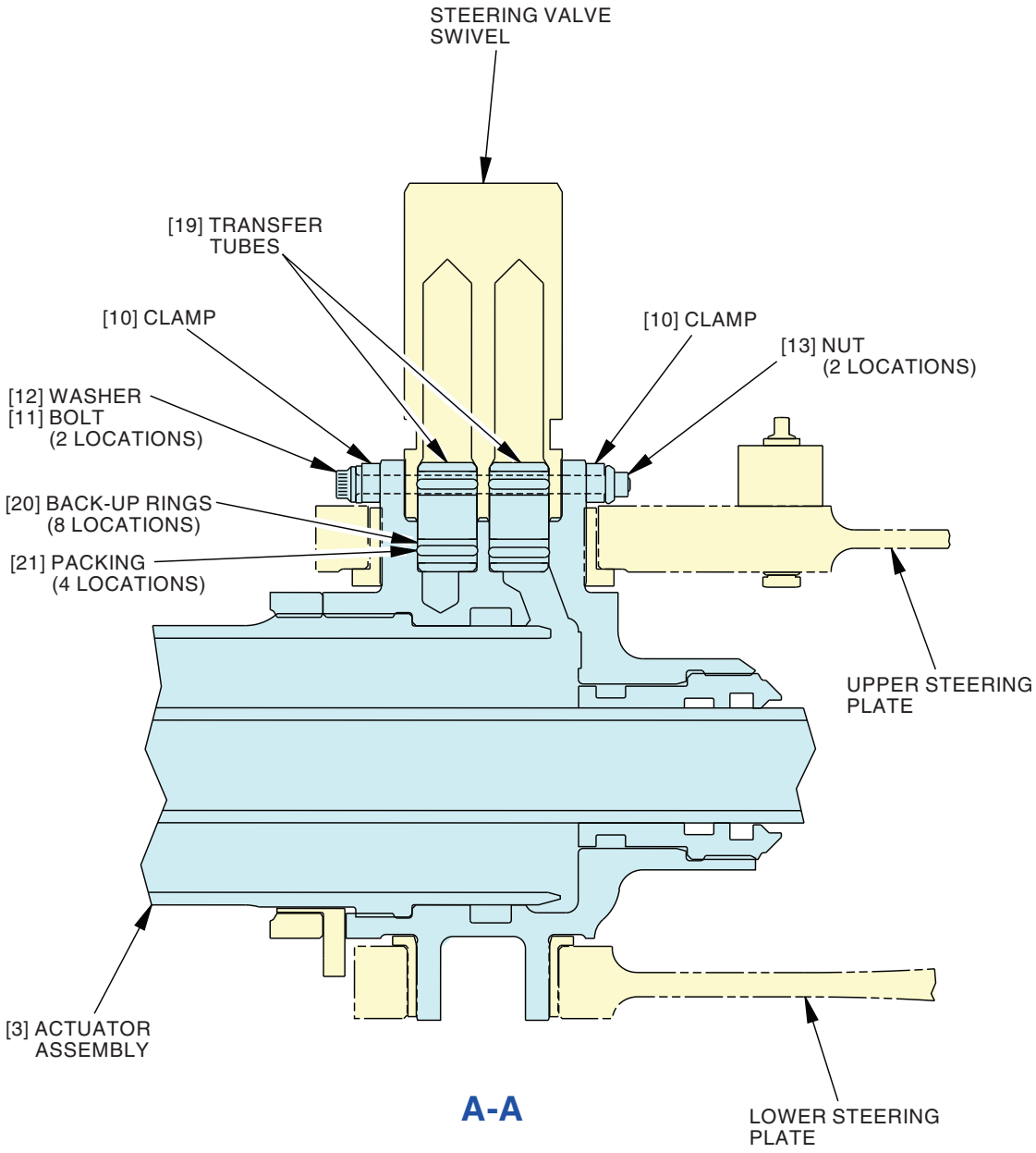
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Nose Gear Steering Actuator Installation
Figure 401/32-51-51-990-801 (Sheet 4 of 5)

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Nose Gear Steering Actuator Installation
Figure 401/32-51-51-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-51-400-801

3. Nose Gear Steering Actuator Installation

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1901	Spanner Wrench - Steering Collar Retainer, NLG Part #: C32040-6 Supplier: 81205 Opt Part #: C32040-1 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Actuator assembly	32-51-51-01-005	PTF ALL
6	Cotter pin	32-51-51-01-215	PTF ALL
21	Packing	32-51-51-01-175	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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F. Prepare for Installation

(Figure 401)

SUBTASK 32-51-51-860-006

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-51-080-001

(2) Remove the caps/plugs, STD-14686, from the open hydraulic ports of the steering valve swivel and the trunnion of the actuator assembly [3].

(a) Make sure the area around the trunnion and the steering valve is clean.

NOTE: When the hydraulic ports are open you can contaminate the trunnion of the actuator assembly [3] and the cavity on the steering valve swivel.

SUBTASK 32-51-51-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Do these steps to prepare the actuator assembly [3] for installation:

(a) Fill the actuator assembly [3] with hydraulic fluid, D00153, until it is full.

(b) Apply hydraulic fluid, D00153, to the transfer tubes [19], back-up rings [20], and new packings [21] in the trunnion of the actuator assembly [3].

(c) Apply a thin layer of Cor-Ban 27L Compound, G50237 (preferred), or corrosion inhibiting material, G50136 (alternate), to the threads, thread reliefs, and mating surfaces of the steering collar retainer nut [17].

(d) Apply a thin layer of Cor-Ban 27L Compound, G50237 (preferred), or corrosion inhibiting material, G50136 (alternate), to the threads, thread reliefs, and mating surfaces of the nut [5], nuts [13], bushings [2], pin [1], bolts [9], bolts [11], bolts [14], and standoffs [8].

G. Nose Gear Steering Actuator Installation

(Figure 401)

SUBTASK 32-51-51-420-001

(1) Put the trunnion of the actuator assembly [3] into the bushed hole in the upper steering sleeve and the cavity of the steering valve swivel.

NOTE: The left hand actuator should be installed on the left side and the right hand actuator should be installed on the right side. (See Figure 401).

(a) Be careful to align the transfer tubes [19] as you lift the actuator assembly [3] into position.

(b) Hold the actuator assembly [3] after the trunnion shoulder is in full contact with the flange of the bushing in the upper steering sleeve.

SUBTASK 32-51-51-410-001

(2) Hold the actuator assembly [3] in its position and raise the lower steering plate [18] up the outer cylinder.

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(a) Put the lower trunnion of the actuator assembly [3] into the hole in the lower steering plate [18].

SUBTASK 32-51-51-420-002

(3) Hold the lower steering plate [18] in its position and move the steering collar retainer nut [17] up the outer cylinder to install it on the threads below the lower steering plate [18].

(a) Tighten the steering collar retainer nut [17] to 62.5 ±12.5 ft-lb (84.7 ±17.0 N·m) with a spanner wrench, SPL-1901.

(b) If it is necessary, loosen the steering collar retainer nut [17] a minimum amount to install the lock tab [16], washers [15], and bolts [14].

(c) Install lockwire on the bolt [14].

SUBTASK 32-51-51-420-003

(4) Install the clamps [10], bolts [11], washers [12], and nuts [13] that attach the trunnion of the actuator assembly [3] to the steering valve swivel.

(a) If it is installed, remove the lockwire from the swivel valves.

(b) Tighten the bolts [11] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m).

SUBTASK 32-51-51-020-009

(5) Install the standoffs [8], bolts [9], and washers [7] that hold the upper sleeve and the lower steering plate [18].

(a) Install lockwire on the bolt [9].

SUBTASK 32-51-51-980-002

(6) Turn the steering collar far enough to permit access to the rod end of the actuator assembly [3] that you will install.

SUBTASK 32-51-51-420-004

(7) Install the anti-rotation washers [22], bushings [2], pin [1], washer [4], and nut [5] to connect the rod end of the actuator assembly [3] to the steering collar.

(a) Tighten the nut [5] to 54 ±4 ft-lb (73 ±6 N·m).

(b) If it is necessary, loosen the nut [5] to the nearest castellation to align the hole in the pin [1].

(c) Install the cotter pin [6].

1) Make sure the ends of the new cotter pin [6] are not closer than 0.10 in. (2.54 mm) to the steering collar when they are bent.

SUBTASK 32-51-51-420-005

(8) Connect the torsion links at the apex join (TASK 32-21-31-400-803).

SUBTASK 32-51-51-860-007

(9) Remove the NLG towing lever pin, SPL-1499, from the towing lever.

H. Nose Gear Steering Actuator Installation Test

SUBTASK 32-51-51-710-001

(1) To do a test on the steering system of the nose wheel, do the parts of this task applicable to the normal steering response test: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801

(a) Examine for leaks.

(b) Make sure that there is no leakage.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-51-000-802

4. Nose Gear Hydraulic Swivel Removal

(Figure 402)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. General

SUBTASK 32-51-51-000-001

(1) This task contains steps to replace either the entire swivel assembly or the individual swivels from the assembly.

D. Prepare for Removal

SUBTASK 32-51-51-940-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Nose Gear Hydraulic Swivel Removal (Individual Swivels)

SUBTASK 32-51-51-020-010

(1) Remove the hydraulic lines [33] from the swivel [30].

(2) Remove the band clamps [34] from the swivel [30].

(3) Remove the swivel [32] from the swivel assembly.

F. Nose Gear Hydraulic Swivel (Assembly) Removal

SUBTASK 32-51-51-020-011

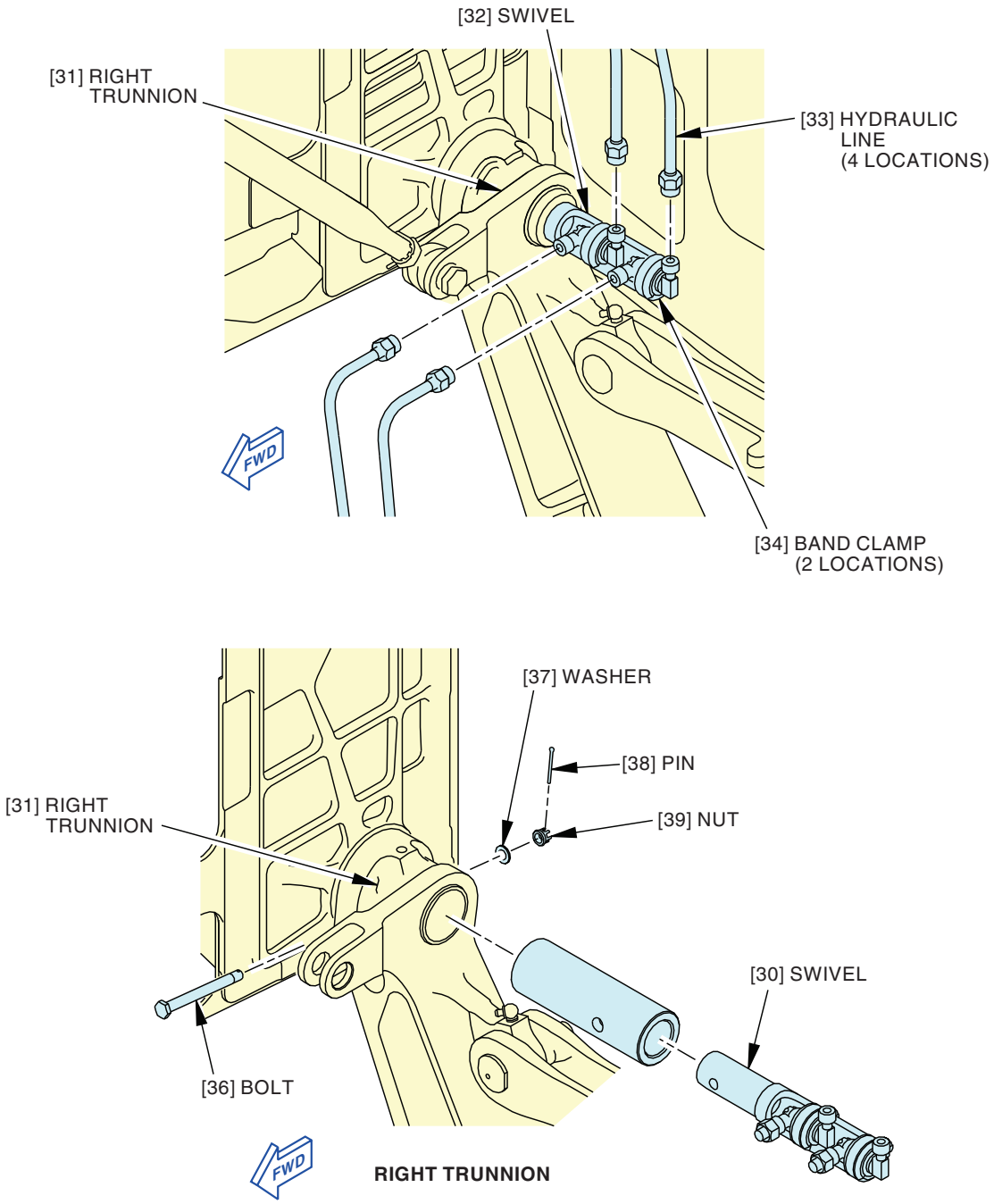
(1) Remove the hydraulic lines [33] from the swivel [30].

(2) Remove the pin [38], nut [39], washer [37], and bolt [36] from the right trunnion [31].

(3) Remove the swivel [30].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Hydraulic Swivel
Figure 402/32-51-51-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-51-400-802

5. Nose Gear Hydraulic Swivel Installation

(Figure 402)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D50180	Grease - Aircraft General Purpose (AeroShell Grease 33)	BMS3-33, MIL-PRF-23827
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
30	Swivel	32-51-51-02-020	PTF ALL

D. Location Zones

Zone	Area
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Nose Gear Hydraulic Swivel (Assembly) Installation.

SUBTASK 32-51-51-640-001

(1) Lubricate the inner diameter of the right trunnion [31] pin and the outer diameter of the swivel [30] bracket with AeroShell Grease 33, D50180.

SUBTASK 32-51-51-420-006

(2) Put the swivel [30] bracket in the right trunnion [31] pin and align the hole for the bolt [36].

SUBTASK 32-51-51-420-007

(3) Install the bolt [36].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin coat of corrosion inhibiting material, G50136, to the shank and threads of the bolt [36].

(b) Install the bolt [36], washer [37], and nut [39].

(c) Tighten the nut [39] to 20 in-lb (2.3 N·m)-24 in-lb (2.7 N·m) above run-on torque and loosen to align the nearest castellation.

(d) Install the pin [38].

SUBTASK 32-51-51-420-008

(4) Connect the hydraulic lines [33].

(a) Tighten the B-nuts on the hydraulic lines [33] to 270 ±14 in-lb (31 ±2 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Nose Gear Hydraulic Swivel Installation

SUBTASK 32-51-51-420-009

(1) Install swivel [32] in the swivel assembly.

(a) Put the swivel [32] in the swivel assembly.

(b) Install the band clamp [34].

(2) Connect the hydraulic lines [33].

G. Nose Gear Hydraulic Swivel Installation Test

SUBTASK 32-51-51-710-002

(1) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(a) Apply the hydraulic pressure for two minutes and make sure there are no leaks.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-51-040-001

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805, if hydraulic power is not necessary.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR STEERING ACTUATOR - INSPECTION/CHECK

1. General

A. This procedure has this tasks:

(1) Nose Gear Steering Actuator Rod End Detailed Inspection

B. There are two identical steering actuators installed on the nose landing gear, just above the wheels.

TASK 32-51-51-200-801

2. Nose Gear Steering Actuator Rod End Detailed Inspection

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-31-000-803	Nose Landing Gear Torsion Link Disconnection (P/B 401)
32-21-31-400-803	Nose Landing Gear Torsion Link Connection (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Detailed Inspection

SUBTASK 32-51-51-860-008

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

(a) Make sure the wheels of the nose landing gear are in the centered position.

(b) Depressurize hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(c) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(d) Install the NLG towing lever pin, SPL-1499 through the towing lever.

NOTE: You use the NLG Towing Lever Lock Pin to hold the towing lever in the TOW position.

E. Actuator Rod End Detailed Inspection

SUBTASK 32-51-51-020-012

(1) If necessary, disconnect the torsion links at the apex joint (TASK 32-21-31-000-803).

SUBTASK 32-51-51-980-003

(2) Turn the steering collar far enough to permit visual access to the rod end that you will inspect.

SUBTASK 32-51-51-020-013

(3) Do an external, non-disassembled, detailed inspection of the nut, washer, bushings, and pin that attach the rod end to the steering collar.

(a) Make sure the rod end is secure and that there is no damage.

(b) Make sure there is no excessive or unusual movement in the rod end bushings and the rod end pin.

SUBTASK 32-51-51-420-010

(4) If necessary, connect the torsion links at the apex joint (TASK 32-21-31-400-803).

SUBTASK 32-51-51-860-009

(5) Remove the NLG towing lever pin, SPL-1499 from the towing lever.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR STEERING SUMMING MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has four tasks:

(1) Removal of the summing mechanism for the steering system for the nose landing gear (without using cable clamp assembly).

(2) Installation of the summing mechanism for the steering system for the nose landing gear (without using cable clamp assembly).

(3) Removal of the summing mechanism for the steering system for the nose landing gear (using cable clamp assembly).

(4) Installation of the summing mechanism for the steering system for the nose landing gear (using cable clamp assembly).

TASK 32-51-52-020-801

2. Summing Mechanism Removal (Without Using Cable Clamp Assembly)

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-51-00-820-801	General Check of the Nose Wheel Steering System Adjustment (P/B 501)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 32-51-52-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-52-860-001

(2) Make sure that the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-52-010-001

(3) Remove the aft access panel from the left sidewall of the nose wheel well to get access to the turnbuckles [3] for the NWS-A cable [1] and NWS-B cable [2] (TASK 53-14-01-020-801).

(a) Open these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-52-480-002

(4) Install these rig pins from the rigging pin kit, SPL-1585 (TASK 32-51-00-820-801):

(a) The rig pin in the Captain's steering tiller.

(b) The rig pin in the rudder pedal interconnect quadrant.

SUBTASK 32-51-52-860-002

(5) Remove the pressure from hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-51-52-860-003

(6) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

SUBTASK 32-51-52-860-004

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Install the NLG towing lever pin, SPL-1499, through the towing lever.

NOTE: You use the NLG Towing Lever Lock Pin to hold the towing lever in the TOW position.

F. Steering Summing Mechanism Removal

SUBTASK 32-51-52-010-002

(1) Remove the screws [5], washers [6], and washers [7] that hold the summing mechanism cover [10] to the support bracket.

SUBTASK 32-51-52-020-001

(2) Remove the clips [4] from the turnbuckles [3].

SUBTASK 32-51-52-020-002

(3) Loosen the NWS-A cable [1] and NWS-B cable [2] at the turnbuckles [3].

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SUBTASK 32-51-52-020-003

(4) Remove the nuts [24], bolts [21], washers [22], bushings [23], and pulley [20] from the pulley brackets on the summing mechanism assembly [11].

SUBTASK 32-51-52-020-004

(5) Remove the nut [15], washers [14], bushing [13], and bolt [12] to disconnect the link assembly [16] from the summing mechanism assembly [11].

SUBTASK 32-51-52-020-005

(6) Remove the nuts [19], washers [28], bolts [17], and washers [18] to disconnect the summing mechanism assembly [11] from the bracket assembly.

SUBTASK 32-51-52-000-001

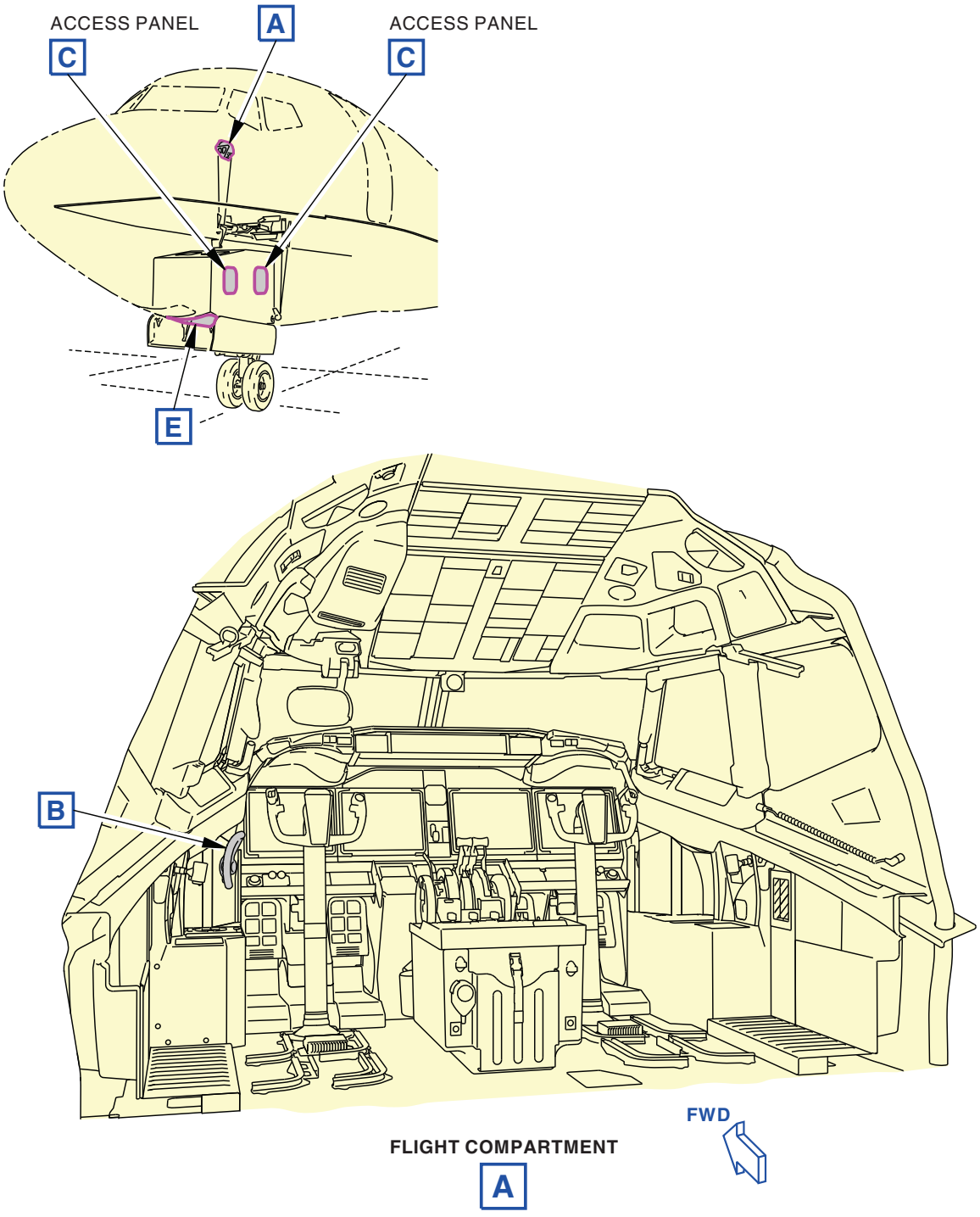
(7) Remove the summing mechanism assembly [11].

SUBTASK 32-51-52-020-006

(8) If it is necessary to remove the pulley bracket assembly from the summing mechanism assembly [11], remove the nuts [27], bushings [26], bolt [25], and washers [14].

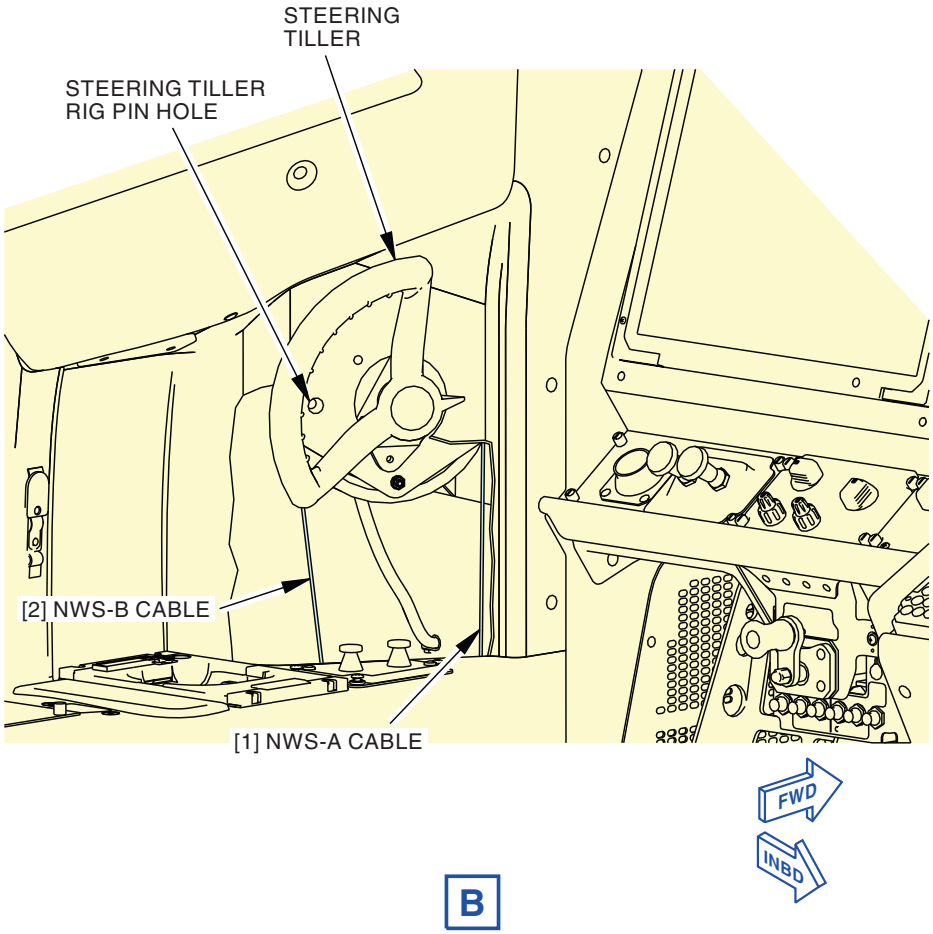
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Steering Summing Mechanism Installation
Figure 401/32-51-52-990-801 (Sheet 1 of 6)

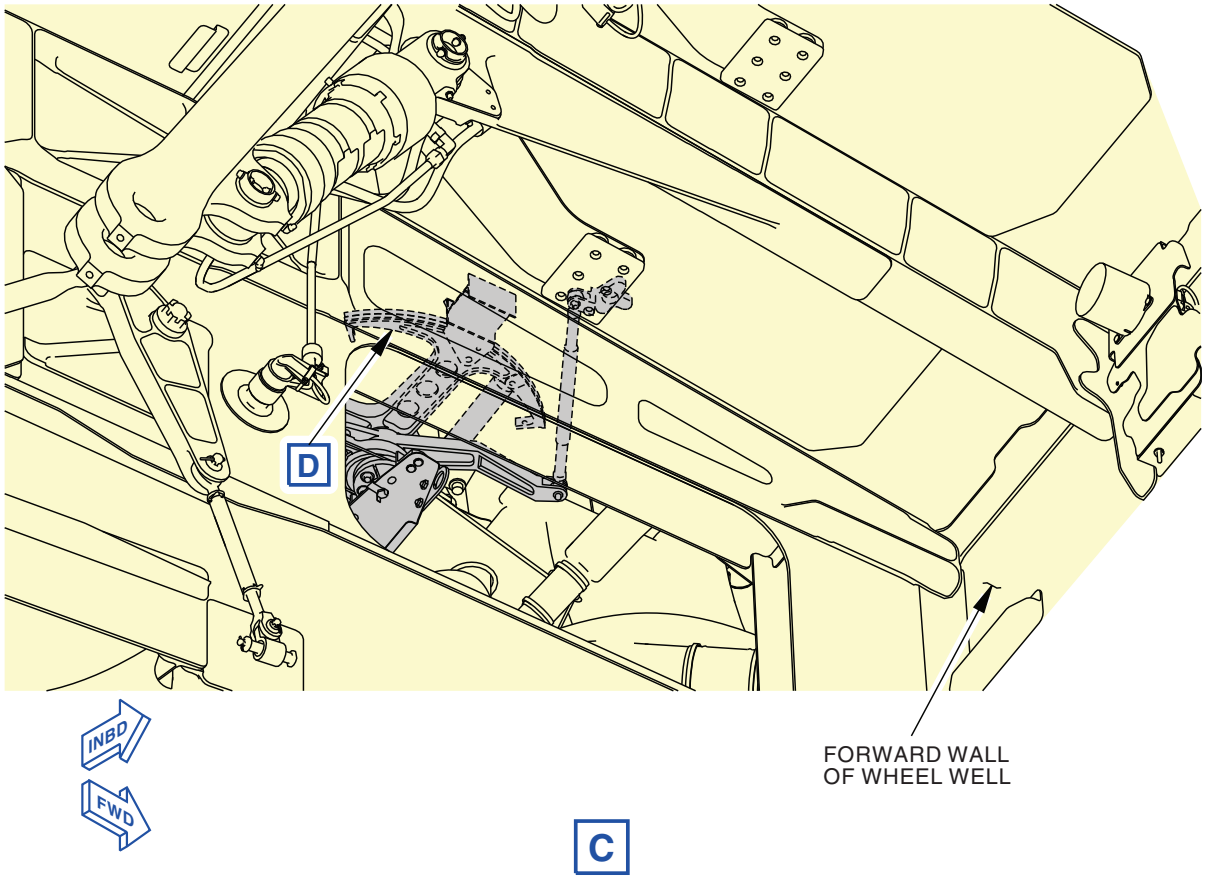
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Landing Gear Steering Summing Mechanism Installation
Figure 401/32-51-52-990-801 (Sheet 2 of 6)

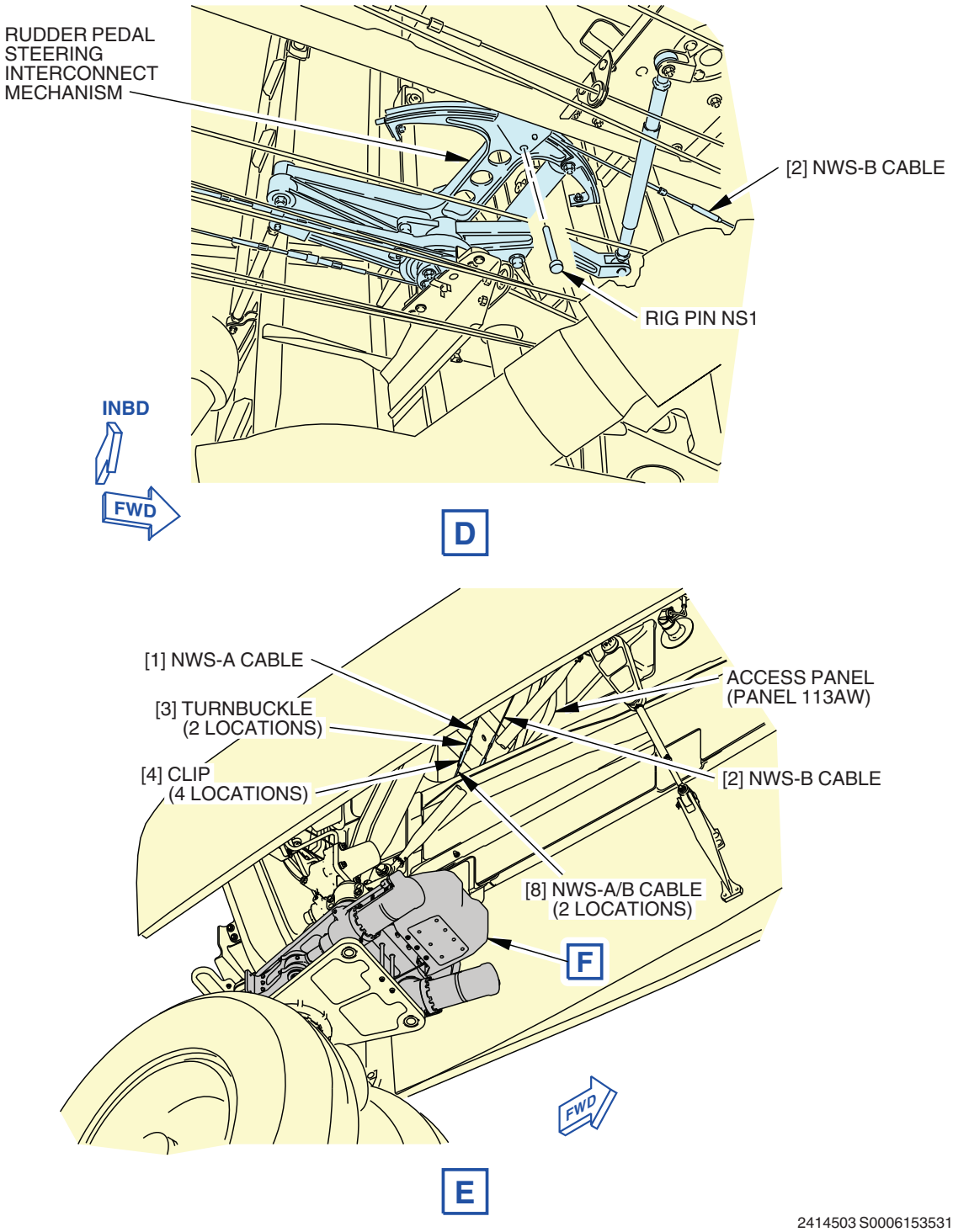
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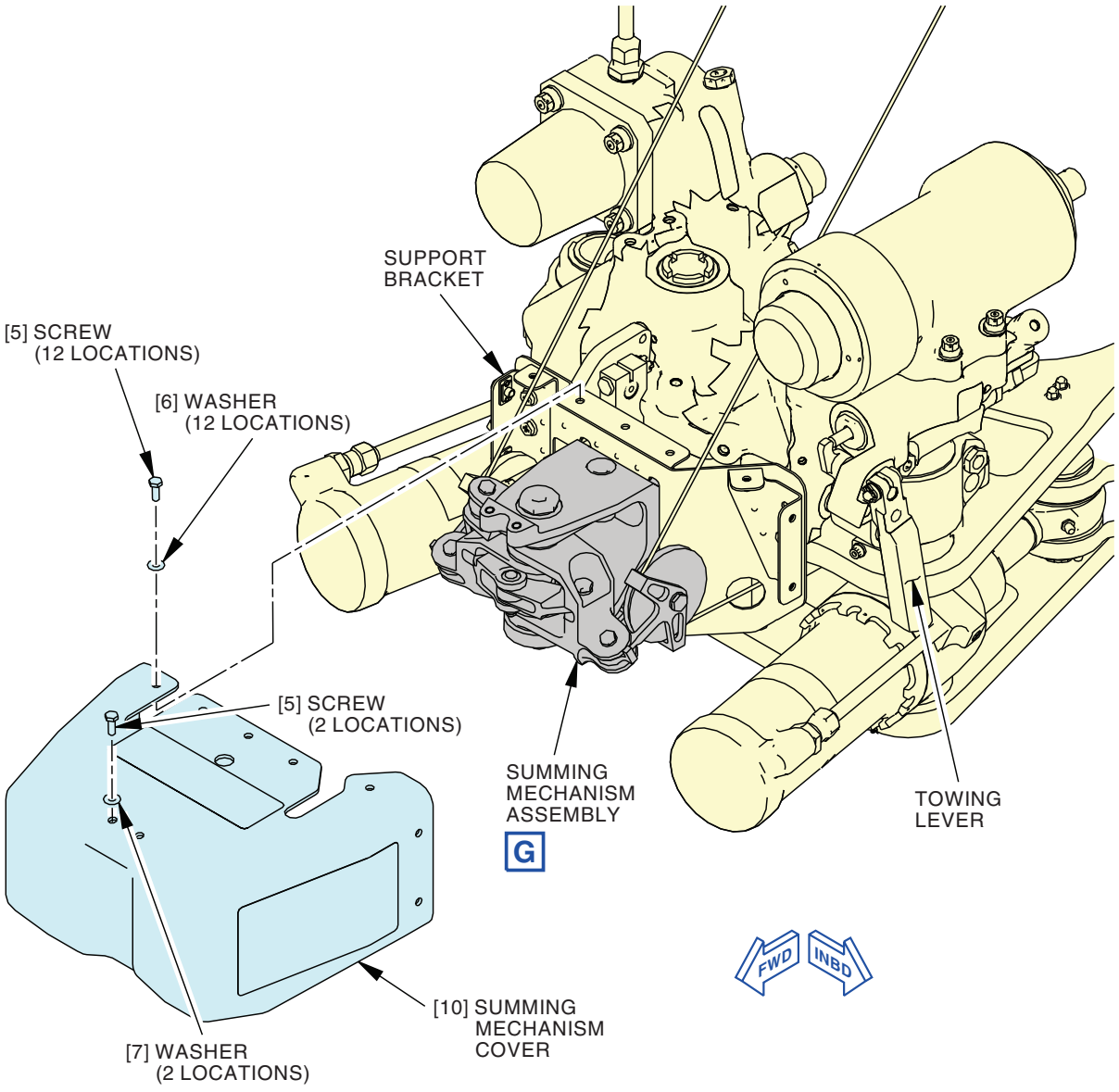
Nose Landing Gear Steering Summing Mechanism Installation
Figure 401/32-51-52-990-801 (Sheet 3 of 6)

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Nose Landing Gear Steering Summing Mechanism Installation
Figure 401/32-51-52-990-801 (Sheet 4 of 6)

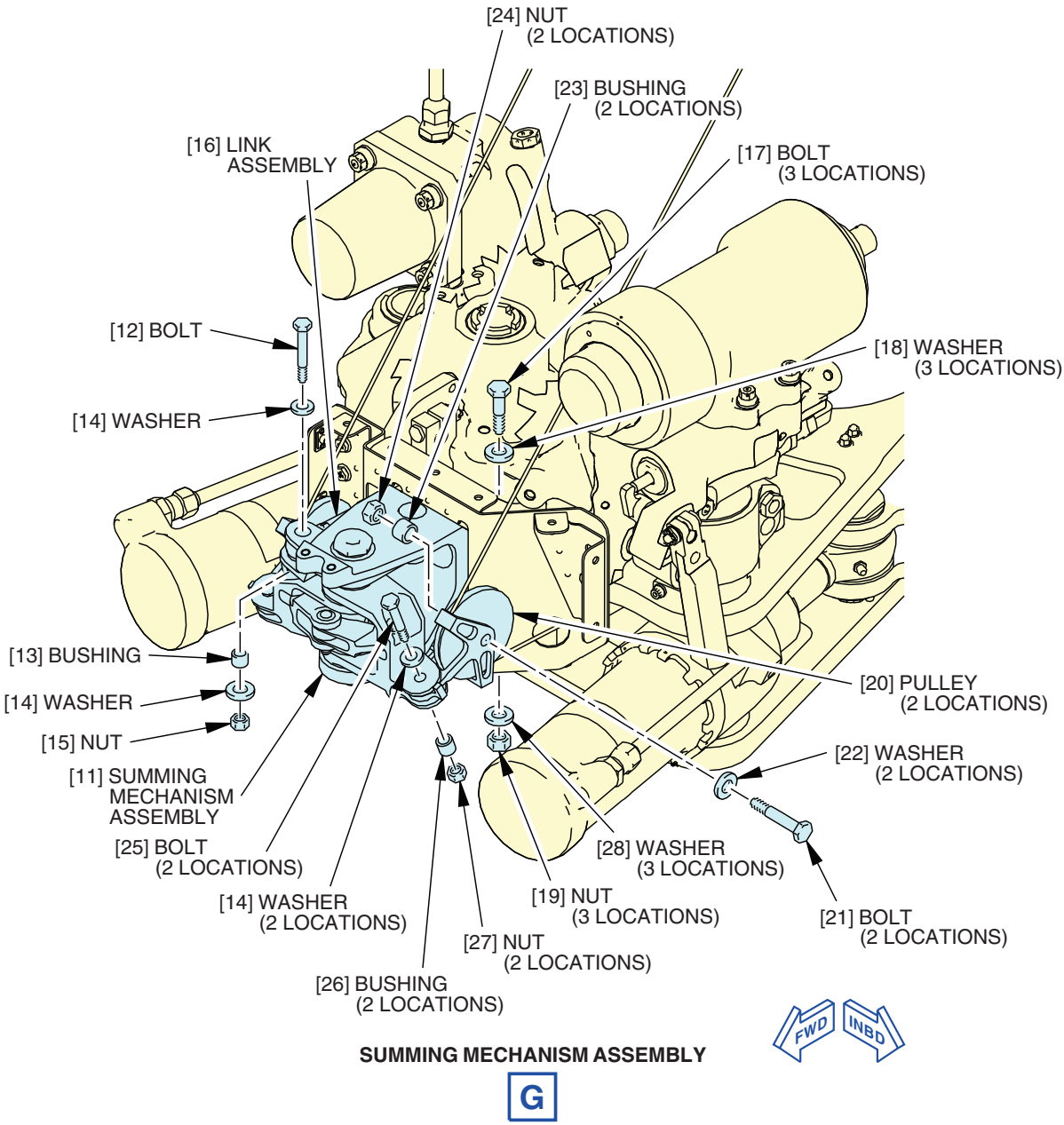
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Nose Landing Gear Steering Summing Mechanism Installation
Figure 401/32-51-52-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Landing Gear Steering Summing Mechanism Installation
Figure 401/32-51-52-990-801 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-52-420-801

3. Summing Mechanism Installation (Without Using Cable Clamp Assembly)

(Figure 401)

A. References

Reference	Title
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-00-820-805	Nose Wheel Steering System Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Summing mechanism assembly	32-51-52-01-175	PTF ALL

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

F. Summing Mechanism Installation

SUBTASK 32-51-52-210-001

(1) Make sure that the wheels of the nose landing gear are in the centered position (Figure 401).

NOTE: The graduated scales on the steering collar and arrows on the lower plate show the angular position of the wheels when the torsion links are connected.

SUBTASK 32-51-52-400-001

(2) Attach the summing mechanism assembly [11] to the bracket assembly:

(a) Align the summing mechanism assembly [11] with the steering plate.

(b) Align the link assembly [16] with the summing mechanism assembly [11].

(c) Install the bolts [17], washers [18], washers [28], and nuts [19] to connect the summing mechanism assembly [11] to the bracket assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-52-420-001

(3) If it is necessary, install the pulley bracket assembly:

(a) Put the pulley bracket assembly in its location.

(b) Install the bushings [26], washers [14], bolts [25], and nuts [27].

(c) Tighten the bolts [25] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m) above the run-on torque.

SUBTASK 32-51-52-420-002

(4) Install the pulleys [20] on the pulley bracket assembly:

(a) Thread the cable over the pulley [20].

(b) Install the bushings [23].

(c) Align the pulleys [20] with the pulley bracket assembly.

(d) Install the bolts [21], washers [22], and nuts [24].

(e) Tighten the bolts [21] to 50 in-lb (5.6 N·m) - 75 in-lb (8.5 N·m) above the run-on torque.

SUBTASK 32-51-52-480-003

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Make sure that the NLG towing lever pin, SPL-1499, is installed through the towing lever.

SUBTASK 32-51-52-480-004

(6) Make sure that these rig pins from the rigging pin kit, SPL-1585, are installed:

(a) The rig pin in the Captain's steering tiller.

(b) The rig pin in the rudder pedal interconnect quadrant.

(c) Install the rig pin to the summing mechanism.

SUBTASK 32-51-52-420-003

(7) Install the bushing [13], bolt [12], washers [14], and nut [15] to connect the link assembly [16] to the summing mechanism assembly [11].

(a) Adjust the link assembly [16] until you can easily install the rig pin in the summing mechanism rig pin hole.

SUBTASK 32-51-52-420-004

(8) Tighten the NWS-A cable [1] and NWS-B cable [2] at the turnbuckles [3] and adjust the tension (TASK 32-51-00-820-805).

SUBTASK 32-51-52-420-005

(9) Install the clips [4] on the turnbuckles [3].

SUBTASK 32-51-52-080-001

(10) Remove these rig pins:

(a) The rig pin in the Captain's steering tiller.

(b) The rig pin in the rudder pedal interconnect quadrant.

(c) Remove the rig pin from the summing mechanism.

G. Summing Mechanism Installation Test

SUBTASK 32-51-52-710-001

(1) Do a test on the steering system of the nose wheel (TASK 32-51-00-700-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-52-410-001

(1) Close the access panels to the nose wheel steering system mechanism and turnbuckles [3] (TASK 53-14-01-420-801).

(a) Close these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-52-420-006

(2) Put the summing mechanism cover [10] in position and install the screws [5], washers [6], and washers [7].

——— END OF TASK ———

TASK 32-51-52-020-802

4. Summing Mechanism Removal (Using Cable Clamp Assembly)

(Figure 402, Figure 403)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-13966	Clamp Assy - Steering Cables, NLG Part #: C32055-18 Supplier: 81205

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-51-52-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-52-860-005

(2) Make sure that the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-52-860-006

(3) Remove the pressure from hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-51-52-860-007

(4) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

SUBTASK 32-51-52-860-008

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Install the NLG towing lever pin, SPL-1499, through the towing lever.

NOTE: You use the NLG Towing Lever Lock Pin to hold the towing lever in the TOW position.

E. Steering Summing Mechanism Removal

SUBTASK 32-51-52-010-003

(1) Remove the screws [5], washers [6], and washers [7] that hold the summing mechanism cover [10] to the support bracket.

SUBTASK 32-51-52-020-007

(2) Use the cable clamp, SPL-13966, to remove the tension in NWS-A cable [1] and NWS-B cable [2] between the trunnion and link assembly (Figure 403).

NOTE: You may need to remove one of the slider bolts in order to install the base plate over the NWS-A cable [1] and NWS-B cable [2].

(a) Put the base plate in position against the trunnion, with NWS-A cable [1] and NWS-B cable [2] positioned in the base plate grooves.

NOTE: Use the lock-trunnion nut to align the base plate into position.

(b) Tighten the clamp bolts evenly to 70 in-lb (7.9 N·m) - 80 in-lb (9.0 N·m) onto the clamp plate.

(c) Turn only the tension bolt until the clamp carrier contacts the head of the slider bolts.

1) Make sure that you do not exceed a maximum of 0.10 in. (2.54 mm) of cable stretch.

SUBTASK 32-51-52-020-008

(3) Remove the nuts [24], bolts [21], washers [22], bushing [23], and pulleys [20] from the pulley brackets on the summing mechanism assembly [11].

SUBTASK 32-51-52-020-009

(4) Remove the nut [15], washers [14], bushing [13], and bolt [12] to disconnect the link assembly [16] from the summing mechanism assembly [11].

SUBTASK 32-51-52-020-010

(5) Remove the nuts [19], washers [28], bolts [17], and washers [18] to disconnect the summing mechanism assembly [11] from the bracket assembly.

SUBTASK 32-51-52-000-002

(6) Remove the summing mechanism assembly [11].

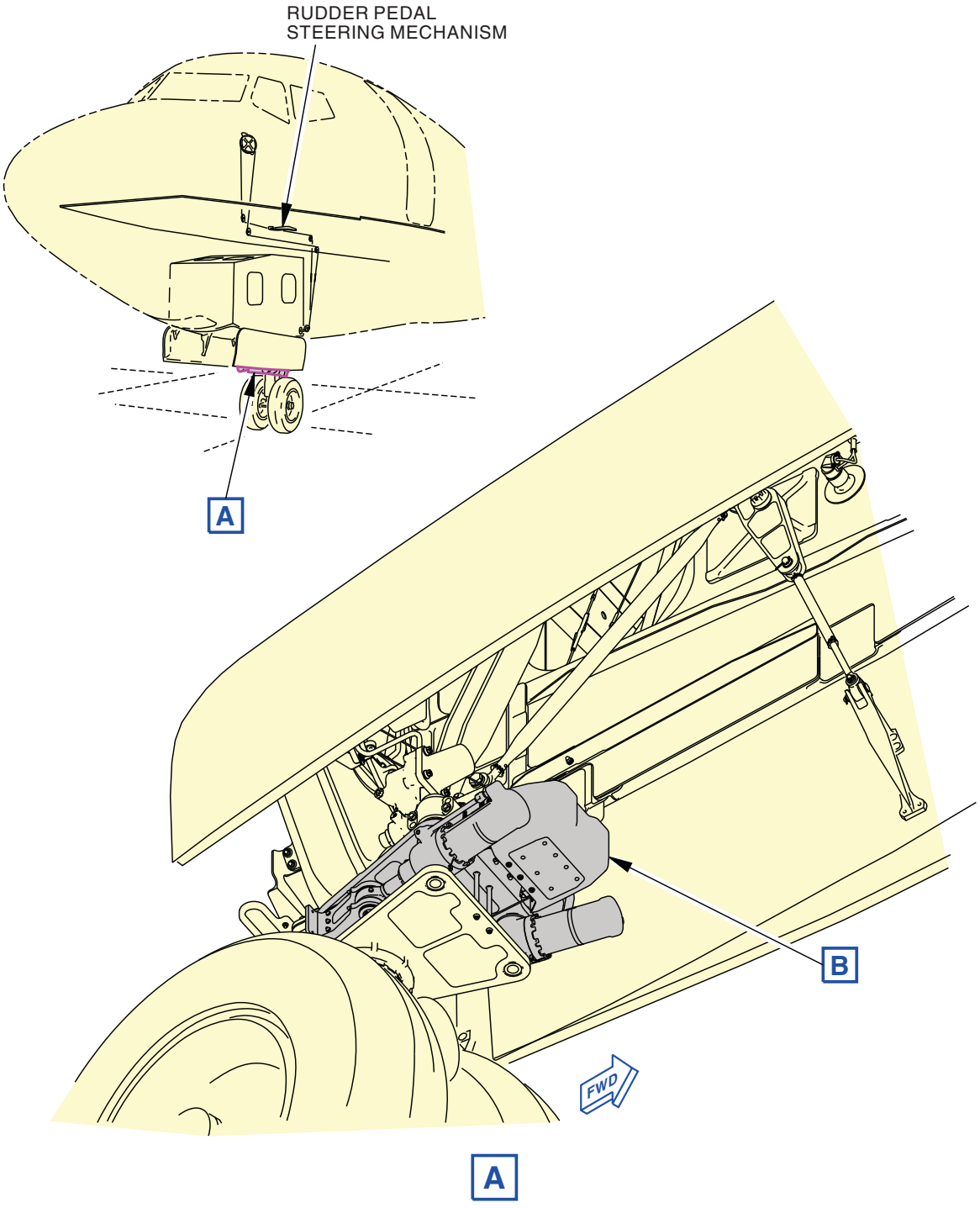
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-51-52-020-011

(7) If it is necessary to remove the pulley bracket assembly from the summing mechanism assembly [11], remove the nuts [27], bushings [26], bolts [25], and washers [14].

——— END OF TASK ———

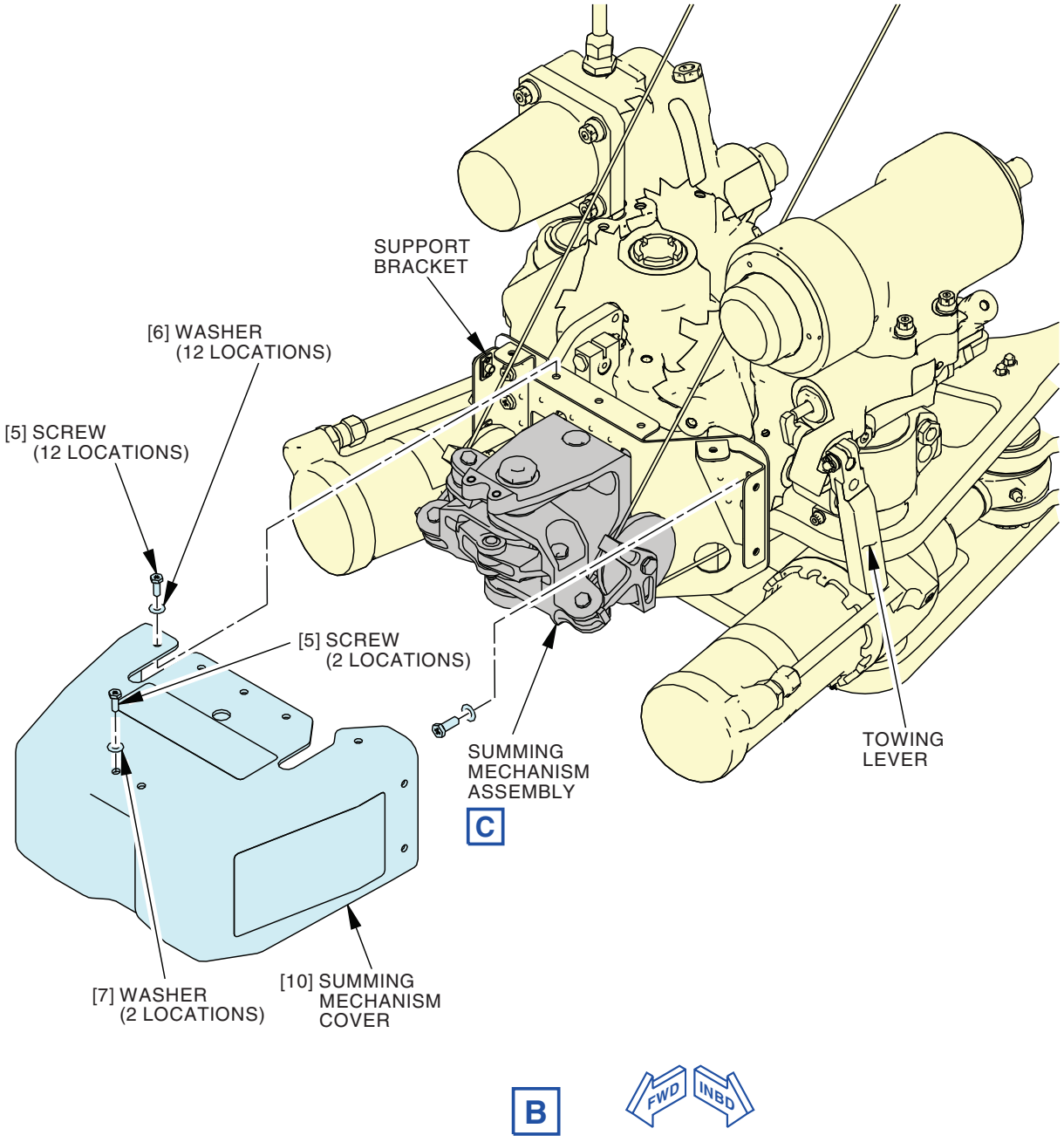
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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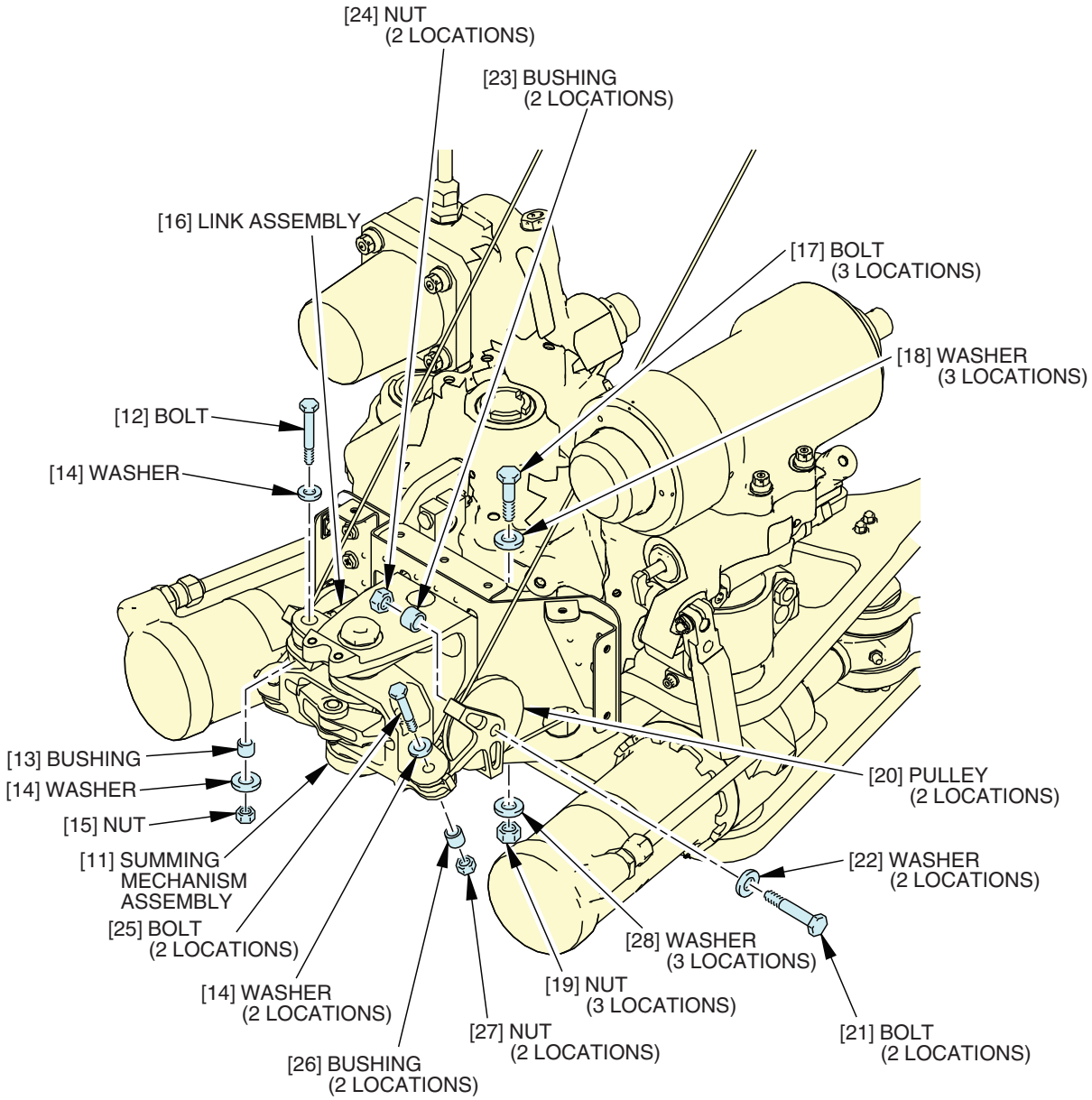
Nose Landing Gear Steering Summing Mechanism Installation
Figure 402/32-51-52-990-802 (Sheet 1 of 3)

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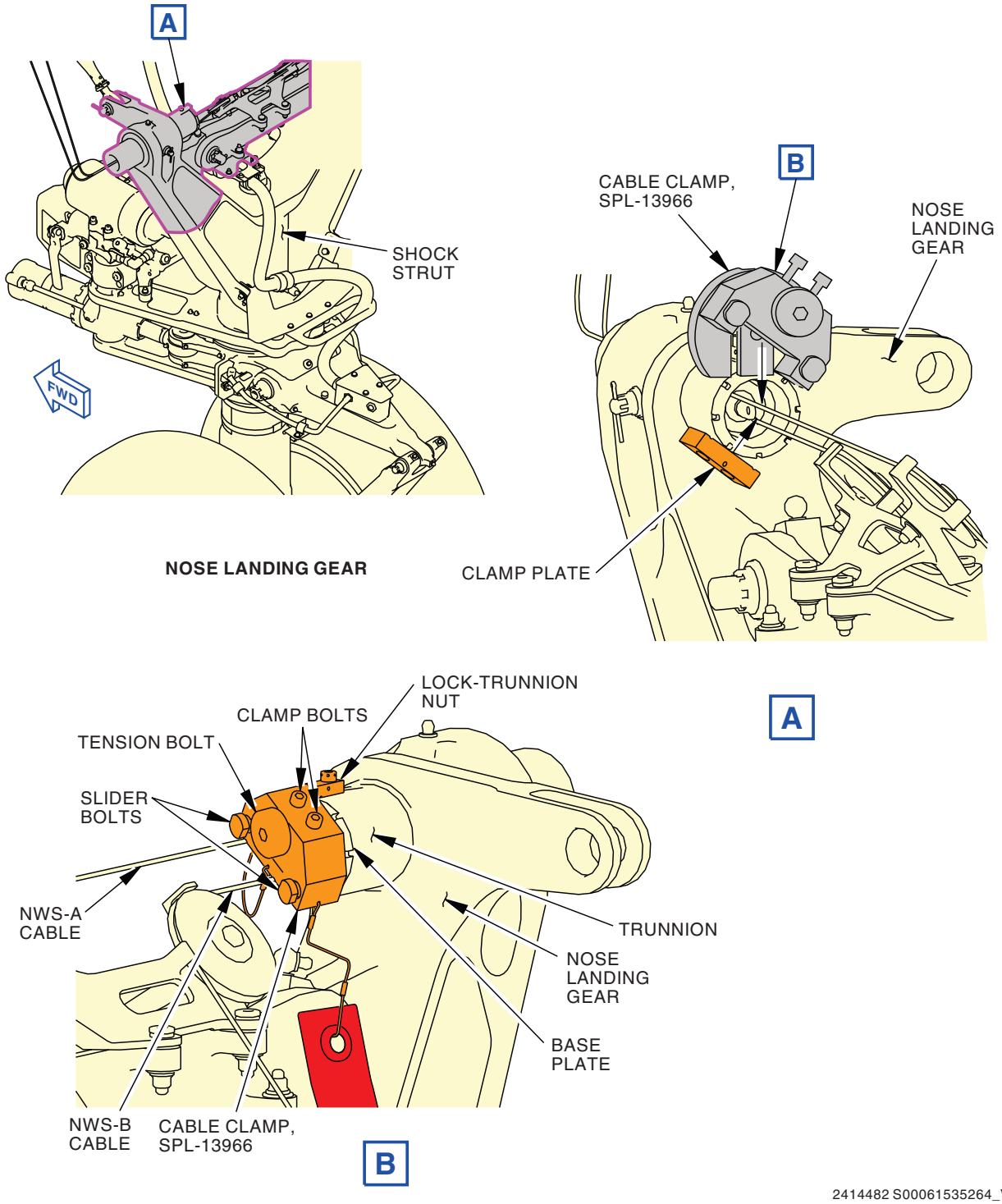
Nose Landing Gear Steering Summing Mechanism Installation
Figure 402/32-51-52-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Steering Summing Mechanism Installation
Figure 402/32-51-52-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Steering Cables Clamp Assembly
Figure 403/32-51-52-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-52-420-802

5. Summing Mechanism Installation (Using Cable Clamp Assembly)

(Figure 402, Figure 403)

A. References

Reference	Title
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1499	Pin - Lock, NLG Towing Lever Part #: A09003-2 Supplier: 81205 Opt Part #: A09003-1 Supplier: 81205
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-13966	Clamp Assy - Steering Cables, NLG Part #: C32055-18 Supplier: 81205

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
11	Summing mechanism assembly	32-51-52-01-175	PTF ALL

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Summing Mechanism Installation

SUBTASK 32-51-52-210-002

(1) Make sure that the wheels of the nose landing gear are in the centered position (Figure 401).

NOTE: The graduated scales on the steering collar and arrows on the lower plate show the angular position of the wheels when the torsion links are connected.

SUBTASK 32-51-52-400-002

(2) Attach the summing mechanism assembly [11] to the bracket assembly:

(a) Align the summing mechanism assembly [11] with the steering plate.

(b) Align the link assembly [16] with the summing mechanism assembly [11].

(c) Install the bolts [17], washers [18], washers [28], and nuts [19] to connect the summing mechanism assembly [11] to the bracket assembly.

SUBTASK 32-51-52-420-007

(3) If it is necessary, install the pulley bracket assembly:

(a) Put the pulley bracket assembly in its location.

(b) Install the bushings [26], washers [14], bolts [25], and nuts [27].

(c) Tighten the bolts [25] to 20 in-lb (2.3 N·m) - 25 in-lb (2.8 N·m) above the run-on torque.

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SUBTASK 32-51-52-420-008

(4) Install the pulleys [20] on the pulley bracket assembly:

(a) Thread the cable over the pulley [20].

(b) Install the bushings [23].

(c) Align the pulleys [20] with the pulley bracket assembly.

(d) Install the bolts [21], washers [22], and nuts [24].

(e) Tighten the bolts [21] to 50 in-lb (5.6 N·m) - 75 in-lb (8.5 N·m) above the run-on torque.

SUBTASK 32-51-52-480-006

WARNING ONLY USE THE CORRECT PIN FOR THE AIRPLANE MODEL. IF YOU USE AN INCORRECT PIN, THE HYDRAULIC STEERING CAN OPERATE. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Make sure that the NLG towing lever pin, SPL-1499, is installed through the towing lever.

SUBTASK 32-51-52-480-007

(6) Install the rig pin NS4 to the summing mechanism.

SUBTASK 32-51-52-420-009

(7) Install the bolt [12], bushing [13], washers [14], and nut [15] to connect the link assembly [16] to the summing mechanism assembly [11].

(a) Adjust the link assembly [16] until you can easily install the rig pin NS4 from the rigging pin kit, SPL-1585, in the summing mechanism rig pin hole.

(b) Remove the rig pin NS4 from the summing mechanism.

(c) Remove the NLG towing lever pin, SPL-1499, from the towing lever.

SUBTASK 32-51-52-080-002

(8) Remove the cable clamp, SPL-13966.

F. Summing Mechanism Installation Test

SUBTASK 32-51-52-710-002

(1) Do a test on the steering system of the nose wheel (TASK 32-51-00-700-801).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-52-420-010

(1) Put the summing mechanism cover [10] in position and install the screws [5], washers [6], and washers [7].

——— END OF TASK ———

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STEERING COLLAR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the steering collar (without using cable clamp assembly)

(2) Installation of the steering collar (without using cable clamp assembly)

(3) Removal of the steering collar (using cable clamp assembly)

(4) Installation of the steering collar (using cable clamp assembly)

(5) Removal of the upper steering sleeve

(6) Installation of the upper steering sleeve.

TASK 32-51-61-000-801

2. Steering Collar Removal (Without Using Cable Clamp Assembly)

(Figure 401)

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-24-000-801	NLG Inner Cylinder Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1901	Spanner Wrench - Steering Collar Retainer, NLG Part #: C32040-6 Supplier: 81205 Opt Part #: C32040-1 Supplier: 81205
STD-837	String

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Access Panels

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

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E. Prepare for the Removal

SUBTASK 32-51-61-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-61-860-001

(2) Make sure the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-61-860-002

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-61-860-003

(4) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

SUBTASK 32-51-61-860-004

(5) Install the rigging pin kit, SPL-1585, through the towing lever.

NOTE: You use the NLG Towing Lever Lock Pin to hold the towing lever in the TOW position.

SUBTASK 32-51-61-410-001

(6) Open the access panels to get access to the nose wheel steering system mechanism and cable turnbuckles.

(a) Open this access panel:

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-61-480-002

(7) Install these rig pins from the rigging pin kit, SPL-1585:

(a) Rig pin in the nose wheel steering tiller

(b) Rig pin in the rudder pedal interconnect quadrant

(c) Rig pin in the summing mechanism.

SUBTASK 32-51-61-020-001

(8) Remove the clips [34] from the turnbuckles [36]t.

SUBTASK 32-51-61-020-002

(9) Loosen the NWS-A/B cable [35] at the turnbuckles [36].

SUBTASK 32-51-61-020-003

(10) Remove the pins [37] to disconnect the NWS-A/B cable [35] from the steering collar [5].

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F. Steering Collar Removal

(Figure 401)

SUBTASK 32-51-61-680-001

WARNING

MAKE SURE THAT YOU DEFLATE THE SHOCK STRUT FULLY BEFORE YOU LOOSEN THE VALVE BODY. THE AIR PRESSURE CAN BLOWOUT THE VALVE BODY WHICH CAN CAUSE INJURY TO PERSONS. MAKE SURE THAT NO PERSONS OR EQUIPMENT ARE BELOW THE AIRPLANE WHILE THE AIRPLANE DESCENDS.

(1) Deflate the shock strut for the nose gear (TASK 12-15-41-610-802).

(a) Remove the valve cap from the charging valve at the top of the shock strut.

(b) Loosen the outer hexnut two turns maximum.

1) Deflate the shock strut slowly to prevent loss of fluid through the air valve.

NOTE: The fluid from the shock strut will begin to foam as soon as the pressure drops to ambient.

SUBTASK 32-51-61-580-001

(2) Lift the nose of the airplane until the shock strut is extended approximately 10 in. (254 mm) (TASK 07-10-21-580-801).

SUBTASK 32-51-61-020-004

(3) Disconnect the box assembly [1] from the upper torsion link [13]:

NOTE: Disconnect the box assembly [1] with the electrical conduits attached.

(a) Remove the nuts [33] and washer [32] from the bolts [30].

(b) Remove the bolts [30] and washers [31]] from the upper torsion link [13].

SUBTASK 32-51-61-020-005

(4) Remove the clamps [23] that hold the electrical conduits [20] to the bracket [24]:

(a) Remove the screws [21] that hold the clamps [23] to the bracket [24].

(b) Remove the clamps [23] and the nut clips [22] from the bracket [24].

SUBTASK 32-51-61-020-006

(5) Disconnect the sensor brackets [15] from the upper torsion link [13]:

(a) Disconnect the sensor brackets [15] with the sensors [14] and the electrical conduits [20] attached.

(b) Remove the nuts [19] and washers [18] from the bolts [16].

(c) Remove the bolts [16] that hold the sensor brackets [15] to the upper torsion link [13].

(d) Remove the washers [17] from the bolts [16].

(e) Isolate the sensor brackets [15] from the upper torsion link [13].

1) If it is necessary, use a string, STD-837, to hold the sensor brackets [15] and attach the string, STD-837, over the top of the outer cylinder [60].

SUBTASK 32-51-61-020-007

(6) Disconnect the upper torsion link [13] from the steering collar [5] and attach it the inner cylinder of the shock strut.

(a) Remove the pin [3], nut [2], and washer [4] from the torsion link pin [6].

(b) Remove the torsion link pin [6] to disconnect the upper torsion link [13] from the steering collar [5].

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SUBTASK 32-51-61-010-001

(7) Do this task: NLG Inner Cylinder Removal, TASK 32-21-24-000-801.

SUBTASK 32-51-61-980-001

(8) Turn the steering collar [5] far enough to permit access to the rod end of the actuator assemblies [47].

SUBTASK 32-51-61-020-008

(9) Remove the cotter pin [50], nut [49], washer [48], bushings [46], anti-rotation washers [65] (not on all airplanes), and pin [45] to disconnect the rod end of the actuator assemblies [47] from the steering collar [5].

SUBTASK 32-51-61-020-009

(10) Remove the bolts [53], washers [51], and standoffs [52] that hold the two front feet of the metering valve assembly to the upper steering sleeve [59] and lower steering plate [54].

SUBTASK 32-51-61-020-010

(11) Remove the bolts [55], washers [56], and lock tab [57] from the steering collar retainer nut [58].

SUBTASK 32-51-61-800-001

(12) Hold the lower steering plate [54] in its position.

SUBTASK 32-51-61-020-011

(13) Loosen the steering collar retainer nut [58] and remove it from the outer cylinder [60] with a spanner wrench, SPL-1901.

SUBTASK 32-51-61-800-002

(14) Hold the steering collar [5] in its position.

SUBTASK 32-51-61-010-002

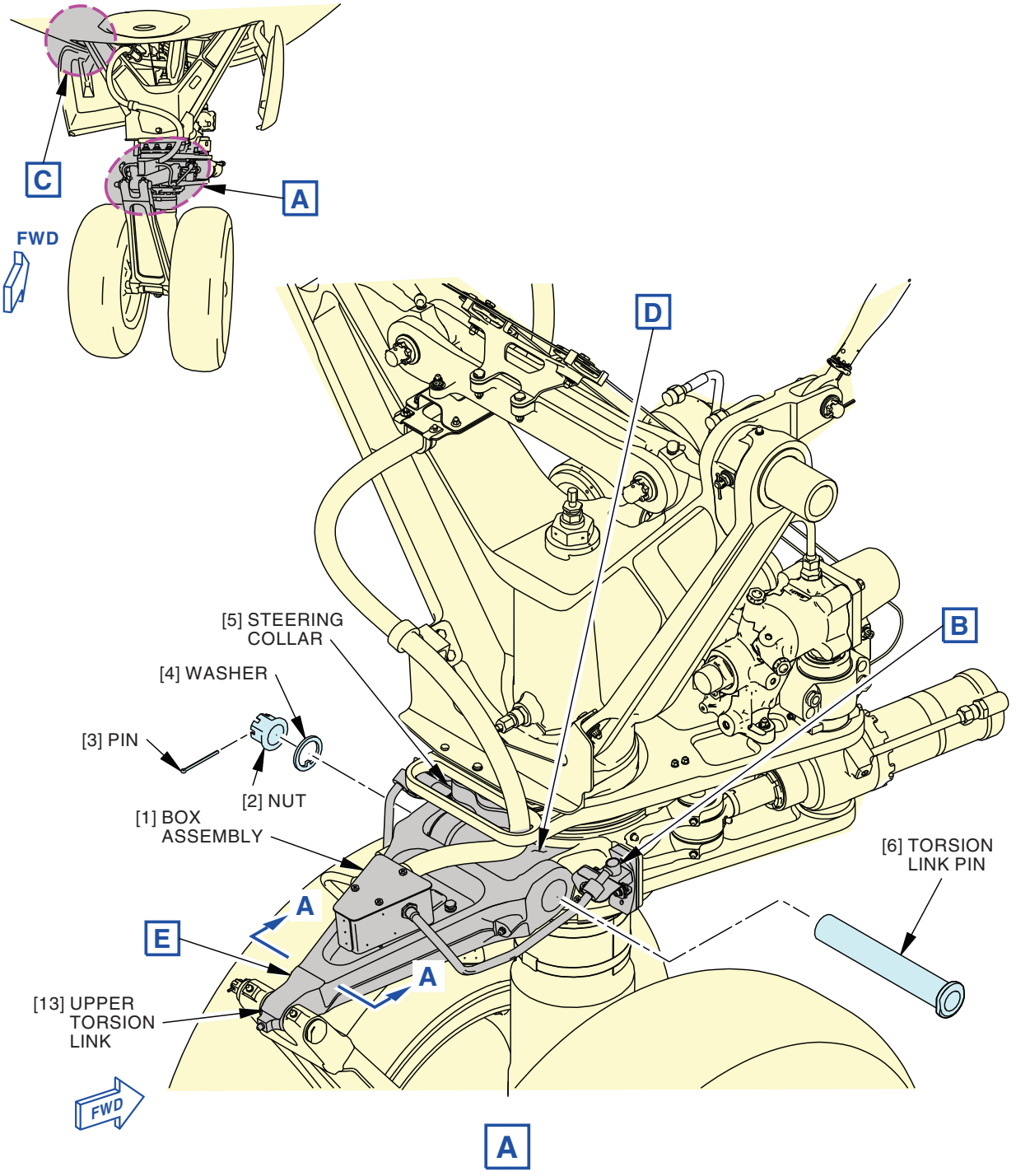
(15) Remove the lower steering plate [54] from the outer cylinder [60].

SUBTASK 32-51-61-020-012

(16) Remove the steering collar [5] from the outer cylinder [60].

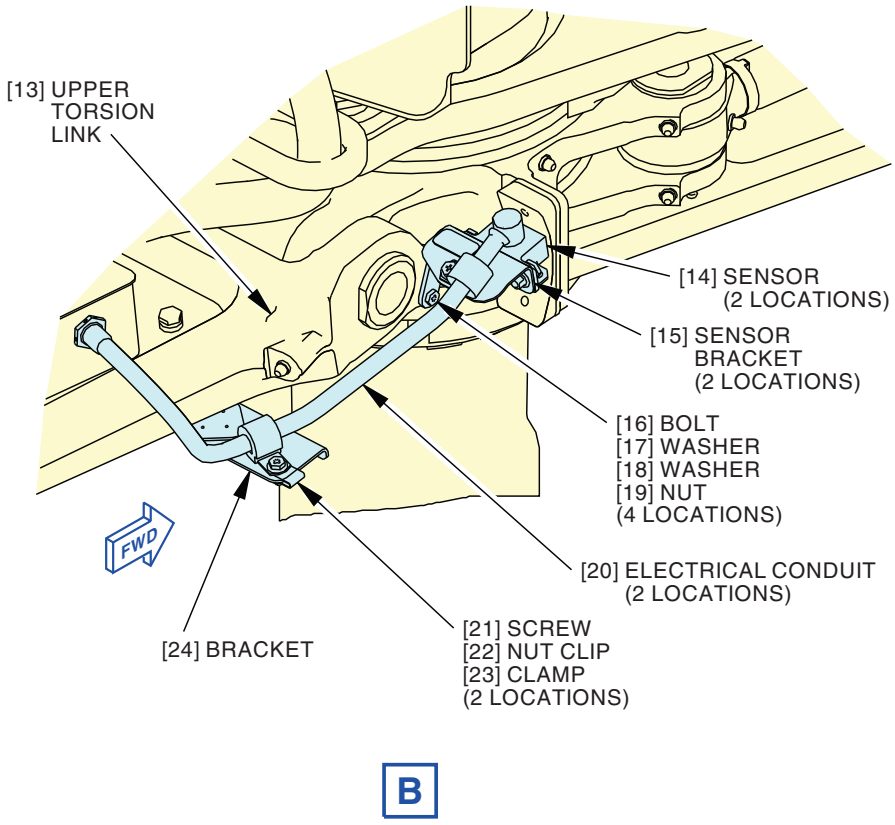
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 1 of 10)

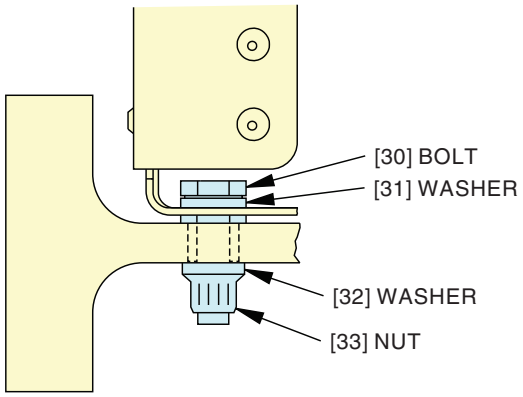
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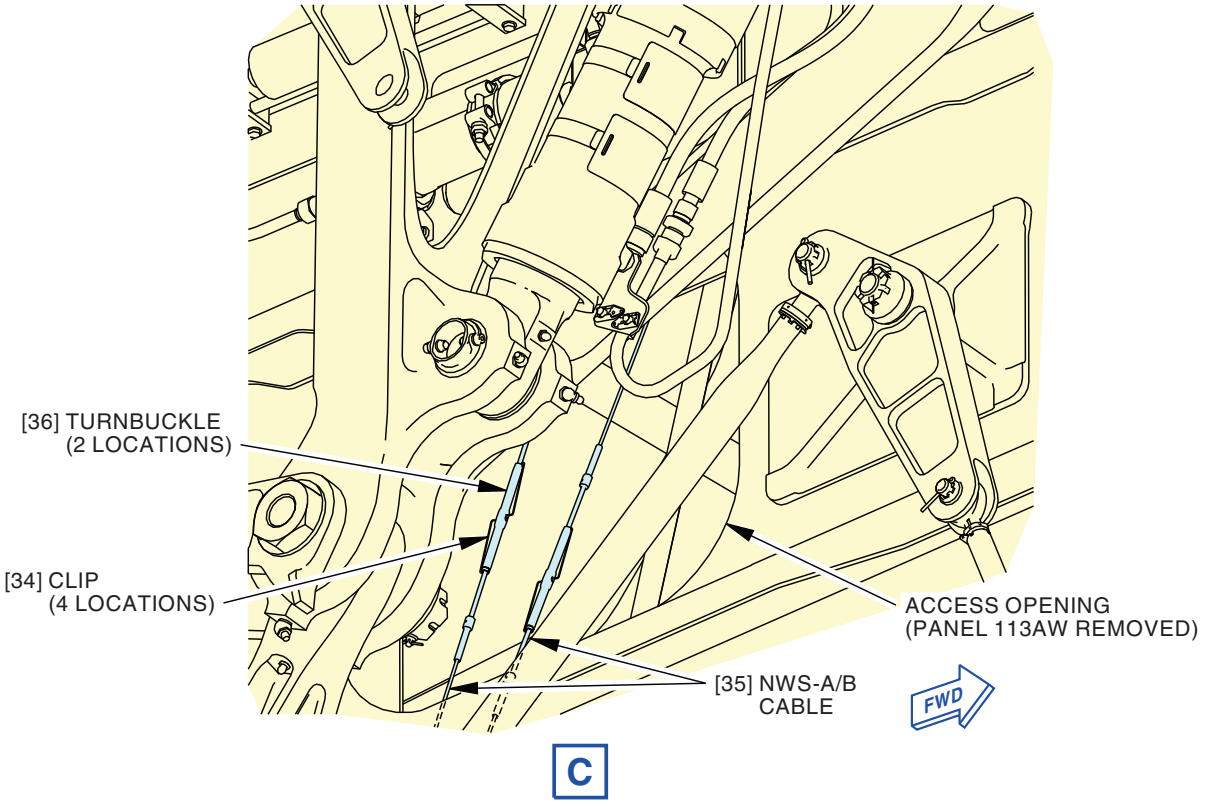
Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 2 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE FORWARD DIRECTION)

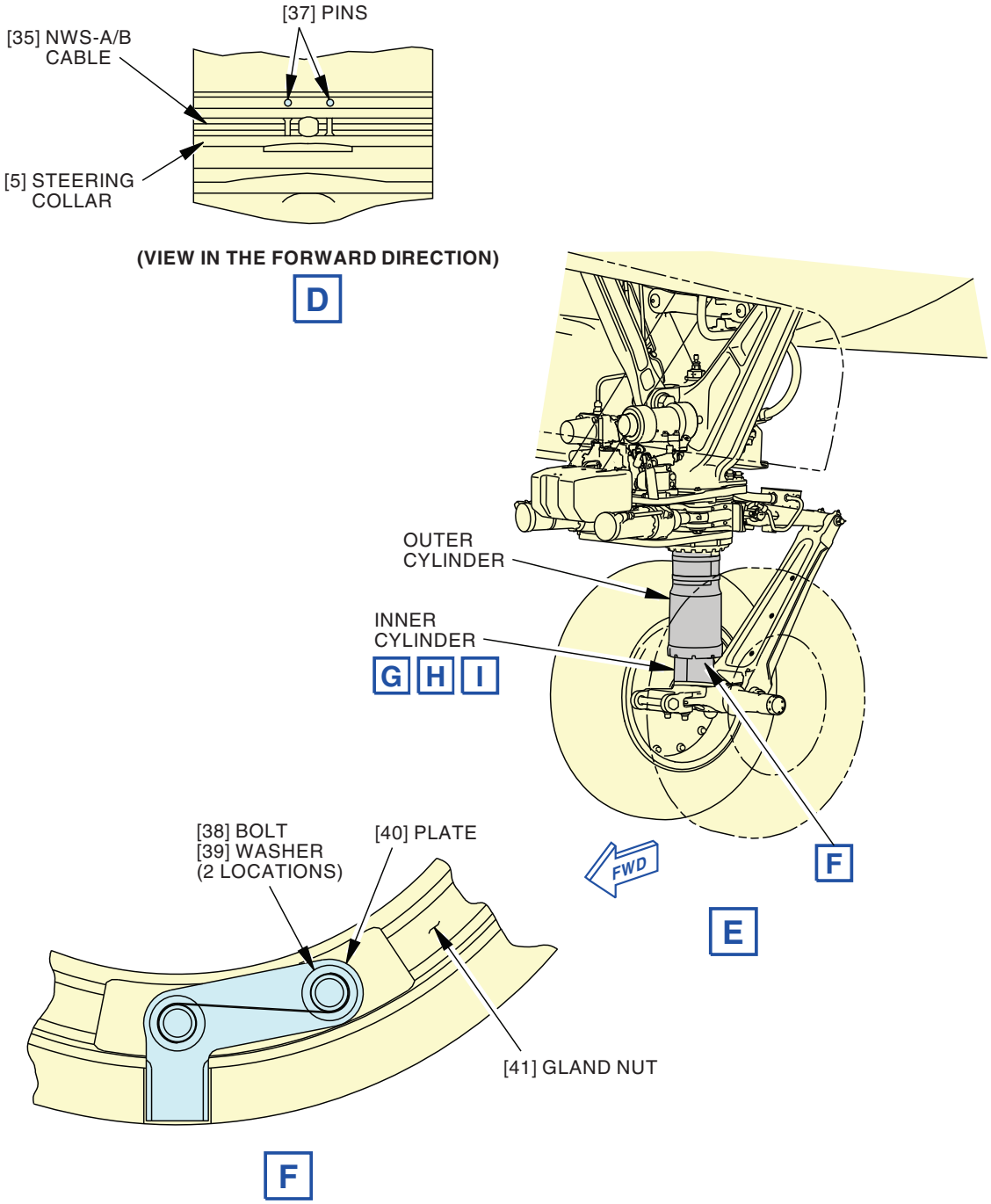
A-A



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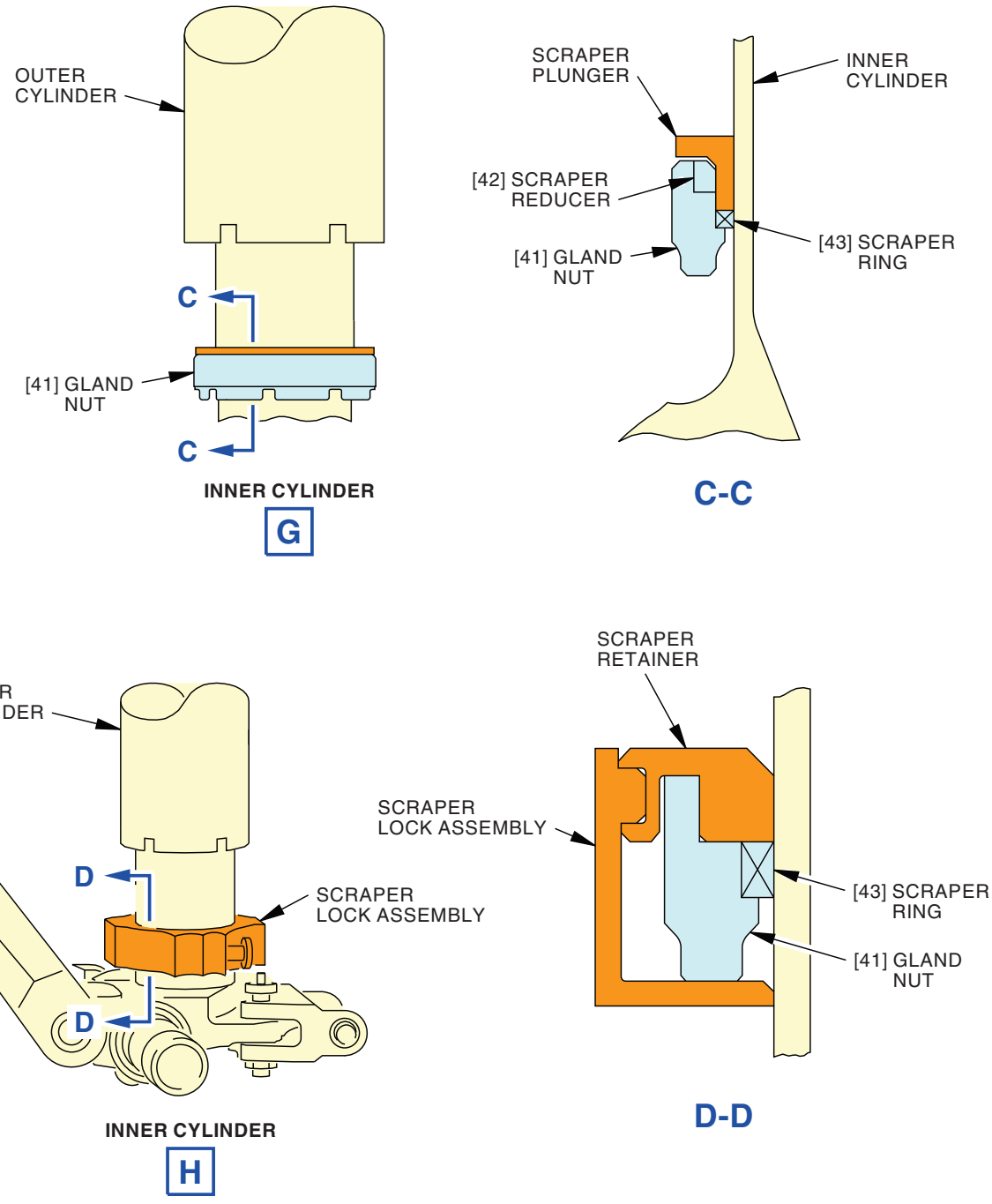
Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 3 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



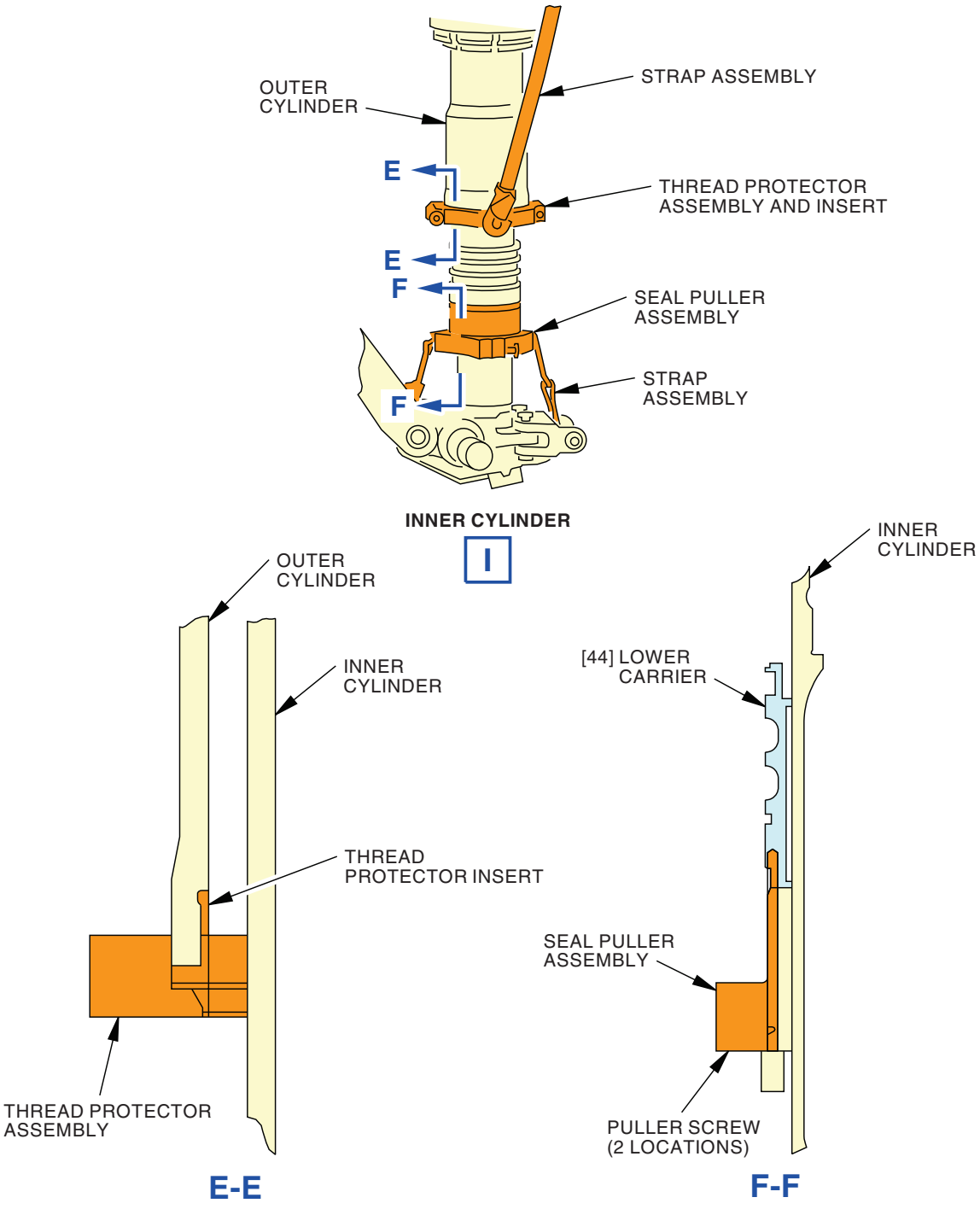
Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 4 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 5 of 10)

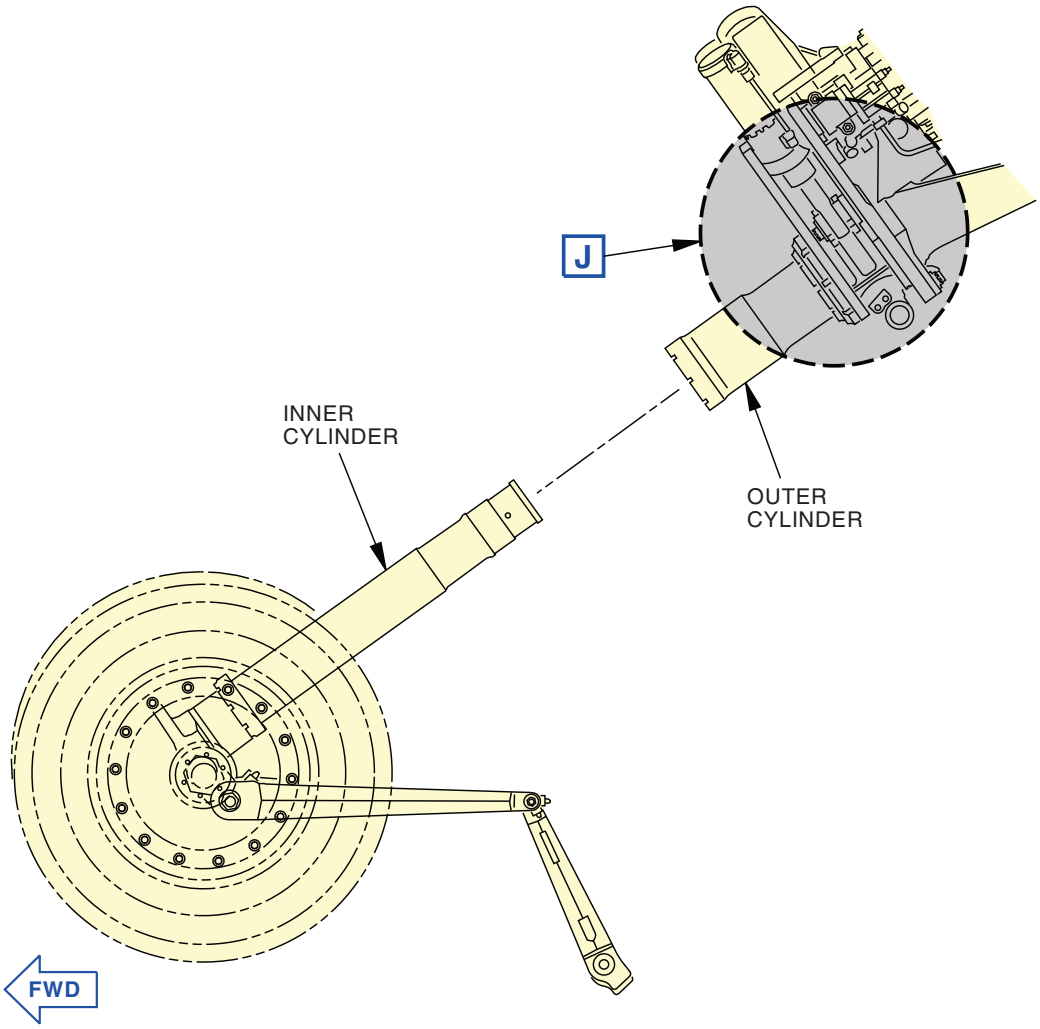
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Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

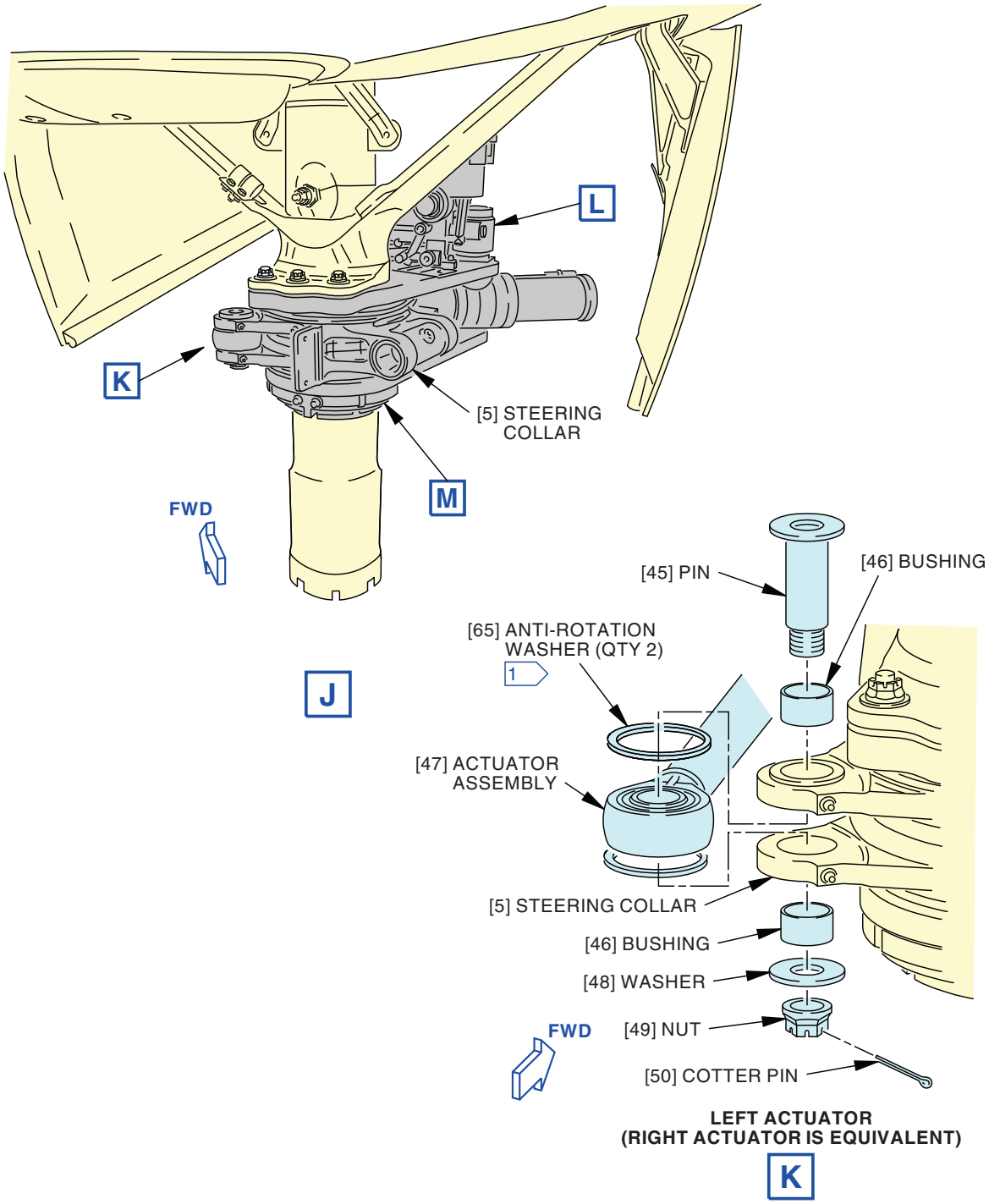


INNER CYLINDER REMOVAL

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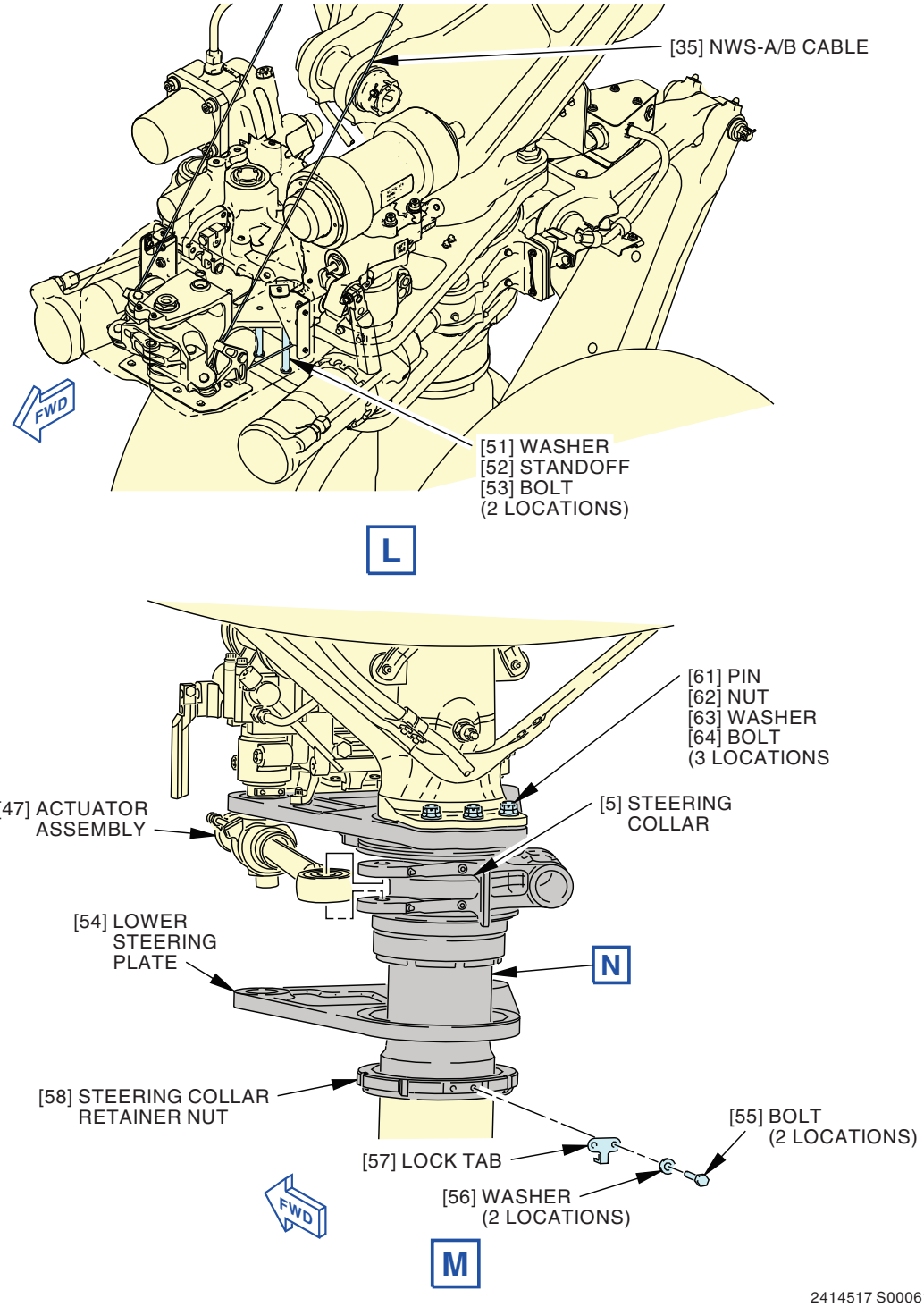
Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 7 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



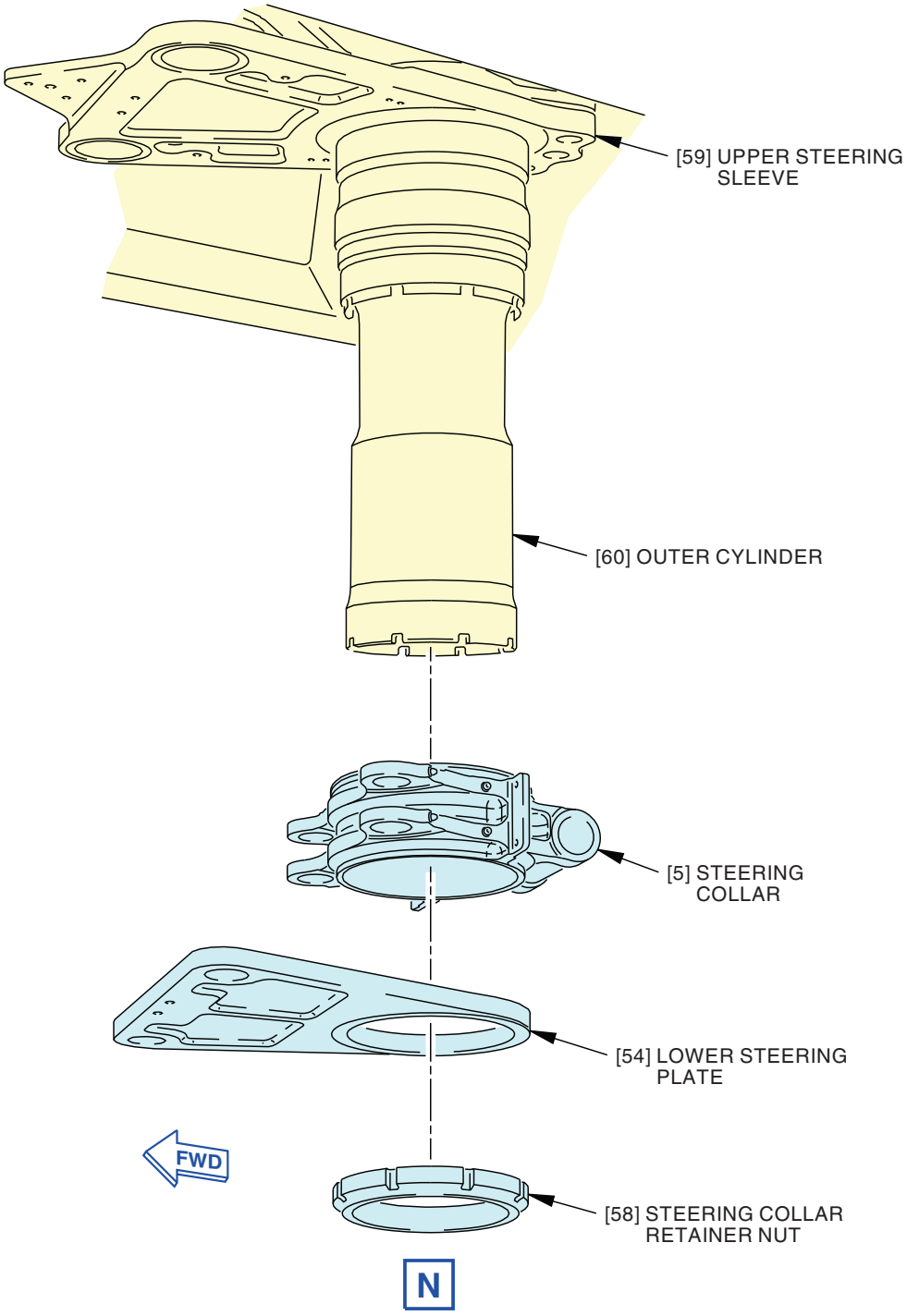
Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 8 of 10)

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Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 9 of 10)

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Nose Gear Steering Collar Installation
Figure 401/32-51-61-990-801 (Sheet 10 of 10)

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TASK 32-51-61-400-801

3. Steering Collar Installation (Without Using Cable Clamp Assembly)

(Figure 401)

A. References

Reference	Title
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-21-24-400-801	NLG Inner Cylinder Installation (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-31-400-802	Nose Gear Steering System Cables Installation (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1901	Spanner Wrench - Steering Collar Retainer, NLG Part #: C32040-6 Supplier: 81205 Opt Part #: C32040-1 Supplier: 81205
STD-837	String

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Steering collar	32-51-61-01-070	PTF ALL
50	Cotter pin	32-51-51-01-215	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

F. Access Panels

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

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G. Steering Collar Installation

NOTE: Make sure that you install the matching steering collar [5] for the steering sleeve installed. Steering collar assembly part number 162A1404-7 can be installed with a steering sleeve assembly part number 162A1405-11 or 162A1405-15 and only steering collar assembly part number 162A1420-2 can be installed with steering sleeve assembly part number 162A1421-3.

SUBTASK 32-51-61-860-005

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-61-600-001

(2) Apply a thin layer of grease, D00633, to the bushing faces and grooves on the steering collar [5].

SUBTASK 32-51-61-420-001

(3) Install the steering collar [5] on the outer cylinder [60].

SUBTASK 32-51-61-420-002

(4) Install the lower steering plate [54] on the outer cylinder [60] and hold it in position.

SUBTASK 32-51-61-600-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a thin layer of corrosion inhibiting material, G50136, to the threads, thread reliefs, and mating surfaces of the steering collar retainer nut [58].

SUBTASK 32-51-61-420-003

(6) Move the steering collar retainer nut [58] up the outer cylinder [60] and install it below the lower steering plate [54].

(a) Tighten the steering collar retainer nut [58] to 62.5 ±12.5 ft-lb (84.7 ±17.0 N·m) with a spanner wrench, SPL-1901.

(b) If it is necessary, loosen the steering collar retainer nut [58] a minimum quantity to install the lock tab [57], washers [56], and bolts [55].

SUBTASK 32-51-61-020-013

(7) Install the standoffs [52], bolts [53], and washers [51] that hold the metering valve assembly to the upper steering sleeve [59] and lower steering plate [54].

SUBTASK 32-51-61-980-002

(8) Turn the steering collar [5] far enough to permit access to the rod end of the actuator assembly [47] that you will install.

SUBTASK 32-51-61-420-004

(9) Install the bushings [46], anti-rotation washers [65] (not on all airplanes), pin [45], washer [48], and nut [49] to connect the rod end of the actuator assembly [47] to the steering collar [5].

(a) Make sure that the flange bushing [46] is positioned face-up in the direction towards the actuator assembly [47].

(b) Tighten the nut [49] to 54 ±4 ft-lb (73 ±6 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) If it is necessary, loosen the nut [49] to align the castellations on the nut [49] with the hole on the pin [45].

(d) Install the new cotter pin [50].

SUBTASK 32-51-61-410-002

(10) Do this task: NLG Inner Cylinder Installation, TASK 32-21-24-400-801.

SUBTASK 32-51-61-420-005

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(11) Connect the upper torsion link [13] to the steering collar [5]:

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The chrome surface on the shank of the torsion link pin [6].

2) The threads of the nut [2].

3) The faces of the washer [4].

4) The mating surfaces of the upper torsion link [13] and the steering collar [5].

(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) Put the upper torsion link [13] in its position on the steering collar [5].

(d) Put the torsion link pin [6] through the upper torsion link [13] and the steering collar [5].

(e) Apply a thin layer of the corrosion inhibiting material, G50136, to the threads and thread reliefs of the torsion link pin [6].

(f) Install the washer [4] and nut [2] on the torsion link pin [6].

(g) Tighten the nut [2] 275 ±25 in-lb (31 ±3 N·m) more than the run-on torque.

(h) If it is necessary, loosen the nut [2] to the nearest castellation to align the holes for the torsion link pin [6].

(i) Install the pin [3] in the torsion link pin [6].

SUBTASK 32-51-61-020-014

(12) Connect the sensor brackets [15] to the upper torsion link [13]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the bolts [16]

2) The threads of the nuts [19]

3) The faces of the washers [17] and washers [18].

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(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) If a string, STD-837, is attached to the sensor brackets [15]], and the outer cylinder [60], remove the string, STD-837.

(d) Put the sensor brackets [15] in their position on the upper torsion link [13].

(e) Install the washers [17] on the bolts [16].

(f) Put the bolts [16] through the upper torsion link [13] and the sensor brackets [15].

(g) Install the washer [18] and nuts [19] on the bolts [16]

(h) Tighten the nuts [19].

SUBTASK 32-51-61-020-015

(13) Connect the box assembly [1] to the upper torsion link [13]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and shank of the bolts [30]

2) The threads of the nuts [33]

3) The faces of the washers [31] and washers [32].

(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) If a string, STD-837, is attached to the box assembly [1], and the outer cylinder [60], remove the string, STD-837.

(d) Put the box assembly [1] in its position on the upper torsion link [13].

(e) Install the washers [31] on the bolts [30].

(f) Put the bolts [30] through the bracket for the box assembly [1] and the upper torsion link [13].

(g) Install the washers [32], and the nuts [33] on the bolts [30].

SUBTASK 32-51-61-410-003

(14) Reconnect the NWS-A/B cable [35] to the steering collar [5] (TASK 32-51-31-400-802).

SUBTASK 32-51-61-980-003

(15) Move the inner cylinder from the fully extended to the fully compressed position.

(a) Make sure the total travel is 15.50 in. (393.70 mm).

SUBTASK 32-51-61-080-001

(16) Remove these rig pins from the rigging pin kit, SPL-1585:

(a) Rig pin in the nose wheel steering tiller

(b) Rig pin in the rudder pedal interconnect quadrant

(c) Rig pin in the summing mechanism.

SUBTASK 32-51-61-860-006

(17) Remove the rigging pin kit, SPL-1585, from the towing lever.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Steering Collar Installation Test

SUBTASK 32-51-61-610-001

(1) Service the shock strut for the nose gear (TASK 12-15-41-610-802).

SUBTASK 32-51-61-710-001

(2) To do a test on the steering system of the nose wheel, do the applicable procedures in this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

Do these procedures:

- General
- Prepare to Test Steering System of the Nose Wheel
- Do the Test of the Normal Steering Response
- Do the Test of Nose Gear Turn Radius and Response to Full Rudder Pedal Input
- Put the Airplane Back to Its Usual Condition

(a) Make sure that there is no leakage.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-61-410-004

(1) Close the access panel to the nose wheel steering system mechanism and cable turnbuckles (TASK 53-14-01-420-801).

(a) Close this access panel:

Number	Name/Location
113BW	Forward Nose Wheel Well Panel

SUBTASK 32-51-61-580-002

(2) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-51-61-640-001

(3) Lubricate the lube fittings on the upper torsion link [13] and the steering collar [5] (TASK 12-21-21-640-802).

——— END OF TASK ———

TASK 32-51-61-000-802

4. Steering Collar Removal (Using Cable Clamp Assembly)

(Figure 402)
(Figure 403)

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-21-24-000-801	NLG Inner Cylinder Removal (P/B 401)

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B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1901	Spanner Wrench - Steering Collar Retainer, NLG Part #: C32040-6 Supplier: 81205 Opt Part #: C32040-1 Supplier: 81205
SPL-13966	Clamp Assy - Steering Cables, NLG Part #: C32055-18 Supplier: 81205
STD-837	String

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-51-61-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-61-860-007

(2) Make sure the wheels of the nose landing gear are in the centered position.

SUBTASK 32-51-61-860-008

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-61-860-009

(4) Move the towing lever, on the forward side of the nose landing gear, to the TOW position.

SUBTASK 32-51-61-860-010

(5) Install the rigging pin kit, SPL-1585, through the towing lever.

NOTE: You use the NLG Towing Lever Lock Pin to hold the towing lever in the TOW position.

SUBTASK 32-51-61-480-004

(6) Install this rig pin from the rigging pin kit, SPL-1585:

(a) Rig pin NS4 in the summing mechanism.

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SUBTASK 32-51-61-020-016

(7) Use the cable clamp, SPL-13966, to remove the tension in steering cables NWS-A and NWS-B between the trunnion and the link assembly (Figure 403).

NOTE: You may need to remove one of the slider bolts in order to install the base plate over the cables.

(a) Put the base plate in position against the trunnion, with the steering cables positioned in the base plate grooves.

NOTE: Use the Lock-Trunnion Nut to align the base plate into position.

(b) Tighten the clamp bolts evenly between 70 in-lb (8 N·m) to 80 in-lb (9 N·m) onto the clamp plate.

(c) Turn only the tension bolt until the clamp carrier contacts the head of the slider bolts.

NOTE: Make sure you do not exceed a maximum of 0.10 in. (0.25 cm) of cable stretch.

SUBTASK 32-51-61-020-017

(8) Remove the pins [37] to disconnect the NWS-A/B cable [35] from the steering collar [5].

E. Steering Collar Removal

(Figure 402)

SUBTASK 32-51-61-680-002

WARNING DO NOT REMOVE THE VALVE BODY UNTIL YOU DEFLATE THE SHOCK STRUT FULLY. THE AIR PRESSURE CAN BLOW THE VALVE BODY OUT AND CAUSE INJURIES TO PERSONNEL.

(1) Deflate the shock strut for the nose gear (TASK 12-15-41-610-802).

(a) Remove the valve cap from the charging valve at the top of the shock strut.

(b) Loosen the outer hexnut two turns maximum.

1) Deflate the shock strut slowly to prevent loss of fluid through the air valve.

NOTE: The fluid from the shock strut will begin to foam as soon as the pressure drops to ambient.

SUBTASK 32-51-61-580-003

(2) Lift the nose of the airplane until the shock strut is extended approximately 10 in. (254 mm) (TASK 07-10-21-580-801).

SUBTASK 32-51-61-020-018

(3) Disconnect the box assembly [1] from the upper torsion link [13]:

NOTE: Disconnect the box assembly [1] with the electrical conduits attached.

(a) Remove the nuts [33] and washers [32] from the bolts [30].

(b) Remove the bolts [30] and washers [31] from the upper torsion link [13].

SUBTASK 32-51-61-020-019

(4) Remove the clamps [23] that hold the electrical conduits [20] to the bracket [24]:

(a) Remove the screws [21] that hold the clamps [23] to the bracket [24].

(b) Remove the clamps [23] and nut clips [22] from the bracket [24].

SUBTASK 32-51-61-020-020

(5) Disconnect the sensor brackets [15] from the upper torsion link [13]:

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(a) Disconnect the sensor brackets [15] with the sensors [14] and electrical conduits [20] attached.

(b) Remove the nuts [19] and washers [18] from the bolts [16].

(c) Remove the bolts [16] that hold the sensor brackets [15] to the upper torsion link [13].

(d) Remove the washers [17] from the bolts [16].

(e) Isolate the sensor brackets [15] from the upper torsion link [13].

1) If it is necessary, use a string, STD-837, to hold the sensor brackets [15] and attach the string, STD-837, over the top of the outer cylinder [60].

SUBTASK 32-51-61-800-003

(6) Attach the electrical conduit [20] and box assembly [1] to the outer cylinder [60] so that it is not in the way.

SUBTASK 32-51-61-020-021

(7) Disconnect the upper torsion link [13] from the steering collar [5] and attach it the inner cylinder of the shock strut.

(a) Remove the pin [3], nut [2], and washer [4] from the torsion link pin [6].

(b) Remove the torsion link pin [6] to disconnect the upper torsion link [13] from the steering collar [5].

SUBTASK 32-51-61-010-003

(8) Do this task: NLG Inner Cylinder Removal, TASK 32-21-24-000-801.

SUBTASK 32-51-61-980-004

(9) Turn the steering collar [5] far enough to permit access to the rod end of the actuator assembly [47].

SUBTASK 32-51-61-020-022

(10) Remove the cotter pin [50], nut [49], washer [48], bushings [46], anti-rotation washers [65] (not on all airplanes), and pin [45] to disconnect the rod end of the actuator assembly [47] from the steering collar [5].

SUBTASK 32-51-61-020-023

(11) Remove the bolts [53], washers [51], and standoffs [52] that hold the two front feet of the metering valve assembly to the upper steering sleeve [59] and lower steering plate [54].

SUBTASK 32-51-61-020-024

(12) Remove the bolts [55], washers [56], and lock tab [57] from the steering collar retainer nut [58].

SUBTASK 32-51-61-800-004

(13) Hold the lower steering plate [54] in its position.

SUBTASK 32-51-61-020-025

(14) Loosen the steering collar retainer nut [58] and remove it from the outer cylinder [60] with a spanner wrench, SPL-1901.

SUBTASK 32-51-61-800-005

(15) Hold the steering collar [5] in its position.

SUBTASK 32-51-61-010-004

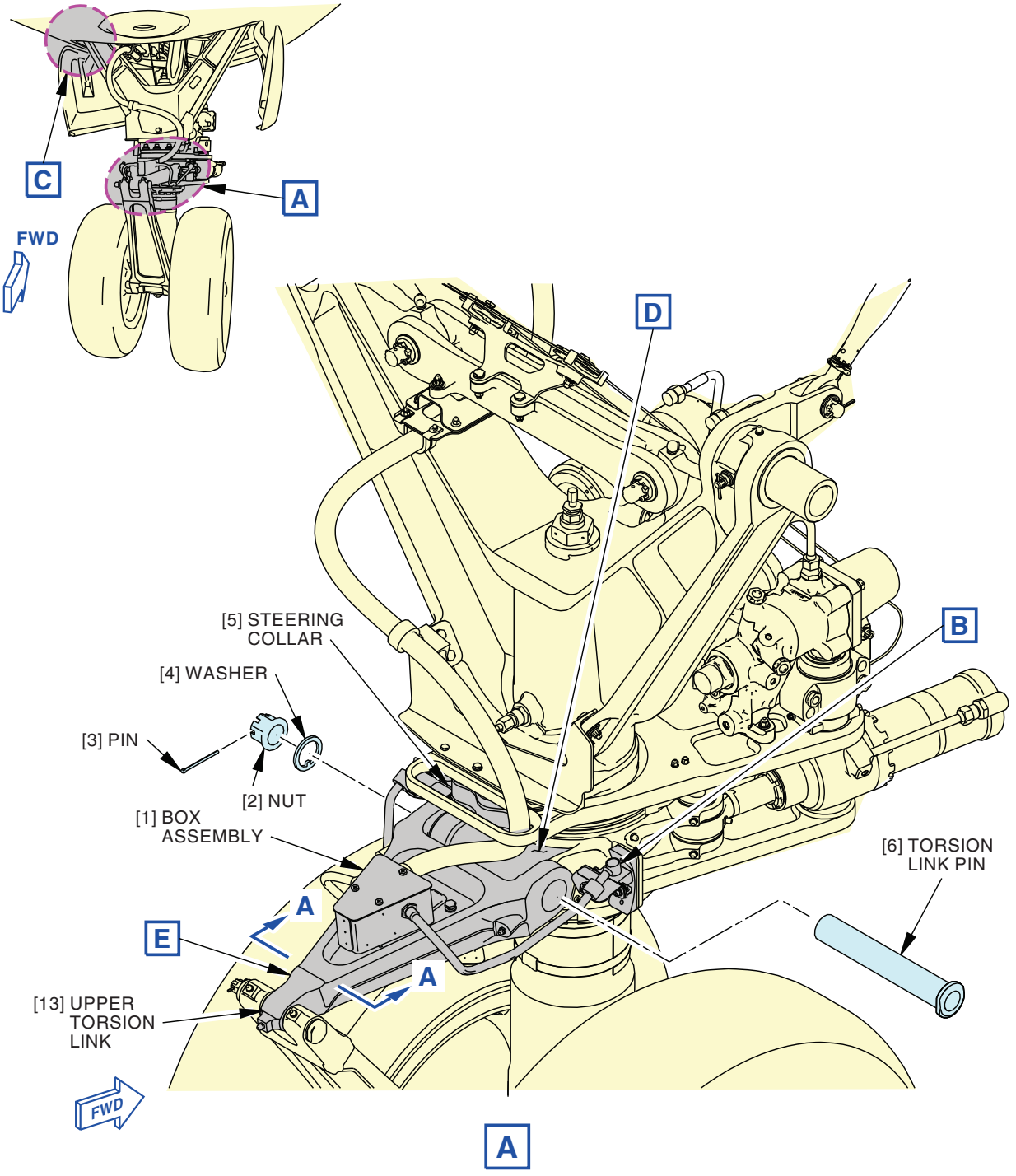
(16) Remove the lower steering plate [54] from the outer cylinder [60].

SUBTASK 32-51-61-020-026

(17) Remove the steering collar [5] from the outer cylinder [60].

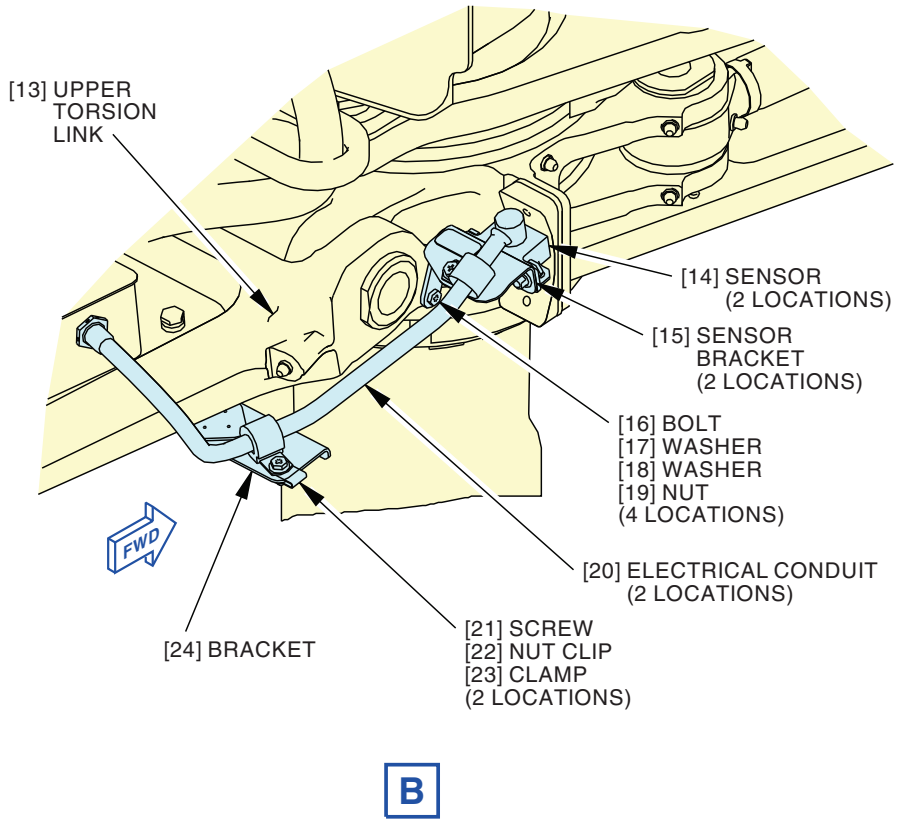
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 1 of 10)

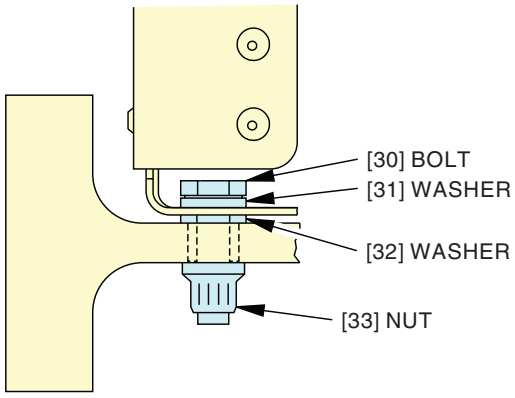
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 2 of 10)

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(VIEW IN THE FORWARD DIRECTION)

A-A

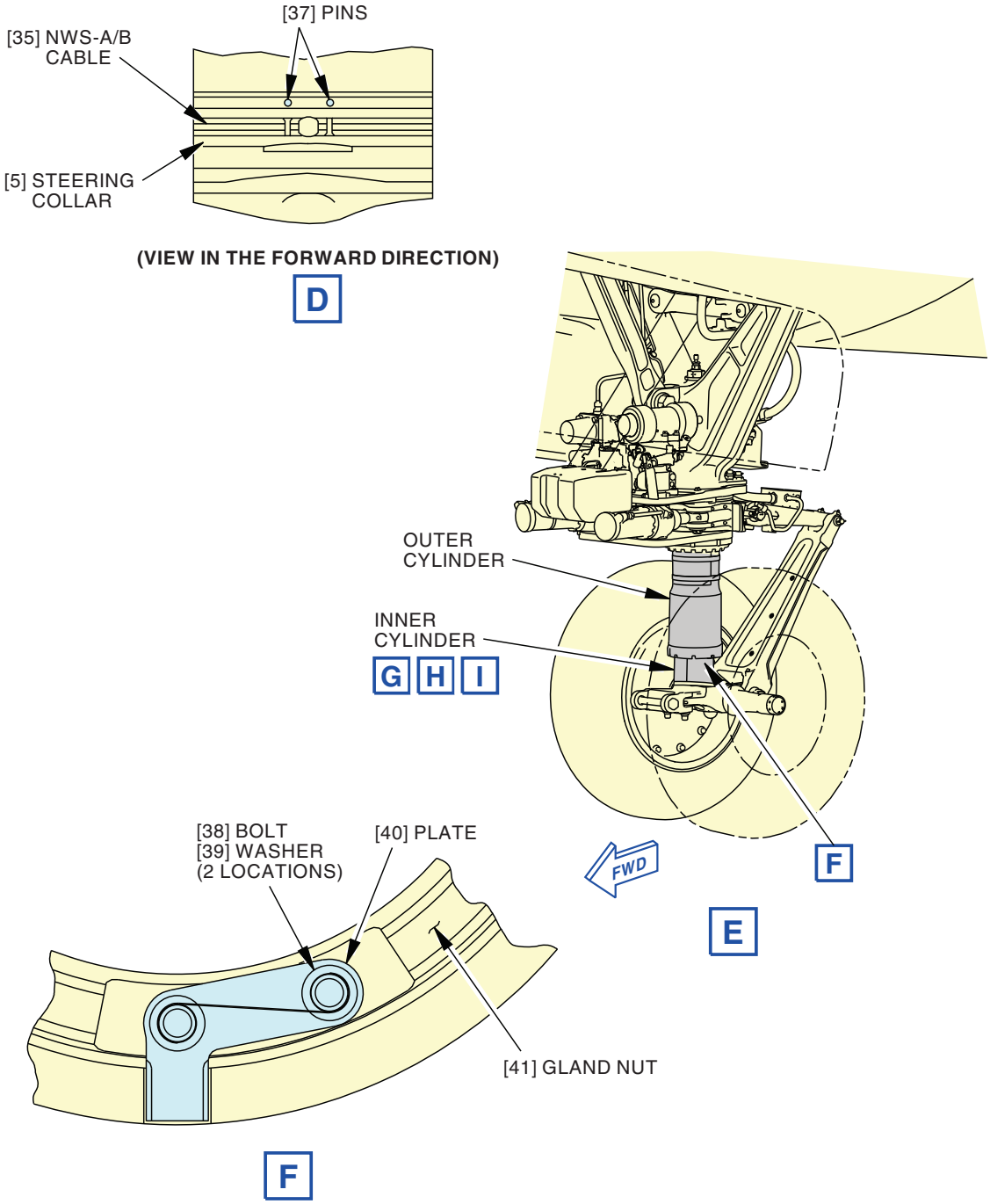
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C

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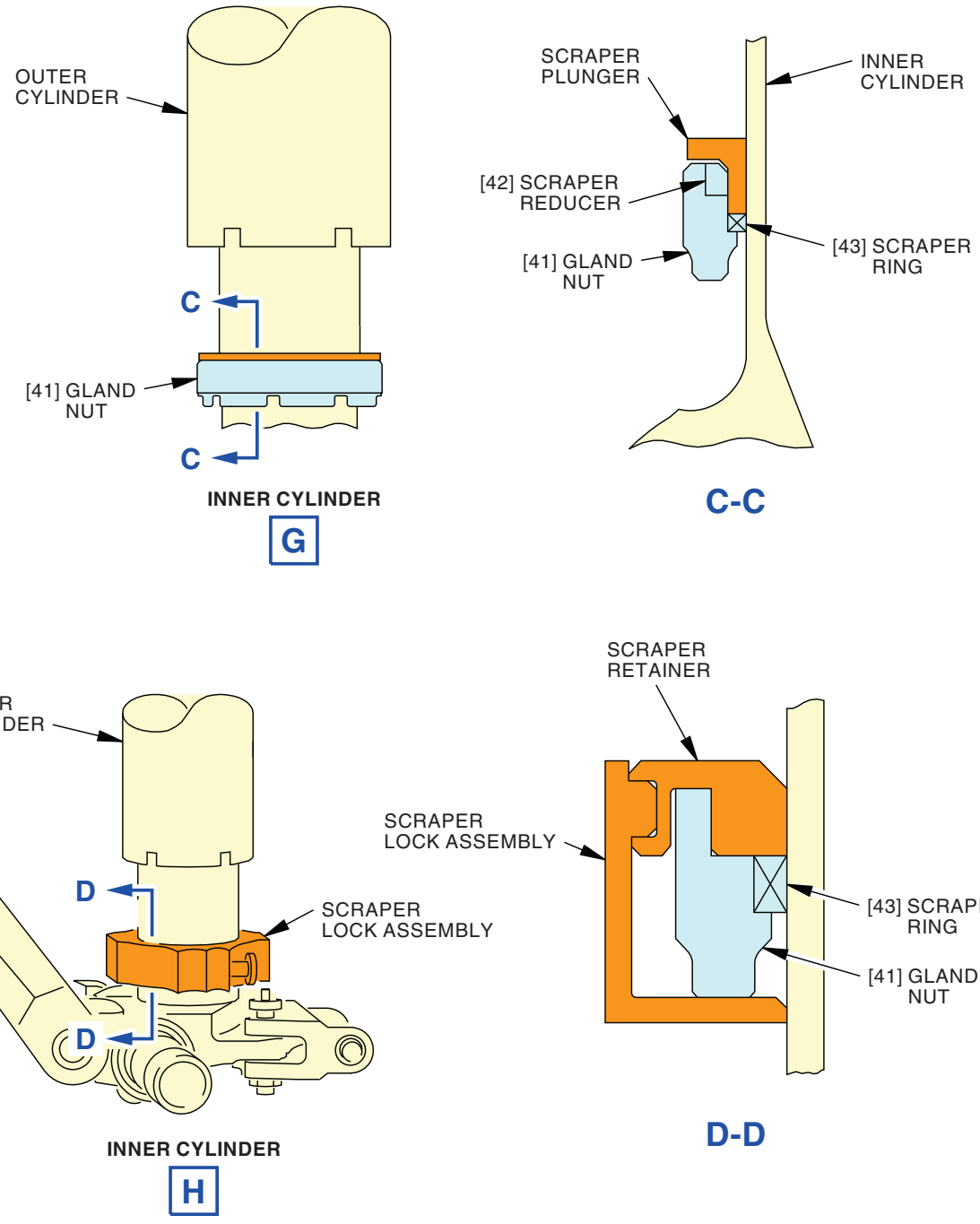
Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 3 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



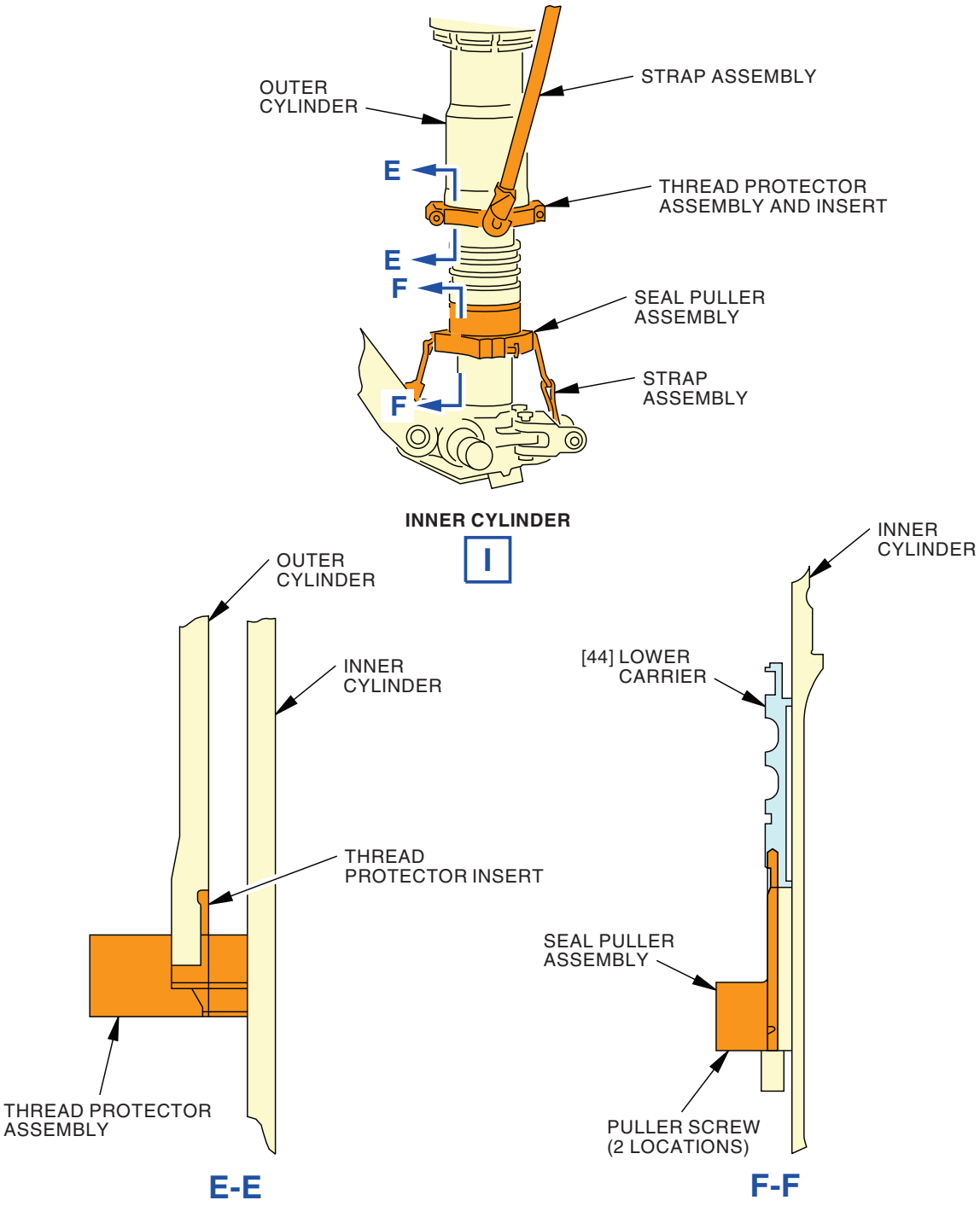
Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 4 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 5 of 10)

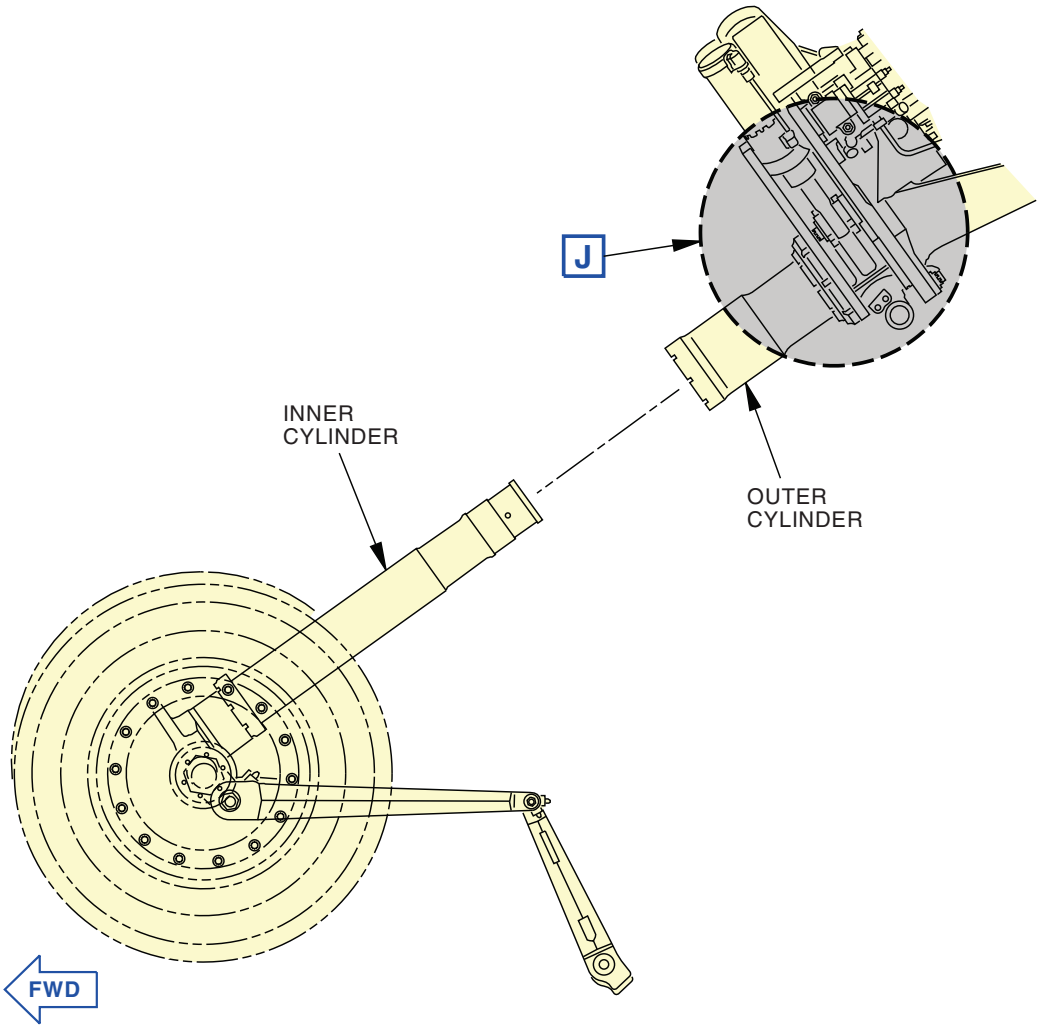
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Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 6 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

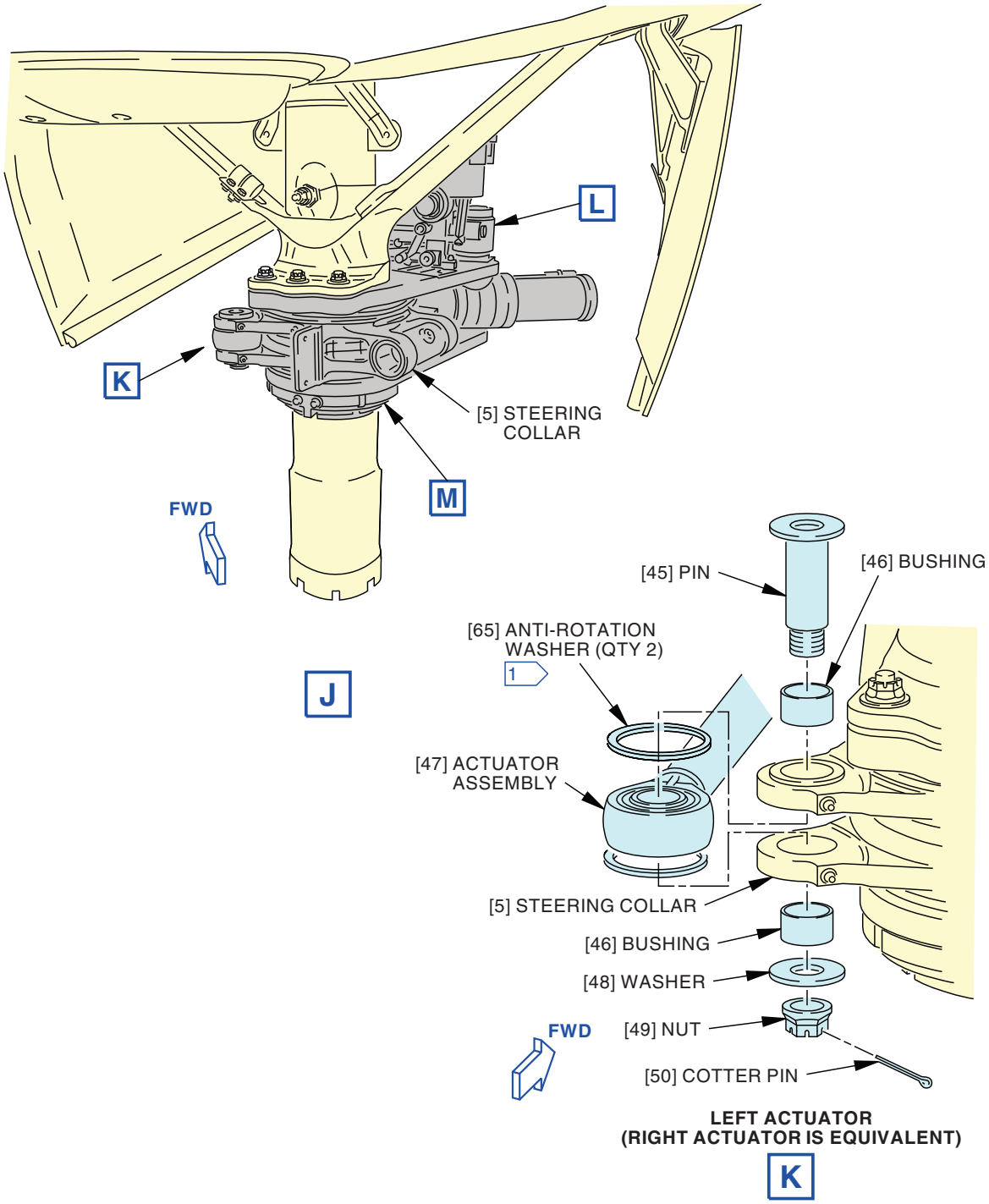


INNER CYLINDER REMOVAL

2414515 S00061535337_V1

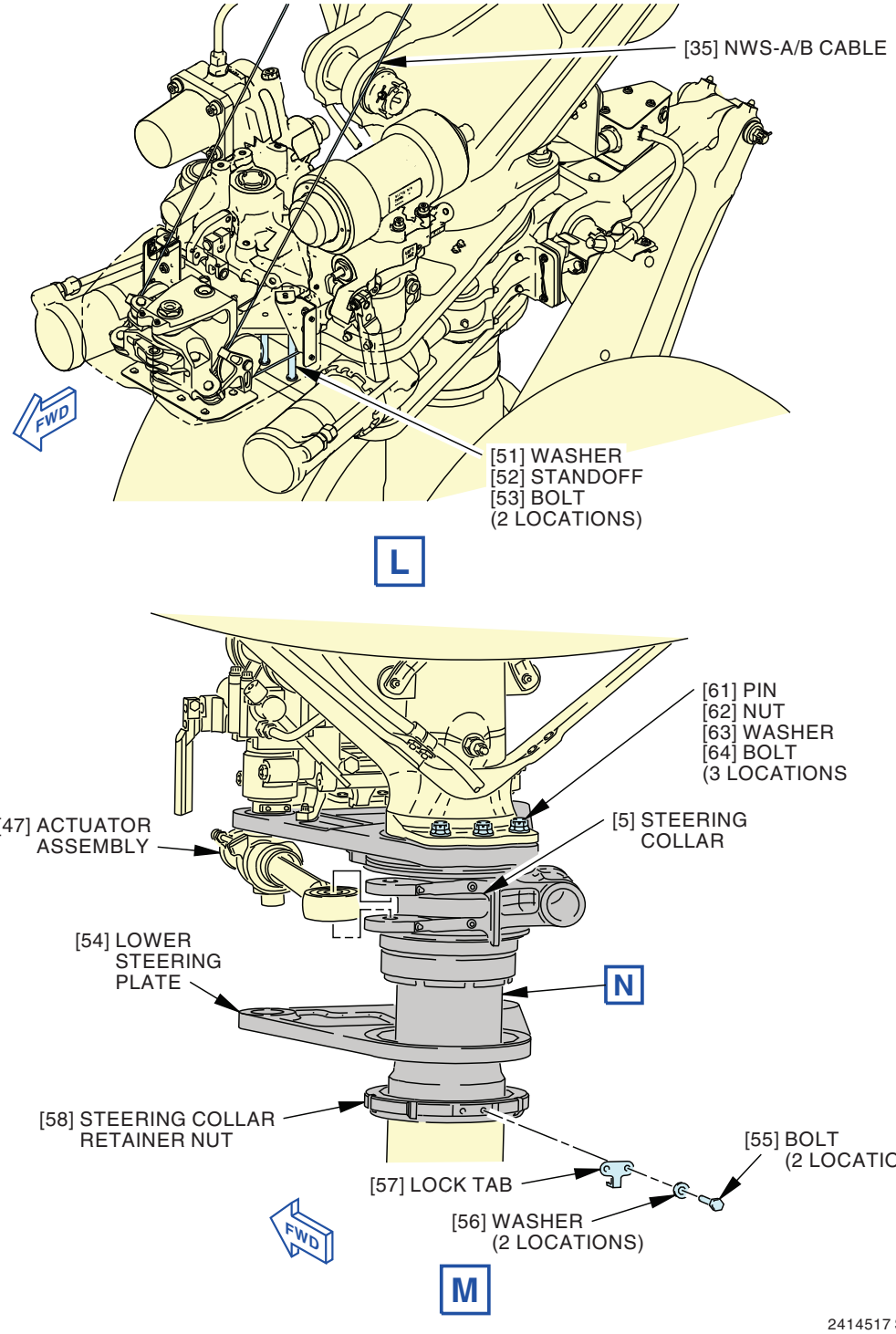
Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 7 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



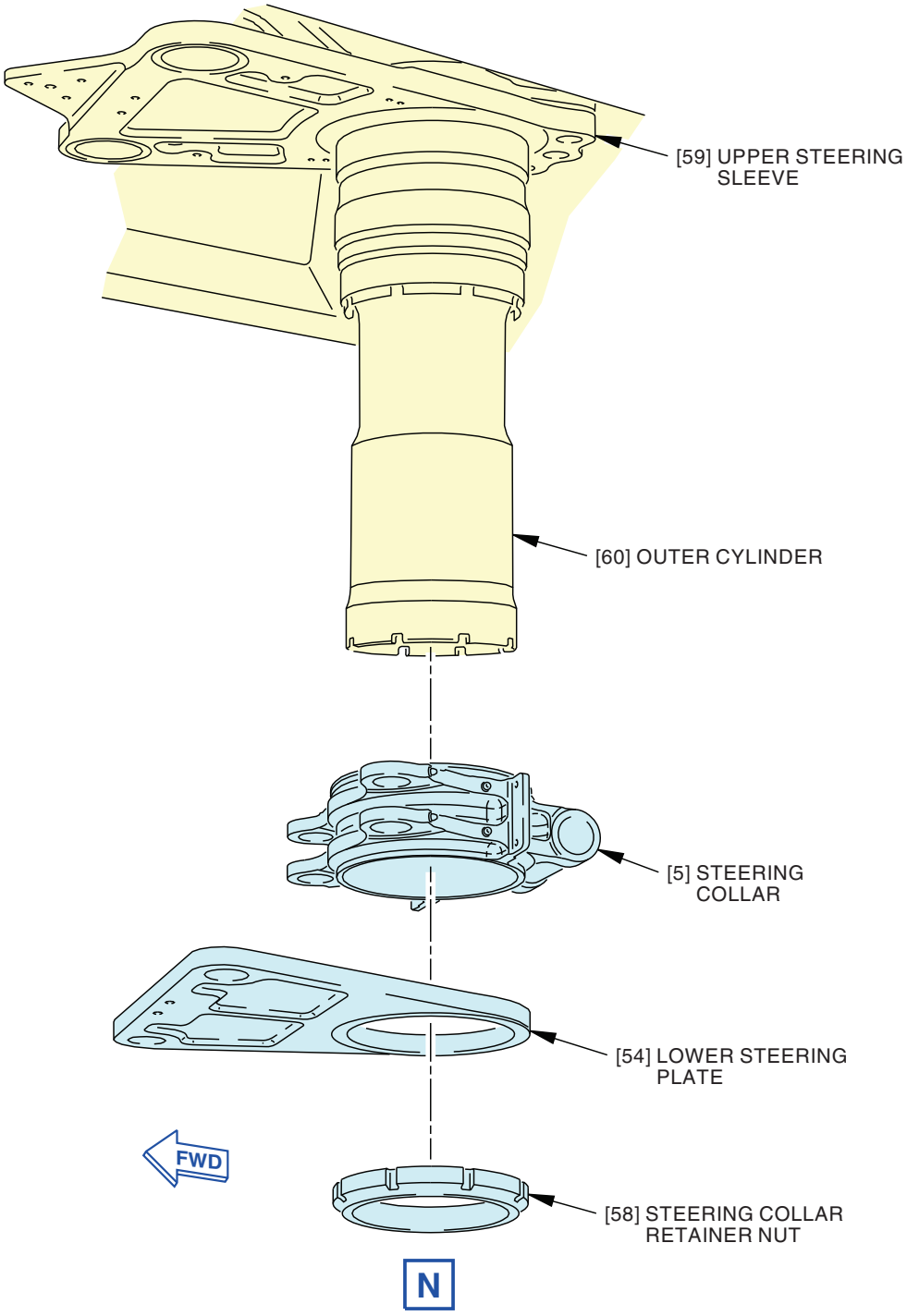
Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 8 of 10)

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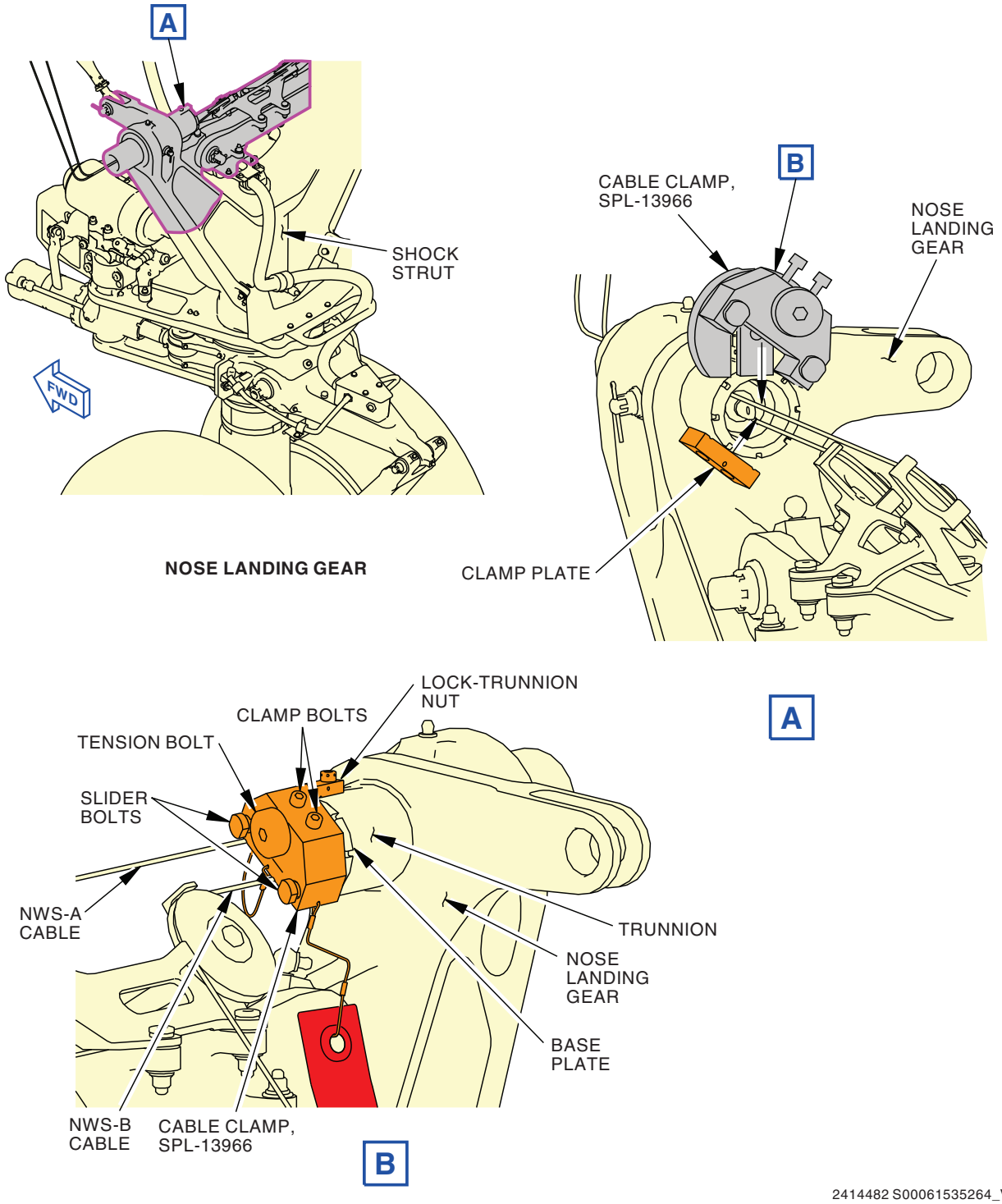
Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 9 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Steering Collar Installation
Figure 402/32-51-61-990-802 (Sheet 10 of 10)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Steering Cables Clamp Assembly
Figure 403/32-51-61-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-61-400-802

5. Steering Collar Installation (Using Cable Clamp Assembly)

(Figure 402,
Figure 403)

A. References

Reference	Title
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-15-41-610-802	Nose Landing Gear Shock Strut Servicing, Airplane on the Ground (P/B 301)
12-21-21-640-802	Nose Landing Gear Lower End Components Servicing (P/B 301)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-21-24-400-801	NLG Inner Cylinder Installation (P/B 401)
32-51-00-700-801	Nose Wheel Steering System Test (Airplane on Ground) (P/B 501)
32-51-31-400-802	Nose Gear Steering System Cables Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-1901	Spanner Wrench - Steering Collar Retainer, NLG Part #: C32040-6 Supplier: 81205 Opt Part #: C32040-1 Supplier: 81205
SPL-13966	Clamp Assy - Steering Cables, NLG Part #: C32055-18 Supplier: 81205
STD-837	String

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Steering collar	32-51-61-01-070	PTF ALL
50	Cotter pin	32-51-51-01-215	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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F. Steering Collar Installation

NOTE: Make sure that you install the matching steering collar [5] for the steering sleeve installed. Steering collar assembly part number 162A1404-7 can be installed with a steering sleeve assembly part number 162A1405-11 or 162A1405-15 and only steering collar assembly part number 162A1420-2 can be installed with steering sleeve assembly part number 162A1421-3.

SUBTASK 32-51-61-860-011

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-51-61-600-003

(2) Apply a thin layer of grease, D00633 to the bushing faces and grooves on the steering collar [5].

SUBTASK 32-51-61-420-006

(3) Install the steering collar [5] on the outer cylinder [60]o.

SUBTASK 32-51-61-420-007

(4) Install the lower steering plate [54] on the outer cylinder [60] and hold it in position.

SUBTASK 32-51-61-600-004

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a thin layer of corrosion inhibiting material, G50136, to the threads, thread reliefs, and mating surfaces of the steering collar retainer nut [58].

SUBTASK 32-51-61-420-008

(6) Move the steering collar retainer nut [58] up the outer cylinder [60] and install it below the lower steering plate [54].

(a) Tighten the steering collar retainer nut [58] 62.5 ±12.5 ft-lb (85 ±17 N·m) with a spanner wrench, SPL-1901.

(b) If it is necessary, loosen the steering collar retainer nut [58] a minimum amount to install the lock tab [57], washers [56], and bolts [55].

SUBTASK 32-51-61-020-027

(7) Install the standoffs [52], bolts [53], and washers [51] that hold the metering valve assembly to the upper steering sleeve [59] and lower steering plate [54].

SUBTASK 32-51-61-980-005

(8) Turn the steering collar [5] far enough to permit access to the rod end of the actuator assembly [47] that you will install.

SUBTASK 32-51-61-420-009

(9) Install the bushings [46], anti-rotation washers [65] (not on all airplanes), pin [45], washer [48], and nut [49] to connect the rod end of the actuator assembly [47] to the steering collar [5].

(a) Make sure that the flange bushing [46] is put in the face-up in the direction, towards the actuator assembly [47].

(b) Tighten the nut [49] to 54 ±4 ft-lb (73 ±6 N·m).

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(c) If it is necessary, loosen the nut [49] to align the castellations on the nut [49] with the hole on the pin [45].

(d) Install the new cotter pin [50].

SUBTASK 32-51-61-410-005

(10) Do this task: NLG Inner Cylinder Installation, TASK 32-21-24-400-801.

SUBTASK 32-51-61-420-010

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(11) Connect the upper torsion link [13] to the steering collar [5]:

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The chrome surface on the shank of the torsion link pin [6].

2) The threads of the nut [2].

3) The faces of the washer [4].

4) The mating surfaces of the upper torsion link [13] and the steering collar [5].

(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) Put the upper torsion link [13] in its position on the steering collar [5].

(d) Put the torsion link pin [6] through the upper torsion link [13] and the steering collar [5].

(e) Apply a thin layer of the corrosion inhibiting material, G50136, to the threads and thread reliefs of the torsion link pin [6].

(f) Install the washer [4] and nut [2] on the torsion link pin [6].

(g) Tighten the nut [2] to 275 ±25 in-lb (31 ±3 N·m) more than the run-on torque.

(h) If it is necessary, loosen the nut [2] to the nearest castellation to align the holes for the torsion link pin [6].

(i) Install the pin [3] in the torsion link pin [6].

SUBTASK 32-51-61-020-028

(12) Connect the sensor brackets [15] to the upper torsion link [13]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and thread reliefs of the bolts [16]

2) The threads of the nuts [19]

3) The faces of the washers [17] and washers [18].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) If a string, STD-837, is attached to the sensor brackets [15] and the outer cylinder [60], remove the string, STD-837.

(d) Put the sensor brackets [15] in their position on the upper torsion link [13].

(e) Install the washers [17] on the bolts [16].

(f) Put the bolts [16] through the upper torsion link [13] and the sensor brackets [15].

(g) Install the washers [18] and the nuts [19] on the bolts [16].

(h) Tighten the nuts [19].

SUBTASK 32-51-61-020-029

(13) Connect the box assembly [1] to the upper torsion link [13]:

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to these items:

1) The threads and shank of the bolts [30]]

2) The threads of the nuts [33]

3) The faces of the washers [31] and washers [32].

(b) Remove all unwanted corrosion inhibiting material, G50136.

(c) If a string, STD-837, is attached to the box assembly [1], and the outer cylinder [60], remove the string, STD-837.

(d) Put the box assembly [1] in its position on the upper torsion link [13].

(e) Install the washer [31] on the bolts [30]].

(f) Put the bolts [30] through the bracket for the box assembly [1] and the upper torsion link [13].

(g) Install the washers [32] and nuts [33] on the bolts [30].

SUBTASK 32-51-61-410-006

(14) Reconnect the NWS-A/B cable [35] to the steering collar [5] (TASK 32-51-31-400-802).

SUBTASK 32-51-61-080-002

(15) Remove the cable clamp, SPL-13966.

SUBTASK 32-51-61-980-006

(16) Move the inner cylinder from the fully extended to the fully compressed position.

(a) Make sure that the total travel is 15.50 in. (393.70 mm).

SUBTASK 32-51-61-080-003

(17) Remove rip pin NS4 from the summing mechanism.

SUBTASK 32-51-61-860-012

(18) Remove the rigging pin kit, SPL-1585, from the towing lever.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Steering Collar Installation Test

SUBTASK 32-51-61-610-002

(1) Service the shock strut for the nose gear (TASK 12-15-41-610-802).

SUBTASK 32-51-61-710-002

(2) To do a test on the steering system of the nose wheel, do the applicable procedures in this task: Nose Wheel Steering System Test (Airplane on Ground), TASK 32-51-00-700-801.

Do these procedures

- General
- Prepare to Test Steering system of the Nose Wheel Well
- Do the Test of the Normal Steering Response
- Do the Test of Nose Gear Turn Radius and Response to Full Rudder Pedal Input
- Put the Airplane Back to Its Usual Condition

(a) Make sure that there is no leakage.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-61-580-004

(1) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-51-61-640-002

(2) Lubricate the lube fittings on the upper torsion link [13] and the steering collar [5] (TASK 12-21-21-640-802).

——— END OF TASK ———

TASK 32-51-61-000-803

6. Upper Steering Sleeve Removal

(Figure 401)

A. References

Reference	Title
32-51-11-000-801	Steering Metering Valve Removal (P/B 401)
32-51-51-000-801	Nose Gear Steering Actuator Removal (P/B 401)

B. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Prepare for the Removal

SUBTASK 32-51-61-010-005

(1) Remove the steering metering valve (TASK 32-51-11-000-801).

SUBTASK 32-51-61-010-006

(2) Remove the two actuator assemblies [47] from the upper steering sleeve [59] (TASK 32-51-51-000-801).

SUBTASK 32-51-61-010-007

(3) Remove the steering collar [5] (TASK 32-51-61-000-801 or TASK 32-51-61-000-802).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Upper Steering Sleeve Removal

(Figure 401)

SUBTASK 32-51-61-010-008

(1) Scrape carefully, to remove the fillet seal of sealant from the upper steering sleeve [59] and the outer cylinder [60].

SUBTASK 32-51-61-010-009

(2) Remove the pins [61], nuts [62], washers [63], and bolts [64] from the upper steering sleeve [59] and outer cylinder [60].

NOTE: You may have to remove sealant from the pins [61], nuts [62] and pins [61] to remove the bolts [64].

SUBTASK 32-51-61-010-010

(3) Remove, by lowering, the upper steering sleeve [59] from the outer cylinder [60].

SUBTASK 32-51-61-010-011

(4) Remove all remaining sealant or grease from the upper steering sleeve [59] and from the outer cylinder [60].

——— END OF TASK ———

TASK 32-51-61-400-803

7. Upper Steering Sleeve Installation

(Figure 401)

A. References

Reference	Title
32-51-11-400-801	Steering Metering Valve Installation (P/B 401)
32-51-51-400-801	Nose Gear Steering Actuator Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A00226	Compound - Tamper-Proof Putty	BMS8-45
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C50075	Coating - Protective Enamel (BAC 707 Gray Color)	BMS10-60 Type II
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Steering collar	32-51-61-01-070	PTF ALL
61	Pin	32-51-61-01-015	PTF ALL

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right

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(Continued)

Zone	Area
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Upper Steering Sleeve Installation

SUBTASK 32-51-61-410-007

(1) Apply grease, D00633, to the chrome-plated (lube fitting) part of the outer cylinder [60].

SUBTASK 32-51-61-410-008

(2) Install the upper steering sleeve [59] on the outer cylinder [60] until the bushings of the upper steering sleeve [59] and outer cylinder [60] align.

SUBTASK 32-51-61-410-009

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Apply a thin layer of corrosion inhibiting material, G50136, to the shanks, thread reliefs, and threads of the bolts [64] and faces of the washers [63].

(a) Remove all unwanted corrosion inhibiting compound.

SUBTASK 32-51-61-410-010

(4) Install the bolts [64], washers [63] and nut [62] through the upper steering sleeve [59] into the outer cylinder [60].

SUBTASK 32-51-61-410-011

(5) Tighten the nuts [62] to 160 in-lb (18 N·m) to 190 in-lb (21 N·m) above the run-on torque.

(a) If it is necessary, loosen each nut [62] to the nearest castellation to align the holes of the pins [61].

SUBTASK 32-51-61-410-012

(6) Install the pins [61] and apply compound, A00226, to fully cover the nuts [62], washers [63] and pins [61].

NOTE: The compound, A00226, for is for protection. If it is tampered, it causes the seal to break.

SUBTASK 32-51-61-410-013

(7) Fill the space between the outer cylinder [60] and upper steering sleeve [59] with sealant, A00247.

(a) Fillet seal around the outer cylinder [60] and upper steering sleeve [59].

(b) Apply coating, C50075, to the sealant and the area around the sealant.

SUBTASK 32-51-61-410-014

(8) Apply a cap seal of sealant, A02315, or sealant, A00247, to the bolts [64], washers [63], nuts [62], and new pins [61] such that the entire end of the assembly, above the flange, is covered with sealant.

NOTE: The purpose of the sealant is to prevent the adjacent conduit from catching on the attachment bolt assembly during nose gear steering activities.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Form the sealant as flat and smooth as possible to let the conduit easily pass over the bolt assembly.

(b) No bumps, ridges or sharp edges more than 0.50 in. (12.70 mm) are permitted.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-51-61-410-015

(1) Install the steering collar [5] (TASK 32-51-61-400-801, or TASK 32-51-61-400-802).

SUBTASK 32-51-61-010-012

(2) Install the two actuator assemblies [47] to the upper steering sleeve [59] (TASK 32-51-51-400-801).

SUBTASK 32-51-61-410-016

(3) Install the steering metering valve (TASK 32-51-11-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

RUDDER PEDAL STEERING ROTARY ACTUATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the rotary actuator for the rudder pedal steering

(2) Installation of the rotary actuator for the rudder pedal steering.

TASK 32-51-81-000-801

2. Rudder Pedal Steering Rotary Actuator Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
117	Electrical and Electronics Compartment - Left

D. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

E. Prepare for the Removal

SUBTASK 32-51-81-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-51-81-480-002

(2) Make sure the tires of landing gear have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-51-81-010-001

(3) Remove the forward access panel from the nose wheel well to get access to the turnbuckles of the NGPP-A and NGPP-B cables, and the rotary actuator [3] (TASK 53-14-01-020-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

SUBTASK 32-51-81-860-001

(4) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

F. Rudder Pedal Steering Rotary Actuator Removal

SUBTASK 32-51-81-020-001

(1) Disconnect the NGPP-A and NGPP-B cables:

(a) Remove the locking clips from the turnbuckles of the NGPP-A and NGPP-B cables.

(b) Loosen the tension at the turnbuckles of the NGPP-A and NGPP-B cables.

(c) Remove the pins [4] that hold the ends of the NGPP-A and NGPP-B cables on each side of the pulley of the rotary actuator [3].

(d) Remove the NGPP-A and NGPP-B cables from the pulley of the rotary actuator [3].

SUBTASK 32-51-81-480-003

(2) Install these rig pins:

(a) Rig pin NS 1, from rigging pin kit, SPL-1585, in the interconnect quadrant [2]

(b) Rig pin NS 5 in the interconnect eccentric drum [1]:

1) Turn the interconnect eccentric drum [1] by hand.

2) Make sure the hole for rig pin NS 5 and the bracket rig pin slot line up.

3) Install rig pin NS 5.

SUBTASK 32-51-81-020-002

(3) Disconnect the electrical connnector [5] from the top of the rotary actuator [3].

SUBTASK 32-51-81-020-003

(4) Remove the rotary actuator [3]:

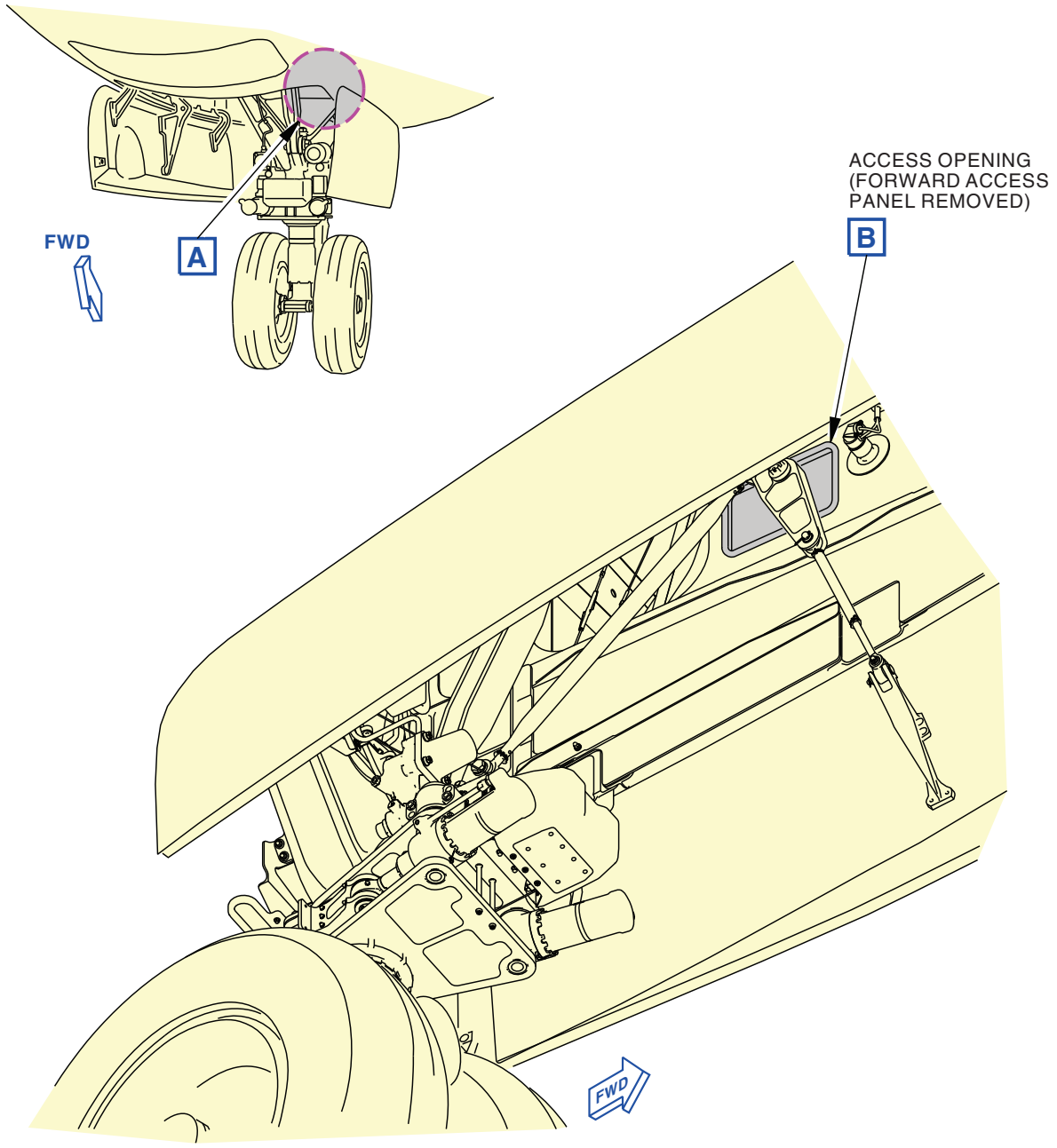
(a) Remove the bolts [6] and washers [7] that hold the rotary actuator [3] to the support bracket.

(b) Remove the washers [7] from the bolts [6].

(c) Remove the rotary actuator [3] from the airplane.

——— END OF TASK ———

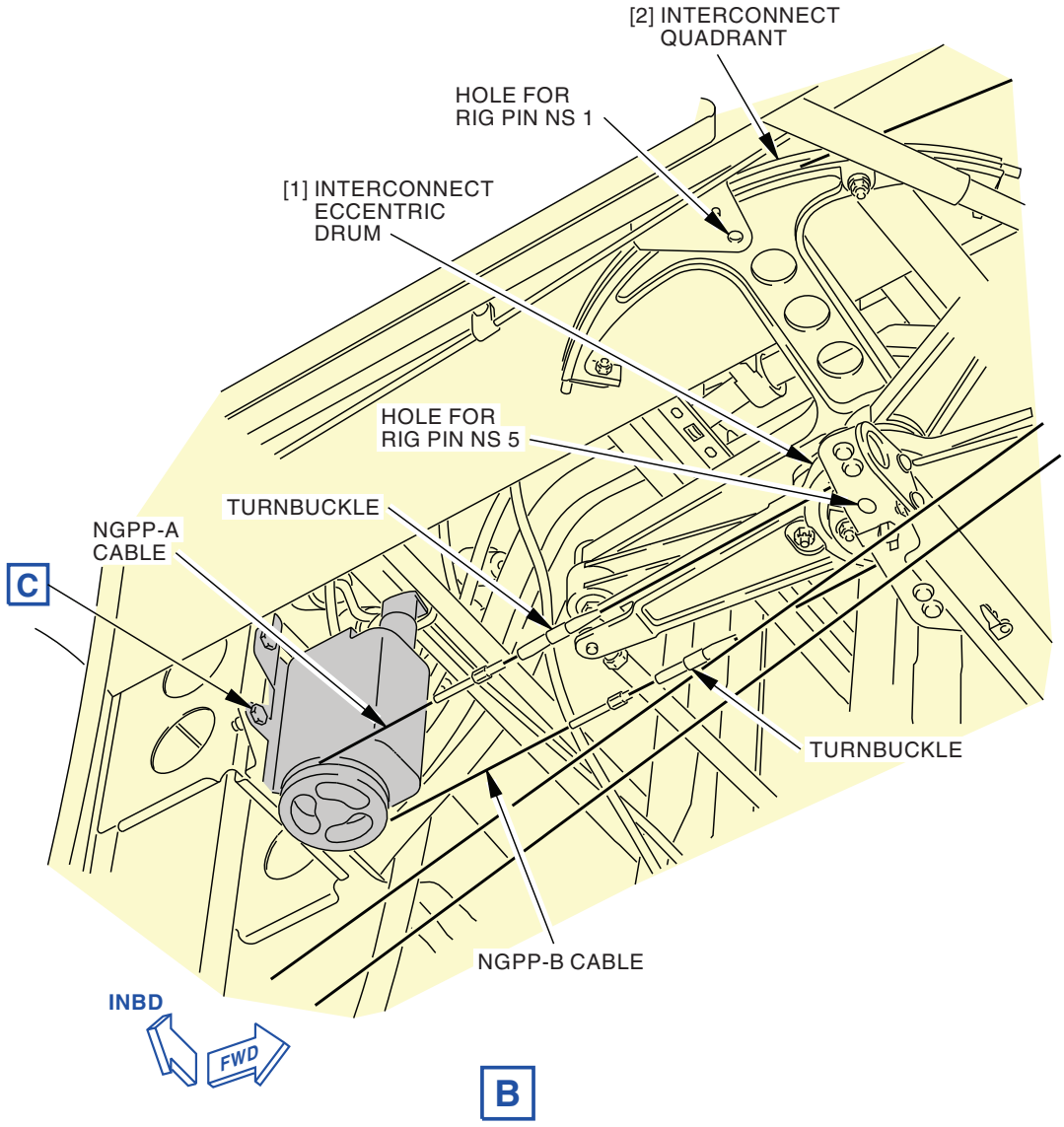
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Rudder Pedal Steering Rotary Actuator Installation
Figure 401/32-51-81-990-801 (Sheet 1 of 3)

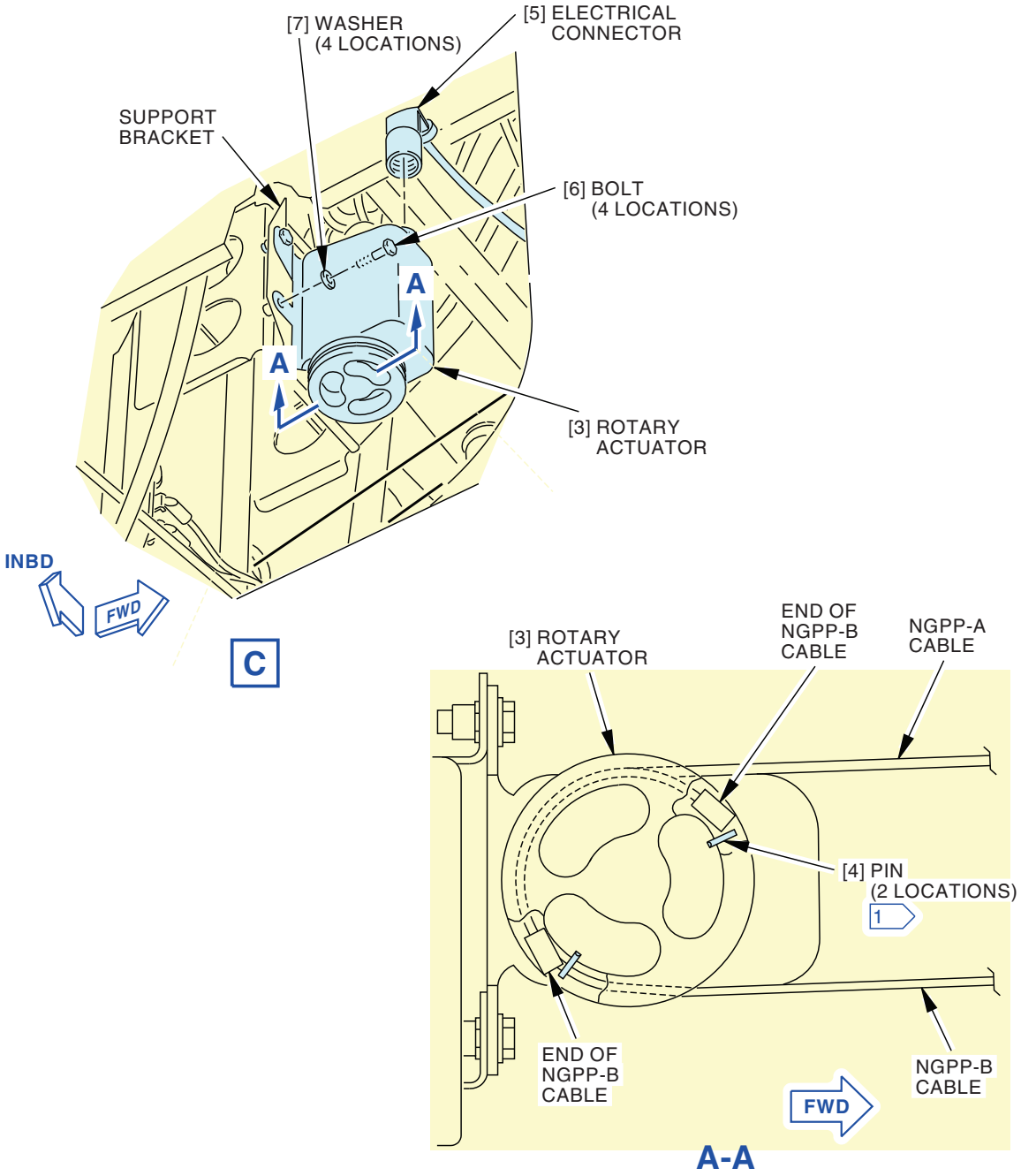
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Rudder Pedal Steering Rotary Actuator Installation
Figure 401/32-51-81-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 CAUTION: MAKE SURE THE ENDS OF THE PINS [4] DO NOT TOUCH THE HOUSING OF THE ACTUATOR [3] IF THE PINS [4] TOUCH THE HOUSING, DAMAGE TO EQUIPMENT CAN OCCUR.

2414525 S00061535357_V1

Rudder Pedal Steering Rotary Actuator Installation
Figure 401/32-51-81-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-51-81-400-801

3. Rudder Pedal Steering Rotary Actuator Installation

(Figure 401)

A. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
32-51-00-820-803	Rudder Pedal Interconnect Mechanism Adjustment (P/B 501)
51-21-41-370-801	Bonderite M-CR 1001 Application Process (P/B 701)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Rotary actuator	32-51-81-02-005	PTF ALL
4	Pin	32-51-31-02-005	PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
117	Electrical and Electronics Compartment - Left

F. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel

G. Prepare for the Installation

SUBTASK 32-51-81-100-001

(1) Do these steps to supply a good electrical bond between the support bracket and the rotary actuator [3]:

(a) Fully clean the mating surfaces of the support bracket and the rotary actuator [3] (TASK 20-10-34-120-801).

(b) Chemically prepare the mating surfaces of the support bracket and the rotary actuator [3] with the coating, C00064 (TASK 51-21-41-370-801).

H. Rudder Pedal Steering Rotary Actuator Installation

SUBTASK 32-51-81-420-001

(1) Install the rotary actuator [3]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply a thin layer of the corrosion inhibiting material, G50136, to all the areas of the fastener holes in the support bracket.

(b) Put the rotary actuator [3] in its position for attachment to the support bracket.

(c) Align the holes in the rotary actuator [3] to the holes in the support bracket.

(d) Install the bolts [6] and washers [7] that attach the rotary actuator [3] to the support bracket.

SUBTASK 32-51-81-760-001

(2) Measure the resistance between the rotary actuator [3] and the support bracket with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

(a) Make sure the resistance is less than 0.0025 ohm (2.5 milliohms).

SUBTASK 32-51-81-410-001

(3) Before you connect the electrical connnector [5] at the top of the rotary actuator [3], examine the electrical connnector [5] for corrosion.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

(b) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connnector [5] (SWPM 20-60-03).

(4) Connect the electrical connnector [5] at the top of the rotary actuator [3].

SUBTASK 32-51-81-410-002

(5) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	17	C01027	LANDING GEAR NOSE GEAR STEER

SUBTASK 32-51-81-860-002

(6) Put airplane into the air mode (TASK 32-09-00-860-801).

SUBTASK 32-51-81-210-001

(7) Make sure marks align on rotary actuator [3].

SUBTASK 32-51-81-420-002

(8) Connect the NGPP-A and NGPP-B cables:

(a) Put the ends of NGPP-A and NGPP-B cables in their position for connection to the pulley of the rotary actuator [3].

CAUTION

DO NOT LET THE ENDS OF THE PINS TOUCH THE HOUSING OF THE ACTUATOR WHEN THE CABLES ARE INSTALLED. DAMAGE TO THE ROTARY ACTUATOR CAN OCCUR.

(b) Install the new pins [4] to hold the ends of the NGPP-A and NGPP-B cables on each side of the pulley of the rotary actuator [3].

SUBTASK 32-51-81-820-001

(9) Adjust the NGPP-A and NGPP-B cables (TASK 32-51-00-820-803).

SUBTASK 32-51-81-080-001

(10) Remove rig pin NS 1, from rigging pin kit, SPL-1585, from the interconnect quadrant [2].

SUBTASK 32-51-81-080-002

(11) Remove rig pin NS 5, from rigging pin kit, SPL-1585, from the interconnect eccentric drum [1].

SUBTASK 32-51-81-840-001

(12) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 32-51-81-480-004

(13) Install the locking clips on the turnbuckles of NGPP-A and NGPP-B cables.

SUBTASK 32-51-81-410-003

(14) Install the forward access panel in the nose wheel well (TASK 53-14-01-420-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Close this access panel:

Number	Name/Location
113AW	Forward Nose Wheel Wall Panel

SUBTASK 32-51-81-860-003

(15) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LANDING GEAR POSITION INDICATING AND WARNING SYSTEM - ADJUSTMENT/TEST

1. General

A. This task does a test of the position indication and warning system for the landing gear. It is necessary to lift the airplane on jacks to let the landing gear retract and extend.

TASK 32-61-00-710-801

2. Landing Gear Position Indicating and Warning System Test

(Figure 501)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
24-22-00-860-803	Supply External Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-10-740-801	Proximity Switch Electronics Unit - Operational Test (P/B 501)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
713	Nose Landing Gear
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Test

SUBTASK 32-61-00-860-001

(1) Do this task: Supply External Power, TASK 24-22-00-860-803.

SUBTASK 32-61-00-860-002

(2) Make sure that the hydraulic systems A and B are not pressurized (TASK 29-11-00-860-805).

SUBTASK 32-61-00-580-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(3) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-00-860-003

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-00-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(5) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-61-00-860-004

WARNING MAKE SURE THAT YOU OPEN THE CIRCUIT BREAKER FOR THE WEATHER RADAR SYSTEM. THE FORWARD MOVEMENT OF A THRUST LEVER CAN CAUSE THE AUTOMATIC OPERATION OF THE SYSTEM. THE OPERATION OF THIS SYSTEM CAN CAUSE SERIOUS INJURY TO PERSONS AND DAMAGE TO EQUIPMENT IN THE AREA OF THE NOSE RADOME.

(6) Move the engine thrust levers to the full forward position.

NOTE: Levers in the advanced position prevent the landing gear warning (red) lights from remaining on after gear is up and locked.

D. Indicator Lights Test

SUBTASK 32-61-00-860-005

(1) Put the landing gear control lever in the DN position.

(a) Make sure that the green landing gear indicator lights above the landing gear lever and on the overhead panel are on for each gear.

NOTE: Auxiliary landing gear position indicator lights are located on the P5 aft overhead panel.

SUBTASK 32-61-00-860-007

(2) Put the LIGHTS test switch on the P2 panel in the DIM position.

(a) Make sure that the green indicator lights go dim.

E. Nose Landing Gear and Indication Test

SUBTASK 32-61-00-860-008

(1) Do these steps to raise the nose landing gear.

(a) Move the landing gear lever to the DN position.

(b) Remove the downlock pin from the nose landing gear only (TASK 32-00-01-080-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Make sure that all persons and equipment are clear of the landing gear area.

(d) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(e) Move the landing gear lever to the UP position.

SUBTASK 32-61-00-210-001

(2) Look at the landing gear indicator lights.

(a) Make sure that the green indicator lights for the nose landing gear go off when the lock braces unlock.

(b) Make sure that the red warning lights for the nose landing gear go off when the gear retracts to the up and locked position.

(c) Make sure that the green indicator lights for the main landing gear stay on.

NOTE: The green down-and-locked lights for the gears that have the downlock pins installed can stay on or can go off depending on the rigging and tolerances.

(d) Make sure that the red warning lights for the main landing gear go on.

SUBTASK 32-61-00-860-009

(3) Lower the nose landing gear.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL EXTEND WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE DN POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(a) Make sure that all persons and equipment are clear of the landing gear area.

(b) Move the landing gear lever to the DN position.

SUBTASK 32-61-00-580-003

(4) Install the downlock pin on the nose landing gear (TASK 32-00-01-480-801).

F. Left Main Landing Gear and Indication Test

SUBTASK 32-61-00-860-010

(1) Do these steps to raise the left main landing gear.

(a) Remove the downlock pin from the left main landing gear (TASK 32-00-01-080-801).

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL RETRACT WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE UP POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(b) Make sure that all persons and equipment are clear of the landing gear area.

(c) Move the landing gear lever to the UP position.

SUBTASK 32-61-00-210-002

(2) Look at the landing gear indicator lights.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the green indicator lights for the left main landing gear go off when the lock braces unlock.

(b) Make sure that the red warning lights for the left main landing gear go off when the gears retract to the up and locked position.

(c) Make sure that the green indicator lights for the nose and right main landing gear stay on.

NOTE: The green down-and-locked lights for the gears that have the downlock pins installed can stay on or can go off depending on the rigging and tolerances.

(d) Make sure that the red warning lights for the nose and right main landing gear go on.

SUBTASK 32-61-00-860-011

(3) Lower the left main landing gear.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL EXTEND WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE DN POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(a) Make sure that all persons and equipment are clear of the landing gear area.

(b) Move the landing gear lever to the DN position.

SUBTASK 32-61-00-580-004

(4) Install the downlock pin on the left main landing gear (TASK 32-00-01-480-801).

G. Right Main Landing Gear and Indication Test

SUBTASK 32-61-00-860-012

(1) Do these steps to raise the right main landing gear.

(a) Remove the downlock pin from the right main landing gear (TASK 32-00-01-080-801).

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL RETRACT WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE UP POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(b) Make sure that all persons and equipment are clear of the landing gear area.

(c) Move the landing gear lever to the UP position

SUBTASK 32-61-00-210-003

(2) Look at the landing gear indicator lights.

(a) Make sure that the green indicator lights for the right main landing gear go off when the lock braces unlock.

(b) Make sure that the red warning lights for the right main landing gear go off when the gears retract to the up and locked position.

(c) Make sure that the green indicator lights for the nose and left main landing gear stay on.

NOTE: The green down-and-locked lights for the gears that have the downlock pins installed can stay on or can go off depending on the rigging and tolerances.

(d) Make sure that the red warning lights for the nose and left main landing gear go on.

SUBTASK 32-61-00-860-013

(3) Lower the right main landing gear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL EXTEND WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE DN POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(a) Make sure that all persons and equipment are clear of the landing gear area.

(b) Move the landing gear lever to the DN position.

SUBTASK 32-61-00-580-005

(4) Install the downlock pin on the right main landing gear (TASK 32-00-01-480-801).

H. All Landing Gear and Indication Test

SUBTASK 32-61-00-860-014

(1) Do these steps to raise all the landing gear:

(a) Remove the downlock pins from the landing gear (TASK 32-00-01-080-801).

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL RETRACT WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE UP POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(b) Make sure that all persons and equipment are clear of the landing gear area.

(c) Move the landing gear lever to the UP position.

SUBTASK 32-61-00-210-004

(2) Look at the landing gear indicator lights.

(a) Make sure that all the green indicator lights for the landing gear go off when the lock braces unlock.

(b) Make sure that the red warning lights for all the landing gear go off when the gears retract to the up and locked position.

SUBTASK 32-61-00-730-001

(3) Put the LIGHTS test switch, on panel P2, in the TEST position.

(a) Make sure that the green and red lights for the landing gear come on.

SUBTASK 32-61-00-730-002

(4) Release the LIGHTS test switch.

(a) Make sure that the green and red lights for the landing gear go off.

SUBTASK 32-61-00-860-016

(5) Lower the landing gear.

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE LANDING GEAR AREA. THE LANDING GEAR WILL EXTEND WHEN THE LANDING GEAR CONTROL LEVER IS PUT IN THE DN POSITION. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO THE EQUIPMENT.

(a) Make sure that all persons and equipment are clear of the landing gear area.

(b) Move the control lever for the landing gear to the DN position.

(c) Make sure that all three red warning lights above the gear lever come on while the gear moves to the down position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Look for these indications when the landing gear goes to the down and locked position:

1) Make sure that the red warning lights go off.

2) Make sure that the green indicator lights above the gear lever and overhead on the P5 panel come on.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-61-00-580-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-00-860-017

(2) If hydraulic power is not necessary, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-61-00-860-018

(3) Move the engine thrust levers to the idle position.

SUBTASK 32-61-00-860-019

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(4) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

SUBTASK 32-61-00-740-001

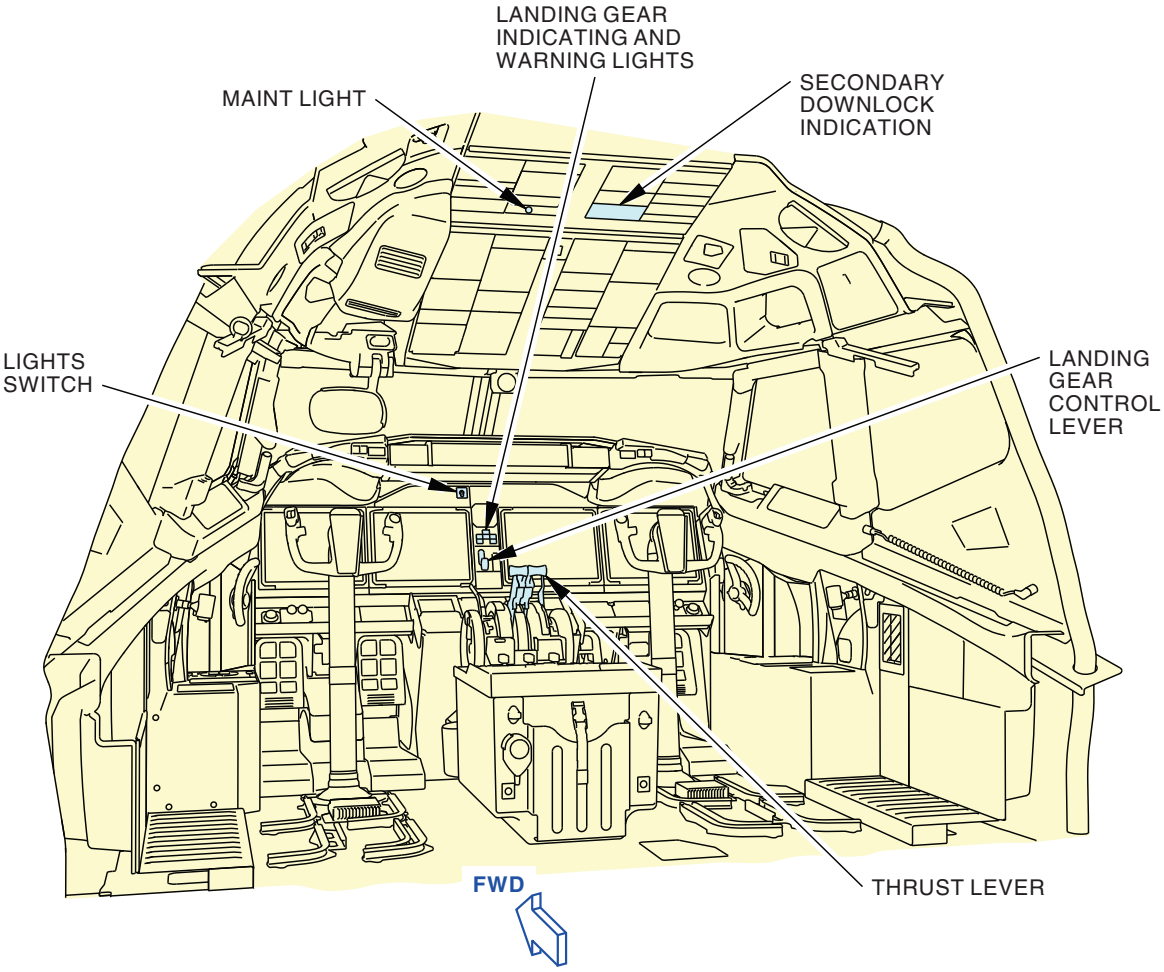
(5) Do a check of the PSEU for stored faults using the SELF-TEST (TASK 32-09-10-740-801).

SUBTASK 32-61-00-860-020

(6) Put the safety-sensitive systems back to their initial conditions (TASK 32-09-00-840-802).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLIGHT COMPARTMENT

2414526 S00061535363_V1

Landing Gear Position Indicating and Warning Test
Figure 501/32-61-00-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR UPLOCK SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Removal of the main landing gear uplock sensor

(2) Installation of the main landing gear uplock sensor.

B. The procedures for the left and right installations are similar.

TASK 32-61-21-020-801

2. Main Landing Gear Uplock Sensor Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
SWPM 20-30-12	Assembly of Splices

B. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

C. Prepare for the Removal

SUBTASK 32-61-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-61-21-480-002

(2) Make sure chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-61-21-840-001

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

SUBTASK 32-61-21-860-001

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Main Landing Gear Uplock Sensor Removal

SUBTASK 32-61-21-020-001

(1) Remove the sensor wires including the splices (SWPM 20-30-12).

SUBTASK 32-61-21-020-002

(2) Do these steps to remove the sensor [5]:

(a) Remove the nuts [4], washers [2], washers [3] and bolts [1] that attach the sensor [5] to the mounting bracket.

(b) Remove the sensor [5] from the mounting bracket.

SUBTASK 32-61-21-210-001

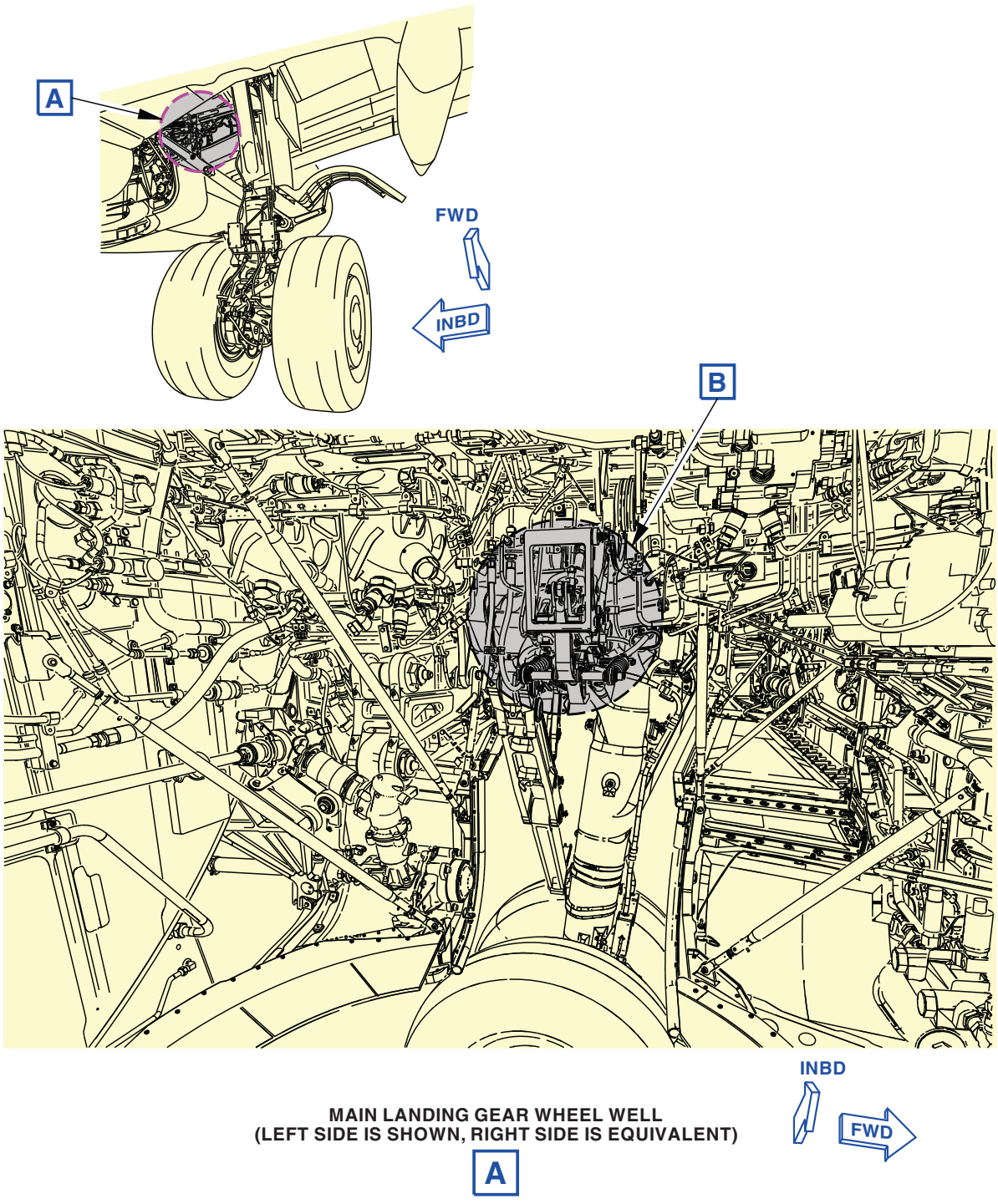
(3) Examine the sensor [5] and adjacent areas for signs of damage.

NOTE: Damage to the sensor [5] can be an indication of an incorrect clearance between the sensor [5] and the target.

(a) If you find damage, make a note of the damage for the installation procedure.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

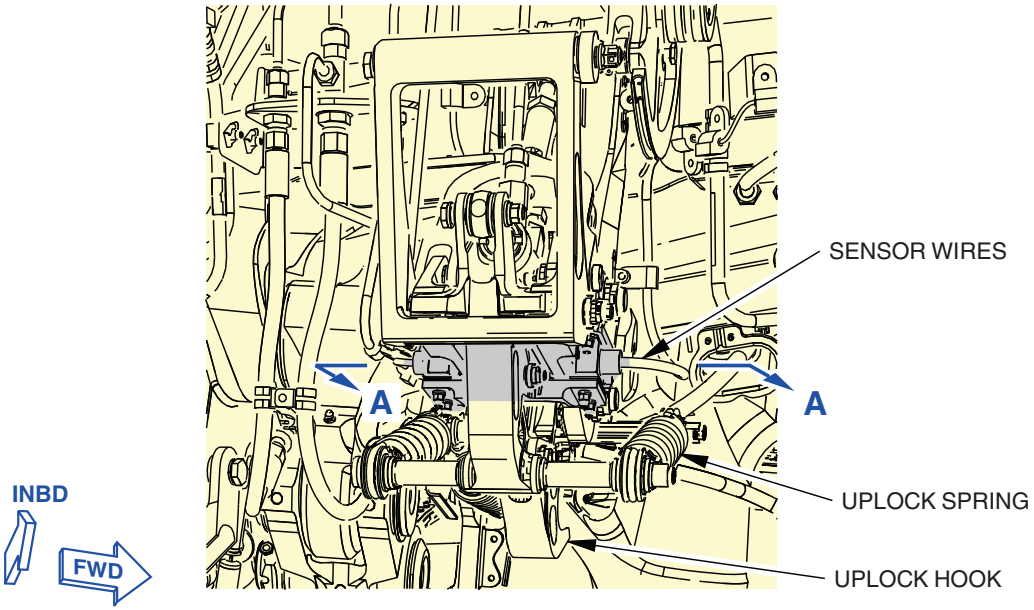


MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

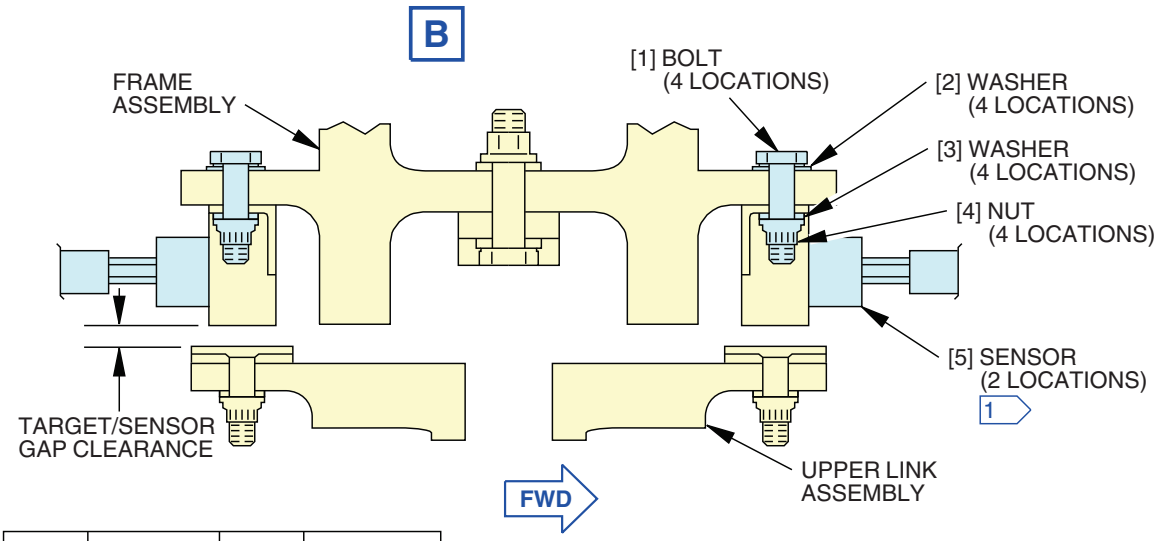
2414527 S00061535367_V1

Main Landing Gear Uplock Sensor Installation
Figure 401/32-61-21-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT MAIN LANDING GEAR UPLOCK
(RIGHT MAIN LANDING GEAR UPLOCK IS EQUIVALENT)



1

ITEM	SENSOR NO.	MLG	LOCATION
[5]	S0072	L	FWD
[5]	S0074	R	FWD
[5]	S1016	L	AFT
[5]	S1017	R	AFT

UP-AND-LOCKED SENSORS
A-A

2414529 S00061535369_V5

Main Landing Gear Uplock Sensor Installation
Figure 401/32-61-21-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-21-400-801

3. Main Landing Gear Uplock Sensor Installation

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-801	Prepare to Put the Airplane in the Air Mode (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-61-21-400-802	Main Landing Gear Uplock Sensor Clearance Measurement (Airplane on Jacks) (P/B 501)
32-61-21-400-803	Main Landing Gear Uplock Sensor Clearance Measurement (Airplane not on Jacks) (P/B 501)
32-61-21-710-801	Main Landing Gear Up-and-Locked Sensor Operational Test (P/B 501)
SWPM 20-30-12	Assembly of Splices

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Sensor	32-61-21-01-040	PTF ALL

D. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

E. Prepare for the Installation

SUBTASK 32-61-21-580-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-21-480-003

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-61-21-860-050

(3) Prepare the safety-sensitive systems for air mode simulation. Do this task: Prepare to Put the Airplane in the Air Mode, TASK 32-09-00-840-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-21-860-003

(4) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

F. Main Landing Gear Uplock Sensor Installation

SUBTASK 32-61-21-410-001

(1) Fay seal the bottom of the sensor with sealant, A00247.

SUBTASK 32-61-21-420-001

(2) Install the sensor [5]:

(a) Position the sensor [5] on the bracket.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin coat of Cor-Ban 27L Compound, G50237, to thread reliefs, threads, and washer faces before the assemble.

(c) Install the bolts [1], washers [2], washers [3], and nuts [4].

NOTE: Use CRES washers below nuts with titanium sensors. Use cad-plated washers below bolt heads.

SUBTASK 32-61-21-420-002

(3) Connect the sensor leads to the airplane:

(a) If it is necessary, cut the sensor leads to the correct length.

(b) Connect the leads with splices (SWPM 20-30-12).

SUBTASK 32-61-21-960-001

(4) If damage to the sensor [5] or the adjacent area during the removal procedure was found, repair or replace all damaged components or areas.

NOTE: Damage to the sensor, the target, or the adjacent areas may be an indication of an incorrect sensor clearance.

G. Main Landing Gear Uplock Sensor Installation Test

SUBTASK 32-61-21-220-005

(1) Measure the sensor [5] clearance, do the applicable task:

- Main Landing Gear Uplock Sensor Clearance Measurement (Airplane not on Jacks), TASK 32-61-21-400-803
- Main Landing Gear Uplock Sensor Clearance Measurement (Airplane on Jacks), TASK 32-61-21-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-21-860-022

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-61-21-860-023

(3) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

SUBTASK 32-61-21-740-004

(4) Do this task: Main Landing Gear Up-and-Locked Sensor Operational Test, TASK 32-61-21-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR UPLOCK SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has four tasks:

(1) Main landing gear uplock sensor clearance measurement (Airplane on Jacks)

(2) Main landing gear uplock sensor clearance measurement (Airplane not on Jacks)

(3) Main landing gear uplock sensor clearance adjustment

(4) Main landing gear up-and-lock sensor operational test.

B. The procedures for the left and right installations are similar.

TASK 32-61-21-400-802

2. Main Landing Gear Uplock Sensor Clearance Measurement (Airplane on Jacks)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
G02020	Clay, Modeling	

C. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

D. Main Landing Gear Uplock Sensor Clearance Measurement

SUBTASK 32-61-21-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(1) Lift the airplane on jacks (TASK 07-10-01-580-801).

SUBTASK 32-61-21-480-004

(2) Attach a minimum of 0.5 in. (12.7 mm) depth clay, G02020, to the entire face of the sensor.

SUBTASK 32-61-21-860-004

(3) Do these steps to raise the main landing gear:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) For the landing gear that the sensor is attached to, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

NOTE: Refer to Figure 501 for the sensor location.

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(c) Put the landing gear lever in the UP position.

SUBTASK 32-61-21-860-005

(4) Put the landing gear lever in the DN position after the main landing gear has retracted fully.

SUBTASK 32-61-21-580-003

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(5) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-21-020-004

(6) Remove the clay, G02020, from the face of the sensor.

SUBTASK 32-61-21-860-006

(7) If hydraulic system A power is not necessary, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-61-21-220-001

(8) Measure the compressed thickness of clay, G02020.

SUBTASK 32-61-21-820-001

(9) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Uplock Sensor Clearance Adjustment, TASK 32-61-21-820-801.

NOTE: Write down the measured clearance so you will know what adjustments must be made.

SUBTASK 32-61-21-840-002

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

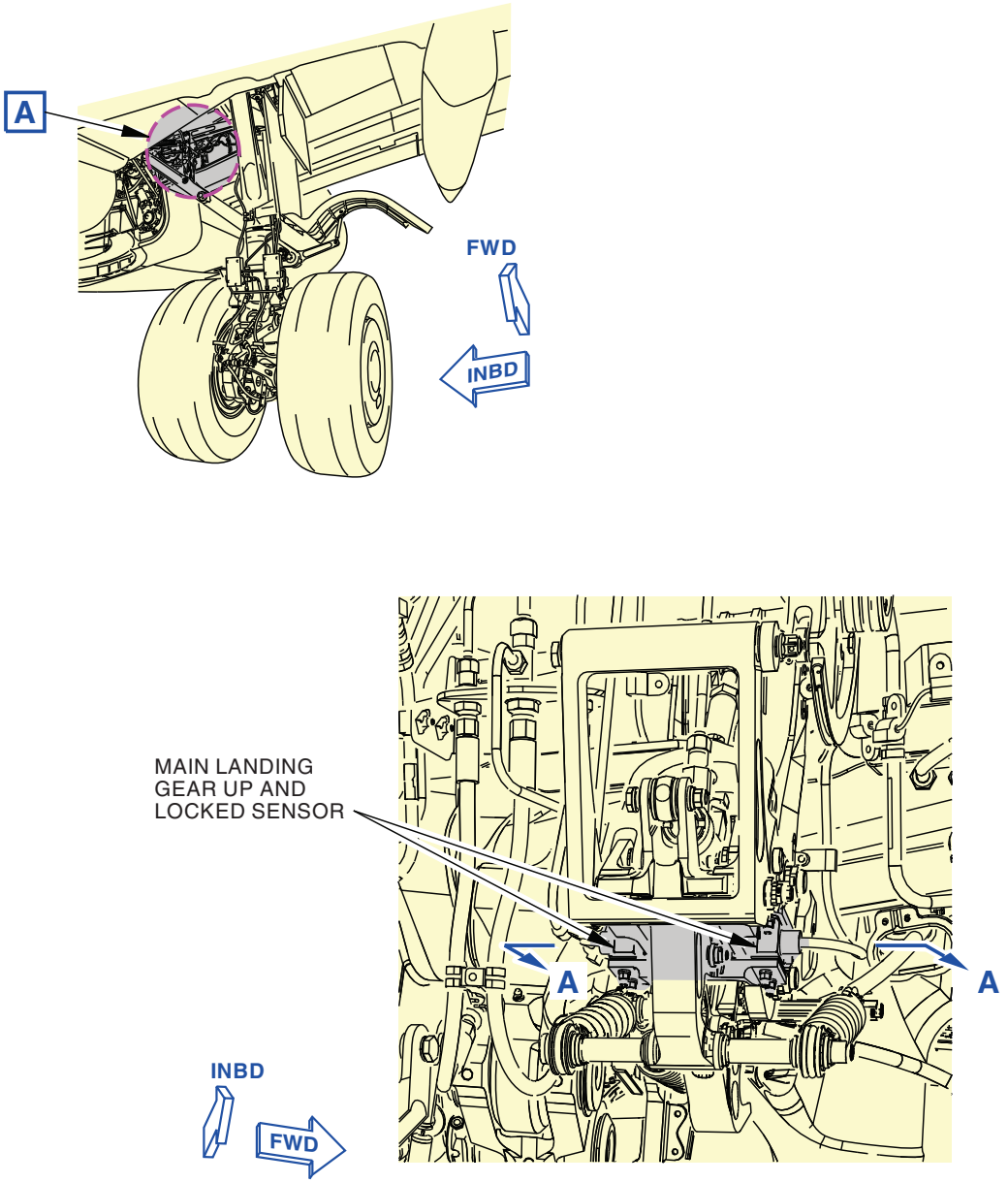
(CAUTION PRECEDES)

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(10) If the sensor clearance is acceptable, do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

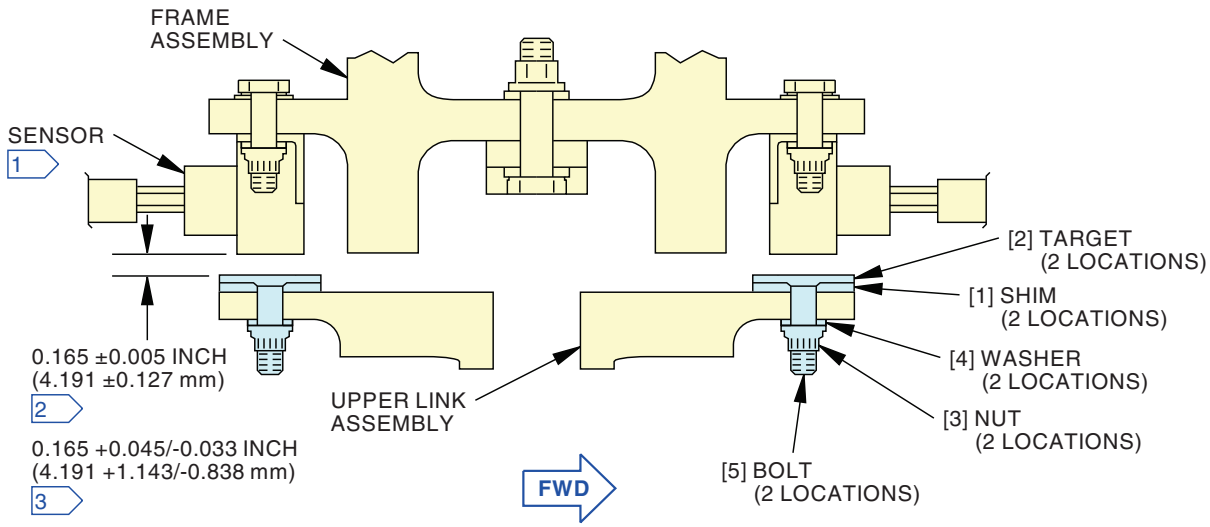
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear UpLock Sensor Adjustment
Figure 501/32-61-21-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



UP-AND-LOCKED SENSORS

A-A

1

SYSTEM NO.	SENSOR NO.	MLG	LOCATION
1	S0072	L	FWD
2	S1016	L	AFT
1	S0074	R	FWD
2	S1017	R	AFT

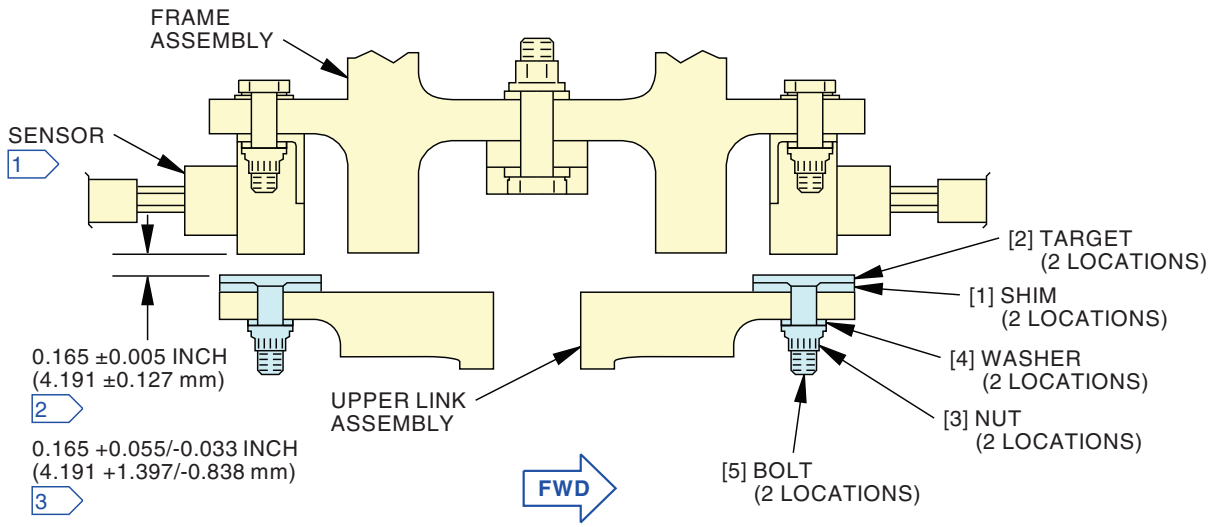
2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

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Main Landing Gear UpLock Sensor Adjustment
Figure 501/32-61-21-990-802 (Sheet 2 of 3)

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UP-AND-LOCKED SENSORS

A-A

1

SYSTEM NO.	SENSOR NO.	MLG	LOCATION
1	S0072	L	FWD
2	S1016	L	AFT
1	S0074	R	FWD
2	S1017	R	AFT

2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2972002 S0000750772_V1

Main Landing Gear UpLock Sensor Adjustment
Figure 501/32-61-21-990-802 (Sheet 3 of 3)

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TASK 32-61-21-400-803

3. Main Landing Gear Uplock Sensor Clearance Measurement (Airplane not on Jacks)

A. References

Reference	Title
24-22-00-860-801	Supply Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4896	Adapter Assembly - Uplock, MLG Part #: C32048-1 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
G02020	Clay, Modeling	

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

E. Prepare for the Measurement

SUBTASK 32-61-21-860-007

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

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SUBTASK 32-61-21-580-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-21-860-008

(3) Make sure that the control lever for the landing gear is in the DN position.

F. Main Landing Gear Up-and-Locked Sensor Clearance Measurement

WARNING MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

SUBTASK 32-61-21-860-024

(1) Make sure that the hydraulic system A power is off (TASK 29-11-00-860-805).

SUBTASK 32-61-21-860-009

(2) Insert the MLG uplock adapter assembly, SPL-4896, in the MLG Uplock hook.

SUBTASK 32-61-21-860-010

(3) Attach a square drive torque wrench or breaker bar to the MLG uplock adapter assembly and turn the MLG Uplock hook until it turns to the latched position.

SUBTASK 32-61-21-220-002

(4) Measure the sensor gap using a feeler gage.

SUBTASK 32-61-21-820-002

(5) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Uplock Sensor Clearance Adjustment, TASK 32-61-21-820-801.

NOTE: Write down the measured clearance so you will know what adjustments must be made.

SUBTASK 32-61-21-860-025

(6) Make sure to put the control lever for the landing gear in the DN position.

SUBTASK 32-61-21-860-026

(7) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-61-21-860-027

(8) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-61-21-860-028

(9) Remove the MLG uplock adapter assembly, SPL-4896.

SUBTASK 32-61-21-860-029

(10) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

G. Main Landing Gear Up-and-Locked Sensor Clearance Measurement (Alternate Method)

SUBTASK 32-61-21-860-030

(1) Make sure that the hydraulic system A power if off (TASK 29-11-00-860-805).

SUBTASK 32-61-21-860-031

(2) Insert the MLG uplock adapter assembly, SPL-4896, in the MLG Uplock hook.

SUBTASK 32-61-21-860-032

(3) Attach a square drive torque wrench or breaker bar to the MLG uplock adapter assembly and turn the MLG Uplock hook until it turns to the latched position.

SUBTASK 32-61-21-200-001

(4) Do this check of the main landing gear uplock sensor:

(a) If it is necessary, supply electrical power (TASK 24-22-00-860-801).

(b) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

(c) Use the Multi Function Display to check for the clearance of the Uplock sensor.

1) Set the GND TESTS Switch, on the P61-4 panel, to Enable position.

2) Select SYS on the MFD Multi Function Display panel.

3) Do these steps to start the check:

a) Select MAINT.

b) Select MAINT DATA PGS.

c) Select 32 Proximity Switch Electronics.

d) Select DISPLAY.

4) Make sure that the MFD shows the sensor name and rigging gap.

NOTE: See the table in Figure 501 for the sensor identification and location.

5) Make sure that the MFD shows the sensor clearance or the acceptable rigging gap range within tolerance.

NOTE: For a description of the sensor clearance display see Figure 501.

NOTE: 1 MIL equals 1/1000th of an inch.

6) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Uplock Sensor Clearance Adjustment, TASK 32-61-21-820-801.

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SUBTASK 32-61-21-860-033

(5) Make sure to put the control lever for the landing gear in the DN position.

SUBTASK 32-61-21-860-034

(6) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-61-21-860-035

(7) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-61-21-860-036

(8) Remove the MLG uplock adapter assembly, SPL-4896.

SUBTASK 32-61-21-860-037

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

H. Main Landing Gear Up-and-Locked Sensor Clearance Measurement (Alternate Method)

SUBTASK 32-61-21-860-038

(1) Make sure that the hydraulic system A power is off (TASK 29-11-00-860-805).

SUBTASK 32-61-21-860-039

(2) Insert the MLG uplock adapter assembly, SPL-4896, in the MLG Uplock hook.

SUBTASK 32-61-21-480-005

(3) Attach a minimum of 0.5 in. (12.7 mm) depth clay, G02020, to the entire face of the sensor.

NOTE: Refer to Figure 501 for the sensor location.

SUBTASK 32-61-21-860-040

(4) Attach a square drive torque wrench or breaker bar to the MLG uplock adapter assembly and turn the MLG Uplock hook until it turns to the latched position.

SUBTASK 32-61-21-860-041

(5) Make sure the control lever for the landing gear is in the DN position.

SUBTASK 32-61-21-860-042

(6) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC

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(Continued)

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

SUBTASK 32-61-21-860-043

(7) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-61-21-860-044

(8) Remove the MLG uplock adapter assembly, SPL-4896.

SUBTASK 32-61-21-860-045

(9) Remove the clay, G02020, from the face of the sensor.

SUBTASK 32-61-21-860-046

(10) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-61-21-220-003

(11) Measure the compressed thickness of the clay, G02020.

SUBTASK 32-61-21-820-003

(12) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Uplock Sensor Clearance Adjustment, TASK 32-61-21-820-801.

NOTE: Write down the measured clearance so you will know what adjustments must be made.

——— END OF TASK ———

TASK 32-61-21-820-801

4. Main Landing Gear Uplock Sensor Clearance Adjustment

(Figure 501)

A. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

B. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door

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(Continued)

Zone	Area
741	Right Main Landing Gear - Outboard Door

C. Main Landing Gear Uplock Sensor Clearance Adjustment

SUBTASK 32-61-21-820-004

(1) Adjust the sensor clearance.

(a) Remove the nuts [3], washers [4], and bolts [5] which hold the target [2] in position.

(b) Remove the targets [2] and shims [1].

(c) Clean the shims [1] and mounting surfaces.

(d) Adjust the shims [1] thickness as required to get the necessary sensor to target clearance, do one of the steps that follow:

1) Remove the laminations from the existing shims [1].

2) Apply additional shims [1].

a) If it is necessary, remove the laminations from the new shims [1].

3) Replace the existing shims [1] with the new shims [1] and adjust the thickness.

a) If it is necessary, remove the laminations from the new shims [1].

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(e) Apply a thin coat of Cor-Ban 27L Compound, G50237, to the thread reliefs, threads, and washer faces prior to assembly.

1) Wipe off the excess.

(f) Put the shims [1] and targets [2] in position on the mounting bracket.

(g) Install the bolts [5], nuts [3], and washers [4] which hold the shim [1] stack.

SUBTASK 32-61-21-220-004

(2) Measure the sensor clearance and compare to the requirement (TASK 32-61-21-400-802 or TASK 32-61-21-400-803).

(a) If the sensor clearance is not correct, do the steps to adjust the sensor clearance again.

SUBTASK 32-61-21-390-001

(3) Seal the shim [1] stack.

(a) Clean the shims [1].

(b) Apply sealant, A00247, on the shim [1] surfaces.

END OF TASK

TASK 32-61-21-710-801

5. Main Landing Gear Up-and-Locked Sensor Operational Test

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)

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(Continued)

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-4896	Adapter Assembly - Uplock, MLG Part #: C32048-1 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for Operational Test

SUBTASK 32-61-21-010-001

(1) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-61-21-860-018

(2) For the operational test, do these steps:

WARNING MAKE SURE THAT THERE IS NO PRESSURE IN THE HYDRAULIC SYSTEM. PRESSURE IN THE HYDRAULIC SYSTEM CAN MOVE THE CONTROL SURFACES, CABLES, LANDING GEARS, AND LANDING GEAR DOORS QUICKLY. THEIR MOVEMENT CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(a) Make sure that the hydraulic system A power is off (TASK 29-11-00-860-805).

(b) Insert the MLG uplock adapter assembly, SPL-4896, in the Main Landing Gear (MLG) Uplock hook.

(c) Attach a square drive torque wrench or breaker bar to the MLG uplock adapter assembly and turn the MLG Uplock hook until it turns to the latched position.

F. Main Landing Gear Up-and-Locked Sensor Operational Test

SUBTASK 32-61-21-740-002

(1) Use the Multi Function Display (MFD) to test the MLG Up-and-Locked sensor.

(a) Set the GND TESTS Switch on the P61-4 panel to Enable position.

(b) Select SYS on the MFD panel.

(c) Do these steps to start the test:

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1) Select MAINT
2) Select MAINT DATA PGS
3) Select 32 PSEU
4) Select DISPLAY.

(d) Make sure that the MFD shows the sensor name and rigging gap.

NOTE: See the table in Figure 501 for the sensor identification and location.

(e) Make sure that the MFD shows the sensor clearance or the acceptable rigging gap range within tolerance.

NOTE: For a description of the sensor clearance display, see Figure 501.

NOTE: 1 MIL equals 1/1000th of an inch.

(f) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Uplock Sensor Clearance Adjustment, TASK 32-61-21-820-801.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-61-21-740-003

(1) Set the GND TESTS switch on the P61-4 Panel to the NORM position.

SUBTASK 32-61-21-840-003

(2) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-61-21-860-047

(3) Make sure the control lever for the landing gear is in DN position.

SUBTASK 32-61-21-860-048

(4) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-61-21-860-019

(5) Remove the MLG uplock adapter assembly, SPL-4896.

NOTE: It is unnecessary to move the MLG Uplock hook to the unlatched position. When the hydraulic system is energized, the MLG Uplock hook will move automatically to the unlatched position.

SUBTASK 32-61-21-860-049

(6) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

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MAIN LANDING GEAR DOWN-AND-LOCKED SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Main landing gear down-and-locked sensor removal

(2) Main landing gear down-and-locked sensor installation.

B. The procedures for the left and right installations are similar.

TASK 32-61-31-020-801

2. Main Landing Gear Down-and-Locked Sensor Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)

B. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

C. Prepare for the Removal

SUBTASK 32-61-31-580-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-31-480-001

(2) Make sure chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-61-31-860-001

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-31-860-002

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

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D. Main Landing Gear Down-and-Locked Sensor Removal

SUBTASK 32-61-31-020-001

(1) Remove the sensor connector from the airplane connector.

SUBTASK 32-61-31-020-002

(2) Remove the sensor contact pins from the connector.

SUBTASK 32-61-31-480-002

(3) Tie a string at least 8 ft (2.4 m) long to the sensor wires.

NOTE: This string will be used to pull wire of the new sensor through the conduit. The string can be tied to the landing gear structure to make sure that the string is not pulled through the conduit.

SUBTASK 32-61-31-020-003

(4) Remove the sensor [4]:

(a) Remove the bolts [1], nuts [2], washers [3], and washers [5] that attach the sensor [4] to the side strut assembly.

(b) Remove the sensor [4] from the side strut assembly.

1) Make sure you do not pull the string through the conduit.

(c) Remove the string from the sensor [4].

SUBTASK 32-61-31-210-001

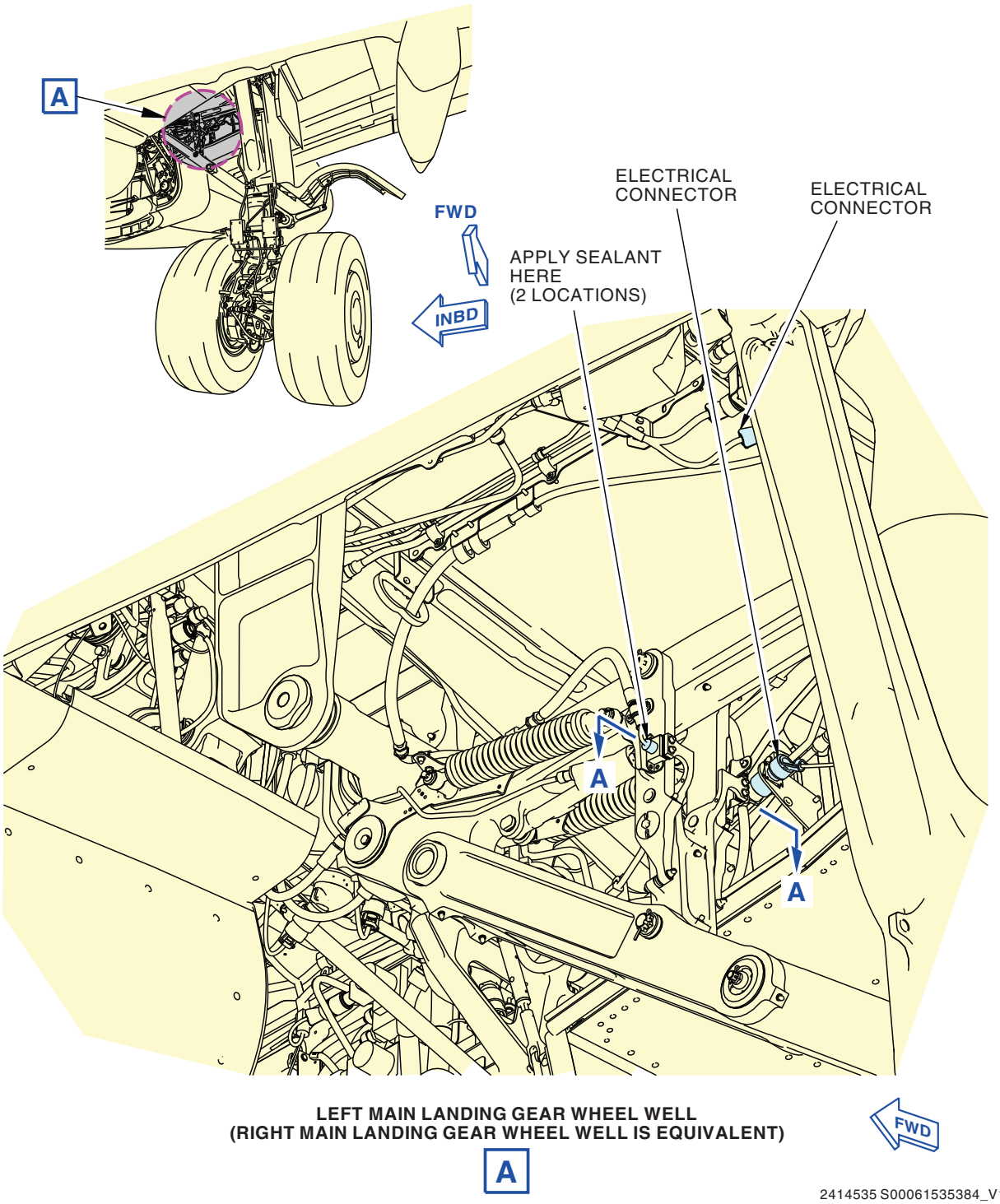
(5) Examine the sensor [4], the side strut assembly, and adjacent areas for signs of damage.

NOTE: Damage to the sensor may be an indication of an incorrect clearance between the sensor and the target.

(a) If you find damage, make a note of the damage for the installation procedure.

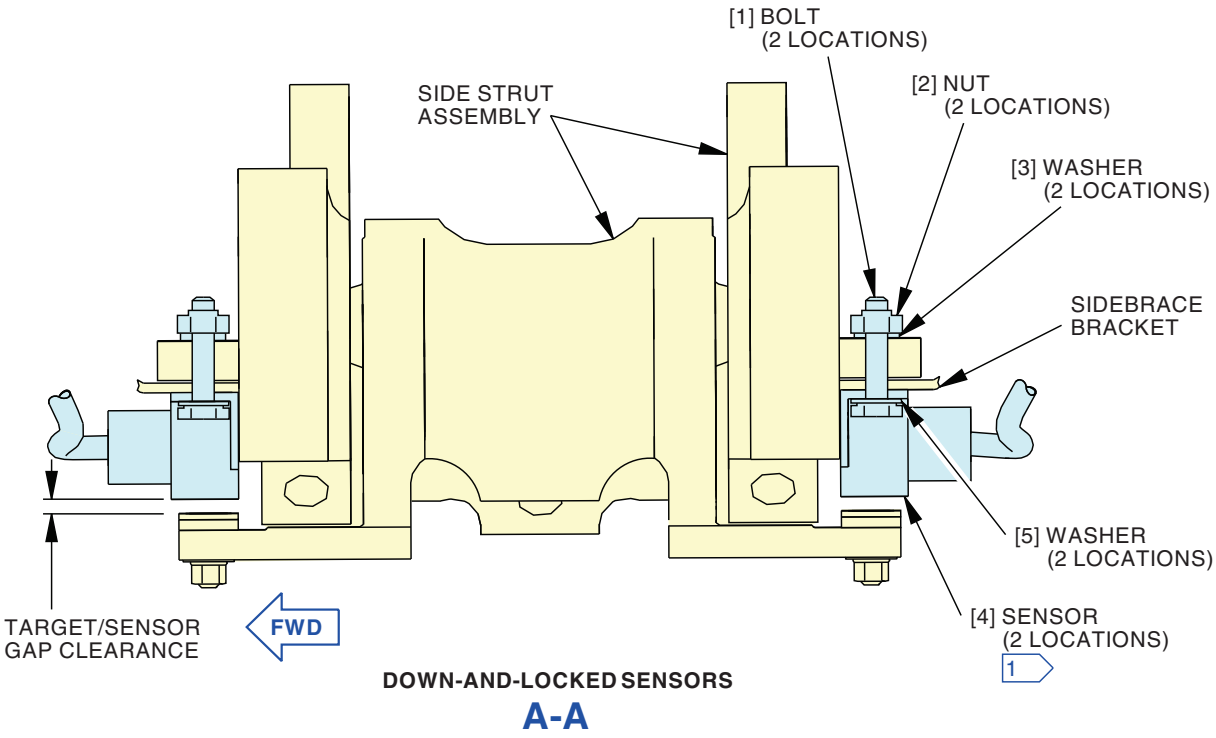
——— END OF TASK ———

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Main Landing Gear Down-and-Locked Sensor Installation
Figure 401/32-61-31-990-801 (Sheet 1 of 2)

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1

ITEM	SENSOR NO.	MLG	LOCATION
[4]	S0071	L	FWD
[4]	S0073	R	FWD
[4]	S0301	R	AFT
[4]	S0302	L	AFT

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Main Landing Gear Down-and-Locked Sensor Installation
Figure 401/32-61-31-990-801 (Sheet 2 of 2)

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TASK 32-61-31-400-801

3. Main Landing Gear Down-and-Locked Sensor Installation

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-61-31-220-801	Main Landing Gear Down-and-Locked Sensor Clearance Measurement (P/B 501)
32-61-31-710-801	Main Landing Gear Down-and-Locked Sensor Operational Test (P/B 501)
SWPM Ch 20	Standard Wiring Practices Manual

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Sensor	32-61-31-01-165	PTF ALL
		32-61-31-01-230	PTF ALL

D. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

E. Prepare for the Installation

SUBTASK 32-61-31-580-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-31-480-003

(2) Make sure chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

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SUBTASK 32-61-31-860-003

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-31-860-004

(4) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

F. Main Landing Gear Down-and-Locked Sensor Installation

SUBTASK 32-61-31-420-001

(1) Prepare the replacement sensor [4] for installation on the airplane.

(a) Cut the sensor leads to the correct length:

1) S0071 - FWD L Gear Down & Locked Sensor: 64 in. (1.6 m)
2) S0073 - FWD R Gear Down & Locked Sensor: 64 in. (1.6 m)
3) S0301 - AFT R Gear Down & Locked Sensor: 52 in. (1.3 m)
4) S0302 - AFT L Gear Down & Locked Sensor: 52 in. (1.3 m)

(b) Attach the contact pins to the sensor leads.

NOTE: Refer to SWPM Ch 20, if it is necessary.

SUBTASK 32-61-31-410-001

(2) Fay seal the bottom of the sensor [4] with sealant, A00247.

SUBTASK 32-61-31-420-002

(3) Install the main landing gear down-and-locked sensor [4].

(a) Attach the string to the new sensor leads.

(b) Pull the sensor leads through the conduit.

(c) Position the sensor [4] and the bracket on the side strut assembly.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(d) Apply a thin coat of Cor-Ban 27L Compound, G50237 to thread reliefs, threads, and washer faces prior to assembly.

(e) Install the bolts [1], washers [3], washers [5], and nuts [2].

NOTE: Use CRES washers below bolt heads with titanium sensors. Use cad-plated washers below nuts.

SUBTASK 32-61-31-420-003

(4) Connect the sensor leads to the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Insert the connector pins into the connector.

(b) Attach the plug side of the connector to the jack side of the connector at the disconnect bracket.

SUBTASK 32-61-31-350-001

(5) If you found damage to the sensor [4] or the adjacent area during the removal procedure, do these steps:

(a) Repair or replace all damaged components or areas.

NOTE: Damage to the sensor, the target, or the adjacent areas may be an indication of an incorrect sensor clearance.

(b) Measure the sensor [4] clearance. To measure the clearance, do this task: Main Landing Gear Down-and-Locked Sensor Clearance Measurement, TASK 32-61-31-220-801.

G. Main Landing Gear Down-and-Locked Sensor Installation Test

SUBTASK 32-61-31-220-001

(1) Do this task: Main Landing Gear Down-and-Locked Sensor Clearance Measurement, TASK 32-61-31-220-801.

SUBTASK 32-61-31-860-005

(2) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-61-31-860-006

(3) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

SUBTASK 32-61-31-740-001

(4) Do this task: Main Landing Gear Down-and-Locked Sensor Operational Test, TASK 32-61-31-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR DOWN-AND-LOCKED SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has three tasks:

(1) Main landing gear down-and-locked sensor clearance measurement

(2) Main landing gear down-and-locked sensor clearance adjustment

(3) Main landing gear down-and-locked sensor operational test.

B. The procedures for the left and right installations are similar.

TASK 32-61-31-220-801

2. Main Landing Gear Down-and-Locked Sensor Clearance Measurement

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Measurement

SUBTASK 32-61-31-580-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-31-480-004

(2) Make sure that chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

D. Main Landing Gear Down-and-Locked Sensor Clearance Measurement

SUBTASK 32-61-31-220-002

(1) Measure the clearance between the sensor [6] and target [1] with a feeler gage.

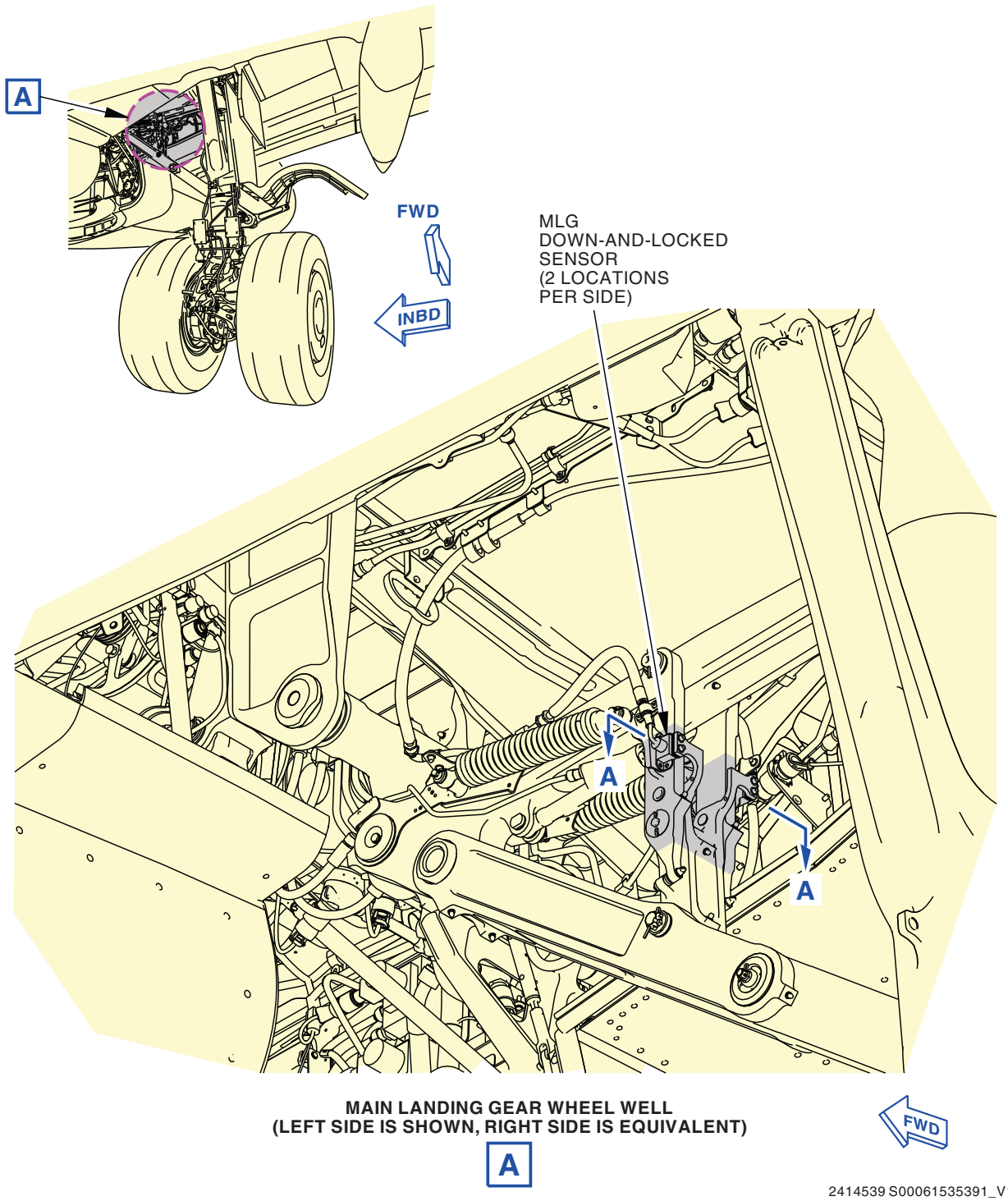
(a) Make a record of the clearance dimension.

SUBTASK 32-61-31-820-001

(2) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Down-and-Locked Sensor Clearance Adjustment, TASK 32-61-31-400-802.

END OF TASK

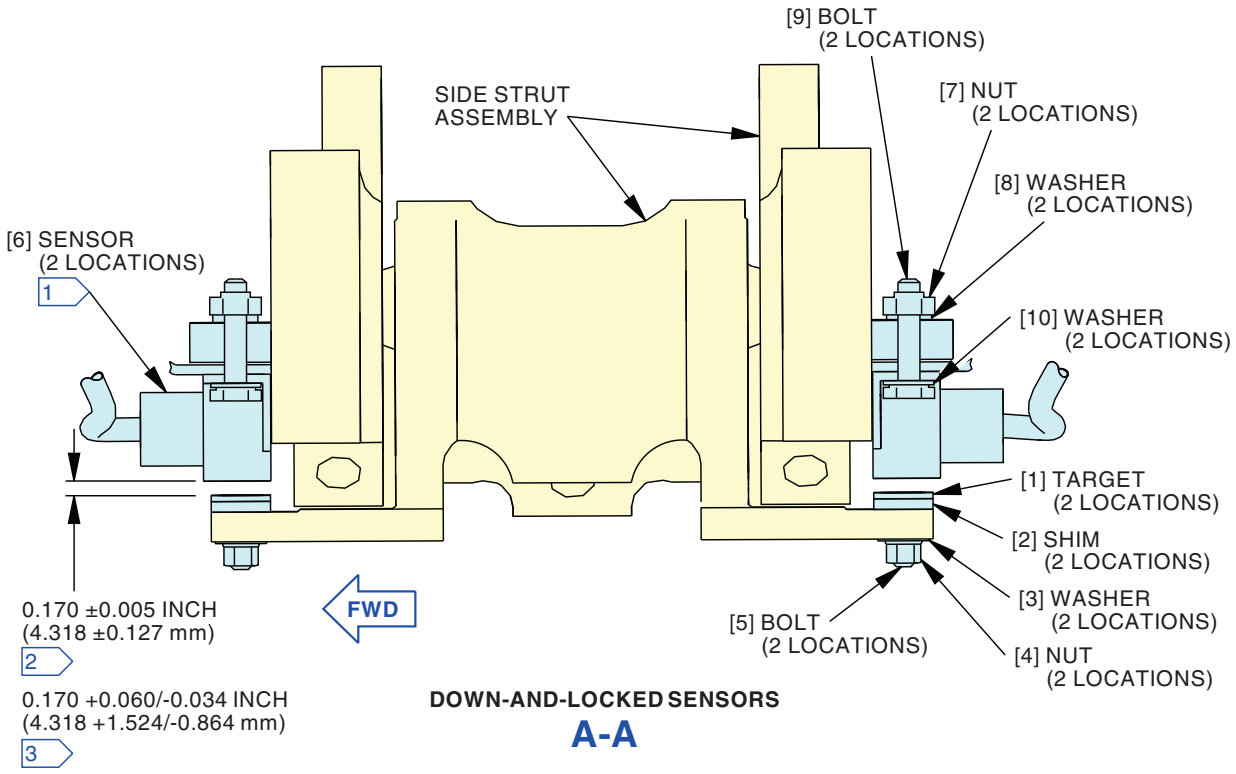
ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

Main Landing Gear Down-and-Locked Sensor Adjustment
Figure 501/32-61-31-990-802 (Sheet 1 of 3)

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1

SYSTEM NO.	SENSOR NO.	MLG	LOCATION	NAME
1	S0071	L	FWD	DN & LKD
1	S0073	R	FWD	DN & LKD
2	S0301	R	AFT	DN & LKD
2	S0302	L	AFT	DN & LKD

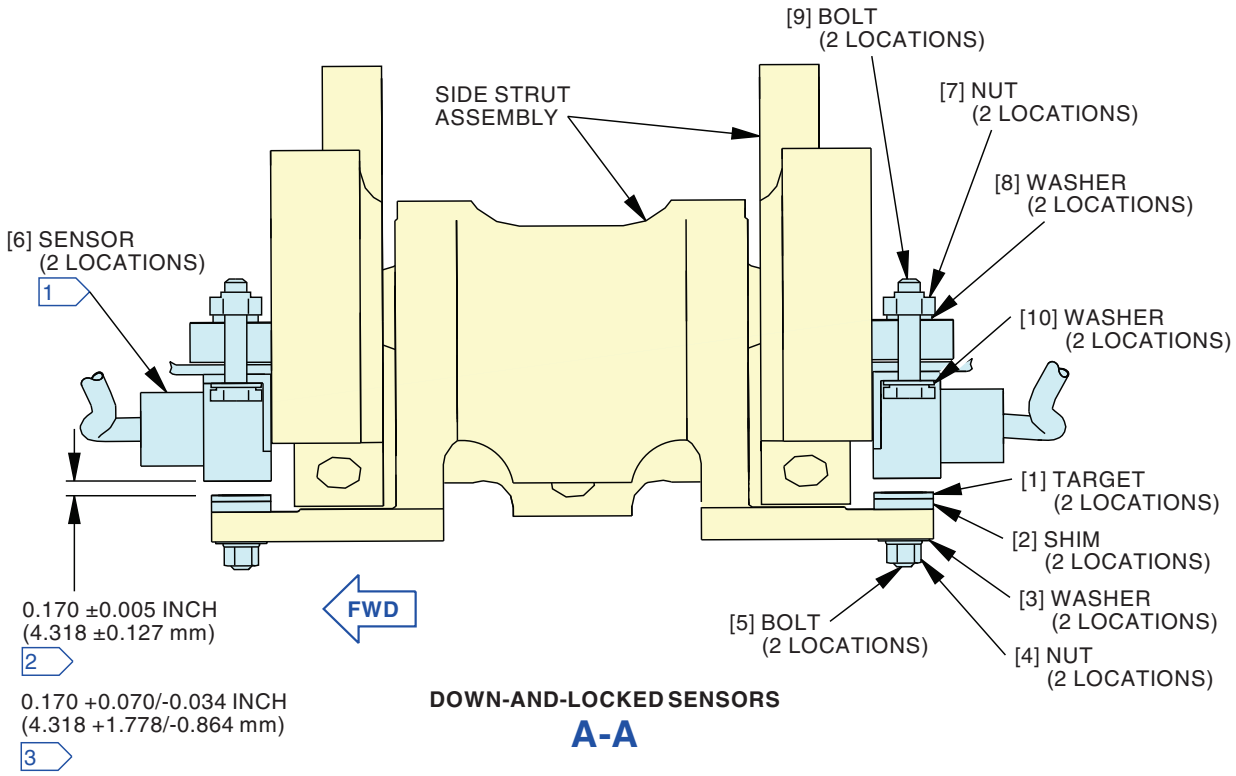
2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2414541 S00061535393_V5

Main Landing Gear Down-and-Locked Sensor Adjustment
Figure 501/32-61-31-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1

SYSTEM NO.	SENSOR NO.	MLG	LOCATION	NAME
1	S0071	L	FWD	DN & LKD
1	S0073	R	FWD	DN & LKD
2	S0301	R	AFT	DN & LKD
2	S0302	L	AFT	DN & LKD

2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2972073 S0000750686_V1

Main Landing Gear Down-and-Locked Sensor Adjustment
Figure 501/32-61-31-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-31-400-802

3. Main Landing Gear Down-and-Locked Sensor Clearance Adjustment

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare to Adjust the Clearance

SUBTASK 32-61-31-580-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-31-480-005

(2) Make sure chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

E. Adjust the Main Landing Gear Down-and-Locked Sensor Clearance

SUBTASK 32-61-31-970-001

(1) Measure the clearance between the sensor [6] and target [1].

SUBTASK 32-61-31-020-004

(2) Do these steps to move the sensor [6] out of the way.

(a) Remove the nuts [7], washers [8], washers [10], and bolts [9] which secure the sensor [6] to the side strut assembly.

(b) Remove the sensor [6] from the side strut assembly.

SUBTASK 32-61-31-020-005

(3) Remove the nuts [4], washers [3], and bolts [5].

SUBTASK 32-61-31-020-006

(4) Remove the target [1] and shims [2].

SUBTASK 32-61-31-110-001

(5) Clean the shims [2] and mounting surfaces.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-31-820-002

(6) Remove or add shims [2] as required to get the necessary clearance.

SUBTASK 32-61-31-620-001

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(7) Apply a thin coat ofCor-Ban 27L Compound, G50237 to the thread reliefs, threads, and washer faces. Wipe off the excess.

SUBTASK 32-61-31-420-004

(8) Assemble the target assembly:

(a) Position the shims [2] and target [1] on the mounting bracket.

(b) Install the bolts [5], washers [3], and nuts [4].

SUBTASK 32-61-31-410-002

(9) Fay seal the bottom of the sensor [6] with sealant, A00247.

SUBTASK 32-61-31-420-005

(10) Do these steps to re-attach the main landing gear down-and-locked sensor to the side strut assembly.

(a) Position the sensor [6] and the bracket on the side strut assembly.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin coat of Cor-Ban 27L Compound, G50237 to thread reliefs, threads, and washer faces prior to assembly.

(c) Install the bolts [9], washers [10], washers [8], and nuts [7].

NOTE: Use CRES washers below bolt heads with titanium sensors. Use cad-plated washers below nuts.

SUBTASK 32-61-31-220-003

(11) Measure the sensor clearance and compare the measurement to Figure 501 (TASK 32-61-31-220-801).

(a) If the sensor clearance is not correct, then do the steps to adjust the sensor clearance again.

SUBTASK 32-61-31-390-002

(12) Seal the shim stack.

(a) Clean the shims.

(b) Apply sealant, A00247 to the shim surfaces.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-31-710-801

4. Main Landing Gear Down-and-Locked Sensor Operational Test

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 32-61-31-580-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-31-480-006

(2) Make sure chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

D. Main Landing Gear Lock Sensor Operational Test

SUBTASK 32-61-31-740-002

(1) Use the Multi Function Display to test the MLG Down-and-Locked Sensor.

(a) Set the GND TESTS Switch on the P61-4 panel to Enable position.

(b) Select SYS on the MFD Multi Function Display panel.

(c) Do these steps to start the test:

1) Select MAINT

2) Select MAINT DATA PGS

3) Select 32 PSEU DISPLAY

(d) Make sure the MFD shows the sensor name and rigging gap.

NOTE: See the table in Figure 501 for the sensor identification and location.

(e) Make sure the MFD shows the sensor clearance or the acceptable rigging gap range within tolerance.

NOTE: For a description of the sensor clearance display see Figure 501.

NOTE: 1 MIL equals 1/1000th of an inch.

(f) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Main Landing Gear Down-and-Locked Sensor Clearance Adjustment, TASK 32-61-31-400-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-61-31-840-001

(1) Set the GND TESTS switch on the P61-4 Panel to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR LOCK SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Nose landing gear lock sensor removal.

(2) Nose landing gear lock sensor installation.

B. The procedures for the number 1 and number 2 installations are similar.

TASK 32-61-41-020-801

2. Nose Landing Gear Lock Sensor Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4938	Insertion/Removal Tool - Contact (size 16, backshell side) Part #: M81969/1-03 Supplier: 11851 Part #: M81969/14-03 Supplier: 11139
STD-837	String

C. Location Zones

Zone	Area
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
713	Nose Landing Gear

D. Prepare for the Removal

SUBTASK 32-61-41-580-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-41-480-001

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-41-860-001

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-41-860-002

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

E. Nose Landing Gear Lock Sensor Removal

SUBTASK 32-61-41-020-001

(1) Disconnect the sensor [4] wires:

(a) Disconnect the plug [8] from the connector on the wheel well wall.

(b) Identify the wires in the plug [8] that go to the sensor [4].

1) Put a label on them or make a drawing to refer to during installation.

(c) Use contact insertion / removal tool, COM-4938, to remove the wires from the plug [8].

(d) Install a protective cap on the plug [8].

(e) Keep the plug [8] safe for the installation.

(f) Attach a string, STD-837, of appropriate length to the sensor [4] wire.

NOTE: The string, STD-837, will be used to pull the wire of the new sensor through the conduit. You can tie the string, STD-837, to the landing gear structure to make sure that the string, STD-837 is not pulled through the conduit.

SUBTASK 32-61-41-020-002

(2) Remove the sensor [4]:

(a) Remove the nuts [3], washers [2], washers [6], bolts [1], which attach the sensor [4] to the mounting bracket.

(b) Remove the sensor [4] from the mounting bracket.

(c) Pull the sensor [4] wires out of the conduit.

1) For system 2 sensor [4], pull the wires through the junction box [7].

2) Make sure that you do not pull the string through the conduit.

(d) Remove the string from the sensor [4] wires.

(e) Remove the sealant from the conduit end.

SUBTASK 32-61-41-210-001

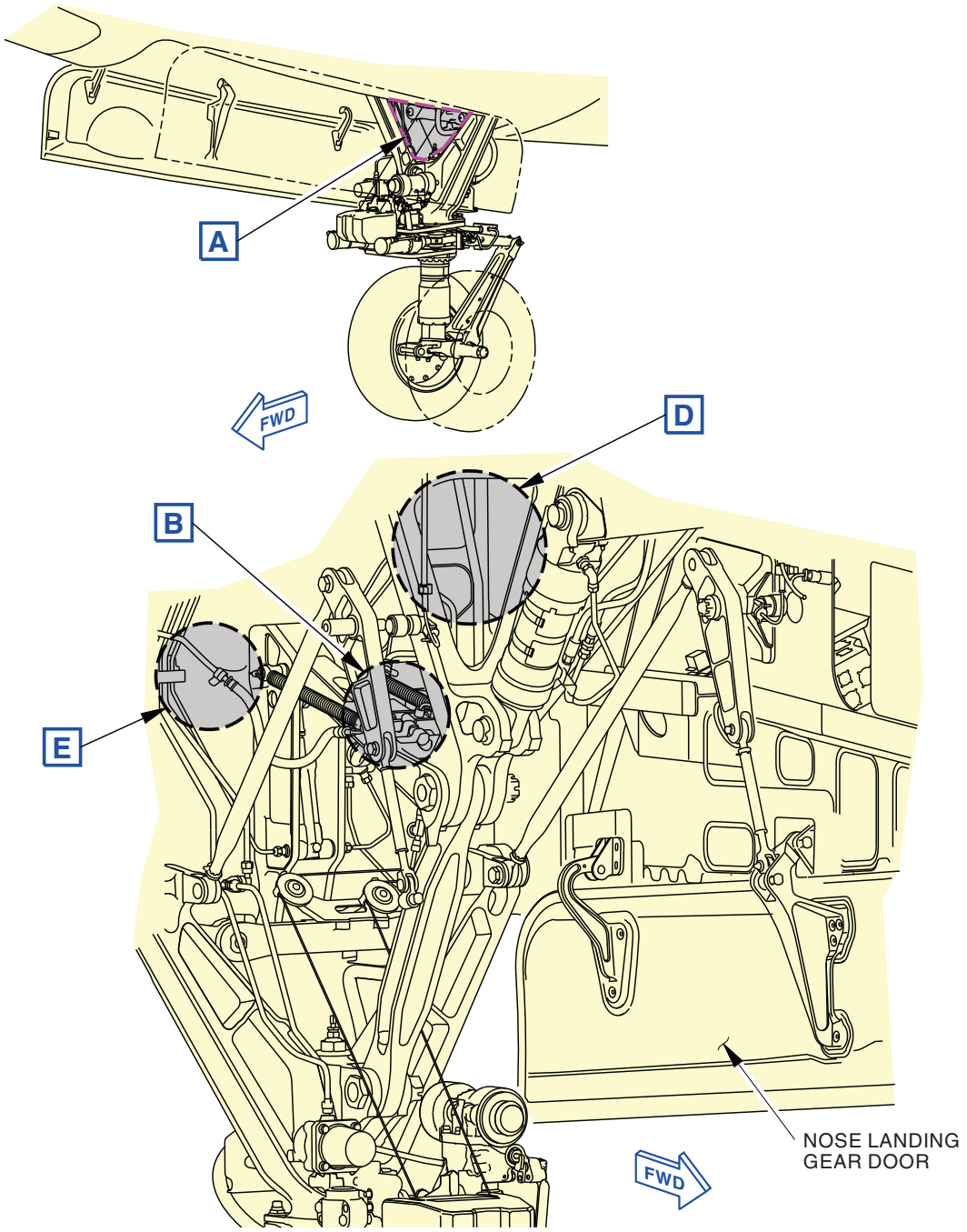
(3) Examine the sensor [4] and adjacent areas for signs of damage.

NOTE: Damage to the sensor may be an indication of an incorrect clearance between the sensor and the target.

(a) If you find damage, make a note of the damage for the installation procedure.

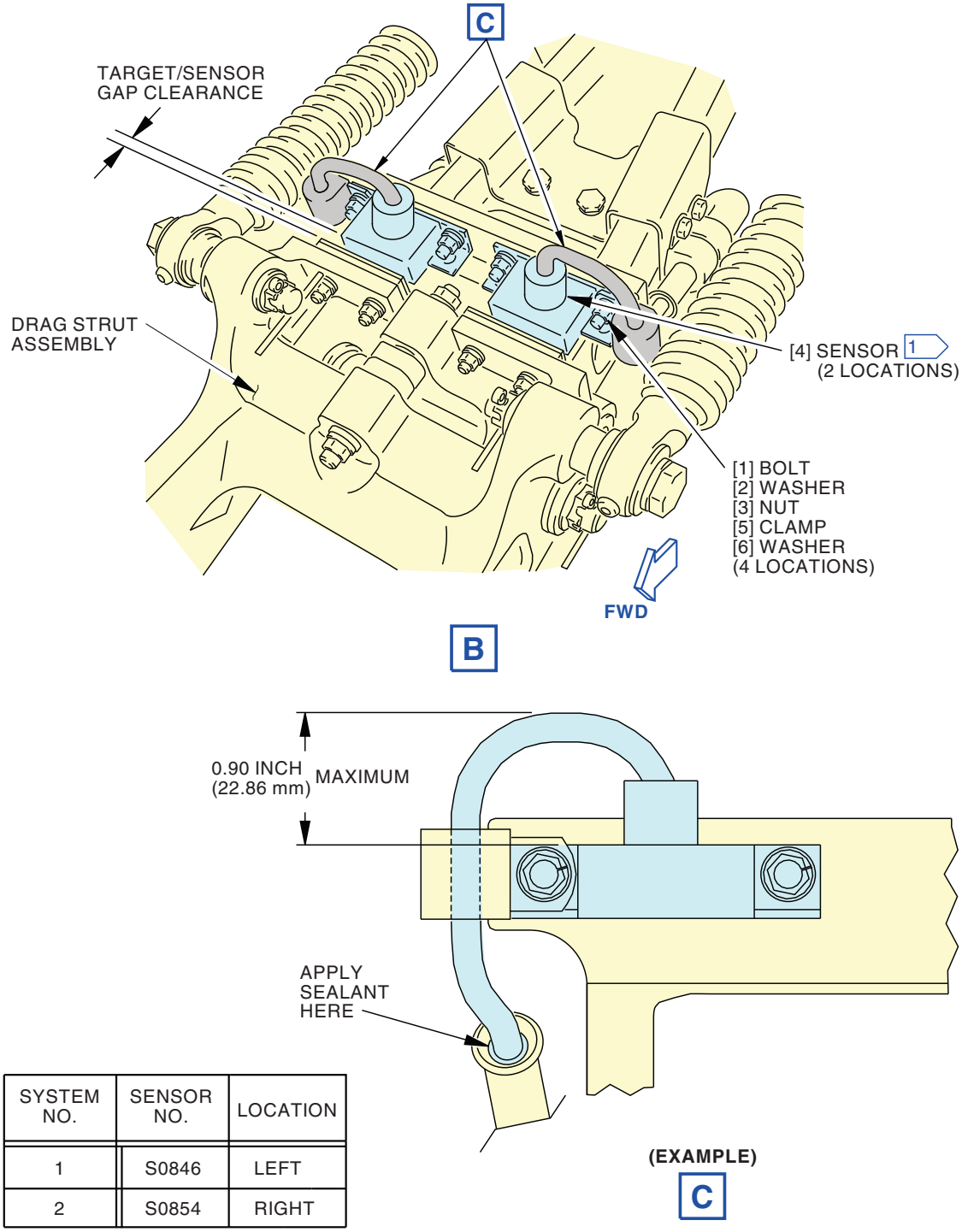
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Lock Sensor Installation
Figure 401/32-61-41-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



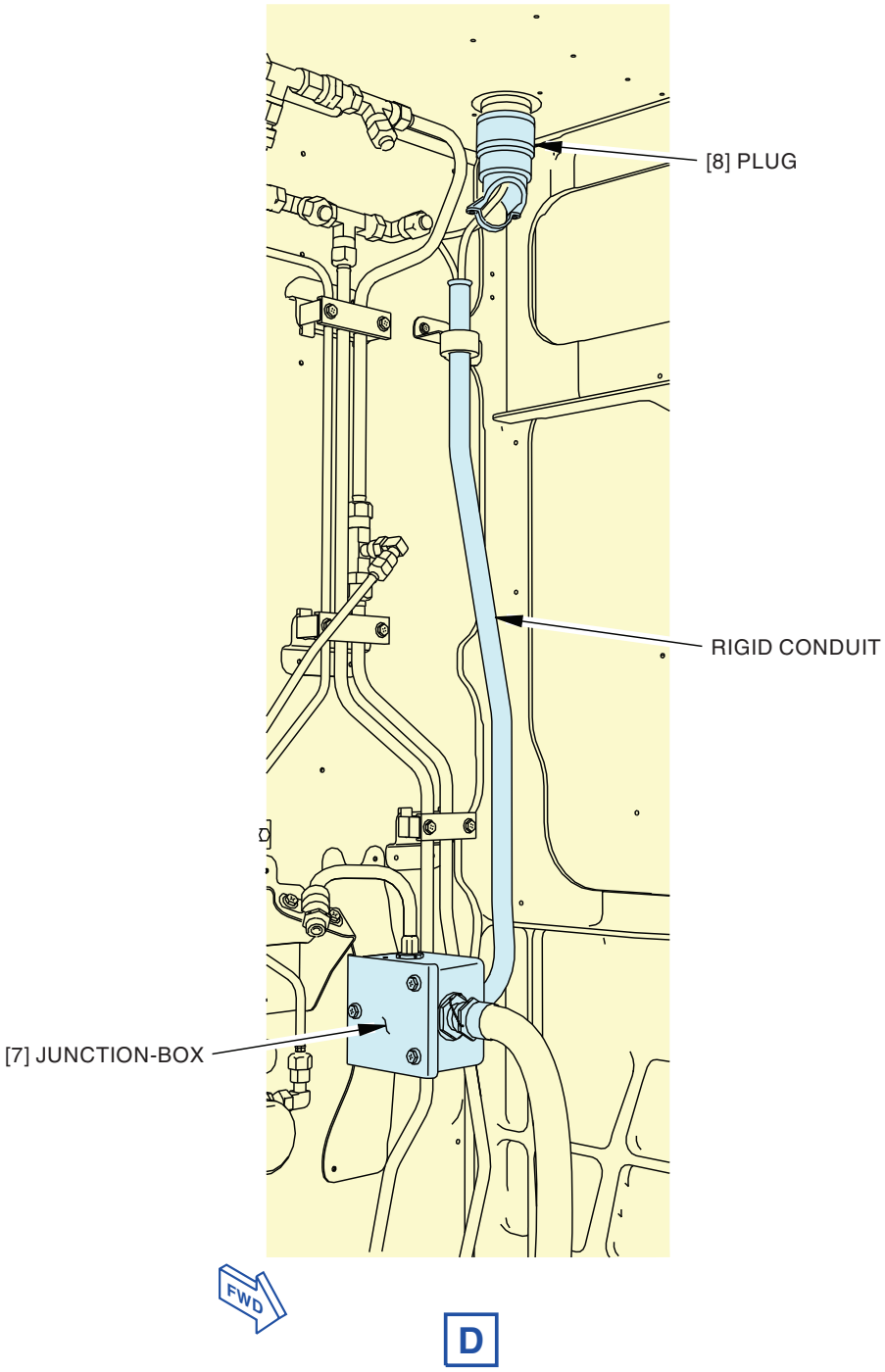
1

SYSTEM NO.	SENSOR NO.	LOCATION
1	S0846	LEFT
2	S0854	RIGHT

2414545 S00061535402_V3

Nose Landing Gear Lock Sensor Installation
Figure 401/32-61-41-990-801 (Sheet 2 of 4)

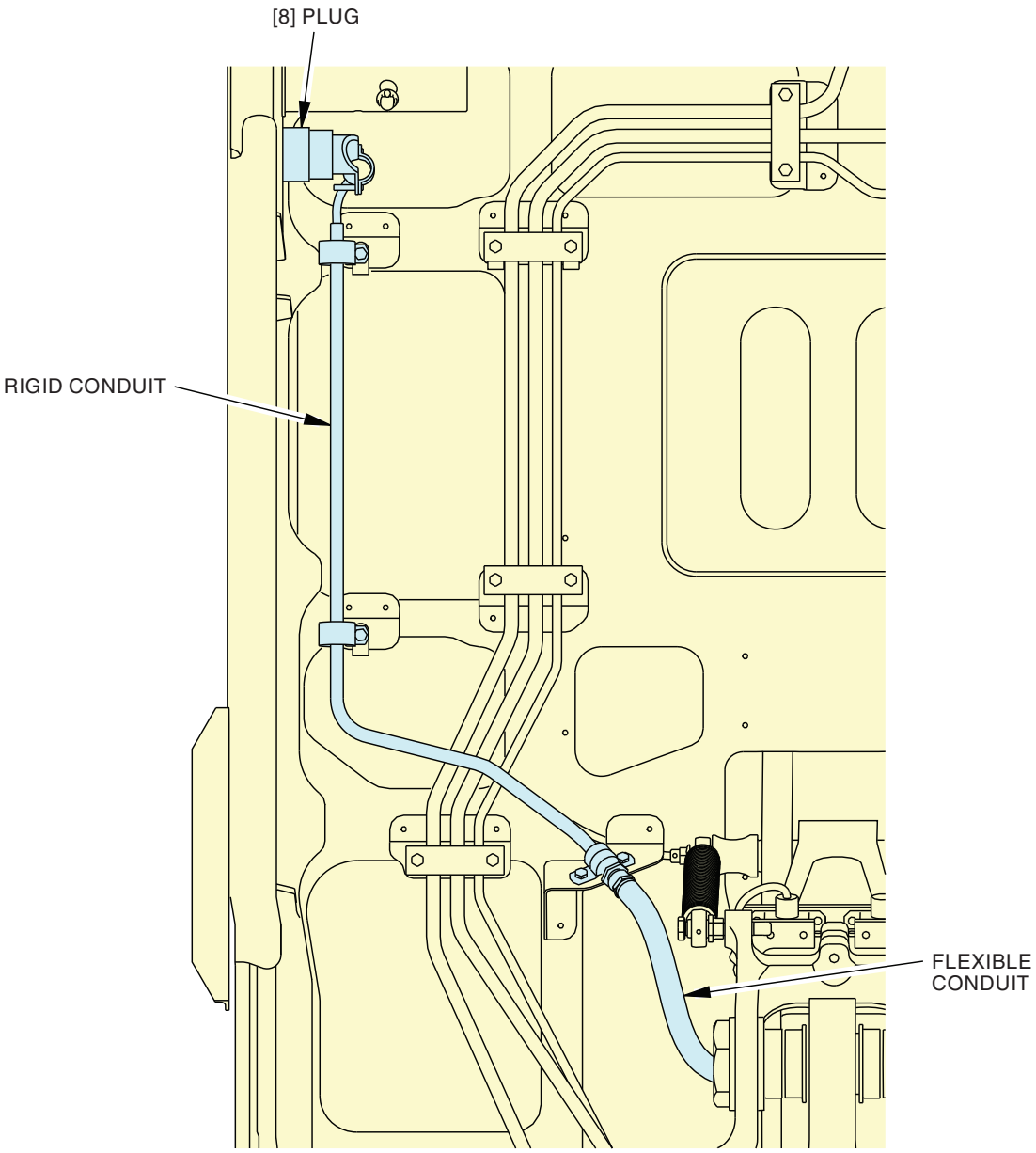
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3031874 S0000802619_V1

Nose Landing Gear Lock Sensor Installation
Figure 401/32-61-41-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW IN THE AFT DIRECTION)

E

3031875 S0000802620_V1

Nose Landing Gear Lock Sensor Installation
Figure 401/32-61-41-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-41-400-801

3. Nose Landing Gear Lock Sensor Installation

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-840-802	Return the Airplane Systems Back to Their Normal on Ground Condition (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-61-41-220-801	Nose Landing Gear Lock Sensor Clearance Measurement (P/B 501)
32-61-41-710-801	Nose Landing Gear Lock Sensor Operational Test (P/B 501)
SWPM 20-83-01/001	Standard Wiring Practices Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4938	Insertion/Removal Tool - Contact (size 16, backshell side) Part #: M81969/1-03 Supplier: 11851 Part #: M81969/14-03 Supplier: 11139
STD-837	String

C. Consumable Materials

Reference	Description	Specification
A00230	Compound - Electrical Insulating Coating	BMS5-37
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Sensor	32-61-41-01-110	PTF ALL

E. Location Zones

Zone	Area
731	Left Main Landing Gear - Outboard Door
741	Right Main Landing Gear - Outboard Door

F. Prepare for the Installation

SUBTASK 32-61-41-580-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

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SUBTASK 32-61-41-480-003

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-61-41-860-007

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-41-860-003

(4) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

G. Nose Landing Gear Lock Sensor Installation

SUBTASK 32-61-41-350-001

(1) Cut the sensor leads to the correct length:

(a) S846 - #1 Nose Landing Gear Lock: 42.5 in. (1.1 m)

(b) S854 - #2 Nose Landing Gear Lock: 37.5 in. (95.3 cm)

SUBTASK 32-61-41-420-001

(2) Attach the contacts to the sensor [4] wires (SWPM 20-83-01/001).

SUBTASK 32-61-41-410-001

(3) Fay seal the bottom of the sensor [4] with sealant, A00247.

SUBTASK 32-61-41-420-002

(4) Install the sensor [4]:

(a) Put the sensor [4] in position on the bracket.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin coat of Cor-Ban 27L Compound, G50237 to the thread reliefs, threads, and washer faces.

1) Wipe off the excess.

(c) Install the bolts [1], washers [2], washers [6], and nuts [3].

NOTE: Use Corrosion Resistant Steel (CRES) washers below nuts with titanium sensors [4]. Use cad-plated washers below bolt heads.

SUBTASK 32-61-41-420-003

(5) Attach the sensor [4] wiring to the landing gear structure:

(a) Put the sensor [4] wiring in the clamp [5].

(b) Loosely attach the clamp [5] to the landing gear structure.

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(c) Make sure that the sensor [4] wiring loop is a maximum of 0.9 in. (22.9 mm) as shown in Figure 401.

(d) Tighten the clamp [5].

SUBTASK 32-61-41-410-002

(6) Attach the sensor [4] wires to the airplane:

(a) Attach the string, STD-837, to the sensor wires.

(b) Pull the sensor [4] wires through the conduit.

1) For system 2 sensor [4], pull the wires through the rigid conduit and the junction box [7].

(c) Remove the string, STD-837, from the sensor [4] wires.

(d) Install the sensor [4] contacts into the plug [8]:

1) Remove the protective cap from the plug [8].

2) Use contact insertion / removal tool, COM-4938, to connect the sensor [4] wires to the plug [8].

a) Use the drawing you made during the removal to connect the correct sensor [4] wire to the correct contact.

(e) Push the plug [8] into the connector on the wheel well wall.

SUBTASK 32-61-41-400-001

(7) If you found damage to the sensor [4] or the adjacent area during the removal procedure, do these steps:

(a) Repair or replace all damaged components or areas.

NOTE: Damage to the sensor, target, or the adjacent areas may be an indication of an incorrect sensor clearance.

(b) Measure the sensor [4] clearance. To measure the clearance, do this task: Nose Landing Gear Lock Sensor Clearance Measurement, TASK 32-61-41-220-801.

H. Nose Landing Gear Lock Sensor Installation Test

SUBTASK 32-61-41-720-001

(1) Do a post installation check of the nose gear lock sensor [4]:

(a) Do this task: Nose Landing Gear Lock Sensor Clearance Measurement, TASK 32-61-41-220-801.

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

(c) Do this task: Return the Airplane Systems Back to Their Normal on Ground Condition, TASK 32-09-00-840-802.

(d) Do this task: Nose Landing Gear Lock Sensor Operational Test, TASK 32-61-41-710-801.

SUBTASK 32-61-41-390-001

(2) Seal the sensor [4] wires in the conduit end.

(a) Fill the conduit end with compound, A00230.

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(b) Make sure that the sealant enters the conduit so that it is visible through the inspection holes on the conduit.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR LOCK SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has three tasks:

(1) Nose landing gear lock sensor clearance measurement

(2) Nose landing gear lock sensor clearance adjustment

(3) Nose landing gear lock sensor operational test.

B. The installation procedures for the #1 and #2 sensor are similar.

TASK 32-61-41-220-801

2. Nose Landing Gear Lock Sensor Clearance Measurement

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
713	Nose Landing Gear

C. Prepare for the Measurement

SUBTASK 32-61-41-580-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-41-480-005

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

D. Nose Landing Gear Lock Sensor Clearance Measurement

SUBTASK 32-61-41-220-001

(1) Measure the clearance between the sensor [6] and target [1] with a feeler gage.

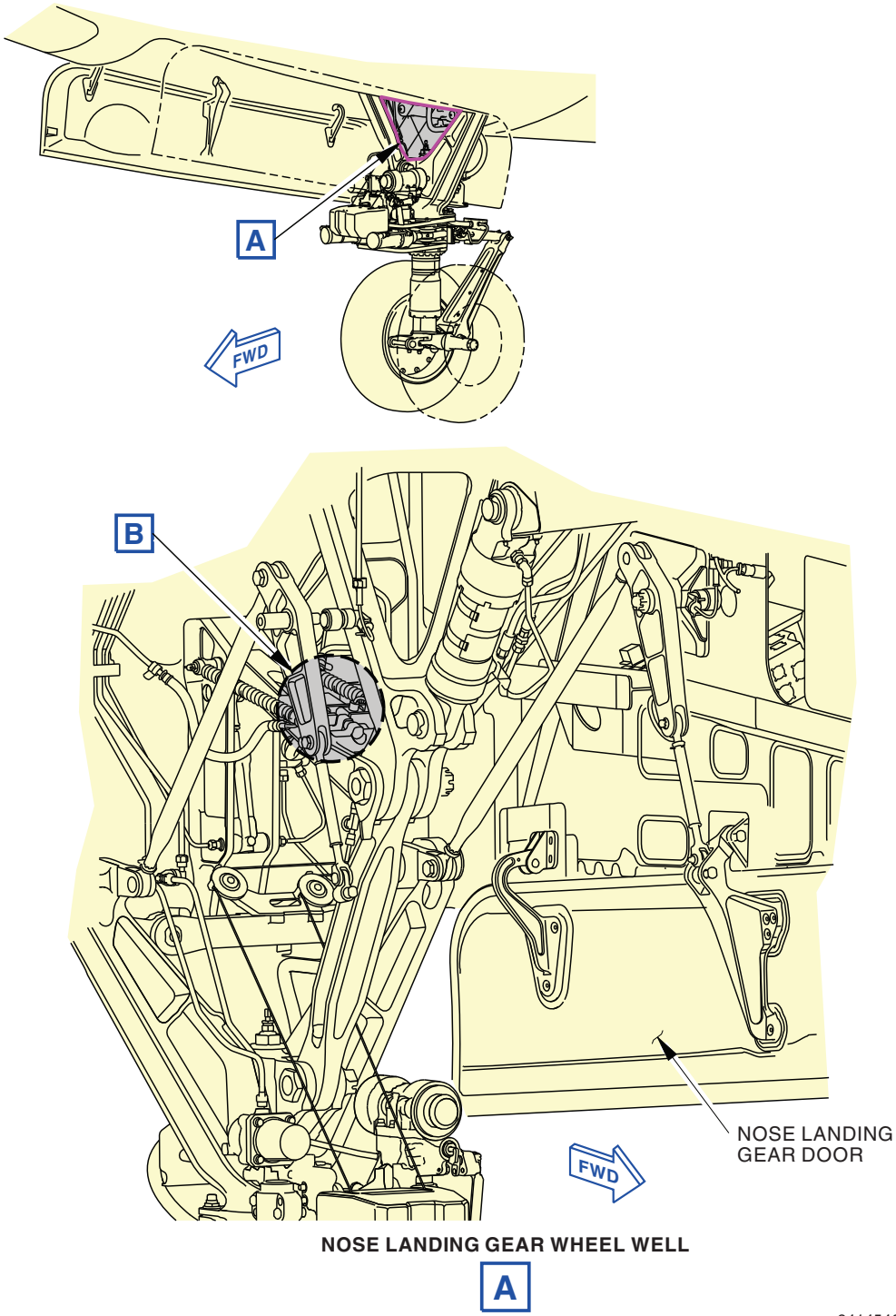
(a) Make a record of the clearance dimension.

SUBTASK 32-61-41-820-001

(2) If the sensor clearance is outside the values shown in Figure 501, do this task: Nose Landing Gear Lock Sensor Clearance Adjustment, TASK 32-61-41-400-802.

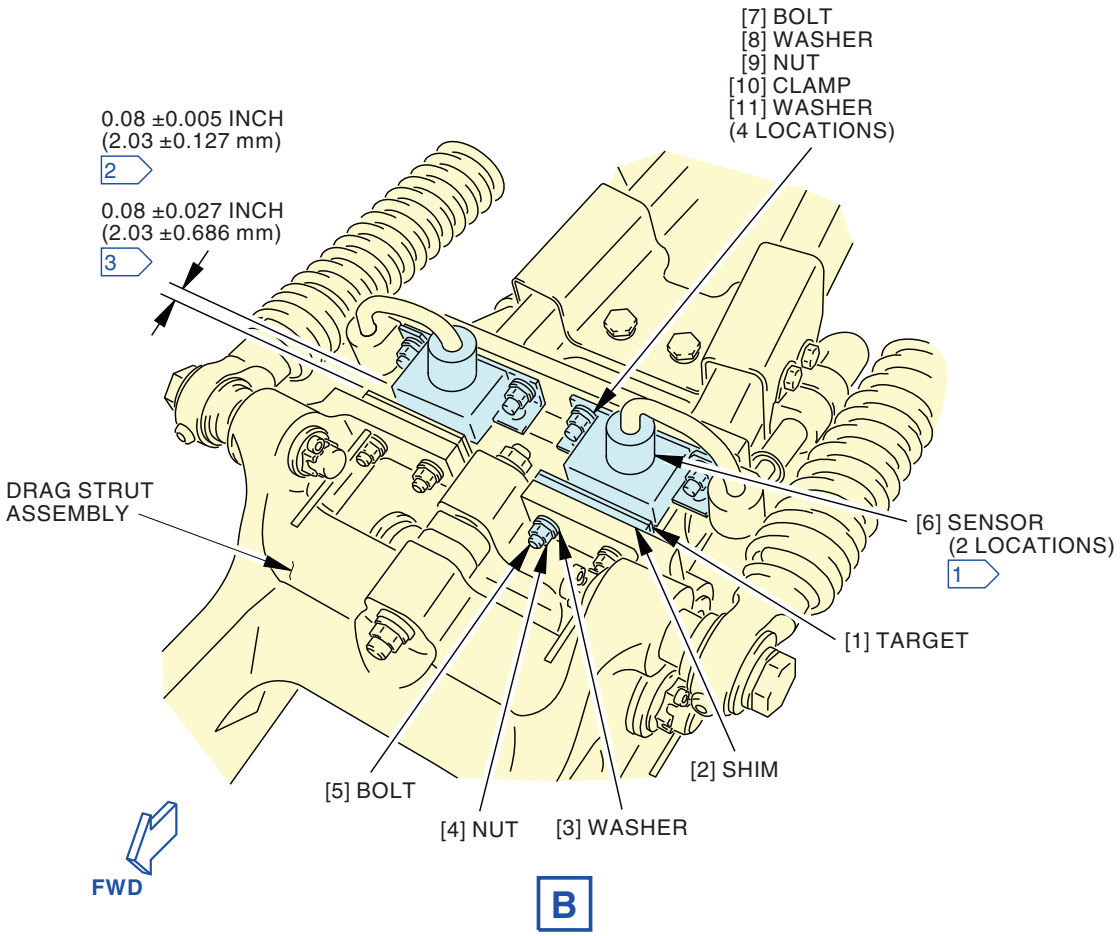
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Lock Sensor Adjustment
Figure 501/32-61-41-990-802 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1

SYSTEM NO.	SENSOR NO.	LOCATION	NAME
1	S0846	L	LKD
2	S0854	R	LKD

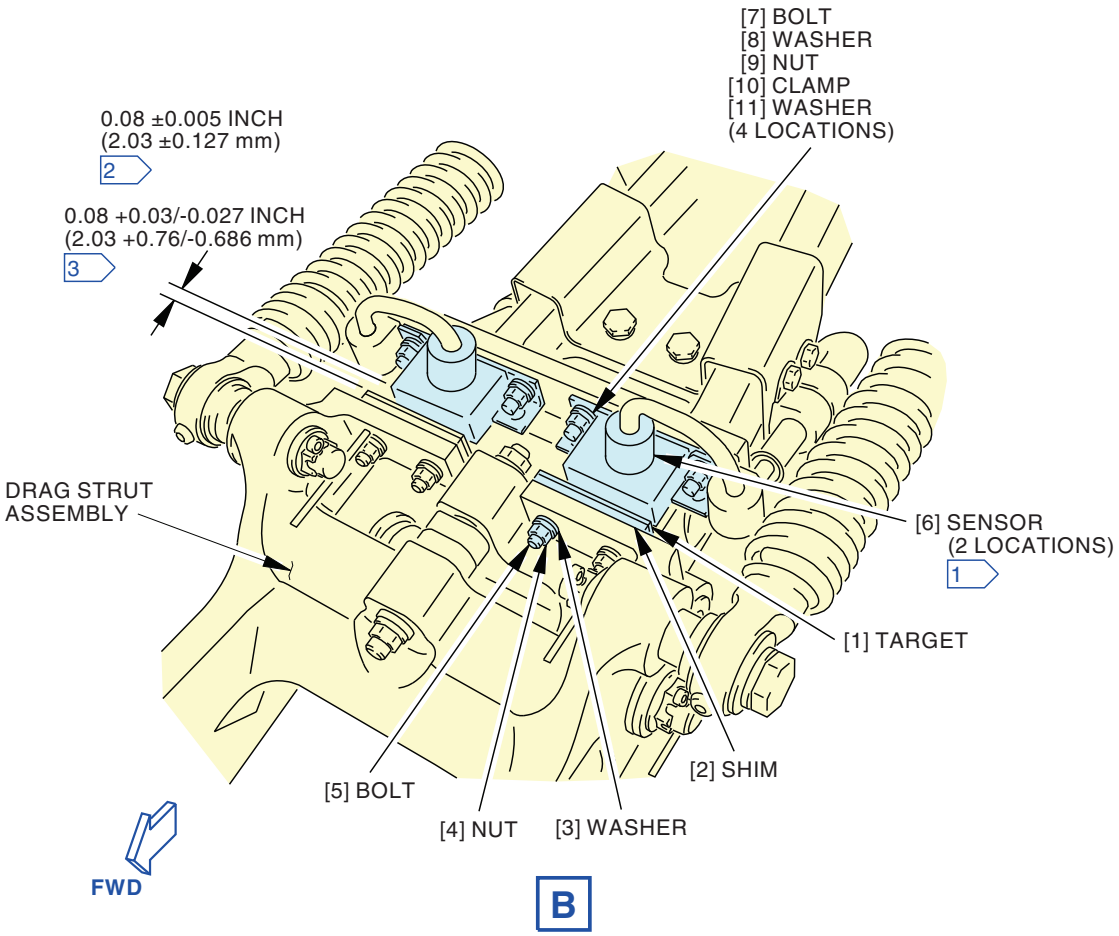
2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2414548 S00061535408_V5

Nose Landing Gear Lock Sensor Adjustment
Figure 501/32-61-41-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1

SYSTEM NO.	SENSOR NO.	LOCATION	NAME
1	S0846	L	LKD
2	S0854	R	LKD

2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2972020 S0000750685_V1

Nose Landing Gear Lock Sensor Adjustment
Figure 501/32-61-41-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-41-400-802

3. Nose Landing Gear Lock Sensor Clearance Adjustment

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Location Zones

Zone	Area
713	Nose Landing Gear

D. Prepare for the Adjustment

SUBTASK 32-61-41-580-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-41-480-007

(2) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

E. Nose Landing Gear Lock Sensor Clearance Adjustment

SUBTASK 32-61-41-020-003

(1) Do these steps to remove the nose landing gear locked sensor [6] from the drag strut:

(a) Remove the bolts [7], washers [8], washers [11], nuts [9], and clamp [10] from the sensor [6].

(b) Move the sensor [6] away from the drag strut.

SUBTASK 32-61-41-020-004

(2) Do these steps to remove the target [1] and shims [2]:

(a) Remove the nuts [4], washers [3], and bolts [5].

(b) Remove the target [1] and shims [2].

SUBTASK 32-61-41-110-001

(3) Clean the shims [2] and mounting surfaces.

SUBTASK 32-61-41-820-002

(4) Do one of the following steps to adjust the shims [2] thickness to get the necessary clearance between the target [1] and sensor [6] (Figure 501):

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(a) Remove the laminations from the existing shims [2].

(b) Apply the new additional shims [2].

1) If it is necessary, remove the laminations from the new shims [2].

(c) Replace the existing shims [2] with the new shims [2].

1) If it is necessary, remove the laminations from the new shims [2].

SUBTASK 32-61-41-620-001

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply thin coat of Cor-Ban 27L Compound, G50237, to the thread reliefs, threads, and washer faces.

(a) Wipe off the excess.

SUBTASK 32-61-41-420-004

(6) Do these steps to assemble the target:

(a) Position the shims [2] and target [1] on the mounting bracket.

(b) Install the bolts [5], nuts [4], and washers [3].

SUBTASK 32-61-41-410-003

(7) Fay seal the bottom of the sensor [6] with sealant, A00247.

SUBTASK 32-61-41-420-005

(8) Do these steps to re-install the sensor [6]:

(a) Position the sensor [6] on the mounting bracket.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply thin coat of Cor-Ban 27L Compound, G50237, to the thread reliefs, threads, and washer faces.

1) Wipe off the excess.

(c) Attach the sensor to the drag strut with the bolts [7], washers [8], washers [11], nuts [9], and clamp [10] attaching hardware.

NOTE: Use CRES washers below nuts with titanium sensors. Use cad-plated washers below bolt heads.

SUBTASK 32-61-41-220-002

(9) Measure the sensor clearance (TASK 32-61-41-220-801).

(a) Make sure that the clearance is correct (Figure 501).

1) If the sensor clearance is not correct, then do the steps to adjust the sensor clearance again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-41-390-002

(10) Do these steps to seal the shims [2] stack:

(a) Clean the shims [2].

(b) Brush primer, C00259, on the shims [2] surfaces.

(c) Apply sealant, A00247, to the shims [2] surfaces.

——— END OF TASK ———

TASK 32-61-41-710-801

4. Nose Landing Gear Lock Sensor Operational Test

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 32-61-41-580-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-41-480-009

(2) Make sure chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

D. Nose Landing Gear Lock Sensor Operational Test

SUBTASK 32-61-41-740-001

(1) Use the Multi Function Display to test the Nose Landing Gear Lock Sensor.

(a) Set the GND TESTS Switch on the P61-4 panel to Enable position.

(b) Select SYS on the MFD Multi Function Display panel.

(c) Do these steps to start the test:

1) Select MAINT

2) Select MAINT DATA PGS

3) Select 32 PSEU DISPLAY

(d) Make sure the MFD shows the sensor name and rigging gap.

NOTE: See the table in Figure 501 for the sensor identification and location.

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(e) Make sure the MFD shows the sensor clearance or the acceptable rigging gap range within tolerance.

NOTE: For a description of the sensor clearance display, see Figure 501.

NOTE: 1 MIL equals 1/1000th of an inch.

(f) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Nose Landing Gear Lock Sensor Clearance Adjustment, TASK 32-61-41-400-802.

E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-61-41-860-005

(1) Set the GND TESTS switch on the P61-4 Panel to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR DOWN POSITION SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Nose landing gear down position sensor removal

(2) Nose landing gear down position sensor installation.

B. The installation procedures for the #1 and #2 sensor are similar.

TASK 32-61-51-020-801

2. Nose Landing Gear Down Position Sensor Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4938	Insertion/Removal Tool - Contact (size 16, backshell side) Part #: M81969/1-03 Supplier: 11851 Part #: M81969/14-03 Supplier: 11139
STD-837	String

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
713	Nose Landing Gear

D. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

E. Prepare for the Removal

SUBTASK 32-61-51-580-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-51-480-002

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-61-51-860-006

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-51-860-001

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

SUBTASK 32-61-51-010-002

(5) To get access to the plug, open these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

F. Nose Landing Gear Down Position Sensor Removal

SUBTASK 32-61-51-020-001

(1) Disconnect the sensor wires:

(a) Disconnect the plug [7] from the connector on the outer side of the pressure seal [6].

(b) Identify the wires in the plug [7] that go to the nose landing gear down sensors.

1) Put a label on them or make a drawing to refer to during installation.

(c) Use contact insertion / removal tool, COM-4938, to remove the wires from the plug [7].

(d) Install a protective cap on the plug.

(e) Keep the plug safe for the installation.

(f) Attach a string, STD-837, of appropriate length to the sensor wire.

NOTE: The string, STD-837, will be used to pull the wire of the new sensor through the conduit. You can tie the string, STD-837, to the landing gear structure to make sure that the string, STD-837, is not pulled through the conduit.

SUBTASK 32-61-51-020-002

(2) Remove the sensor [5]:

(a) Remove the bolts [1], washers [2], washers [4], and nuts [3] that hold the sensor [5] to the mounting bracket.

(b) Remove the sensor [5] from the bracket.

(c) Pull the sensor [5] wires out of the conduit.

(d) Make sure that you do not pull the string, STD-837, through the conduit.

(e) Remove the string, STD-837, from the sensor [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-61-51-210-001

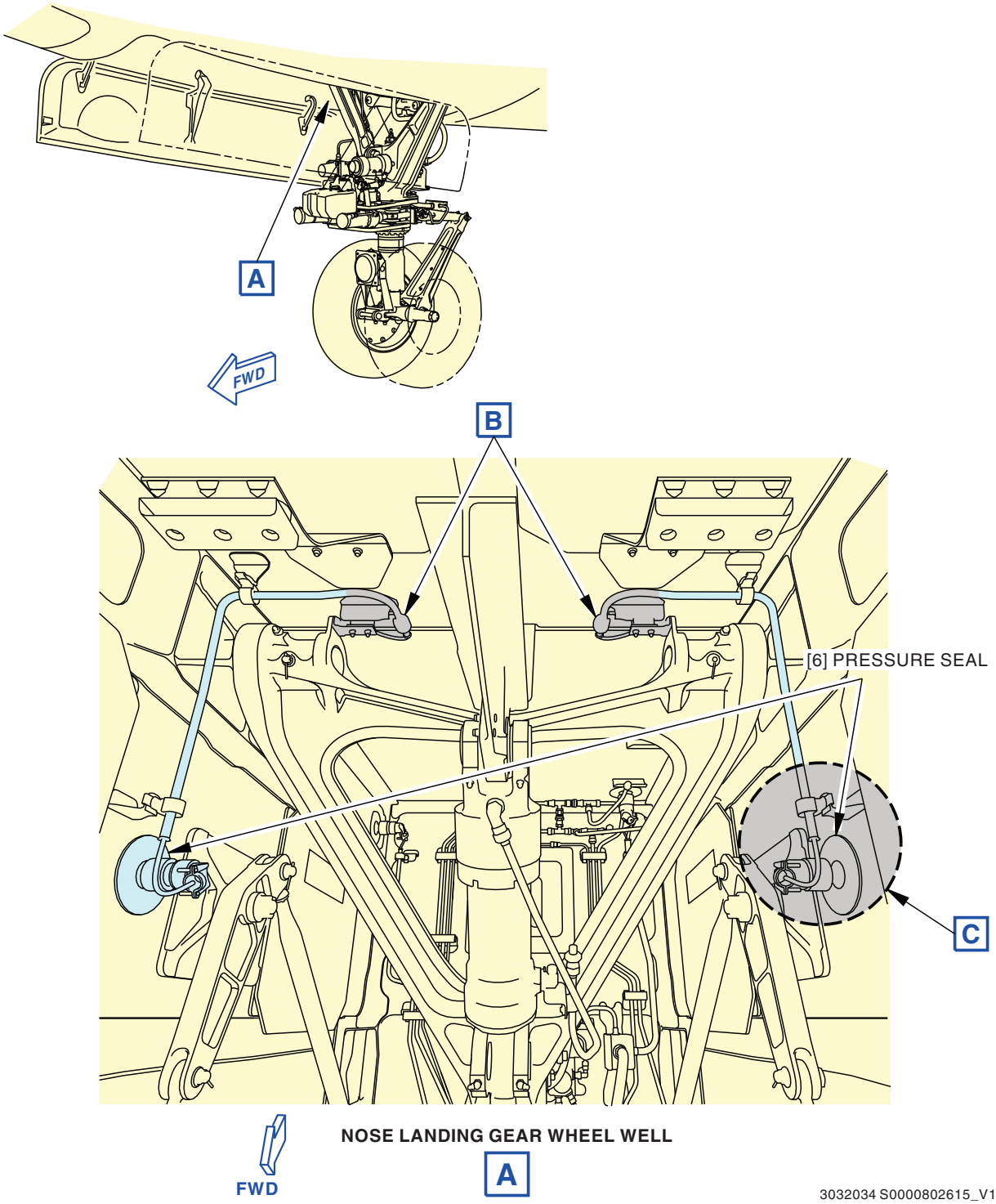
(3) Examine the sensor [5] and adjacent areas for signs of damage.

NOTE: Damage to the sensor can be an indication of an incorrect clearance between the sensor and the target.

(a) If you find damage, make a note of the damage for the installation procedure.

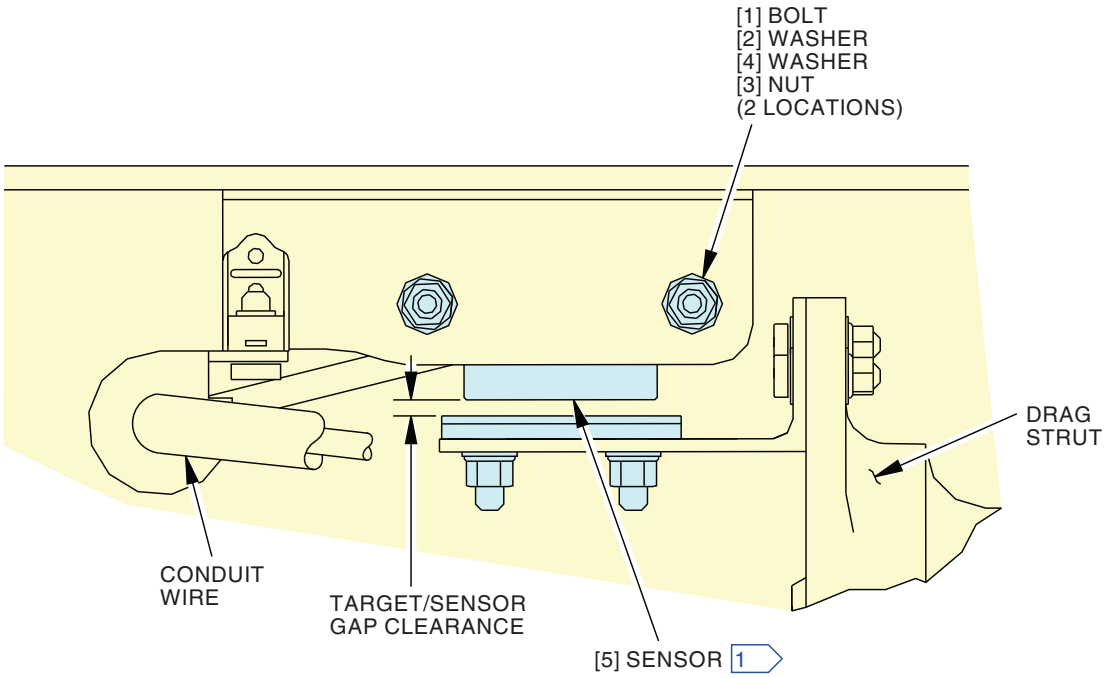
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Down Sensor Installation
Figure 401/32-61-51-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

B

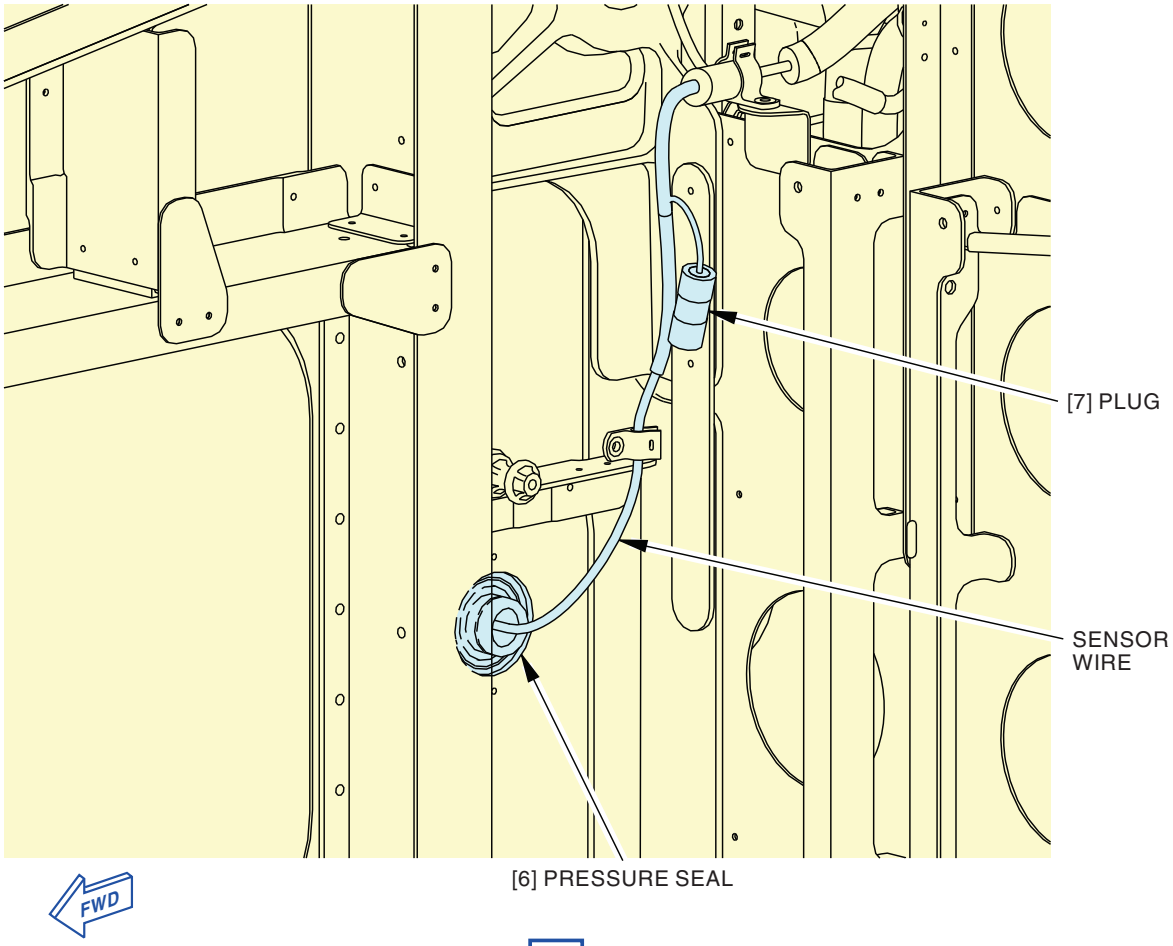
1

SYSTEM NO.	SENSOR NO.	LOCATION
1	S0845	L
2	S0853	R

2414551 S00061535416_V4

Nose Landing Gear Down Sensor Installation
Figure 401/32-61-51-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3043678 S0000810664_V1

Nose Landing Gear Down Sensor Installation
Figure 401/32-61-51-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-51-400-801

3. Nose Landing Gear Down Position Sensor Installation

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-61-51-220-801	Nose Landing Gear Down Position Sensor Clearance Measurement (P/B 501)
32-61-51-710-801	Nose Landing Gear Down Position Sensor Operational Test (P/B 501)
SWPM 20-10-20	Standard Wiring Practices Manual
SWPM 20-83-01/001	Standard Wiring Practices Manual

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-4938	Insertion/Removal Tool - Contact (size 16, backshell side) Part #: M81969/1-03 Supplier: 11851 Part #: M81969/14-03 Supplier: 11139
STD-837	String

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Sensor	32-61-51-01-005	PTF ALL

E. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
713	Nose Landing Gear

F. Access Panels

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Installation

SUBTASK 32-61-51-580-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-51-480-004

(2) Make sure that chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-61-51-860-007

WARNING OBEY THE PROCEDURE THAT PUTS THE AIRPLANE IN THE AIR MODE. IF YOU DO THE PROCEDURE INCORRECTLY, INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Prepare the safety-sensitive systems for air mode simulation (TASK 32-09-00-860-801).

SUBTASK 32-61-51-860-002

(4) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

H. Nose Landing Gear Down Position Sensor Installation

SUBTASK 32-61-51-410-001

(1) Apply a fay seal to the bottom of the sensor [5] with sealant, A00247.

SUBTASK 32-61-51-420-001

(2) Attach the sensor [5] to the mounting bracket:

(a) Put the sensor [5] in its position on the mounting bracket.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin coat of Cor-Ban 27L Compound, G50237, to the thread reliefs, threads, and washer faces.

1) Remove all excess material.

(c) Install the washers [2], washers [4], nuts [3], and bolts [1].

NOTE: Use Corrosion Resistant Steel (CRES) washers below bolt heads with titanium sensor.

SUBTASK 32-61-51-410-002

(3) Attach the sensor [5] wires to the airplane:

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(a) Attach the string, STD-837, to the sensor wires.

(b) Pull the wires through the conduit and pressure seal [6].

(c) Remove the string, STD-837, from the sensor [5] wires.

SUBTASK 32-61-51-350-001

(4) Cut the sensor [5] wires to the correct length.

SUBTASK 32-61-51-420-002

(5) Attach the contacts to the sensor [5] wires (SWPM 20-83-01/001).

SUBTASK 32-61-51-420-003

(6) Install the sensor contacts into the plug [7].

(a) Remove the protective cap from the plug [7].

(b) Use contact insertion / removal tool, COM-4938, to connect the sensor wire to the plug [7].

1) Use the drawing you made during the removal to connect the correct wires to the contacts.

(c) Push the plug [7] from the sensor wire into the connector on the other side of the pressure seal [6].

SUBTASK 32-61-51-410-007

(7) If necessary, apply the pressure seal [6] again (SWPM 20-10-20).

SUBTASK 32-61-51-400-001

(8) If you found damage to the sensor [5] or the adjacent area during the removal procedure, do these steps:

(a) Repair or replace all damaged components or areas.

NOTE: Damage to the sensor, target, or adjacent areas can be an indication of an incorrect sensor clearance.

(b) Measure the sensor [5] clearance, do this task: Nose Landing Gear Down Position Sensor Clearance Measurement, TASK 32-61-51-220-801.

I. Nose Landing Gear Down Position Sensor Installation Test

SUBTASK 32-61-51-410-008

(1) Close these access panels:

Number	Name/Location
113AW	Forward Nose Wheel Well Panel
114AW	Forward Nose Wheel Well Panel

SUBTASK 32-61-51-720-001

(2) Do a post installation check of the nose landing gear down position sensor:

(a) Do this task: Nose Landing Gear Down Position Sensor Clearance Measurement, TASK 32-61-51-220-801.

(b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do this task: Nose Landing Gear Down Position Sensor Operational Test, TASK 32-61-51-710-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE LANDING GEAR DOWN POSITION SENSOR - ADJUSTMENT/TEST

1. General

A. This procedure has three tasks:

(1) Nose landing gear down position sensor clearance measurement

(2) Nose landing gear down position sensor clearance adjustment

(3) Nose landing gear down position sensor operational test.

B. The procedures for both installations are similar.

TASK 32-61-51-220-801

2. Nose Landing Gear Down Position Sensor Clearance Measurement

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

C. Prepare for the Clearance Measurement

SUBTASK 32-61-51-480-013

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-51-480-006

(2) Make sure chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

D. Measure the Nose Landing Gear Down Position Sensor Clearance Measurement

SUBTASK 32-61-51-220-001

(1) Measure the clearance between the sensor and target.

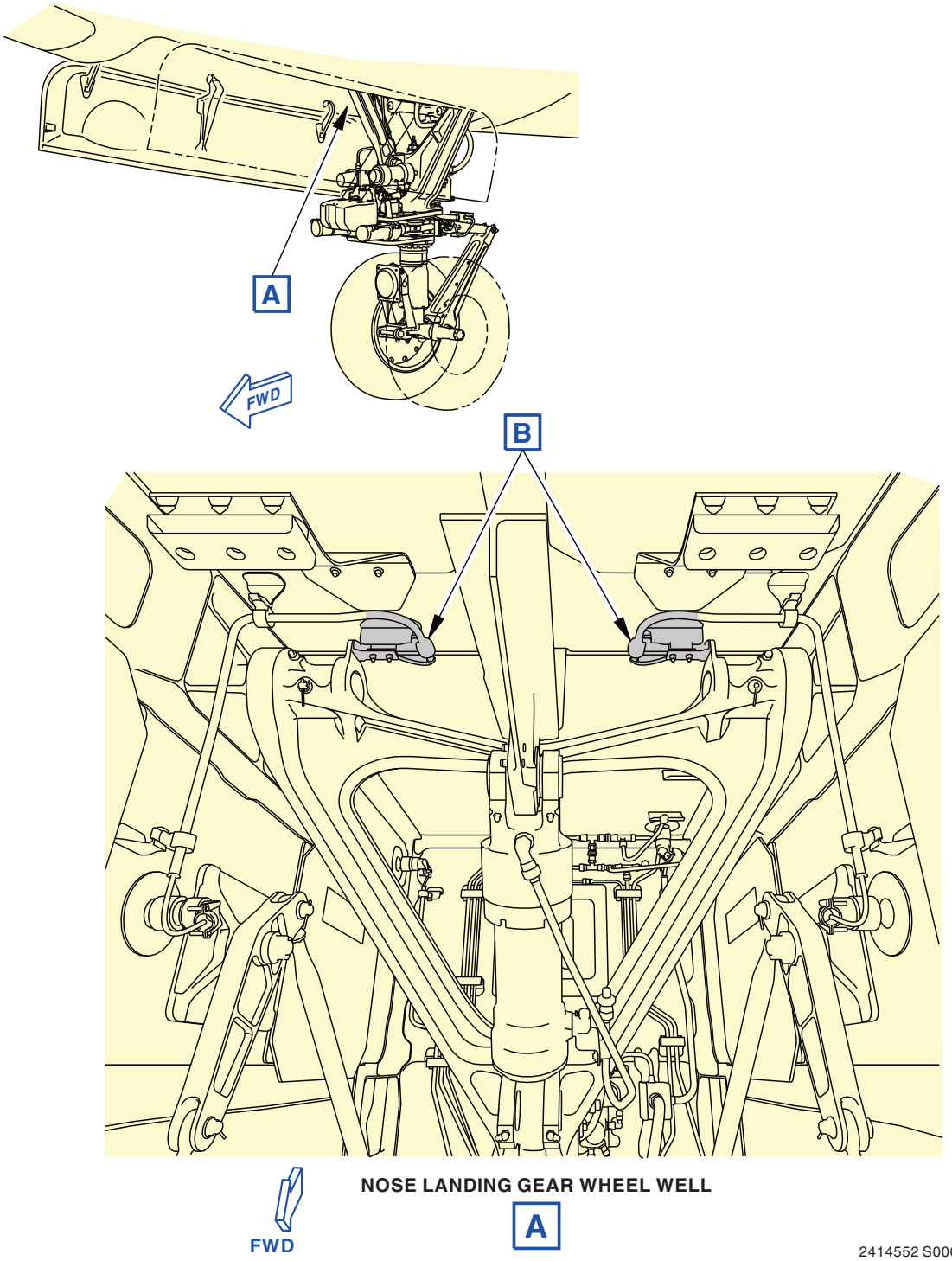
SUBTASK 32-61-51-820-001

(2) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Nose Landing Gear Down Position Sensor Clearance Adjustment, TASK 32-61-51-400-802.

(a) Make a written record of the measured clearance so you will know what adjustments must be made.

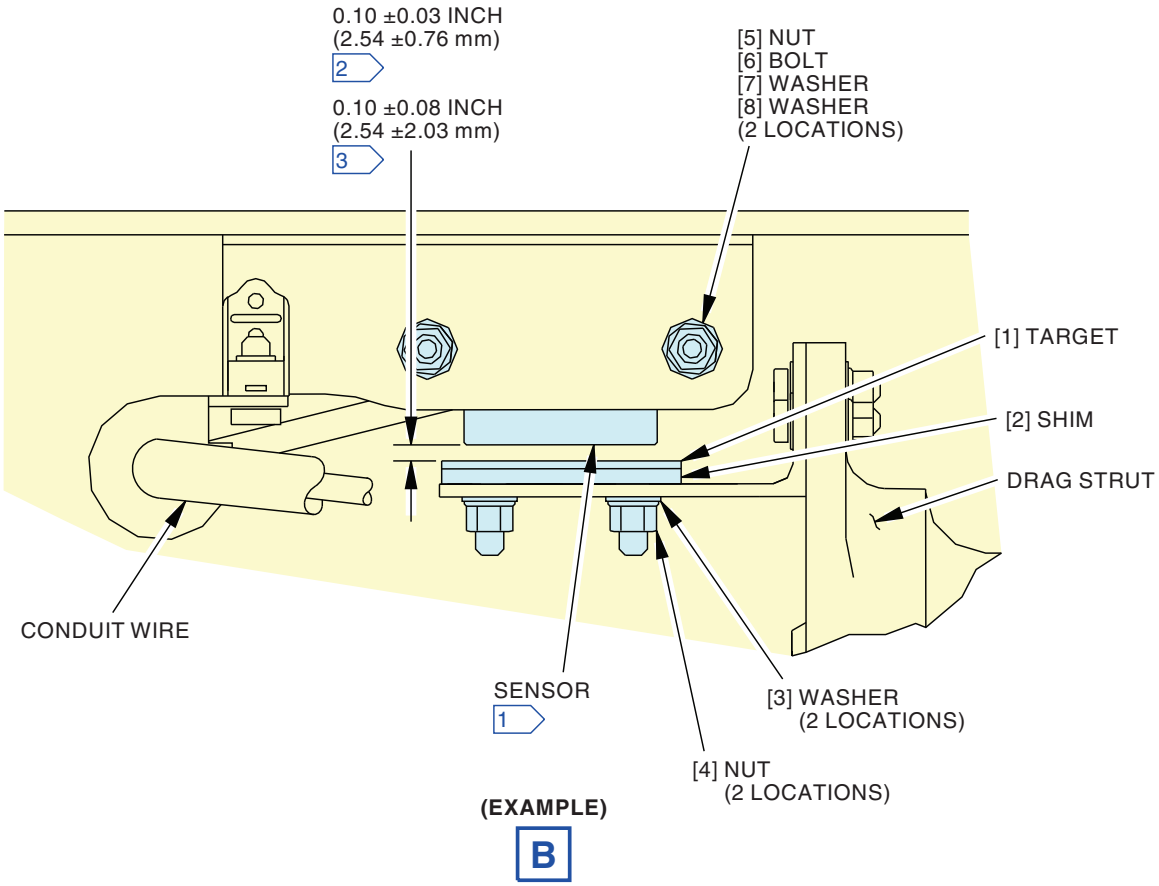
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Down Position Sensor Adjustment
Figure 501/32-61-51-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1

SYSTEM NO.	SENSOR NO.	LOCATION	NAME
1	S0845	L	DN
2	S0853	R	DN

2 PHYSICAL GAP TOLERANCE

3 PSEU TOLERANCE

2414554 S00061535422_V4

Nose Landing Gear Down Position Sensor Adjustment
Figure 501/32-61-51-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-51-400-802

3. Nose Landing Gear Down Position Sensor Clearance Adjustment

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

C. Location Zones

Zone	Area
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
713	Nose Landing Gear

D. Prepare for the Adjustment

SUBTASK 32-61-51-480-012

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-51-480-008

(2) Make sure that chocks are installed at the wheels (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

E. Nose Landing Gear Down Position Sensor Clearance Adjustment

SUBTASK 32-61-51-010-001

(1) To loosen the sensor from the mounting bracket, remove the nuts [5], bolts [6], washers [7], and washers [8].

SUBTASK 32-61-51-020-003

(2) Remove the nuts [4] and washers [3].

SUBTASK 32-61-51-020-004

(3) Remove the target [1] and shims [2].

SUBTASK 32-61-51-110-001

(4) Clean the shims and mounting surfaces.

SUBTASK 32-61-51-820-002

(5) Adjust the shims [2] thickness as required to get the necessary sensor to target clearance, do one of the steps that follow:

(a) Remove the laminations from the existing shims [2].

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(b) Apply additional shims [2].

1) If it is necessary, remove the laminations from the new shims [2].

(c) Replace the existing shims [2] with the new shims [2] and adjust the thickness.

1) If it is necessary, remove the laminations from the new shims [2].

SUBTASK 32-61-51-620-001

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(6) Apply a thin coat of Cor-Ban 27L Compound, G50237, to the thread reliefs, threads, and washer faces.

(a) Wipe off the excess of compound.

SUBTASK 32-61-51-420-004

(7) Assemble the target assembly.

(a) Position the shims [2] and target [1] on the mounting bracket.

(b) Install the nuts [4] and washers [3].

SUBTASK 32-61-51-410-005

(8) Fay seal the bottom of the sensor with sealant, A00247.

SUBTASK 32-61-51-410-006

(9) Attach the sensor to the mounting bracket.

(a) Put the sensor in its position on the mounting bracket.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin coat of Cor-Ban 27L Compound, G50237, to the thread reliefs, threads, and washer faces.

1) Wipe off the excess of compound.

(c) Install the washers [7], washers [8], nuts [5], and bolts [6].

NOTE: Use CRES washers below bolt heads with titanium sensors. Use cad-plated washers below nuts.

SUBTASK 32-61-51-220-002

(10) Measure the sensor clearance and compare to the requirement (TASK 32-61-51-220-801).

(a) If the sensor clearance is not correct, do the steps to adjust the sensor clearance again.

SUBTASK 32-61-51-390-001

(11) Seal the shim stack.

(a) Clean the shims.

(b) Apply sealant, A00247, to the shim surfaces.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-61-51-710-801

4. Nose Landing Gear Down Position Sensor Operational Test

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Prepare for the Operational Test

SUBTASK 32-61-51-480-011

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-61-51-480-010

(2) Make sure chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

D. Nose Landing Gear Down Position Sensor Operational Test

SUBTASK 32-61-51-740-001

(1) Use the Multi Function Display to test the Nose Landing Gear Down Position Sensor.

(a) Set the GND TESTS Switch on the P61-4 panel to Enable position.

(b) Select SYS on the MFD Multi Function Display panel.

(c) Do these steps to start the test:

1) Select MAINT

2) Select MAINT DATA PGS

3) Select 32 PSEU DISPLAY

(d) Make sure the MFD shows the sensor name and rigging gap.

NOTE: See the table in Figure 501 for the sensor identification and location.

(e) Make sure the MFD shows the sensor clearance or the acceptable rigging gap range within tolerance.

NOTE: For a description of the sensor clearance display, see Figure 501.

NOTE: 1 MIL equals 1/1000th of an inch.

(f) If the sensor clearance is outside the tolerance shown in Figure 501, do this task: Nose Landing Gear Down Position Sensor Clearance Adjustment, TASK 32-61-51-400-802.

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E. Put the Airplane Back to Its Usual Condition

SUBTASK 32-61-51-860-004

(1) Set the GND TESTS switch on the P61-4 Panel to the NORM position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TAILSKID ASSEMBLY - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Tailskid Energy Absorber Cartridge Replacement

(2) Tailskid Shoe Replacement

(3) Tailskid Warning Decal Replacement.

TASK 32-71-00-960-801

2. Tailskid Energy Absorber Cartridge Replacement

(Figure 201)

A. General

(1) This procedure has steps to replace the tailskid energy absorber cartridge.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-71-00-000-803	Tailskid Cartridge Assembly Removal (P/B 401)
32-71-00-400-803	Tailskid Cartridge Assembly Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Prepare for the Tailskid Energy Absorber Cartridge Replacement

SUBTASK 32-71-00-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-71-00-020-014

(2) Remove the tailskid cartridge assembly, do this task: Tailskid Cartridge Assembly Removal, TASK 32-71-00-000-803.

F. Tail Skid Energy Absorber Cartridge Replacement

SUBTASK 32-71-00-020-002

(1) Do these steps to remove the energy absorber cartridge [3]:

(a) Remove the bolts [41], washers [42], and nuts [43] that hold the lower end fitting cartridge [5] to the energy absorber cartridge [3].

(b) Remove the lower end fitting cartridge [5] from the energy absorber cartridge [3].

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(c) Remove the bolts [41], washers [42], and nuts [43] that hold the upper end fitting cartridge [6] to the energy absorber cartridge [3].

(d) Remove the upper end fitting cartridge [6] from the energy absorber cartridge [3].

(e) Discard the energy absorber cartridge [3].

SUBTASK 32-71-00-210-001

CAUTION

DO NOT USE DAMAGED TAIL SKID PARTS. DAMAGE TO THE AIRCRAFT CAN OCCUR.

(2) Do these steps to visually inspect the lower end fitting cartridge [5] and upper end fitting cartridge [6]:

(a) Inspect the lower end fitting cartridge [5], upper end fitting cartridge [6], and upper and lower spherical bearings for damage or excessive wear.

(b) Install the lower end fitting cartridge [5] onto the upper end fitting cartridge [6].

1) Make sure that the two move freely.

SUBTASK 32-71-00-420-001

(3) Do these steps to install the new energy absorber cartridge [3]:

(a) Make sure to assemble the upper end fitting cartridge [6] to a new energy absorber cartridge [3].

1) Install bolts [41], washers [42], and nuts [43] wet with Cor-Ban 27L Compound, G50237 and tighten to 24.0 in-lb (2.7 N·m) - 26.0 in-lb (2.9 N·m) per SRM SUBJECT 51-40-04.

(b) Make sure to assemble the lower end fitting cartridge [5] to a new energy absorber cartridge [3].

1) Install bolts [41], washers [42], and nuts [43] wet with Cor-Ban 27L Compound, G50237 and tighten to 24.0 in-lb (2.7 N·m) - 26.0 in-lb (2.9 N·m) per SRM SUBJECT 51-40-04.

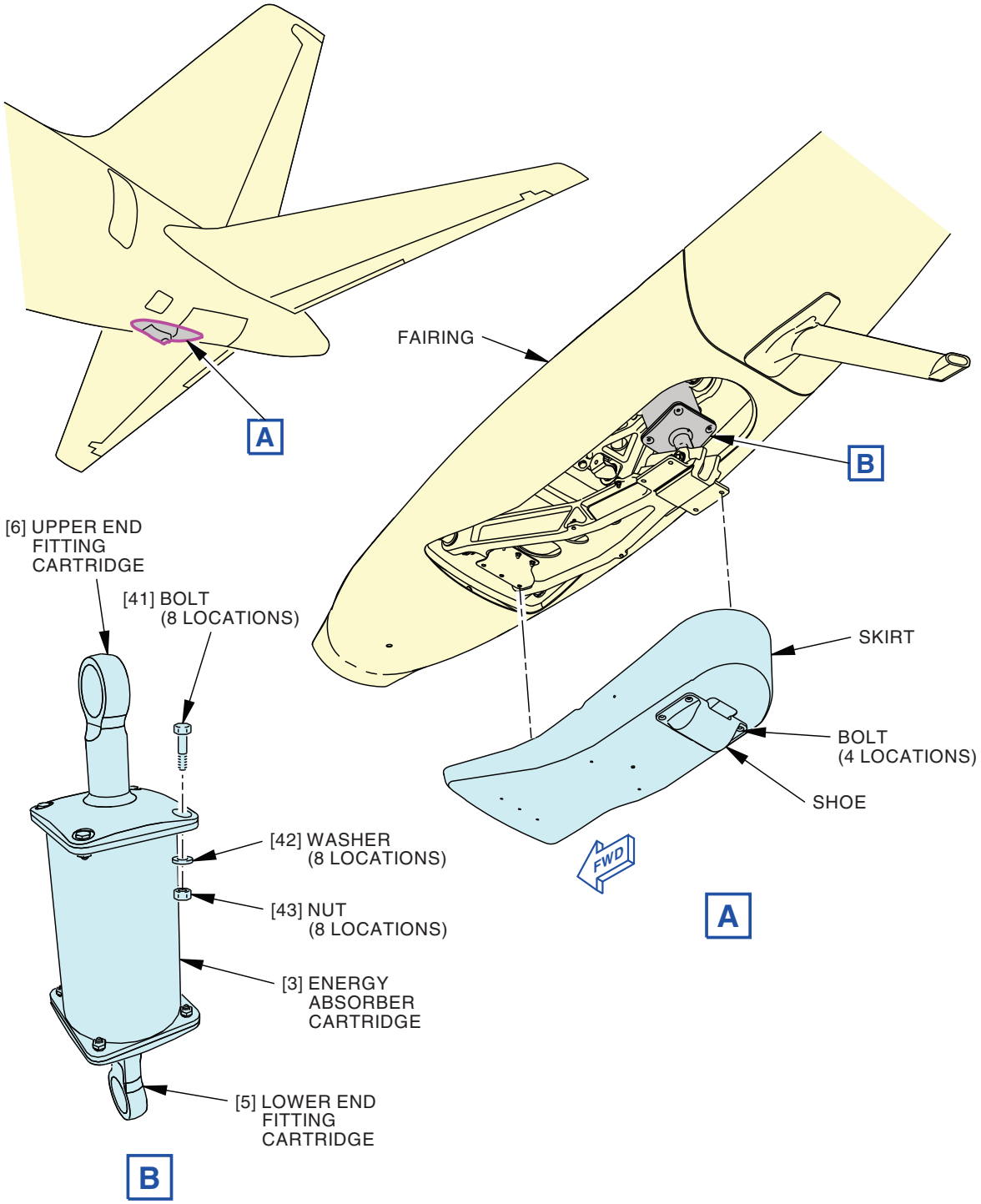
G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-71-00-420-018

(1) Install the Tailskid Cartridge Assembly, do this task: (Tailskid Cartridge Assembly Installation, TASK 32-71-00-400-803).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2522935 S0000592679_V2

Tailskid Cartridge Assembly Replacement
Figure 201/32-71-00-990-808

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-960-802

3. Tailskid Shoe Replacement

(Figure 202)

A. General

(1) This procedure has steps to replace the tailskid shoe.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Prepare for the Tailskid Shoe Replacement

SUBTASK 32-71-00-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

E. Tail Skid Shoe Replacement

SUBTASK 32-71-00-020-003

(1) Remove the shoe [9]:

(a) Remove the four bolts [8] from the skirt [1] and drag lever [2].

(b) Discard the shoe [9].

SUBTASK 32-71-00-420-003

(2) Install a new shoe [9]:

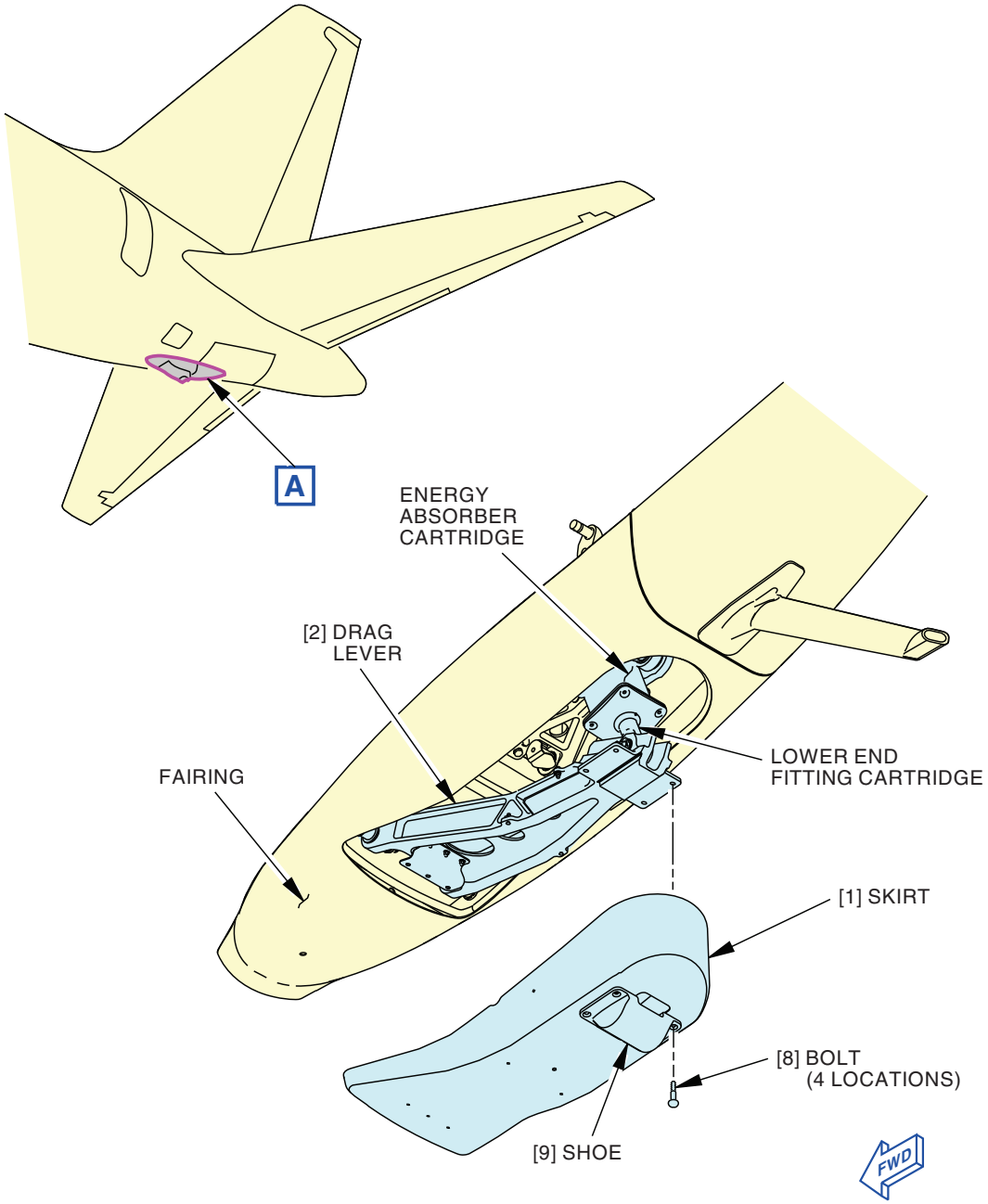
(a) Put the shoe [9] in position on the skirt [1].

(b) Install four bolts [8] to the skirt [1], drag lever [2] and shoe [9].

(c) Tighten the bolts [8].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TAILSKID ASSEMBLY

A

2414556 S00061535431_V3

Tailskid Shoe Replacement
Figure 202/32-71-00-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-960-803

4. Tailskid Warning Decal Replacement

(Figure 203)

A. General

(1) This task gives instructions to replace the Tailskid Warning Decal.

B. References

Reference	Title
20-10-11-400-801	Pressure-Sensitive Decal Installation (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tailskid Warning Decal Replacement

SUBTASK 32-71-00-480-031

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-71-00-020-017

(2) If it is necessary, remove the shoe [9]:

(a) Remove four bolts [8] that hold the shoe [9] to the skirt [1] and the drag lever [2].

(b) Remove the shoe [9] from the skirt [1].

SUBTASK 32-71-00-020-018

(3) If it is necessary, remove the skirt [1]:

(a) Remove the screws [7] that hold the skirt [1] to the drag lever [2].

(b) Remove the skirt [1] from the drag lever [2].

SUBTASK 32-71-00-420-021

(4) Install a new tailskid warning decal [22] on the aft side of the skirt [1] (TASK 20-10-11-400-801).

(a) Make sure that the upper edge of the tailskid warning decal [22] is in line with the fairing (Figure 203).

SUBTASK 32-71-00-420-022

(5) If removed, install the skirt [1]:

(a) Put the skirt [1] in its position on the drag lever [2].

(b) Install the screws [7] that attach the skirt [1] to the drag lever [2].

SUBTASK 32-71-00-420-023

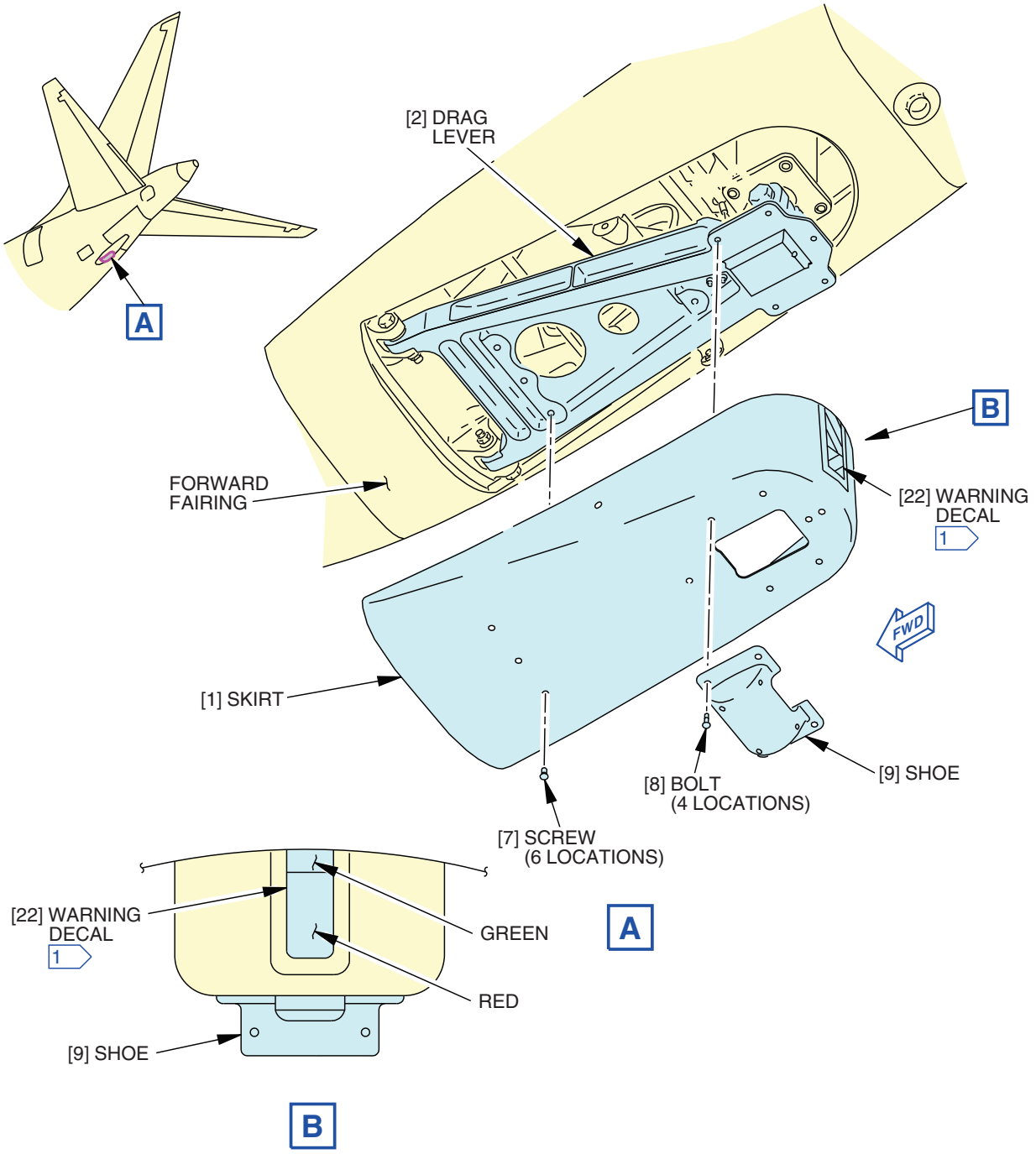
(6) If removed, install the shoe [9]:

(a) Put the shoe [9] in its position on the skirt [1].

(b) Install the four bolts [8] that attach the shoe [9] to the drag lever [2] and skirt [1].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

1 LOCATE GREEN EDGE OF THE DECAL IN LINE WITH FAIRING OML APPROX STA 1088

2554601 S0000608969_V3

Tailskid Warning Decal Replacement
Figure 203/32-71-00-990-814

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TAILSKID ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Tailskid Skirt Removal

(2) Tailskid Skirt Installation

(3) Tailskid Drag Lever Removal

(4) Tailskid Drag Lever Installation

(5) Tailskid Cartridge Assembly Removal

(6) Tailskid Cartridge Assembly Installation.

TASK 32-71-00-000-801

2. Tailskid Skirt Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the Tailskid Skirt.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-71-00-210-801	Tailskid Cartridge Assembly Inspection/Check (P/B 601)
32-71-00-211-801	Tailskid Shoe Inspection (P/B 601)

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Prepare for the Tailskid Skirt Removal

SUBTASK 32-71-00-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-71-00-480-007

(2) Use a ladder or equivalent to get access to the tailskid assembly.

SUBTASK 32-71-00-210-002

(3) Inspect the tailskid shoe (TASK 32-71-00-211-801).

SUBTASK 32-71-00-210-003

(4) Inspect the tailskid cartridge assembly (TASK 32-71-00-210-801).

E. Tailskid Skirt Removal

SUBTASK 32-71-00-020-004

(1) Do these steps to remove the skirt [1]:

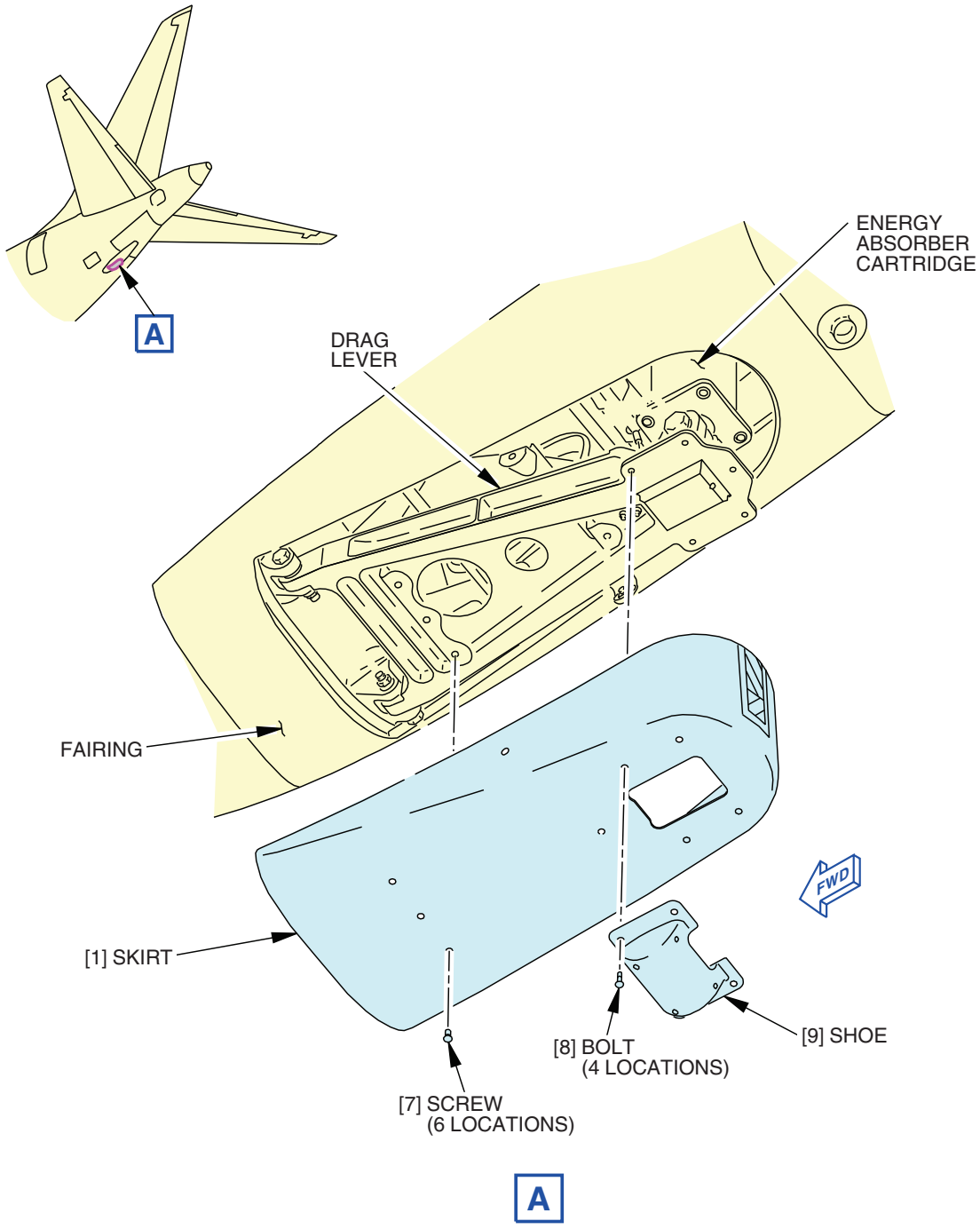
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the four bolts [8] from the shoe [9].

(b) Remove the screws [7] from the skirt [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3119832 S0000855033_V1

Tailskid Skirt Installation
Figure 401/32-71-00-990-805

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-400-801

3. Tailskid Skirt Installation

(Figure 401)

A. General

(1) This procedure has steps to install the Tailskid Skirt.

B. Consumable Materials

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Skirt	32-71-00-04-235	PTF ALL

D. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

E. Prepare for the Tailskid Skirt Installation

SUBTASK 32-71-00-480-009

(1) Use a ladder or equivalent to get access to the tailskid assembly.

SUBTASK 32-71-00-420-004

CAUTION

BE CAREFUL WHEN INSTALLING THE TAILSKID CARTRIDGE ASSEMBLY TO THE AIRCRAFT. DAMAGE TO THE ASSEMBLY CAN OCCUR.

(2) If necessary, install the tailskid cartridge assembly (TASK 32-71-00-400-803).

SUBTASK 32-71-00-420-005

(3) If necessary, install the tailskid drag lever (TASK 32-71-00-400-802).

F. Tailskid Skirt Installation

SUBTASK 32-71-00-420-006

(1) Do these steps to install the skirt [1]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply corrosion inhibiting material, G50136 to faying surfaces of the skirt [1] and drag lever [2].

(b) Install the screws [7] to hold the skirt [1] to the drag lever [2].

(c) Install the four bolts [8] to the shoe [9], skirt [1], and drag lever [2].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-000-802

4. Tailskid Drag Lever Removal

(Figure 402)

A. General

(1) This procedure has steps to remove the Tailskid Drag Lever.

B. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

C. Prepare for the Tailskid Drag Lever Removal

SUBTASK 32-71-00-480-010

(1) Use a ladder or equivalent to get access to the tailskid assembly.

SUBTASK 32-71-00-020-005

(2) Remove the tailskid skirt, do this task: Tailskid Skirt Removal, TASK 32-71-00-000-801.

D. Tailskid Drag Lever Removal

SUBTASK 32-71-00-020-015

(1) Do these steps to remove the fairing [21]:

(a) Remove the screws [22] from the fairing [21].

(b) Remove the fairing [21] from the airplane.

SUBTASK 32-71-00-020-012

(2) Do these steps to remove the drag lever [2]:

(a) Remove the nut [11], washers [12], and bolt [14].

(b) Disconnect the lower end fitting cartridge [5] from the drag lever [2].

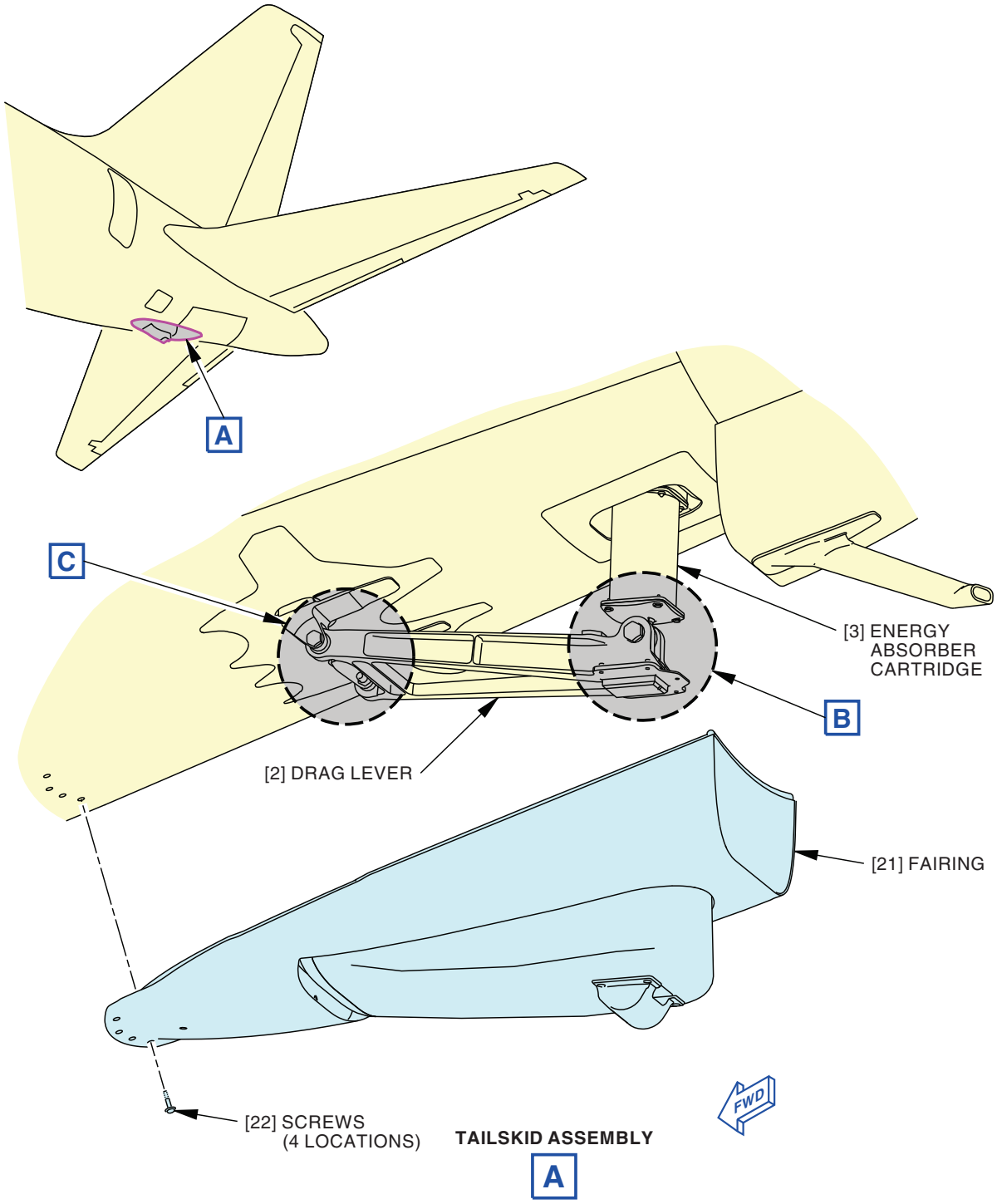
(c) Remove the nuts [17], washers [18], and shoulder bolts [20] from the tailskid support structure.

(d) Disconnect the drag lever [2] from the tailskid support structure.

(e) Remove the drag lever [2] from the airplane.

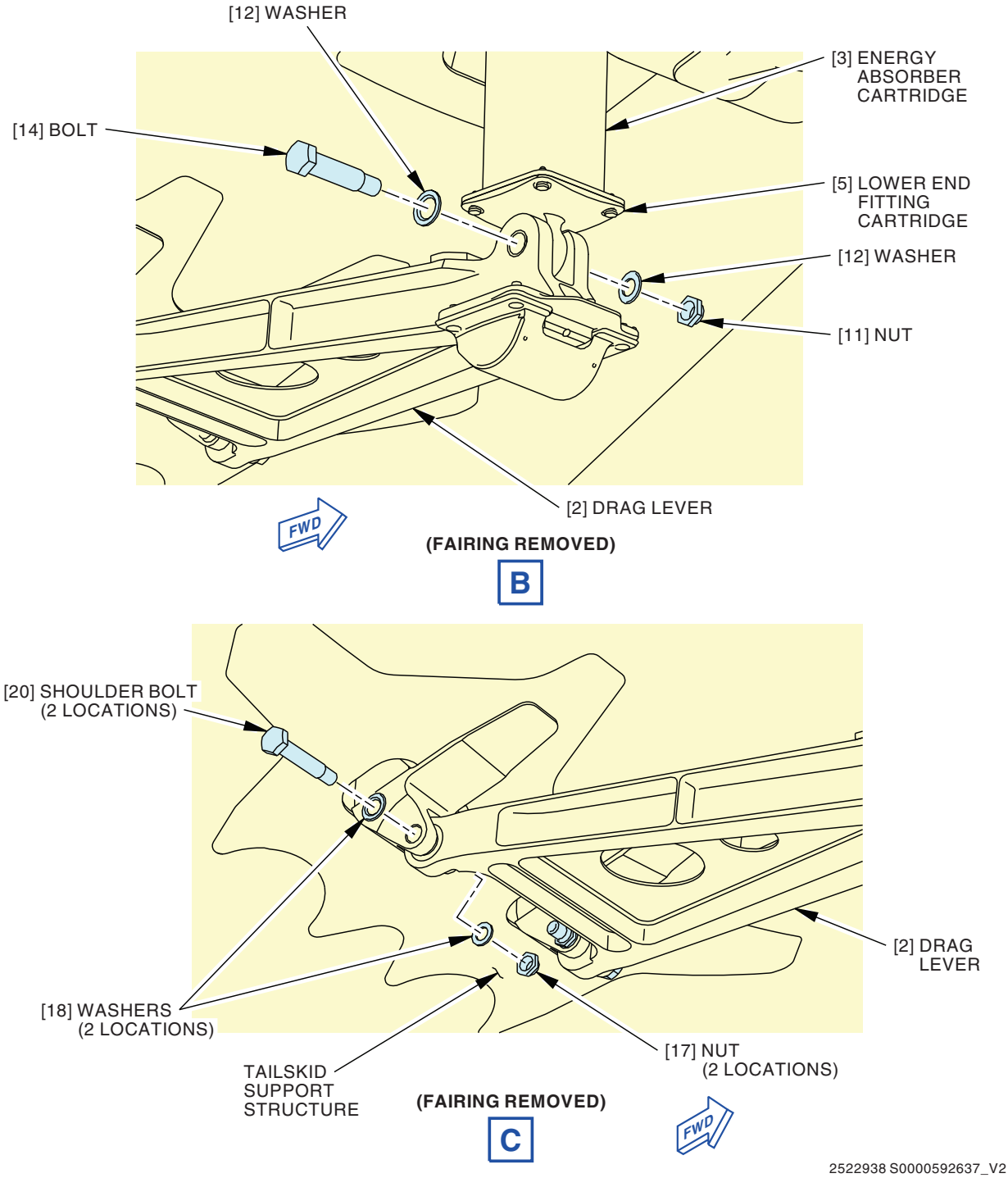
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Position Tailskid Drag Lever Installation
Figure 402/32-71-00-990-811 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Position Tailskid Drag Lever Installation
Figure 402/32-71-00-990-811 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-400-802

5. Tailskid Drag Lever Installation

(Figure 402)

A. General

(1) This procedure has steps to install the Tailskid Drag Lever.

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Drag lever	32-71-00-04-232	PTF ALL

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Prepare for the Tailskid Drag Lever Installation

SUBTASK 32-71-00-480-011

(1) Use a ladder or equivalent to get access to the tailskid assembly.

SUBTASK 32-71-00-420-007

CAUTION

BE CAREFUL WHEN INSTALLING THE TAILSKID CARTRIDGE ASSEMBLY TO THE AIRCRAFT. DAMAGE TO THE ASSEMBLY CAN OCCUR.

(2) If it is necessary, install the tailskid cartridge assembly (TASK 32-71-00-400-803).

E. Tailskid Drag Lever Installation

PTF ALL; AIRPLANES WITH FIXED POSITION TAIL SKID

SUBTASK 32-71-00-420-015

(1) Do these steps to install the drag lever [2]:

(a) Align the drag lever [2] with the tailskid support structure holes.

(b) Install the shoulder bolts [20], washers [18], and nuts [17] to the tailskid support structure.

(c) Align the lower end fitting cartridge [5] and the drag lever [2] hole.

(d) Install the bolt [14], washers [12], and nut [11].

PTF ALL

SUBTASK 32-71-00-420-019

(2) Put the fairing [21] in position on the airplane.

SUBTASK 32-71-00-420-020

(3) Install the screws [22] to the fairing [21].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-71-00-420-009

(1) Install the tailskid skirt, do this task: Tailskid Skirt Installation, TASK 32-71-00-400-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-000-803

6. Tailskid Cartridge Assembly Removal

(Figure 403)

A. General

(1) This procedure has steps to remove the tailskid cartridge assembly (referred to as energy absorber cartridge [3]).

B. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

C. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

D. Prepare for the Tailskid Cartridge Assembly Removal

SUBTASK 32-71-00-480-012

(1) Use a ladder or equivalent to get access to the tailskid assembly.

SUBTASK 32-71-00-020-007

(2) Remove the tailskid skirt (TASK 32-71-00-000-801).

SUBTASK 32-71-00-020-023

(3) Remove the screws [22] from the fairing [21].

SUBTASK 32-71-00-020-024

(4) Remove the fairing [21] from the airplane.

SUBTASK 32-71-00-010-003

(5) Open this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Tailskid Cartridge Assembly Removal

SUBTASK 32-71-00-020-013

(1) Do these steps to remove the energy absorber cartridge [3]:

(a) Remove the nut [11], washer [12], and bolt [14].

(b) Disconnect the lower end fitting cartridge [5] from the drag lever [2].

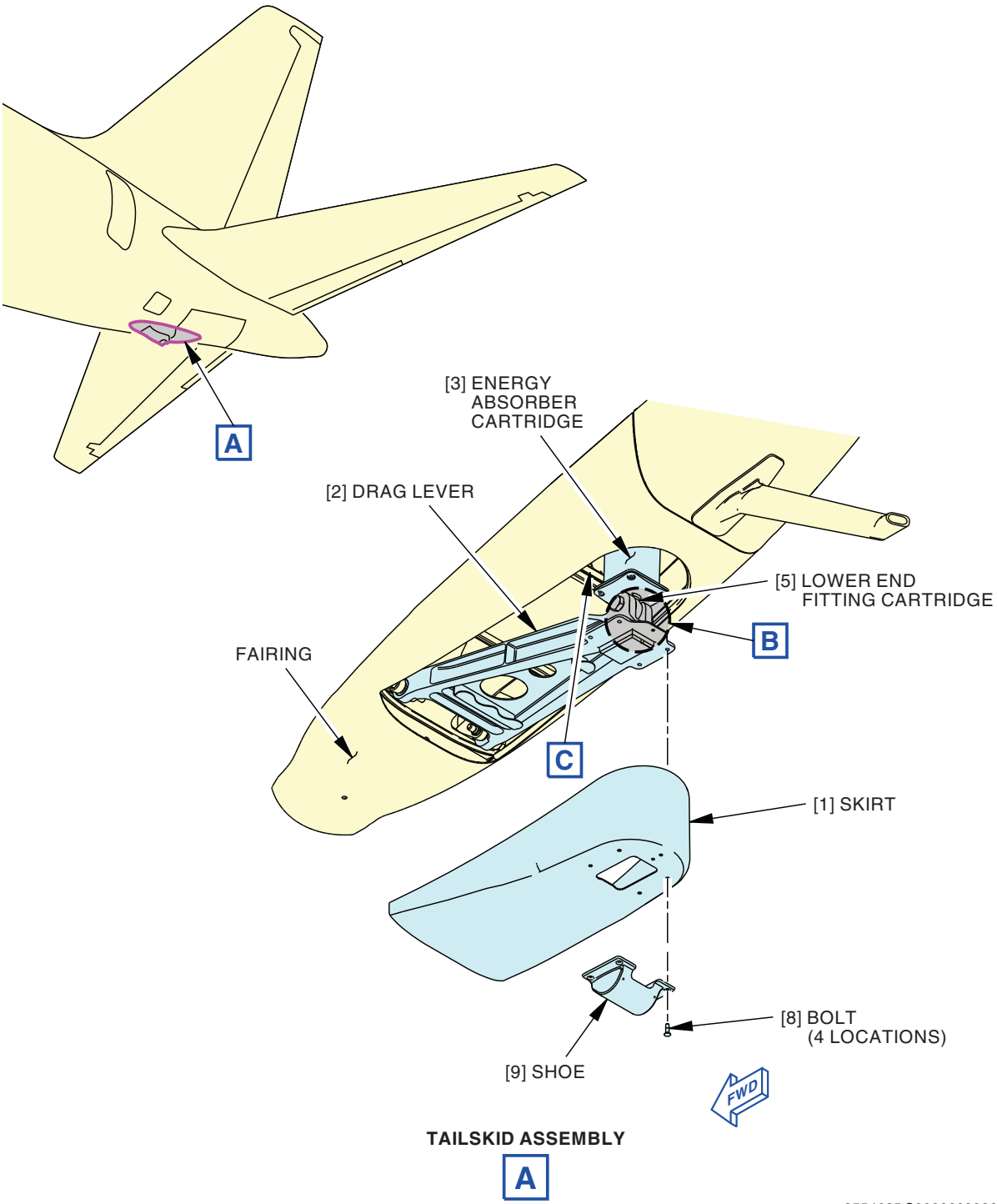
(c) Remove the nut [31], washers [32], and bolt [34].

(d) Disconnect the upper end fitting cartridge [6] from the pivot linkage [27].

(e) Remove the energy absorber cartridge [3] from the airplane.

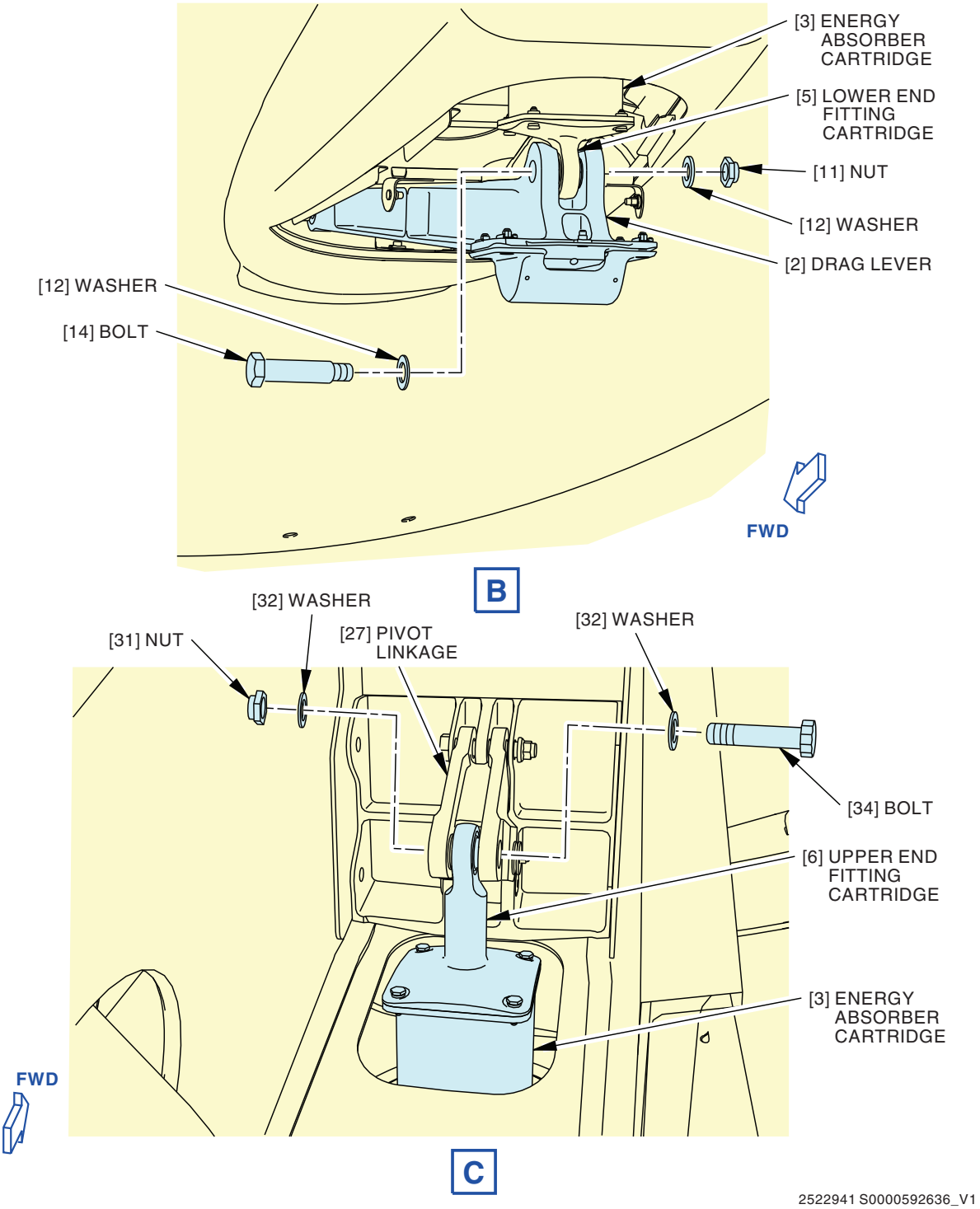
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Tail Skid Cartridge Assembly Installation
Figure 403/32-71-00-990-812 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fixed Tail Skid Cartridge Assembly Installation
Figure 403/32-71-00-990-812 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-00-400-803

7. Tailskid Cartridge Assembly Installation

(Figure 403)

A. General

(1) This procedure has steps to install the tailskid cartridge assembly (referred to as energy absorber cartridge [3]).

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Energy absorber cartridge	32-71-00-04-095	PTF ALL

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Prepare for the Tailskid Cartridge Assembly Installation

SUBTASK 32-71-00-480-013

(1) Use a ladder or equivalent to get access to the tailskid assembly.

SUBTASK 32-71-00-420-013

(2) If it is necessary, install the tailskid drag lever (TASK 32-71-00-400-802).

F. Tailskid Cartridge Assembly Installation

CAUTION

BEFORE THE PIN IS INSTALLED, MAKE SURE THE LINKAGE ASSEMBLY DOES NOT HIT THE AIRCRAFT. DAMAGE TO THE AIRCRAFT CAN OCCUR.

PTF ALL; AIRPLANES WITH FIXED POSITION TAIL SKID

SUBTASK 32-71-00-420-016

(1) Do these steps to install the energy absorber cartridge [3]:

(a) Make sure that the energy absorber cartridge [3] is securely assembled, before installing the assembly

(b) Align the upper end fitting cartridge [6] and pivot linkage [27] bolt hole.

(c) Install the bolt [34], washer [32], and nut [31].

(d) Align the lower end fitting cartridge [5] and drag lever [2].

(e) Install the bolt [14], washer [12], and nut [11].

PTF ALL

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-71-00-410-003

(1) Close this access panel:

Number	Name/Location
311BL	Stabilizer Trim Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-71-00-420-024

(2) Put the fairing [21] in position on the airplane.

SUBTASK 32-71-00-420-025

(3) Install the screws [22] to the fairing [21].

SUBTASK 32-71-00-420-014

(4) Install the tailskid skirt (TASK 32-71-00-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TAILSKID - INSPECTION/CHECK

1. General

A. Do an inspection/check of the tailskid assembly after there has been a tail strike.

B. This procedure has these tasks:

(1) Tailskid Cartridge Assembly Inspection/Check

(2) Tailskid Shoe Inspection

C. This task does not remove or install the tail skid assembly.

TASK 32-71-00-210-801

2. Tailskid Cartridge Assembly Inspection/Check

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to visually check the Tailskid Cartridge Assembly.

B. References

Reference	Title
05-51-32-210-801	Tail Strike Inspection (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-71-00-000-803	Tailskid Cartridge Assembly Removal (P/B 401)
32-71-00-400-803	Tailskid Cartridge Assembly Installation (P/B 401)

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Prepare for the Tailskid Cartridge Assembly Inspection/Check

SUBTASK 32-71-00-480-014

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-71-00-410-006

(2) Do a visual check of the tailskid energy absorber cartridge for evidence of tail strike.

NOTE: The cartridge assembly warning placard must be examined as soon as possible after the tail strike. The tailskid skirt fairing can re-extend due to gravity as time passes, resulting in reading error on the warning placard decal.

NOTE: The tailskid assembly is in normal position if the upper edge of the warning decal is in line with the fairing.

SUBTASK 32-71-00-480-023

(3) Use a ladder or equivalent to get access to the tailskid assembly.

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E. Tailskid Cartridge Assembly Inspection/Check

SUBTASK 32-71-00-210-007

(1) Check the warning decal [3] on the aft side of the skirt [1]:

(a) If there is evidence of a tail strike or damage to the airplane, do this task: Tail Strike Inspection, TASK 05-51-32-210-801.

(b) If any of the green part of the warning decal [3] is visible, then the energy absorber cartridge is still acceptable.

(c) If the green part of the warning decal [3] is not visible, then replace the energy absorber cartridge.

1) Do this task: Tailskid Cartridge Assembly Removal, TASK 32-71-00-000-803.

2) Do this task: Tailskid Cartridge Assembly Installation, TASK 32-71-00-400-803.

——— END OF TASK ———

TASK 32-71-00-211-801

3. Tailskid Shoe Inspection

(Figure 601)

A. General

(1) This procedure has steps to inspect the Tailskid Shoe.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-71-00-960-802	Tailskid Shoe Replacement (P/B 201)
51-21-99-300-802	Decorative Exterior Paint System Repair (P/B 701)

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Prepare for the Tailskid Shoe Inspection

SUBTASK 32-71-00-480-015

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-71-00-480-025

(2) Use a ladder or equivalent to get access to the tailskid assembly.

E. Tailskid Shoe Inspection

SUBTASK 32-71-00-210-008

(1) Do these steps to inspect the tailskid shoe [2] after a tail strike:

NOTE: The tailskid assembly is in normal position if the upper edge of the warning decal is in line with the fairing.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do a check of the tailskid shoe [2] for cracks or chips:

1) If there are cracks in the tailskid shoe [2] or pieces of the tailskid shoe [2] are gone, outboard of the bolt holes, the tailskid shoe [2] is serviceable.

2) If there are cracks that start at the bolt holes, replace the tailskid shoe [2] (TASK 32-71-00-960-802).

(b) If the tailskid shoe [2] is worn to the WEAR DIMPLES, replace the tailskid shoe [2] (TASK 32-71-00-960-802).

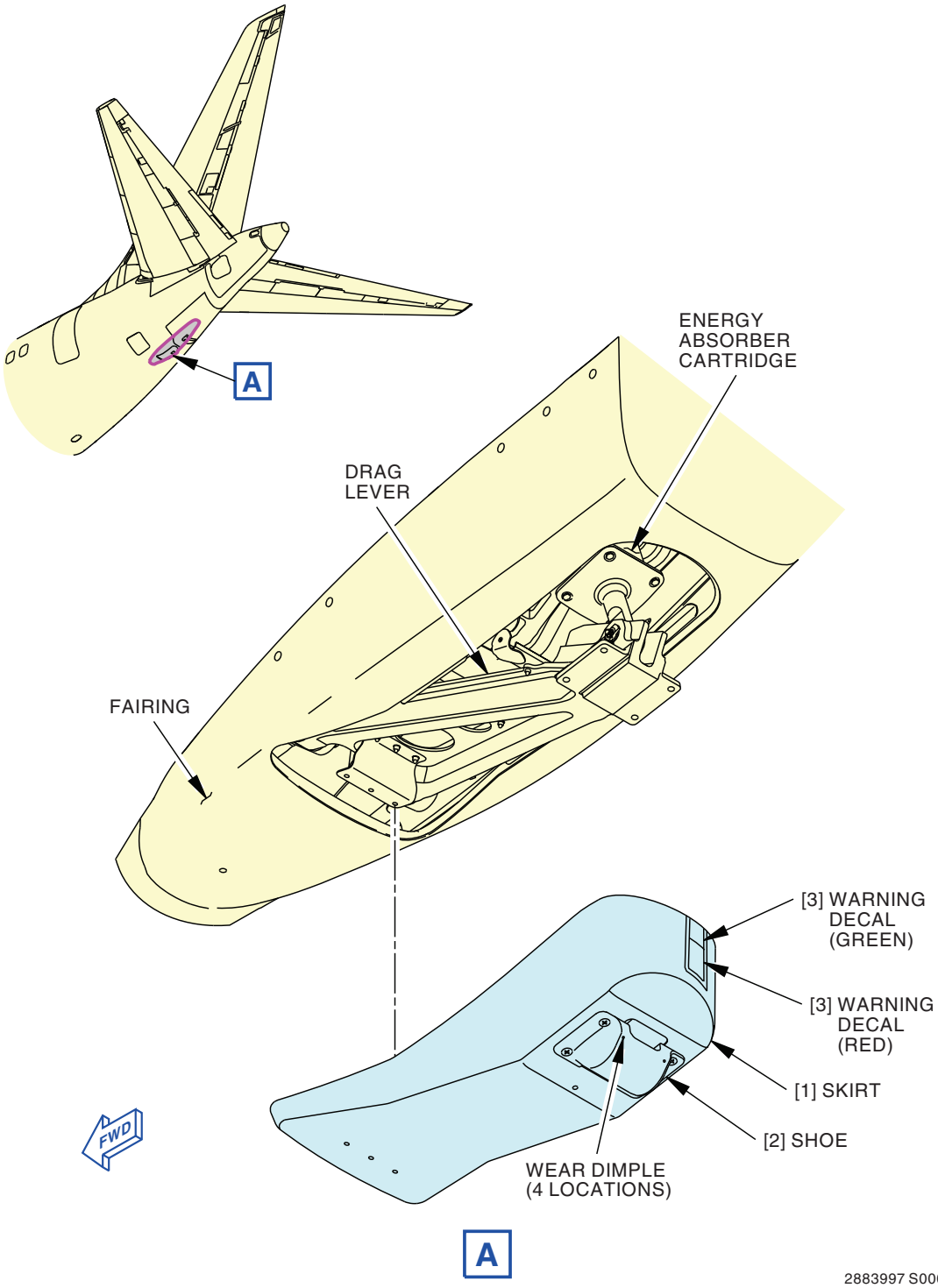
(c) Make sure the tail skid shoe [2] is within tolerance and the paint finish on the lower surface of the tail skid shoe [2] has not been gouged or scratched by contact with the runway.

1) To paint the skid shoe [2], do this task: Decorative Exterior Paint System Repair, TASK 51-21-99-300-802.

NOTE: If the tail skid shoe [2] scratch or gouge damage is not painted over, a person may interpret the damage as new and request an unnecessary inspection. The addition of paint can help to determine when new tail strikes have occurred.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tailskid Inspection
Figure 601/32-71-00-990-822

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TAIL SKID PIVOT LINKAGE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Tailskid Pivot Linkage Removal

(2) Tailskid Pivot Linkage Installation.

TASK 32-71-21-000-801

2. Tailskid Pivot Linkage Removal

(Figure 401)

A. General

(1) This procedure has steps to remove the Tailskid Pivot Linkage (referred to as pivot linkage [1]).

B. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

C. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

D. Prepare for the Tailskid Pivot Linkage Removal

SUBTASK 32-71-21-010-001

(1) Open this access door for the stabilizer trim:

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Tailskid Pivot Linkage Removal

SUBTASK 32-71-21-020-001

CAUTION

REMOVE THE TAILSKID CARTRIDGE ASSEMBLY FROM THE AIRCRAFT. DAMAGE TO THE ASSEMBLY CAN OCCUR.

(1) Disconnect the cartridge assembly [2] from the pivot linkage [1]:

(a) Remove the nut [13] and washer [12] from the bolt [10].

(b) Remove the bolt [10] that holds the female telescope rod side of the cartridge assembly [2] to the pivot linkage [1].

(c) Remove the washer [11] from the bolt [10].

SUBTASK 32-71-21-020-002

(2) Remove the pivot linkage [1]:

(a) Remove the nut [4], and the washer [3] from the shoulder bolt [19].

(b) Remove the shoulder bolt [19] to disconnect the pivot linkage [1] from the support fitting.

(c) Remove the nut [9], and the washer [8] from the shoulder bolt [15].

(d) Remove the shoulder bolt [15] from the fuse pin [16].

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(e) If necessary, remove the bushings [17] and washer [14] from the shoulder bolt [15].

NOTE: The bushings [17] are plastic spacers intended to center the bolt in the fuse pin.

(f) Remove the fuse pin [16] to disconnect the pivot linkage [1] from the fuse link [7].

(g) Remove the pivot linkage [1] from the airplane.

SUBTASK 32-71-21-020-003

(3) If necessary, remove the fuse link [7]:

(a) Remove the nut [6], and the washer [5] from the shoulder bolt [18].

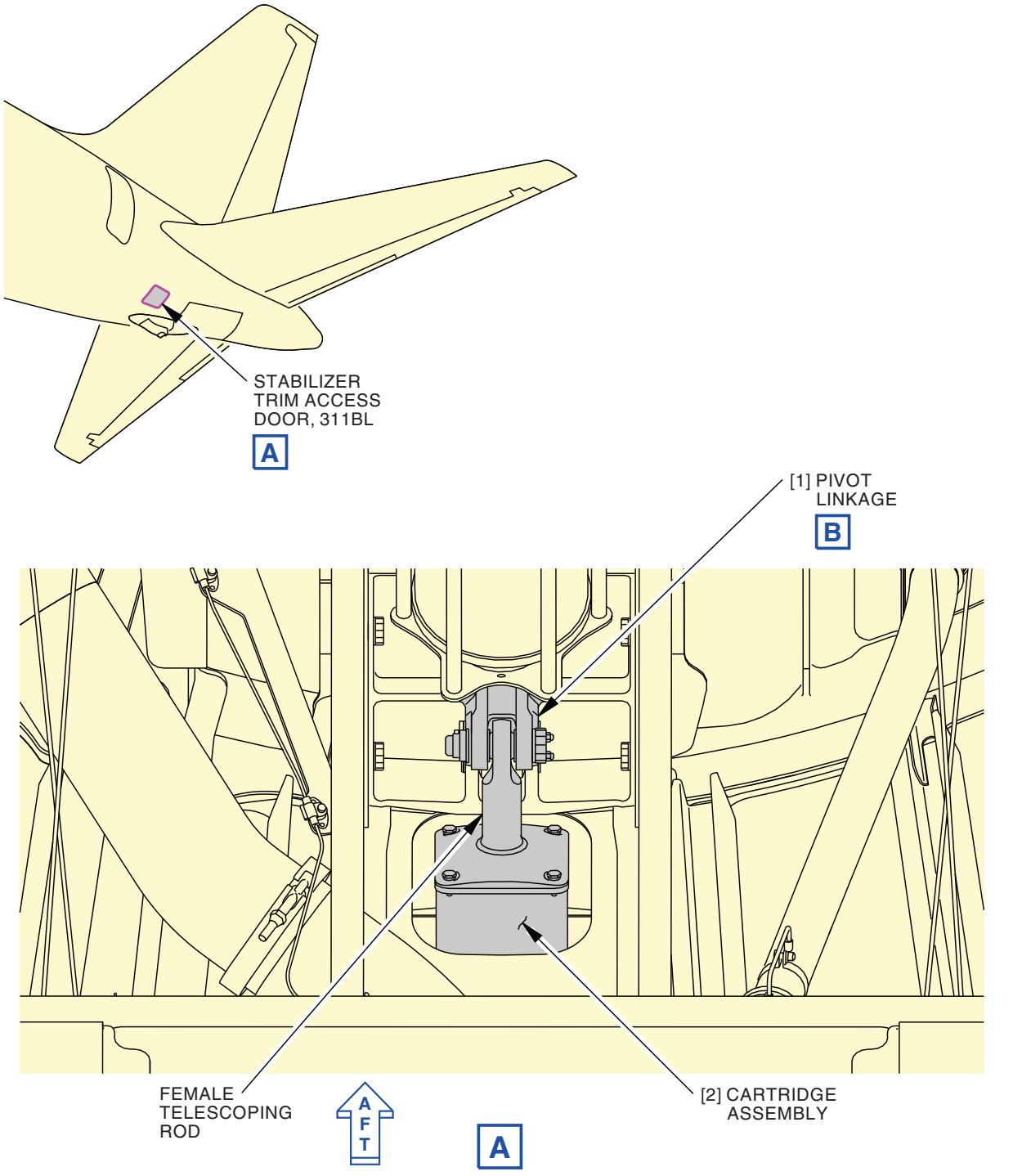
(b) Remove the shoulder bolt [18] from the shear pin [20].

(c) Remove the shear pin [20] to disconnect the fuse link [7] from the support fitting.

(d) Remove the fuse link [7] from the airplane.

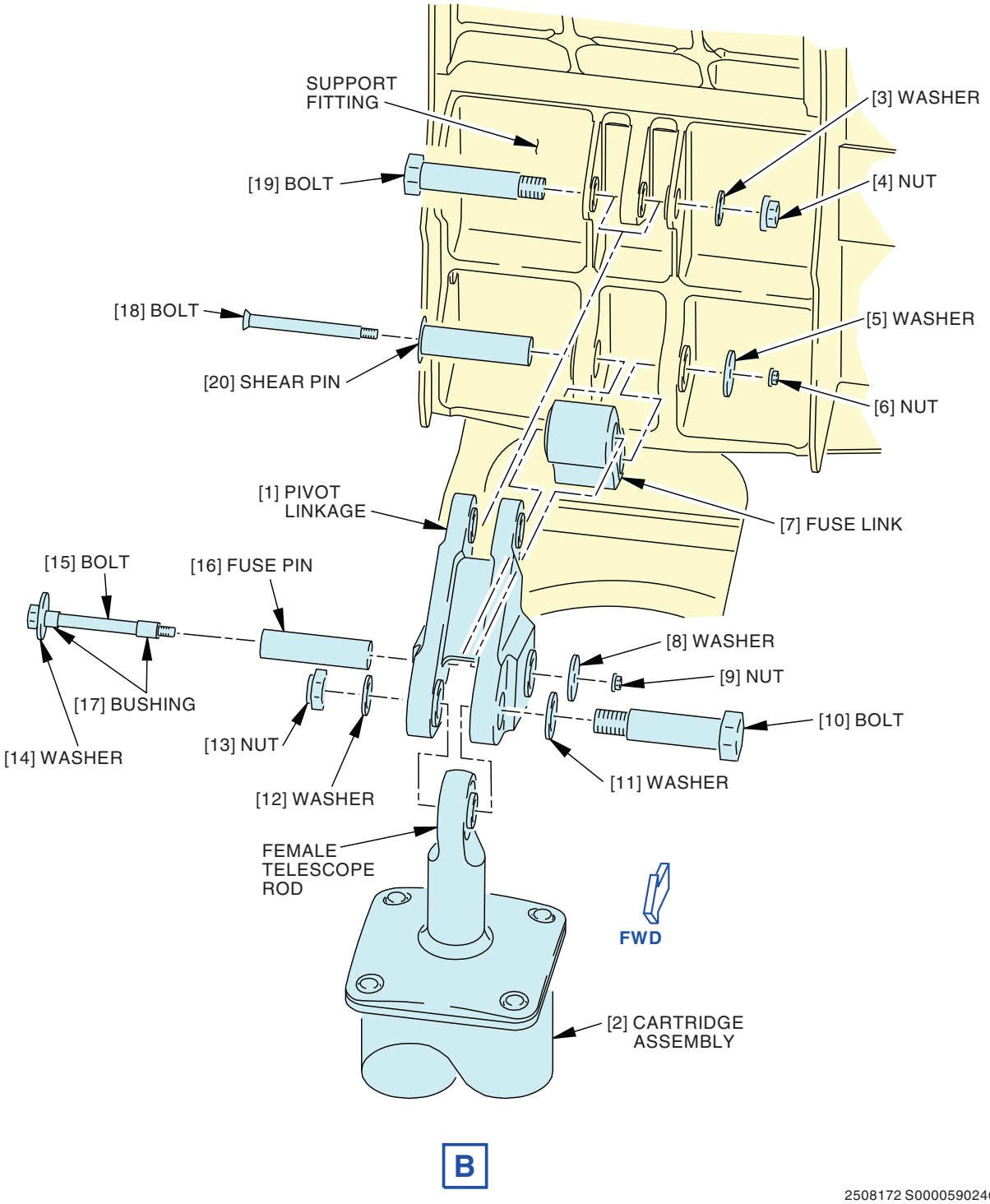
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tailskid Pivot Linkage Installation
Figure 401/32-71-21-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Tailskid Pivot Linkage Installation
Figure 401/32-71-21-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-71-21-400-801

3. Tailskid Pivot Linkage Installation

(Figure 401)

A. General

(1) This procedure has steps to install the Tailskid Pivot Linkage (referred to as pivot linkage [1]).

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Location Zones

Zone	Area
311	Area Aft of Pressure Bulkhead - Left
312	Area Aft of Pressure Bulkhead - Right

D. Access Panels

Number	Name/Location
311BL	Stabilizer Trim Access Door

E. Tailskid Pivot Linkage Installation

SUBTASK 32-71-21-420-001

(1) If necessary, install the fuse link [7]:

(a) Put the fuse link [7] in its position on the support fitting.

(b) Align the holes in the fuse link [7] to the holes in the support fitting.

(c) Put the shear pin [20] through the support fitting, and the fuse link [7].

(d) Put the shoulder bolt [18] through shear pin [20].

(e) Install the washer [5], and the nut [6] on the shoulder bolt [18].

(f) Tighten the nut [6] to 20.00 in-lb (2.26 N·m) - 25.00 in-lb (2.82 N·m).

SUBTASK 32-71-21-420-002

(2) Install the pivot linkage [1]:

(a) Put the pivot linkage [1] in its position on the support fitting.

(b) Align the upper hole in the pivot linkage [1] to the hole in the support fitting.

(c) Put the shoulder bolt [19] through support fitting, and the pivot linkage [1].

(d) Install the washer [3], and the nut [4] on the shoulder bolt [19].

(e) Tighten the nut [4] to 240.00 in-lb (27.12 N·m) - 300.00 in-lb (33.90 N·m).

(f) Align the lower hole in the pivot linkage [1] to the hole in the fuse link [7].

(g) Put the fuse pin [16] through the pivot linkage [1], and the fuse link [7].

1) Make sure not to damage the dry film lubrication of the surfaces of the fuse pin [16], shear pin [20], and the bolt [19].

2) Install the shear pin [20] last.

NOTE: If there is not sufficient clearance between the fuse link [7] and the support fitting, you can smooth the surface of the fuse link [7]. Make sure there is a finish of 32AA MICROINCHES on the surface.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(h) If removed from the shoulder bolt [15], install the washer [14] and bushing [17] on the shoulder bolt [15] use wet sealant, A00247.

(i) Apply grease, D00633 to all surfaces of the washer [14] and bushings [17] on the bolt [15], and pack the cavity between bushings.

(j) Put the shoulder bolt [15] through the fuse pin [16].

(k) Install the washer [8], and the nut [9] on the shoulder bolt [15].

(l) Tighten the nut [9].

SUBTASK 32-71-21-420-003

CAUTION

BE CAREFUL WHEN INSTALLING THE TAILSKID CARTRIDGE ASSEMBLY TO THE AIRCRAFT. DAMAGE TO THE ASSEMBLY CAN OCCUR.

(3) Connect the cartridge assembly [2] to the pivot linkage [1]:

(a) Align the female telescope rod side of the cartridge assembly [2] to the forward hole in the pivot linkage [1].

(b) Install the washer [11] on the bolt [10].

(c) Put the bolt [10] through the pivot linkage [1], and the female telescope rod side of the cartridge assembly [2].

(d) Install the washer [12], and the nut [13] on the bolt [10].

(e) Tighten the nut [13].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-71-21-410-001

(1) Close this access door for the stabilizer trim:

Number	Name/Location
311BL	Stabilizer Trim Access Door

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details